Section 6.1 / A quantitative conditional window

Suppose a genuine source-dependent RG construction supplied, on a controlled local region, an oscillation bound

$$\delta \le Cg(d)^{\eta} B^{r} (H/a')^{s}, \qquad \eta > 0, \quad 0 \le r, s < \infty, \tag{8}$$

together with correct observable transport, covariance comparison, background control and a separately bounded bad-field contribution. Let $t = \log(1/d)$ in fixed physical units and suppose $g(d) \asymp t^{-1/2}$. Choose

$$B = t^{p}, \qquad H = dt^{p}, \qquad a' = dt^{-p}, \qquad 0 < p < \frac{\eta}{2(r+2s+4)}.$$

Then

$$\begin{aligned} \delta (d/a')^{4} &\le Ct^{-\eta/2 + p(r+2s+4)} \longrightarrow 0, \\ (1+B^{2})(d/H)^{4} &= O(t^{-2p}), \\ (1+B^{4})(d/H)^{8} &= O(t^{-4p}), \end{aligned}$$

and the ultraviolet heat-kernel error also vanishes. Polynomially many cutoff tests of canonically normalized Gaussian variables with uniformly bounded variances have tails dominated by $e^{-cB^{2}}$. For example $p = \eta/[4(r+2s+4)]$ makes the density error $O(t^{-\eta/4})$.

The order of limits is essential. The scale a' is an intermediate spacing after integrating the original bare lattice $a_0 \ll a'$. It does not replace removal of a_0 at fixed physical separation. The hypothesis must hold uniformly after arbitrarily many preceding ultraviolet steps. A bound established only when $a_0 = a'$ would not construct a continuum theory.

Section 6.1 / A quantitative conditional window

A fourth-moment bound on an arbitrary coarse field is not a uniform action-oscillation bound over its unbounded values. Either (8) must hold uniformly for the reference being used, or one must control a good coarse event and its complement in the actual interacting law. Gaussian tail estimates cannot simply be assigned to that actual law.

Section 9.1 / Contact singularities do not disappear under centering

For a free four-dimensional scalar with covariance $G(x) = 1/(4\pi^2|x|^2)$, a cutoff Wick square satisfies

$$\operatorname{Var}(:\phi_a^2:(f)) = 2\iint f(x)f(y)G_a(x-y)^2\,\mathrm{d}x\,\mathrm{d}y.$$

For nonzero nonnegative f this diverges logarithmically as $a \to 0$. Maxwell action density has the degree-eight singularity (6), which produces a power divergence in the ordinary smeared self-variance. Subtracting the mean removes neither divergence. Wightman composites can nevertheless exist; their Euclidean continuations require the proper contact treatment or off-diagonal test domain. An ordinary positive random-distribution moment bound for this composite is an unjustified stronger target.

Section 9.2 / A non-Gaussian scalar observable need not signal interaction

For a Gaussian field X with covariance C_{ij} and $O_i =: X_i^2 :$, Wick contraction gives

$$\kappa_4(O_1, O_2, O_3, O_4) = 16\,(C_{12}C_{23}C_{34}C_{41} + C_{12}C_{24}C_{43}C_{31} + C_{13}C_{32}C_{24}C_{41})\,.$$

It is generically nonzero although the underlying field is free. Thus a connected four-point function of $\operatorname{tr} F^2$ is not by itself an interaction certificate.

Even a logarithmically weakened two-point singularity can be reproduced without Yang–Mills. Take a generalized-free scalar with nonnegative spectral density

$$\rho(s) = \frac{s^2}{[\log(e+s)]^2}.$$

Its separated Euclidean covariance is

$$C(r) = \frac{1}{2\pi^2 r^8} \int_0^\infty \frac{t^6 K_1(t)}{[\log(e + t^2/r^2)]^2}\,\mathrm{d}t.$$

The interval $t \in [1,2]$ gives a lower bound of order $r^{-8}/\log^2(1/r)$. For the upper bound, $t < \sqrt{r}$ contributes $O(r^3)$ inside the integral, while for $t \geq \sqrt{r}$ the denominator is at least $\log^2(1/r)$ and $t^6 K_1(t)$ is integrable. Hence

$$C(r) \asymp \frac{r^{-8}}{\log^2(1/r)}.$$

Polynomial spectral growth gives a positive tempered generalized-free Wightman field, and its higher truncated functions vanish. A scalar two-point asymptotic therefore cannot identify the gauge dynamics.

Section 9.3 / Fixed Euclidean heat flow is not a local reconstruction shortcut

Let a stationary real process have even covariance k that is C^2 near zero. Reflection positivity applied to the positive-time difference at $s, s+h$ requires

$$k(2s) - 2k(2s+h) + k(2s+2h) \geq 0.$$

Taking $h \to 0$ and then $s \to 0$ gives $k''(0) \geq 0$. Stationarity gives $k''(0) \leq 0$. Equality forces mean-square increments to be $o(h^2)$; dividing a fixed interval into n pieces and applying Cauchy–Schwarz shows that every increment has zero variance. Such a process is constant in mean square.

For a spatially smeared free field at positive four-dimensional heat-flow time τ, the temporal covariance is

$$k_\tau(s) = \int \frac{\mathrm{d}^3 p \, \mathrm{d}\omega}{(2\pi)^4} |\widehat{f}(p)|^2 \frac{e^{i\omega s} e^{-2\tau(\omega^2 + |p|^2)}}{\omega^2 + |p|^2 + m^2}.$$

For $f \neq 0$, $k_\tau''(0) < 0$, giving an explicit negative reflected difference norm. Fixed positive flow time does not directly reconstruct the desired local field. Removing that smoothing remains part of the construction.

Section 12.2 / An all-degree reserve for primitive Wilson marks

Give $\mathfrak{A}_N$ the coefficient norm $\|\sum_A c_A z_A\|_r = \sum_A r_A |c_A|$. For $O_i = \sum_p f_i(p) a_p$,

$$\left\| e^{\sum_i z_i O_i} \right\|_r = \prod_i (1 + r_i |O_i|) \le \exp\left[\sum_p \left(\sum_i r_i |f_i(p)| \right) a_p \right].$$

With disjoint supports the pointwise load is at most $\rho = \max_i r_i \, \|f_i\|_\infty$, independent of N. Hence a fixed real action reserve pays for every finite source degree at once.

The same observation applies to stored history sites when the primitive action has a bound

$$\sum_{p \text{ stored}} a_p \le C N_{\text{stored}}.$$

For example, on a regular layer with mesh η, the bounds $\|1 - U_p\| \le C\eta^2$ and $a_p \le C\eta^4$ compensate the $O(\eta^{-4})$ fine plaquettes per stored scale-site; on a bare layer use $a_p \le 2$. Thus the primitive source costs at most $e^{C\rho N_{\text{stored}}}$. An existing estimate with a spare weight $e^{c\sum_j N_j \log x_j}$ can absorb this cost for large enough x_j. This implication proves absorption of the primitive marks; it does not establish the requisite ledger or the locality of every generated source response.

The source here shifts the inverse coupling additively. Marking βa_p instead introduces an extra factor β and requires a corresponding real action reserve. Gaussian normalization constants cause no new difficulty at first order: a term $-(D/2)\log x$ has source derivative $D/(2x)$. A general cluster vacuum derivative still requires a marked local integration estimate.

Section 13.1 / A depth-independent extension on a layered current domain

The independent-current formalism can be used through an explicit extension of the primitive source current. Let the physical fine spacing be η and the assigned local layer scale be ξ_b. In the physical pairing $\eta^4 \sum_b$,

$$J_{\mathrm{phys}}(U;b) = \eta^{-3} \sum_{p \ni b} D_b a_p(U).$$

Suppose the admitted field domain has

$$|D_b a_p(U)| \le C\alpha\eta^2 \xi_p^{-2}, \qquad \xi_p \ge c_* \xi_b.$$

The curvature bound $|1 - U_p| \le \alpha\eta^2\xi_p^{-2}$ implies the first inequality. It also holds in a smaller complex first-derivative field domain. For $U_b = e^{i\eta A'_b} U_b^{\mathrm{real}}$ require both $|A'_b| \le \alpha_1 \xi_b^{-1}$ and $|\nabla_{U^{\mathrm{real}}} A'| \le \alpha_1 \xi_b^{-2}$. These imply curvature bounded by $C(\alpha + \alpha_1 + \alpha_1^2)\eta^2\xi_p^{-2}$, while the links and inverse links have norm at most $e^{C\alpha_1}$. Thus the finite transports cost a depth-independent constant, with α enlarged by the indicated domain constants. A bound on A' alone would not control rapidly oscillating curvature.

The scale comparison follows for the buffered interior layers used in the construction [20]. If

$$\mathrm{dist}(\Omega_j^c, \Omega_{j+1}) \ge \kappa \xi_j, \qquad \xi_j = L^j \eta,$$

with $\kappa L > c_{\mathrm{geom}}$, two anchors within $c_{\mathrm{geom}}\eta$ cannot have layer indices differing by two: one would lie in Ω_{j+1}^c and the other in Ω_{j+2}, contradicting the buffer. Thus $c_* = L^{-1}$ works for incident interior bonds and plaquettes. Outer-boundary terms retain their boundary assignment.

Section 13.1 / A depth-independent extension on a layered current domain

Set

$$E_f^{\mathrm{phys}}(U;b) = \eta^{-3} \sum_{p \ni b} (f_p - f_b) D_b a_p(U).$$

Using $|f_p - f_b| \le C\eta \,\mathrm{Lip}(f)$, every power of the fine spacing cancels:

$$\xi_b^3 |E_f^{\mathrm{phys}}(b)| \le C\alpha \,\mathrm{Lip}(f) \xi_b^3 \sum_{p \ni b} \xi_p^{-2} \le C\alpha c_*^{-2} \xi_b \,\mathrm{Lip}(f). \tag{30}$$

For an independent pair (U, J) with $|J|_{(-3)} = \sup_b \xi_b^3 |J_b| \le \alpha$, define

$$J_f^{\mathrm{ext}}(U, J) = fJ + E_f^{\mathrm{phys}}(U).$$

This is local, analytic, gauge covariant and affine in J. On the graph $J = J_{\mathrm{phys}}(U)$ it equals the physical weighted Wilson current. On a region X with local scale at most ξ_X,

$$|J_f^{\mathrm{ext}} - f_* J|_{(-3),X} \le C\alpha \left(\sup_{X^+} |f - f_*| + \xi_X \,\mathrm{Lip}(f) \right).$$

This supplies a legitimate primitive direction in the layered pair domain. It does not identify the derivative of an already integrated activity with another Wilson current; the normalized covariance term in (43) still has to be controlled.

Section 14.2 / The source remainder fits a fixed local allowance

Assign plaquettes to fixed-size step blocks with finite multiplicity. For one block Y put

$$F_{Y,w}(B) = \sum_{p \text{ assigned to } Y} w(p) a_p(e^{\Phi(B)}U).$$

If $\sup_p \|w(p)\|_r \le \rho$, the local chart and analytic plaquette action give $\sup_{\|B\|_\infty \le r_L} \|F_{Y,w}(B)\|_r \le C_L\rho$. There is no factor equal to the number of source labels. With $B = gb$, $\|b\|_\infty \le P_0$ and $gP_0 < r_L/4$, Cauchy's formula along a line gives

$$F_{Y,w}(gb) = F_{Y,w}(0) + gDF_{Y,w}(0)[b] + \tfrac{1}{2}g^2D^2F_{Y,w}(0)[b,b] + \mathcal{R}_{Y,w}(b), \tag{32}$$

$$\|\mathcal{R}_{Y,w}(b)\|_r \le C_L\rho g^3 P_0^3, \tag{33}$$

$$\|\mathcal{R}_{Y,w}(b)\|_r \le C_L\rho g^3 P_0 \sum_{b' \subset Y^+} |b(b')|^2. \tag{34}$$

The last estimate uses the bounded number of coordinates in the determining block and the third-derivative bound. It also holds on a slightly larger complex box with fixed room. A growth norm based on $1 + \sum |b|^2$ therefore receives a source remainder of size $C_L\rho g^3 P_0$. This tends to zero when P_0 is any fixed power of $\log g^{-2}$.

The quadratic source size is $O(\rho g^2)$, with its true constrained Hessian. The linear size is controlled by the current commutator; at a full effective saddle it also contains the old-force term from the preceding section. If a clean unmarked expansion has an absolute numerical cluster majorant with spare potential room, these estimates put the primitive source in that room. The same proof works in the finite-source coefficient algebra because its norm is submultiplicative. A merely bounded scalar holomorphic map, without its absolute majorant, would not justify a source-degree-independent norm bound.

Section 16 / A connected-history criterion and the remaining scale estimate

The missing connected estimate can be derived from a precise local coefficient ledger, if an exact polymer representation is obtained. This is weaker than postulating the final correlation bound, and does not require a complex source neighborhood.

Theorem 16.1 (Finite-source connected polymer bound)**.** *Let polymers P be connected subsets of a finite lattice graph, with overlap as incompatibility, and suppose the partition function has the exact representation*

$$Z(z) = \sum_{\Gamma \text{ compatible}} \prod_{P \in \Gamma} a_P(z), \qquad a_P(z) = \sum_A a_{P,A} z_A \in \mathfrak{A}_N.$$

Let $\widehat{a}_P = \sum_A r_A |a_{P,A}|$ and assume, for $a_*, \mu > 0$,

$$\sup_x \sum_{P \ni x} e^{(a_* + \mu)|P|} \widehat{a}_P \le a_*. \tag{45}$$

Suppose $a_{P,A} = 0$ unless P meets S_i for every $i \in A$. If $T(S_A)$ is the minimum number of edges of a connected subgraph meeting every indicated support, then for nonempty A,

$$r_A |[z_A] \log Z(z)| \le a_* \min_{i \in A} |S_i| \, e^{-\mu T(S_A)}. \tag{46}$$

Proof. For a polymer tuple, let $\phi^T(P_1,\dots,P_k)$ be the alternating sum over connected spanning subgraphs of its incompatibility graph. Deletion–contraction proves

$$|\phi^T(P_1,\dots,P_k)| \le \sum_{T\ \text{tree}} \prod_{uv\in E(T)} \mathbf{1}_{P_u\cap P_v\neq\varnothing}.$$

Indeed, multiplying the alternating connected-subgraph sum by $(-1)^{k-1}$ turns its deletion–contraction recurrence into an addition recurrence. Induction bounds it by the spanning-tree count: the single-vertex value is one, disconnected graphs contribute zero, and a loop cancels the alternating sum while leaving a nonnegative tree count.

Put $b_P = e^{\mu|P|}\widehat{a}_P$ and $A(P) = a_*|P|$. The site ledger implies

$$\sum_{Q:Q\cap P\neq\varnothing} b_Q e^{A(Q)} \le A(P).$$

Rooted-tree sums are bounded by $e^{A(P)}$. To see this directly, iterate

$$T_0(P) = 1, \qquad T_{h+1}(P) = \exp\left(\sum_{Q\cap P\neq\varnothing} b_Q T_h(Q)\right).$$

The exponential enumerates the unordered children of the root. Induction gives $T_h(P) \le e^{A(P)}$, and monotone convergence covers every finite tree. Combining this with the tree bound gives the absolute rooted cluster estimate.

The connected expansion is consequently absolutely convergent in each finite-source coefficient:

$$\log Z(z) = \sum_{k\ge1} \frac{1}{k!} \sum_{P_1,\dots,P_k} \phi^T(P_1,\dots,P_k) \prod_{j=1}^{k} a_{P_j}(z).$$

For an identity proof, multiply all activities by a scalar τ. Near zero this is the finite exponential formula; the absolute bounds continue it to $|\tau| \le 1$, proving the identity at $\tau = 1$ and unmarked nonvanishing there.

A contributing connected union meets all the source supports, so $T(S_A) \le \sum_j |P_j|$. The weights $e^{\mu|P_j|}$ therefore pay the displayed separation factor. Choose one label $i \in A$. It occurs at exactly one vertex of each nonzero square-free coefficient assignment. Root there; the k choices replace $1/k!$ by $1/(k-1)!$. The rooted bound leaves at most

$$e^{-\mu T(S_A)} \sum_{P\cap S_i\neq\varnothing} e^{(a_*+\mu)|P|} \sum_{B\ni i} r_B |a_{P,B}| \le a_*|S_i| e^{-\mu T(S_A)}.$$

Minimize over i. □

If the exact activities arise from component integrals at a retained field V, a sufficient input is

$$L_{P,A} = \sup_V \sum_h e^{cv_h} \int |[z_A] I_h(\xi; V, z)| \, \mathrm{d}\mu_h(\xi; V),$$

with (45) for $\sum_A r_A L_{P,A}$. The supremum stays outside the integral. Compatible scalar activities may share the same fixed retained V; their product can still be bounded by their individual uniform bounds. This observation requires an exact compatible-product identity after the relevant internal variables have been integrated.

Section 16 / A connected-history criterion and the remaining scale estimate

For the unresolved raw history terms, reconstructed backgrounds couple different components, and an upper bound on whole labels is not an equality expressing a local polymer gas. The later continuation, Section 20, restores the original factorization at a completed large-field step after its cross-component dependence has been localized. The remaining task is the marked local activities, their source-support rule, and the coefficient ledger on the actual domains. The primitive action reserve helps pay these coefficients; it does not by itself establish the localization.

The distinction can be checked in a small counterexample. Let a hidden variable choose one of two arbitrarily small probabilities p_+, p_-. Conditional on it, place independent Bernoulli marks at every site. Then

$$\Pr(\text{all sites in } S \text{ marked}) \le p_{\max}^{|S|}, \qquad \operatorname{Cov}(I_x, I_y) = \operatorname{Var}(p_H) > 0$$

at every separation. Thus excellent marked-animal upper bounds alone do not prove connected decay. Keeping the hidden variable as a coarse variable moves, rather than removes, its contribution to the terminal terms in (24).

For geometric physical scales ℓ, a sufficient bound for the actual exact coefficients Γ_ℓ is

$$|\mathbb{E}\Gamma_\ell(\{1,\dots,n\};f_1,\dots,f_n)| \le C^n (n!)^p g_\ell^{2n} \int \prod_i |f_i(x_i)|\, \ell^{-4n} \times e^{-c\ \mathrm{tree}(x_1,\dots,x_n)/\ell}\, \mathrm{d}x_1 \cdots \mathrm{d}x_n. \tag{47}$$

The constants must be uniform in bare cutoff and volume; terminal coarse-field terms require their own bound. Equation (47) has not been proved for the full Yang–Mills construction in this note.

Its sufficiency for separated estimates can be checked explicitly. If every pair of supports has separation at least d, the spanning-tree length is at least $(n-1)d$. Then

$$\sum_{\ell\ \text{geometric}} \ell^{-4n} e^{-c(n-1)d/\ell} \le C_1^n d^{-4n}, \qquad n \ge 2.$$

For $\ell \ge d$, sum a geometric series. For $\ell = dL^{-r} < d$, the factor is $d^{-4n}\exp(4nr\log L - c(n-1)L^r)$; its superexponential tail and bounded maximum give C_1^n. Thus a proved coefficient ledger with the physical $g_\ell^{2n}\ell^{-4n}$ readout would produce the earlier factorial flat-test bounds.

At order two the corresponding kernel is

$$C g_\ell^4 \ell^{-8} e^{-c|x-y|/\ell}.$$

For bounded tests of size and separation d, the integration volume contributes $O(d^8)$. Scales below d are exponentially suppressed; scales above d are suppressed by $(d/\ell)^8$. Along a controlled logarithmic coupling trajectory, split the latter scales when g_ℓ^{-2} reaches half of g_d^{-2}. Before that point $g_\ell^4 \le 4g_d^4$; after it the geometric suppression beats the possible polynomial growth. The summed scale contribution is $O(g_d^4)$. For the same conclusion about the total, require a shrinking-support bound on the terminal term: a uniformly bounded fixed-scale two-point kernel gives $O(d^8) = o(g_d^4)$. A terminal form merely bounded in the test sup norm would not suffice.

The new proofs therefore establish actual bare Wilson control, primitive source admissibility, a clean-step source expansion, exact multi-insertion transport, and a connected-polymer implication. The unresolved work is transporting the already formed marked activities and their physical field/current readout through the full coupled history construction with a uniform local coefficient ledger. The scale sum itself is no longer the unknown step.

Section 17.1 / The insertion is a temperature tangent

For normalized hidden Haar measure, write

$$Z_\beta(y) = \int e^{-\beta S(x,y)}\,\mathrm{d}\nu(x), \qquad F_\beta(y) = -\beta^{-1}\log Z_\beta(y), \qquad M_\beta(y) = \mathbb{E}_\beta[S \mid y].$$

If $D_\beta(y)$ denotes the relative entropy of the conditional hidden law with respect to ν, then exactly

$$M_\beta = \partial_\beta(\beta F_\beta) = F_\beta - \beta^{-1} D_\beta, \qquad D_y D_\beta = \beta^2 \operatorname{Cov}_\beta(S, D_y S \mid y). \tag{48}$$

The entropy sign follows by inserting the conditional density $e^{-\beta S}/Z_\beta$ in its definition. Differentiating the normalized expectation proves the second identity. In particular, preservation of an upper Hessian bound for F_β cannot be differentiated into a bound for M_β.

If the effective density at level j has

$$-\log \rho_j = c_j(\beta_0) + x_j(\beta_0) A_j + R_j(\beta_0),$$

its transported constant insertion, up to its overall centering constant, is

$$x_j'(\beta_0) A_j + \partial_{\beta_0} R_j. \tag{49}$$

All old remainders have to be differentiated. The original bare parameter β_0 cannot simply be relabelled as the current running coupling.

Section 17.2 / A stronger estimate on an actual regular constraint chart

Proposition 17.1 (Regular-block normalized source expansion)**.** *Fix a finite gauge-fixed Wilson block, or a finite-dimensional regular nonlinear averaging fiber. On a compact set of exterior data y, assume a common chart $a \in B_r \subset \mathbb{R}^d$ centered at a smooth constrained minimum, with positive analytic density $J_y(a)\,\mathrm{d}a$, independent of β. Suppose*

$$S(a,y) = S_*(y) + \tfrac{1}{2} a^{\mathsf{T}} H_y a + O(|a|^3), \qquad D_a^2 S \geq m^2 I,$$

on that chart, with uniform finite derivative bounds, including one exterior derivative. Let Z_{reg} be its restricted integral. For sufficiently large real β, $Z_{\mathrm{reg}}(z,y)$ is nonzero on $|z-\beta| \leq \beta/2$, and

$$\log Z_{\mathrm{reg}}(z,y) = -zS_*(y) - \tfrac{d}{2}\log z + \log a_0(y) + r(z,y), \qquad \sup_{|z-\beta|\leq\beta/2} \|r(z,\cdot)\|_{C^1} \leq C\beta^{-1}, \tag{50}$$

where $a_0(y) = (2\pi)^{d/2} J_y(0)(\det H_y)^{-1/2}$. *Consequently*

$$\mathbb{E}_{\mathrm{reg}}[S \mid y] = S_*(y) + \frac{d}{2\beta} + O_{C^1}(\beta^{-2}), \tag{51}$$

$$\left\| \kappa_n(S \mid y) - \frac{d}{2}(n-1)!\beta^{-n} \right\|_{C^1} \le C\, 4^n n!\, \beta^{-n-1}, \qquad n \ge 2. \tag{52}$$

Constants may depend on this fixed block and chart.

Proof. Set $g = \beta^{-1/2}$, $a = gb$ and $t = z/\beta$. The exponent is $tQ_y(b) + tgT_{3,y}(b) + O(g^2\,|b|^4)$, with $Q_y = \frac{1}{2} b^{\mathsf{T}} H_y b$ and T_3 cubic. The density expands as

$$J_y(gb) = J_y(0)\{1 + g j_{1,y}(b) + O(g^2\,|b|^2)\},$$

where j_1 is linear. Both first corrections are odd. They integrate to zero against the even Gaussian e^{-tQ_y} on the symmetric chart. Since $\Re t \ge 1/2$, the Hessian lower bound supplies uniform Gaussian domination. A Taylor expansion with integral remainder, and a Gaussian estimate for the chart boundary, give

$$e^{zS_*} Z_{\mathrm{reg}}(z, y) = z^{-d/2} a_0(y)\{1 + O_{C^1}(\beta^{-1})\}.$$

One exterior derivative has the same bound: the differentiated quadratic term is even, while the first differentiated corrections remain odd. The boundary error is a polynomial in β times $e^{-c\beta r^2}$. For fixed dimension the relative error is less than $1/2$ at large β, proving nonvanishing and a coherent logarithm. Finally $\kappa_n = (-1)^n \partial_\beta^n \log Z_{\mathrm{reg}}$ and Cauchy's formula on a smaller temperature disc give the two conclusions. □

These assumptions hold on a fixed sufficiently small regular Wilson box after removal of gauge zero modes. At the identity the curl quadratic form has only gauge directions in its kernel on a contractible box; tree gauge removes them. Finite-dimensional positivity, the implicit-function theorem, and the regular coarea chart then give the stated neighborhood. This argument does not make its constants uniform over arbitrary layered fibers.

For an elementary contractible finite box in full tree gauge, there is a useful stronger corollary. At identity exterior data, the identity is the only zero-action configuration: flatness on every plaquette and trivial tree links force every other link to be trivial. Compactness gives a positive action gap outside the regular chart. That gap persists for sufficiently small retained non-tree link data, while the unique inner minimum persists by the implicit-function theorem. The complement of the chart is therefore exponentially small relative to its saddle integral, including one exterior derivative. In this specific product-Haar setting, (50)–(52) hold for the *full* fixed-block integral. This does not establish a global coarea or gap statement for general nonlinear averaging fibers.

The determinant $\det H_y$ can depend on the exterior field. Nevertheless it is independent of z in (50), and hence drops out of every constant-source temperature derivative. The universal term $d/(2\beta)$ is field independent. This is the cancellation that is lost by estimating the entire hidden entropy in absolute value.

An exact real identity makes that cancellation visible. In canonical coordinates write $\beta(S - S_*) = Q + R$ and let $\mathcal{B}$ be the normalized outward flux of $be^{-Q-R} J_y(gb)$ at the chart boundary. Integration by parts gives

$$\mathbb{E}_{\mathrm{reg}} S = S_* + \frac{d}{2\beta} + \frac{1}{\beta}\mathbb{E}_{\mathrm{reg}}\left[R - \tfrac{1}{2} b \cdot \nabla R + \tfrac{1}{2} b \cdot \nabla \log J_y(gb)\right] - \frac{\mathcal{B}}{2\beta}.$$

The order-g term in brackets is $(j_1 - T_3)/2$ and is odd. Interpolating from the even Gaussian makes its expectation $O(g^2)$, giving (51) directly.

Section 17.2 / A stronger estimate on an actual regular constraint chart

On a uniformly massive clean fiber with local moment bounds, the covariance localization theorem above permits the same Euler correction to be estimated *per rooted local term.* If $R = \sum_v r_v$, $r_v = gT_{3,v} + O(g^2)$, then

$$|\mathbb{E}T_{3,v}| \le \int_0^1 \sum_w |\mathrm{Cov}_t(T_{3,v}, r_w)| \, \mathrm{d}t \le Cg \sum_w e^{-\alpha d(v,w)} \le C'g.$$

For an exterior derivative rooted at u, the localized covariance identity gives a field-response error $O(\beta^{-3/2})$ without a total hidden-volume factor. Its hypotheses include a common massive interpolation, local moments, and localized exterior derivatives. A volume-uniform improvement to β^{-2} additionally needs a rooted three-cumulant estimate. Both $\beta^{-3/2}$ and β^{-2} are summable along a logarithmic ultraviolet coupling trajectory; summability alone does not prove preservation of this class.

The original completed **R** operation already contains a useful factorization. Bałaban localizes cross-component dependence into $V(Y, U_k)$, then factors the component integrations before equation (1.71), and uses a Mayer expansion in (1.90)–(1.91) [8]. This supplies a route that does not require a supremum over all reconstructed backgrounds inside every raw stored-variable integral.

At a fixed retained background and fixed still-live exterior variables B, the relevant form is

$$Z(B, z) = \sum_{I \text{ admissible}} \int \prod_{i \in I} K_i(\xi_i; B, z)\, \mathrm{d}\mu_i(\xi_i) \exp\left(\sum_Y V_Y(\xi_{I \cap Y}; B, z) \right). \tag{58}$$

The component measures are fixed positive measures; any complex parameter dependence is included in the kernels. Core supports are disjoint, admissibility factors over disconnected supports, and each bridge includes in its assigned support all variables on which it depends. Live components remain in B or in the local operator state. These are exact representation conditions, not consequences of a probability bound on marked sets.

Expand $e^{\sum_Y V_Y} = \prod_Y (1 + f_Y)$, $f_Y = e^{V_Y} - 1$, in the finite source algebra. Join cores and selected bridges whenever their supports overlap or a bridge depends on a core variable. Fubini factors each term over connected components. Conversely their compatible union reconstructs the original term, so one obtains an exact polymer gas. Its activities satisfy

$$\|a_P(B, z)\|_r \leq \sum_{\substack{(I, \mathcal{D}) \text{ connected} \\ \text{support}(I, \mathcal{D}) = P}} \prod_{i \in I} M_i \prod_{Y \in \mathcal{D}} b_Y,$$

Section 20 / The localized large-field representation and finite sources

where $M_i = \sup_B \int \|K_i(\xi_i; B, z)\|_r \, \mathrm{d}\mu_i$ and $b_Y = \sup_{B,\xi} \left\| e^{V_Y} - 1 \right\|_r$. The supremum stays outside the now genuinely separate component integral. Source support passes termwise to the resulting polymers, so the earlier marked polymer theorem applies if this numerical majorant meets its criterion.

The argument does not turn a raw integral $\int \prod_i K_i(\xi_i; p_i(\xi)) \prod_i \mathrm{d}\mu_i$ into a product. The background substitutions must first have been localized as in (58).

Section 25.1 / The spatial marginal map is nonlinear already at tree level

Consider two scalar bonds with endpoints 0 and B, hidden midpoint u, and positive stiffnesses $lambda_1$, $lambda_2$. The action is

$$S_\lambda(u,B) = \tfrac{1}{2}\lambda_1 u^2 + \tfrac{1}{2}\lambda_2 (B-u)^2.$$

Eliminating u gives

$$\min_u S_\lambda(u,B) = \frac{1}{2}\frac{\lambda_1\lambda_2}{\lambda_1+\lambda_2}B^2 = \frac{1}{4}C(\lambda_1,\lambda_2)B^2, \qquad C(\lambda_1,\lambda_2) = \frac{2\lambda_1\lambda_2}{\lambda_1+\lambda_2}.$$

The normalization makes $C(x,x) = x$. On $\lambda = (x+t, x-t)$,

$$C(x+t,x-t) = x - \frac{t^2}{x}, \qquad \frac{d^2}{dt^2}C(x+t,x-t)\big|_{t=0} = -\frac{2}{x}. \tag{4.1}$$

This is exact. A desired bound of order $x^{-1-\epsilon}$, $\epsilon > 0$, for all second local-coupling derivatives cannot hold even in this Gaussian example. Likewise $C(\lambda) = R\lambda + E(\lambda)$, R the arithmetic average, cannot

Section 25.1 / The spatial marginal map is nonlinear already at tree level

have $DE = O(x^{-1-\epsilon})$ throughout a fixed relative spatial product box. At $|t| = \rho x$ its remainder is $-\rho^2 x$.

The conditional theorem remains a valid implication of its hypotheses. The example shows that those hypotheses cannot simply be identified with the general spatial Wilson coupling map. The exact homogeneous classical map must be retained, or its source-gradient operators must be included in an enlarged state. Constant-coupling running is unaffected: $C(x + c, x + c) = x + c$.

Section 25.9 / A sharp nonexpansion theorem for the complex Schur energy ball

The bound in Section 8 can be sharpened substantially. The stronger form also applies when a running scalar counterterm is subtracted between coarse steps, where ordinary Schur associativity alone does not apply.

25.9.1 9.1 The contractive block identity

Let $\mathcal{H} = \mathcal{C} \oplus \mathcal{V}$ be an orthogonal decomposition, and let

$$T = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \qquad \|T\| \le 1.$$

For $|z| < 1$ put

$$\mathcal{F}_T(z) = A + zB(I - zD)^{-1}C. \tag{4.22}$$

Then $\|\mathcal{F}_T(z)\| \le 1$. This assertion needs no self-adjointness or positivity of T. Indeed, for c in the coarse space set $v = z(I - zD)^{-1}Cc$. Then

$$T(c, v) = (\mathcal{F}_T(z)c, v/z).$$

Contraction of T gives

$$\|\mathcal{F}_T(z)c\|^2 + (|z|^{-2} - 1)\|v\|^2 \le \|c\|^2. \tag{4.23}$$

The case z=0 follows directly from $\|A\| \le 1$.

For $x > r \ge 0$ and $H = xI + rT$, the hidden block $xI + rD$ is invertible and its Schur complement is

$$S(H) = xI + r\left[A - (r/x)B(I + (r/x)D)^{-1}C\right] = xI + r\mathcal{F}_T(-r/x).$$

Consequently

$$\boxed{\|S(H) - xI\| \le r.} \tag{4.24}$$

This is sharper than the elementary r/(1-r/x) estimate obtained by bounding the three off-diagonal factors separately. The improvement uses their membership in one contractive block matrix.

25.9.2 9.2 Arbitrary real coarse maps and physical energy weights

Let H_0 be positive definite, Q a real surjective linear map, and $S_0 = (QH_0^{-1}Q^*)^{-1}$. The same real maps are used under complex continuation. Suppose

$$H = xH_0 + E, \qquad \|H_0^{-1/2} E H_0^{-1/2}\| \le r < x.$$

Then H is accretive, its constrained Gaussian Schur precision exists, and

$$\boxed{\left\| S_0^{-1/2}[S_Q(H) - xS_0]S_0^{-1/2} \right\| \le r.} \tag{4.25}$$

To prove this, set $C_0 = QH_0^{-1}Q^*$ and $J = H_0^{-1/2}Q^*C_0^{-1/2}$. Then J is an isometry. Whiten the fine variable by H_0 and decompose it orthogonally into $\operatorname{ran} J$ and $\ker J^*$. A prescribed coarse variable b has whitened coordinate $c = C_0^{-1/2}b$. The normalized coarse precision is exactly the Schur complement of $xI + H_0^{-1/2}EH_0^{-1/2}$ in this decomposition. Apply (4.24).

The proof holds for arbitrary finite dimensions with constant one. It also holds on a common nullspace quotient when H_0 and every admitted E annihilate that nullspace, and Q induces a surjective map between the quotients. This is the appropriate flat gauge-theory interpretation after gradients and harmonic zero modes have been removed. It is not an assertion that a gauge Hessian is positive on its unquotiented space.

For a scalar or matrix-valued curvature coefficient h with $\|h\|_\infty \le r$, $E = D^*\Lambda_h D$ and $H_0 = D^*D$ meet the hypothesis: $DH_0^{-1/2}$ is an isometry on the positive subspace, so the relative energy norm of E is at most r. Complex coefficients are allowed. In particular no real-part comparison alone is substituted for a complex operator-norm bound.

25.9.3 9.3 A running scalar subtraction does not enlarge the source ball

Let $Q_1, \dots, Q_n$ be fixed real coarse maps and define their reference precisions recursively by

$$H_j = S_{Q_j}(H_{j-1}), \qquad j = 1, \dots, n.$$

Let $x_0, \dots, x_n$ be positive real numbers, with $x_j = x_{j-1} - b_j$; the individual b_j may have either sign. Start from $\widehat{H}_0 = x_0H_0 + E_0$ and form

$$\widehat{H}_j = S_{Q_j}(\widehat{H}_{j-1}) - b_jH_j. \tag{4.26}$$

If $\|H_0^{-1/2}E_0H_0^{-1/2}\| \le r < \min_j x_j$, then every step is well defined and

$$\|H_j^{-1/2}(\widehat{H}_j - x_jH_j)H_j^{-1/2}\| \le r \quad (0 \le j \le n). \tag{4.27}$$

At each step (4.25) gives a perturbation bounded by r relative to H_j around $x_{j-1}H_j$. Subtracting b_jH_j changes the center to x_jH_j and leaves that same perturbation. This proves the induction. There is no product of n constants and no sum of r^2/x_j losses.

More generally, if one adds an extra quantum/source correction W_j with $\|H_j^{-1/2}W_jH_j^{-1/2}\| \le \epsilon_j$, the same proof gives a radius $r_j \le r_0 + \sum_{l \le j} \epsilon_l$, as long as the running center stays above this radius. If the verified correction instead obeys $\epsilon_j \le q_j r_{j-1}$, then $r_j \le r_0 \prod_{l \le j}(1 + q_l)$. Thus a summable sequence of actual source-dependent quantum corrections suffices. Such a sequence must be proved from the RG; it is not part of (4.25).

25.9.4 9.4 Uniform analyticity and source jets

If E_0 is a holomorphic family on a source polydisc with the preceding bound r, (4.26) stays holomorphic there, since all hidden precisions remain accretive. The output normalized source precision is bounded by the same r on the whole polydisc. Mixed Cauchy estimates therefore give depth-independent jets on any strictly smaller polydisc.

For example, if the initial source is linear, $E_0(z) = D^*\Lambda_{\sum z_i f_i} D$, choose positive radii t_i with $\sum t_i \|f_i\|_\infty \le r$. For distinct variables,

$$\|H_n^{-1/2}\partial_{z_1}\cdots\partial_{z_m}\widehat{H}_n(0)H_n^{-1/2}\| \le r\prod_i t_i^{-1}. \tag{4.28}$$

Taking $t_i = r/(m\|f_i\|_\infty)$, and using $m^m \le e^m m!$, gives the usual factorial bound $e^m m! r^{1-m}\prod_i \|f_i\|_\infty$, uniformly in n. Repeated-variable derivatives have their usual multi-index factorials. These conclusions concern the actual complex function, not a termwise absolute sum of an unresummed perturbation expansion.

25.9.5 9.5 Scope for the interacting construction

The flat field Hessian of a constrained classical Wilson critical value is precisely a Schur complement of its flat quadratic action. Consequently (4.25) is relevant to its whole spatial, potentially anisotropic quadratic source sector. It does not require scalar coefficients after coarse-graining; the retained perturbation may be a nonlocal quadratic operator.

Three further assertions do not follow from this theorem:

1. a pointwise or Hölder bound on the minimizing field;
2. a rooted spatial decay bound on the absolute coefficients of all source derivatives;
3. a summable bound on the additional source-dependent quantum corrections W_j in the actual Wilson/history construction.

The first has the rough-coefficient obstruction in Sections 6–7. The second needs localization in addition to energy control. The third is the remaining matching task for the non-Gaussian birth terms and age ledger. The theorem removes an artificial accumulation in the quadratic sector without asserting those distinct estimates.

The retained quadratic state should not be extended to a layered background by pretending that a flat Fourier formula is a nonlinear minimizer. A direct gauge-invariant completion gives the required analytic function whenever its connected regular carrier has already been obtained by localization.

25.11.1 11.1 Definition and exact flat Hessian

Fix a connected lattice carrier X, a finite set of oriented plaquettes based at vertices in X, and paths $gamma_pq$ from the base of q to the base of p, contained in X. Reverse paths are used for reversed pairs. Let k_X(p,q;z) be scalar color-invariant kernels, holomorphic in the source variables z (or with values in the finite coefficient algebra). Their orientation indices may form a full tensor; no scalar rotation assumption is imposed. For G=SU(N) set

$$C_N(V) = \frac{V - V^{-1}}{2} - \frac{\mathrm{Tr}(V - V^{-1})}{2N} I, \qquad T_{pq}(U) = U_{\gamma_{pq}}.$$

Use a fixed invariant bilinear color pairing, with the corresponding positive real norm on the anti-Hermitian real Lie algebra. Define

$$\mathcal{Q}_X(U; z) = \sum_{p,q} k_X(p, q; z) \, \langle C_N(U_p), \mathrm{Ad}(T_{pq}(U)) C_N(U_q) \rangle \, . \tag{4.36}$$

This function is holomorphic in all SL(N,C) link variables and in z. Inversion is holomorphic on that group; no plaquette logarithm branch is used. Endpoint gauge transformations cancel in the invariant pairing, so the function is gauge invariant even when the paths wind around a regular hole. For real symmetric kernels and reversed paths it is real on SU(N) fields. A finite average of path conventions preserves these properties and can impose the desired lattice isometry covariance.

At the identity, the derivative of $C_N(U_p)$ is the linear curvature of A at p, while $T_{pq} = I + O(A)$. Therefore

$$\mathcal{Q}_X(e^A; z) = \sum_{p,q} k_X(p, q; z)\langle (dA)_p, (dA)_q \rangle + O(A^3). \tag{4.37}$$

Thus (4.36) has precisely the prescribed quadratic coefficient. If the precision is written with a factor one-half, insert that same factor in (4.36). On the ACTUAL physical coarse-field composition with its own current, subtracting a completion of the full physical Hessian leaves a germ whose field degree is at least three, after its vacuum and linear jets have also been removed. This does not remove independent AJ or JJ jets on the analytic paired (U, J) domain. Those jets remain in the paired state, with their original current weights, and are estimated through the actual own-current readout. Subtracting only the zero-momentum marginal coefficients likewise does not remove the full physical Hessian. Semisimplicity removes an invariant linear pair jet; the vacuum value must still be subtracted explicitly. These statements distinguish the analytic pair germ from its physical composition instead of identifying them at one saddle.

25.11.2 11.2 The layered analytic bound

Suppose on the actual regular carrier that $\|C_N(U_p)\| \le \epsilon_p$, and each transported path satisfies

$$\|\operatorname{Ad} T_{pq}(U)\| \le \exp(\tau\, \ell(\gamma_{pq})). \tag{4.38}$$

Then directly from (4.36),

$$\|\mathcal{Q}_X(U; z)\|_{\mathcal{A}_N} \le \sum_{p,q} \|k_X(p, q; z)\|_{\mathcal{A}_N} \epsilon_p \epsilon_q e^{\tau \ell(\gamma_{pq})}. \tag{4.39}$$

For example, if paths have weighted length at most c d(X), all $\epsilon_p \le \epsilon$, and the kernel has row/column allowance a_X, the right side is bounded by $C a_X |X| \epsilon^2 e^{c\tau d(X)}$. A small fixed τ and a spare locality exponent absorb both this transport factor and the polynomial |X|. The actual plaquette powers in $epsilon_p$ are retained; one does not replace them by a uniform order-one bound before the old-scale root count.

Condition (4.38) has a simple gauge-independent content. Real SU(N) transports are isometries. If a complex link is a real unitary link times a perturbation of size at most τ times its physical edge length, the product of adjoint norms is at most the exponential of C τ times the total weighted path length. The same bound holds after passing to any admitted regular frame. Thus this step does not need a common small gauge around a noncontractible hole. It does need the actual near-unitary analytic link margins along the whole chosen path; endpoints alone would be insufficient.

The completion is independent of the auxiliary current variable J, so it extends to the original paired (U,J) domain without imposing a new equation on J. Terms involving J in the remainder retain their original dimension-three current normalization. The proof does not change that normalization or claim that every current is a scalar observable source.

25.11.3 11.3 Domination by the primitive Wilson action on real wild fields

There is also a useful bound on every real SU(N) configuration, whether or not its plaquettes are small. Let $a_p = 1 - N^{-1} \operatorname{Re} \operatorname{Tr} U_p$. For eigenvalues $e^{i\theta_l}$,

$$\left\| \frac{U_p - U_p^{-1}}{2} \right\|_{\mathrm{HS}}^2 = \sum_l \sin^2 \theta_l \le 2 \sum_l (1 - \cos \theta_l) = 2N a_p.$$

Traceless projection decreases that Hermitian norm. Real transports are isometries. Cauchy–Schwarz and $2uv \le u^2 + v^2$ imply

$$|\mathcal{Q}_X(U;z)| \le C_G \sum_{p,q} |k_X(p,q;z)|\,(a_p + a_q). \tag{4.40}$$

The same statement holds for the source coefficient norm. If the sum of row and column kernel allowances of all retained carriers meeting a plaquette is at most d, then

$$\left\| \sum_X \mathcal{Q}_X(U;z) \right\|_{\mathcal{A}_N} \le C_G d \sum_p a_p. \tag{4.41}$$

Consequently their exponential source load can be paid by a fixed additive cold shift C_G d of the inverse Wilson coefficient. This matches the additive-shift comparison used in the R cancellation: subtracting the same number from every stored inverse coupling leaves the differences of stored and current coefficients unchanged.

This estimate does not authorize charging the same plaquette action at every historical birth. The kernel allowance d must be the allowance of the actual retained state, with old quadratic terms either absorbed into it or assigned once to their stored variables. That assignment is part of the live-component identity.

25.11.4 11.4 What the lemma supplies to the induction

The result provides an explicit analytic extension and positive-action majorant for a localized curvature quadratic source state. It applies on a connected regular carrier with specified paths, including an annular carrier. It does not assert that every gauge-invariant quadratic form on a punctured region is a curvature quadratic form. Pure cycle/holonomy coordinates need their own retained boundary state. Likewise a path crossing an unregularized wild core cannot be treated by the regular complex bound (4.39); its real dependence remains inside the island state and uses the action bound (4.41).

For the flat regular constrained harmonic lifts, the connected curvature decomposition is supplied by the chain-homotopy construction in the clean-map section: its local closed rows are filled by plaquettes, and the resulting paths retain the original exponential locality. The completion then supplies a literal subtraction of that physical quadratic kernel. A punctured carrier with independent cycle data or a live island still retains the separate boundary state described above. The completion itself is explicit and does not require an additional existence theorem for a source-dependent minimizing field.

Section 26.11 / The cumulative clean one-loop coefficient

Define the massive current-free Wilson determinant functional

$$\mathcal{L}(z;U) = \log\det(\Delta_U + z) - \frac{1}{2}\log\det(K_U + z),$$

normalized at $U = 1$. Apply (89) to its vector and site parts and include the coarse Schur factors. The explicit part is $\mathcal{L}(1) - \mathcal{L}'(1) + \frac{1}{2}\mathcal{L}''(1)$; the remaining quadratic coefficient is uniformly bounded by (91) and (95). Returning from physical spacing η to the unit bare lattice sends a unit mass to $z = \eta^2$ and supplies the corresponding η^2, η^4 on the derivatives. Associativity, current cancellation, and affine coefficient preservation give the actual cumulative identity

$$\sum_{\ell=1}^{k} b_\ell^0 = b_m(\eta^2) - \eta^2 b_m'(\eta^2) + \frac{1}{2}\eta^4 b_m''(\eta^2) + O(1), \qquad \eta = L^{-k}. \tag{97}$$

Here b_ℓ^0 is extracted from the fresh Gaussian normalizer in the actual small-field construction, and b_m is extracted from $\mathcal{L}$. This is not a newly chosen surrogate sequence.

The actual Wilson one-loop computation of Lüscher–Weisz [26], with background conventions from [27], gives the logarithmic numerator. The mass/external-momentum regulator comparison established earlier in this report gives, in our convention,

$$b_m(z) = \frac{11N^2}{24\pi^2}\log(1/z) + O(1), \qquad |z b_m'(z)| + |z^2 b_m''(z)| \le C. \tag{98}$$

The derivative bounds come from differentiated lattice integrals, not from differentiating a bare $O(1)$ statement. That comparison subtracts the constant and linear external-momentum Taylor terms in the summed integrand before the low/high split; Ward cancellation is not incorrectly applied separately in each region. A fixed alternate precision normalization replaces z by a fixed positive multiple and changes only the bounded term.

Section 26.11 / The cumulative clean one-loop coefficient

Theorem 26.4 (Bounded cumulative clean Gaussian error)**.** *For the actual uniformly blocked regular Wilson saddle,*

$$\left|\sum_{\ell=1}^{k} b_\ell^0 - kc_0\right| \le C_s, \qquad c_0 = \frac{11N^2}{12\pi^2}\log L > 0, \tag{99}$$

with C_s independent of depth and volume.

Proof. Insert (98) in (97) and use $\log(\eta^{-2}) = 2k\log L$. The two mass-derivative terms and the constrained remainder are bounded uniformly. □

This theorem does not assert geometric convergence of the individual increments. It also does not estimate localized seam determinants or stored large-field factors. Its construction-level identification is the Gaussian normalizer of [4, (1.4)–(1.5), (2.10)–(2.12)], with the actual mean of that paper's (0.10)–(0.12).

Section 27.1 / Hypotheses and conclusions

Sections 2–9 prove actual fixed-block geometric and Gaussian results, including a constraint chart, a hidden curl gap, complex inverse kernels and local coefficient response. Their constants are not asserted uniform when the number of fine variables inside ONE blocking step tends to infinity. The nonlinear classical map is kept exactly. Sections 11–14 then use the published clean background/localization inputs and the companion generic cluster proof to construct an all-depth smooth-source class on the original fixed clean paired cores, with actual origin ledgers and ordinary complex source domains.

Sections 15–16 describe the vacuum, local contact counterterms and physical partition interface, using the separately stated original-map large-field matching. Sections 17–20 establish the coupling-weighted

Section 27.1 / Hypotheses and conclusions

two-label refinement, retained-field bounds, incoming scalar normalization and an accumulated PURE CLEAN Gaussian comparison on a finite ultraviolet window. Section 21 gives the fixed-torus physical comparison from the companion original-map unnormalized large-field, prefix-state and preparation estimates; its finite-window defect interpolation avoids a false union-derivative shortcut. The fixed clean cores of I–II and the shrinking cores of large-field-origin members are kept distinct. No claim of a complete Yang–Mills construction is made. The original source-free preparation inputs are imported explicitly, and the remaining continuum, reconstruction and full observable-algebra statements require their own theorems.

Section 27.3 / An actual local coordinate chart for one Wilson averaging step

Work on the current fine lattice with raw bond angles and counting measure. Fix an odd blocking factor L and a sufficiently large torus so that the determining double blocks have unambiguous local lifts. The gauge group is $SU(2)$; the same argument works for a fixed compact matrix group in a faithful representation. All constants in this note may depend on L, the group, and a fixed later grouping size, but not on the number of blocks.

Fix an internal spanning tree in each block, rooted at its coarse site, and set its links to the identity using gauge transformations equal to the identity at the block roots. Use the actual one-step path-log/Federbush average from Section 17, (9.1). Its paths cross exactly one fine face corresponding to their coarse row. The tree gauge is an exact gauge choice for the path average with fixed block-root endpoints; no gauge-dependent determinant is introduced by removing these tree variables with Haar measure.

Choose one pivot bond $p(c)$ in each coarse face F_c. At the flat configuration, the derivative of the coarse row with respect to that pivot is

Section 27.3 / An actual local coordinate chart for one Wilson averaging step

$$D_{p(c)}q_c(0) = L^{-3}I, \qquad D_{p(c')}q_c(0) = 0 \quad (c' \neq c). \tag{2.1}$$

The factor L^{-3} follows because each face bond is used by L of the L^4 straight paths. Node factors and internal contour means depend only on internal bonds, so they do not create additional pivot dependence. The same support statement is the one used in the distributed-face right inverse

Let u denote the non-tree, non-pivot fine coordinates, and let v be the target coarse logarithm. The analytic implicit-function theorem can be applied **row by row** to

$$q_c(u, p_c) = v_c.$$

It gives a common sup-norm neighborhood and an analytic chart

$$A = \Phi(v, u), \qquad q(\Phi(v, u)) = v, \tag{2.2}$$

with the following properties:

- the free coordinates of Φ are exactly u, and tree coordinates are zero;
- each pivot coordinate depends only on its own coarse row and the free variables in a determining double block;
- $\Phi(0, 0) = 0$, and every fixed derivative has a bound depending only on L;
- the radius does not shrink with the number of blocks.

For example, after decreasing the radius so that the pivot Jacobian differs from $L^{-3}I$ by at most $L^{-3}/2$, its inverse has norm at most $2L^3$. The remaining bounds follow from the finite-patch analytic action and the same quantitative contraction argument as

The real coarea density in this chart is positive. Near the flat point the oriented pivot determinant has fixed positive sign, so its reciprocal, the local Haar factors, and the coarse delta-coordinate factor have a holomorphic product extension. Write this density as $J(v, u)$. Its logarithm is a sum of fixed-range analytic terms; all its first fixed-order local derivative row sums are bounded by constants depending on L. The chart and this density do not depend on the Wilson coefficients.

Section 27.7 / The exact nonlinear classical response

For real positive λ in this neighborhood, let

$$\mathcal{E}(\lambda, v) = \min_{u \text{ in the regular chart}} \sum_p \lambda_p a_p(e^{\Phi(v,u)}).$$

The local analytic continuation is defined using the critical point just constructed. If $A_f = \sum_p f_p a_p$, put

$$F_f = D_u A_f(u_*, v), \qquad H_\lambda = D_u^2 \sum_p \lambda_p a_p(u_*, v).$$

The envelope theorem and differentiation of stationarity give the exact formulas

Section 27.7 / The exact nonlinear classical response

$$D_\lambda \mathcal{E}[f] = A_f(u_*, v), \qquad D^2_\lambda \mathcal{E}[f,k] = -F_f^{\mathsf{T}} H_\lambda^{-1} F_k. \tag{6.1}$$

This is the nonlinear classical contribution that cannot be called power-irrelevant. At a homogeneous coefficient $\lambda = \beta$, the force F_1 vanishes exactly. The one-step constraint-current identity therefore gives, for a test varying slowly across a determining block,

$$\|F_f\|_{\mathrm{local}} \le C_L \alpha \, \mathrm{osc}_{\mathrm{local}}(f), \tag{6.2}$$

or $C_L \alpha \ell \, \mathrm{Lip}(f)$ in physical step units. Combining (5.5), (6.1), and (6.2) gives two endpoint oscillation factors and the genuine β^{-1} second-source scaling. The kernel between separated endpoint forces is exponentially decaying. Higher classical derivatives retain the exact homogeneous degree-one scaling in λ; one must not replace it by a fictitious additional coupling power.

For the Gaussian action this mechanism is exactly the Schur-complement formula including its open-chain source coefficients. The following Gaussian determinant gives closed chains instead.

Section 27.8 / The Gaussian determinant has an actual local cycle expansion

At a flat reference, write $H(h) = H_0 + \sum_p h_p B_p$. In the smaller current-step product box where the weighted norm of $H_0^{-1} \sum_p h_p B_p$ is less than $1/2$,

$$\log \frac{\det H(h)}{\det H_0} = \sum_{m \geq 1} \frac{(-1)^{m+1}}{m} \operatorname{Tr} \left[H_0^{-1} \sum_p h_p B_p \right]^m . \tag{7.1}$$

Every coefficient is a cyclic product of the inverse kernels from (4.4) and the bounded-range plaquette matrices. Its support is connected after each inverse-kernel edge is assigned a fixed lattice path. The weighted row sums (4.5) and finite incidence give a summable rooted cycle majorant per current block. The factor $1/m$ and the absolute geometric series bound the sum uniformly in volume after the vacuum trace is localized per root.

For the regular background Hessian, use (5.5) and its local analytic dependence in the same way. The background-dependent coefficients are obtained from the convergent rooted-tree expansion of the minimizer, and adjoining those trees to the determinant cycle preserves connectedness. This is a direct locality construction for the Gaussian logarithm; no final connected-source estimate has been assumed.

All gauge and coarea determinant factors must still be retained with their actual signs. Exact associativity and multilevel telescoping are separate hypotheses; (7.1) concerns the explicitly defined hidden-coordinate Hessian.

Section 27.11 / Interface after the current-step result

The calculations above close actual geometric and Gaussian subtheorems for a current-step Wilson integral. By themselves they do not assert the complete source-free effective-density induction. The later sections supply the clean source and age interfaces described here.

The generic localized Gaussian/Mayer allowance is proved in The construction directly from the original CMP 116 (1988) construction. It accepts the total analytic potential family with the linear/quadratic load in (8.2), keeps the exact classical/Gaussian contribution, and maps the nonlinear and old potential families into a connected localized logarithm.

An additional distinction remains after that one-step map is established: a bounded derivative in the old interaction norm need not be a strict irrelevant contraction. The actual construction uses the birth-age bounds of Sections 11–12 and the two origin ledgers of Section 14, rather than an assumed Markov contraction of arbitrary irrelevant activities. Scalar-only extraction is insufficient for arbitrary spatial coefficients.

For deep layered backgrounds, the current-step local constants cannot replace the actual physical field/current estimates. Smooth-source or smooth-patch domains may be necessary, as in elliptic regularity. The present current-step theorem is a genuine missing local ingredient, with its range of validity explicitly bounded.

27.13.1 12.1 Restoring the volume count

A fixed present-scale region contains $O(r^{-4})$ possible birth roots. Summing (11.9) over domains and roots gives

$$CB_j \left[\rho_{j,k}^{\cdot 2} r^{\zeta} + \rho_{j,k}^{\cdot 3} r^2\right] \tag{12.1}$$

per present-scale cell. Suppose the born field-dependent activity is normalized as

$$B_j \le R_j^2 s_j. \tag{12.2}$$

Then (12.1) becomes

$$CA_k^2 s_j \left[r^{\zeta} + \frac{A_k}{R_j} r^2\right]. \tag{12.3}$$

For the running radii $R_j = x_j^{-1/2} (\log x_j)^q$, $q \ge 0$, a logarithmic trajectory $x_j \asymp x_k + c(k-j)$ makes A_k/R_j at worst polynomial in the age $k-j$, when $A_k \le CR_k$. The geometric factors in (12.3) absorb that polynomial. Thus there is an actual summable memory kernel

$$w_n = C(1+n)^a L^{-\zeta' n}, \qquad \sum_{n \ge 0} w_n < \infty, \tag{12.4}$$

with a fixed $0 < \zeta' \le \zeta$. This is the relevant invariant ledger; no fictitious rescaling contraction for one Markov coordinate has been used.

27.13.2 12.2 Why the normalized born prefactor is available on a fixed additive source domain

On a fixed additive spatial-source ball $\sup_p |w_p| \le r_*$, the primitive fluctuation growth load on a fixed larger external-field core of radius R_0 is

$$O(r_* g R_0 + r_* g^2).$$

It is small at weak coupling. The generic local-potential allowance can therefore be applied on that fixed larger field core, without demanding a relative spatial coupling disc there. If a vacuum-subtracted born term has norm b_j on that core, gauge invariance gives its zero linear field jet. Directional Taylor/Cauchy on the inner radius $R_j \le R_0/2$ yields

$$B_j \le C b_j (R_j/R_0)^2. \tag{12.5}$$

Hence (12.2) holds with $s_j = Cb_j/R_0^2$. Here the radial path in both the physical potential and the independently normalized physical current must lie in the actual common near-zero coarse-coordinate core; a mere scalar bound at critical pairs would not justify it. On the original clean core this is the coarse-coordinate extension used in the source-free Taylor argument. This works in a finite source

coefficient algebra on the same fixed pointwise source load. It does not require an arbitrary perturbed fine-scale minimizer to remain in a Hölder domain: all background readouts are frozen source-free maps.

The zero-fluctuation classical source is a separate birth member. After retaining its full quadratic tensor its remaining field dependence can be $O(r_* R_0^3)$ on the fixed larger core, independently of g. Its smallness is the field-radius gain on the smaller core, or its ordinary age decay; it is not a fresh $O(g)$ fluctuation contribution. This distinction is preserved in the induction below.

Large constant-coupling discs can still be used separately for the homogeneous tangent and tuning. A constant multiplicative change of the primitive Wilson action does not move its constrained minimum. Mixing these two analytic domains into one unnecessarily large spatial domain would reintroduce the linear-force and regularity obstructions.

27.13.3 12.3 An old-member difference has a genuinely small factor

The old activity enters the next fluctuation exponent as a difference between two nearby admitted background readouts. Suppose the source-free interpolation of those readouts has size δ_k and an available analytic radius α_k, with fixed spare room. The one-variable Cauchy/Schwarz estimate on that interpolation gives, for the renormalized old member,

$$\|R_j(\text{new}) - R_j(\text{old})\| \le C \frac{\delta_k}{\alpha_k} \sup_{\text{admitted interpolation}} \|R_j\|. \tag{12.6}$$

Apply the age estimate throughout that actual common interpolation domain. Its small factor is therefore

$$\theta_k = C\delta_k/\alpha_k,$$

not a bound asserted from endpoint values alone. This is the mechanism of the original localized field-direction differentiation. It is preserved coefficientwise because the interpolation does not depend on the source. The ordinary hierarchy chooses δ_k/α_k small; with logarithmic radii it tends to zero when the background logarithmic power exceeds the fluctuation power.

The generic local Gaussian/Mayer map has a bounded derivative in the added potential norm on a fixed allowance, obtained by an auxiliary scalar Cauchy estimate. Therefore the born norm obeys a memory inequality of the form

$$s_{k+1} \le C b_{k+1}^{\text{fresh}} + C\theta_k \sum_{j \le k} w_{k-j} s_j. \tag{12.7}$$

If $C \sup_k \theta_k \sum_n w_n \le 1/2$, induction gives

$$\sup_{j \le k} s_j \le 2C \sup_{j \le k} b_j^{\text{fresh}}. \tag{12.8}$$

The same argument preserves a monotone power-small envelope for the non-Gaussian fresh terms when it is present. The exact classical and Gaussian source contributions are retained in their separate state; they are not forced into the power-small envelope.

27.13.4 12.4 Scalar homogeneous extraction and anisotropic source defects

For constant source values, the clean family retains its Euclidean symmetries. The sum of the curvature tensors in (11.7), with a covariant pointing, reduces to the scalar Yang–Mills tensor (one coefficient for $SU(2)$). The extraction and its age remainder commute with every constant-source coefficient. This gives the original scalar coupling member its own source coefficient, rather than borrowing the coefficient of a different activity.

For a spatial source, work with the actual pointed source-local member rooted at y, and freeze every test direction at $f_i(y)$. Source locality and multilinearity give the exact decomposition into its constant-value part and terms containing at least one $f_i - f_i(y)$. On a birth domain their norms gain

$$C\,\ell_j\,\mathrm{Lip}(f_i)\,\mathrm{diam}(X),$$

where ℓ_j is the physical birth scale. Polynomial domain diameter is paid by the spare locality exponent. The constant-value family uses the same scalar coefficient as above. The remaining quadratic tensors need not be isotropic; they are retained in the defect ledger with their genuine source-oscillation factor. Gauge invariance alone gave their engineering order r^4, so the volume count leaves the additional geometric factor ℓ_j, which is summable over older birth scales. If full tensor extraction is preferred, (11.7) supplies it explicitly and its defect coefficients have the same summable gain.

For multiple source labels the subset expansion has finitely many terms at each fixed order, or is estimated directly in the coefficient algebra. The nonlinear classical Gaussian terms have the stronger two-endpoint oscillation mechanism proved in the parent note. The field-independent local contact terms and the final retained-field cumulants must still be handled by their actual bookkeeping; this age estimate does not discard them.

27.13.5 Exact range of the age conclusion

Equations (11.9)–(12.8) prove the deterministic coefficientwise old-member ledger on clean admitted domains, using ordinary source-free field/current regularity and an actual common interpolation. The generic cluster allowance and wall localization provide its bounded formation operations. Applying it to every modified core or every domain around a wild hole still requires the corresponding source-free domain identification and marked-boundary reserve. Those cases must not be included by calling all old terms “irrelevant.”

Section 27.14 / All-depth locality of the exact Gaussian spatial source

This section uses the actual uniform flat blocking, rather than iterating a crude fixed-step norm. Its source-free ingredient is proved in the stated results, §§7–10: the physical Bloch-strip inverse and coarse Schur bounds for the actual averages. They imply a depth-independent exponential cell-block bound on the curvature harmonic lift. This norm must be distinguished from an energy-normalized coarse kernel.

27.14.1 13.1 Curvature bands and uniform block estimates

Let the physical coarse spacings be $\ell_j = L^j \ell_0$. Use raw plaquette angles for the fine curvature Hilbert space, with counting measure. In dimension four this is equivalent to physical curvature with physical

volume measure. Coarse link angles also use counting measure. Let F_j be the fine curvature of the actual flat constrained harmonic lift from level j. The source-free strip bounds give

$$\|\chi_a^{(j)} F_j \chi_b^{(j)}\|_{2\to 2} \le C e^{-c d_j(a,b)}, \qquad \|F_j\|_{2\to 2} \le C. \tag{13.1}$$

The output restriction is to fine plaquettes in a level-j physical cell; the input restriction is to finitely many coarse links. Constants do not depend on the number of fine plaquettes inside that output cell. The transpose below uses the corresponding physical pairings.

For clarity, (13.1) follows from the actual flat formula $H_j = \Xi_j^{-1} Q_j^\top (Q_j \Xi_j^{-1} Q_j^\top)^{-1}$, followed by $F_j = dH_j$, with opposite-momentum transpose on the complex strip. The energy inverse controls $d\Xi_j^{-1}$, the uniformly bounded coarse Schur inverse controls the last factor, and the common strip gives exponential cell-block kernels. This is one direct depth-j estimate; no product of j constants occurs.

The harmonic curvature subspaces $\mathcal{E}_j = \operatorname{ran} F_j$ are nested. Let Γ_j project orthogonally onto $\mathcal{E}_j \ominus \mathcal{E}_{j+1}$. Equivalently,

$$\Gamma_j = F_j C_j F_j^\top, \qquad C_j = T_j (T_j^\top S_j T_j)^{-1} T_j^\top, \qquad S_j = F_j^\top F_j, \tag{13.2}$$

where T_j is a gauge-fixed injection of the kernel of the next average. Thus $F_j C_j F_j^\top$ is the orthogonal projector onto $\operatorname{ran}(F_j T_j)$. For a straightened average, the residual gauge parameter used to impose internal tree gauge has BLOCK MEAN zero, $R\lambda = 0$; starting with a root-fixed tree-killing parameter, a constant in each block adjusts its mean without changing its internal tree differences. This preserves Q_{next}, since $Q_{\mathrm{next}} d\lambda = d_c R\lambda = 0$. The constrained least-energy characterization identifies that image with the complement of the next harmonic space.

The required coarse curl lower bound follows from an additional exact chain identity; a massive Schur bound alone would not prove it. At flat field write $n = L^j$, A_0 for normalized disjoint box averaging, and $L_\mu = n^{-1} \sum_{r=0}^{n-1} T_\mu^r$. The straightened average and its two-form companion are $Q_\mu = A_0 L_\mu$ and $Q_{\mu\nu}^{(2)} = A_0 L_\mu L_\nu$. Discrete Stokes gives $d_c Q = Q^{(2)} d_\eta$. Jensen's inequality with the physical weights gives $\|Q^{(2)}\| \le 1$. Since $QH_j = I$, this proves $\|d_c B\| \le \|F_j B\|$, up to the fixed action convention. Centering adds only phases. Section 3 now gives a fixed-step hidden gap on the next-average kernel, uniformly in j. The exponentially localized precision and the weighted inverse argument of Section 4 give an exponential kernel for C_j. Substituting in (13.2) yields

$$\|\chi_a^{(j+1)} \Gamma_j \chi_b^{(j+1)}\|_{2\to 2} \le C e^{-c d_{j+1}(a,b)}. \tag{13.3}$$

This is a bound for the actual orthogonal fluctuation curvature bands. Fine dimensions inside cells are harmless because these are block operator norms. The projections are mutually orthogonal, have norm one, and obey $\Gamma_j F_n = 0$ for $j < n$.

27.14.2 13.2 Hölder commutators and final-scale weights

Let M_f multiply fine plaquettes by a scalar profile at their physical centers. For $0 < \alpha \le 1$, (13.3) implies

$$\|[\Gamma_j, M_f]\|_{2\to 2} \le C \ell_{j+1}^\alpha [f]_{C^\alpha}. \tag{13.4}$$

Here is a block proof that never counts fine coordinates. In each level-$j+1$ cell subtract the scalar value at its center. Each within-cell error has norm at most $C\ell_{j+1}^\alpha [f]_{C^\alpha}$. Between centers the difference

is at most $C\ell_{j+1}^{\alpha}(1+d_{j+1}(a,b))^{\alpha}[f]_{C^{\alpha}}$. Multiply by (13.3) and use the scalar row and column Schur sums over cells.

At final scale n, put $W_{n,y}(p)=\exp[\kappa\,\mathrm{dist}(p,y)/\ell_n]$, with fixed small $\kappa>0$. The same proof gives

$$\begin{aligned}\|W_{n,y}\Gamma_j W_{n,y}^{-1}-\Gamma_j\| &\le C\kappa\ell_{j+1}/\ell_n,\\ \|W_{n,y}[\Gamma_j,M_f]W_{n,y}^{-1}\| &\le C\ell_{j+1}^{\alpha}[f]_{C^{\alpha}}, \qquad j<n.\end{aligned} \tag{13.5}$$

For the first inequality use $|e^t-1|\le|t|e^{|t|}$ and the Lipschitz property of distance. Both estimates hold also with W^{-1}. The ratios ℓ_{j+1}/ℓ_n are summable, which prevents a constant raised to the depth.

27.14.3 13.3 Running scalar subtractions and the exact resolvent

Let x_j be real reference inverse couplings in a weak tube, and put $\underline{x}_n=\min_{0\le j\le n}x_j>0$. Pulling all source-independent scalar Gaussian counterterms back to the fine space gives the diagonal band precision proved in the parent note. Its hidden curvature covariance at level n is

$$C_n^{\mathrm{h}}=\sum_{j<n}x_j^{-1}\Gamma_j, \qquad (C_n^{\mathrm{h}})^{1/2}=\sum_{j<n}x_j^{-1/2}\Gamma_j. \tag{13.6}$$

The square root is fixed at source zero. In (13.7)–(13.13), first replace every bound involving x_n by the same bound involving $\underline{x}_n$. Orthogonality and (13.5) then prove them exactly as written in the monotone case. The actual weak tube need not be monotone: its bounded-partial-sum one-loop law gives $x_j\ge x_n-2C_s+c(n-j)$, and $x_n\ge 4C_s$ implies $\underline{x}_n\ge x_n/2$. Absorb the fixed powers of two into C^m to obtain the displayed x_n-bounds. For readability, those bounds are

$$\|(C_n^{\mathrm{h}})^{1/2}\|\le\sqrt{2}\,x_n^{-1/2}, \qquad \|W_{n,y}(C_n^{\mathrm{h}})^{1/2}W_{n,y}^{-1}\|\le Cx_n^{-1/2}. \tag{13.7}$$

For the second bound, retain the unweighted orthogonal sum as one operator of norm at most $\underline{x}_n^{-1/2}$, and sum only its conjugation differences. Their sum is at most $C\kappa\sum_{j<n}x_j^{-1/2}\ell_{j+1}/\ell_n\le C_L\kappa\underline{x}_n^{-1/2}\le\sqrt{2}C_L\kappa x_n^{-1/2}$. Summing the unweighted projection norms separately would create an artificial depth loss.

For an additive source h, define

$$q_n(h)=(C_n^{\mathrm{h}})^{1/2}M_hF_n, \qquad T_n(h)=(C_n^{\mathrm{h}})^{1/2}M_h(C_n^{\mathrm{h}})^{1/2}. \tag{13.8}$$

The exact classical coarse quadratic energy is

$$\begin{aligned}E_n(h;B)=\frac{x_n}{2}\|F_nB\|^2+\frac{1}{2}\langle F_nB,M_hF_nB\rangle\\ -\frac{1}{2}\langle q_n(h)B,(I+T_n(h))^{-1}q_n(h)B\rangle.\end{aligned} \tag{13.9}$$

Pairings in the holomorphic continuation are bilinear. Completing the hidden quadratic form proves (13.9). It includes all interleaved running scalar subtractions; it is not the bare-temperature formula reused with an informal running coupling.

The resolvent is analytic for $\|h\|_\infty<\underline{x}_n$. On a fixed smaller ball $\|h\|_\infty<cx_n$ it remains bounded under all final-scale conjugations in (13.7). Moreover,

$$q_n(f) = \sum_{j<n} x_j^{-1/2} [\Gamma_j, M_f] F_n \tag{13.10}$$

because $\Gamma_j F_n = 0$. The geometric scale sum and (13.1), (13.5) yield

$$\|q_n(f)\|_{\kappa,n} \le C x_n^{-1/2} \ell_n^{\alpha} [f]_{C^\alpha}, \qquad \|T_n(f)\|_{\kappa,n} \le C x_n^{-1} \|f\|_\infty. \tag{13.11}$$

This notation means all the displayed final-scale conjugated operator norms between the relevant fine and coarse spaces, with both signs of the weight. For the unweighted endpoint bound orthogonality improves the triangle sum to the square root of $C \sum_{j<n} x_j^{-1} \ell_{j+1}^{2\alpha} [f]_{C^\alpha}^2$; either bound proves (13.11).

27.14.4 13.4 All orders with two actual oscillations

For $m \ge 2$, differentiating (13.9) at zero gives the quadratic form

$$D^m E_n(0)[f_1, \dots, f_m] = \frac{(-1)^{m-1}}{2} \sum_{\sigma \in S_m} q_n(f_{\sigma_1})^\top T_n(f_{\sigma_2}) \cdots T_n(f_{\sigma_{m-1}}) q_n(f_{\sigma_m}). \tag{13.12}$$

Consequently,

$$\begin{aligned} \|D^m E_n(0)[f_1, \dots, f_m]\|_{\kappa,n} \le C^m (m-2)!\, x_n^{1-m} \ell_n^{2\alpha} \\ \times \sum_{i \ne j} [f_i]_{C^\alpha} [f_j]_{C^\alpha} \prod_{a \ne i,j} \|f_a\|_\infty. \end{aligned} \tag{13.13}$$

The coarse blocks at anchors u, v have the additional factor $e^{-c\,\mathrm{dist}(u,v)/\ell_n}$. A constant endpoint vanishes exactly; a coefficient with at most one nonconstant direction vanishes for $m \ge 2$. This is a genuine all-depth Gaussian locality bound with the running denominator.

At first order the exact operator is $F_n^\top M_f F_n/2$. A gauge-invariant local freezing replaces f on each fine plaquette in a level-n cell by its value at that cell's center. Its error has weighted coarse cell-block norm at most $C\ell_n^\alpha [f]_{C^\alpha}$. The frozen pieces are the exponentially localized quadratic forms $F_n^\top \chi_y^{(n)} F_n$ times $f(y)$. This avoids multiplying a final coarse precision on just one side by a spatial scalar, which would not preserve its gauge null space.

These are ordinary coarse block operator norms and physical curvature norms. They do not assert that conjugation by massless $S_n^{-1/2}$ preserves locality. The independent energy-ball theorem controls that different norm. The two results supply amplitude and locality without conflating them.

27.14.5 13.5 Joint source flags and a final-cell tree bound

The preceding two-oscillation estimate can be filed locally together with source support. This statement is at the FINAL CELL resolution; it does not assert an absolute fine-coordinate coefficient-density estimate.

For final cells A, B, retain the exact operator block

$$K_{AB}^{j,f} = \chi_A^{(n)} [\Gamma_j, M_f] \chi_B^{(n)}.$$

It depends only on f restricted to $A \cup B$. Applying the fine-$(j+1)$-cell proof of (13.4) to those restrictions, and using the final-scale weight based on distance to A, gives

$$\|K_{AB}^{j,f}\| \le C\ell_{j+1}^{\alpha}[f]_{C^\alpha} e^{-cd_n(A,B)}. \tag{13.14}$$

No zero-extension of $f|_A$ is made: the differences are estimated BEFORE the restriction. More explicitly, on two smaller cells a, b the block commutator is the sum of

$$\Gamma_{ab}(M_f - f(b)) - (M_f - f(a))\Gamma_{ab} + (f(b) - f(a))\Gamma_{ab}.$$

Each difference has the Hölder bound on those very cells. Scalar cell row/column sums give (13.14), independently of their internal dimensions.

File $K_{AB}^{j,f}$ as a two-cell source flag. It is zero if f vanishes on both cells. Sum its endpoint band label with $x_j^{-1/2}$; (13.14) retains the factor $Cx_n^{-1/2}\ell_n^{\alpha}[f]_{C^\alpha}$. Each middle $T_n(f)$ is filed using the exponentially localized source-zero blocks of $(C_n^{\mathrm{h}})^{1/2}$, one multiplication block $M_{f|C}$, and the next square-root block. It has its one marked cell C.

Every product in (13.12) is consequently a connected block path joining its output cells u, v and at least one cell meeting every source support. Keep half its exponential for that marked tree and sum the other half over all intermediate cells. The scalar Schur sums cost only C^m. If $\tau_n(u, v; K_1, \dots, K_m)$ is the minimum current-cell tree length joining u, v and one point of each source support K_i, this proves

$$\begin{aligned}\|\chi_u D^m E_n(0)[f_1, \dots, f_m]\chi_v\| \le{}& C^m(m-2)!\, x_n^{1-m}\ell_n^{2\alpha} e^{-c\tau_n(u,v;K_1,\dots,K_m)} \\ &\times \sum_{i\neq j}[f_i]_{C^\alpha}[f_j]_{C^\alpha} \prod_{a\neq i,j}\|f_a\|_\infty.\end{aligned} \tag{13.15}$$

This is a bound on the EXACT multilinear profile operator. Both oscillatory endpoint flags are present in the filing, rather than recovered by multiplying two unrelated estimates. For sufficiently small fixed additive sources the corresponding connected block-path expansion converges normally.

27.14.6 13.6 Local curvature factorization, including torus harmonic modes

The curvature completion used on regular carriers needs curvature kernels, not only a potential kernel. They can be obtained without a massless inverse.

Fix a final output cell A. Write $J_A(b)$ for the row block of F_n at coarse bond b. Gauge covariance gives

$$F_n d_c = 0.$$

Exponential kernels extend this identity to polynomially growing gauge parameters on the infinite lattice. On a torus, F_n also annihilates every constant coarse one-form: a compatible constant fine one-form has zero curl and hence zero minimum quadratic action. Thus the row current J_A has zero discrete divergence AND zero harmonic flux.

Choose coordinate-ordered coarse paths $\gamma_A(x)$ from the anchor of A to x. A bond b defines the closed chain

$$C_b = \gamma_A(b_-) + b - \gamma_A(b_+).$$

On the infinite lattice, fill it by commuting the finitely many coordinate segments. The resulting plaquette chain S_b has at most $C(1 + d(A, b))^2$ plaquettes and stays within distance $C(1 + d(A, b))$ of A. Since J_A is divergence free, its path terms cancel:

$$\sum_b J_A(b)C_b = \sum_b J_A(b)b.$$

Taking the fillings proves the exact factorization

$$F_n = \mathcal{K}_n d_c, \qquad \mathcal{K}_n(A,q) = \sum_b J_A(b) S_b(q). \tag{13.16}$$

For a periodic lattice choose coordinate representatives in a fundamental cube centered at A. A path loop crossing a remote cut can have winding h_b. Subtract its fixed fundamental cycles before filling: $\partial S_b = C_b - \sum_\mu h_b^\mu \gamma_\mu^{\mathrm{fund}}$. The coefficient of a fundamental cycle in the total is the flux of J_A through that cut. Zero divergence makes this flux independent of the cut, and zero pairing with the constant one-form makes it zero. Hence (13.16) is still exact.

A bond that does not approach a cut has the local filling just described. Every bond whose filling uses a remote cut has distance at least cN from A, where N is the relevant coarse period; its filling has only a fixed polynomial cost in N. The row bound e^{-cN} pays that cost and every allowed smaller exponential locality weight. The same argument with the shortest relevant period handles rectangular tori. Small tori have bounded size and are included by increasing the fixed constant. Thus

$$\sum_q e^{c' d_n(A,q)} \|\mathcal{K}_n(A,q)\| \le C \tag{13.17}$$

with a matching column bound and constants independent of depth and period.

The polynomial SIZE of a filling is used only to sum its numerical coefficients. It is not filed as a field carrier of area d^2. After the coefficients for one curvature leg q are summed, that term depends on the curvature at q, so it is filed on a path from A to q of length $O(d(A,q)+1)$. This distinction preserves exponential TREE locality.

Insert (13.16) on both legs of (13.12) and of the first-order source kernel. This gives connected exponentially local COARSE CURVATURE kernels with the same source flags, rates and oscillation factors, after a fixed rate loss. Each kernel can therefore be completed by the gauge-invariant plaquette/transport formula of the parent note on a connected regular carrier. Its flat quadratic form is exact. The completion does not imply that mixed A, J or J, J jets vanish on an independent-current pair domain; those jets retain their own higher-dimensional readout. Wild-crossing carriers remain inside their stored island operator.

Neither (13.15) nor (13.17) bounds a trace over all fine modes. Vacuum source loops require their birth-by-birth scalar localization and volume count, treated separately below.

Section 27.15 / A clean smooth-source induction with two origin ledgers

The previous estimates give an all-step clean source result on the original uniform small-field domains. The useful spatial class is a fixed Hölder ball of bare profiles, not an arbitrary fine-plaquette product box. There is no source-dependent minimizing map. The source is retained in the density, and every chart, test, current assignment and window map is the one at source zero.

27.15.1 14.1 The precise unmarked inputs and the source norm

Fix the blocking and localization geometry and one admissible exponent $0 < \alpha \leq 1$. Use the original clean local member domains for independent pairs (U, J), with their fixed positive margins. Let R_0 be a fixed sufficiently small physical potential/current radius inside those domains. The inputs used here are the following concrete ordinary clean facts.

1. The original window/background maps have the curvature, Hölder and current estimates (11.2), the full coarse-coordinate analytic extension, and the field-direction disc of radius cR_0/δ for a physical fluctuation of size δ. Inner windows form their own constrained background and current before an old member is evaluated; this is not an estimate for an arbitrary outer current treated as a critical one. Outer and removed windows retain the original exponentially small distant-field estimates.
2. Frozen walk-wall localization and the Gaussian/Mayer formation map act on arbitrary gauge-invariant analytic local potentials in a fixed small growth allowance. They preserve a common spare locality exponent and have a bounded first difference in that allowance. The coefficient-algebra proof is in the companion marked-bridges note. This is a statement about the actual formed local integrals, not an assumed bound on their final scalar logarithm.
3. In the SAME fixed-physical-cutoff scheme, a constant additive bare source c has a holomorphic clean family on $|c| < r_{\mathrm{hom}}$, uniformly in its number of steps. Write

$$Y_k(c) = x_k(c) - x_k(0), \qquad Y_k(0) = 0.$$

   The constant-source result of The construction gives, after decreasing the disc by a fixed factor,

$$\sup_k |Y_k(c)| \le C|c|, \qquad \sum_j \sup_{|c|\le r_{\mathrm{hom}}/2} |\partial_c(b_j(c) - b_j(0))| \le C. \tag{14.1}$$

   Every homogeneous birth member has its original field-domain norm bounded uniformly on this disc, and keeps its OWN scalar counterterm. The Gaussian coefficient and all cutoff tails are included in that same scheme. No independently recomputed R history is identified with it.
4. The pointed-member and window-first convention is retained. At one birth anchor, the complete normally convergent pointed sum uses a single scalar parameter and its own extracted coefficient before its window estimate is taken. Assigning a different parameter to each summand after extracting a coefficient would not justify the age cancellation.

The first two are the source-free inputs used in the original generation/cluster proof, with their coefficient-algebra extension already established. The third is the actual homogeneous analytic induction, not a spatial-source hypothesis. These facts are now sufficient for the spatial induction below.

There are TWO different domain classes in the original papers, and the fixed radius here belongs only to the PURE CLEAN class. CMP 109 (1987), (1.18), printed p. 263, explicitly assumes and its Theorem 3/CMP 116 proof preserves positive absolute radii α_0, α_1, independent of the birth index j, for the clean functions $E^{(j)}$. The radius R_0 is a fixed smaller radius in this class. In contrast, CMP 119 (1988), (2.28), uses $\alpha_{0,j} = g_j C_0 (\log g_j^{-2})^{q_0}$ and a corresponding shrinking current radius in the complete large-field induction. A newly born R-member is NOT granted a fixed-radius extension by (1.18). Every connected formation term containing an R-origin is kept in the separately tagged R-class, with its actual shrinking-core Cauchy losses and superpolynomially small birth amplitude. Only terms with no such origin enter the fixed-core clean family proved here. Consequently the full $E + R$ density is not asserted to have a uniform fixed-core extension. The companion marked-bridges note's $\alpha_j^{-2} e^{-cP_0(x_j)}$ ledger pays the regular R readout and its descendants; it is not replaced by an R_0^{-2} estimate.

The precise original full-domain chain is worth recording. In CMP 109 (1987), (3.10)–(3.17), the physical current is explicitly replaced by an independent J, the troublesome second-order response is rewritten using its Landau equation and the covariant Laplacian, and (3.13)–(3.16) give the reduced-current analytic extension and its Cauchy disc on the independent-(U, J) domain. Inner windows

instead use Lemma 4, (3.53), which constructs the old box background and its OWN current from the inner datum. In CMP 116 (1988), (1.21) controls the response in all required regularity norms, (1.22)–(1.23) give the amplified field-direction disc and its Cauchy derivative, and Lemma 1, (1.33)–(1.36), states the localized function on the full independent-(U, J) space times the original physical fluctuation box. Those are the actual substitutions used here. A generic off-shell pair is never silently declared to be a critical pair.

Let $\mathcal{A}_N$ be the commutative finite source algebra with $z_i^2 = 0$ and norm

$$\Big\| \sum_{I \subset \{1,\dots,N\}} a_I z_I \Big\|_{\boldsymbol{r}} = \sum_I \|a_I\| \prod_{i \in I} r_i.$$

For physical profiles f_i, set

$$w(p) = \sum_i z_i f_i(p), \qquad t = \sum_i r_i \|f_i\|_\infty, \qquad \sigma = \sum_i r_i [f_i]_{C^\alpha}. \tag{14.2}$$

Assume $t + \sigma \le r_*$, where $r_* > 0$ is fixed, smaller than $r_{\mathrm{hom}}/8$, and chosen within the fixed local-potential allowance. Constants below are independent of N, the fine cutoff and the volume. Take the final physical spacing to be at most one.

Scalar homogeneous functions are evaluated in $\mathcal{A}_N$ by their absolutely convergent Taylor series. If F is bounded on $|c| < r_{\mathrm{hom}}$, then $\|F(w(y))\|_{\boldsymbol{r}} \le \sup |F|/(1 - t/r_{\mathrm{hom}})$. The same argument on a smaller fixed disc bounds its c-derivative. Nilpotence makes every such series finite, while the geometric bound is independent of N. This supplies the dimension-free scalar composition used below.

For a vacuum-subtracted gauge-invariant defect birth family D_j, define its norm a_j by its ORIGINAL complete local (U, J) domain:

$$a_j = R_0^{-2} \sup_y \sum_{X \ni y} e^{\kappa d_j(X)} \sup_{\mathcal{U}_j(X)} \|D_j(X; U, J)\|_{\boldsymbol{r}}. \tag{14.3}$$

The near-zero R_0 ball is used for Taylor readout; it does not replace the full member domain in this definition. Fixed pointing factors or a supremum norm with an equivalent entropy allowance give the same proof. Each defect is kept with its birth scale and its original source anchors. Its full curvature-quadratic tensor is retained, either by the extraction of Section 11 or by a gauge-invariant path completion of its localized quadratic kernel. All 21 symmetric $SU(2)$ curvature types are allowed.

27.15.2 14.2 An unsubtracted old member has a uniform bound, not an age gain

The following simpler estimate is needed for the spatial defects. It does not require Euclidean invariance or the subtraction of a marginal coefficient.

Old-member lemma. For a gauge-invariant birth family with zero flat vacuum value and norm (14.3), its complete readout in one present-scale window is bounded by

$$C R_0^2 a_j. \tag{14.4}$$

Its fluctuation difference on the actual common field-direction interpolation is bounded by

$$C R_0 \delta_k a_j. \tag{14.5}$$

Both bounds hold after the ordinary rooted localization and include the coefficient norm.

Proof. Semisimplicity gives zero first field/current jet. On a contractible birth patch, tree gauge and the actual old-level inner-window regularity give link coordinates of order $R_0 r^2$ and its own raw current of order at most $R_0 r^2$, where $r = \ell_j/\ell_k$, with polynomial domain-diameter factors. The stronger own-current bound in (11.2) is available there but is not needed for this lemma. For the independent outer current, use the original reduced-current extension and the separate outer/removed-window estimates specified above. Taylor's formula therefore bounds an individual inner member by $CR_0^2 a_j r^4 e^{-\kappa' d_j(X)}$. Its $O(r^{-4})$ birth roots per present cell exactly cancel that engineering factor. No strictly positive age power remains, and none is claimed. Polynomial factors are paid from the spare locality exponent. Long, wrapping and removed domains contribute at most $CR_0^2 a_j r^{-4} e^{-c/r}$, also uniformly bounded; a small carrier around a genuine wild hole is excluded from this clean lemma.

Use the very same bound throughout the genuine window field-direction disc $|\nu| < cR_0/\delta_k$, where the complete fields and their own currents remain of size CR_0. Cauchy's formula for ∂_ν, followed by the window integral, gives (14.5). For outer and removed terms use their actual response decay and common core margin before summing. The argument is coefficientwise and uses the actual interpolation, not only its endpoints. □

A sum of defects consequently enters the new fluctuation potential with norm

$$CR_0\delta_k M_k, \qquad M_k = \sum_{j\le k} a_j. \tag{14.6}$$

Their zero-fluctuation values are carried forward unchanged. They are not generated anew at each step. This exact separation is the reason that the next estimate contains a small difference factor, not a repeated constant multiplying every old birth.

27.15.3 14.3 Rooted freezing creates a geometric birth factor

At each birth anchor y, evaluate the whole homogeneous pointed member at the single scalar $c = w(y)$, and subtract that member at $c = 0$. Its own counterterm difference is evaluated with those same two scalars. Denote the resulting locally frozen SOURCE-EXCESS reference action by $\mathcal{H}_k[w]$. Explicitly, it consists of the current pointed action with coefficient profile $Y_k(w(y))$, plus the sum of homogeneous remainder differences $\mathcal{R}_{j,y}^{w(y)} - \mathcal{R}_{j,y}^{0}$. Every remainder includes its own scalar counterterm before this difference and is read with its assigned birth anchor and ordinary source-free background. Thus $\mathcal{H}_k[0] = 0$.

When a prospective output carrier Y is rooted at y, freeze all the scalar values occurring inside it to $w(y)$. Any anchor z retained in that carrier obeys

$$\|w(z) - w(y)\|_{\boldsymbol{r}} \le C\sigma \ell_{k+1}^{\alpha}(1 + \operatorname{diam}_{k+1} Y)^{\alpha}. \tag{14.7}$$

All old source anchors remain in the carrier: this is part of the exact wall/union filing. Remote old anchors therefore cost a correspondingly long carrier; they are not dropped when their field argument is localized.

For the old homogeneous remainders, differentiation in c commutes with their own extraction. Apply the ordinary age estimate to this derivative, then sum their ages. For the current leading coefficient use the bounded derivative of Y_k. When pointed action weights are moved between scales, their source-dependent counterterm differences use the summable increment bound in (14.1). The source-zero counterterms cancel before this comparison; an $O(1)$ coefficient per old scale is not multiplied by the source oscillation and summed.

These estimates show that the difference of the locally frozen input and the input with one constant $w(y)$ has total normalized local size at most

$$C\sigma\ell_{k+1}^{\alpha}. \tag{14.8}$$

A direct proof inserts an interpolation in every scalar argument from $w(y)$ to its own anchored value. Differentiate once along it. The complete pointed old remainder is differentiated before its age bound, and every resulting derivative carries (14.7). The uniform scalar disc bounds those derivatives. Rooted wall localization and the fixed generic formation allowance then sum the local contributions. The polynomial in (14.7) is absorbed by a fixed spare decay. No independent fine-plaquette coefficient domain, and no perturbed fine minimizer, is used.

The zero-fluctuation classical discrepancy is included in (14.8) as an analytic birth member. It need not be $O(g)$ on the fixed field core. Its quadratic Gaussian part may instead be retained by the exact resolvent of Section 13. That part has one oscillation at first order and two endpoint oscillations at every order at least two; specifically its higher coefficients have the $\ell_{k+1}^{2\alpha}x_{k+1}^{1-m}$ bound (13.13). This stronger estimate is compatible with, but not necessary for, the one-oscillation ledger (14.8).

27.15.4 14.4 Exact definition of the defects and the closing induction

At finite volume let $\Phi_k[w]$ be the SOURCE INCREMENT of the analytic effective logarithm: minus the sourced logarithm plus its source-zero logarithm, with exactly the same frozen transfer and with common sharp external tests kept outside that analytic factor. Keep the corresponding source increment of its local field-independent logarithm in $v_k[w]$. Thus $\Phi_k[0] = v_k[0] = 0$. Subtract the source-excess family $\mathcal{H}_k[w]$ defined above. The remainder is represented as

$$\Phi_k[w] - v_k[w] = \mathcal{H}_k[w] + \sum_{j\le k} \mathcal{T}_{j\to k} D_j[w], \tag{14.9}$$

where $\mathcal{T}_{j\to k}$ is the ACTUAL source-free old-background readout, including its fixed node convention. Equation (14.9) defines the bookkeeping, with $D_j[0] = 0$. It is an equality of finite source algebras and finite lattice functions. The retained quadratic source jet is a record of these same terms; when used in a completed Gaussian it is subtracted from the potential in the same exact step, so it is never inserted twice.

At the initial scale the primitive Wilson source has this form exactly, or with the harmless cell-sampling defect $a_0 \le C\sigma\ell_0^{\alpha}$. In the next transfer, split every old defect into its value at the new background and its fluctuation difference. The first part is the old term in (14.9); the second part is (14.6). Subtract the locally frozen homogeneous birth family at the new anchor. Define the resulting connected local logarithm, together with the zero-fluctuation freezing discrepancy, to be D_{k+1}. This is an exact definition by the convergent Mayer/connected expansion; it is not an approximation of the transfer.

The bounded first difference of the generic formation map, (14.6), and (14.8) prove

$$\boxed{a_{k+1} \le C_1\sigma\ell_{k+1}^{\alpha} + \epsilon_k M_k, \qquad \epsilon_k \le C_2\delta_k/R_0.} \tag{14.10}$$

The finite source algebra introduces no additional dimension factor: the input source load is (14.2), multiplication is submultiplicative, and the cluster construction already has its ordinary scalar auxiliary parameters. Gauge invariance and vacuum subtraction give zero first field jet to the new defect. Its quadratic tensor is recorded rather than discarded.

In the original scheme $\delta_k = \epsilon_1$ is the fixed PHYSICAL fluctuation cutoff. Select it, once and before the weak run, so that

$$\epsilon_* := C_2\epsilon_1/R_0 < \tfrac{1}{2}(L^{\alpha} - 1). \tag{14.11}$$

The ordinary parameter hierarchy allows this smaller positive cutoff. This proves the induction without replacing the scheme by a running logarithmic cutoff. The inserted-and-removed inner-cutoff estimate can improve ϵ_k to a quantity tending to zero, but is not needed here.

Since $M_{k+1} = M_k + a_{k+1}$, (14.10) implies

$$M_{k+1} \le (1+\epsilon_*)M_k + C_1\sigma\ell_{k+1}^{\alpha}.$$

As $\ell_{k+1} = L\ell_k$, elementary induction gives

$$\boxed{M_k \le B\sigma\ell_k^{\alpha}, \qquad B \ge \frac{C_1}{1-(1+\epsilon_*)L^{-\alpha}},} \tag{14.12}$$

after increasing B to cover the initial sampling term. Shrink r_* once so that Br_* fits the fixed allowance. This is independent of the ultraviolet length.

This closes two distinct origin ledgers. The homogeneous, correctly subtracted old members use the ordinary summable age kernel. The spatial defects retain their marginal tensor and use their geometric PHYSICAL BIRTH factor, with the small old-member difference in (14.10). They are not asserted to be ordinary irrelevant operators. Arbitrary rough spatial boxes have no bound like (14.7), and are not covered by this conclusion.

27.15.5 14.5 All-order consequence and exact scope

On any fixed final clean cell, the locally frozen homogeneous part has a uniformly bounded source norm by the ordinary age sum and (14.1). The defect part is bounded by (14.4), (14.12). Hence the field-dependent sourced effective action obeys

$$\|\Phi_k[w] - v_k[w]\|_{\mathrm{loc}} \le CR_0^2(t+\sigma) \tag{14.13}$$

for $\ell_k \le 1$, on the admitted final core. Choosing r_i proportional to $r_*/(N\|f_i\|_{C^\alpha})$ extracts every fixed mixed coefficient and gives

$$\|D^N\{\Phi_k - v_k\}(0)[f_1,\dots,f_N]\|_{\mathrm{loc}} \le C^N N! r_*^{1-N} R_0^2 \prod_i \|f_i\|_{C^\alpha}. \tag{14.14}$$

Repeated directions follow by polarization. The constants are uniform in the fine cutoff and volume. The separate exact Gaussian coefficients retain (13.13), which is stronger than this generic analytic estimate.

This theorem concerns the CLEAN source transport on the original complete member domains. It does not bound arbitrary vacuum contact counterterms by (14.13): $v_k[w]$ was retained exactly. Mixed coefficients for separated supports require the connected source-anchor/bridge estimate before their volume count, rather than the one-oscillation factor alone. It also does not erase the final retained-field cumulants.

For literal R matching use local atoms or a convergent connected local quadratic-kernel expansion, not one global Schur operator justified only in energy norm. The exact Gaussian state can be recorded as the sum of the quadratic jets in (14.9), with its local gauge-invariant completion; it is not an additional factor multiplying the same density a second time. In a live island, keep the original local source atoms and any uneliminated auxiliary Gaussian coordinates in the unnormalized operator. The primary source-free layered map must place every completed reading in the ORIGINAL member core of (14.3), with its spare radius. Once that actual inclusion is used, all the preceding born functions extend coefficientwise because their source coefficients never change the field maps. This does not certify an independently modified prepared domain or a source-dependent R run.

27.15.6 14.6 Ordinary complex polydiscs and local logarithms

The same result holds on genuine ordinary complex source domains. Fix any finite collection of profiles and radii with

$$\sum_i r_i \|f_i\|_{C^\alpha} < r_*, \qquad \mathcal{D}_{\boldsymbol{r}} = \{z : |z_i| < r_i\}. \tag{14.15}$$

Replace $\mathcal{A}_N$ throughout the proof by the Banach algebra of bounded holomorphic functions on $\mathcal{D}_{\boldsymbol{r}}$, with the supremum norm. On smaller closed polydiscs this algebra has exactly the submultiplicativity used in every estimate. The bounds in (14.2) then hold with the same t, σ. Scalar composition $E^c|_{c=w(y)}$ is ordinary holomorphic composition inside the genuine homogeneous disc.

At the initial step the finite-cutoff density is the actual entire factor $\exp[-A(w,U)]$. Assume the old local functions have their proved holomorphic domains and bounds. The frozen field maps and fixed tests introduce no source dependence. The generic local Gaussian/Mayer series converges absolutely and locally uniformly on (14.15), because (14.10)–(14.12) keep its full complex potential in its fixed allowance. The connected series defines the branch of the LOCAL logarithm agreeing with the source-zero one. Its scalar part is retained in $v_{k+1}[w]$; its field-dependent part and locally frozen subtraction define the ordinary holomorphic D_{k+1}. This proves the same induction directly for the original integrals. It is not a reconstruction of arbitrary finite jets without a convergence argument.

The homogeneous old-age sum is dominated by its fixed summable age kernel; the defect sum is dominated by the geometric solution of (14.10). Both dominate uniformly on every smaller closed polydisc, so all their compatible age sums converge normally. For a finite run the constants are independent of its length. This does not by itself identify coefficients from DIFFERENT cutoff trajectories or prove their cross-cutoff convergence; it supplies the common analytic bounds required for that later compactness argument.

The finite nilpotent formulas are the Taylor coefficients of these ordinary holomorphic local functions. Conversely, their dimension-free coefficient estimates follow by ordinary polydisc Cauchy inequalities, giving (14.14) with fixed constants after one decrease of r_*.

The conclusion is a uniform source domain for the local effective logarithm and its field-dependent density on the clean admitted cores. The field-independent $v_k[w]$ can be extensive and can contain divergent local contact coefficients as the cutoff is removed. It has not been bounded by a cutoff-independent scalar constant. Nor is a zero-free domain for the final partition function, after the remaining field/history integration, inferred from these local logarithms.

27.15.7 14.7 Uniform product discs for smooth patch amplitudes

The sum of the individual profile norms in (14.15) is only a convenient sufficient condition. The proof uses the sharper absolute coefficient-profile loads

$$\begin{aligned} T_{\boldsymbol{r}} &= \sup_x \sum_i r_i |f_i(x)|, \\ S_{\boldsymbol{r}} &= \sup_{x \neq y} \frac{\sum_i r_i |f_i(x) - f_i(y)|}{\operatorname{dist}(x,y)^\alpha}. \end{aligned} \tag{14.16}$$

Thus $T_{\boldsymbol{r}} + S_{\boldsymbol{r}} < r_*$ suffices, with t, σ in the whole proof replaced by $T_{\boldsymbol{r}}, S_{\boldsymbol{r}}$. Indeed $\|w(x)\|_{\mathcal{A}_N} \le T_{\boldsymbol{r}}$ and $\|w(x) - w(y)\|_{\mathcal{A}_N} \le S_{\boldsymbol{r}} \operatorname{dist}(x,y)^\alpha$. These are the only source-profile inequalities used in scalar composition, local freezing, or formation. The ordinary polydisc statement follows with the same loads.

Section 27.15 / A clean smooth-source induction with two origin ledgers

For example, let ψ_v be smooth patch functions of fixed physical size a, with

$$\sup_x \sum_v |\psi_v(x)| \le C_0, \qquad \sup_x \sum_v |\nabla\psi_v(x)| \le C_1/a. \tag{14.17}$$

The sum may contain arbitrarily many patches. For $w = \sum_v z_v\psi_v$, $|z_v| < r$, the fundamental theorem of calculus and the first bound give

$$\sum_v r|\psi_v(x) - \psi_v(y)| \le r\min\{2C_0, (C_1/a)|x-y|\}.$$

Consequently

$$T_{\boldsymbol{r}} + S_{\boldsymbol{r}} \le Cr(1 + a^{-\alpha}). \tag{14.18}$$

A product disc of radius $cr_*/(1 + a^{-\alpha})$ is therefore admissible independently of the NUMBER of patches or the volume. At a fixed physical patch scale, this is a genuine spatial product domain. If patches shrink to the bare mesh, its radius shrinks like a^α; no rough fine-coefficient regularity conclusion has been reintroduced.

Reflected copies supported away from patch seams satisfy the same bounded-overlap and summed-gradient conditions. Thus this version provides ordinary uniform patch-amplitude discs for localization or chessboard arguments while retaining a smooth physical source class.

Section 27.16 / Vacuum bridges and the physical partition estimate

A field-dependent source estimate is not a vacuum-energy estimate. This section keeps every regular scalar vacuum birth, proves its separated-source bound BEFORE summing its physical roots, and then combines the clean theorem with the original-map R interface.

27.16.1 15.1 The universal local scalar family

At birth scale j, let $\nu_j(X;w)$ be the SOURCE EXCESS of the scalar logarithm assigned to a connected current-scale carrier X. Its sign is the one with which it enters the logarithm of the partition function. Include:

- the current-step Gaussian count and normalization constants;
- the flat value of every regular clean E birth;
- the flat value of every regular R birth.

The last item is required. A tiny R norm such as $e^{-c(\log x_j)^p}$ does not pay the physical root count ℓ_j^{-4} for its vacuum value. Its field-dependent quadratic readout estimate applies only after that value has been removed.

The universal R family and its flat subtraction are part of the original induction: III (2.30), (2.32)–(2.33), and its closure in R-II. The ordinary complex-source extension is given in Section 15 of the companion marked-bridges note. Internal healed records are summed within a carrier; unsummed live components remain external or are included whole in that carrier. Thus $\nu_j(X;w)$ is an actual local function of the source restricted to the carrier (with its fixed collar), not a coefficient that still reads an unsummed exterior history.

On any ordinary source polydisc of Sections 14.6–14.7, after using a fixed amount of source-radius room, the following rooted norm holds:

$$\sup_{j,Q} \sum_{X \ni Q} e^{\kappa_0 d_j(X)} \sup_{\mathcal{D}_{\boldsymbol{r}}} |\nu_j(X;w)| \le C(T_{\boldsymbol{r}} + S_{\boldsymbol{r}}). \tag{15.1}$$

Here and below the finitely many scalar birth types are included in the sum.

To see that (15.1) is an actual formation bound, consider one current step. Its Gaussian integral has a fixed number N_L of current fluctuation coordinates per current cell. The source-zero constrained precision has a uniform hidden gap. The source perturbation has the localized operator/atom bounds of Sections 8 and 13, and its normalized size is $O(r_*/x_j)$. The normally convergent trace expansion of the determinant therefore gives connected current-cell cycles. For a pinned trace the factor is at most N_L, not the number of fine plaquettes inside a deep harmonic lift. This is a CURRENT-STEP trace; it is not a trace of the full fine operator estimated by its operator norm. The actual Gaussian telescoping identity allows precisely this birth-by-birth representation.

The non-Gaussian part is the ordinary generic local Gaussian/Mayer map before its vacuum subtraction. It has the same local bound on a fixed larger source allowance. Its source excess vanishes at zero; one auxiliary scalar source parameter and Schwarz's lemma give a linear factor $T + S$, with constants fixed by that larger allowance. The old homogeneous terms are age-summed before applying the map, and old defects use (14.6), (14.12). A regular R birth has the stronger marked bound from the companion theorem before its flat value is removed, so the same source-excess argument applies to it. All these constructions retain exact rooted source carriers and spare locality. This proves (15.1).

For finite cutoff and torus define

$$\mathcal{E}_K[w] = \sum_{j \le K} \sum_X \nu_j(X; w). \tag{15.2}$$

It is a finite-cutoff holomorphic scalar functional. It is generally extensive and may diverge as the cutoff is removed. In the sign convention of (14.9), its clean part equals $-v_K[w]$. No absolute ultraviolet sum of (15.2) is asserted.

27.16.2 15.2 Separated source derivatives are summable

Let $f_1, \dots, f_m$ be compactly supported C^α profiles, $m \ge 2$, and suppose some two of their supports are separated by a positive distance d. Write $B_i = \|f_i\|_{C^\alpha}$. Choose ordinary polydisc radii $r_i = r_*/(2mB_i)$, omitting a zero direction. Cauchy's inequalities applied to (15.1) imply

$$\sup_Q \sum_{X \ni Q} e^{\kappa_0 d_j(X)} |\partial_{z_1} \cdots \partial_{z_m} \nu_j(X; \textstyle\sum z_i f_i)|_{z=0} \le C^m m! r_*^{1-m} \prod_i B_i. \tag{15.3}$$

Exact source support is now decisive. The derivative for a carrier vanishes if any f_i vanishes on that carrier and its fixed collar. A carrier contributing to the displayed mixed derivative must therefore meet both separated supports. Its tree length is at least $cd/\ell_j - C$. Spend half the spare exponential on this bridge and use the remaining half for the pinned sum.

If one selected support lies in a fixed physical ball of radius R, the number of its intersecting birth cells is at most $C_R \ell_j^{-4}$ for $\ell_j \le 1$. Therefore

$$\sum_X |\partial_1 \cdots \partial_m \nu_j(X; 0)| \le C_R C^m m! r_*^{1-m} \ell_j^{-4} e^{-cd/\ell_j} \prod_i B_i. \tag{15.4}$$

The geometric scale sum satisfies

$$\sum_{\ell_j \le 1} \ell_j^{-4} e^{-cd/\ell_j} \le C_{c,L} d^{-4}. \tag{15.5}$$

For example put $u_j = d/\ell_j$; the resulting sum of $u_j^4 e^{-cu_j}$ over a geometric progression is uniformly bounded. Thus

$$|D^m \mathcal{E}_K(0)[f_1, \dots, f_m]| \le C_R C^m m! r_*^{1-m} d^{-4} \prod_i \|f_i\|_{C^\alpha}, \tag{15.6}$$

uniformly in the fine cutoff and in tori containing the supports with their required separation. A full marked tree connecting all supports can be retained instead of a single pair distance.

The proof first selects CONNECTED carriers bridging separated supports and only then counts their roots. The factor ℓ_j^{α} from smooth-source freezing, by itself, would not pay ℓ_j^{-4}. One-point vacuum derivatives and general diagonal contact coefficients remain in the explicit local functional $\mathcal{E}_K$.

27.16.3 15.3 The exact common-vacuum factor and D1-type bounds

Let V_K be the source-zero sum of all the scalar births just listed, including regular R vacua. In a particular unfolded history h, only a subfamily I_h of them appears outside the stored islands. The exact identity

$$e^{\sum_{b\in I_h}\Delta v_b[w]} = e^{\mathcal{E}_K[w]}e^{-\sum_{b\notin I_h}\Delta v_b[w]} \tag{15.7}$$

places every missing term in its proper stored operator with its ORIGINAL birth label. Here $b=(j,X,\text{type})$, with the fixed ownership convention, and Δv_b is its source excess. This is the vacuum version of the literal action completion in R-II (1.70).

By (15.1), each missing term can be assigned once to the first actual birth-scale record its carrier meets, giving

$$\sum_{b\notin I_h}\sup|\Delta v_b| \le Cr_*\sum_j N_j(h). \tag{15.8}$$

The $N_j(h)$ count actual stored birth records, not the latest-scale swallowed volume. The original cold-action bank already pays a fixed charge per such record, with its larger logarithmic reserve. Section 15 of the marked-bridges note proves this transfer together with the exact failed/merge/heal identities and the ordinary-polydisc extension. The source-free part of the vacuum is already in V_K, and is not charged a second time.

Consequently the exact physical finite-cutoff source functional has the representation

$$Z_K[w] = e^{V_K+\mathcal{E}_K[w]}\,\mathcal{I}_K[w], \tag{15.9}$$

where $\mathcal{I}_K[w]$ still integrates the FINAL retained field, all surviving live operators, and all remaining history data. It is not replaced by its value at a deterministic reference field.

The original terminal stability estimates and the marked absolute-history majorant give

$$\boxed{|Z_K[w]| \le \exp\{V_K + \operatorname{Re}\mathcal{E}_K[w] + C_+N_K\}, \qquad Z_K[0] \ge \exp\{V_K - C_-N_K\}.} \tag{15.10}$$

Here N_K is the number of terminal physical cells; $C_\pm$ depend on the fixed weak endpoint and the source/reserve budgets, not on the fine cutoff or number of earlier scales. The source domain is the ordinary patch/profile polydisc proved above.

For the upper bound, retain a positive fraction of the real primitive Wilson action after the absolute source and quadratic-completion estimates, pay (15.8) from the recorded reserve, and apply the same original component/cluster majorant. All regular field-dependent terms have their uniform per-terminal-cell bounds, so the remaining integral is at most $e^{C_+N_K}$. No individual sourced island mass is inverted.

For the lower bound use the positive COMPLETED branch with no surviving terminal large-field island. It includes the summed healed earlier records; it is not restricted to histories with no large-field event at any fine scale. This distinction is necessary because V_K includes regular R vacuum constants. On a fixed positive-Haar-volume terminal regular core the leading action and the vacuum-subtracted

regular lists are bounded by CN_K. The positive branch therefore contributes at least $e^{V_K - C_- N_K}$. This is the source-zero lower estimate only.

Equations (15.2), (15.6), (15.9)–(15.10) give the requested literal D1-type interface for the ORIGINAL construction. They do not claim that $\mathcal{E}_K$ is a uniformly bounded scalar, or that the full partition function has a volume-independent zero-free complex disc.

27.16.4 15.4 The retained-field cumulants remain explicit

At every fixed cutoff, $\mathcal{I}_K[0] > 0$. The ordinary analytic germ at zero therefore satisfies

$$\log \frac{Z_K[w]}{Z_K[0]} = \mathcal{E}_K[w] + \log \frac{\mathcal{I}_K[w]}{\mathcal{I}_K[0]}. \tag{15.11}$$

The second term is not discarded. It contains fluctuations of the final retained field and all conditional/history mixing.

One can differentiate it without normalizing any individual live state. If $I = \{1, \dots, m\}$, then

$$D_I \log \mathcal{I}_K(0) = \sum_{\pi \vdash I} (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{B \in \pi} \frac{D_B \mathcal{I}_K(0)}{\mathcal{I}_K(0)}. \tag{15.12}$$

Every numerator is the integral of the corresponding marked, UNNORMALIZED final operators over the displayed retained field and histories. The only denominator is the positive whole source-zero functional. If a conditional description is used, (15.12) is the usual total-cumulant formula, including the covariance of conditional means.

For a fixed terminal torus, (15.10) and Cauchy give finite source-derivative bounds with the factor $e^{(C_+ + C_-)N_K}$. Removing that extensive factor for local physical observables uses reflection positivity/chessboard or a separate terminal clustering argument. The partition comparison alone does not make that step automatic. The separated vacuum derivative in (15.6) is already uniform and is available for that argument.

The fixed-radius Cauchy bound (14.14) is too weak for a sharp two-point comparison. It does not remember that a connected second insertion must use two fluctuation factors. The following finite-jet refinement retains that fact. It concerns the PURE CLEAN fixed-core family of I–II; the distinction from shrinking-core tagged R-members is made explicitly in Section 14.1. All source-free background maps, sharp tests and old parameters are frozen as before.

27.18.1 17.1 Two marked fluctuations in one actual clean integral

Use the two-label algebra $z_1^2 = z_2^2 = 0$. Write the incoming source increment, after its scalar vacuum has been separated, as

$$z_1 O_1 + z_2 O_2 + z_1 z_2 O_{12}. \tag{17.1}$$

Every O_I is gauge invariant and has zero constant and linear field/current jets on its original pair domain. Constants of lower source degree stay in the scalar vacuum ledger. Let μ_g be the normalized current CLEAN fluctuation law at source zero, including the actual unmarked old potentials, and let $\Psi_g(U, J; B)$ denote its actual analytic field/current substitution. Up to the fixed sign convention for the effective action, the exact coefficients are

$$W_i = \langle O_i \circ \Psi_g \rangle_{\mu_g}, \qquad W_{12} = \langle O_{12} \circ \Psi_g \rangle_{\mu_g} - \mathrm{Cov}_{\mu_g}(O_1 \circ \Psi_g, O_2 \circ \Psi_g). \tag{17.2}$$

These formulas include the quotient terms from the source-zero normalization. They are not unnormalized moment identities. Subtracting the background value in each covariance argument is EXACT.

Section 27.18 / A two-label refinement with the actual powers of the coupling

On a fixed smaller pair core of radius R_0, Taylor's formula for the ORIGINAL substitution gives

$$\Delta O_i := O_i \circ \Psi_g - O_i \circ \Psi_0 = gL_i(U,J)[B] + g^2 Q_i(U,J;B), \tag{17.3}$$

where, in the rooted local norms with spare carrier decay,

$$|L_i(U,J)[B]| \le Ca_i R_0 |B|_{\mathrm{loc}}, \qquad |Q_i(U,J;B)| \le Ca_i |B|_{\mathrm{loc}}^2, \qquad L_i(1,0) = 0, \tag{17.4}$$

provided $\|O_i\| \le R_0^2 a_i$ on the fixed larger core. The bounds hold for $|gB| \le \varepsilon_1$, with constants depending on the fixed core margins. The first variation of the field/current substitution is uniformly local, and its second variation has the displayed quadratic bound. The semisimple color action removes the entire first pair jet at $(1,0)$, including the independent-current direction. No assertion that an arbitrary outer current is a constrained current is used.

Here and below $|B|_{\mathrm{loc}}$ may denote a fixed exponentially weighted sum over the fluctuation blocks meeting a carrier. All its fixed polynomial moments are uniformly bounded in the rooted connected expansion. To see this without an integral-of-a-supremum shortcut, leave a fixed positive quadratic Gaussian reserve at each distinguished vertex. For a polynomial of degree r,

$$|B|^r \le C_{r,\gamma} e^{\gamma |B|^2} \tag{17.5}$$

uses a sufficiently small fixed part of that reserve. The Gaussian conditioning and the original small quadratic-growth tilt then integrate it. A connected term with two such vertices has a tree connecting both vertices and its output carrier; summing the remaining unmarked trees uses the unchanged convergent cluster allowance. Polynomial carrier factors are absorbed by a fixed loss of locality. This proves a rooted connected polynomial-moment estimate, uniform in volume and ultraviolet depth, for the actual clean local logarithm. It is stronger than bounding a total normalized integral by a volume-dependent constant.

Apply this estimate to the product in (17.3). On the fixed core,

$$|\operatorname{Cov}(\Delta O_1, \Delta O_2)|_{\mathrm{loc}} \le Ca_1 a_2 (g^2 R_0^2 + g^4). \tag{17.6}$$

At the flat pair, both linear terms vanish, and hence

$$|\operatorname{Cov}(\Delta O_1, \Delta O_2)(1,0)|_{\mathrm{loc}} \le Cg^4 a_1 a_2. \tag{17.7}$$

The covariance is gauge invariant. After its flat value is removed it again has zero first pair jet. Take the weak ceiling small enough that $g \le cR_0$. Equations (17.6)–(17.7), followed by radial field Cauchy on the fixed larger core, therefore give

$$\begin{aligned} \|\text{new mixed field member}\|_{R_0} &\le Cg^2 R_0^2 a_1 a_2, \\ \|\text{same member on a physical core } \alpha\| &\le Cg^2 \alpha^2 a_1 a_2, \\ |\text{new mixed vacuum}| &\le Cg^4 a_1 a_2. \end{aligned} \tag{17.8}$$

The coefficient of a connected carrier carries, in addition, its tree decay to BOTH original source anchors. This follows from the two distinguished vertices just used; no absolute fine-coordinate row estimate is inferred.

For an OLD mixed member born at j, assume $\|O_{12,j}\| \le g_j^2 R_0^2 a_{12,j}$. Its zero-fluctuation value is passed unchanged. At the current flat pair its fluctuation difference starts quadratically in $g_s B$, so its newly generated scalar is bounded by

$$Cg_s^2 g_j^2 a_{12,j} \le Cg_s^4 a_{12,j}, \qquad j \le s. \tag{17.9}$$

The last inequality uses $x_j \ge x_s/2$, not pointwise monotonicity. The ordinary old-member field interpolation gives the same small difference factor as (14.5), multiplied by g_j^2. Thus a second-label member retains its correct running power under old transport.

27.18.2 17.2 Closing the spatial weighted birth ledger

Write $t_i = \|f_i\|_\infty$, $\sigma_i = [f_i]_{C^\alpha}$, and use the rescaled physical units in which the stopping scale is at most one. The first-source spatial defects have the norm and bound of Section 14:

$$\|D_{i,j}\| \le R_0^2 a_{i,j}, \qquad M_{i,s} := \sum_{j\le s} a_{i,j} \le C\sigma_i \ell_s^\alpha. \tag{17.10}$$

For the mixed source defects define the WEIGHTED norm by

$$\|D_{12,j}\| \le g_j^2 R_0^2 a_{12,j}, \qquad M_{12,s} = \sum_{j\le s} a_{12,j}. \tag{17.11}$$

The homogeneous mixed source reference is retained with its own counterterm. Its uniform bare disc at the current weak endpoint gives $Y_s''(0) = O(g_s^3 \operatorname{polylog} x_s)$; its constant-source field births have at least the same admissible g_s^2 weight after increasing the weak floor. The exact spatial classical Gaussian precision of Section 13 is retained without approximation; its mixed coefficient is already $O(g_s^2)$ with two endpoint oscillations.

Now subtract the fully locally frozen homogeneous coefficient in (17.2), before estimating. Every remaining first-source discrepancy carries either an anchored profile difference or a prior first-source defect. The bounded current first-source reference has size $CR_0^2 t_i$. Bilinearity of the covariance and (17.8) therefore yield, after the SAME rooted freezing used in (14.7),

$$\begin{aligned} a_{12,s+1} \le{}& C\{t_1\sigma_2 + t_2\sigma_1 + \sigma_1\sigma_2\}\ell_{s+1}^\alpha \\ &+ C\epsilon_* \sum_{j\le s} (g_j^2/g_{s+1}^2) a_{12,j}. \end{aligned} \tag{17.12}$$

In detail, the covariance difference is the sum of a homogeneous first coefficient paired with one discrepancy and the product of two discrepancies. The first terms cost $t_1 M_{2,s} + t_2 M_{1,s}$; the last costs $M_{1,s}M_{2,s}$. Insert (17.10), use $\ell_s^{2\alpha} \le \ell_s^\alpha$, and absorb the source-diameter polynomials into spare decay. The zero-fluctuation discrepancy of the homogeneous mixed reference uses its OWN paired counterterm and the same scalar derivative bounds; it has the displayed g_s^2 weight. Old mixed fluctuations give the second line of (17.12). Existing scalar vacuum terms cancel from both covariance arguments, so no old vacuum constant enters this recurrence.

The running tube bounds the ratio in (17.12) by a fixed constant. Reduce the original fixed physical fluctuation cutoff once so that this constant times ϵ_* is less than $(L^\alpha - 1)/2$. Exactly the induction of (14.12) proves

$$\boxed{M_{12,s} \le C\{t_1\sigma_2 + t_2\sigma_1 + \sigma_1\sigma_2\}\ell_s^\alpha.} \tag{17.13}$$

Thus the second source degree has field norm $g_s^2 R_0^2$, and on the canonical physical core $\alpha_s = g_s P_{\mathrm{bg}}(x_s)$ it has norm $g_s^4 P_{\mathrm{bg}}(x_s)^2$. Its new scalar vacuum coefficients have weight g_s^4 by (17.7),(17.9). The corresponding first-degree weights are R_0^2, $g_s^2 P_{\mathrm{bg}}^2$, and g_s^2, respectively. No growing ordinary source disc is used to obtain these powers: they follow from the exact two-label quotient formula and actual fluctuation factors.

For a prefix stopped at physical scale a', the same argument uses $t_i + (a')^\alpha \sigma_i$. Hence a smooth source varying on scale $d \gg a'$ has a uniform local first-source load $O(t_i)$ at that stopping scale. There is no factor counting all finer bare roots. If two such sources are separated by d, every mixed birth below a' has a carrier bridging that separation; the retained mixed local family at a' is bounded by a fixed polynomial in d/a' times $e^{-cd/a'}$, with its $g_{a'}^2 R_0^2$ field weight. Connected carriers are selected before the old-root volume count.

This is a two-label CLEAN transport result. It is not a comparison of the interacting physical density with a Gaussian density. In particular it has not discarded any terminal first-source covariance or terminal mixed-source expectation.

27.18.3 17.3 A fresh Gaussian-subtracted gain at a flat background

There is a further genuine weak-coupling gain in a NEW clean covariance birth. Freeze the entire old clean state and its source insertions while varying only the current scalar g. At the flat pair, Taylor's integral remainder gives

$$O_i \circ \Psi_g - O_i \circ \Psi_0 = g^2\{Q_i(B) + gC_i(B) + \cdots\}. \tag{17.14}$$

The quotient by g^2 has uniformly bounded first two real current-g derivatives in the rooted polynomial-Gaussian norm. This follows directly from its Taylor integral: each derivative adds a fixed power of B, paid by (17.5), while all field Cauchy radii are the fixed clean radii. For the unmarked interaction, the actual cubic Taylor remainder has $\partial_g V = O(|B|^3)$, $\partial_g^2 V = O(|B|^4)$, and the old activity/Jacobian differences have the corresponding bounded polynomial derivatives. The original independent-current extension has cancelled the dangerous saddle linear term before these formulas are extended off shell. The source-free Gaussian quadratic form is independent of this current g.

Differentiating a normalized expectation produces connected covariance/cumulant expressions. The same rooted polynomial bound therefore controls the first two derivatives of

$$R(g) = \mathrm{Cov}_{\mu_g}\left(g^{-2}\Delta O_1, g^{-2}\Delta O_2\right).$$

The substitution and all physical tests are unchanged under $(g, B) \mapsto (-g, -B)$, so R is even. For the original fixed PHYSICAL cutoff, derivatives of its rescaled boundary are polynomial powers of g^{-1} times $e^{-c\varepsilon_1^2/g^2}$; they are flat at zero and have the same rooted bounds. Consequently

$$\boxed{\mathrm{Cov}_{\mu_g}(\Delta O_1, \Delta O_2)(1,0) = g^4\,\mathrm{Cov}_\gamma(Q_1, Q_2) + O(g^6 a_1 a_2),} \tag{17.15}$$

with the cutoff tail included in the error. Here γ is the ACTUAL current-step centered Gaussian. If another normalization leaves source-dependent linear/quadratic terms at $g = 0$, their complete Gaussian completion must be included in this baseline before the remainder is said to vanish. The formula is not obtained by differentiating a moving sharp test and omitting its boundary term.

On the fixed nonflat core the analogous leading mixed term is $g^2\langle L_1, CL_2\rangle$, the exact nonlinear classical critical-value response. Keeping its full quadratic coarse-field jet in the exact Gaussian sector, the remaining classical physical-field term begins at degree three and has size $Cg^2\alpha^3$ on a physical core of radius α, with fixed core constants. The genuinely integrated Gaussian-subtracted field term has size $Cg^4\alpha^2$, and its flat scalar starts at g^6. The first-source analogues have sizes $C\alpha^3$ for the classical higher-field part and $Cg^2\alpha^2$ for the integrated field part; its Gaussian-subtracted flat scalar starts at g^4. At $\alpha = gP_{\mathrm{bg}}$, these are positive powers of g, up to the displayed background logarithms, relative to the respective first/second Gaussian source weights.

The assertion about the degree-three classical remainder concerns the ACTUAL physical coarse-field composition with its own current and its COMPLETE flat coarse-field Hessian removed. It does not follow from subtracting only the 21 zero-momentum marginal coefficients, and it does not assert that independent AJ and JJ pair jets vanish. These qualifications are necessary when the local kernel is completed on layered domains.

Rooted freezing differences in these fresh estimates acquire the same $\sigma\ell^\alpha$ factor as before, by differentiating the complete bounded clean member in its source profile before filing the connected carrier.

Section 27.18 / A two-label refinement with the actual powers of the coupling

The exact old Gaussian mixed vertices must still be transported. Formula (17.15) is a fresh one-step comparison; multiplying it by a large window-volume factor without bounding the old insertion kernels would not prove the interacting comparison in (G24) of the Gaussian-window note.

Section 27.20 / Matching a scalar first source to its classical incoming kernel

There is a useful refinement of the one-label version of Section 14. Define

$$\omega_s = x_s^{-1/2}(\log x_s)^A \tag{19.1}$$

with a fixed exponent A large enough to cover the homogeneous bare-disc estimates. The weak floor is chosen beyond the turning point of this decreasing function. The actual running tube then gives $\omega_j/\omega_s \le C$ for $j \le s$. The homogeneous clean result implies, for a run stopped at its OWN value x_s,

$$|Y_s'(0) - 1| \le C\omega_s. \tag{19.2}$$

The statement uses the entire bare tangent, not just differentiation with respect to the last coupling.

For one scalar profile f, let $\mathcal{C}_s[f]$ be the exact zero-fluctuation transport of the primitive Wilson insertion through the clean hierarchy. Associativity of the source-free constrained backgrounds identifies it with the primitive action evaluated on the actual constrained background U_s, with the original fine profile still present. Thus its flat physical coarse-field Hessian is exactly

$$F_s^* M_f F_s, \tag{19.3}$$

in the action normalization of Section 13. Both its local fixed-core norm and its rooted localization are bounded by $CR_0^2\|f\|_\infty$: the plaquette bound contributes the fine-volume factor η^4, and the ordinary background/localizer estimates sum the carrier. This is the same primitive Wilson-atom estimate used at the start of the marked construction.

Subtract $\mathcal{C}_s[f]$ and the scalar vacuum from the actual first source coefficient. Write the remaining field-dependent first source as $\mathcal{Q}_s[f]$. For a constant profile, differentiating the SAME fixed-cutoff homogeneous action gives the term $(Y_s'(0) - 1)A(U_s)$ plus the derivatives of the old homogeneous remainders with their OWN counterterms. The first term is bounded by (19.2). Cauchy on each birth's uniform additive bare disc of radius comparable to $x_j/\operatorname{polylog} x_j$ bounds its differentiated fixed-core member by $Cg_j^2 \operatorname{polylog} x_j$; its already paired age estimate is summable. Since $g_j^2 \operatorname{polylog} x_j \le C\omega_s$ for $j \le s$, the complete constant-source quantum reference has local field norm at most $CR_0^2\omega_s|f|$.

Section 27.20 / Matching a scalar first source to its classical incoming kernel

For a spatial profile, freeze this COMPLETE reference at the birth root before extracting or summing it. The exact zero-fluctuation classical family $\mathcal{C}_s[f]$ has already been removed. A new fluctuation-generated first-source difference has the actual $O(g_s^2)$ bound proved by the rooted current-g argument of the tensor-insertion note, with old parameters fixed. The local source-freezing difference contributes $\sigma\ell_{s+1}^{\alpha}$, $\sigma = [f]_{C^\alpha}$. Differences of the constant-source quantum reference instead carry its bound $C\omega_s$, which is larger and covers both effects. Consequently the quantum spatial birth coefficients can be written with norms $\|Q_j\| \le R_0^2\omega_j q_j$, and satisfy

$$q_{s+1} \le C\sigma\ell_{s+1}^{\alpha} + C\epsilon_* \sum_{j\le s} (\omega_j/\omega_{s+1}) q_j. \tag{19.4}$$

This is the same bounded first-difference formation estimate as (14.10), now applied after an exact classical subtraction. The old quantum zero-fluctuation value is carried, not reborn. The scalar reference's marginal increments are differentiated with their own counterterms before the ordinary age bound; otherwise the uniform estimate just used would fail.

The ratio in (19.4) is uniformly bounded, so the already available small-cutoff choice and geometric induction prove

$$\boxed{\|\mathcal{Q}_s[f]\|_{\mathrm{loc}} \le CR_0^2\omega_s\{\|f\|_\infty + \ell_s^{\alpha}[f]_{C^\alpha}\}.} \tag{19.5}$$

This is an all-depth CLEAN first-source estimate. It uses the scalar homogeneous normalization; it is not an assertion that every independent tensor insertion has a bare tangent near one.

On the physical coarse-field composition with its own current, remove the full Hessian (19.3) from $\mathcal{C}_s[f]$. Analyticity on the fixed clean core and its zero constant and linear jets imply that the remainder has size $C\alpha_s^3\{\|f\|_\infty + \ell_s^{\alpha}[f]_{C^\alpha}\}$. The quantum part restricted from (19.5) has size $C\omega_s\alpha_s^2$ times the same source load. Thus, at a prefix stopped at a' and for a profile varying on scale d, the incoming scalar first insertion is its exact classical quadratic kernel plus a canonical physical error with relative size

$$C\{\omega_{a'} + \alpha_{a'}\}\{1 + (a'/d)^{\alpha}\}. \tag{19.6}$$

The error is recorded as two classes: a small fixed-core quantum member and a classical member with zero full physical quadratic jet. In a subsequent canonical Gaussian moment the latter gains its extra physical factor gB; it is not declared small on an unrelated fixed field core. Source anchors and exponential carrier decay are retained in both classes. Terms born from an R-origin still require their separate shrinking-core ledger and cannot be added to (19.5) without that comparison.

The remaining comparison uses a finite number of marked coefficients, not a Cauchy radius measured in a global physical C^5 norm of the shrinking test functions. Such a radius would introduce a power of d^{-1} and would destroy the large-field estimate. The inputs in this section are the ORIGINAL-map unnormalized marked interface, the single-lineage estimate, and the fixed-clean/shrinking-tagged separation in Sections 14–16 and 19 of the companion marked-bridges note. In particular this argument imports the original source-free prepared-domain inclusion and complete stored-action mass estimates identified there; it is a finite-order extension of those inputs, not an independent proof of those source-free preparation theorems. The fixed source-zero torus and weak endpoint are kept throughout; constants may depend on them. The finite-window interpolation below is kept unnormalized, including at complex interpolation parameter.

27.22.1 21.1 The prefix supplies physical marked states at their own scale

Stop first at scale a'. Decompose a bump f_i, supported in a region of diameter Cd, by a smooth scale-a' cell partition:

$$f_i = \sum_{q \in A_i} f_{i,q}, \qquad \|f_{i,q}\|_\infty + (a')^\alpha [f_{i,q}]_{C^\alpha} \le C t_i, \qquad |A_i| \le C(d/a')^4. \tag{21.1}$$

Only a bounded number of pieces overlap. Apply the prefix source construction in units of a' to these pieces. It gives the actual finite-jet state, with the universal prefix vacuum separated. On regular members the first coefficients have their proved own-cell norm; mixed members retain the degree-two weight of Section 17 and their original source anchors. The pure-clean first quadratic part is (19.3). Tagged regular pieces use their genuine shrinking cores and the superpolynomial envelope of the companion Section 19.

Inside a live component, the state instead retains the ORIGINAL source factors and all uneliminated old variables under its unnormalized kernel. It is not defined by extending a regular analytic function into a wild core. The primitive source load is bounded by the corresponding stored Wilson action; after differentiation it is a polynomial of fixed degree in that stored action and the actual coordinate count. Missing vacuum terms are assigned to their actual birth records as in (15.8). Thus (21.1)

supplies globally carried physical density coefficients, with regular analytic representatives only where those representatives are admitted.

The two-source mixed prefix pieces have either a connected regular carrier spanning the distance d, hence an ultraviolet bridge factor, or live/boundary source factors attached to the actual component carrying them. Both possibilities remain in the state. In particular the prefix theorem does not discard the second possibility merely because a clean norm has been proved.

27.22.2 21.2 A finite marked rooted norm after the prefix

Here is the fixed-order propagation needed for the physical estimate. Use three squarefree labels: two source labels z_1, z_2 and a history flag u. After vacuum subtraction every healed scalar coefficient is gauge invariant and has zero first pair jet, with no colored spectator. For a birth on its actual core α_j, normalize its TOTAL rooted norm by α_j^{-2}. There is no translation-root count: the source roots in (21.1) and the flag root are fixed. A marked quadratic coefficient is never promoted to an unanchored change of the global coupling. Keep its full pointed quadratic member, or keep any extracted quadratic counterterm paired with the SAME original root and carrier. The source-free marginal flow is fixed. This condition is necessary for the one-root gain below; it is different from the promotion of a translation-summed unmarked R family in Section 19 of The construction.

The regular readout kernel is bounded by

$$w_m = C(1+m)^q L^{-4m}, \qquad m \geq 1. \tag{21.2}$$

A new regular birth supplies the small actual fluctuation/core factor. At a first encounter with a new wild component, no decay in the elapsed regular age is assumed. Instead the new encounter at scale v costs $\delta_v \leq C x_v^q e^{-cP_0(x_v)}$; subsequent live dwell has an exponential age factor. The corrected nonstationary kernel in the companion (16.7a–c) therefore has total linear ℓ^1 norm at most

$$q_{\mathrm{mark}} \leq C\epsilon_* \sum_{m\geq 1} w_m + \frac{C}{1-e^{-B_1}} \sum_v \delta_v < \frac{1}{2}. \tag{21.3}$$

The actual running tube makes the second sum uniformly small at a sufficiently weak fixed endpoint. This bound does not invent an exponential gain across a late first wild encounter.

For completeness, the same linear bound controls the coefficient of any fixed nonempty subset of the three labels. Its nonlinear forcing is a sum of products of strictly lower-degree coefficients. In the REGULAR connected local expansion, a term with two marked vertices has a tree joining them; reserve part of that tree decay to sum its unmarked branches. The product's time/kernel norm is bounded by the product of the lower-degree rooted ℓ^1 norms. The same argument applies to three vertices.

The live case requires a different, joint estimate. One must not multiply separately integrated island masses that depend on the same hidden variables. Before integration, inherited regular functions are bounded by their actual stored-domain sup norms. Retained physical Wilson sources and missing-vacuum constants are jointly bounded by a fixed-degree polynomial in the SAME stored action and actual record count. For a nonnegative stored action A, a separate fraction $\theta x A$ of the cold reserve gives

$$A^m e^{-\theta x A} \leq \left(\frac{m}{e\theta x}\right)^m, \qquad m = 1, 2. \tag{21.3a}$$

Gaussian energy polynomials are paid by the same fixed quadratic reserve and cost a fixed power of the actual coordinate count. Apply the marked R absolute majorant ONCE to that common cold

integrand, retaining the single flag once. This proves the mixed operator coefficient bound without an L^1-product assertion. After healing, its scalar coefficient joins the ordinary rooted Volterra norm. Induction over the seven nonempty label subsets and the resolvent of (21.3) now bound every coefficient by a fixed polynomial in the incoming one-root norms. Scalar coefficients are recorded when born and are not generated again at every later scale.

One can state this with weighted finite-algebra norms. Give the two source labels weights

$$r_i = \frac{c}{(1+x_{a'})^q N_i t_i}, \qquad N_i = C(d/a')^4, \tag{21.4}$$

omitting a zero source. The fixed power covers the actual core-radius conversions at the first stored reading. The incoming TOTAL marked norm is then within the ordinary small allowance. This source weight has no factor d^α. It is permissible because the subsequent construction propagates finite rooted coefficients, not a fresh global Hölder source on every later lattice.

For a flag at (j, Q) with live age at least a, reserve a disjoint small fraction of its creation and live-volume credits. The companion single-lineage argument permits the additional flag weight

$$r_u = c(1+x_j)^{-q} \exp[c_1 P_0(x_j) + Ba]. \tag{21.5}$$

Any fixed polynomial in stored energy, coordinate count, or old-core radius is paid from a separate fixed part of the same reserve. For genuinely retained, uneliminated physical variables, a source record m levels older costs at most a fixed power of L^m, and its inverse coupling is at most a polynomial in $x_j + m$. Its record age is bounded by the relevant live lineage age, so the independently chosen live-age rate pays both. An earlier HEALED regular origin is instead bounded by its stored analytic sup norm and, on first entry into a new hole, by the nonstationary encounter factor in (21.3). No live-age decay is assigned to the intervening regular years. Every record is charged once at its own scale; no fine physical volume is relabeled as a current-scale volume.

Equations (21.2)–(21.5), with the unchanged source-free stability majorant, prove ordinary source-zero coefficient inequalities with at most two source labels and one flag. They do not require a source-dependent live-island inverse. Only the FINAL source-zero physical partition function will be divided out.

27.22.3 21.3 The full common vacuum cancels in the flag quotient

First let H be a SINGLE lineage flag. Write the exact physical generating numerator and partition function as

$$Z(z) = e^{V+\mathcal{E}(z)}\widetilde{Z}(z), \qquad Z_H(z) = e^{V+\mathcal{E}(z)}\widetilde{H}(z). \tag{21.6}$$

The same full universal $E + R$ vacuum is used in both. Its equality follows from the per-history completion identity of Section 15 before the absolute majorant is taken. The flag is inserted at its displayed source-free history label and changes neither a chart nor a source factor. Its connected scalar outputs stay with the flag, while the source-only universal vacuum is unchanged. Thus the pure source vacuum does not get estimated by its ultraviolet-extensive absolute value. Pointwise domination of an arbitrary union selector by a sum of flags would NOT, by itself, justify domination of its vacuum-subtracted derivatives. The finite-window case is proved by the separate interpolation below.

Set

$$M_i = C(1+x_{a'})^q N_i t_i, \qquad M_{12} = C(1+x_{a'})^q N_1 N_2 t_1 t_2.$$

The finite marked bound above and the actual source-zero stability lower bound give

$$\begin{aligned} |\widetilde{Z}_i| &\le e^{CN_K} M_i, & |\widetilde{Z}_{12}| &\le e^{CN_K}(M_1M_2 + M_{12}), \\ |\widetilde{H}_0| &\le e^{CN_K}\delta_H, & |\widetilde{H}_i| &\le e^{CN_K}\delta_H M_i, \\ |\widetilde{H}_{12}| &\le e^{CN_K}\delta_H(M_1M_2 + M_{12}), & \widetilde{Z}_0 &\ge e^{-CN_K}. \end{aligned} \tag{21.7}$$

All derivatives are at source zero. For a single flag, $\delta_H \le C(1+x_j)^q e^{-cP_0(x_j)-Ba}$. For a linear sum of flags, add its individual δ_H's. Fixed powers of the terminal lower bound only change the constant on this ONE fixed physical torus.

The normalized flagged expectation is $p(z) = Z_H(z)/Z(z) = \widetilde{H}(z)/\widetilde{Z}(z)$. Its exact second derivative is

$$p_{12} = \frac{\widetilde{H}_{12}}{\widetilde{Z}_0} - \frac{\widetilde{H}_1\widetilde{Z}_2 + \widetilde{H}_2\widetilde{Z}_1 + \widetilde{H}_0\widetilde{Z}_{12}}{\widetilde{Z}_0^2} + \frac{2\widetilde{H}_0\widetilde{Z}_1\widetilde{Z}_2}{\widetilde{Z}_0^3}. \tag{21.8}$$

Consequently

$$|p_{12}| \le C_T\delta_H(M_1M_2 + M_{12}), \qquad |p_i| \le C_T\delta_H M_i, \qquad 0 \le p_0 \le C_T\delta_H. \tag{21.9}$$

The flag-connected scalar vacuum terms remain in $\widetilde{H}$; only the source-only universal vacuum cancels. The denominators in (21.8) are global positive source-zero masses, not sourced or source-free live-island masses. No global complex zero-free assertion is used.

27.22.4 21.4 A finite-window defect interpolation, with no selector shortcut

Choose a source-centered observation neighborhood with radius BH, where B exceeds the original collars and is a fixed power of $\log x_d$. Let $\mathcal{B}$ be the finite set of selected scale-cell pairs needed to test whether a live region meets the required interior window at some scale $a' \le \ell_j \le H$, including the actual collars. Its size is at most

$$N_{\mathrm{bad}} \le C(1+n)(H/a')^4 \operatorname{polylog} x_d \tag{21.10}$$

and write $H_b = \mathbf{1}_{\{Q\subset Z_j\}}$ for $b = (j,Q) \in \mathcal{B}$. Introduce an ordinary complex parameter u, distinct from the squarefree flag bookkeeping above, by the EXACT history multiplier

$$W_u(\omega) = \prod_{b\in\mathcal{B}}(1 - uH_b(\omega)), \qquad |u| \le 2. \tag{21.10a}$$

Then $W_0 = 1$ and $W_1 = \mathbf{1}_{\mathrm{good}}$. The factors depend on frozen history labels. They do not modify field tests, charts or the reference marginal trajectory. For complex u,

$$|W_u| \le 3^{\sum_b H_b}, \qquad \partial_u W_u = -\sum_b H_b \prod_{c\in\mathcal{B}\setminus\{b\}}(1 - uH_c). \tag{21.10b}$$

The exponent counts selected LIVE visits, not all ultraviolet creation events. Each such cell is present in the actual stored layer at that visit. A disjoint fixed fraction of the own-layer volume reserve pays its factor three. This leaves the cold action, the single distinguished-flag rarity, and the future live-age reserves available after a fixed weakening of constants. In particular differentiation in (21.10b) requires no division by $1 - uH_b$, even at $u = 1$.

All new u-dependent regular and vacuum births have a distinguished selected-cell origin. They retain that root, including any pointed quadratic counterterm. Their total direct load is bounded by

$$D_{\mathcal{B}} = C\sum_{(j,Q)\in\mathcal{B}}(1+x_j)^q e^{-cP_0(x_j)} \le P(x_d, H/a')e^{-c'P_0(x_d)}. \tag{21.10c}$$

The age sum is taken before the cell sum. On this window $x_j \asymp x_d$, so $D_{\mathcal{B}} \to 0$. The ordinary-u closure must distinguish a live genealogy from an already integrated regular activity. Two selected visits in the SAME live component have $H_{b_1}H_{b_2} = H_{b_1}$ when their indicators agree; they cost one rarity factor, not its square. Keep all factors $(1-u)^m$ of that genealogy grouped inside its single unintegrated kernel. The complete factor is paid by 3^m in (21.10b). No product of integrated rarity norms is used.

Here is a linear distinguished-root proof. Work provisionally in a fixed small ball $\mathcal{A}(u) \le A_*$ for the actual regular u-excess family $\mathfrak{E}(u)$, where $\mathcal{A}(u) = \|\mathfrak{E}(u)\|_{\mathrm{root}}$. Differentiate once in u; the product rule in (21.10b) assigns one selected visit as the derivative root. Let R_s be its newly formed regular rooted norm, and let L_s be its FIRST complete marked-healing or final-exit flux at scale s. Intermediate live copies are not entries in L_s. A partial dropout keeps the root in the residual operator until its entire marked carrier is regular. At an exact split, only the product-rule owner carries this derivative root.

The actual one-root regular estimate and the grouped live cold bound give

$$\begin{aligned} R_s &\le d_s^R + \epsilon_{\mathrm{eff}} \sum_{t<s} w_{s-t} R_t + C L_s, \\ L_s &\le C \sum_{v \le s} e^{-B(s-v)} \left(d_v^L + \delta_v \sum_{t<v} R_t \right), \qquad \sum_s (d_s^R + d_s^L) \le C D_{\mathcal{B}}. \end{aligned} \tag{21.10c1}$$

The u-dependent masks other than the distinguished visit are prepaid inside the complete live kernel. Genuine regular activities can be multiplied by their stored-domain sup bounds; their provisional excess A_* merely changes the linear formation constant. A new wild encounter still has the nonstationary factor δ_v; no dwell decay is assigned before that encounter. Summing (21.10c1), with the original weakened reserve, bounds the differentiated state and scalar outputs by $B_u \le C D_{\mathcal{B}} + (q_0 + C\epsilon_{\mathrm{src}} + C A_*) B_u$. Here $q_0 < 1/2$, and the source allowance and A_* are chosen once so that the parenthesis is less than $3/4$. Therefore $B_u \le 4 C D_{\mathcal{B}}$. Since $\mathfrak{E}(0) = 0$, integration along the radius from 0 to any $|u| < 2$ gives $\mathcal{A}(u) \le 8 C D_{\mathcal{B}} < A_*/2$ when $D_{\mathcal{B}} \le D_0$ is sufficiently small. Finite-prefix analyticity and this first-exit argument close the ordinary disc. This is the grouped-kernel argument of companion Section 22.3; a live $\mathcal{A}^2$ rarity estimate would be invalid.

The same distinguished-root recursion, differentiated at the two finite source labels and estimated by their joint cold polynomial norm, gives

$$\|\partial_u \mathfrak{E}\| \le C D_{\mathcal{B}}, \qquad \|\partial_u D_i \mathfrak{E}\| \le C D_{\mathcal{B}} M_i, \qquad \|\partial_u D_{12} \mathfrak{E}\| \le C D_{\mathcal{B}} (M_1 M_2 + M_{12}). \tag{21.10c2}$$

For the last estimate the forcing partitions the two source labels between at most two other distinguished vertices. Every live product uses (21.3a) on the SAME complete kernel, with its one derivative root. The same bounds hold for scalar outputs, recorded once. Their source-free total is $O(D_{\mathcal{B}})$, so exponentiating them costs at most $e^{C D_{\mathcal{B}}}$. The estimates are uniform for $|u| < 2$. They establish local-series holomorphy and coefficient bounds; they do not require an analytic logarithm of the full partition function.

Write the unnormalized partition function with multiplier W_u as

$$Z_u(z) = e^{V + \mathcal{E}(z)} \widetilde{Z}_u(z). \tag{21.10d}$$

Here $V, \mathcal{E}$ are the ORIGINAL $u = 0$ source-free and source-only vacua. All u-excess scalar outputs remain in $\widetilde{Z}_u$; their bounded total has just been estimated. There is no assertion that Z_u is nonzero. For $I = \varnothing, \{1\}, \{2\}, \{1,2\}$, put $\mathcal{M}_\varnothing = 1$, $\mathcal{M}_i = M_i$, and $\mathcal{M}_{12} = M_1 M_2 + M_{12}$. The marked absolute bound gives

$$\left| \partial_u D_I \widetilde{Z}_u(0) \right| \le e^{C N_K + C D_{\mathcal{B}}} D_{\mathcal{B}} \mathcal{M}_I. \tag{21.10e}$$

Integrate along the real interval $0 \le u \le 1$. The differences $\widetilde{H}_I = \widetilde{Z}_{0,I} - \widetilde{Z}_{1,I}$ therefore satisfy (21.7) with $\delta_H = CD_{\mathcal{B}}$. They are exactly the derivatives of the bad-event numerator, since $1 - W_1 = \mathbf{1}_{\mathrm{bad}}$ in the ORIGINAL positive history representation. This proves the union estimate by an unnormalized finite-window defect expansion, rather than by differentiating a pointwise union bound.

Use only the positive original mass $\widetilde{Z}_0(0)$ in the quotient formula (21.8). It yields

$$|p_{12}^{\mathrm{bad}}| + |p_1^{\mathrm{bad}} p_2^{\mathrm{bad}}| \le C_T t_1 t_2 \, P(x_d, H/a') e^{-cP_0(x_d)}, \tag{21.11}$$

where P is a FIXED polynomial (the coarse count alone is at most power twelve). The scalar probability tends to zero by the same estimate. Neither the polynomial nor its degree depends on the bare cutoff.

For sufficiently small physical d, $p_0^{\mathrm{bad}} < 1/2$. Let $Z_{\mathrm{good}} = Z - Z_{H_{\mathrm{bad}}}$. The exact identity $Z_{\mathrm{good}} = Z(1 - p^{\mathrm{bad}})$ gives

$$|D_{12} \log Z - D_{12} \log Z_{\mathrm{good}}| \le 2|p_{12}^{\mathrm{bad}}| + 4|p_1^{\mathrm{bad}} p_2^{\mathrm{bad}}|. \tag{21.12}$$

This compares physical covariances directly. It does not bound an uncentered divergent bare energy on a bad event. Multiplying and dividing the right side by $g_d^4 = x_d^{-2}$ only inserts the fixed polynomial x_d^2. Thus the bad-window error is

$$\boxed{C_T g_d^4 t_1 t_2 \, P_1(x_d, H/a') e^{-cP_0(x_d)}.} \tag{21.13}$$

27.22.5 21.5 Tagged regular terms and the good physical window

The permanent-origin construction in the companion Section 19 keeps every descendant containing an R origin on its actual shrinking core, including its own marginal after the PRIMARY promotion time. Its normalized envelope is

$$F(x) = C(1+x)^q e^{-c(\log(1+x))^p}, \qquad p > 1. \tag{21.14}$$

For a prefix stopped at a', its first and mixed source coefficients have this envelope with a fixed additional polynomial loss, using the own-scale loads (21.1). Their zero flat values are removed before the old-root count. For regular mixed vacuum terms below a', complete the entire carrier and choose a cell meeting the FIRST source as the root before counting it. The remaining connected carrier must span the distance d. Consequently their sum is bounded by

$$Ct_1 t_2 \sum_{\ell_j \le a'} (d/\ell_j)^4 (1+x_j)^{q'} F(x_j) e^{-cd/\ell_j} \le Ct_1 t_2 P(x_{a'}, d/a') F'(x_{a'}) e^{-c'd/a'}. \tag{21.14a}$$

Here F' has the same superpolynomial form with a smaller exponent constant. To verify the sum, set $\ell_j = a' L^{-m}$. The weak tube bounds x_j above and below by $x_{a'} + O(m+1)$ with a positive linear lower drift toward the ultraviolet. Absorb the fixed powers of x_j into a slightly weakened decreasing envelope F', and use $\sum_{m \ge 0} (DL^m)^4 e^{-cDL^m} \le CD^4 e^{-c'D}$ for $D = d/a' \ge 1$. This is an own-cell root count, with no intrinsic factor d^{-4}.

This bridge applies only to a completed REGULAR carrier. A relative boundary tree can cross a live hole at zero relative distance and does not prove (21.14a). Such a term stays in the original unnormalized live state. If its live prefix component meets the observation region, it is already included in the bad event at a'; the exterior alternative is handled by the boundary bridge below. A component that has healed below a' is included in full in its regular birth carrier before (21.14a) is applied.

Inside a good window, the difference between the full local formation and the no-R-origin formation has at least one tagged vertex:

$$\mathfrak{F}(E_{\mathrm{cl}} + R) - \mathfrak{F}(E_{\mathrm{cl}}) = \int_0^1 D\mathfrak{F}(E_{\mathrm{cl}} + tR)[R] \, dt. \tag{21.15}$$

Section 27.22 / Fixed-torus physical two-point matching through the original R state

Apply the actual-domain connected coefficient bound with the two already rooted source marks. The tagged core losses are fixed powers of x_j, paid by (21.14); no fixed R_0 extension of R is used. Summing the window scales and coarse source roots costs only a fixed polynomial in $x_d, H/a'$. Thus the regular tagged correction to the two-source coefficient is bounded by (21.13), with a possible decrease of its exponent constant. The tagged scalar marginal is included in the FULL reference trajectory; it is not silently removed from the running inverse coupling.

The original layered background maps and relative-tree localization identify an interior regular carrier with the pure-clean carrier up to a boundary connection reaching distance BH. Reserve its connecting exponential before summing the external histories. The resulting error is $C_T t_1 t_2 P(x_d, H/a') e^{-cB}$. One may choose the observation collar B, without changing the RG algorithm, as a sufficiently large power of $\log x_d$; its cell count is still polynomial-logarithmic and this tail is absorbed in (21.13). This uses the original prepared-domain inclusion and the unnormalized boundary norm, not a gauge-field extension through a hole.

Conditioned on the good window, the hidden clean integrals have exactly the frozen coordinate tests of the matched clean family. Their remaining source-zero coarse/history law is positive. The estimates (18.8) are uniform on its admitted physical core and therefore bound its retained two-source terms without assuming that law is Gaussian. Future replacements preserve the old variables and source factors under positive unnormalized kernels; they do not turn a bounded admitted-core insertion into an arbitrarily chosen function on a wild coarse reference field.

Combining (20.7), (18.8), the exact Gaussian off-diagonal comparison, the ultraviolet bridge, and (21.13) proves the scalar separated two-point comparison on the fixed physical reference torus. In the window $a' = dt^{-\varepsilon}$, $H = dt^{\varepsilon}$, $x_d \asymp t$, choose $0 < \varepsilon < 1/32$. The clean relative error is a negative power of t times logarithms; the retained terms decay as in Section 18; and every tagged/bad error is a fixed polynomial times $e^{-c(\log t)^p}$. All vanish as $d \downarrow 0$, uniformly as the bare cutoff is removed below a'.

This is a fixed-torus scalar physical comparison using the original-map marked and preparation inputs stated at the start of this section. It does not assert a volume-uniform local rare-event bound. The companion physical chessboard/sign-averaging argument transfers a fixed-reference scalar covariance to compatible larger periodic volumes. Higher composite families, all-order normalization and the final reconstruction hypotheses remain separate questions.

Section 28.1 / Scope and precise scope

The argument develops the frozen-density route through a completed large-field R operation. It does not attempt to preserve a regular minimizer domain under arbitrary independently rough spatial Wilson coefficients. The source-free backgrounds, averaging charts, cutoffs, propagators and reference replacement laws are fixed. Finite spatial source jets are local functions of the fields produced by those maps.

The advance over the argument is a quantitative localization and small-potential theorem, not only an algebraic connected regrouping. The nonlinear source-free localization construction has an exact wall-component property. Applying its fundamental-theorem-of-calculus expansion to an algebra-valued local function produces marked bridges with a bound for each term and exact source support. Primitive Wilson atoms satisfy the required input bound on the original regular complex domains. The same statement applies to inherited analytic activities with their actual incoming coefficient norm. A direct Mayer/Gaussian proof below establishes the generic small-potential allowance that was previously left as an assumption.

Live islands are kept as unnormalized operator states. A frozen positive replacement kernel gives an exact source-preserving extension of the source-free R operation. This avoids dividing by a sourced island mass, for which a uniform source disc can genuinely fail. It does not remove the island or the terminal coarse-field dependence.

Sections 11–12 prove the constant-source clean family and its uniform bare disc. Sections 13–14 match the marked completion interface to the ORIGINAL large-field construction, importing its published source-free preparation and domain theorems. The clean smooth spatial class is constructed in the stated results. Section 19 closes the tagged class on its actual shrinking cores, retaining its own paired marginal coefficients even after the primary construction relabels it as regular. It also closes an ordinary homogeneous bare-source disc for the full real run. Sections 15–18 give the partition, finite-reference-history, and physical transfer consequences. Sections 20–21 deduce all scalar separated-moment bounds and a uniform scalar insertion form from the strictly local source normalizer. Sections 22–23 prove the exact finite-window mask bound with one live derivative owner, and extend it to every fixed number of scalar source labels on the same bare sequence. Section 24 gives the corresponding fixed-order physical cumulant transfer, with a positive d^2 gain at order four for $\alpha = 1/2$. A log-small local source quadratic form is not declared power-irrelevant. The physical ultraviolet Gaussian leading terms and the full continuum observable identification remain separate, and no completed Yang–Mills existence proof is claimed.

Section 28.2 / Source-free inputs used from the primary construction

Primary references:

- T. Balaban, *Renormalization group approach to lattice gauge field theories. II. Cluster expansions*, CMP 116 (1988), 1–22, DOI 10.1007/BF01239022.

Section 28.2 / Source-free inputs used from the primary construction

article

- T. Balaban, *Large field renormalization. II. Localization, exponentiation, and bounds for the R operation*, CMP 122 (1989), 355–392, DOI 10.1007/BF01238433.

The 1988 construction localizes an arbitrary local analytic function after a nonlinear minimizer substitution. Its formulas (1.2)–(1.5) express the nonlinear maps as fixed points depending on propagators. Random-walk parameters yield exact component separation in (1.9)–(1.10). The bounds (1.11)–(1.21) allow independent parameter discs of radius e^{κ}, with fixed losses $e^{16\kappa}$ and $e^{32\kappa}$. Section 2 supplies the Mayer, cutoff, conditioned-Gaussian and coarse-localization construction. In the 1989 paper, (1.61)–(1.72) performs the corresponding large-field relocations and factorization. The extra normalized-island representation (1.103)–(1.104) is optional.

The arguments below keep those source-free maps. They do not cite a source theorem from either paper. The parameter extension is proved here.

28.3.1 2.1 A fixed-point estimate with explicit operator dependence

Let $\mathcal{E}$ be one of the actual regular potential/first-difference Banach spaces. The following elementary estimate isolates the part of the original nonlinear construction that matters for adding coefficient algebras.

Suppose $C : \mathcal{E} \to \mathcal{E}$ is holomorphic near zero and

$$\|C(a)\| \le c\|a\|^2, \qquad \|DC(a)\| \le c\|a\|. \tag{2.1}$$

Let $\|H\| \le K$, $K \ge 1$. For $\|a\| \le r$, consider

$$d = C(a - Hd). \tag{2.2}$$

If $4cKr \le 1$, the right side maps the ball $\|d\| \le 4cr^2$ into itself: its argument has norm at most $2r$. If $2cKr \le 1/2$, it is a contraction on that ball. Uniform Picard iteration therefore produces a unique solution, holomorphic jointly in a, H, with $\|d\| \le 4cr^2$. These statements remain true on any set of operator parameters for which the same bound K holds.

The next equation has the form

$$a_0 = -G\,N_H(H_0 b + a_0). \tag{2.3}$$

Suppose N_H is holomorphic and satisfies quadratic and derivative bounds of the form (2.1), with a constant c_K, uniformly for the admitted H, and $\|G\|, \|H_0\| \le K$. Put $u = K\|b\|$. If u is smaller than a fixed multiple of $(c_K K)^{-1}$, the same proof gives

$$\|a_0\| \le 4Kc_K u^2, \qquad \|H_0 b + a_0 - Hd(H_0 b + a_0)\| \le C_K\|b\|. \tag{2.4}$$

This proves a common holomorphic domain; it does not merely prove existence separately for each real parameter.

In the actual constraint construction, C is the nonlinear remainder of the local averaging map. N_H is the nonlinear remainder after the quadratic fluctuation form has been removed. Their source-free bounds are the bounds used in the original fixed-point proof. Only the scalar observable evaluated after these maps will acquire source coefficients.

28.3.2 2.2 Random-walk parameters and the size of the common domain

Write a propagator as a sum of localized walk terms. A walk term is multiplied by the product of s_Δ over all parameter cubes met by its carrier. On independent discs

$$|s_\Delta| < R = e^{\kappa} \tag{2.5}$$

the original walk estimate gives $\|H(s)\|, \|G(s)\|, \|H_0(s)\| \le K_\kappa$, with $K_\kappa \le Ce^{16\kappa}$. Long walks pay their parameters from their exponential length factor once the localization cube size M is chosen larger than a fixed multiple of κ. The bounded number of cubes meeting at one corner accounts for the fixed exponential loss.

Insert K_κ into (2.1)–(2.4). Choosing the physical fluctuation radius small enough, in particular of order $e^{-C\kappa}$, establishes the same fixed-point domains for every parameter choice in (2.5). In the original notation the sufficient inequalities include an $e^{32\kappa}$ loss. Since κ is chosen first, a fixed weak-coupling ceiling makes $gP(g)$ smaller than that radius.

For an observable requiring a regular field radius α, one also retains the explicit inclusion

$$C_\kappa gP(g) < c\alpha. \tag{2.6}$$

For radii $\alpha = g(\log x)^{p_{\mathrm{bg}}}$ and $P = (\log x)^{p_{\mathrm{fl}}}$, this follows at weak coupling when $p_{\mathrm{bg}} > p_{\mathrm{fl}}$. No spatial source coefficient enters (2.1)–(2.6), because the maps remain source independent.

28.3.3 2.3 Exact component locality of the nonlinear solutions

Set $s_\Delta = 0$ outside a union Y of parameter cubes together with the fixed seed collar X_0. Every surviving walk carrier lies within one wall-connected component of that union. Its kernel vanishes near the corresponding boundary. The local functions C and N_H and the separated propagators consequently give separate fixed-point equations in those components. Every Picard iterate on one component depends only on the variables and parameters there. Its limit has the same property.

Thus a local function supported in the seed depends only on the wall-component containing that seed. Differentiating a parameter in another component gives zero. This is an exact identity, not an exponentially small approximate independence statement. Gauge transformations and local changes of chart used in the construction must have their assigned support included in the seed collar; they cannot be dropped at a seam.

Section 28.4 / A coefficient-algebra localization theorem

Let

$$\mathcal{A}_N = \mathbb{C}[z_1, \dots, z_N]/(z_1^2, \dots, z_N^2), \qquad \|F\|_{\boldsymbol{r}} = \sum_A r_A |F_A|. \tag{3.1}$$

This is a commutative normed algebra with a submultiplicative norm. The same proof works for any finite-dimensional coefficient algebra with that property.

Let $F_{X_0}(s,z)$ be a local function composed with the frozen nonlinear maps above. Assume it is holomorphic in the independent s discs, with

$$\sup_{s,p,\xi} \|F_{X_0}(s,z)\|_{\boldsymbol{r}} \le A_{X_0}. \tag{3.2}$$

Here p ranges over the assigned complex core, and ξ over fixed real stored variables or fixed real cutoff coordinates. The common core and cutoff sets are independent of s, z. If an integrand is defined only on its selected cutoff set, it is set to zero outside that set.

For $Y \supset X_0$, put $K = Y \setminus X_0$, set the seed parameters to one and all parameters outside Y to zero, and define

$$(\mathcal{L}_{Y,X_0}F)(z) = \int_{[0,1]^K} \prod_{\Delta\in K} \partial_{s_\Delta} F_{X_0}(1_{X_0} + s_K, z)\, ds_K. \tag{3.3}$$

28.4.1 Theorem 3.1

At finite volume,

$$F_{X_0}(1,z) = \sum_{\substack{Y\supset X_0 \\ Y \text{ rooted wall–connected}}} \mathcal{L}_{Y,X_0}F(z), \tag{3.4}$$

with the precise seeded component convention of section 2.3. Every term is local in the fields on Y and satisfies

$$\|\mathcal{L}_{Y,X_0}F\|_{\boldsymbol{r}} \le A_{X_0}(R-1)^{-|Y\setminus X_0|}. \tag{3.5}$$

If a coefficient of F_{X_0} contains a source label only when its support meets X_0, the corresponding coefficient of every term in (3.4) vanishes unless its final domain meets that source support.

Proof. Apply the finite identity $f(1) = f(0) + \int_0^1 f'(t)\,dt$ independently to every parameter outside the seed. This gives (3.3) summed over all subsets. Section 2.3 kills terms with a differentiated component disjoint from the seed. Heredity identifies each survivor with its local version on Y. For each differentiated parameter take a Cauchy circle centered at its real value $t \in [0,1]$, with radius increasing to $R-1$. Each parameter is differentiated once, so there is no factorial. Take the coefficient norm inside the Cauchy integrals to obtain (3.5). The source-support assertion holds term by term because neither interpolation nor differentiation in s introduces a new z variable. QED.

The same argument works with a uniform integrated bound $\sup_s \int \|K_0(s,\xi)F(s,\xi,z)\|_{\boldsymbol{r}}\, d\mu(\xi) \le A_{X_0}$, using a fixed positive reference measure. Fubini and Cauchy then give (3.5) for the integrated term. It is not necessary to bound an integral of a supremum over s.

28.4.2 3.2 Inherited analytic activities

For an inherited coefficient-algebra activity $E_X(z)$, use its actual incoming holomorphic norm, not the scalar bound on $E_{X,0}$. Composition with the already admitted source-free maps preserves that bound.

Theorem 3.1 then applies with that incoming norm as A_{X_0}, after adding the prescribed collar to X.

A useful explicit norm estimate is as follows. Work in connected cube domains and suppose the collar has size at most $c_b|X|$. Suppose

$$\|E_X\|_{\boldsymbol{r}} \le a e^{-\lambda_{\mathrm{in}}|X|}, \qquad \|\mathcal{L}_{Y,X} E_X\|_{\boldsymbol{r}} \le a e^{-\lambda_{\mathrm{in}}|X|} e^{-\kappa_c |Y \setminus X^\sharp|}. \tag{3.6}$$

For $t = \lambda_{\mathrm{out}} + \log 2 + 1$, take $\lambda_{\mathrm{in}} \ge c_b t$ and $\kappa_c \ge t$. Since $|Y| \le c_b|X| + |Y \setminus X^\sharp|$, each summand is at most $a e^{-t|Y|}$. There are at most $2^{|Y|}$ possible subsets $X \subset Y$. Consequently

$$\left\| \sum_{X \subset Y} \mathcal{L}_{Y,X} E_X \right\|_{\boldsymbol{r}} \le a e^{-(\lambda_{\mathrm{out}}+1)|Y|}. \tag{3.7}$$

A fixed finite number of origin types changes the constant. Infinitely many older levels must first be combined using their actual summable incoming norm; this estimate does not manufacture that age summability. For the relative tree norm used near wild cores, the core volumes are charged to their separate activity reserve, and the same argument is applied to the regular connecting cubes.

Thus the extension to inherited activities is a bounded linear map on the actual coefficient norm, with an explicit spare-locality cost. It is not an inference from an unmarked value or from holomorphy in one constant coupling.

Section 28.6 / A direct generic potential allowance

This section proves the coefficient-algebra version of the ordinary small-field cluster allowance. The analytic inputs are the literal source-free local Gaussian functionals and the frozen localization maps, not a sourced theorem or an assumed bound on a logarithm.

28.6.1 5.1 The Gaussian estimate used by the expansion

Let the real reference covariance C_0 on n local coordinates satisfy $K^{-1}I \le C_0 \le KI$. Let a complex symmetric covariance C satisfy $\|C - C_0\| \le \epsilon$, with ϵ small depending only on K. A Neumann series

for the inverse and the series for $\operatorname{Tr}\log(I + C_0^{-1/2}(C - C_0)C_0^{-1/2})$ give

$$|d\nu_C(B)| \le e^{C_K \epsilon n}\, e^{C_K \epsilon |B|^2}\, d\nu_{C_0}(B). \tag{5.1}$$

For a complex mean m, write $A = C^{-1}$, $A_R = \operatorname{Re} A$ and complete the real square in $-B^T A_R B/2 + \operatorname{Re}(Am)^T B$. The resulting real Gaussian has mean $\mu = A_R^{-1}\operatorname{Re}(Am)$, with $|\mu| \le C_K|m|$. Its normalization differs from the modulus of the complex Gaussian normalization by at most $e^{C_K\epsilon n + C_K|m|^2}$. The exact exponential-moment formula for that real Gaussian then gives the estimate below. A uniformly bounded mean per coordinate therefore costs $e^{C_K\|m\|_\infty^2 n}$; there is no reciprocal power of the failed-test tilt in this cost.

If v distinct coordinate blocks fail $|B_b| < P$, then $\sum_{\text{failed}} |B_b|^2 \ge P^2 v$. Reserving a fixed extra Gaussian tilt $\gamma > 0$ yields

$$\int \prod_{b \in P_0} 1_{\{|B_b| \ge P\}}\, e^{\tau|B|^2} |d\nu_{C,m}(B)| \le \exp\{C_K(\epsilon + \tau + \gamma + \|m\|_\infty^2)n\} e^{-\gamma P^2 v}, \tag{5.2}$$

provided the total tilt is below a fixed covariance-dependent threshold. This estimate does not shift a contour through a sharp test. It allows the exact full Gaussian determinant/Schur expression to be extracted, while the cutoff correction remains exponentially small in P^2.

For the actual conditioned Gaussian localization, its real conditional mean depends linearly on an exterior auxiliary Gaussian field. It is not a uniformly bounded deterministic m, and that mean must not be inserted in (5.2) with a supremum over the exterior field. Instead retain the real conditioned joint Gaussian as the reference. In (2.15)–(2.25) of the primary paper, the three error forms R_1, R_2, R_3 satisfy an exponentially decaying matrix bound with coefficient

$$\epsilon_{\text{loc}} \le Ce^{-cM} + C(\alpha_0 + \alpha_1)$$

uniformly on the chosen coarse discs. Their mixed form is bounded by local quadratic forms in B and the auxiliary X. Integrate the real conditional B Gaussian first. At zero error its induced X-quadratic form cancels exactly the compensating form from conditioning. The remaining form is $O(\epsilon_{\text{loc}})$ on the coordinates in Z, and integrating those X coordinates costs $\exp(C\epsilon_{\text{loc}}(LM)^4|Z|)$.

Equivalently, this last bound follows from

$$\mathbb{E}\exp\{\epsilon|P_Z u|^2\} = \det(I - 2\epsilon\operatorname{Cov}(P_Z u))^{-1/2} \le e^{C_K\epsilon\operatorname{rank}(P_Z)}$$

for a real reference Gaussian with bounded local covariance, where $\operatorname{rank}(P_Z) \le C(LM)^4|Z|$. The corresponding quadratic tilts in (5.6) and the failed-test estimate are included before this integration. Thus there is no factor counting the whole exterior auxiliary dimension. Formula (5.2) also permits a separate bounded deterministic complex shift, if one is introduced by an exact source Gaussian completion; it is an additional allowance and does not replace this conditioning cancellation.

28.6.2 5.2 Local potentials and the literal term estimate

Suppose the total finite collection of potential types satisfies, on its assigned field boxes and complex pair domains,

$$\|V_Y(B,z)\|_{\boldsymbol{r}} \le ae^{-\lambda_V d_f(Y)} \left(1 + M^{-4}\sum_{b \subset Y} |B_b|^2\right). \tag{5.3}$$

Older origins and source profiles are combined inside a fixed number of types; their number is not counted as a new combinatorial factor. For a selected family $\mathcal{D}$, retain its required real cutoff set Y_0, and let P_0 be a set of distinct failed bonds outside it. The integrand is exactly

$$(-1)^{|P_0|}\chi_{Y_0}\prod_{b\in P_0}(1-\chi_b)\prod_{Y\in\mathcal{D}}(e^{V_Y}-1). \tag{5.4}$$

Let $Z_0^\sharp$ be the assigned coarse collar of this support. For a final coarse domain Z, coarse localization differentiates the $h=|Z\setminus Z_0^\sharp|$ independent parameters once. Write $r=|\mathcal{D}|$, $v=|P_0|$, $N=|Z|$.

Choose λ_0 above the fine-domain entropy threshold and a small Gaussian budget $W>0$. For every selected potential introduce an independent scalar multiplier ζ_Y, with radius

$$\mathcal{R}_Y=(W/a)e^{(\lambda_V-\lambda_0)d_f(Y)}. \tag{5.5}$$

Assume $a\le W/2$. The rooted domain sum gives

$$\sum_{Y\in\mathcal{D}}|\zeta_Y|\|V_Y(B,z)\|_{\boldsymbol{r}}\le CWL^4N+CWM^{-4}|B|^2. \tag{5.6}$$

The coefficient norm of each Mayer factor is at most $2e^{|\zeta_Y|\|V_Y\|_{\boldsymbol{r}}}$. Choose W so that the quadratic tilt in (5.6), together with the fixed failed-test reserve, fits (5.2). The integrated expression on the enlarged multiplier discs is bounded by $2^re^{C_0N}p^v$, where $p=e^{-\gamma P^2}$. It vanishes when any selected $\zeta_Y=0$. Repeated scalar Schwarz therefore divides the bound by $\prod_Y\mathcal{R}_Y$ at $\zeta_Y=1$. Mixed coarse Cauchy on discs of radius $R_c>2$ then proves the literal bound

$$\|T(Z,\mathcal{D},P_0)\|_{\boldsymbol{r}}\le e^{C_0N}(2a/W)^re^{-(\lambda_V-\lambda_0)\sum_{Y\in\mathcal{D}}d_f(Y)}p^v(R_c-1)^{-h}. \tag{5.7}$$

All Cauchy and Schwarz operations are in auxiliary scalar variables. They apply to the coefficient norm by the Banach-valued Cauchy formula, or by duality. No source zero-free disc is used.

28.6.3 5.3 Explicit resummation

The local collar geometry gives

$$N\le C_h\left(r+L^{-1}\sum_{Y\in\mathcal{D}}d_f(Y)+v+h\right). \tag{5.8}$$

It follows by covering each selected fine domain by a spanning walk, splitting that walk into length-L pieces, and counting the bounded number of coarse cubes met by each piece. A failed bond has a fixed-size collar. There is no surrounding-volume term.

Fix a desired activity exponent Λ, and set $K_*=C_0+\Lambda+1$. Choose the spare decay and coarse radius so that, for a sufficiently small fixed $\delta>0$,

$$\begin{aligned}(2a/W)e^{K_*C_h} &\le \delta,\\ pe^{K_*C_h} &\le \delta,\\ (R_c-1)^{-1}e^{K_*C_h} &\le \delta,\\ \lambda_V-\lambda_0 &\ge K_*C_h/L+\lambda_1,\end{aligned} \tag{5.9}$$

Here λ_1 is above the fine-domain entropy threshold. The following explicit hierarchy resolves both the Gaussian-domain dependence and the M-dependence of the entropy budget.

First fix $L,\Lambda,C_h,\lambda_0,\lambda_1$. The rooted lattice entropy constants are absolute in cube units. For each sufficiently large allowed M, set

$$\delta_M=c_{\mathrm{ent}}(LM)^{-4},\qquad \kappa_c(M)\ge(\Lambda+2)C_h+\log(2/\delta_M)+2,\qquad R_c=e^{\kappa_c(M)}. \tag{5.9a}$$

The fixed constant c_{ent} is small enough for both entropy sums below. Increase κ_c, if needed, by a fixed lower bound required by the already chosen fine localization exponent; in the primary paper this is the inequality $\frac{1}{2}(\kappa_1-1)\ge 2L\kappa$. Thus $\kappa_c=O(\log M)+O_{\Lambda,L,\lambda_V}(1)$.

The coarse expansion is arranged with one localization domain covering the whole root support Z_0 and a collar of width proportional to M; the other localization domains stay away from that root. Consequently every walk that changes a matrix entry relevant to the conditioned root Gaussian travels a distance proportional to M. The same walk counting as in section 2 pays the factor $e^{16\kappa_c}$ from the complex parameters using a part of that length. Explicitly, the argument preceding (2.16) in the primary paper assumes

$$e^{-\delta_0 M/3}e^{16\kappa_c}<1$$

and leaves the bound $Ce^{-\delta_0 M/3}$ for the matrix error in (2.16). Choose M, among the allowed lattice powers, so large that this inequality holds with slack and

$$C_K(LM)^4e^{-\delta_0 M/3}\le 1/20. \tag{5.9b}$$

This choice exists because the left cost in the first condition is linear in M, whereas $\kappa_c(M)$ grows logarithmically. No later step increases κ_c or M.

Now reduce the complex background comparison radii, the potential tilt budget W, the failed-test tilt $\gamma>0$, and any additional deterministic mean allowance m_*, until

$$C_K(LM)^4(\alpha_0+\alpha_1+\gamma+m_*^2)+C_KL^4W\le 1/10. \tag{5.9c}$$

All these choices are fixed numbers, independent of the fine cutoff and the eventual weak-coupling scale. The local Gaussian comparison then gives $C_0\le 1$, after an immaterial further reduction of the right sides of (5.9b),(5.9c). Use $K_*=\Lambda+2$ throughout (5.9). Choose the incoming spare exponent λ_V before (5.9a) to satisfy its last line. Finally choose the physical potential ceiling a and the weak-coupling ceiling so that its first two lines hold. Since $P(g)\to\infty$, a fixed, possibly small γ still makes $p=e^{-\gamma P(g)^2}\to 0$.

This is the analogue of the primary paper's $\alpha_5=O(e^{-\delta_0 M/3})+O(\alpha_0+\alpha_1)+O(\alpha_4)+\gamma^2$ and the restrictions $(LM)^4\alpha_i,(LM)^4\gamma^2=O(1)$ following (2.37). The parameter discs are paid from actual long-walk collars before the weak-coupling ceiling is chosen. They are not posited to remain harmless on arbitrarily large radii.

Multiplying (5.7) by $e^{(\Lambda+1)N}$ and using (5.8) leaves at most $\delta^{r+v+h}\prod_Y e^{-\lambda_1 d_f(Y)}$. The empty Mayer/failure selection contributes no nonempty activity, because the normalized local Gaussian functional sends one to one. Thus $r+v\ge 1$ in every nonzero term. Extract one factor bounded by $e^{K_*C_h}(2a/W+p)$.

Section 28.6 / A direct generic potential allowance

For fixed Z, the remaining selected-domain sum is at most $\exp(C_f \delta L^4 N)$, using the rooted fine-domain entropy sum. The failed-bond sum is at most $\exp(C_b \delta (LM)^4 N)$. Dropping the remaining δ^h only enlarges the sum. Choose δ so that these exponentials are at most e^N. We obtain

$$\|H(Z,z)\|_{\boldsymbol{r}} \le C(a+p)e^{-\Lambda|Z|}. \tag{5.10}$$

The fixed constant includes the extracted δ^{-1} and geometric factors; it is independent of source degree, fine cutoff, volume and number of older origin labels.

There are exponentially many rooted connected cube domains, so choosing Λ with a fixed additional margin makes (5.10) satisfy the weighted hard-core polymer criterion. Its convergent connected series defines the local logarithm in the coefficient algebra. Thus a genuine fixed small generic potential allowance has been proved from the ordinary Gaussian/localization data. The proof provides source-support filing at the same time: every source coefficient remains attached to a selected local potential whose support is contained in its final connected polymer.

The amplitude in (5.10) is the actual a. If a is log-small, the result is log-small. A power-small nonlinear remainder can be isolated by introducing its own scalar multiplier and repeating Schwarz with the other, small linear/quadratic terms already inside the allowed budget.

28.7.1 6.1 A local identity

Let b denote the variables retained outside a replacement region. For a source-free local history h, let $q_{h,0}(\xi, b) \geq 0$ be its input density on the old internal variables and let $r_{h,0}(y, b) \geq 0$ be the reference replacement density on the new internal variables. Put

$$D_{h,0}(b) = \int r_{h,0}(y, b)\, dy > 0, \qquad \nu_{h,0}(dy \mid b) = D_{h,0}(b)^{-1} r_{h,0}(y, b)\, dy. \tag{6.1}$$

For any coefficient-algebra input density $q_h(\xi, b, z)$, define

$$(\mathcal{R}_0 q_h)(y, b, z) = \frac{r_{h,0}(y, b)}{D_{h,0}(b)} \int q_h(\xi, b, z)\, d\xi. \tag{6.2}$$

Then

$$\int (\mathcal{R}_0 q_h)(y, b, z)\, dy = \int q_h(\xi, b, z)\, d\xi \tag{6.3}$$

coefficient by coefficient. The map is linear and positive on its scalar component. At source zero, it is exactly the original reference replacement. Summing the source-free history partition gives the corresponding identity for the full local density.

Formula (6.2) is the frozen extension of the numerator/reference denominator structure of the original R operation. In the actual construction, fixed coarea coordinates, gauge-fixing factors and stored fields are included in ξ, b, h. Gauge removal is an equivalence for gauge-invariant observables; Wilson action marks have that invariance. If history branches have different internal variable spaces, use their disjoint union rather than identifying their coordinates.

Neither the denominator nor the replacement field law acquires a source. A physical source evaluated on the old field stays under the ξ integral. It is not reinterpreted as a source evaluated on the replacement variable y.

28.7.2 6.2 Exact iterated source identity

Each frozen elementary step gives a probability kernel on an extended state containing the currently stored histories. Composing these kernels produces the source-free joint law of old fields, histories and new fields. For every finite prefix,

$$\int \rho_j(B,z)\,dB = \int \rho_0(U)\exp\left(\sum_i z_i A(f_i,U)\right) dU. \tag{6.4}$$

The right side is the original finite-lattice source functional. This is a direct consequence of (6.3), not a differentiation of an equality whose kernels secretly change with the source.

Where the source-free marginal $\rho_j(B,0)$ is positive, $\rho_j(B,z)/\rho_j(B,0)$ is the conditional finite-jet moment generating function under that joint law. Thus all higher logarithmic coefficients, including the terminal coarse coefficients, are retained. A new source-dependent effective action can be extracted by algebra, but no such extraction is used to justify (6.4).

To retain a chosen coarse Wilson source explicitly, multiply and divide by its exponential and localize the exact difference of the old and new weighted actions using section 4. This is an identity. The zero-fluctuation action is not discarded as a vacuum term.

28.7.3 6.3 What analytic normalization is still needed

The source-free denominators $D_{h,0}(b)$ still require their ordinary analyticity and nonvanishing statement on the external complex core. The original completed-R construction supplies this for the local denominators on which its comparison is made. Freezing the source removes a new sourced denominator problem; it does not prove a missing source-free complex comparison.

If one instead chooses a source-dependent local normalization, the small-cell inverse theorem in the stated results applies after the zero-fluctuation action has been factored out. There is no need to choose that convention for a live island.

28.9.1 8.1 A fractional-action bound valid for all source degrees

On a real SU(2) field, $a_p \geq 0$. If the source load obeys $\rho_p \leq \theta x_p$, $0 < \theta < 1$, then

$$e^{-\sum_p x_p a_p} \left\| \exp\left(\sum_p w(p) a_p \right) \right\|_{\boldsymbol{r}} \leq e^{-\sum_p (1-\theta) x_p a_p}. \tag{8.1}$$

This follows from $\| \exp(\sum_p w(p) a_p) \|_{\boldsymbol{r}} \leq \exp(\sum_p \rho_p a_p)$. It is uniform in source degree and does not count disjoint supports by a sum of their separate suprema. It uses the action coefficient of the layer on which that action is retained.

For a fixed additive load ρ, the earlier per-stored-site bound $e^{C\rho N(h)}$ is sufficient. For radii growing like $x/(\log x)^M$, use (8.1); the logarithmic ledger alone would not pay that action.

There is a useful explicit rule for a zero-fluctuation subtraction. If the retained real core field is U, while U_* is a regular reference on the same plaquettes, then

$$e^{-\sum_{p\in\Lambda} x_p a_p(U)} \left\| e^{A_w(U;\Lambda) - A_w(U_*;\Lambda)} \right\|_{\boldsymbol{r}} \leq e^{-\sum_{p\in\Lambda} (x_p - \rho_p) a_p(U)} e^{\sum_{p\in\Lambda} \rho_p |a_p(U_*)|}. \tag{8.1a}$$

The last exponent is at most $C\rho\alpha^2 N_\Lambda$, using the regular plaquette estimate and its physical plaquette count. The same estimate handles the opposite sign of the subtraction. Thus the regular reference action is retained as a small extensive source cost. It is not accidentally cancelled with a source-free normalization denominator. In a transition which splits a plaquette action with a fixed cutoff $0 \leq \zeta_p \leq 1$, apply (8.1) to $\zeta_p a_p$ and the regular estimate to the remainder only where the ordinary construction actually guarantees regularity. No action fraction is available for a mark on an unrelated stored field.

28.9.2 8.2 Exact dimension accounting

Suppose a reference normalized Gaussian block has n_j independent real coordinates at its own stored level. Replacing its positive action part by at least a fraction $1 - \theta_j$ of the original quadratic form changes its integral by at most

$$(1-\theta_j)^{-n_j/2} \le \exp(C\theta_j n_j), \qquad 0 \le \theta_j \le 1/2. \tag{8.2}$$

Gauge-fixing positive terms that are not changed only improve this comparison. Compact group-variable integrals are still bounded using their normalized Haar measure. Each independent Gaussian coordinate is counted once in its original integration block. The fixed number of bond/color coordinates per stored site gives

$$n_j \le CN_j(h), \qquad \prod_j e^{C\theta_j n_j} \le \exp\left(C\sum_j \theta_j N_j(h)\right). \tag{8.3}$$

There is no additional factor η_j^{-4} for a regular background reconstructed from those coordinates, and no repeated count of an old integration block at later ages.

The extra cost in (8.3), as well as fixed local interaction and normalizer costs, is paid by $v(h) = \sum_j (\log x_j) N_j(h)$ at weak coupling. Source-free reference denominators in (6.1) remain unchanged.

28.9.3 8.3 Large-field credits after the fractional loss

The ordinary large-field estimate uses positive Wilson action, positive fluctuation forms and the failed tests to create credits of the form $c(\log x)^{2p_0}$, and renewal credits whose logarithmic power strictly exceeds the power needed for the final p_0 factor. Replacing the retained positive action by its fraction $1-\theta$ replaces the corresponding positivity constants by positive constants, and adds the dimension cost (8.3).

The strict inequalities among the logarithmic exponents are unchanged. For example, the renewal comparison in the original proof uses $2p_1 - (d+5)r_0 > p_0$. Multiplying its leading positive constant by $1-\theta$ still leaves it larger than the required p_0 cost for a sufficiently weak fixed coupling ceiling. Likewise a creation credit of order p_0^2 still dominates a fixed per-site logarithmic charge and the prescribed polynomial-logarithmic waiting cost.

The stability of the numerical induction can be checked explicitly; the following is the version needed here. Write

$$P_0(x) = (\log x)^{p_0}, \qquad R(x) \asymp (\log x)^{r_0}.$$

Fix the usual geometric constants, the desired locality exponent, and a fixed positive coefficient c in the history ledger. Let $N_j(Z)$ count the genuinely stored coordinates/cells at level j. The elementary volume estimate gives

$$N_j(Z) \le C(MR_j)^d (d'_j(Z)+1), \qquad \log x_j \le CR_j, \tag{8.3a}$$

with $r_0 \ge 1$. All new costs in (8.1a)–(8.3), the factor $e^{cv(h)}$, and the ordinary interaction/history-counting costs are consequently bounded by a charge of the form

$$w_j(Z) = C_w M^d R_j^{d+1} (d'_j(Z)+1). \tag{8.3b}$$

The coefficient C_w is enlarged once. It is independent of source degree, age and fine cutoff. Older origin labels in a regular activity have already been summed in its incoming norm; they do not appear again in C_w.

Section 28.9 / Paying primitive marks with the retained real action

Let S be the original enlargement/rescaling operation on a live domain and let $K_j(Z)$ be its waiting time until it can next be tested for removal. Define its future charge

$$\mathcal{F}_j(Z) = \sum_{n=j+1}^{j+K_j(Z)} w_n(S^{n-j}Z). \tag{8.3c}$$

The geometric estimate used in (1.81) gives

$$\mathcal{F}_j(Z) \le CC_w M^d L^{d+1} R_j^{d+2}(d'_j(Z)+1). \tag{8.3d}$$

Its proof uses the geometric decay of the rescaled linear size until the component fits inside a fixed number of MR_n cubes, followed by at most R_j additional waiting steps. Multiplying the local charge by a fixed constant merely multiplies the same bound by that constant.

Keep a reserve $B_j = 2P_0(x_j)$ at every birth or renewal, separate from a bank $K(Z)$ paying (8.3c). The primitive creation credit is at least $c_\theta P_0(x_j)^2(d'_j(Z)+1)$. The renewal credit, before replacing it by P_0, has logarithmic power $2p_1-(d+5)r_0 > p_0$. Thus for a sufficiently weak fixed ceiling both credits pay their birth reserve and their future bank. A sufficient explicit hierarchy is

$$p_0 > (d+2)r_0, \qquad 2p_1-(d+5)r_0 > p_0. \tag{8.3e}$$

The finite constants M, L, C_w, c_θ^{-1} are absorbed by the weak-coupling ceiling. These inequalities are compatible with the original choice of sufficiently large cutoff exponents.

There are only three numerical updates:

- If there is no new failure, $Z_{\text{new}} = S(Z_{\text{old}})$. Subtract the current charge $w(Z_{\text{new}})$ from the bank. The future sum (8.3c) loses exactly that first charge, so the bank remains sufficient. Neither the old source exponent nor its old Gaussian coordinates are recharged.
- At a renewal in a small eligible component, use the new failed-test credit for its new reserve and future bank. The strict second inequality in (8.3e) permits this with the reduced positivity coefficient c_θ.
- At a merger, add the child banks and reserves and the credits of newly born pieces, then subtract the new local charge and retain one reserve for the merged component. For a binary merge, the other child's reserve is at least $c_{\text{run}}P_0(x_j)$, by the original coupling-comparability rule. The geometric overlap argument in (1.86)–(1.88) bounds the extra future charge, beyond the two child sums, by $CC_wM^dR_j^{d+2}$. The first inequality in (8.3e) makes the available reserve pay that excess. A spanning tree reduces a merger of any finite number of components to these binary updates; each child occurs once.

At healing, the waiting bank is nonnegative and the remaining reserve gives the small completed mass. A prescribed extra geometric weight is included by increasing C_w at the start. The ordinary history summation is included in the local charge, so the estimate applies to the sum over histories, not just to one history. This proves the same form of bound

$$\sum_h e^{cv(h)} \int |K_{X,h,0}|\, \|M_{X,h}\|_{\boldsymbol{r}}\, d\mu_h \le e^{-c_\theta p_0(x)-\kappa d(X)} \tag{8.4}$$

for completed components, with a possibly changed positive c_θ, once the source load and reserve fractions have been fixed before the weak-coupling ceiling. This conclusion is an extension of the explicit credit recurrence, under its stated action and coordinate assignments. It would not follow from the numerical inequality for $\mathcal{T}_X(0)1$ alone.

Sections 13–14 check the retained-action allocation at the original drop-out and swallowed-history transitions. The state rules in section 7 specify which action and which coordinates are charged. They do not allow the same positive action to be spent twice.

28.9.4 8.4 A common conservative source radius for old layers

Fix a finite prefix ending at inverse coupling x_*, and put $a = x_* - 2C_s$. The actual running tube proved in section 11 gives $x_j \geq a$ on every retained older layer. For $M > 0$ and $a > e^M$, $x/(\log x)^M$ is increasing. Thus one can choose

$$\rho = c\, a/(\log a)^M \tag{8.5}$$

and obtain $\rho/x_j \leq c/(\log a)^M$ on every old layer. This supplies a common fractional-action allowance. If the regular radius obeys $\alpha_j^2 \leq C x_j^{-1} (\log x_j)^{2p_{\mathrm{bg}}}$, then for $a > e^{2p_{\mathrm{bg}}}$ and $M > 2p_{\mathrm{bg}}$,

$$\rho \alpha_j^2 \leq C c (\log a)^{2p_{\mathrm{bg}} - M}. \tag{8.6}$$

Therefore regular zero-fluctuation subtraction terms and boundary source loads have a small per-stored-site bound, while the unrestricted positive primitive action uses (8.1). This radius is conservative. A larger clean radius is possible if the classical/Gaussian factors remain exact, but such an optimization is not needed for this history estimate.

Section 28.10 / One completed-R parameter-extension statement

Consider the literal source-free completed-R construction on its original admitted domains, with a weak-coupling run, fixed frames and positive reference kernels. Assume the incoming marked data consist of:

1. the retained primitive Wilson source with the load in (8.1);
2. local analytic regular and boundary activities with their actual coefficient norms and spare locality as in section 3.2;
3. unnormalized live-island states with the reserve and support assignments of sections 7–8.

Then one completed R operation has the following marked extension:

- The frozen replacement maps preserve the original source functional exactly, coefficient by coefficient.
- Its regular background relocations produce marked bridges by (3.3), with the termwise bound (3.5), the primitive Wilson bound (4.5), and the inherited-activity bound (3.7).
- The actual small-field Gaussian/Mayer construction acts on the combined coefficient norm with the bound (5.10). Thus a sufficiently small generic incoming potential budget is a theorem of that construction, not an additional source-specific logarithm assumption.
- Its completed core integrals retain a small absolute mass by (8.1)–(8.4). Source-dependent inverses of full island masses are never used.
- Every source label remains attached to a local anchor or to the retained island containing that anchor. A resulting connected term has support containing all of those anchors.
- Surviving islands remain states or retained-field multipliers. Healing produces scalar activities only after their internal variables are integrated.

Section 28.10 / One completed-R parameter-extension statement

The exact compatible connected regrouping is then the one proved in the stated results.
In regions where the original construction exponentiates a small regular polymer gas, the same coefficient-norm majorant permits its coefficient-algebra logarithm. In a sector conditioned on a particular live island, no separate uniform logarithm of its source mass is asserted.

This is a local induction step. Survival of the incoming source class under regular RG formation and classical/marginal extraction is a distinct statement. It is supplied for the pure-clean class in and for the tagged shrinking-core class by section 19 below. It is not inferred from the local completion theorem alone.

Section 28.11 / Scope of the local completion theorem

The argument marked-localization gap is addressed by an actual construction on the original admitted domains: marked localization can be performed on the frozen original nonlinear maps, with quantitative auxiliary-parameter bounds. It is unnecessary to infer a common envelope from a small operator norm of the full reconstructed background, and unnecessary to assume a source-dependent island denominator has a uniform zero-free disc.

The scope distinctions used in the later sections are:

- Every modified paired-field core and every seam in the candidate implementation must match the ordinary domains used in section 2. The primary theorem does not validate a later change of average, gauge, or node convention.
- The incoming coefficient norm must include the exact classical and quadratic spatial-source responses. A homogeneous temperature tangent does not control those responses. They must be retained or extracted by the correct Schur/Gaussian formulas before power-irrelevant estimates are claimed.
- Sections 13–14 perform the retained-action and independent-coordinate matching for the ORIGINAL live-island transitions. A scalar final mass inequality alone would have been insufficient.
- The original clean fixed-core family must be kept separate from shrinking-core large-field descendants, including descendants relabeled as regular in III. Section 19 proves the cutoff matching and the paired marginal estimates used by the full source induction.
- The terminal coarse/history functional remains in the exact identity (6.4). Conditional ultraviolet locality does not remove its cumulants.

No completed Yang–Mills existence proof is claimed here.

Section 28.12 / A narrower extension: independent constant couplings at birth scales

This section concerns the clean coupling functions, rather than arbitrary spatial source coefficients. It gives a useful all-history-of-scales analytic domain without varying the coefficient from plaquette to plaquette. Its inputs are the actual clean construction and the bounds already used above.

28.12.1 11.1 The original fixed physical cutoff has local complex discs

Use the original fixed physical cutoff in the first primary paper, (2.9). Fix a real finite weak run $g_s > 0$, and write $x_s = g_s^{-2}$. These are reference numbers for the coordinates below. The physical fluctuation chart and the fixed physical cutoff ϵ_1 are kept unchanged as a coupling parameter z_s varies near g_s.

The constrained Wilson minimum is independent of a spatially constant coefficient multiplying the action. Rescale the physical fluctuation using the reference g_s, not the varying z_s:

$$a = g_s C_s B - h_s D_s(g_s C_s B), \qquad t_s = (g_s/z_s)^2, \qquad P_s^{\mathrm{out}} = \epsilon_1/g_s. \tag{11.1}$$

Here C_s is the original linear constraint chart. Its nonlinear correction changes only the eliminated pivot coordinates, so the fixed tests on the independent coordinates are $|B_b| < P_s^{\mathrm{out}}$. The whole integration domain in these coordinates is independent of z_s.

The current classical Wilson action is extracted with its actual coefficient z_s^{-2}. The quadratic precision is $t_s(C^{(s)})^{-1}$, including the original soft-gauge convention. Equivalently the canonical covariance is

$$C_{t_s}^{(s)} = t_s^{-1} C^{(s)}. \tag{11.2}$$

The Gaussian integral changes by the vacuum factor $t_s^{-n_s/2}$. Its background-dependent determinant/-coarea part, hence its F^2 coefficient b_{s+1}^0, is unchanged. If the gauge Gaussian is normalized separately, its compensating normalization is included in this same vacuum identity. For $|z_s - g_s| \le c_* g_s$, c_* fixed and small, $|t_s - 1| \le C c_*$ and $\operatorname{Re} t_s \ge 1/2$. Thus section 5 applies to this actual complex Gaussian, uniformly on its enlarged field/localization domains.

After these classical and Gaussian factors are removed, all remaining vertices are holomorphic in z_s on this disc. For example the Wilson cubic coefficient is $z_s^{-2} g_s^3 = t_s g_s$; the old interactions occur as their exact fluctuation differences in the fixed physical chart. No singularity at $z_s = 0$ is being removed or crossed. This is consistent with the primary statement that the fixed-cutoff family is smooth at zero, without being analytic there.

Introduce only for estimates an intermediate fixed canonical box

$$P_s = (\log x_s)^p, \qquad p > 1/2, \qquad P_s < P_s^{\mathrm{out}}/2. \tag{11.3}$$

This smaller cutoff is also independent of z_s. It is inserted and removed by an exact tail-flag expansion below; it does not change the RG scheme. Its two useful properties are $g_s P_s \to 0$ and a tail bound $p_s = e^{-cP_s^2}$ smaller than every fixed power of x_s^{-1}.

The analytic scaled-cutoff alternative mentioned immediately after I (2.9) would give another route. It is not needed for the local positive-coupling discs constructed here.

28.12.2 11.2 The inherited activity bound is uniform on the product

The relevant input from the second primary paper is more precise than a bound on a final scalar partition function. Lemma 1, (1.33)–(1.36), localizes the entire sum of old fluctuation differences. With each old birth activity bounded by E_0 in its assigned analytic locality norm, its bound is

$$C E_0 \epsilon_1 e^{-\lambda d_s(Y)} \tag{11.4}$$

on the physical fluctuation domain $\|g_s B\| \le \epsilon_1$. Its constants depend on the fixed field radii, geometry and spare decay, but not on the number of old birth indices. The old-index summation is already performed there: the text preceding Lemma 1 uses the factor $L^j \eta$ to sum the old ages. Thus (11.4) may not be replaced by a bound for one old activity and then multiplied by the number of scales.

The proof of that lemma uses linear analytic operations, the fixed common field domains, the Ward cancellations with each birth activity's own extracted marginal coefficient, and absolute bounds on the birth activities. It does not use that their coupling parameters are real. In particular it applies pointwise

to a holomorphic family of old activities with the same uniform E_0 bound. Spatially constant scale parameters preserve gauge and lattice Euclidean symmetries, so the same scalar marginal extraction and Ward identities apply. This statement would fail as a justification for arbitrary spatial couplings.

The old state must include the pair $(E^{(u)}, \beta_u)$, with $\beta_u = \mathcal{B}_u E^{(u)}$ its own fixed linear marginal extraction. The β_u's are not independent unconstrained inputs. Evaluation at the central configuration, subtraction of the vacuum, and this marginal extraction are bounded linear operations on the assigned analytic/locality norms, with a fixed spare-radius/decay loss. The Ward cancellations in the old-age sum occur before taking absolute values and are algebraic identities for this pair. They therefore hold for the complex family as well. Estimating the separate marginal counterterms by their absolute sum would destroy the age bound and is not part of this argument. Each birth object is kept in its original assigned norm after birth. A later fluctuation difference starts from that same object and its original analytic margin; it is not obtained by iterating a shrinking evaluation radius at every age. The coarse rescaling/decay gain in the final step of the original cluster proof restores the prescribed locality exponent for the newly born object. Thus the fixed losses just described do not accumulate with the number of historical parameters.

On the full outer box, the local non-Gaussian potential family has the quadratic-growth budget

$$a_s^{\mathrm{out}} \le C E_0 \epsilon_1. \tag{11.5a}$$

The bounded scalar factor t_s does not change this order. On the inner box replace the physical fluctuation ceiling in the proof of (11.4) by $C_\kappa g_s P_s$, where the fixed factor C_κ pays the auxiliary localization discs. The resulting budget is

$$a_s^{\mathrm{in}} \le C E_0 g_s P_s, \qquad C_\kappa g_s P_s < \epsilon_*. \tag{11.5b}$$

The Wilson cubic remainder has the same bound, since its coefficient is $t_s g_s$ and one field factor is bounded by P_s. Constraint and measure remainders obey the same growth allowance. These are estimates of the same actual interaction on two domains, not a change to the interaction. If a chart requires a fixed higher polynomial power P_s^m, replace P_s by P_s^m in subsequent bounds; none of the summability conclusions changes.

We need the following quantitative comparison of the two cutoffs. Suppose a local interaction family V has outer-box budget a^{out} below a fixed fraction of the allowance in section 5. Let

$$Z_{\mathrm{out}} = \int \prod_b \chi_b^{\mathrm{out}} e^V d\nu, \qquad Z_{\mathrm{in}} = \int \prod_b \chi_b^{\mathrm{in}} e^V d\nu, \quad \chi_b^{\mathrm{in}} = 1_{\{|B_b|<P\}}, \tag{11.6a}$$

where $P < P^{\mathrm{out}}$, and let the complex covariance satisfy the already established local comparison. Their localized logarithms obey

$$\| \log Z_{\mathrm{out}} - \log Z_{\mathrm{in}} \|_{\mathrm{loc}} \le C e^{-cP^2}. \tag{11.6b}$$

Here and below constants independent of the background are subtracted in the usual local way if the norm omits the vacuum.

To prove (11.6b), insert the exact factor

$$\chi_b^{\mathrm{in}} + \lambda(\chi_b^{\mathrm{out}} - \chi_b^{\mathrm{in}}) \tag{11.6c}$$

at every bond. It interpolates between the two integrals at $\lambda = 0, 1$. Expand the product in tail flags. A selected flag is bounded in absolute value by $1_{\{|B_b|\ge P\}}$, while the outer cutoff still controls every potential assigned to that variable. Reserve the same Gaussian tilt as in (5.2). A term with v flags

acquires $(|\lambda|e^{-cP^2})^v$. To apply the expansion relative to the original Gaussian without dividing by an inner conditioned partition function, one may equivalently use the identity

$$\chi^{\mathrm{in}} + \lambda(\chi^{\mathrm{out}} - \chi^{\mathrm{in}}) = 1 - (1 - \chi^{\mathrm{out}}) - (1 - \lambda)\chi^{\mathrm{out}}(1 - \chi^{\mathrm{in}}).$$

There are two failure types. An outer failure is disjoint from the support where an outer-box potential requires its cutoff. An inner-tail failure may lie inside that support, but its field is still in the outer box, where the same potential bound holds. The Gaussian proof only uses that the flagged coordinates are distinct and exceed P; it does not require these inner-tail flags to avoid potential supports. The tail weight is at most $(1 + |\lambda|)e^{-cP^2}$. Thus the same support counting and locality bound apply. Keep the outer potential multiplier budget fixed and small. The literal majorant (5.7), its resummation, and the connected polymer criterion then converge uniformly for

$$|\lambda| < c_{\mathrm{flag}} e^{cP^2}. \tag{11.6d}$$

The constant $c_{\mathrm{flag}} > 0$ is chosen within the unused Gaussian/polymer budget. This proves a holomorphic local logarithm on that disc; it does not assume a lower bound on a large-volume partition function. Subtract its value at $\lambda = 0$, and apply scalar Schwarz, or retain the mandatory tail flag in its connected series. Either procedure gives (11.6b). In a polymer connected term the mandatory flag stays on that term's actual support. All estimates are uniform in the old parameter product.

Applying the inner version of section 5 and then (11.6b) gives the bound for the original outer-cutoff fresh interaction:

$$\|E_{s+1}^{\mathrm{new}}\|_{\mathrm{loc}} \le C\{E_0 g_s P_s + e^{-cP_s^2}\}. \tag{11.6}$$

The Gaussian cutoff-only term is included in the tail on this right side. It may depend on z_s through t_s; it is not declared constant or zero. Its local coupling derivatives are controlled by the same product Cauchy estimate below.

For clarity, here is the induction that produces that domain. Let E_G bound the source-free Gaussian birth activity in the same locality norm and fix $E_0 > 2E_G$. Define

$$\mathcal{D}_j = \prod_{s<j} \{z_s : |z_s - g_s| < R_s\}, \qquad R_s = \frac{c_* g_s}{(\log x_s)^M}, \quad c_* \le 1/4. \tag{11.7}$$

Off the running-coupling recursion, the triangular extension is defined explicitly: at step s, put the independent coefficient z_s^{-2} in front of the current Wilson action and carry all previously constructed pairs $(E^{(u)}, \mathcal{B}_u E^{(u)})$ in the original renormalized form. Compute the next fresh activity from that input. This is a well-defined formal input to the ordinary one-step map on every point of the product. On a prefix satisfying the actual coupling recursion it equals the physical forward state. It need not be the pushforward of one fixed bare measure off that recursion; the analytic extension of the triangular coupling function is all that is used by Cauchy and the subsequent chain rule.

Choose c_* first small enough for the local complex Gaussian comparison and choose the fixed physical ϵ_1 so that

$$CE_0\epsilon_1 < (E_0 - E_G)/2 \tag{11.8a}$$

and the outer-box potential budget has the spare allowance required in (11.6d). Then choose the weak ceiling so that

$$C_\kappa g_s P_s < \epsilon_*, \qquad P_s < P_s^{\mathrm{out}}/2, \qquad Ce^{-c(P_s^{\mathrm{out}})^2} < (E_0 - E_G)/2 \tag{11.8}$$

at every reference level. At the initial level there are no older nonlinear birth activities. If all older ones are holomorphic and bounded by E_0 on their respective products, they have the same bounds on $\mathcal{D}_{s+1}$, since the later coordinates do not enter them. The outer budget (11.5a), the fixed Gaussian

part, (11.8a), and (11.8) give that same E_0 bound for the next birth activity. Its sharper bound (11.6) is then available from the two-box argument. Uniform convergence of the local Mayer and connected series gives joint holomorphy. This proves the induction for every finite number of scale parameters, with constants independent of that number.

The radius loss in (11.7) is conservative for this clean argument: a fixed small relative radius also works when only the clean birth activities are considered. The smaller radius is compatible with the retained-action and regular-boundary source allowances of sections 8.1 and 8.4.

This argument invokes the actual old-age sum of the primary construction. It does not infer that sum from a homogeneous scalar estimate, and it does not assume a new spatial product domain for the minimizing field.

There are no new accumulating history denominators in this induction. The Gaussian scalar factors $t_s^{-n_s/2}$ are explicit vacuum factors. The normalization at background one subtracts the corresponding local logarithmic constants; it does not feed the reciprocal of a full past partition function into the next activity. Each new logarithm is constructed by the uniformly convergent local connected series. The renormalization counterterm is its bounded linear extraction. All vacuum factors remain in the exact density identity when that identity, rather than only the coupling coefficient, is being used. A live-island mass is not divided out.

28.12.3 11.3 A scalar row-Cauchy estimate and the bare tangent

Let $\mathcal{B}$ be the fixed linear extraction of the scalar F^2 coefficient. Exponential locality and two fixed field Cauchy derivatives make it a bounded functional on the above birth norm. The same exponential majorants control the second spatial moment in this extraction and the finite-volume limit, uniformly on compact subproducts of the coupling discs. Consequently that limit commutes with the parameter Cauchy integrals. Write the clean increment as

$$\beta_{s+1} = b^0_{s+1} + e_{s+1}, \qquad |e_{s+1}| \le A_s := C\{g_s P_s + p_s\}, \qquad p_s = e^{-c_{\mathrm{tail}} P_s^2}. \tag{11.9}$$

Here b^0 is the full Gaussian coefficient identified by the determinant calculation. The bound holds uniformly on $\mathcal{D}_{s+1}$. The cutoff-only and tail-comparison terms are retained in e_{s+1}; their coupling derivatives are not silently set to zero.

For a scalar holomorphic function f bounded by A on a product of discs centered at g, one has

$$\sum_{u \le s} R_u |\partial_{g_u} f(g)| \le A. \tag{11.10}$$

To prove it, choose phases ω_u such that $\omega_u \partial_{g_u} f(g) \ge 0$, and apply the one-variable Cauchy formula to $f(g + \zeta(R_u \omega_u)_u)$. A limit from smaller discs removes any boundary issue. This is a scalar row estimate; no unsupported sum of Banach-valued coordinate bounds is used. Applying it to (11.9) yields

$$\sum_{u \le s} R_u |\partial_{g_u} e_{s+1}| \le A_s. \tag{11.11}$$

Use now the actual, possibly nonmonotone, real trajectory. Suppose

$$s_j = \sum_{v=1}^{j} (b^0_v - c_0), \qquad |s_j| \le C_s, \qquad |e_{j+1}| \le \delta < c_0, \qquad c = c_0 - \delta > 0. \tag{11.11a}$$

These are the cumulative Gaussian estimate and the real weak remainder bound used by the shooting theorem. For $u \le s$, the exact recursion gives

$$\begin{aligned} x_u - x_s &= c_0(s-u) + (s_s - s_u) + \sum_{v=u}^{s-1} e_{v+1} \\ &\ge c(s-u) - 2C_s. \end{aligned} \tag{11.11b}$$

Thus if the weak floor is at least $4C_s$, then $x_u \ge x_s/2$. No sign of an individual Gaussian increment has been assumed. Take the floor also above $2e^M$. Since $(\log x)^M/x$ is decreasing for $x > e^M$, (11.7),(11.11) imply

$$\sum_{u\le s} g_u^3 |\partial_{g_u} e_{s+1}| \le \left(\max_{u\le s} \frac{g_u^3}{R_u}\right) A_s \le C \frac{(\log x_s)^M}{x_s} \{x_s^{-1/2} P_s + p_s\}. \tag{11.12}$$

For $d_u = \partial x_u/\partial x_0$, $dg_u/dx_0 = -\frac{1}{2} g_u^3 d_u$. Differentiating the actual triangular coupling recursion therefore gives

$$|d_{s+1} - d_s| \le a_s \max_{u\le s} |d_u|, \qquad a_s = C\left[x_s^{-3/2}(\log x_s)^{M+p} + x_s^{-1}(\log x_s)^M e^{-c_{\mathrm{tail}}(\log x_s)^{2p}}\right]. \tag{11.13}$$

Only the full Gaussian b^0 term has zero derivative in this calculation. The explicit second term pays the cutoff tails. It is superpolynomially small for $p > 1/2$ and cannot create an order-one historical derivative. The argument differentiates the full old-parameter dependence of the fresh correction; it does not replace the derivative by its current-parameter part.

If the actual terminal value is $x_K = x_*$, (11.11b) also gives $x_s \ge a + c(K-s)$, where $a = x_* - 2C_s$. For a sufficiently large fixed a, the two functions in (11.13) are decreasing, and the tail term is bounded by the first one. Consequently

$$\sum_{s<K} a_s \le \frac{C}{c} \int_a^\infty x^{-3/2}(\log x)^{M+p}\, dx \le \frac{4C}{c}\, a^{-1/2}(\log a)^{M+p}. \tag{11.14}$$

For the last inequality it is enough to take $\log a \ge 4(M+p)$; integration by parts absorbs the remaining integral. Thus (11.14) is an explicit bound for the actual nonmonotone run derived from its bounded cumulative Gaussian error. Iteration of (11.13), with $d_0 = 1$, gives

$$\max_s |d_s| \le e^{\sum_s a_s}, \qquad \max_s |d_s - 1| \le e^{\sum_s a_s} \sum_s a_s. \tag{11.15}$$

It is sufficient to choose the fixed weak stopping threshold so that the right side of (11.14) is at most $1/4$. Then $e^{1/4}/4 < 1/2$, and $d_s \ge 1/2$ throughout that finite run, uniformly in its ultraviolet length. At finite depth the product-domain holomorphy gives the ordinary local chain rule for the actual triangular recursion, so these are its genuine bare derivatives.

All coordinates and tests were fixed using the reference positive coupling only as a change-of-variables device. For nearby real parameters they still represent the original fixed physical cutoff integral exactly. The auxiliary logarithmic inner cutoff is inserted and removed, not recomputed as part of the definition of the flow. Therefore no unaccounted moving-cutoff surface term is hidden in this derivative.

The theorem concerns the actual clean coupling recursion on the original common field domains. It does not prove arbitrary spatial-source transport, nor identify the tangent of a different cutoff convention without its boundary terms. Matching a physical source to a particular frozen-history representation still requires the exact state identity of sections 6–7.

Section 28.14 / Literal matching with the readable full-density construction

This section verifies the marked extension of the ORIGINAL completed-R construction, importing the published source-free preparation and domain identities. It distinguishes those inputs from a later modification of the construction. The additional primary source used here is

T. Balaban, *Convergent renormalization expansions for lattice gauge theories*, CMP 119 (1988), 243–285, DOI 10.1007/BF01217741:

The first large-field paper, CMP 122 (1989), 175–202, is referred to below as R-I; the readable second paper, pp. 355–392, as R-II. Its detailed formulas (1.64)–(1.70), (1.89), and (1.99)–(1.102) therefore enter only where R-II explicitly states the corresponding result. No unseen formula from R-I is treated as stated. This is a bibliographic limitation on the analysis, not an additional mathematical hypothesis about sources: the published source-free preparation theorem is an imported input for the original map.

28.14.1 13.1 What the actual stored operators contain

III (2.19)–(2.22) gives the following concrete organization. For disjoint components of Z_k, the operators $T_k(X)$ commute and factor. Each is an ordered composition of one-step integrals. At a stored step j the integration variables are the actual group field V_j on the indicated large-field region and the actual fluctuation field A_j on the indicated overlap with the small-field region. In (2.21), a quadratic form couples only fluctuation variables in the SAME connected component. Its other terms are moved into the boundary action B. A preceding R operation changes the characteristic functions and can empty a fluctuation domain, but preserves this form.

In particular, an ordinary $T_k(X)1$ need not carry the Wilson action suppression. The Wilson action remains in the function to which T_k is applied. The distinction in section 7 is therefore forced by the actual formula, rather than being a choice of notation.

At the end of R-II, (1.71) composes the new local B integral with the old stored operator. The text immediately before that formula explicitly states that these integrations factor in components of Z. Formula (1.72) then keeps the history summation inside the interaction terms $V(Y)$ which depend on that history. The pure one-component terms $V(X^{\sim})$ are inside $T'_k(X)$, while terms joining components remain in the final exponential.

There is an important order of dependence here. The new determining set in R-II (1.33) removes the old determining variables in a replaced component and uses the current-scale field there. The new background in (1.35) is formed from this new datum. After the localization (1.61)–(1.65), it is an external argument of each completed component integral. The old internal variables occur in the component's stored integrals and in the explicitly localized bridges. One must perform this rewrite before using component bounds. A bound for a component written with the old, shared reconstructed background is not the bound required here.

28.14.2 13.2 Finite source algebras commute with the actual composition

Let $\mathcal{A}_N$ be the coefficient algebra of section 3. For every elementary integration in III (2.20), keep the original scalar kernel, chart and tests, and let it act coefficientwise on $\mathcal{A}_N$-valued functions. For a history h, retain all source factors in a multiplier M_h under the corresponding old-variable integrals. Integration, finite history summation, substitution into a fixed source-free background map, and multiplication are then operations in one commutative algebra. Thus their associativity and the original support factorization remain exact.

An infinite local polymer sum is handled after its absolute coefficient majorant has been established; section 3 supplies that majorant for the regular interpolation. No interchange of an integral with a supremum over exterior configurations is made.

Here are the precise transition identities. Write L_h for an unfolded scalar history kernel and $W \geq 0$ for the Wilson part which still sits in its carrying function. If $0 \leq \chi \leq 1$ is a frozen preparation test, then

$$L_h[M_h e^{-W} F] = L_h[\chi M_h e^{-W} F] + L_h[(1-\chi) M_h e^{-W} F]. \tag{13.1}$$

The failed branch has exactly the same source multiplier and the same action; there is no source derivative of χ. If it is renamed as a drop-out operator, choose an actual plaquette partition $W = W_X + W_{\mathrm{rest}}$, with nonnegative parts, and define

$$D_h^J F = L_h[(1-\chi)e^{-W_X} M_h F]. \tag{13.2}$$

The carrying factor is then $e^{-W_{\mathrm{rest}}}$, not the old e^{-W}. In particular, the small factor is not attributed to $L_h(1-\chi)1$. The partition can use the same smooth cutoffs $0 \le \phi \le 1$ as the Wilson decomposition, since $A(w,U)$ is exactly linear in w.

For swallowing, unfold the product of the child operators before integrating:

$$L_{\mathrm{parent}}\left[\prod_{a=1}^{m}\left\{\int L_{h_a}(d\xi_a)\, M_{h_a}(\xi_a)\right\} M_{\mathrm{new}}\, F\right]. \tag{13.3}$$

After the actual wall localization, all remaining cross-component dependence is in the displayed bridges. Fubini and submultiplicativity can now be applied to the separate child variables. Each child multiplier occurs once. No child scalar mass is divided out and multiplied back in. The new multiplier is the product of the old multipliers and the genuinely new local factors.

A coordinate record is indexed by its integration step, component and coordinate, not only by its physical position. Overlapping physical regions at different stored scales contain different integration records. Conversely, reusing one record in a later readout does not create another integration dimension or another primitive source charge. The ordered product (2.20) and the composition (1.71) supply exactly this record convention.

For healing, R-II first partitions components into its small-field and large-field classes. The extra characteristic functions are removed only on the class where they are identically one. Gauge-fixing delta functions are removed by the reversed Faddeev–Popov procedure only after integration over the corresponding current-scale gauge variables. The paper explicitly calls this equivalence, not equality of the displayed densities. Gauge-invariant source multipliers can be kept under the same integral, so this equivalence holds coefficientwise for them. Only the completed integral then becomes a scalar polymer activity. The large-field class remains an operator state.

Equations (13.1)–(13.3) verify the bookkeeping for the elementary moves that appear in the readable construction. They do not assert that an independently modified seam gauge has the same integration identity.

28.14.3 13.3 A fixed additive cold shift preserves the Wilson cancellation

There is a sharper way to check the denominator issue than using one relative fraction on every scale. Let $x_j = g_j^{-2}$, and let $d \ge 0$ be an upper bound for the absolute source load in the Wilson coefficients on every stored layer. Assume

$$d \le \tfrac{1}{4} \inf_j x_j. \tag{13.4}$$

For a real field U and coefficient-algebra plaquette weight w with $\|w(p)\|_{\boldsymbol{r}} \le d$,

$$e^{-A(x,U)} \left\| e^{A(w,U)} \right\|_{\boldsymbol{r}} \le e^{-A(x-d,U)}. \tag{13.5}$$

This uses the positivity of each real Wilson plaquette. The comparison shifts every inverse coupling by the same ADDITIVE number d; it is not a new physical RG run. All charts, backgrounds, tests and denominators stay frozen.

The first Wilson split in R-II (1.14) is

$$A(x_k(\cdot), U) = x_k A(U) + A(x_k(\cdot) - x_k, U). \tag{13.6}$$

Replacing every x_j by $x_j - d$ leaves the second weight $x_k(\cdot) - x_k$ exactly unchanged. This is the essential advantage of an additive comparison.

The source-free denominator in R-II (1.2)–(1.10) extracts the saddle factor $\exp[-x_k A(\zeta_0, U_0)]$, where U_0 minimizes the UNWEIGHTED Wilson action with the specified exterior constraints, as stated in (1.3). The scalar x_k does not change this minimizer. The remaining denominator is the same source-free local Gaussian/analytic normalization used throughout the proof.

For clarity, collect the Wilson manipulations (1.14)–(1.48) in the following notation. W_{in} is the incoming explicit Wilson energy, W_{ref} is the Wilson energy in the replacement prefactor, and $W_{\text{sad}} = x_k A(\zeta_0, U_0)$ is the extracted denominator energy. After the stated expansions,

$$W_{\text{in}} + W_{\text{ref}} - W_{\text{sad}} = W_{\text{out}} + W_{\text{core}} + Q_{\text{new}} + E. \tag{13.7}$$

Here W_{core} is the real retained Wilson action in (1.48), $Q_{\text{new}} \geq 0$ is the retained leading quadratic form in (1.21), and E collects the controlled Wilson expansion remainders. These names merely group the actual positive and remainder terms of the displayed primary identities.

The same algebra can be performed with the coefficients $x_j - d$, on exactly the same fields. Write a superscript d for those linear-coefficient expressions. Since the frozen denominator and replacement prefactor still use their original coefficients,

$$\begin{aligned} -W^d_{\text{in}} - W^0_{\text{ref}} + W^0_{\text{sad}} = {} & - W^d_{\text{out}} - W^d_{\text{core}} - Q^d_{\text{new}} - E^d \\ & - dA_{\text{ref}} + dA(\zeta_0, U_0). \end{aligned} \tag{13.8}$$

This identity is obtained by adding and subtracting W^d_{ref} and W^d_{sad}. For real reference fields, $A_{\text{ref}} \geq 0$, so its negative contribution can be discarded in an upper bound. The only new denominator cost is therefore

$$e^{dA(\zeta_0, U_0)}. \tag{13.9}$$

It is not a reciprocal of a sourced island integral. In particular there is no factor depending on how small the original island mass is.

The remainder estimates used for E^d are the ordinary ones with coefficients decreased by at most one quarter. All difference weights in (13.6) are unchanged, and the other leading coefficients lie between $3x_j/4$ and x_j. Positivity estimates for the retained real energies lose at most a fixed factor. Current-scale quadratic terms obtained from the scalar Wilson part are multiplied by $1 - d/x_k$. Frozen positive gauge-fixing energies may also be relaxed by this same factor for purposes of a bound.

28.14.4 13.4 The new costs use the actual retained dimensions

The regular saddle in (13.9) satisfies the original layered curvature bounds. On a layer with fine spacing η relative to its own stored unit scale, write the bound as

$$\|1 - U_0(\partial p)\| \leq C\alpha_j \eta^2.$$

For $SU(2)$, the determinant identity of section 4 implies

$$A(\zeta, U_0)\big|_{\text{one stored cell}} \leq C\alpha_j^2, \tag{13.10}$$

because the $O(\eta^{-4})$ plaquettes cancel their η^4 factor. The prepared saddle has $\alpha_j \le C g_j (\log x_j)^{p_R}$, with a fixed power p_R determined by the original buffers, waiting times and regularity estimates. Consequently the new local saddle cost is bounded by

$$\exp\Big[C d\, x_j^{-1} (\log x_j)^{2p_R} N_j\Big], \tag{13.11}$$

where N_j counts the cells or independent site records in that actual stored layer. It does not count a new fine-lattice copy of a reconstructed background.

For a normalized Gaussian on n_j actual free coordinates with positive precision H_j,

$$\frac{\int e^{-(1-d/x_j)\langle B, H_j B\rangle/2}\, dB}{\int e^{-\langle B, H_j B\rangle/2}\, dB} = (1-d/x_j)^{-n_j/2} \le e^{C(d/x_j)n_j}. \tag{13.12}$$

The actual tree/delta constraints only decrease n_j, and $n_j \le C N_j$ in the local coordinates of III (2.21) and R-II (1.71). Compact group integrations use normalized Haar measure. Their old normalization constants are already the $O(\log x_j) N_j$ costs in R-II p. 380 and (1.79).

On the nonmonotone weak run of sections 11–12, $x_j \ge a = x_* - 2C_s$. If

$$d \le c\, a/(\log a)^M, \qquad M > 2p_R, \tag{13.13}$$

then, for large a,

$$\sup_j \frac{d}{x_j} \le \frac{c}{(\log a)^M}, \qquad \sup_j d\, x_j^{-1} (\log x_j)^{2p_R} \le C c (\log a)^{2p_R - M}. \tag{13.14}$$

Only the eventual decrease of $x^{-1}(\log x)^{2p_R}$ is used. There is no monotonicity assumption on the actual run. A fixed small spatial source radius is automatically allowed once the weak floor is large. The potentially larger constant-source radius must still satisfy (13.13).

Thus the literal denominator and Gaussian comparison add a fixed bounded cost per actual stored record. Any prescribed extra history weight $\exp[c_{\mathrm{res}} \sum_j (\log x_j) N_j]$ changes only the constant C_w in the bank calculation of section 8.3. The original creation, waiting, renewal and merging power inequalities continue to pay it. The failure factors used in R-II (1.76)–(1.78) come from precisely the retained Wilson or quadratic energies in (13.7); the cold comparison keeps at least three quarters of each energy. All the original creation/renewal exponents therefore keep a positive fixed fraction.

This checks the action and dimension account at a completed original R branch and at its ordinary failed-test branch, without first assuming a weighted version of the final scalar mass bound. The completed mass estimate has the form

$$\sum_{h \to X} e^{c_{\mathrm{res}} v_h} \int |K_{h,0}(\xi; U, J)| \|M_h(\xi; U, J)\|_{\boldsymbol{r}}\, d\xi \le e^{-c_1 P_0(g_k) - \kappa_1 d_k(X)}, \tag{13.15}$$

on the original admitted external domain, after the same locality resummation as R-II. Its constants depend on the fixed source and reserve budgets. The exterior pair is fixed in this inequality; it is not an integral of a supremum over pairs. For a large general region the final tree term is retained as in the primary general-region bound.

28.14.5 13.5 The actual analytic-domain inclusion and the marked bridges

The needed inclusion is stated in R-II (1.65), on p. 375. The localized background obtained by successively substituting (1.61), (1.62), and (1.63) is an analytic function on the original external

layered pair space. Its values lie in the prepared space defined in R-I (1.64)–(1.69), with $n = N$. The following paragraph explicitly places that range inside the analyticity domain of EVERY term of the effective action, including the old terms. The proof there imports the prepared-space construction from R-I; it does not assert this inclusion for an arbitrary different average or node convention.

For original-domain analytic source members this is exactly the required composition theorem. The source coefficients are scalars in $\mathcal{A}_N$, so they do not change the map or its range. Inherited homogeneous members use their own paired counterterm and original age estimate. The spatial defects use their full tensor and cumulative birth norm. A retained quadratic operator must be represented by local source atoms, or by connected local kernels with an analytic gauge-invariant completion on the regular carrier. A bound for one global energy-normalized matrix is not sufficient.

More explicitly, R-II (1.59)–(1.60) first subtracts the old member's value at each successive localized background. The walk expansion has a root in its original support and must reach a changed component; other terms vanish when the connecting parameters are zero. The same operation is repeated for the four backgrounds. Section 3's coefficient-algebra Cauchy estimate applies at each repetition with the original spare analytic margin. The additional walk exponential and the old support exponential give the connected relative length defined by R-II (1.67):

$$d_{k,Z}(Y) = M^{-1}\{\text{length of a shortest tree in } Y \text{ meeting all } M\text{-cubes of the components of } Y \setminus Z\}. \tag{13.16}$$

For a marked term the tree also contains its retained source anchors, since these are never erased when a field argument is localized. The ordinary counting proof following (1.66) therefore gives the same exponential with the coefficient norm:

$$\|V^J(Y)\|_{\boldsymbol{r}} \le C b_{\mathrm{in}} e^{-(1+3\beta)\kappa d_{k,Z}(Y)} \quad \text{for a noninternal bridge.} \tag{13.17}$$

Here b_{in} is the actual aggregate incoming rooted norm, including the homogeneous age sum, the spatial-defect birth sum, and the new local atoms. It is not the number of old levels. For a term entirely inside one component, the corresponding bound is the extensive own-scale volume bound of (1.69); it is included inside the operator and paid by (13.15).

The action completion in (1.70) is also valid coefficientwise. Add the missing exterior source member and subtract exactly that same member from the internal bridge. Its constant part, when history dependent, stays in the bridge. A history-dependent vacuum value is not discarded as a common scalar. This removes a possible double counting at swallowing and at healing.

Consequently, for an incoming spatial class that already has these ORIGINAL full member-domain norms, the readable completed-R construction preserves the marked local representation, its source support, and the weighted island bound. Together with III's ordinary component structure, this supplies the marked completion interface to the clean smooth-source induction.

28.14.6 13.6 Source-free normalization is only an integrated identity

With the scalar replacement density $r_0(y, b)$ and its actual reference normalization $D_0(b) = \int r_0(y, b)\, dy$, the marked numerator is retained exactly as

$$\frac{r_0(y, b)}{D_0(b)} \int q_h(\xi, b) M_h(\xi, b)\, d\xi. \tag{13.18}$$

The source-free denominator is unchanged. R-II's analysis of (1.2)–(1.10), and its small complex comparison (1.73)–(1.75), concern those original normalizers. No inverse of the marked numerator is

introduced. Integrating y proves the mass-preserving identity coefficient by coefficient. It also defines an enlarged joint law which retains the old variables when an old physical observable is inserted.

This does not identify the current reference field y with a deterministic coarse readout of the original physical field. In particular, the normalization identity does not imply $R\rho(V) = \rho(V)$ pointwise, or equality of their current-field observables. The statement in R-II p. 378 about removing gauge delta functions is explicitly integrated equivalence. The original observable must remain in the old-variable numerator unless a further observable transport identity has been proved.

28.14.7 13.7 Quantitative limits at modified seams

The common-domain requirement is substantive. The construction changes inward-bond and node conventions in several places. Its own seam example can be checked directly by the following elementary identity.

Let G be a compact group with normalized Haar measure and let $f : G \to (0, \infty)$ be nonconstant with $\int f\, dU = 1$. For fixed transports p, s, a residual seam density

$$\Phi(V, s) = f(Vp^{-1}s^{-1}) \tag{13.19}$$

is not invariant under an independent left gauge change $V \mapsto hV$. Its gauge average and its integral over s are both one:

$$\int dh\, \Phi(hV, s) = 1, \qquad \int ds\, \Phi(V, s) = 1. \tag{13.20}$$

Thus a procedure which removes the residual node variable by gauge averaging or integration preserves the relevant integral, but changes the displayed conditional density. No argument from ordinary gauge invariance makes $\Phi(V, s)$ identically one.

The mismatch need not be a power-small boundary interaction. For example, take the normalized soft factor

$$f_x(U) = z_x^{-1} e^{-xa(U)}, \qquad z_x = \int_G e^{-xa(U)}\, dU. \tag{13.21}$$

Along $U = e^{itT}$ with a fixed nonzero Lie-algebra generator,

$$\log f_x(e^{itT}) - \log f_x(1) = -c_T x t^2 + O(xt^4), \qquad c_T > 0. \tag{13.22}$$

At a usual regular threshold $t = gP(g)$, the magnitude is of order $P(g)^2$, not a constant tending to zero. For imaginary $t = iu$, its real part has the opposite sign at leading order. A complex-domain bound for this factor therefore needs the explicit quadratic form, its own normalization and the actual node constraints. A generic assertion that every seam term has a small analytic potential norm does not cover it.

There is also a distinct derivative issue. A complex variation with bounded potential and first difference need not have a bounded second-difference current. This is why the original domain uses an independent J and imposes bounds on the reconstructed backgrounds of its averages. A modified constraint current, a new seam row, or a changed datum normalization has to satisfy that clause on a full complex neighborhood, not only at its real critical point.

For the original admitted histories, R-II (1.65) provides the required source-free range statement and the present coefficient extension adds no new field-domain hypothesis. The detailed source-free preparation theorem is imported, not reproved. For the candidate's MODIFIED histories, the corresponding seam gauge identity and full paired-domain comparison remain unverified. That distinct issue is not a missing hypothesis for the original-domain result stated next.

28.15.1 14.1 Precise statement

Use the original averaging, determining-set, gauge, characteristic-function, and replacement conventions of III and R-I/R-II. Import their published source-free background/preparation and analytic-domain theorems. Fix a real weak run and the original geometric parameters. Fix a reserve weight $c_{\mathrm{res}} \geq 0$.

An incoming smooth source is a finite jet

$$w = \sum_i z_i f_i, \qquad \sum_i r_i \|f_i\|_{C^\alpha} \leq r_{\mathrm{sp}}, \quad 0 < \alpha \leq 1. \tag{14.1}$$

Choose $r_{\mathrm{sp}} > 0$ below the fixed generic formation allowance and below a fixed fraction of the homogeneous bare disc of section 12. The coefficient class at a given level consists of:

1. the original homogeneous pointed birth families evaluated at their assigned source anchors, each with its own coupling counterterm;
2. the spatial defects with their full curvature quadratic tensors, original source anchors, original analytic member domains, and the cumulative birth bound of Section 14 of the stated results;
3. the localized R activities and boundary activities in the original III classes, with the absolute coefficient norm in place of the scalar modulus;
4. the actual unnormalized live operators, including their stored physical marks, with the weighted cold-action account of section 13.

The quadratic state is counted once: either as local atoms still inside a stored integral, or as a connected local kernel after that integral has been performed. On a regular carrier its gauge-invariant analytic completion is part of the source member. It is not additionally multiplied into the same density a second time.

Original-map marked interface theorem. For a sufficiently weak ceiling, chosen after the fixed source and reserve budgets, one full original preparation/completion step maps this class to the same class. It preserves the exact finite source functional after integration over the displayed current field and histories. The new regular R activities obey

$$\|R^{(k),J}(X)\|_{\boldsymbol{r}} \leq C e^{-cP_0(g_k)} e^{-\kappa' d_k(X)}. \tag{14.2}$$

The boundary activities have the corresponding original relative-tree decay and rooted source support, and the live operators obey (13.15). All constants are independent of the ultraviolet length, the volume, and the number of source generators. No marked-island inverse is used. This is an interface theorem for the stated incoming class. The companion clean induction supplies the pure-clean spatial estimates. Its use in the full construction also requires the clean/tagged distinction and marginal pairing in section 19; the theorem does not enlarge the analytic domain of a tagged member. No additional spatial minimizing-background theorem is posited.

28.15.2 14.2 Proof through the actual state transitions

First apply the original preparation maps with all scalar kernels and tests frozen. The preparation consists of the original integrations, changes of gauge or coordinates, and decompositions of unity. By (13.1), a failure retains its old mark and its carrying Wilson weight. By (13.2), moving that weight inside a named drop-out operator removes exactly the same weight from its exterior carrying action. Thus failure does not spend the action twice or discard it.

Section 28.15 / Full marked interface for the original construction

The source-free domain theorem identifies the prepared backgrounds and their complex extensions. The ranges are those used in R-II (1.65). Consequently every old analytic coefficient can be substituted into every prepared field on the same original domain. The coefficient algebra changes no gauge orbit, chart, delta constraint, or cutoff. Ordinary gauge substitutions remain valid because all the inserted Wilson sources and inherited members are gauge invariant.

The preparatory small-field integrals use the generic formation bound in section 5. The homogeneous old members retain their own paired marginal cancellation. On a stored layer of scale n, the spatial defects that are actually evaluated there have cumulative bound

$$\sum_{j\le n} a_j \le C r_{\mathrm{sp}} \ell_n^{\alpha}. \tag{14.3}$$

Their complete unextracted readout in a layer-n cell is at most $CR_0^2 \sum_{j\le n} a_j$. Their fluctuation difference has the additional physical fluctuation factor. Contributions meeting a different layer are assigned by the original localized window/cover, and the spare exponential pays the boundary connections. This is exactly the mixed-scale cover construction of III pp. 275–276, with the old coefficient norm carried through it. It does not insert the number of historical scales in place of (14.3). Inside a component being prepared, an extensive internal term is retained in that component and paid once per actual stored record.

Next perform the Wilson completion with its original denominator. The absolute coefficient norm of a real primitive Wilson mark is majorized by (13.5). Equations (13.6)–(13.9) show that this majorization survives the actual saddle cancellation. The only added saddle term is $dA(\zeta_0, U_0)$; the only Gaussian relaxation costs are (13.12). Equations (13.11)–(13.14) put these in the same own-scale volume account as the original normalizers. The bank recurrence in section 8.3 therefore gives (13.15). By decreasing the weak ceiling, the fractional energy loss can be made smaller than any fixed part of the original spare exponent.

Apply the four localization expansions in R-II (1.59)–(1.65) to each marked member. The coefficient version of the wall estimate was proved in section 3. The original domain inclusion just checked is its composition hypothesis. The relative tree estimate and root filing are (13.17). Swallowed states are composed as in (13.3), with each old source multiplier and each integration record occurring once. Source-dependent field-independent terms that vary with the history remain inside that state.

Now separate the healed and live classes exactly as in R-II p. 378. Perform the Mayer expansion and the connected regrouping (1.90)–(1.91) with values in $\mathcal{A}_N$. In this representation, the LIVE Y_i operators remain outside the expansion. They are fixed holes for the first connectedness relation; no normalization by their marked masses occurs. The scalar proof of the activity bound (1.92) explicitly uses only (1.68), (1.73), and (1.89). Its marked version uses their coefficient-norm counterparts, namely (13.17), the fixed exterior complex comparison, and (13.15). The same tree counting and KP expansion therefore converge. They give (14.2) for the terms disjoint from the surviving large-field region, and the original boundary class for the remaining terms.

The optional further normalization and exponentiation of the LIVE operators in R-II (1.103)–(1.104) is not used. The source-free paper itself introduces that step as another possible representation after its theorem has already been proved. Leaving the live states unnormalized therefore changes no required induction statement.

Finally, each frozen replacement has the integrated identity (13.18). Composition and the original integrated gauge equivalences give the exact source identity for the full step. This proves the theorem. Its proof uses the published unmarked preparation identities as inputs, and proves the source estimates that do not follow from those identities alone. □

28.15.3 14.3 Newly born R activities do not accumulate an unsummed marginal error

The R class is separate from the ordinary homogeneous E class. The original III induction gives it vacuum subtraction, but does not require a new scalar coupling subtraction. This remains a valid convention for the marked class.

Indeed the bound (14.2), with the original logarithmic P_0 exponent larger than one, beats every fixed inverse-coupling power. First apply the unextracted gauge-invariant old-member estimate to a REGULAR member, with no colored spectator insertion. Its zero flat vacuum and linear jets give the engineering factor r^4, where $r = \ell_j/\ell_n$, on a layer-n readout. This cancels the actual r^{-4} number of old roots in a present cell. The raw number of roots is exponential in the age and is not being called polynomial in x_j. After that cancellation, Cauchy readout on the original shrinking pair core costs α_j^{-2}; this and the remaining coordinate and diameter factors cost at most a fixed polynomial x_j^q, with fixed logarithms included. Thus the regular R birth ledger can be taken as

$$a_j^R \le C\alpha_j^{-2} e^{-cP_0(g_j)}. \tag{14.3a}$$

Boundary members crossing a genuine wild hole stay in the separate relative-tree and own-scale cold reserve. No clean near-flat Ward readout is applied across that hole. On a weak trajectory with $x_j \ge a + c(K-j)$ up to the already absorbed bounded partial-sum error,

$$\sum_{j\le K} x_j^q e^{-c_1 P_0(g_j)} \le C \int_{a-C}^{\infty} (1+t)^q e^{-c_2(\log(1+t))^{p_0}}\, dt. \tag{14.4}$$

For $p_0 > 1$, the integral is finite and tends to zero as $a \to \infty$. A fixed change in its constants covers the first few cells of the discrete sum. This is the summable original R budget which feeds the later clean integrations and boundary resummations. It applies to the full quadratic tensor of a marked R member. Such a tensor is not set to zero, and need not be called strictly irrelevant at one individual birth.

28.15.4 14.4 What this theorem does and does not identify

The theorem establishes the missing marked localization/large-field interface FOR THE ORIGINAL RG CONSTRUCTION, conditional only on its ordinary published background and preparation theorems and the stated incoming source class. The pure-clean and tagged pieces of that class require the distinct domains and paired marginal treatment in section 19 and in The construction. No new uniform nonlinear minimizer theorem for rough coefficients, no integral-of-supremum estimate, and no source-dependent island zero-free disc is assumed.

It retains the terminal field and all live-history dependence. It gives neither reflection positivity of the reference field nor an identification of that field with a deterministic physical coarse observable. Physical observables stay under the old-variable numerators of the frozen kernels. Separate work is still needed to construct the intended continuum observable law and to transfer its positivity, locality and nontriviality. It also does not certify The construction different seam and retained-node conventions.

28.16.1 15.1 The source algebra need not be nilpotent

Let D be a finite-dimensional open complex polydisc and let $\mathcal{A} = H^\infty(D)$, with its supremum norm. Use uniform limits on smaller closed polydiscs if an explicit complete boundary-continuous

algebra is preferred. Every estimate in sections 2–14 holds with this algebra in place of the finite jet algebra. The relevant properties are completeness, submultiplicativity, and the triangle inequality for a positive-reference integral. No step uses nilpotency.

For example, if $w(p;z)$ is holomorphic and $\sup_{z\in D}|w(p;z)|\le d$, then for real fields

$$\sup_{z\in D}\left|e^{-A(x,U)+A(w(z),U)}\right|\le e^{-A(x-d,U)}. \tag{15.1}$$

The integration majorants of sections 5 and 13 are independent of z. Dominated integration therefore preserves holomorphy. Wall derivatives are estimated by scalar Cauchy inequalities uniformly in z; the resulting sums converge normally. The original source-free normalizers are nonzero on their original admitted background domains. They remain the same normalizers. In particular, the argument does not divide by a source-dependent island mass.

Thus the original-map marked interface of section 14 is an ordinary-polydisc theorem whenever its incoming clean members are bounded and holomorphic on that polydisc with the specified uniform spatial/source norms. For smooth patches these are the actual pointwise and Hölder assembly loads of the companion clean construction, rather than the sum of the separate supremum norms of disjoint patches.

28.16.2 15.2 All regular vacuum births must be included

At a birth scale j, let $v_b(z)$ denote a local scalar vacuum coefficient, with $b=(j,X,\text{type})$. The types include the Gaussian count constants and the flat values of BOTH the regular E and regular R members. Fix a deterministic ownership rule for their local supports and put

$$V_N=\sum_b v_b(0),\qquad \mathcal{E}_N(z)=\sum_b\big(v_b(z)-v_b(0)\big). \tag{15.2}$$

These are finite sums at finite cutoff and volume. The source-free signs in v_b are the signs with which the constants occur in the logarithm of the exact density.

The inclusion of R vacua is essential. A bound $e^{-c(\log x_j)^{p_0}}$ per birth cell does not make $\ell_j^{-4}e^{-c(\log x_j)^{p_0}}$ summable in physical volume as $\ell_j\downarrow 0$. The engineering r^4 gain of section 14.3 applies only AFTER the flat value has been removed.

The primary induction has the needed universal regular family: III (2.30) explicitly subtracts the flat values of $R^{(j)}$, and III (2.32)–(2.33) defines the family $R^{(j)}(X,\cdot)$ on all admissible whole-lattice domains. R-II proves that same induction. At the level of its construction, the first history sums in (1.91) sum the healed components and internal records contained in the carrier. A component not contained in the carrier must be included whole in the seed, or it remains an external live hole. Exact wall-component locality then shows that a regular member on a fixed carrier reads only the source data and the internal summed records there. It does not read an unsummed exterior history.

This argument extends coefficientwise to the frozen source construction: sources change no admissibility test or carrier identity. It establishes the universal regular source family required in (15.2). Boundary members crossing a live hole remain history-dependent and are not inserted into that universal scalar sum.

28.16.3 15.3 Exact completion and payment of missing vacuum factors

Let I_h be the multiset of regular vacuum births actually present outside the stored islands in a fixed unfolded history h. Multiplicities can instead be fractional ownership weights when a fixed partition

of unity is used; the same identity holds. Write $\Delta v_b(z) = v_b(z) - v_b(0)$. Then

$$e^{\sum_{b\in I_h} \Delta v_b(z)} = e^{\mathcal{E}_N(z)} \exp\left[- \sum_{b\notin I_h} \Delta v_b(z) \right]. \tag{15.3}$$

The second factor stays inside the corresponding history operators, with its actual birth labels. This is an exact algebraic identity, not a comparison of two different normalizations.

For the original construction, the missing terms have localizations meeting the relevant large-field region. R-II (1.70) performs precisely this action completion, and the text immediately following defines $E_k(Z)$ as the normalization and vacuum terms whose localizations meet Z. Each missing birth term is assigned to the first stored birth-scale cell it meets. If the universal source-excess vacuum family has rooted bound

$$\sup_{j,Q} \sum_{\substack{b=(j,X,\mathrm{type}) \\ X\ni Q}} \|\Delta v_b\|_{H^\infty(D)} e^{\kappa_0 d_j(X)} \le C r_{\mathrm{sp}}, \tag{15.4}$$

the assignment and the original finite overlap give

$$\sum_{b\notin I_h} \|\Delta v_b\|_{H^\infty(D)} \le C r_{\mathrm{sp}} \sum_j N_j(h). \tag{15.5}$$

Here $N_j(h)$ is the number of the actual records at birth scale j. It is not the volume of the final swallowed region measured at the latest scale. Terms with long carriers are paid by the exponential in (15.4); a carrier is assigned once.

Consequently the absolute value of the correction in (15.3) is at most $\exp[C r_{\mathrm{sp}} \sum_j N_j(h)]$. This is a fixed additional charge in the bank of section 8.3 and is strictly weaker than its allowed $\exp[c_{\mathrm{res}} \sum_j (\log x_j) N_j(h)]$ charge. The ordinary vacuum terms at zero source already belong to the original normalizer ledger. Their ultraviolet extensive part remains in V_N, and is not paid again.

Thus factoring out ONE common $\exp[V_N + \mathcal{E}_N(z)]$ is compatible with all surviving islands, provided every universal regular vacuum birth in (15.2) is included. The uniform vacuum formation bound (15.4) is the incoming local cluster bound; for the clean spatial family it is proved by the ordinary analytic formation estimates, and for a new R family it follows from (14.2) before vacuum subtraction.

After this exact refactoring, the original terminal stability estimates give constants C_+, C_-, depending on the fixed weak endpoint and source budgets, such that

$$|Z_N(z)| \le \exp[V_N + \operatorname{Re} \mathcal{E}_N(z) + C_+ N_K], \qquad Z_N(0) \ge \exp[V_N - C_- N_K]. \tag{15.6}$$

The lower bound uses the source-free completed branch with no surviving terminal large-field island and its positive Haar-volume core. This branch still includes the sums of healed fine-scale records encoded in the regular R family. It is not the event that no large-field test ever failed. Replacing it by that event would omit the potentially ultraviolet-extensive R vacuum contribution in V_N. The upper bound uses the full absolute history majorant and the cold comparison. Here N_K is the number of retained terminal cells. The constants do not count fine sites. Equation (15.6) is a partition estimate for the exact integrated physical source functional. It does not identify the terminal reference density with a physical pushforward.

Section 28.17 / A single historical flag and finite-torus rarity

This section concerns the positive law on the ORIGINAL bare Wilson field and the auxiliary history variables. Physical insertions remain functions of the old variables under the unnormalized kernels. No terminal reference plaquette is identified with a bare observable.

Fix a finite run ending at K, with N_K terminal cells and weak endpoint x_*. All elementary replacement laws are source-free positive normalized laws. At finite cutoff, unfolding the transformations therefore gives a positive joint history law whose bare marginal is the original Wilson law. The normalization of that law is $Z_N(0)$.

28.17.1 16.1 A global birth mark would change the vacuum

It is NOT valid to multiply every creation at every fine scale by a fixed factor and then use the source-zero vacuum V_N in the numerator's stability bound. Such a modification changes the regular R vacuum births. Even a change of order $e^{-cP_0(x_j)}$ per scale-j cell can be extensive in the number of fine cells.

Instead fix one level-j cell Q, and attach ONE squarefree source z to its unique live component, when that component exists. Let $A_j(Q)$ be the age of that component's live lineage; set $A_j(Q) = 0$ on the complementary event for notation only. For a specified $a \geq 0$, the flag is

$$H_{j,Q,a} = \mathbf{1}_{\{Q \subset Z_j,\ A_j(Q) \geq a\}}, \qquad z^2 = 0. \tag{16.1}$$

The integrand is multiplied by $1 + zH_{j,Q,a}$. The zero-degree coefficient, and hence every unmarked vacuum birth in V_N, is unchanged.

At the time of marking, H is a function of the displayed history label. It is constant on that label's integration variables. Thus it does not create a new analyticity requirement for a sharp field test. After a merge the parent carries the single flag. After healing its coefficient is a local analytic insertion, with the entire old source anchor included in its carrier. Terms still attached to a wild hole remain boundary/operator insertions. A flag is never copied into two independent descendants.

28.17.2 16.2 The paid root and age factors

Choose in advance an age rate $B > 0$. Increase the fixed charge coefficient in section 8.3 and reserve a fixed small part of each creation/renewal credit. On the event in (16.1), its lineage contains at least one creation at a level $n \leq j$, and contains a nonempty live region on each of at least a successive levels. The actual coupling estimates give

$$x_n \geq x_j + c(j - n) - 2C_s. \tag{16.2}$$

For a sufficiently weak floor this implies $P_0(x_n) \geq c_0 P_0(x_j)$. A small fixed multiple of $P_0(x_j)$ is therefore paid from that creation's credit. The factor e^{Ba} is paid from the own-scale volume record on each of these live levels. The same argument pays a prescribed extra exponential in the subsequent number of live levels before the tagged state heals.

This use of the ledger does not charge a reconstructed fine background as a new integration block. The record is $(n, \text{component}, \text{coordinate})$, as in section 13.2. A different stored scale is a different record; rereading one stored coordinate is not.

There is an additional geometric count when the first creation is older than the root. At age $m = j - n$, possible ancestral positions over one level-j cell have at most CL^{4m} times polynomial diameter and buffer factors. The latter are paid by the spare tree exponent and the polynomial-logarithmic part of the volume charge. Choose the paid age rate larger than $B + 4\log L + 1$, including those fixed polynomial factors. The resulting anchored history sum is bounded by

$$C \exp[-cP_0(x_j) - Ba] \tag{16.3}$$

before later regular propagation. A positive fraction of the original final reserve and locality exponent is still left. This is the same three bank updates as section 8.3 with a larger fixed charge; all its strict exponent inequalities are unchanged.

For a tag which remains inside an unhealed operator, (16.3) is an integrated mass statement on the original admitted external pair domain. There is no normalized marked-island expectation in this statement.

28.17.3 16.3 Why a healed scalar flag does not acquire a volume density

Here is the needed one-root formation estimate. Let $D_t(X)$ be the gauge-invariant scalar coefficient of a healed flag born at level t, after its flat vacuum value has been separated. Its source anchor is the specified root y. Use the TOTAL rooted norm

$$b_t = \sum_X e^{\kappa\ \mathrm{tree}_t(y,X)} \sup_{\mathcal{U}_t(X)} |D_t(X)|, \qquad A_t = \alpha_t^{-2} b_t. \tag{16.4}$$

The carrier convention includes the fixed collar and every retained root/bridge. This is not a supremum over all possible translates of y.

Gauge invariance and semisimplicity give the zero first link/current jet. The actual source-free inner-window readout, in tree gauge, has link and own-current size at most $C\alpha_s r^2$, where $r = L^{t-s}$. Taylor/Cauchy on the original birth core therefore gives for a contractible old member

$$C b_t (\alpha_s/\alpha_t)^2 L^{-4(s-t)}. \tag{16.5}$$

Long, wrapping and removed domains use their ordinary exponential tail. The running-radius ratios are at most polynomial in $s-t$, and the exponential tail absorbs that polynomial. A carrier crossing a genuine wild hole is not treated by this tree-gauge argument.

The $L^{-4(s-t)}$ in (16.5) remains: there is only one old marked root. The $L^{4(s-t)}$ translation count of a spatial density is absent. Summing the output windows of a localization produces an $O(1)$ sum near that root, and an exponentially summable response kernel away from it.

On the actual common field-direction interpolation, Cauchy adds the factor $\theta_s = C\delta_s/\alpha_s$ to the old fluctuation difference. The derivative of the generic local Gaussian/Mayer map is bounded on its fixed allowance. This follows either by a scalar Cauchy estimate in the added potential or directly by its connected expansion: at first order in z, every connected term has one distinguished marked vertex; summing trees attached to that vertex costs its norm times a fixed polynomial of its carrier size, absorbed in spare locality. There is no unrestricted additional source root.

Consequently the new regular flag births satisfy

$$A_s \le d_s + \varepsilon_* \sum_{t<s} C(1+s-t)^q L^{-4(s-t)} A_t, \tag{16.6}$$

where d_s is the directly healed-island input. The same bound controls the TOTAL newly produced scalar vacuum coefficient. In particular it does not multiply that scalar by the number of fine cells. The zero-fluctuation value of an old member is carried forward; it is not generated anew at every step.

For clarity about wild encounters, keep a marked term in the unnormalized boundary/island class whenever its carrier meets a live hole. Its bounded old member is then evaluated at its own stored layer, without applying (16.5) across the hole. The positive mass bound pays its LIVE dwell time by the additional age charge of section 16.2. When it becomes regular again, its next birth is entered into (16.4).

Section 28.17 / A single historical flag and finite-torus rarity

An old regular member of origin t can first meet a new wild component at a much later scale v. Its own-layer bound alone gives no geometric gain in $v-t$. It would be incorrect to put an $e^{-B_1(v-t)}$ dwell factor on those REGULAR levels. Instead retain the nonstationary encounter factor from the actual component mass:

$$\delta_v = C(1+x_v)^{q'}e^{-cP_0(x_v)}. \tag{16.7a}$$

The polynomial includes the fixed radius, coordinate, and rooted bridge/attachment costs. The scalar member's radius conversion is harmless: $\alpha_t/\alpha_v \le C$ for $t \le v$ on the weak tube, after a fixed enlargement of the constant. The rooted sum of possible components touching a carrier is paid by its spare tree weight. This is an absolute component estimate from (13.15), not a conditional probability given that the component is present.

Allocate disjoint fractions of the island credit to (16.7a), its LIVE dwell weight, and its ordinary bank. Creation has the higher-power credit and renewal has the strict power surplus of (8.3e), so all three allocations are possible by fixing their constants before the weak ceiling. No division by that component's mass is used. The first encounter at v, followed by its regular return at $s > v$, is therefore bounded by $Ce^{-B_1(s-v)}\delta_v \sum_{t\le v} A_t$. An immediate return is allowed by replacing the strict inequality with the corresponding one-step convention.

Write $w_m = C(1+m)^q L^{-4m}$. The combined formation inequality is

$$A_s \le d_s + \varepsilon_* \sum_{t<s} w_{s-t}A_t + C\sum_{v<s} e^{-B_1(s-v)}\delta_v \sum_{t\le v} A_t. \tag{16.7}$$

The actual nonmonotone trajectory, and the same integral comparison as (14.4), give

$$\sum_v \delta_v \le C\int_{x_*-2C_s-C}^{\infty} (1+u)^{q'}e^{-c'(\log(1+u))^{p_0}}\,du. \tag{16.7b}$$

This tends to zero as the weak floor tends to infinity. Thus the new-hole contribution is summable over ENCOUNTER SCALES even without an age gain before entry.

Choose the ceiling so that

$$\varepsilon_*\|w\|_1 + \frac{C}{1-e^{-B_1}}\sum_v \delta_v \le \tfrac{1}{2}. \tag{16.7c}$$

Summing the nonnegative Volterra majorant then gives

$$\sum_s A_s \le 2\sum_s d_s. \tag{16.8}$$

For the original root flag, (16.3) with a further fixed future-age charge gives

$$d_t \le C(1+x_j)^q e^{-cP_0(x_j)-Ba}e^{-B_1(t-j)}. \tag{16.9}$$

The factor α_t^{-2} is included here: for $t \ge j$, the actual nonmonotone trajectory has $x_t \le x_j + 2C_s$, up to a fixed enlargement of the constant. Thus this radius cost is at most a fixed polynomial in x_j. Equations (16.8)–(16.9) bound every subsequent marked scalar vacuum and the final regular readout by

$$C(1+x_j)^q e^{-cP_0(x_j)-Ba}. \tag{16.10}$$

This proves the one-root extension needed here. It uses the original regular member domains and the actual unmarked background interpolation; it requires no Hölder regularity of a history flag. The flag is a scalar gauge-invariant insertion, with no colored spectator.

28.17.4 16.4 The literal numerator and denominator

Let $Z_N[H]$ be the positive unfolded numerator with the event (16.1), retaining its old physical variables. The coefficient of z in the completed expansion has zero-degree vacuum V_N. Its own scalar vacuum coefficient and regular terms obey (16.10), and its still-live terms obey (16.3). The terminal absolute majorant therefore gives

$$0 \le Z_N[H_{j,Q,a}] \le C(1+x_j)^q e^{V_N + C_+(x_*,B)N_K} e^{-cP_0(x_j)-Ba}. \tag{16.11}$$

The denominator is the original unsourced partition function, with the SAME V_N:

$$Z_N(0) \ge e^{V_N - C_-(x_*)N_K}. \tag{16.12}$$

These are real numerator/denominator inequalities. Combining them, and absorbing the fixed power of x_j into $P_0(x_j) = (\log x_j)^{p_0}$, with $p_0 > 1$, proves

$$\boxed{\mathbb{P}_{\text{phys,hist}}(Q \subset Z_j,\ A_j(Q) \ge a) \le C_B \exp[C_B(x_*)N_K - c_B P_0(x_j) - Ba]\,.} \tag{16.13}$$

The probability is that of the original Wilson measure augmented by its normalized history choices. It is uniform in the fine cutoff, but the displayed terminal-volume factor is real and is not suppressed.

For any fixed $q_0 \ge 0$, choose $B > q_0 \log L$. Then

$$\sum_{a \ge 0} L^{q_0 a} \mathbb{P}_{\text{phys,hist}}(Q \subset Z_j,\ A_j(Q) \ge a) \le \frac{C_B e^{C_B(x_*)N_K - c_B P_0(x_j)}}{1 - e^{q_0 \log L - B}}. \tag{16.14}$$

The same argument admits a fixed extra polynomial in the actual stored volume or energy, paid by a slightly larger reserve. The label age is bounded by the live lineage age. An age defined from ALL past creations at a physical location, including ones healed long ago, is not covered by (16.14).

28.17.5 16.5 Consequence for a finite physical window

For a union W of observation neighborhoods, the union bound and (16.13) give

$$\mathbb{P}_{\text{phys,hist}}(Z_j \cap W \ne \varnothing) \le C\, \#_j(W^+)\, e^{C(x_*)N_K - cP_0(x_j)}. \tag{16.15}$$

Here W^+ includes the fixed buffer/large-field cell collars, and $\#_j$ counts cells in their own level-j units. The age-weighted version replaces the probability on the left by the corresponding sum over touching cells in (16.14).

For example take a window of physical diameter $O(d)$, choose $\ell_j = a'$ with

$$d/a' \asymp (\log x_j)^R \tag{16.16}$$

and choose that room large enough to contain the original polylogarithmic collars. The cell count is at most a fixed polynomial in $\log x_j$. More generally a fixed power of x_j is allowed. Since

$$x_j \ge x_* - 2C_s + c(K-j), \qquad K - j \asymp \log(1/a'), \tag{16.17}$$

the right side of (16.15) tends to zero as $d \downarrow 0$, on each FIXED physical torus. All estimates remain uniform as the bare cutoff is removed below a'.

Section 28.17 / A single historical flag and finite-torus rarity

This is the finite-volume large-field input for a UV-window comparison. If the transported first and second source jets in that window have the usual own-scale bounds, namely polynomial-logarithmic costs times L^{4A} per first insertion and L^{8A} for a product, (16.14) makes their bad contribution at most

$$Ce^{C(x_*)N_K}\exp[C(\log x_j)^{r_0}]\,(1+x_j)^q e^{-cP_0(x_j)}\|f_1\|_\infty\|f_2\|_\infty. \tag{16.18}$$

The source jets here are the old-variable jets of the exact transported density after its common vacuum has been factored out. They are not jets of an arbitrary terminal reference plaquette. The primitive stored-action part of this count uses $a_p \le 2$, the actual number of old stored cells, and the regular η^4 plaquette gain where such regular reconstruction is used. Polynomial Gaussian energy insertions are handled by the cold-action majorant, not by a pointwise bound on an unbounded Gaussian variable.

For $p_0 > r_0$ the factor in (16.18) is $o(x_j^{-m})$ for every fixed m, at fixed N_K. In particular it is smaller than a nonzero Gaussian F^2 signal of order $x_j^{-2}\|f_1\|_\infty\|f_2\|_\infty$. The actual clean-window Gaussian coefficient, the transported source normalization, and the stated source-jet bounds still have to be proved in that comparison; (16.13)–(16.18) do not replace those tasks.

The threshold obtained in this way depends on the physical torus volume through $C(x_*)N_K$. It cannot by itself prove a volume-uniform local probability, or noncollapse on $\mathbb{R}^4$. Removing that dependence requires a separate physical local comparison, or a valid reflection-local witness construction.

Apply reflection positivity only to ordinary local physical Wilson sources, whose support is unambiguous. This gives a different route from the fixed-reference Gaussian window to larger volumes. It does not apply reflection positivity to a history event. The argument below checks the finite-volume part of the candidate ET comparison using the actual C^α source class.

28.19.1 18.1 An exact projected periodic transfer identity

In a fixed lattice direction let U denote the spatial links on one finite periodic transverse slice. Put

$$k_\beta(g) = e^{-\beta a(g)}, \qquad C_\beta(U, U') = \prod_e k_\beta(U_e U_e'^{-1}), \qquad b(U) = e^{-\beta S_{\mathrm{spatial}}(U)/2}. \tag{18.1}$$

The Wilson character expansion has nonnegative coefficients. Hence the convolution C_β is positive and trace class on product Haar L^2: on each Peter–Weyl block its eigenvalue is the character coefficient divided by the representation dimension, and the sum of eigenvalues with multiplicities is $k_\beta(1) = 1$. Tensor products preserve trace class. Thus $T = M_b C_\beta M_b$ is positive and trace class.

Let P average over the slice's site gauge transformations. It is an orthogonal projection and commutes with T. Retain the temporal links in the periodic Wilson integral. Their integration averages the endpoint of each layer kernel over its slice gauge transform. Its exact kernel is therefore TP, and kernel composition gives

$$Z_n = \mathrm{Tr}(T^n P). \tag{18.2}$$

Normalized Haar gives no gauge-volume constant. This derivation retains the periodic temporal holonomy integration; it does not set all periodic temporal links equal to one. Gauge-invariant sources commute with P.

The primary transfer-matrix construction is Lüscher, DESY 76/54, pure-gauge factors (17), (20), (21) and Proposition 1: Identity (18.2) is the direct projected periodic integral.

Place a bounded complex gauge-invariant source inside a slab, away from its two cutting faces. Reflect the slab and conjugate the source in the neighboring slab. On PL^2, if the first slab has operator A, the reflected one has A^*. The pair is the positive trace-class operator $Y = A^*A$. For every integer $q \geq 1$,

$$0 \leq \operatorname{Tr} Y^q \leq (\operatorname{Tr} Y)^q. \tag{18.3}$$

This follows by summing its nonnegative eigenvalues. Temporal plaquette sources strictly inside the slab are included in A; positivity is required only of the source-free crossing kernels. All plaquette orientations are therefore allowed.

28.19.2 18.2 Exact nonwrapping vacuum companions

Use one common coupling run and the same volume-independent marginal extraction and local auxiliary choices on compatible tori. The marginal coefficients are extracted from the infinite local family, as in the original construction, and are not recomputed separately from each torus's wrapping terms.

This convention is literal in I (1.20)–(1.22): after constructing the finite-volume polarization tensor, the author takes its infinite-volume limit and only then defines the marginal coefficient. III (3.62)–(3.64) identifies its pointed extraction with that definition. For terms with a retained large-field origin the same infinite local family and its own paired extraction must be used; relabeling a term does not change this convention.

A wall-localized term whose COMPLETE carrier fits into a box shorter than a fixed fraction of the torus has a unique infinite-lattice lift. With walls outside that carrier set to zero, all its propagator walks, fixed-point equations, local trees, internal histories and integration domains are identical to those of the lift. Their coefficients agree exactly. A winding walk or history cannot be contained in such a carrier because its whole connected support is retained.

There are three actual ingredients behind this assertion. II (1.6)–(1.10) uses kernels supported away from their local-domain boundaries; when exterior walk parameters vanish, both nonlinear fixed-point equations split exactly into component equations. Their convergent Picard solutions on a lifted component therefore agree. III (3.16)–(3.20) leaves only fluctuation-coordinate cutoffs on an interior regular component, and the paragraph following (3.47) explicitly identifies the interior coefficients with their whole-lattice versions. Finally R-II (1.71)–(1.72) states that every expression in a component integral is localized in that component, while its bridge coefficients depend only on determining histories in components contained in their carrier. The connected construction in (1.90)–(1.98) retains the union of those component and bridge carriers.

Consequently the induction proving equality of lifted coefficients includes their internal determining sets, axial trees, integration variables, normalizers, and cutoffs, not just the final external field arguments. A history whose hull or determining collar winds around the torus belongs to the wrapping remainder. All partitions and deterministic ownership choices are chosen as periodizations of one infinite-lattice choice, and the reference torus is large enough and commensurate with the finitely largest current-scale block sizes.

This applies to the universal regular E and R vacuum births of section 15. Boundary histories are first completed by (15.3); they are not themselves called universal scalar coefficients. The current-step Gaussian count constants are proportional to the current cell count. For instance the physical hidden dimension of a full L^4 block, after residual gauge directions are removed, is $(d-1)\dim G\,(L^4-1)$. Indeed the local pivot constraints remove dN_c coordinates from dN_f, and the residual gauge space has dimension $(N_f-N_c)\dim G$; its block-mean-zero condition excludes the constant gauge mode. Their difference is $(d-1)(N_f-N_c)\dim G$. The normalized soft gauge factor compensates exactly its residual Gaussian power. Global retained gauge/holonomy variables do not create a new cutoff-dependent bulk count. This count is made before taking a logarithm or comparing volumes.

Rooting each vacuum carrier by fixed ownership and using the locality/entropy surplus consequently gives

$$\left|\ell^{-4}V_{N,\ell}-v_N^{\infty}\right|\le C\sum_{s=0}^{N}L^{4s}e^{-c\ell L^s}. \tag{18.4}$$

The common density v_N^{∞} may diverge with the fine cutoff. For $\ell\ge\ell_*>0$,

$$\sum_{s\ge0}L^{4s}e^{-c\ell L^s}\le e^{-c\ell/2}\sum_{s\ge0}L^{4s}e^{-(c\ell_*/2)L^s}\le C_{\ell_*}e^{-c\ell/2}. \tag{18.5}$$

Thus two compatible vacuum densities differ by a bounded amount δ_{vac}, uniformly in the fine cutoff. The exact nonwrapping identity is used BEFORE the ultraviolet roots are summed. Equal numerical activity bounds on two tori would not prove this statement.

28.19.3 18.3 One holomorphic cell normalizer

Fix a reference side ℓ_*, a cell side l_c, and an even integer $n_*=\ell_*/l_c$. Put the physical source support in an interior ball of one cell, at a fixed positive distance from the faces. On each $\ell=q\ell_*$, reflect it into all cells and conjugate its coefficients in odd cells. Call the resulting source W_z. Its ordinary patch C^α assembly norm is bounded by a fixed multiple of the single-cell norm, independently of the number of cells. The vanishing collar is used here.

Let $M_\ell(z)$ be the physical Wilson generating function of the original SINGLE-cell source, normalized by $Z_\ell(0)$. Finite chessboard gives, with $n_\ell=\ell/l_c$,

$$|M_\ell(z)|\le[Z_\ell(W_z)/Z_\ell(0)]^{1/n_\ell^4}. \tag{18.6}$$

The reflected/conjugated numerator is nonnegative. Apply (18.3) in each coordinate direction, using rectangular intermediate tori. Because the sources repeat in reflected slab pairs,

$$Z_\ell(W_z)\le Z_*(W_z)^{(\ell/\ell_*)^4}. \tag{18.7}$$

D1 on the reference torus, the source-free floor on the larger torus, and (18.4) give

$$|M_\ell(z)|\le\exp\left[n_*^{-4}\operatorname{Re}\mathcal{E}_*(W_z)+(C_++C_-+\delta_{\mathrm{vac}})l_c^4\right]. \tag{18.8}$$

Keep only source-excess vacuum carriers lying entirely inside one reference cell. In an even cell use their value. In an odd cell conjugate their value once more, so it has the holomorphic form $\overline{\phi(\bar z)}$. Average these sums over the n_*^4 cells, obtaining $\mathcal{E}^\circ(z)$, with $\mathcal{E}^\circ(0)=0$. Its real part equals the averaged real part of the kept terms. This definition needs no covariance of individual localized terms under cell reflections.

An omitted source-excess carrier meets a cell face and the source support. At spacing L^{-s} its length is at least $cL^s - C$. Its root count is at most $C(1+L^s)^4$, so the spare locality bounds the total per cell by the finite series

$$C_{\mathrm{face}} \le C \sum_{s\ge 0} (1+L^s)^4 e^{-c' L^s}. \tag{18.9}$$

The finitely many coarsest scales are bounded separately. Hence ONE common holomorphic normalizer serves all compatible tori:

$$\boxed{|e^{-\mathcal{E}^\circ(z)} M_\ell(z)| \le Q, \qquad \ell = q\ell_*.} \tag{18.10}$$

The constant Q depends on the fixed reference torus, cell collar, weak endpoint and source budget, but not on the larger volume or the fine cutoff. At a fixed cutoff the argument to every weak limit of those periodic Wilson measures, since the single-cell physical observable is bounded and continuous. Extension to EVERY DLR state would need a separate physical domination statement; periodic subsequence limits require no such claim.

28.19.4 18.4 Covariance differences gain $d^{4-2\alpha}$

For two compatible measures, including the reference torus and a periodic subsequence limit, put

$$D(f,h) = \mathrm{Cov}_\rho(O(f),O(h)) - \mathrm{Cov}_\tau(O(f),O(h)). \tag{18.11}$$

In the logarithmic second derivative the common $\mathcal{E}^\circ$ cancels. If $\|F_i\|_{C^\alpha} = B_i$, the normalized functions in (18.10) equal one at zero and have bound Q on $\sum_i |z_i| B_i < r_0$. Schwarz along complex rays gives radii $cr_0/[(Q+1)B_i]$ on which they lie within $1/2$ of one. Their local logarithms are bounded. Cauchy then gives

$$|D(F_1,F_2)| \le C(Q,r_0) B_1 B_2. \tag{18.12}$$

Only this smaller local logarithm is used, not a global zero-free source theorem.

Take fixed-shape bumps f, h of scale d, contained in a ball of radius Cd, with $\|f\|_{C^\alpha} \le Cd^{-\alpha}\|f\|_\infty$, and similarly for h. Pack $N_d \ge cd^{-4}$ translates of the pair inside one fixed interior cell ball, separated by distances comparable to d. Choose all translations on the common bare grid. For independent signs ϵ_v, use the SAME sign on the two entries of each pair:

$$F^\epsilon = \sum_v \epsilon_v \tau_v f, \qquad H^\epsilon = \sum_v \epsilon_v \tau_v h. \tag{18.13}$$

Their C^α norms are at most $Cd^{-\alpha}\|f\|_\infty$ and $Cd^{-\alpha}\|h\|_\infty$, with no factor depending on the number of copies. Translation invariance and sign averaging give $\mathbb{E}_\epsilon D(F^\epsilon, H^\epsilon) = N_d D(f,h)$. Thus

$$\boxed{|D(f,h)| \le Cd^{4-2\alpha}\|f\|_\infty \|h\|_\infty.} \tag{18.14}$$

For $\alpha = 1$ this is the d^2 gain in the candidate ET argument. Every $0 < \alpha \le 1$ gives a positive physical power. The covariance is that of the physical Wilson observable.

If the Gaussian window on the ONE reference torus is independently proved with main term $cx(d)^{-2}\|f\|_\infty\|h\|_\infty$, its relative transfer error is at most $Cd^{4-2\alpha}x(d)^2$. Since the weak run has $x(d) = O(\log(1/d))$, this tends to zero uniformly in the larger compatible volume. This can transfer finite-reference noncollapse without the historical witness theorem of section 17. It does not itself prove the interacting Gaussian reference window.

28.19.5 18.5 Multicell chessboard at arbitrary even cell counts

The physical chessboard bound used here has no power-of-two restriction. Let a periodic lattice be partitioned into $m_1 \times \cdots \times m_4$ congruent cells, with each m_ν an even integer. Let F_c be a bounded gauge-invariant complex function supported strictly inside cell c. Reflection includes complex conjugation. Write $\mathcal{D}_c F_c$ for its reflected dissemination to every cell of the torus, and $N = \prod_\nu m_\nu$. Then

$$\left|\left\langle \prod_c F_c \right\rangle\right| \le \prod_c \langle \mathcal{D}_c F_c \rangle^{1/N} . \tag{18.15}$$

The factors on the right are nonnegative. Empty cells have $F_c = 1$ and contribute one.

For completeness, slice first in one coordinate. The product of all functions in slab i, including its source-free half-boundary factors, defines an operator A_i on the gauge-invariant slice Hilbert space. The numerator is $\mathrm{Tr}(A_1 \cdots A_m)$. The source-free margins ensure each A_i is in the required Schatten class. Generalized trace Hölder gives

$$|\,\mathrm{Tr}(A_1 \cdots A_m)| \le \prod_{i=1}^{m} \|A_i\|_m = \prod_{i=1}^{m} [\mathrm{Tr}(A_i^* A_i)^{m/2}]^{1/m}. \tag{18.16}$$

Since m is even, the last trace is exactly the partition integral with alternating reflected and conjugated copies of that slab on the same torus. Divide by the common source-free partition function. Its powers on the right sum to one. Repeat this argument in the remaining three coordinates. Coordinate reflections commute, and their exponents multiply to $1/N$, proving (18.15). Odd blocking factor L causes no difficulty: begin with an even number of terminal cells, so every finer count is even.

There is a useful exact-local specialization. Suppose a holomorphic scalar functional $\mathcal{L}_N[J]$ is real under conjugation, covariant under the cell reflections, and additive for profiles with disjoint interior cell supports. If

$$|e^{-\mathcal{L}_N[J]} Z_N[J]/Z_N[0]| \le e^{C\,\mathrm{Vol}(T)} \tag{18.17}$$

holds on the admitted source domain, then disseminating a cell profile yields $\mathcal{L}_N[\mathcal{D}_c J] = N\ \mathrm{Re}\,\mathcal{L}_N[J]$. Therefore

$$\left|\left\langle \prod_{c\in\mathcal{C}} e^{-\mathcal{L}_N[J_c]} e^{O(J_c)} \right\rangle\right| \le e^{C\,\mathrm{Vol(cell)}\,|\mathcal{C}|}. \tag{18.18}$$

The normalizer need not agree between two different volumes for this within-volume statement. Strict locality and reflection covariance, rather than an exponentially decaying approximate additivity, give the exact cancellation in (18.18).

If the construction supplies (18.17) with a different local normalizer for each translated or reflected choice of the auxiliary grid, average those local normalizers over the finite symmetry group. The physical partition function transforms covariantly. Averaging the inequalities for $\log|Z_N[J]/Z_N[0]|$ preserves (18.17), and the averaged normalizer has the required covariance. This does not average physical measures or replace their observable by a reference-field observable.

28.20.1 19.1 The primary parameters are different

Three different uses of a small radius must not be identified.

Object	Actual parameter	Consequence
Clean integral in I (2.9) and II (1.19)–(1.36)	A fixed absolute physical fluctuation bound ϵ_1; after scaling, $\lvert B\rvert \le \epsilon_1/g_k$	The clean birth functions are analytic on fixed (U, J) radii α_0, α_1, as explicitly stated in I (1.18).
Full construction's real regularity and fluctuation tests	$\varepsilon_k = A_0 g_k(\log x_k)^{p_0}$, and a smaller physical fluctuation threshold proportional to $\delta_k = A_1 g_k(\log x_k)^{p_0}$	These are the moving, source-free real thresholds in III (1.1), (1.8), and (3.16), frozen at the chosen reference run. They are not the fixed ϵ_1 of I.
Full construction's analytic member domains	$\alpha_{i,k} = C_i g_k(\log x_k)^{q_i}$, III (2.28)	The generic E/R/B representation is asserted only on these shrinking cores. In particular, a large-field origin cannot be differentiated using a fixed absolute field radius.

The fixed physical cutoff of section 11 constructs the clean family and its scalar analytic extension. It does not, by itself, say that the entire full effective action is analytic on the same fixed core.

There is a further literal point in III. On printed p. 271, immediately before (3.26), sufficiently old R terms are included in E once the original block partition has become compatible with the current lattice Euclidean group. Thus the symbol E in III can include descendants of R origins. Its name does not give those descendants the stronger fixed-core property of the pure-clean I–II family.

28.20.2 19.2 The original smaller clean cutoff has a fixed-core extension

For an interior regular domain the needed cutoff matching is supplied explicitly by III, not by assuming analyticity of a sharp moving test. After (3.15), the author introduces (3.16) precisely to replace the approximate-fluctuation tests, which depend on the new external field, by tests on the independent fluctuation coordinates alone. Equations (3.17)–(3.19) prove that the previous background-dependent test equals one in this regular sector. Equation (3.20) and the following paragraph state that the remaining product on Λ_{k+1} is $\chi^{(k)}(\Lambda_{k+1})$, formed only from the coordinate tests in (3.16). Boundary exceptions stay in the boundary class.

Define the pure-clean whole-lattice kernel using exactly this smaller canonical cutoff, with the reference value of its threshold frozen. It is a fixed box of size

$$P_k^{\mathrm{III}} \asymp (\log x_k)^{p_0} \tag{19.1}$$

in the independent scaled fluctuation coordinates. The original pivot linearization changes only eliminated coordinates; this is the reason for the starred sets in III (1.8)–(1.9), and is stated explicitly after (1.14).

The fixed I–II background maps and Gaussian comparison are valid on a fixed larger field/current core for every physical fluctuation below a fixed ϵ_1. Since $g_k P_k^{\mathrm{III}} \to 0$, this smaller box lies there. Its indicator does not change under a complex variation of the background or the source. The section 5 construction consequently gives an analytic pure-clean birth on that fixed core, with the usual E allowance, and a sharper fresh non-Gaussian bound

$$C\{g_k(P_k^{\mathrm{III}})^m + e^{-c(P_k^{\mathrm{III}})^2}\}. \tag{19.2}$$

The exponent m is fixed by the local chart. Uniform holomorphy follows from dominated Gaussian integration and the same absolutely convergent localized series as in section 11.

This is a construction with the actual smaller cutoff, not an identity between two different cutoff integrals. If a fixed-ϵ_1 clean family is used as a reference instead, the exact tail-flag interpolation (11.6a)–(11.6d) gives their local logarithmic difference, including its vacuum value and its marginal extraction. Its norm is $Ce^{-c(P_k^{\mathrm{III}})^2}$. That correction must be retained. There is no derivative of a source- dependent characteristic function in this argument.

In particular, a homogeneous source family using frozen tests need not equal the family obtained by recomputing every real threshold after varying the bare coupling. Both are exact physical decompositions when their counterterms and tail terms are retained. Their coordinate derivatives cannot be identified just from source-free equality at one parameter value.

28.20.3 19.3 Permanent origin tags commute with the exact operations

Refine the state by an auxiliary origin flag. A new term with no large-field input is pure-clean. Every term containing at least one R birth or a completed large-field history retains that origin, even when III later places it inside E. An optional cutoff-comparison term may be assigned a separate tail origin with the same bookkeeping.

This split is exact in the convergent connected series. Equivalently, for the local formation map $\mathfrak{F}$, on its actual common smaller domain,

$$\mathfrak{F}(E_{\mathrm{cl}} + R) - \mathfrak{F}(E_{\mathrm{cl}}) = \int_0^1 D\mathfrak{F}(E_{\mathrm{cl}} + tR)[R]\, dt. \tag{19.3}$$

Every term on the right has a distinguished R vertex; all its other vertices are summed with the ordinary absolute majorant. The same identity is valid in the source Banach algebra. Wall localization, original gauge substitutions, and unnormalized live transport do not erase a flag. The pure-clean term on the left is evaluated by its fixed-core construction. Terms on the right use only their actual shrinking common core.

The primary promotion time and marginal extraction must also be retained. If an origin born at j becomes Euclidean covariant at k, extract its own scalar marginal at that time and keep the counterterm paired with that origin. The extraction is linear, so

$$\beta_{k+1} = \beta_{k+1}^{\mathrm{cl}} + \beta_{k+1}^{\mathrm{tag}}. \tag{19.4}$$

The tagged term on the right contributes to the leading scalar coupling whenever the primary scheme performs that subtraction. Dropping it, or leaving all tagged quadratic terms permanently unrenormalized, changes the scheme. In particular the recurrence $M_{k+1} = (1+\epsilon)M_k + \delta_k$ would not be controlled merely by $\delta_k = e^{-c(\log x_k)^p}$ at arbitrarily old origins.

For a spatial source first freeze the constant homogeneous family, including its own $\beta^{\mathrm{tag}}(c)$, at the pointed source anchor. Only the remaining heterogeneous difference is assigned the full nonscalar quadratic tensor. That difference gains the source oscillation at its own birth scale. There is no Euclidean scalar-marginal identity for an arbitrary spatial tensor.

28.20.4 19.4 The paired tagged age estimate and its quantitative closure

The estimates can be proved without restoring a fixed field radius. Let a vacuum- subtracted tagged birth at scale j have rooted norm b_j on its original core of radius α_j, and put

$$u_j = \alpha_j^{-2} b_j. \tag{19.5}$$

The factor α_j^{-2} is essential. Gauge invariance and the zero first joint jet give the regular readout factor

$$b_j(\alpha_k/\alpha_j)^2 L^{-4(k-j)}. \tag{19.6}$$

For a translation-summed family the root count is $O(L^{4(k-j)})$, which cancels precisely the fourth power in (19.6). Thus an unpromoted quadratic term gives no extra age decay. A fluctuation difference supplies the additional factor $C\delta_k/\alpha_k$, on a common spare-radius neighborhood.

After the primary promotion and subtraction of the paired scalar marginal, the ordinary I–II Ward/-Taylor argument supplies a strict age gain. Written with explicit radii, its kernel has the form

$$\begin{aligned} &W_{k,j} \le C, && k-j \le h_j, \\ &W_{k,j} \le C(1+k-j)^r(\alpha_k/\alpha_j)^r L^{-\sigma(k-j-h_j)}, && k-j > h_j, \end{aligned} \tag{19.7}$$

with fixed r, $\sigma > 0$, and promotion delay

$$h_j \le C_h\{1 + \log\log x_j\}. \tag{19.8}$$

Here is the required domain and the analysis. III (2.30) explicitly assigns R the properties (i)–(iii) stated after (2.27), including the SAME analytic pair space and the same gauge action. R-II (1.65) places each old member's prepared readout inside that space. The source-free minimizing map and its normalized own current are independent of the origin label. On an inner regular window their linear response annihilates affine constant-curvature data: the centered averages reproduce the affine potential, and the quadratic Wilson Euler operator d^*d annihilates it, so its constrained multiplier/current is zero.

The original local analytic response bounds, applied to the Hölder error in the transported datum, then give

$$J^{\mathrm{own}} = O(\alpha_k r^{2+\zeta}) + O(\alpha_k^2 r^4), \qquad r = L^{j-k}. \tag{19.7a}$$

The remote-window term has the ordinary exponential response bound. This is the source-free normalized-current estimate, not an estimate of an arbitrary outer independent current. III's proof of Theorem 2, (3.48)–(3.54), explicitly applies the original Ward/Taylor construction at the actual stored layer, replacing its scale by L^{-n}.

In tree gauge, the link coordinate is $r^2\mathcal{A}F + O(\alpha_k r^{2+\zeta} + \alpha_k^2 r^4)$. Its joint Taylor expansion, (19.7a), and the zero first jet give, after removing the appropriate quadratic curvature tensor,

$$Cb_j\left[(\alpha_k/\alpha_j)^2 r^{4+\zeta} + (\alpha_k/\alpha_j)^3 r^6\right]. \tag{19.7b}$$

Polynomial carrier sizes use the spare locality exponent. The r^{-4} root count leaves $\alpha_k^2 u_j[r^\zeta + (\alpha_k/\alpha_j)r^2]$. After promotion the symmetry-allowed tensor is the scalar one, and its subtraction is the paired marginal subtraction in (19.4). Before promotion only (19.6) is used. The fixed few near ages use the spare birth-core margin directly; later ages satisfy $\alpha_k r^2 < c\alpha_j$ by the running tube. Outer or removed windows retain the original response/locality tail. A boundary crossing a genuine wild hole remains an operator term and is not covered by (19.7b).

This proves (19.7), with a fixed finite near-age enlargement absorbed in h_j. The delay follows directly from the primary criterion that the birth partition of side MR_j, transported to the current lattice, fit inside an L block. Formula (19.7) is the radius-explicit form of the ordinary renormalized old-member estimate; it must be applied to the paired member, not separately to its marginal counterterm.

The actual running tube gives

$$x_k - 2C_s + c(k-j) \le x_j \le x_k + 2C_s + C(k-j), \qquad j \le k. \tag{19.9}$$

For x_k above the fixed monotonicity threshold of $x^{-1/2}(\log x)^q$, $\alpha_k/\alpha_j \le C(1+k-j)^{1/2}$. Moreover the sum of the kernels (19.7) satisfies

$$\sum_{j\le k} W_{k,j} \le C(1+\log\log x_k)^{r+1}. \tag{19.10}$$

To see this with the variable delays, split at $k-j = C'(1+\log\log x_k)$. The near part has that many terms, each bounded by a fixed polynomial in the split size. In the far part, (19.9) and $\log\log(x_k + C(k-j)) = o(k-j)$ imply $h_j \le (k-j)/2$, after increasing the fixed split constant; the remaining polynomial times $L^{-\sigma(k-j)/2}$ is summable.

The derivative form (19.3) and the actual generic formation bound therefore give

$$u_{k+1} \le d_{k+1} + C\frac{\delta_k}{\alpha_k}\sum_{j\le k} W_{k,j}u_j. \tag{19.11}$$

Here d_{k+1} contains newly completed large-field origins and any explicitly retained cutoff-comparison tail. A fixed polynomial radius loss is included in d; it is not repeated as an unexplained loss at every later age.

Choose the permitted domain hierarchy so that $\delta_k/\alpha_k \le C(\log x_k)^{-\tau}$ with $\tau > 0$. This is an explicit hierarchy condition: the analytic field radius must exceed the physical fluctuation ceiling, including the fixed localization loss. Then

$$\sup_k C(\delta_k/\alpha_k)(1+\log\log x_k)^{r+1} < \tfrac{1}{2} \tag{19.12}$$

for a sufficiently weak floor. A decreasing superpolynomial envelope

$$F(x) = (1+x)^q e^{-c(\log(1+x))^p}, \qquad p > 1, \tag{19.13}$$

absorbs the direct birth bound. Since $F(x_j) \le CF(x_k)$ for $j \le k$ by (19.9), enlarge the fixed constant in (19.12) by this envelope-comparability constant before choosing the weak floor. That constant need not be one: the real running coefficients need not be pointwise monotone. With this choice, induction in (19.11) gives

$$u_k \le CF(x_k), \qquad \sum_k u_k \le C\int_{a-C}^{\infty} F(u)\,du. \tag{19.14}$$

The integral tends to zero with the weak floor. Any fixed additional polynomial Cauchy loss can be included in the envelope before choosing a. The linear marginal functional has norm bounded by $C\alpha_j^{-2}$ after its locality moment sum, so $|\beta_j^{\mathrm{tag}}| \le Cu_j$. Here the bound is first per promoted origin. At an actual promotion step k, its marginal can be the sum of the origins j whose promotion time is k. Their ages are $O(1+\log\log x_k)$, so the sum costs at most a fixed polynomial in $1+\log\log x_k$. Absorb this multiplicity into the envelope before using (19.18). It remains superpolynomially small and summable on the running tube. If the tagged family is bounded on the ordinary bare source

disc, its source derivatives satisfy the corresponding Cauchy bounds on a smaller disc and remain summable.

The hierarchy in (19.12) is permitted by the original domain parameters: choose the fixed analytic logarithmic powers above the physical fluctuation power, then choose the weak ceiling. In R-II (1.73) the resulting complex-field cost is a fixed power of logarithms times g_k; increasing a fixed logarithmic power does not change its tendency to zero. The source cold radius is decreased once to accommodate this choice, as in (13.13). No analytic radius is lost repeatedly with the age.

28.20.5 19.5 The real trajectory used by the full frozen-source family

The reference run in this section is the FULL real recursion, including the promoted tagged marginal coefficient:

$$x_{s+1} = x_s - b^0_{s+1} - e^{\mathrm{cl}}_{s+1} - \beta^{\mathrm{tag}}_{s+1}. \tag{19.15}$$

The Gaussian sequence is the same volume- independent sequence as in the clean calculation. Write $S_n = \sum_{v\le n}(b^0_v - c_0)$, with $|S_n| \le C_s$. The pure-clean error is uniformly small, and the real tagged error is bounded by (19.14). Choose the weak floor so that their sum in absolute value is at most $\delta < c_0/2$. The continuous shooting argument applies to (19.15) itself: subtracting the bounded S_n reduces it to a strictly decreasing real recursion. It gives a full real run with prescribed weak terminal value. Its tube is

$$x_* + (c_0 - \delta)(N - s) - 2C_s \le x_s \le x_* + (c_0 + \delta)(N - s) + 2C_s. \tag{19.16}$$

No claim that a clean shooting root already has the same terminal value for the full scheme is used.

28.20.6 19.6 One ordinary homogeneous bare disc, before derivative estimates

Use one homogeneous bare source parameter c, with every physical test, chart and replacement law frozen at the real run (19.15). Write $w_s(c) = x_s(c) - x_s(0)$, $w_0(c) = c$. The pure-clean triangular functions are holomorphic on the history product of sections 11–12, including the actual smaller cutoff of 19.2.

Put $a = x_* - 2C_s$, and let M exceed all fixed logarithmic powers required by the clean source and cold-action estimates. There is a fixed $c_{\mathrm{add}} > 0$, independent of the ultraviolet length, such that

$$d_{\mathrm{add}} = c_{\mathrm{add}} a/(\log a)^M \tag{19.17}$$

is an admissible absolute additive coefficient allowance on every old level. The monotonicity of $x/(\log x)^M$, together with (19.16), puts this allowance inside all clean half-history discs after decreasing c_{add} once. The estimate (13.14) places it inside the own-scale cold account. Fix $r = d_{\mathrm{add}}/16$, and use the provisional coefficient bound $\sup_s |w_s| < 2r$.

At a finite step the original coefficient- algebra operations define the tagged state holomorphically in c whenever the previous states and leading coefficients are in this input class. Sections 13–14 and (19.11)–(19.14) give the absolute bound

$$|\beta^{\mathrm{tag}}_{s+1}(c)| \le F_s, \qquad \sum_s F_s = \epsilon_R(a). \tag{19.18}$$

It holds uniformly on the provisional source disc and also at $c = 0$. In applying the one-step theorem the leading coefficient variation consumes at most $2r$ of d_{add}; the remaining allowance pays the regular

source and boundary terms. All denominators are source-free. No historical product theorem for sourced island masses is needed.

The pure-clean row-Cauchy estimate on the half-history box gives

$$|\beta^{\mathrm{cl}}_{s+1}(x+w)-\beta^{\mathrm{cl}}_{s+1}(x)|\le a_s\max_{u\le s}|w_u|,\qquad \sum_s a_s=\epsilon_{\mathrm{cl}}(a). \tag{19.19}$$

The Gaussian part b^0 has no such variation. By the actual tube, $\epsilon_{\mathrm{cl}}(a)\to 0$ as $a\to\infty$, and $\epsilon_R(a)$ is superpolynomially small. Choose a so that $\epsilon_{\mathrm{cl}}<1/8$ and $\epsilon_R<r/8$.

Inductively on the finite step, suppose the old functions are holomorphic on $|c|<r$ and bounded by $2r$. The exact triangular recursion yields

$$\max_{u\le s+1}|w_u(c)|\le r+2r\epsilon_{\mathrm{cl}}+2\epsilon_R<2r. \tag{19.20}$$

Thus one and the same ordinary source disc is available at every step. This closes the holomorphic induction BEFORE taking tagged derivatives. Since $w_s(0)=0$, Schwarz's lemma gives

$$|w_s(c)|\le 2|c|,\qquad |w_s^{(n)}(0)|\le 2n!\,r^{1-n}. \tag{19.21}$$

On $|c|\le r/2$, ordinary Cauchy with radius $r/2$ gives the analogous uniform higher-derivative estimates. In particular

$$\sum_s|\partial_c\beta^{\mathrm{tag}}_{s+1}(0)|\le\epsilon_R/r, \tag{19.22}$$

with a factor two for estimates on a fixed smaller interior disc.

The same recursion also gives the stronger ordinary-disc bound

$$\sup_{|c|<r}|w_s(c)-c|\le 2r\epsilon_{\mathrm{cl}}+2\epsilon_R. \tag{19.22a}$$

Consequently

$$|w_s''(0)|\le 4\epsilon_{\mathrm{cl}}/r+4\epsilon_R/r^2. \tag{19.22b}$$

If the prefix is stopped at its own weak endpoint x_s, use the same argument with that endpoint's radius and tail sums. The clean estimates then give $O(x_s^{-3/2}\operatorname{polylog} x_s)$ for (19.22b), and the tagged term is smaller than any prescribed power. A bound with the fixed terminal radius alone would not give this sharper own-endpoint weight.

If $d_s=\partial_c x_s(0)$, differentiate the triangular recursion only now. With $M_d=\max_s|d_s|$,

$$|d_s-1|\le\epsilon_{\mathrm{cl}}M_d+\epsilon_R/r. \tag{19.23}$$

Hence $M_d\le(1+\epsilon_R/r)/(1-\epsilon_{\mathrm{cl}})$ and $d_s\ge 1/2$ after the once-chosen weak-floor increase. The analytic implicit function theorem supplies local analytic tuning of the full terminal coefficient. This tangent is for the exact frozen-source family. It is not identified with the coordinate derivative of a scheme in which every sharp real cutoff is recomputed. The original physical source identity requires no such identification.

28.20.7 19.7 Smooth heterogeneous tagged sources

For a pointed tagged birth use its constant homogeneous family at the anchor value of the source, INCLUDING its paired $\beta^{\mathrm{tag}}(c)$. Holomorphy on the disc just proved and exact carrier locality make

its difference from a spatial profile an actual source-direction Cauchy difference. For a C^α profile with fixed budget r_{sp}, the normalized direct defect bound is

$$d_j^{\mathrm{het}} \le C r_{\mathrm{sp}} \ell_j^\alpha F(x_j). \tag{19.24}$$

Carrier-diameter factors use spare locality. All full quadratic tensor components stay in this defect. Only the frozen constant homogeneous part uses a scalar marginal.

Let $M_k = \sum_{j\le k} u_j^{\mathrm{het}}$. The unextracted readout (19.6), its exact root-count cancellation, and the ordinary fluctuation difference give

$$M_{k+1} \le (1 + C\theta_k) M_k + C r_{\mathrm{sp}} \ell_{k+1}^\alpha \widehat{F}_{k+1}, \quad \theta_k = \delta_k/\alpha_k, \qquad \widehat{F}_k = \max_{j\le k} F(x_j). \tag{19.25}$$

The sequence $\widehat{F}_k$ is nondecreasing, and (19.9) gives $\widehat{F}_k \le CF(x_k)$. Since $\ell_{k+1} = L\ell_k$, choosing $1 + C\sup\theta_k < L^\alpha$ proves

$$M_k \le C r_{\mathrm{sp}} \ell_k^\alpha F(x_k). \tag{19.26}$$

This is the same geometric source-oscillation argument as for clean spatial defects, with the true shrinking-radius normalization and the extra tagged birth envelope.

Consequently the marked interface uses two distinct incoming classes: the pure-clean fixed-core source family, and the promoted/unpromoted tagged family on its original shrinking cores, with its own paired marginal coefficients. The published source-free preparation/current estimates remain imports. No fixed-core extension of the whole full-R action is asserted. With this distinction the ordinary source norm used in sections 15–18 is stable; the physical Gaussian window and continuum reconstruction are separate conclusions.

Section 28.21 / Strict local counterterms, one grid, and all scalar moments

This section records a consequence of the ordinary source theorem, not a new assertion about a replacement measure. Work throughout with the physical periodic Wilson measure at fixed bare mesh a. The source is the original scalar Wilson action insertion $O_a(f)$, transported exactly when the RG representation is used. Use the strictly local contact functional constructed from the fourth spatial source jet in Section 16 of the stated results. After the symmetry average in 18.5, denote it by $\mathcal{L}_{a,T}$. Its properties are

$$\begin{aligned} \mathcal{L}_{a,T}[0] = 0, \qquad \overline{\mathcal{L}_{a,T}[J]} = \mathcal{L}_{a,T}[\overline{J}], \\ \left| e^{-\mathcal{L}_{a,T}[J]} \left\langle e^{O_a(J)} \right\rangle_{a,T} \right| \le \exp\{C \ \mathrm{Vol}(T)\}, \qquad \|J\|_{C^5} < r_0. \end{aligned} \tag{20.1}$$

The constants $C, r_0 > 0$ are independent of the bare cutoff and of the compatible torus volume. The functional is covariant under the lattice translations and coordinate reflections. It is a sum of holomorphic functions of the finite jet at one point. Hence, for sources supported in disjoint cell interiors, its additivity is exact. All flat E and regular R vacuum births enter its construction; it is not only the pure-clean vacuum.

28.21.1 20.1 The cell norm uses only one source at a time

Let a cell have side ℓ, with even cell counts in each direction on the ambient torus. If f vanishes on a neighborhood of its faces, its alternating reflected dissemination has the same C^5 norm as f, up to a

fixed convention for the norm. In particular this norm has no cell-count factor. For

$$D_{a,T;f}(z) = \exp\{zO_a(f) - \mathcal{L}_{a,T}[zf]\}, \qquad |z|\,\|f\|_{C^5} < r_0, \tag{20.2}$$

the exact cancellation in (18.18) gives the chessboard seminorm bound

$$p_{a,T,\ell}(D_{a,T;f}(z)) \le e^{C\ell^4}. \tag{20.3}$$

Complex conjugation on reflected cells is part of dissemination. Thus (20.3) uses (20.1) for the single disseminated profile and the identity $\mathcal{L}[\mathcal{D}(zf)] = N_{\mathrm{cell}}\,\mathrm{Re}\,\mathcal{L}[zf]$.

For distinct occupied cells and functions f_i supported inside them, multicell chessboard yields

$$\left|\left\langle \prod_{i=1}^{n} D_{a,T;f_i}(z_i)\right\rangle_{a,T}\right| \le e^{Cn\ell^4}. \tag{20.4}$$

This holds on the product of the individual discs $|z_i|\|f_i\|_{C^5} < r_0$. It does NOT require $\|\sum_i z_i f_i\|_{C^5} < r_0$: each chessboard factor is estimated by its own one-cell source theorem. The original finite-cutoff physical product is well-defined there. Strict locality supplies the product of its cell normalizers.

Put $B_i = \|f_i\|_{C^5}$, omit zero entries, and use $r_i = r_0/(2B_i)$. The mixed first Cauchy estimate in (20.4) gives

$$\left|\left\langle \prod_{i=1}^{n}\{O_a(f_i) - \mathcal{L}'_{a,T}[0](f_i)\}\right\rangle_{a,T}\right| \le C^n \prod_i B_i, \qquad 0 < \ell \le 1. \tag{20.5}$$

The normalizer can contain nonlinear contact coefficients; only its first derivative occurs here because each distinct cell is differentiated once.

Let $m_{a,T}(f) = \langle O_a(f)\rangle_{a,T}$. The one-cell version of (20.5) bounds $|m_{a,T}(f) - \mathcal{L}'_{a,T}[0](f)| \le CB$. Expanding the product after this scalar change of center, and applying (20.5) to each subset of occupied cells, therefore gives

$$\left|\left\langle \prod_{i=1}^{n}\{O_a(f_i) - m_{a,T}(f_i)\}\right\rangle_{a,T}\right| \le C^n \prod_i B_i. \tag{20.6}$$

The factor 2^n is absorbed in C^n. This centering is linear in f, real, and translation covariant.

28.21.2 20.2 A single admissible grid for all small profiles

Suppose the n source anchors have mutual Euclidean distances at least h, $0 < h \le 1$. Let each profile be supported in an axis-parallel cube of side at most

$$\rho = \frac{h}{KLn}. \tag{20.7}$$

Here K is a sufficiently large fixed constant. Choose an admissible side length $\ell \in [h/(16L), h/16]$, a power of the odd blocking factor times the bare mesh. On a torus with an even terminal count the corresponding number of cells is even.

Consider all offsets of this grid on the bare lattice. In one coordinate the fraction for which one prescribed profile lies within distance ρ of a cell face is at most $C(\rho + a)/\ell$. This follows by counting

the bad residue classes modulo ℓ/a; it includes the endpoint rounding error. A union bound over the four coordinates and all n profiles gives a bad fraction at most

$$Cn(\rho + a)/\ell. \tag{20.8}$$

For $a \le \rho/10$ and fixed sufficiently large K, this is less than one half. Thus ONE bare-lattice grid offset puts every profile strictly inside a cell, with a common positive margin. The $O(a)$ enlargement in the count includes the physical plaquette's link support, not just its chosen center. The cells are distinct because their diameters are less than the distance between anchors. The same grid is used in every factor of (20.4). Independently choosing a grid for each profile would not justify that application of chessboard.

Consequently (20.6) applies to all these profiles. If $f_i(x) = t_i\phi_i((x - y_i)/\rho)$ with $\|\phi_i\|_{C^5} \le 1$, then

$$\left|\left\langle \prod_i \{O_a(f_i) - m_{a,T}(f_i)\} \right\rangle_{a,T}\right| \le C^n (n!)^5 h^{-5n} \prod_i |t_i|. \tag{20.9}$$

Indeed $B_i \le C|t_i|\rho^{-5}$ and $n^{5n} \le e^{5n}(n!)^5$. This estimate is uniform once $a \le ch/n$. For every fixed n, h it therefore holds eventually along a continuum sequence. It is not a simultaneous all-n assertion at one fixed nonzero mesh.

For general smooth tests with supports of diameter at most R and mutual distance at least h, first use a smooth partition into cubes of side $O(\rho)$. The number of patches per test is at most $C_R\rho^{-4}$, and each patch has C^5 norm at most $C\rho^{-5}\|f_i\|_{C^5}$. Multilinearity permits applying the preceding common-grid argument to each chosen n-tuple of patches. Thus a convenient coarse bound is

$$\left|\left\langle \prod_i \{O_a(f_i) - m_{a,T}(f_i)\} \right\rangle_{a,T}\right| \le C_R^n (n!)^9 h^{-9n} \prod_i \|f_i\|_{C^5}. \tag{20.10}$$

The constants may be enlarged for R, h in fixed compact ranges. This deliberately coarse factorial power is finite. It supplies uniform local C^5 continuity and fixed-order translation equicontinuity on separated supports. No estimate of a history event by a bare local witness was used.

28.21.3 20.3 The local cube bound is uniform in every cutoff

There is an elementary complement when the mesh is too coarse for the common-grid construction. Suppose each test is supported in a cube of diameter at most C_0h, and the cubes have mutual distance at least h, $0 < h \le 1$. The original scalar source has the literal form

$$O_a(f) = \sum_p f(x_p)a(U_p), \qquad 0 \le a(U_p) \le 2, \tag{20.11}$$

up to the fixed convention for plaquette orientations. The number of plaquettes whose chosen centers lie in the support cube is at most $C_{C_0}(1 + h/a)^4$. Thus for EVERY configuration,

$$|O_a(f)| \le C_{C_0}(1 + h/a)^4 \|f\|_\infty. \tag{20.12}$$

Physical centering at any probability state costs at most a factor two. If $a > ch/n$, the product of n centered insertions is consequently bounded by

$$C_{C_0}^n n^{4n} \prod_i \|f_i\|_\infty. \tag{20.13}$$

This estimate needs neither a cell margin nor a source logarithm.

For the complementary meshes $a \le ch/n$, partition each source cube into patches of side $O(h/(KLn))$. There are only $C_{C_0} n^4$ patches per source, since its original diameter is $O(h)$. Every patch has C^5 norm bounded by $C(n/h)^5 \|f_i\|_{C^5,h,c_i}$, where $\|f_i\|_{C^5,h,c_i} := \|f_i(c_i + h\,\cdot)\|_{C^5}$ and c_i is the original cube center. The product rule proves this: a derivative on the patch cutoff costs $O(n/h)$, and a derivative on f_i costs h^{-1} in the scaled norm. In any chosen n-tuple of patches, its anchors are separated by at least a fixed multiple of h; enlarge the fixed grid constant to absorb the patch diameters. Equations (20.6)–(20.8) then give $C_{C_0}^n n^{9n} h^{-5n} \prod_i \|f_i\|_{C^5,h,c_i}$. Since $n^{9n} \le e^{9n}(n!)^9$, combining this with (20.13) proves

$$\boxed{\left| \left\langle \prod_{i=1}^{n} \{O_a(f_i) - m_{a,T}(f_i)\} \right\rangle_{a,T} \right| \le C_{C_0}^n (n!)^9 h^{-5n} \prod_i \|f_i\|_{C^5,h,c_i}.} \tag{20.14}$$

It is uniform in ALL bare cutoffs and compatible volumes. The same statement holds in their periodic infinite-volume limits. Thus the local product-cube estimate has derivative order five, separation exponent $5n$, and factorial exponent nine. In the coarse-mesh case a plaquette may meet a cell face; the compact group bound (20.12) covers that case directly. No near-diagonal interchange of a cutoff limit with a nonuniform estimate is required.

Section 28.22 / A raw reflection proof of uniform scalar insertion strength

The all-n estimates alone do not bound an insertion uniformly in arbitrary outside polynomials. This section obtains that stronger conclusion directly from the cell seminorm. The argument is on raw polynomial forms until boundedness has been proved; it does not presume that multiplication by a cell observable is already a bounded operator on a reflection-positive quotient.

28.22.1 21.1 Fixed-mesh infinite-volume cell normalization

First fix the mesh $a > 0$. Choose a cofinal sequence of compatible periodic tori and a weak limit of their physical Wilson measures. Call the resulting translation-invariant, coordinate-reflection- positive state ρ_a. It inherits both site- and link-plane reflection positivity.

For a fixed cell profile f, the local holomorphic functions $\mathcal{L}_{a,T}[zf]$ are bounded on every smaller source disc, uniformly as the torus grows. This statement is at FIXED mesh: the number of birth scales is finite, each local vacuum birth has the ordinary rooted source bound, and the number of roots where the fourth jet of f is nonzero is finite. The bound may diverge as $a \downarrow 0$, which causes no problem here. The complete-carrier companion identity in 18.2 gives the same local coefficient limit. Alternatively, normal-family compactness supplies a locally uniform limit along a further subsequence of the chosen periodic measures. Denote one such limit by $E_{a,f}(z)$, with $E_{a,f}(0) = 0$.

Consequently

$$D_{a,f}(z) = \exp\{zO_a(f) - E_{a,f}(z)\} \tag{21.1}$$

is one fixed bounded local function at each fixed a, z. The finite-volume difference of normalizers is a scalar tending to zero locally uniformly in z. Passing (20.3) along the same cofinal sequence gives

$$q_{a,\ell}(D_{a,f}(z)) := \limsup_T p_{a,T,\ell}(D_{a,f}(z)) \le e^{C\ell^4}. \tag{21.2}$$

This order of operations is important. The constant used to normalize a cell is fixed before repeating it an arbitrarily large number of times in ρ_a. A different untracked normalizer at every repetition

would not define the operator used below. The final centered insertion estimate will be independent of which limit $E_{a,f}$ was chosen.

28.22.2 21.2 Formal adjoints and log-convex moments on the raw form

Fix a cell of side $\ell = L^j a$, centered at a lattice site. Its lower time face t is a link reflection plane because L is odd. The middle time plane $t + \ell/2$ is a site reflection plane. Let D be a bounded gauge-invariant function supported in its interior. On bounded positive-time polynomials above t, use the raw semidefinite form

$$\langle B, A\rangle_t = \rho_a(\overline{\theta_t B}\, A), \qquad T_D A = D\, \tau_{\ell e_4} A, \qquad D^* = \overline{\theta_{t+\ell/2} D}. \tag{21.3}$$

Translation invariance and the elementary identity $\theta_t \theta_{t+\ell/2} = \tau_{-\ell e_4}$ give

$$\langle B, T_D A\rangle_t = \langle T_{D^*} B, A\rangle_t. \tag{21.4}$$

This is a formal adjoint identity on the raw algebra, requiring no bounded operator or quotient.

Put $K = T_{D^*} T_D$ and $m_j = \langle A, K^j A\rangle_t$. Then

$$m_{2r} = \|K^r A\|_t^2, \qquad m_{2r+1} = \|T_D K^r A\|_t^2, \qquad m_j^2 \le m_{j-1} m_{j+1}. \tag{21.5}$$

The two parity cases of the last inequality are Cauchy–Schwarz in the raw reflection form, using (21.4). If $m_0 = 0$, raw Cauchy–Schwarz already gives $m_1 = 0$. Otherwise log convexity, or its continuous version if some moment vanishes, gives

$$m_1^N \le m_0^{N-1} m_N. \tag{21.6}$$

The physical integral for m_N contains two fixed endpoint polynomials and a vertical column of $2N$ alternating D, D^* cells. Choose $\ell' = L^r \ell$ large enough that a fixed number of ℓ'-cells contain the supports of both endpoint polynomials. Take N through multiples of L^r. The endpoint cells absorb at most a fixed number c_A of the column factors. Bound each entire endpoint-cell function by its supremum norm. This produces a finite constant $C(a, A, D)$, independent of N.

For a remaining ℓ'-cell, its observable is a product of ℓ'/ℓ fine-cell factors. Disseminate that coarse cell and apply the fine-cell multicell inequality (18.15). The fine-cell seminorms are all the same as that of D, by translations, coordinate reflections and conjugation. There are exactly ℓ'/ℓ active fine cells per coarse cell. Taking the coarse-cell root proves the refinement bound

$$p_{\ell'}\left(\prod_{\text{column}} D_{\text{fine}}\right) \le p_\ell(D)^{\ell'/\ell}. \tag{21.7}$$

All other fine cells have the identity observable and norm one. This is the required refinement estimate; no inequality of the form $p(AD) \le \|A\| p(D)$ is used for an endpoint intersecting many cells.

Apply coarse-cell chessboard, then (21.7), on the finite periodic tori. For each FIXED N, pass first to their chosen infinite-volume limit. The definition of the limsup seminorm gives, for every $\epsilon > 0$,

$$0 \le m_N \le C(a, A, D)\{q_{a,\ell}(D) + \epsilon\}^{2N - c_A}. \tag{21.8}$$

The constant may depend on the fixed endpoint polynomials; its Nth root tends to one. Let $N \to \infty$ along the specified multiples, then $\epsilon \downarrow 0$. Combining with (21.6) yields the raw estimate

$$\|T_D A\|_t \le q_{a,\ell}(D) \|A\|_t. \tag{21.9}$$

The argument also covers $q = 0$. It proves that null vectors are preserved. Only NOW may T_D be descended and extended to a bounded operator on the Hilbert completion. Raw Cauchy–Schwarz gives, for arbitrary outside polynomials,

$$|\rho_a(\overline{\theta_t B}\, D\, \tau_{\ell e_4} A)| \le q_{a,\ell}(D)\|B\|_t\|A\|_t. \tag{21.10}$$

28.22.3 21.3 Cauchy produces one insertion, uniformly in outside fields

Apply (21.10) to (21.1) on the disc $|z| < r_0/(2\|f\|_{C^5})$, with $0 < \ell \le 1$. The scalar function on its left is holomorphic. Equations (21.2) and Cauchy's estimate therefore give

$$\begin{aligned} &|\rho_a(\overline{\theta_t B}\{O_a(f) - E'_{a,f}(0)\}\tau_{\ell e_4} A)| \\ &\qquad \le C\|f\|_{C^5}\|B\|_t\|A\|_t. \end{aligned} \tag{21.11}$$

The constant is independent of the mesh, cell side, endpoint supports, and number of outside fields.

For use in changing the scalar center, time translation is a contraction on the raw form. For $u \ge 0$ an admitted lattice translation, reflection and translation invariance give

$$\|\tau_{ue_4} A\|_t^2 = \langle A, \tau_{2ue_4} A\rangle_t \le \|A\|_t\|\tau_{2ue_4} A\|_t. \tag{21.12}$$

Iteration bounds the second factor at time $2^n u$ by the fixed finite-cutoff supremum norm of A; its exponent is 2^{-n}. Letting $n \to \infty$ proves contraction, including preservation of null vectors.

Set $A = B = 1$ in (21.11). If $m_a(f) = \rho_a(O_a(f))$, then $|m_a(f) - E'_{a,f}(0)| \le C\|f\|_{C^5}$. The constant insertion in (21.10) has norm at most one by (21.12). Thus, with $\widehat{O}_a(f) = O_a(f) - m_a(f)$,

$$|\rho_a(\overline{\theta_t B}\, \widehat{O}_a(f)\, \tau_{\ell e_4} A)| \le C\|f\|_{C^5}\|B\|_t\|A\|_t. \tag{21.13}$$

The chosen local vacuum normalizer has disappeared from the conclusion. The estimate concerns the centered original physical Wilson observable.

28.22.4 21.4 Small balls near a mirror need no further margin power

Fix $0 < \eta \le 1$. Suppose f is supported in a ball of radius $c_L\eta$, centered at y with $|y_4| \le \eta/8$. Choose $c_L > 0$ sufficiently small in terms of the fixed blocking factor. For sufficiently fine mesh $a \ll c_L\eta$, choose an odd-L cell length $\ell \asymp \eta$ and a nearby lattice-site center so that the ball lies strictly inside the cell and its time faces satisfy

$$-\eta/4 < t < t + \ell < \eta/4. \tag{21.14}$$

For example take the largest admissible $\ell \le \eta/16$; then $\ell > \eta/(16L)$, and $c_L < 1/(256L)$, with the once-smaller mesh condition, provides the required geometric margin.

Let F, G be arbitrary positive-time polynomials with all entries strictly above time $\eta/4$. For the continuum passage these mean the admitted separated-test algebra and, subsequently, its proved flat-test completion. No unrenormalized product of overlapping same-side scalar tests is inferred. In (21.13) take

$$B = \tau_{2(t-\bar{t})e_4} F, \qquad A = \tau_{-\ell e_4} G. \tag{21.15}$$

Here $\bar{t}$ is a link mirror within $a/2$ of zero. The numbers $t - \bar{t}$ and ℓ are integer multiples of the mesh. Both polynomials lie above the raw mirror t, by (21.14), for sufficiently small a, and $\theta_t B = \theta_{\bar{t}} F$, $\tau_{\ell e_4} A = G$. After the admitted common translation from t to $\bar{t}$,

$$\|B\|_t = \|\tau_{(t-\bar{t})e_4} F\|_{\bar{t}}, \qquad \|A\|_t = \|\tau_{(\bar{t}-t-\ell)e_4} G\|_{\bar{t}}. \tag{21.16}$$

Choose u_a, an integer multiple of a, tending to $\eta/4$ from above. The shifts in (21.16) exceed $-u_a$, and all the pulled-down polynomials still have strictly positive support gaps for sufficiently fine mesh. Time contraction bounds their norms by $\|\tau_{-u_a e_4} F\|_{\bar{t}}$ and its G analogue. No half-mesh translation invariance is used. The link mirror $\bar{t}$ tends to the desired continuum mirror zero. Consequently the limiting scalar functional satisfies

$$\boxed{|\sigma(\theta\overline{F}\cdot(f)\cdot G)| \le C\|f\|_{C^5}\|[\tau_{-\eta e_4/4}F]\|\,\|[\tau_{-\eta e_4/4}G]\|.} \tag{21.17}$$

The stated small-ball input has margin exponent $p = 0$. One may weaken it to any fixed $p > 0$ on $0 < \eta \le 1$. The scaling cost of a narrow test is already in its C^5 norm. A partition of a macroscopic slab is unnecessary for this small-ball statement.

Here is the precise limiting qualification. Begin with any subsequence for which all centered separated moments converge. For each fixed finite collection of tests, (20.10) supplies local C^5 equicontinuity. Adjusting a mirror or test translation by $O(a)$ therefore changes its moments by $o(1)$, since translations are continuous in C^5 for compactly supported C^5 tests. All support gaps remain positive. Pass (21.13) and its reflected norms along that subsequence, then apply the continuum time-contraction limit of (21.12). This proves (21.17) for arbitrary finite polynomials in that admitted algebra, including null vectors, with one constant independent of their degrees and coefficients. The separate fixed-order moment bounds justify convergence; their field-number constants do not enter (21.17).

28.22.5 21.5 The resulting conditional analytic-density input

The conclusions of this section require the physical source bound (20.1), the literal strict locality of its counterterm, and physical coordinate reflection positivity. They do not require that an adaptive RG history be reflection-local. They hold for the periodic infinite-volume limits just constructed; an assertion about every DLR state would require an additional argument.

Together with the independently proved spectral cube cone, (21.17) is exactly the uniform form hypothesis in the stated results, with $s = 5, p = 0$. Its compact Δ^5 parametrix then gives an analytic, time-regularized scalar insertion density of norm at most $C\eta^{-9}$, and the corresponding L^1 plate bound. Using the weaker $p = 4$ version gives $C\eta^{-13}$. This deduction is scalar. It does not identify the complete tensor observable algebra or prove its rotational covariance.

Section 28.23 / A finite-window history defect and its literal majorant

This section closes a distinction needed in a physical window comparison. For a fixed single flag, the marked bounds of section 16 apply. For an arbitrary selector $0 \le H \le \sum_b H_b$, that pointwise inequality does NOT by itself dominate the derivatives of a vacuum-normalized source numerator. The cancellations that formed its source vacuum may have used histories which the selector removes. The following finite-window interpolation retains those histories until their actual algebraic weights have been inserted.

28.23.1 22.1 The exact mask uses only displayed original labels

Let $\mathcal{B}$ be a finite set of scale-cell pairs $b = (j, Q)$, with $a' \le \ell_j \le H$, chosen so that the absence of every indicated live cell is the desired good-window event. All geometric collars are included in this definition. In the ORIGINAL unfolded positive history representation set

$$H_b(\omega) = \mathbf{1}_{\{Q \subset Z_j(\omega)\}}, \qquad m_{\mathcal{B}}(\omega) = \sum_{b \in \mathcal{B}} H_b(\omega), \qquad W_u(\omega) = \prod_{b \in \mathcal{B}} (1 - u H_b(\omega)). \tag{22.1}$$

Then $W_u = (1-u)^{m_{\mathcal{B}}}$ on each history, $W_0 = 1$, and $W_1 = \mathbf{1}_{\text{good}}$. All physical sources, charts, tests, reference laws, and real running coefficients remain fixed. The parameter u changes only this displayed polynomial history weight.

Each factor is filed when its scale-cell label is present. For one component at one scale, the factor is $(1-u)^{m(j,\text{component})}$, with m the number of selected cells in that component. It is a scalar constant on the component's integration variables. If the component survives, its polynomial stays in the unnormalized state. If it heals, its complete carrier retains the selected cell origins. A merge multiplies the two polynomials on the joint state; it does not make independent copies of a selected visit. No locality of H_b as a function of the bare field is asserted or needed.

The grouping is by the entire maximal live or completed component, including its retained genealogy. ALL selected visit factors in that component stay in ONE unnormalized kernel before its variables are integrated. Two cells in the same island do not give two separately small operator activities. Its full polynomial may have high u-degree, but its differentiated mass has one distinguished visit.

For $|u| \le 2$, the literal pointwise bounds are

$$|W_u| \le 3^{m_{\mathcal{B}}}, \qquad \partial_u W_u = -\sum_{b\in\mathcal{B}} H_b \prod_{c\neq b}(1-uH_c), \qquad |\partial_u W_u| \le \sum_{b\in\mathcal{B}} H_b\, 3^{m_{\mathcal{B}}-1}. \tag{22.2}$$

When there is no selected visit, the last sum is zero. These are polynomial identities, valid also at $u = 1$. Neither W_u nor a component kernel is divided by $1-u$, and no logarithm of $1-u$ is introduced.

28.23.2 22.2 Payment from actual own-layer records

In the original ledger, a selected cell belongs to one live component at its own scale. Thus, with the fixed cell/block conversion included,

$$m_{\mathcal{B}}(\omega) \le C_{\text{vol}} \sum_{\text{actual live layer records } r} N_r. \tag{22.3}$$

This counts current cells in Z_j, including its prescribed collars. It does not count every fine link in a reconstructed background. A repeated visit at a different actual scale is a new layer record; rereading one stored coordinate is not. The volume in (22.3) is precisely the live-volume quantity charged in sections 8.3 and 13, not a newly postulated charge.

Increase that fixed charge by $C_{\text{vol}} \log 3 + 1$ before choosing the weak ceiling. The original creation and renewal inequalities have the strict surplus needed for this fixed increase. Reserve disjoint parts for the original unmarked estimates, the physical source cold-action comparison, the selected-mask factor $3^{m_{\mathcal{B}}}$, and the distinguished root and live-age factors of section 16. After this payment, the rooted bound for ONE distinguished visit at (j, Q) is still

$$d_{j,Q} \le C(1+x_j)^q e^{-cP_0(x_j)}, \tag{22.4}$$

with a decreased but positive c. Prescribed fixed polynomial coordinate costs and live-age exponentials are paid by separate once-chosen increases as in (16.14). All these choices are independent of $|\mathcal{B}|$, the number of ultraviolet scales, and the window size.

For two source derivatives inside a live component, apply a single joint cold majorant. Inherited regular members are bounded on their actual stored pair domains. Original Wilson source factors and missing-vacuum constants give a fixed-degree polynomial in the SAME stored action and record count. In particular

$$A^m e^{-\theta x A} \le \left(\frac{m}{e\theta x}\right)^m \quad (A \ge 0,\ m = 1, 2). \tag{22.5}$$

Gaussian energy polynomials use the analogous fixed quadratic reserve, with a fixed power of the actual dimension. The mask estimate (22.2) and these source polynomials are inserted before integrating the common kernel. This is not a product estimate for separately integrated island masses sharing hidden variables. The degree m in (22.5) counts physical source factors; it does not supply a second rarity factor for a second selected visit in the same island.

The sum over possible ancestors of one visit is the same own-scale age sum as in (16.3), with the weakened reserve. Sum that age FIRST, then sum the selected roots. The resulting total direct load is

$$D_{\mathcal{B}} = C \sum_{(j,Q)\in\mathcal{B}} (1+x_j)^q e^{-cP_0(x_j)}. \tag{22.6}$$

There is one remaining rarity factor for the distinguished visit, even if many selected cells belong to the same component. Such correlated cells are not assigned independent rarity factors.

28.23.3 22.3 Ordinary u, finite source jets, and a rooted bootstrap

The coefficient algebra is now

$$H^\infty(\{|u|<2\})[z_1,z_2]/(z_1^2,z_2^2), \tag{22.7}$$

with the weighted coefficient norm for z_1, z_2. Choose source weights such that the already integrated prefix state at a' has a small TOTAL rooted norm:

$$r_i M_i \le \epsilon, \qquad r_1 r_2 M_{12} \le \epsilon. \tag{22.8}$$

Here the physical prefix supplies actual regular member functions on admitted cores and original source densities inside its unnormalized live states. A clean-core norm alone is not an extension of an insertion into a wild core. For the scalar bumps in the companion window calculation one may take $M_i = C(1+x_{a'})^q N_i t_i$, with $N_i = O((d/a')^4)$, and the corresponding mixed budget.

Every u-excess regular term, scalar vacuum, or boundary operator contains a selected visit. Keep that origin in all connected unions. A rooted marked quadratic term is retained in its pointed member. If it is extracted for bookkeeping, its compensating counterterm keeps the same root and support. It is never turned into an unanchored global change of the scalar coupling.

The original source-free domain inclusion and the zero first pair jet therefore give the one-root regular kernel $C(1+m)^q L^{-4m}$ of (16.5)–(16.7). After a new wild encounter use the small nonstationary factor δ_v, followed only by the ACTUAL live dwell decay. Thus the linear rooted propagation has ℓ^1 norm $q_0 < 1/2$, after the once-chosen weak-floor adjustment. The bounds are uniform in $|u|<2$ by (22.2)–(22.5).

There are three different cases at an island encounter. If differentiation hits a newly selected visit, it gives a direct root in (22.6). If it hits a previously healed REGULAR function, that function's stored-domain sup norm may multiply the jointly integrated new-island majorant; this is the nonstationary $\delta_v \sum_{t\le v} A_t$ term of (16.7). If it hits an already LIVE operator, keep that root through the entire merged genealogy and use the original one-lineage creation/dwell bound. No product of separately integrated rarity factors is used in this last case. Every other mask factor is absorbed by (22.3).

Here is a linear distinguished-root bootstrap which makes the ordinary-parameter closure precise. Provisionally suppose that the total normalized rooted norm of the REGULAR u-excess state is at most $\mathcal{A}_*$ on a finite prefix of the remaining run. Choose $\mathcal{A}_*$ inside the fixed spare regular allowance. Differentiate the exact component polynomials FIRST. Let $\mathcal{B}_u$ denote the total rooted norm of the derivative state, including its completed direct births from still-live lineages, in the finite source algebra. There is only ONE distinguished visit in every term.

Section 28.23 / A finite-window history defect and its literal majorant

More explicitly, let R_s be the total derivative norm of new regular births at s, and let L_s be the FIRST complete healing/exit flux of distinguished live epochs into that scale. The live dwell has already been resummed in the second inequality below:

$$\begin{aligned} R_s &\le d_s^R + \epsilon_{\mathrm{eff}} \sum_{t<s} w_{s-t} R_t + C L_s, \\ L_s &\le C \sum_{v \le s} e^{-B(s-v)} \left\{ d_v^L + \delta_v \sum_{t<v} R_t \right\}, \qquad \sum_s (d_s^R + d_s^L) \le C D_{\mathcal{B}}. \end{aligned} \tag{22.8a}$$

Here ϵ_{eff} is the original regular response coefficient plus $C\epsilon + C\mathcal{A}_*$. A terminal-survival output is bounded by the same live convolution. Immediate healing is assigned to either d^R or the $s = v$ term of L, once, not to both.

The ownership convention is essential. A partial dropout which leaves marked old variables or a marked carrier inside a wild component stays in its residual unnormalized operator. No u-excess regular birth is emitted from that partial operation and independently counted while the same marked ancestor remains live. On complete healing, its entire analytic coefficient is filed once in R_s, and that live epoch is closed. Subsequent entry of that regular member into a new island starts a new live epoch through the $\delta_v R_t$ term. These are exact linear state transfers, with the source-free normalizations kept as in section 13.

Summing (22.8a) gives the regular kernel norm and $C(1 - e^{-B})^{-1} \sum_v \delta_v$, precisely the corrected one-root coefficient q_0. In particular there is no $L \cdot L$ term or a rarity-product assumption at a merger.

In a regular formation, products with that differentiated vertex are products of actual analytic functions on the same admitted core. Their sup norms are submultiplicative. The connected expansion sums the other regular vertices on its spare allowance, giving an additional $C\epsilon + C\mathcal{A}_*$ in the linear response norm. Its rooted locality and birth-time kernels are the ones in (16.7). In a live merger use the preceding three-case classification and the single joint cold integral. Thus the actual derivative majorant is

$$\mathcal{B}_u \le C D_{\mathcal{B}} + (q_0 + C\epsilon + C\mathcal{A}_*) \mathcal{B}_u. \tag{22.9}$$

There is NO quadratic estimate between integrated live operator norms in this inequality. In particular the false implication from $\mathbb{E}H = \delta$ to $\mathbb{E}H^2 \le C\delta^2$ is never used.

Choose $\epsilon, \mathcal{A}_* > 0$ once so that the coefficient in parentheses is below $3/4$. Then $\mathcal{B}_u \le 4 C D_{\mathcal{B}}$. Every regular excess vanishes at $u = 0$. Integrating its derivative along the segment from zero to u, and summing the nonnegative rooted norms, gives

$$\mathcal{A}(u) \le \int_0^1 |u| \, \mathcal{B}_{tu} \, dt \le 8 C D_{\mathcal{B}}, \qquad |u| < 2. \tag{22.10}$$

Take D_0 so small that $8CD_0 < \mathcal{A}_*/2$. A first-exit argument on each finite causal prefix closes the provisional ball, uniformly in the number of scales and selected roots. At finite cutoff every initial mask is a polynomial, and each regular formation is holomorphic on the spare allowance; the strict bound prevents a loss of that domain. This proves the ordinary-disc extension before using any Cauchy estimate in u. It applies when $D_{\mathcal{B}} \le D_0$, not for an arbitrary large selected region with uncontrolled total load.

The same argument records each new scalar vacuum output ONCE. Its total excess and its u-derivative are $O(D_{\mathcal{B}})$. Neither a source vacuum constant nor an already recorded marked scalar is generated anew at every later level. This explains why the estimate contains $e^{C D_{\mathcal{B}}}$, rather than $e^{C|\mathcal{B}|}$ or an ultraviolet volume.

28.23.4 22.4 Exact original-vacuum normalization and the physical difference

Let $Z_u(z)$ be the original unnormalized partition function with the multiplier W_u. Use the ORIGINAL $u = 0$ full source vacuum from section 15:

$$Z_u(z) = e^{V+\mathcal{E}(z)}\widetilde{Z}_u(z). \tag{22.11}$$

The factor contains the Gaussian, pure-clean, and regular R source-only vacuum births. At a displayed history, its completion and missing-birth compensation commute with the polynomial multiplier W_u. In subsequent connected formations, split each coefficient into its value at $u = 0$ and its u-excess. The latter contains a selected origin by the exact difference formula (19.3). Its flat value stays among the bounded marked scalar outputs just counted. It is not silently incorporated into $\mathcal{E}$.

Terminal source-zero stability and the preceding rooted bound give, for $I = \varnothing, \{1\}, \{2\}, \{1,2\}$,

$$|\partial_u D_I \widetilde{Z}_u(0)| \le e^{CN_K + CD_{\mathcal{B}}} C D_{\mathcal{B}}\, \mathcal{M}_I, \qquad |u| \le 1, \tag{22.12}$$

where $\mathcal{M}_\varnothing = 1$, $\mathcal{M}_i = M_i$, and $\mathcal{M}_{12} = M_1M_2 + M_{12}$. No assertion that $Z_u(z)$ is nonzero has been used. Integrating from $u = 0$ to $u = 1$ bounds the coefficients of

$$Z_0(z) - Z_1(z) = Z_{\mathrm{bad}}(z), \tag{22.13}$$

which is the EXACT physical bad-event numerator. Only after this operation divide by the positive real mass $\widetilde{Z}_0(0) \ge e^{-CN_K}$. The elementary finite-jet quotient formula then gives

$$|D_{12}(Z_{\mathrm{bad}}/Z_0)(0)| \le C_T D_{\mathcal{B}}(M_1M_2 + M_{12}), \tag{22.14}$$

and the analogous first and zeroth bounds. These are the bounds used in the companion window comparison.

For its window, $|\mathcal{B}| \le C(1+n)(BH/a')^4$, B is polylogarithmic, and $x_j \asymp x_d$ on the selected levels. Hence

$$D_{\mathcal{B}} \le P(x_d, H/a')e^{-cP_0(x_d)} \longrightarrow 0. \tag{22.15}$$

The smallness condition $D_{\mathcal{B}} \le D_0$ is therefore eventually satisfied uniformly in the bare cutoff below a'. The physical torus is fixed in this statement; its factor C_T is not a volume-uniform local probability estimate.

The algebraic multiplier, joint cold estimate, and rooted finite-algebra propagation are explicit here. Their application to Wilson histories uses the original published preparation/domain inclusion, the actual own-layer volume ledger already matched in sections 13–14, and the physical prefix coefficient class in (22.8). It does not replace those inputs by abstract component masses or by an assumed monotonicity of vacuum-normalized selectors.

Section 28.24 / The finite-window estimate with any fixed number of scalar labels

The argument of section 22 is not restricted to two physical source labels. This section gives the extension for a fixed integer m, in particular $m = 4$. Its constants and the required weak ceiling may depend on m. They do not require retuning the bare sequence: subsection 23.3 explains an earlier fixed physical analysis stop on that SAME sequence. No uniform factorial estimate in m is inferred here; the all-order scalar moment bound in section 20 has a different proof.

The source-free preparation and stored-domain inputs are exactly those stated in sections 13–14 and 22. The only new issue is to check all the mixed source coefficients, their live-state ownership, and the final connected quotient. There is still only ONE differentiated history visit.

28.24.1 23.1 Own-scale prefix packets and their coefficient algebra

Write $[m]=\{1,\dots,m\}$, $z_S=\prod_{i\in S} z_i$, and $D_S=\prod_{i\in S}\partial_{z_i}|_{z=0}$. Use

$$\mathcal{C}_m=H^\infty(\{|u|<2\})[z_1,\dots,z_m]/(z_1^2,\dots,z_m^2),\qquad \|F\|_{\boldsymbol{r}}=\sum_{S\subset[m]} r_S\sup_{|u|<2}\|F_S(u)\|,\quad r_S=\prod_{i\in S} r_i. \tag{23.1}$$

For regular coefficients the inner norm is the actual stored analytic member norm. Before a live integration it is the pointwise coefficient majorant on its unnormalized kernel. These are different realizations of the same finite algebra. The coefficient norm is submultiplicative before integration. No multiplication of integrated live masses is asserted. Since every label appears at most once, $D_SF=F_S$; no factorial is missing from this convention.

Stop the original source construction at a'. Decompose each source by the own-scale smooth partition

$$f_i=\sum_{q\in A_i} f_{i,q},\qquad \|f_{i,q}\|_\infty+(a')^\alpha[f_{i,q}]_{C^\alpha}\le Ct_i,\qquad N_i=|A_i|\le C(1+d/a')^4. \tag{23.2}$$

The supports have bounded overlap. At fixed selected cells $(q_i)_{i\in S}$, the existing ordinary source prefix family, in units of a', supplies all these coefficients on a fixed own-cell source disc. Factor its COMPLETE universal source vacuum first. Cauchy on that disc bounds the remaining regular packet by $C_m\prod_{i\in S}t_i$, apart from the fixed powers of the actual current core radius and coupling. The pure-clean part uses its fixed core. The tagged part uses its shrinking core and its superpolynomial birth envelope, as in section 19.

Every nonempty coefficient retains at least one of the selected source-cell origins; choose its least labelled origin as the root and keep every other origin in the carrier. Summing the choices of these ACTUAL cells therefore gives at most $\prod_{i\in S}N_i$. It is not a count of all fine plaquettes in the support. The fine-scale vacuum has already been factored, and the prefix's field-dependent age sums are the ones in its proved ordinary source norm. Consequently the coarse bound

$$M_S\le C_m(1+x_{a'})^{q_m}\prod_{i\in S}N_it_i,\qquad \varnothing\ne S\subset[m], \tag{23.3}$$

holds uniformly in the earlier bare cutoff. This deliberately discards useful powers of $g_{a'}$ and connected bridge decay. Those sharper estimates are needed for the clean leading term, not for the complementary-history bound.

A prefix packet inside a live component is represented differently: it retains the original physical source factors and all uneliminated variables under ONE common cold kernel. The next subsection proves the same coarse bound for that packet. It is not an analytic continuation of a regular packet into a wild core. Missing source-vacuum constants are assigned to their actual origin records before taking this majorant. The source-prefix construction used here is thus the exact one on both kinds of state, not only its clean restriction.

Choose $P_m(x)=C_m(1+x)^{Q_m}$ large enough that

$$\lambda_i=P_m(x_{a'})N_it_i,\qquad M_S\le\prod_{i\in S}\lambda_i. \tag{23.4}$$

Remove a zero source rather than assigning it a radius. For the others set $r_i=\eta_m/\lambda_i$. Then

$$\sum_{\varnothing\ne S\subset[m]} r_SM_S\le(1+\eta_m)^m-1. \tag{23.5}$$

Choose $\eta_m > 0$ once so that this is below the spare source allowance ϵ_m. The source radius has no factor d^α and no global physical C^5 cost. It is a weight on the finite incoming rooted packets, after the prefix has actually formed them.

The Cauchy argument above requires only the ordinary own-scale source prefix theorem already used in (22.8). Its extension to m parameters is direct: its input norm is the sum of the m weighted own-cell loads, and all its local formation estimates are estimates in a commutative Banach algebra. It does not require a new product domain for independently rough fine-plaquette coupling constants.

28.24.2 23.2 Fixed-degree cold bounds and one live owner

Allocate a fixed positive fraction θ of the stored action reserve to ALL physical source polynomials together. For every $0 \le r \le m$,

$$A^r e^{-\theta x A} \le \left(\frac{r}{e\theta x}\right)^r \quad (A \ge 0), \tag{23.6}$$

with the $r = 0$ right side interpreted as one. A product of $r \le m$ local source energies is bounded by a constant times the r-th power of their common stored action, multiplied by the profile amplitudes. Thus the SAME reserve, rather than a new reserve for each coefficient, pays every source subset in (23.1).

The Gaussian version uses its common nonnegative quadratic energy. Polynomial Gaussian normalizations and the scalar compensations assigned to missing birth records have degree bounded in terms of m, and cost a fixed power of the ACTUAL integrated dimension and stored record count. The exponential of the original vacuum-compensation identity has, at degree at most m, just the corresponding finite Bell polynomials in those record constants. These are paid inside the same joint cold majorant. No separate conditional expectation or sourced island denominator occurs.

For a retained coordinate of live age a, these degree bounds cost at most

$$C_m(1 + x_j + a)^{q_m} L^{D_m a} \tag{23.7}$$

at its actual own-layer reading, for a fixed D_m. For instance the raw coordinate-count contribution has exponent at most $4m$; the fixed chart and core conversions can be included by increasing D_m. Choose the original live-age reserve $B(m) > D_m \log L + B_{\text{spare}}$ before fixing the weak ceiling. Polynomial powers of a are then summable. This choice is uniform in the number of scales. An already healed regular origin uses its stored analytic norm and a later encounter factor δ_v; none of (23.7) is assigned to its intervening regular years.

After these payments, one distinguished selected visit still has direct load

$$d_{j,Q}^{(m)} \le C_m(1 + x_j)^{q_m} e^{-c_m P_0(x_j)}, \qquad D_{\mathcal{B},m} = C_m \sum_{(j,Q)\in\mathcal{B}} (1 + x_j)^{q_m} e^{-c_m P_0(x_j)}. \tag{23.8}$$

The constants have weakened, but $c_m > 0$. The mask cost $3^{m_{\mathcal{B}}}$ is still paid by (22.3), independently of the number of source labels. Its derivative still distinguishes one visit by (22.2). Two source labels, or two selected cells, in the same live genealogy do not give two independent rarity factors.

All z-coefficients of a live genealogy stay in its one operator-valued polynomial. A merge forms the common kernel before integration. A partial dropout retains every coefficient whose marked physical variables or carrier still meet the residual wild state. At a genuine factorization, a product-rule term has one owner of the u-derivative; source labels may be partitioned between the factors in (23.1). Until

such a factorization is exact, it is not replaced by a product of integrated norms. Complete healing emits the entire coefficient once. These are the same state transitions as section 22; the additional source labels do not change them.

For a healed coefficient with any nonempty source subset, gauge invariance and absence of a colored spectator give zero first pair jet after its scalar vacuum is removed. Its normalization is α_j^{-2} on its actual birth core, ONCE for the whole coefficient. It is not $\alpha_j^{-2|S|}$. Its full pointed quadratic tensor and source origins remain with it. Therefore the one-root regular kernel $w_a = C(1+a)^q L^{-4a}$ applies at every fixed degree just as it did at degree two.

28.24.3 23.3 The coupled derivative estimate at fixed degree

First fix the same bare sequence as in the original scalar construction. Let its original terminal spacing be ℓ_{end}, and its terminal weak parameter be x_*. The source-free running tube has constants $c_-, c_+ > 0$ and C_{run}, independent of the fine cutoff, such that k levels before that endpoint,

$$x_* + c_- k - C_{\mathrm{run}} \le x_{N-k} \le x_* + c_+ k + C_{\mathrm{run}}. \tag{23.9a}$$

Let X_m be the weak threshold sufficient for the fixed-degree estimates, and choose a fixed integer

$$k_m \ge \max\left(0, \frac{X_m - x_* + C_{\mathrm{run}}}{c_-}\right), \qquad \ell_m = \ell_{\mathrm{end}} L^{-k_m}. \tag{23.9b}$$

Stop the MARKED ANALYSIS at $N - k_m$. Every step used by that analysis is in its required weak range. Its terminal parameter is bounded both above and below by constants depending on m, uniformly in the bare cutoff. The physical torus now has

$$N_{K,m} \le C N_K L^{4k_m} \tag{23.9c}$$

terminal cells. This number can be large, but it is fixed independently of d and of the finer cutoff. For d small enough the whole window $[a', H]$ lies below ℓ_m.

The charts, tests, masks, counterterm assignments and coupling values before this stopping point are exactly those of the original run. The parameter m changes neither a Wilson measure nor its tuning. Complete the remaining retained-field integral at this earlier scale by the original finite-torus stability bound. Its source-zero floor and its marked upper bound cost $\exp[C_m N_{K,m}]$, which is an admissible fixed-reference constant. Surviving marked islands remain their joint cold kernels in this final integral; their contribution is the terminal-exit flux in (23.9), counted once. They are not forced into a regular analytic core.

Only numerical reserves have been increased. The fixed spare tree decay absorbs each fixed polynomial diameter loss with an m-dependent constant. The larger live-age rate is paid by the larger weak creation/renewal credits in (23.9a), keeping the original localization geometry unchanged. Thus no order-dependent change of the underlying R construction is hidden in this stopping argument. The base regular response coefficient is the original one; the m-dependent source response is made small by the weights in (23.5).

Run the proof of (22.8a) in the norm (23.1). Let $R_s^{(m)}$ be the total norm of new regular u-derivative births and $L_s^{(m)}$ their FIRST complete marked healing or final-exit flux. With the same ownership

convention,

$$\begin{aligned} R_s^{(m)} &\le d_s^R + \epsilon_{\mathrm{eff,m}} \sum_{t<s} w_{s-t} R_t^{(m)} + C_m L_s^{(m)}, \\ L_s^{(m)} &\le C_m \sum_{v \le s} e^{-B(m)(s-v)} \left(d_v^L + \delta_{v,m} \sum_{t<v} R_t^{(m)} \right), \\ \sum_s (d_s^R + d_s^L) &\le C_m D_{\mathcal{B},m}, \qquad \delta_{v,m} \le C_m (1+x_v)^{q_m} e^{-c_m P_0(x_v)}. \end{aligned} \tag{23.9}$$

Here $\epsilon_{\mathrm{eff,m}}$ is the ordinary regular formation coefficient plus $C_m\epsilon_m + C_m\mathcal{A}_{*,m}$, where the last term is the provisional total REGULAR u-excess allowance. There is no $L^{(m)}L^{(m)}$ term. Products of genuine regular functions are bounded by their stored sup norms. All products sharing live variables were handled before integration by (23.6)–(23.7).

The weak running tube makes $\sum_v \delta_{v,m}$ as small as needed for fixed m. After the once-chosen weak-floor and reserve choices, sum (23.9) to obtain

$$\mathcal{B}_{u,m} \le C_m D_{\mathcal{B},m} + (q_m^{\mathrm{lin}} + C_m\epsilon_m + C_m\mathcal{A}_{*,m})\mathcal{B}_{u,m}, \qquad q_m^{\mathrm{lin}} < \tfrac{1}{2}. \tag{23.10}$$

Choose the two spare allowances so that the displayed coefficient is below $3/4$. The finite-prefix, radial first-exit argument then gives

$$\sup_{|u|<2} \mathcal{B}_{u,m} \le C_m D_{\mathcal{B},m}, \qquad \sup_{|u|<2} \mathcal{A}_m(u) \le 2C_m D_{\mathcal{B},m}. \tag{23.11}$$

It applies when $D_{\mathcal{B},m} \le D_{0,m}$. All scalar u-excess outputs are recorded once, and obey the same differentiated bound.

This proof may equivalently be organized by induction over the $2^m - 1$ nonempty source subsets. A regular formation partitions a subset between its marked vertices; there are finitely many such partitions. Using the single weighted norm (23.1) sums exactly those products. It makes clear that this extension introduces no new infinite hierarchy or unproved factorization of live operators.

28.24.4 23.4 The original-vacuum bound and the finite quotient

Keep the full ORIGINAL source vacuum $V + \mathcal{E}(z)$ of the performed run through the fixed analysis stop ℓ_m. It includes every Gaussian, clean and regular-R flat birth at those scales. All unperformed coarser integrations remain in the exact terminal integral; they are not silently discarded. Thus the symbols $V, \mathcal{E}$ here refer to the common normalizer at that stop, rather than requiring a different marked estimate for the finitely many later scales of the original longer analysis. The physical partition function itself is unchanged. Define

$$Z_u(z) = e^{V+\mathcal{E}(z)} \widetilde{Z}_u(z), \qquad \widetilde{H}(z) = \widetilde{Z}_0(z) - \widetilde{Z}_1(z). \tag{23.12}$$

The exact mask identity says that $H = Z_0 - Z_1$ is the physical bad-event numerator. The u-excess scalar outputs just bounded remain in $\widetilde{Z}_u$; they are not moved into a different vacuum.

Terminal stability, (23.11), and coefficient extraction give, for every $S \subset [m]$,

$$\begin{aligned} |D_S \widetilde{Z}_0(0)| &\le C_m e^{C_m N_{K,m}} \lambda_S, \\ |\partial_u D_S \widetilde{Z}_u(0)| &\le C_m e^{C_m N_{K,m} + C_m D_{\mathcal{B},m}} D_{\mathcal{B},m} \lambda_S, \qquad |u| \le 1, \\ \widetilde{Z}_0(0) &\ge e^{-C_m N_{K,m}}, \qquad \lambda_S = \prod_{i \in S} \lambda_i, \quad \lambda_\varnothing = 1. \end{aligned} \tag{23.13}$$

Fixed factors $\eta_m^{-|S|}$ from (23.4)–(23.5) are absorbed into C_m. The zeroth source coefficient of the u-excess scalar exponential is bounded by $e^{C_m D_{\mathcal{B},m}}$; its higher coefficients are bounded by the finite source-algebra norm. The source-only vacuum is never estimated by its ultraviolet-extensive absolute value.

Integrate the second line from $u = 0$ to one and set

$$p(z) = \frac{H(z)}{Z_0(z)} = \frac{\widetilde{H}(z)}{\widetilde{Z}_0(z)}. \tag{23.14}$$

Only the positive original value $\widetilde{Z}_0(0)$ is used as a denominator. To verify every mixed coefficient explicitly, if $T \neq \varnothing$, then

$$D_T(\widetilde{Z}_0^{-1})(0) = \sum_{\pi \in \mathcal{P}(T)} (-1)^{|\pi|} |\pi|!\, \widetilde{Z}_0(0)^{-|\pi|-1} \prod_{B \in \pi} D_B \widetilde{Z}_0(0). \tag{23.15}$$

For $T = \varnothing$ it is simply $\widetilde{Z}_0(0)^{-1}$. Multiply this formula by $D_A \widetilde{H}(0)$ and sum over $A \subset S$. The number of partitions and the largest denominator power are bounded in terms of m. Equations (23.13)–(23.15) prove

$$|D_S p(0)| \leq C_{m,T} D_{\mathcal{B},m} \lambda_S, \qquad S \subset [m], \tag{23.16}$$

where one may take $C_{m,T} \leq C_m \exp[C_m N_{K,m}]$ on the fixed terminal torus. It is independent of the finer cutoff and the observation scale d. No inverse or nonvanishing theorem for $Z_u(z)$ at $u \neq 0$ has been used.

28.24.5 23.5 Connected cumulants and the four-label consequence

For a sufficiently small window, $0 \leq p(0) \leq C_{m,T} D_{\mathcal{B},m} < 1/2$. The good-event partition function is $Z_1(z) = Z_0(z)(1 - p(z))$. For every nonempty S, the difference of the connected source derivatives is therefore the finite-jet identity

$$\begin{aligned} D_S \log Z_1(0) - D_S \log Z_0(0) &= D_S \log(1-p)(0) \\ &= -\sum_{\pi \in \mathcal{P}(S)} \frac{(|\pi|-1)!}{(1-p(0))^{|\pi|}} \prod_{B \in \pi} D_B p(0). \end{aligned} \tag{23.17}$$

Subtracting the separate constants $\log Z_i(0)$ does not affect these nonempty derivatives. The formula also applies to the cumulants in the physically normalized full and good-event laws. Every summand contains at least one factor of $D_{\mathcal{B},m}$; products of additional such factors can be dropped in an upper bound once it is small. Hence

$$|\kappa_S^{\mathrm{good}} - \kappa_S^{\mathrm{full}}| \leq C_{m,T} D_{\mathcal{B},m} \prod_{i \in S} \lambda_i. \tag{23.18}$$

This is cancellation in an EXACT quotient. It neither differentiates a pointwise union inequality nor takes an absolute value of an uncancelled source vacuum.

On the prescribed window, $x_j \asymp x_d$, $N_i \leq C(1 + d/a')^4$, and $|\mathcal{B}| \leq C(1+n)(BH/a')^4$. Thus, after possibly increasing a polynomial depending on m,

$$|\kappa_{[m]}^{\mathrm{good}} - \kappa_{[m]}^{\mathrm{full}}| \leq C_{m,T} \mathcal{Q}_m(x_d, H/a') e^{-c_m P_0(x_d)} \prod_{i=1}^{m} t_i. \tag{23.19}$$

Section 28.24 / The finite-window estimate with any fixed number of scalar labels

The factor B and the number of window levels are included in $\mathcal{Q}_m$ for the chosen polylogarithmic window. This bound is uniform in the bare cutoff below a'.

For $m = 4$, the natural Gaussian connected four-point scale of a quadratic curvature scalar is $g_d^8 = x_d^{-4}$. Multiplying (23.19) by x_d^4 only changes its fixed polynomial. More generally the same observation applies to x_d^m at every FIXED order m. With the chosen H/a' polynomial in x_d and $P_0(x_d)$ growing faster than $\log x_d$, the relative bad-history error tends to zero. This statement does not assert that the cold estimate itself contains the leading Gaussian power, or that a clean m-point asymptotic has already been proved.

In particular a separately proved nonzero fourth connected ultraviolet limit, together with the fixed-order comparison above, could exclude a generalized free scalar field. It would still concern this scalar observable sector. It would not identify the complete gauge-invariant Yang–Mills field family.

Section 29 / An actual multiscale Gaussian short-distance window

This section strengthens the earlier abstract Gaussian comparisons. The covariance is built from the flat harmonic lifts and fluctuation bands of the actual block constraint. Its conclusions concern that Gaussian model. Applying the comparison to the interacting law still requires the local density, insertion, and exceptional-history estimates stated at the end.

Section 30 / Fixed-order scalar cumulants and the first nonzero Gaussian tests

The two-label clean estimate has a useful fixed-order extension. It follows from finite Taylor jets and connected Gaussian graphs, without assuming convergence of a perturbation series. This section proves that one-step statement and computes the Gaussian higher-point targets. It does not assert an all-scale physical higher-point comparison merely from the already derived two-point comparison.

Section 31.1 / Scope, state, and finite-order inputs

This note extends the clean two-label argument through four fixed source labels. The source-free maps, field/current charts and physical tests are frozen. Pure-clean activities use the fixed paired core of the original small-field construction. A large-field-origin term is not assigned that larger core; its matching is stated separately below.

The analytic inputs are the actual old-member readout, the rooted Gaussian/cluster allowance, and the fixed-order current-coupling derivative estimates used in the stated results, Sections 17–20. Here derivatives through order ten suffice. These are finite derivatives of the actual fixed-core integrals, paid by polynomial Gaussian reserves. No convergent power series in the coupling is assumed. The

original smaller frozen canonical box differs from the full Gaussian by the retained tail

$$\tau_s \le C(1+\log x_s)^q e^{-cP_s^2}. \tag{FL1}$$

All Gaussian formulas below use the full Gaussian; (FL1) is included in each comparison remainder.

Let I be a nonempty subset of $\{1,2,3,4\}$, and write $z_I = \prod_{i\in I} z_i$, with $z_i^2 = 0$. The incoming source increment is

$$\Phi(z,A) = \sum_{I\neq\varnothing} z_I\{\nu_I + O_I(A)\}. \tag{FL2}$$

Every O_I has zero flat value and zero first pair jet. Its full physical coarse-field Hessian, with the actual own-current composition made first, is $K_I = \mathsf{P}O_I$. All these quadratic tensors are retained. A marginal extraction is only a bookkeeping decomposition of the same source coefficient. Each extracted source counterterm keeps its original source-cell roots and carrier. It is not promoted to an unanchored change of the reference real running coupling.

Fix four test supports of diameter at most Cd, with all pairwise distances between d and Cd. Their amplitudes are t_i. Stop the ultraviolet prefix at $a' \ll d$, and the subsequent window at $H \gg d$. Put

$$D = d/a', \qquad n = 1 + \#\{s : a' \le \ell_s \le H\}, \qquad g_d^2 = x_d^{-1}. \tag{FL3}$$

The actual running tube gives g_s/g_d bounded above and below through this window, provided $1+\log(H/a') \ll x_d$. Let $N_i \le CD^4$ be the number of own-a'-scale source cells, and set

$$M_I = \prod_{i\in I} N_i t_i, \qquad \omega = x_{a'}^{-1/2}\,\mathrm{polylog}\, x_{a'}, \qquad \epsilon_{\mathrm{w}} = g_d^2 + \max_s \tau_s. \tag{FL4}$$

The fixed powers of R_0 in core norms are absorbed into constants. The main grading is

$$O_I :\ g_d^{2(|I|-1)} R_0^2, \qquad \nu_I :\ g_d^{2|I|}. \tag{FL5}$$

Constants in this note may depend on the fixed label number four and the admitted core margins, but not on the ultraviolet cutoff or n.

Section 31.8 / Four-point comparison and the separate physical matching

In the window (FL28), the vacuum error (FL21), divided by $g_d^8 \prod t_i$, is at most

$$Ct^{-1/2+32\varepsilon} \operatorname{polylog} t + \text{superpolynomial tails.} \tag{FL30}$$

Thus any fixed $0 < \varepsilon < 1/64$ is sufficient; choosing $\varepsilon < 1/128$ leaves a convenient margin. The ultraviolet prefix (FL15) and the tail (FL1) vanish faster than this power.

The Gaussian fourth cumulant is a sum of three four-edge cycles. Every edge joins distinct source supports. Hence the actual off-diagonal covariance comparison of the Gaussian-window note can be applied to each edge and telescoped through the product. There is no self-variance loss. Let $\kappa_4^{G,\mathrm{win}}$ denote the actual running Gaussian window cumulant, and let κ_4^{Max} denote the continuum unit-strength Maxwell curvature-square cumulant evaluated on the same four profiles, written $\mathbf{f} = (f_1, f_2, f_3, f_4)$. The precise estimate is the ABSOLUTE normalized bound

$$\begin{aligned} &\frac{\left|\kappa_4^{G,\mathrm{win}}(\mathbf{f}) - g_d^8 \kappa_4^{\mathrm{Max}}(\mathbf{f})\right|}{g_d^8 \prod_i t_i} \\ &\le C\left\{\frac{1+\log(H/a')}{x_d} + (d/a')^4 e^{-cd/a'} + (a/d)^2 + (d/T)^4 + (d/H)^4\right\}. \end{aligned} \tag{FL31}$$

Here $t_i > 0$; a zero input gives a zero mixed cumulant. The denominator is the fixed-shape scale $g_d^8 \prod_i t_i$, NOT the value of the Gaussian fourth cumulant. The estimate is therefore uniform also for configurations where $\kappa_4^{\mathrm{Max}} = 0$, including any cancellation between the three cycles. No positivity or nonvanishing of the comparison target was used in the edge-by-edge operator-norm estimate. The first four-label Gaussian state at a' in this comparison is its exact all-depth state, including every mixed Hessian. Associativity of the actual Gaussian Schur steps identifies the cycle covariance with the running fluctuation bands.

Equations (FL20), (FL26), and (FL31) therefore give the complete PURE-CLEAN four-label window comparison, not merely its one-step power count. For four sufficiently small nonnegative profiles near four distinct collinear points, the limiting Maxwell fourth cumulant is strictly positive: all four-edge reflection products are the identity on the six curvature components. Its value is the Gaussian target computed in the stated results.

For the PHYSICAL fixed-torus statement, the remaining history interface uses only coarse packet bounds

$$M_S^{\mathrm{hist}} \le C_4(1+x_{a'})^{q_4} \prod_{i\in S} N_i t_i, \qquad \varnothing \neq S \subset \{1,2,3,4\}. \tag{FL32}$$

They follow from the own-scale prefix disc and fixed mixed derivatives after the common full vacuum is removed. Regular packets are actual admitted-core functions. Inside a live island the original source factors remain in its unnormalized kernel; their product of degree at most four is bounded JOINTLY by stored action and coordinate polynomials. For example $A^r e^{-\theta x A} \le (r/(e\theta x))^r$, $r \le 4$. No d^{-5} global profile radius and no product of integrated same-island masses is used.

The fixed-degree grouped-mask/one-owner argument in companion history Section 23 then bounds the physical bad-window and tagged corrections by

$$C_T \prod_i t_i \ \mathrm{poly}_4(x_d, H/a') e^{-c_4 P_0(x_d)}. \tag{FL33}$$

Dividing by $g_d^8 = x_d^{-4}$ costs only the fixed polynomial x_d^4. The R-origin scalar counterterms remain in the full real trajectory. A regular tagged prefix mixed vacuum uses its completed source bridge; a live alternative stays in the marked numerator. The complete source-only vacuum cancels before any quotient estimate, as in the two-label case.

Consequently, WHEN those original-map preparation and fixed-degree history interfaces are supplied, (FL28)–(FL33) give a fixed-torus physical fourth cumulant whose leading term is g_d^8 times the integrated free-curvature-square cycle kernel, with an $o(g_d^8 \prod_i t_i)$ error even when that leading term vanishes. The nonzero value for the collinear profiles above would prove that the scalar field is not generalized free. It would not by itself identify a full interacting Yang–Mills observable algebra: a composite of free fields can also have a nonzero fourth cumulant. This note retains that distinction and does not assert a mass-gap theorem.

The physical finite-reference transfer in companion history Section 24 then moves this fixed-torus conclusion to compatible larger periodic volumes. It uses the SAME universal source normalizer. For four profiles, its packed-sign projection gives the difference bound $Cd^{4-4\alpha} \prod_i t_i$. Taking $\alpha = 1/2$ makes it $Cd^2 \prod_i t_i = o(g_d^8 \prod_i t_i)$. This additional conclusion uses that separately proved physical-source transfer; it is not a volume-uniform rare-history assertion.

Section 33.4 / A clean first-insertion estimate and finite center normalization

There is a more concrete result for the 21 local curvature-bilinear types in the clean I–II construction. Use their pointed constant-curvature jets, including the pseudoscalar, before any global sum. The independent-current analytic extension and its field-direction margins are supplied by [4, (3.10)–(3.17), Lemma 4] and [9, (1.21)–(1.36)]. They concern the full original paired core. Its fixed radius is an explicit hypothesis preserved by [4, (1.18), Theorem 3]: the constants α_0, α_1 are independent of the birth scale and carrier. This clean core is distinct from the shrinking full layered/R radii in [10, (2.28)]. R-origin terms keep their own radius losses and small-amplitude ledger, even when a later notation merges them into an E family. Inner windows form their own current; arbitrary independent outer currents retain their separate response estimates.

Freeze an admitted old I–II clean state and let $\mathcal{T}_g O$ be the actual clean first-insertion transfer. In canonical fluctuation coordinates its centered Gaussian is independent of the current g, and the chart depends on gB. After the classical and quadratic parts are removed, a local action remainder has the Taylor-integral form

$$V_g(B) = g \int_0^1 c(t) D^3 S(tgB)[B, B, B] \, \mathrm{d}t.$$

On the fixed physical fluctuation box, analytic Cauchy gives $|\partial_g V_g| \le C|B|^3$ and $|\partial_g^2 V_g| \le C|B|^4$; use $g|B| \le \varepsilon_1$ for the apparent higher powers. Old activity and measure derivatives have degree at most one and two, respectively, times their analytic norm. The linear tangent term was cancelled before the independent-current extension was defined, so no unaccounted $g^{-1} J \cdot B$ is reintroduced [4, (2.10)–(2.13)].

A fixed number of local polynomial Gaussian insertions is bounded using

$$|B_X|^m \le \left(\frac{m}{2e\tau} \right)^{m/2} e^{\tau |B_X|^2}$$

with a fixed small tilt. In normalized derivatives the vacuum components cancel; the remaining connected terms meet the insertion root. The ordinary Gaussian/localization estimate and its spare exponent sum the other roots. In particular

$$\begin{aligned} \partial_g \mathcal{T}_g O &= \mathcal{T}_g O' - \mathrm{Cov}_g(O, V'), \\ \partial_g^2 \mathcal{T}_g O &= \mathcal{T}_g O'' - 2\,\mathrm{Cov}_g(O', V') - \mathrm{Cov}_g(O, V'') + \kappa_g(O, V', V') \end{aligned}$$

have local norm at most $C\|O\|$ through second order. The original physical cutoff contributes only rooted boundary tails bounded by fixed powers of g^{-1} times $e^{-c\varepsilon_1^2/g^2}$; these are flat at zero.

The actual construction is invariant under $(g, B) \mapsto (-g, -B)$ with the old state fixed. The first Taylor coefficient is odd in the centered Gaussian, and $\mathcal{T}_0 O$ is its classical central readout. Thus

$$\|\mathcal{T}_g O - \mathcal{T}_0 O\|_{\mathrm{loc}} \le C g^2 \|O\|_{\mathrm{loc}}. \tag{139}$$

This is a real smooth estimate, not a holomorphic disc through zero for a sharp cutoff and not a derivative through the past coupling history.

Lemma 33.1 (Fresh two-insertion Gaussian comparison)**.** *Freeze an admitted old clean I–II state and two scalar gauge-invariant insertions O_i, with their flat constants removed and fixed-core norms at most $R_0^2 a_i$. At flat external pair, let $\Phi(gB)$ be the actual fluctuation/window readout and μ_g its normalized current-step law. If Q_i is the exact flat quadratic insertion and μ_0 the centered constrained Gaussian, then, in the connected local norm,*

$$\mathrm{Cov}_{\mu_g}(O_1 \circ \Phi(gB), O_2 \circ \Phi(gB)) = g^4 \,\mathrm{Cov}_{\mu_0}(Q_1, Q_2) + O(g^6 a_1 a_2). \tag{140}$$

For $Q_i(B) = \frac{1}{2} B^T K_i B$, the Gaussian covariance equals $\frac{1}{2}\,\mathrm{Tr}(C K_1 C K_2)$.

Proof. At $(U, J) = (1, 0)$, every first pair jet is an invariant linear functional on an adjoint representation. Semisimplicity makes it zero, including the independent-current directions. Since $\Phi(0) = (1, 0)$, Taylor's formula gives

$$O_i \circ \Phi(gB) = g^2 L_i(g, B), \qquad L_i(g, B) = \int_0^1 (1 - t) D^2 (O_i \circ \Phi)(tgB)[B, B]\, \mathrm{d}t.$$

The fixed core yields $|\partial_g^r L_i| \le C a_i |B|^{r+2}$ for $r = 0, 1, 2$. Put $R(g) = \mathrm{Cov}_{\mu_g}(L_1, L_2)$. Differentiating twice creates only finitely many polynomial Gaussian vertices of fixed degree. The rooted connected estimate above gives $\|R''(g)\| \le C a_1 a_2$; physical-cutoff boundary terms are flat Gaussian tails. Simultaneous $(g, B) \mapsto (-g, -B)$ makes R even, so $R(g) = R(0) + O(g^2 a_1 a_2)$. Multiplication by g^4 proves the claim, and Wick's quadratic covariance identity gives the displayed trace. □

This lemma compares a fresh step with its exact Gaussian part. Old mixed Gaussian insertions remain in the leading Gaussian recursion. They cannot be discarded as an error of order g^4, the same order as the signal. At a nonflat external pair the corresponding two-label field birth has bound $C g^2 R_0^2 a_1 a_2$. An old mixed member with fixed-core norm $g_j^2 R_0^2 a_{12}$ has flat mean of order $g_s^2 g_j^2 a_{12}$ at a later step, which is at most $C g_s^4 a_{12}$ on the actual weak running tube.

Affine reproduction preserves the pointed curvature-bilinear coefficient of the central readout. For the zero-vacuum, zero-full-quadratic remainder, the actual own-current analytic response kills affine constant curvature. Together with the original regularity estimates this leaves a strict positive age gain. This must use the normalized own-current map: a raw plaquette Hölder difference alone can lose an internal fine spacing. Outer and removed windows use their exponential response bounds instead.

Store these remainders with their birth norms and weight age m by $L^{-\sigma m}$, with σ below the strict age exponent. Pure aging is a contractive shift; the newborn difference from old remainders has the small physical-cutoff factor. Taking the original block geometry and then the cutoff sufficiently small gives a linear first-insertion bookkeeping

$$\begin{aligned} c_{j+1} &= (I + A_j) c_j + B_j r_j, & \|A_j\| + \|B_j\| &\le \epsilon_j = C/x_j, \\ r_{j+1} &= C_j c_j + D_j r_j, & \|C_j\| &\le C_0, \qquad \|D_j\| \le q < 1. \end{aligned} \tag{141}$$

The center is the 21-dimensional pointed tensor space, not a scalar homogeneous action. This is the linear first coefficient of the frozen transfer; no nonlinear tensor-source disc is presumed.

Lemma 33.2 (Finite normalization of the insertion plane)**.** *For* (141)*, choose* $K \ge \max(\|\Gamma_0\|, 2C_0/(1-q)+1)$ *and suppose* $\epsilon_j(1+K) \le (1-q)/8$*. The evolved insertion plane is a bounded graph* $r_j = \Gamma_j c_j$*, and its center factors are invertible. More precisely,*

$$M_j = I + A_j + B_j\Gamma_j, \qquad \Gamma_{j+1} = (C_j + D_j\Gamma_j)M_j^{-1}, \qquad \|\Gamma_j\| \le K. \tag{142}$$

For $Z_{n:0} = M_{n-1}\cdots M_0$*,*

$$\|Z_{n:0}\| + \|Z_{n:0}^{-1}\| \le 2\exp\left(C_K \sum_{j<n} \epsilon_j\right). \tag{143}$$

Proof. If $\|\Gamma_j\| \le K$, then $\|M_j - I\| \le \epsilon_j(1+K) < 1$ and

$$\|\Gamma_{j+1}\| \le \frac{C_0 + qK}{1 - \epsilon_j(1+K)} \le K.$$

This proves the graph and invertibility by induction. For two graphs,

$$\Gamma_1' - \Gamma_2' = (D_j - \Gamma_2' B_j)(\Gamma_1 - \Gamma_2)M_1^{-1},$$

so the graph map is contractive after a further fixed smallness choice. The Neumann bound and $1 + t \le e^t$ bound each center factor and inverse by $e^{C_K\epsilon_j}$; multiply them to obtain (143). □

Multiply the bare insertion basis by $Z_{n:0}^{-1}$. Its terminal center is exactly the identity and its terminal remainder stays bounded by K. Along the tuned run, $x_j \ge a + c(n-j)$, so

$$\|Z_{n:0}\| + \|Z_{n:0}^{-1}\| \le C(1 + cn/a)^{C_K C/c}.$$

Strict engineering decay dominates this polynomial age growth. This supplies a clean normalized first-insertion family; it does not yet bound all mixed physical moments after the full R history is summed.

Section 33.5 / The remaining all-order distribution interface

Finite Gaussian polynomial moments have controlled order growth: a total degree m costs at most $C^m\sqrt{m!}$ in a fixed local covariance norm. For n insertions of degree at most D, this is at most $C_D^n(n!)^{D/2}$ up to a fixed larger factorial power. Old histories and renormalization factors must still be counted before transferring that estimate to the interacting theory.

For example, using only a completed-island bound $e^{-c(\log x)^p}$ to pay x^{Cn} gives an optimized constant of order $e^{C'n^{p/(p-1)}}$. It is larger than $C^n(n!)^q$ for every fixed q. Thus proving each fixed-order bound by increasing the history-reserve constant with that order does not establish the corrected OS growth condition. Separated source supports can improve the count: sufficiently fine records meet at most a fixed number of sources, and a multi-mark component must retain the tree connecting their supports. That assignment must be proved before the history sum.

A precise sufficient interface uses product tests, without a pointwise correlation kernel. For each fixed finite dimension-filtered operator list, require constants independent of the cutoff such that on separated cubes of size h,

$$|S_{\alpha_1\dots\alpha_n,a}(f_1\otimes\cdots\otimes f_n)| \le A_D B_D^n (n!)^{p_D} h^{-b_D n}\prod_i \|f_i(c_i+h\,\cdot)\|_{C^{s_D}}. \tag{144}$$

Section 33.5 / The remaining all-order distribution interface

The separated-product-test lemma turns this into a flat Schwartz bound

$$|S_{\alpha_1\dots\alpha_n,a}(F)| \le A_D(C_D)^n(n!)^{p'_D}p_{k_Dn}(F), \qquad F \in \mathcal{S}_{\Delta,n}. \tag{145}$$

Its proof uses a Fourier projective-tensor expansion in each separated chart, a dyadic separation partition with derivative costs $n^{O(n)}$, and flatness of F to pay the separation powers and grid count. The derivative order is linear in n. No ordinary probability moment bound for a singular local composite is substituted for this condition.

Diagonal extraction can include countably many species after their normalizations and finite-list constants have been fixed. Joint reflection positivity passes through finite Gram matrices. Local real counterterms in multiple source species preserve those matrices by diagonal congruence when the reflected source includes the tensor/parity action; separated-support inclusion–exclusion cancels their contact contributions. Neither assertion requires a uniform nonzero-source disc for a higher composite.

To reconstruct a genuine local family one must also supply full Euclidean covariance, the required vacuum or cluster property, and appropriate nonzero reflected forms, then apply the corrected reconstruction framework [12]. A common mixed hierarchy is essential; constructing species separately would not determine the cross Gram matrices. Identification as Yang–Mills further requires the renormalized operator correspondence and the short-distance structure in [3]. The compact Wilson-loop system and the finite insertion lemmas above give concrete building blocks, but they do not assert these remaining interacting estimates.

Section 34.5 / Scope of the new result

The result is a finite clean extension. Large-field-origin cubics have their genuine shrinking-radius Cauchy costs; fixed R_0 cannot be used for them. Physical integrated dimension-six insertions have factor ℓ^{-2}. A healed odd member's leading degree-six readout can cancel that factor under engineering rescaling, and a long live lineage has its actual age reserve. But the full mixed live/regular filing and all-order separated bounds still require proof. A scalar island mass is not enough.

Tensor rotations and a common distributional reconstruction remain to be constructed. There is no odd dimension-six Lorentz scalar to which the scalar angular argument applies. Higher dimensions also contain genuine derivative operators and lower-dimensional mixing, so the sharp-face obstruction remains. The new content is the local three-curl factorization, the strict cubic own-current remainder, and the 56-dimensional clean normalization; it is not full Yang–Mills existence.

Section 35.1 / Definitions and what remains fixed

Let $\mathcal{P}_2$ be the 21-dimensional curvature-quadratic tensor family, and let $\mathcal{P}_3^-$ be the 56-dimensional charge-conjugation-odd family

$$d^{abc}F_I^a F_J^b F_K^c, \qquad I \le J \le K. \tag{JC1}$$

The latter requires $SU(N)$, $N \ge 3$. Write $p_i = 2$ or 3, and $D_i = 2p_i$, for the curvature degree and engineering dimension of the operator assigned label i.

A globally defined lattice representative can be built from the traceless part of $(U_p - U_p^{-1})/(2i)$. Its integrated physical insertion is

$$O_{i,a}(f_i) = a^{4-D_i} \sum_y f_i(y)\, P_i(X_p(y)), \tag{JC2}$$

with its previously fixed first-insertion normalization and counterterms. In particular, a cubic has the factor a^{-2}. Formula (JC2) does not have a uniform ordinary complex source disc as $a \downarrow 0$. Finite squarefree source coefficients are sufficient for the following argument.

For every nonempty label set I, put

$$m_I = |I|, \quad P_I = \sum_{i\in I} p_i, \quad \sigma_I = \sum_{i\in I}(D_i - 4), \quad \omega_I = \sigma_I + 4 = \sum_{i\in I} D_i - 4(m_I - 1). \tag{JC3}$$

For cubics only, $\sigma_I = 2m_I$ and $\omega_I = 2m_I + 4$.

The source-zero Wilson trajectory, constraints, prepared backgrounds, physical tests, and all source-free counterterms remain fixed. Choose the first-insertion normalization matrices for $\mathcal{P}_2$ and $\mathcal{P}_3^-$ once on that sequence. Any contribution of large-field origin to those FIRST matrices must be included when they are defined; it cannot be added later in a way depending on a source tuple. Every coefficient carrying two or more labels is instead a pointed multi-source coefficient. Extracting one of its quadratic or cubic jets does not change the source-zero trajectory or either first-insertion matrix.

Section 35.2 / The exact joint recursion

Use the finite algebra $z_i^2 = 0$, and write the incoming source increment

$$\Phi_j(z) = \sum_{I \neq \varnothing} z_I \{\nu_{I,j} + O_{I,j}\}, \qquad O_{I,j}(0) = 0. \tag{JC4}$$

Here the zero denotes the flat physical pair, and all current variables are interpreted in the original paired domain. The physical field jets are taken only after the actual own-current substitution. The constants $\nu_{I,j}$ remain in a scalar ledger.

Let μ_j be the actual normalized source-zero regular fluctuation law, and Ψ_j its frozen analytic substitution. The exact local effective logarithm has coefficients

$$W_{I,j} = \sum_{\pi \in \mathrm{Part}(I)} (-1)^{|\pi|-1} \kappa_{\mu_j}(O_{B,j} \circ \Psi_j : B \in \pi). \tag{JC5}$$

The inherited $\nu_{I,j}$ is added separately. There is no factorial missing from (JC5): the factor $1/r!$ in the logarithm cancels the $r!$ orders of an unordered partition into r blocks. Constants can be removed separately from every argument of a cumulant with $r \geq 2$. Consequently the zero-fluctuation subtraction used for two labels remains exact at every fixed order.

The term with one block in (JC5) transports the OLD mixed member. Every other term joins proper label subsets. The source-zero normalization in μ_j supplies all quotient terms. An unnormalized product-moment formula would not give (JC5).

After computing (JC5), separate its flat value and retain the complete remaining analytic function. Any further field-jet extraction is an identity inside that function, with the same complete label set and source carrier. Scalar outputs are recorded once and do not re-enter later connected cumulants.

For a Gaussian step with actual linear harmonic mean HA and hidden covariance g^2D, the complete finite-source identity is

$$e^{-W^G(z,A)} = \exp\left(\frac{g^2}{2}D : \partial_\xi^2\right) e^{-\Phi(z,HA+\xi)}\Big|_{\xi=0}. \tag{JC6}$$

At each fixed squarefree source coefficient the exponential in the source is finite. For polynomial incoming coefficients its field Taylor series is finite as well. Nested Gaussian conditioning makes (JC6) associative. It therefore retains all old mixed vertices; it is not a prescription to reset the source state to its primitive linear terms at each scale.

Quadratic source coefficients may equivalently be completed by the full Gaussian Schur formula first. The remaining cubic and higher vertices then use that formal covariance and mean. Expanding either convention in the same finite source algebra gives (JC6), so this reorganization cannot insert the quadratic state twice.

35.3.1 3.1 Connected degree counting

Start with primitive homogeneous polynomials of degrees p_i. A coefficient with all labels I and coarse-field degree q is a sum of connected Wick multigraphs on those primitive vertices. Self-pairings are allowed when their tensor contraction is nonzero. If there are E hidden pairings, then

$$2E = P_I - q, \qquad E \geq m_I - 1. \tag{JC7}$$

Thus the coefficient has power g^{P_I-q}, and can be nonzero only when

$$q \equiv P_I \pmod 2, \qquad 0 \leq q \leq P_I - 2(m_I - 1). \tag{JC8}$$

The cumulant partition identity cancels every graph whose primitive source vertices split into two disconnected sets. This proves the connectedness used in (JC7), rather than assuming it from a diagram name.

For m cubic labels, the maximum Gaussian field degree is $m+2$. The possible degrees begin as follows; entries forbidden by gauge or charge-conjugation invariance are omitted:

Cubic labels	Gaussian field sectors and coupling powers
1	degree 3, g^0
2	degree 4, g^2; degree 2, g^4; vacuum, g^6
3	degree 5, g^4; degree 3, g^6
4	degree 6, g^6; degree 4, g^8; degree 2, g^{10}; vacuum, g^{12}

Charge conjugation is odd precisely when the number of cubic labels is odd. In that case the complete coefficient has zero ordinary pair jets of degrees zero, one, and two. For an even number of cubic labels, only the linear jet is automatically zero after the vacuum is removed. In particular the mixed coefficient of two cubics must not be assigned a cubic readout factor.

35.3.2 3.2 An explicit two-cubic calculation

Write $P_i(a) = T_i(a,a,a)/6$, with T_i symmetric. At the flat reference background the hidden covariance is color diagonal. Hence the symmetric color tensor d^{abc} gives $\mathrm{Tr}_D\, T_i = 0$, by $\sum_a d^{aac} = 0$. There are no Gaussian self-pairings of either cubic.

Let $T_i(a)$ be the matrix with entries $T_i(a, e_b, e_c)$. Direct Wick contraction gives

$$\begin{aligned}\mathrm{Cov}\big(P_1(a+gB), P_2(a+gB)\big) = {}& \frac{g^2}{4}\langle T_1(a,a,\cdot), DT_2(a,a,\cdot)\rangle \\ &+ \frac{g^4}{2}\,\mathrm{Tr}\big(T_1(a)DT_2(a)D\big) \\ &+ \frac{g^6}{6}\langle T_1, D^{\otimes 3}T_2\rangle.\end{aligned} \tag{JC9}$$

Here B has covariance D. The mixed effective action is minus (JC9), in addition to the transfer of any old O_{12}. The factors are respectively $(1/2)^2$, $(1/2)^2 \cdot 2$, and $(1/6)^2 \cdot 6$. This displays the quartic, quadratic, and scalar sectors explicitly. All three have the same two source anchors.

For an actual source tensor whose Gaussian internal trace is nonzero, (JC6) includes the extra self-pairings. Dropping them would change both the first-insertion normalization and later mixed traces.

Section 35.8 / The physical history interface and the precise remaining work

The clean proof never multiplies independently integrated live-island masses. Inside an unresolved component, all marked primitive factors and missing-vacuum terms stay in the same original unnormalized kernel. A product of physical cubic sources has the joint bound by the corresponding stored-action polynomial, and one cold reserve integrates that whole polynomial. The factor ℓ^{-2} per cubic label must be paid at its actual source record or by the complete live lineage age. It cannot be moved to a later scale without that record accounting.

At FIRST complete marked healing, a one-label cubic output is an actual scalar gauge-invariant charge-conjugation-odd member on its original pair domain. Only then does its zero degree-zero/one/two jet justify the cubic Cauchy factor $\alpha_{\mathrm{birth}}^{-3}$ and the degree-six own-current readout. Partial dropout or an open boundary kernel is not separately such a member. An even multi-label output instead uses the generic zero-vacuum degree-two readout and retains its complete source bridge, as in (JC22)–(JC24).

The necessary physical extension has the following concrete form: apply the original marked R majorant once to the joint stored polynomial; preserve every source anchor until its carrier is wholly regular;

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charge a single distinguished history flag once per live genealogy; and transfer every completed mixed output into the rooted norm used in (JC21). Below the separation scale, a completed regular mixed carrier uses (JC24). A live carrier connecting separated supports must pay its actual complete spanning component or a retained boundary bridge; a relative tree that crosses a live hole for free is insufficient.

The companion history work is establishing this interface together with the first cubic R-center correction. If the latter is $M_j = I + O(x_j^{-1}) + O(\mathrm{poly}(x_j)e^{-cP_0(x_j)})$, its inclusion changes the fixed first normalization by a summable perturbation. It does not introduce an order-dependent normalization. Once the physical interface is proved in the indicated joint norm, the finite-order argument (JC21)–(JC25) applies coefficientwise without changing the bare sequence.

For quantitative bounds in all m, ordinary source analyticity is not the appropriate shortcut. A cubic Gaussian polynomial already has moment growth of order $C^m(m!)^{3/2}$. The squarefree norm with weights

$$w_I = \frac{\prod_{i\in I} r_i}{\sqrt{|I|!}} \tag{JC28}$$

is submultiplicative, and its exponential is controlled by a fixed Gaussian quadratic reserve for a small sum of cubic loads. This is the useful Gevrey coefficient algebra being developed in the companion history note. However a fixed final radius does not make the fine primitive $r_i\ell^{-2}$ small below separation. Those one-label branches must first use the normalized linear cubic cocycle; only genuine multi-label joins and their descendants use the once-paid engineering bound (JC25a)–(JC25b). That reserve, the marked cluster constants, and the joint R ledger must all be included before claiming a uniform all-order Gevrey index.

Thus the new resolved point is the exact joint recursion, its finite interacting grade, and the clean fixed-order separated induction with lower-degree outputs retained. The remaining physical assertion is a specific marked-history norm transfer, while the general higher-dimensional single-insertion problem remains a separate derivative-filtration problem.

35.11.1 11.1 Fix an exactly covariant microscopic representative

The six forward-quadrant plaquette loops based at a vertex are not, by themselves, an exact same-vertex two-form under lattice reflections: a reflected loop can occupy another quadrant. For the newly proposed tensor family choose a clover representative ONCE, before selecting a profile, a grid, or a tuple.

Let $U_{\epsilon\mu,\eta\nu}(x)$ be the loop based at x with successive directions $\epsilon\mu, \eta\nu, -\epsilon\mu, -\eta\nu$, and let $X(U)$ be the traceless Hermitian plaquette variable in (JC2). Define

$$\mathcal{F}_{\mu\nu,a}(x) = \frac{1}{4} \sum_{\epsilon,\eta=\pm1} \epsilon\eta \, X(U_{\epsilon\mu,\eta\nu}(x)). \tag{JC45}$$

It is antisymmetric in μ, ν, gauge covariant at x, globally bounded on compact real links, and charge-conjugation odd. Its leading smooth jet is $a^2 F_{\mu\nu}(x)$. If a signed permutation sends e_μ to $s_\mu e_{\pi(\mu)}$,

relabeling ϵ, η gives the EXACT identity

$$\mathcal{F}_{\mu\nu,a}(rU)(x) = s_\mu s_\nu \mathcal{F}_{\pi(\mu)\pi(\nu),a}(U)(rx), \tag{JC46}$$

with the pullback convention $(rU)(b) = U(rb)$. All loops on both sides have their common base point at the displayed vertex; no unmatched gauge transport is left.

Use the 21 quadratic polynomials in these six components and the 56 symmetric-color cubic polynomials to define the bare tensor representatives. They carry the fixed finite representations

$$\rho_2 = \mathrm{Sym}^2(\Lambda^2 r), \qquad \rho_3 = \mathrm{Sym}^3(\Lambda^2 r). \tag{JC47}$$

Choose the fixed component inner products so that these finite signed-permutation matrices are orthogonal. This is a fixed microscopic choice for the new family. It is not an assertion of equality, at finite cutoff, with the earlier unsymmetrized forward-corner polynomial. The previous center and age proofs apply because the clover representatives have the same leading pointed curvature jets and the same bounded local analytic and cold estimates.

35.11.2 11.2 The actual blocking map has the required symmetry

The map symmetry is part of the actual source-free construction, not inferred merely from symmetry of the bare Wilson action. In Balaban I, printed pp.252–255, (0.4), (0.7), and (0.11)–(0.16) replace an earlier nonsymmetric path choice by averages over all shortest-contour permutations and a compatible averaged gauge fixing. Printed p.269, (2.17)–(2.18), gives the change of fluctuation variables for lattice Euclidean symmetries, including bond reversal and the background adjoint action. It is orthogonal and preserves the characteristic restriction. This verifies covariance of the COMPLETE transfer integral.

To see the marked consequence, write the first-insertion operator abstractly as

$$(T_j O)(V) = \frac{\int K_j(V,U)\rho_j(U)O(U)\,dU}{\int K_j(V,U)\rho_j(U)\,dU}. \tag{JC48}$$

The denominator is source zero. The same statement holds on the augmented original R state before any optional normalization of a live component. Change variables $U \mapsto rU$, including the original covariant gauge change in the fluctuation chart. Haar measure, the source-free density, the averaging kernel and its frozen restrictions have the same transformation. Gauge invariance removes the compensating gauge transformation in the inserted observable. Consequently

$$T_j \mathcal{R}_r = \mathcal{R}_r T_j \tag{JC49}$$

on arbitrary gauge-invariant pointed insertions, where $\mathcal{R}_r$ acts on their complete field/current arguments, source roots and history records. This proves an intertwining statement for insertions, rather than just invariance of the scalar source-zero action.

At a fixed intermediate stage the transformation must preserve the current block partition, or the partition must be carried to its transformed copy as part of the augmented state. Universal matrices below use the complete coefficient only after its original covariance-promotion stage. A freshly healed, still partition-dependent local coefficient is not already a universal translation-covariant matrix entry.

A deterministic decomposition into local carriers may obscure (JC49). The complete coefficient must be formed before an arbitrary asymmetric choice of root is used for a center. One may make the LINEAR filing map equivariant once by averaging its conjugates over the finite lattice group. Its total represented function is unchanged, because the complete transfer already satisfies (JC49); carrier

supports and norms increase by at most one fixed finite factor. For a live state this averaging is performed on the complete represented first-source coefficient, with the transformed history variables and all compensating terms, not on a sourced island mass. Equivalently retain the finite orbit of filings as part of the state until the complete coefficient is summed. No final normalization matrix is averaged.

35.11.3 11.3 Equivariant pointed extraction and the normalization matrices

Let B_q be the POINTED constant-curvature jet map after the actual own-current substitution, for $q = 2$ or 3. On affine curvature data, a change of origin alters the linear potential only by a gauge gradient. The Ward identity removes that gradient. Thus

$$B_q \mathcal{R}_r = \rho_q(r) B_q. \tag{JC50}$$

The target is the full local polynomial space, including the quadratic pseudoscalar direction. No translated-root sum or global constant-source derivative is taken in this definition.

A completion map can safely be made equivariant by finite-group averaging at this stage. If $B_q C_q = I$, then

$$\overline{C}_q = \frac{1}{|\mathcal{G}|} \sum_{r \in \mathcal{G}} \mathcal{R}_r^{-1} C_q \rho_q(r) \quad \text{satisfies} \quad B_q \overline{C}_q = I. \tag{JC51}$$

Its locality and bounds change only by a fixed factor. The stable projection $I - \overline{C}_q B_q$ is now equivariant. The same is true of the origin-tagged remainder/age norm. These choices are fixed on the original RG sequence.

Let V_j be the actual first-source fundamental matrix obtained by applying the complete linear marked transfer to this fixed initial basis and then applying B_q. Equations (JC49)–(JC51) give

$$V_j \rho_q(r) = \rho_q(r) V_j. \tag{JC52}$$

The center/stable normalization argument proves invertibility. Therefore every successive center matrix $V_{j+1} V_j^{-1}$, its accumulated normalization, and its inverse commute with the same $\rho_q(r)$. There is no cutoff-dependent conjugation of the reflection matrix and no use of the condition number of V_j. Averaging an already accumulated invertible matrix would not have proved this: such an average can be singular.

The same conclusion follows from the unique bounded graph transform, since its map and initial data are equivariant. Wrapping terms remain separately tagged; universal local first matrices are not recomputed when the physical torus is replicated.

35.11.4 11.4 The full 21-dimensional first matrix includes R origins

The physical quadratic family requires the analogue of the cubic R correction, not merely its clean first matrix. At FIRST complete marked healing, subtract and record the scalar vacuum of the gauge-invariant quadratic-type coefficient. Its first pair jet vanishes by semisimplicity. Hence the chain rule for its physical second jet is

$$D^2(O \circ \mathcal{H}_j)(0) = D^2 O(0)[D\mathcal{H}_j(0)]^{\otimes 2}. \tag{JC53}$$

There is no nonlinear reconstruction term. Cauchy on its actual shrinking birth core and the localized linear reconstruction give the complete two-curvature kernel bound $C\alpha_j^{-2}\|O\|_{\mathrm{loc}}$. Extract ALL 21

pointed constant-curvature coefficients with (JC50), retain their fixed compact completions, and pair the remainder with that same origin.

This local extraction may be performed at complete healing. Entry into the UNIVERSAL 21-dimensional fundamental matrix waits until the original compatible covariance promotion. During the finite delay $h_s = O(\log\log x_s)$, the local center, its completion, and its compensating remainder retain their partition/origin tag. The promotion wait is not live residence and receives no new live-age credit. Its finite polynomial/logarithmic readout costs, and the factor e^{bh_s} needed when writing an exponential promoted-row estimate, are absorbed into the weakened envelope F_2, exactly as for the cubic family.

The zero-center remainder has the ordinary quadratic Ward/own-current gain, while the density root count cancels its degree-four leading readout. There is no cubic ℓ^{-2} engineering excess in this case. A genuine live entry and its first complete healing use the same one-source excursion norm as the cubic proof, with the smaller core loss α_j^{-2}. Intervening regular time in the center is represented by its CURRENT center value; it is not charged as live age.

Under the same original first-healing/preparation account, the additional first-center rows are bounded by

$$CF_2(x_j)e^{-b(j-t)}(1+j-t)^q, \qquad F_2(x) = (1+x)^{Q_2}e^{-cP_0(x)}, \tag{JC54}$$

where t is the actual live-entry scale, or the birth of an already center-subtracted stable remainder. The memory bootstrap used for the 56-dimensional family now gives

$$M_j^{(2)} = I + O(x_j^{-1}) + O(F_2(x_j)), \qquad M_j^{(3)} = I + O(x_j^{-1}) + O(F_3(x_j)). \tag{JC55}$$

Both fixed first normalizations in the joint PHYSICAL theorem include these R-origin corrections. Gauge invariance, charge conjugation, and the actual finite lattice symmetries are applied to the complete first-healing coefficient before extraction. No independent normalization is assigned to a partial dropout.

35.11.5 11.5 Consequence for lattice reflection positivity

The fixed normalized clover family obeys the same matrices (JC47) at every cutoff. A lattice reflection of a smeared component therefore replaces it by the corresponding fixed finite linear combination at the reflected profile. Since the physical Wilson measure has the stated lattice reflection positivity, the required tensor reflection form is the original physical reflection form with those fixed coefficients. Repeated chessboard reflections introduce only bounded finite component matrices, uniformly in the cutoff and the number of tiles.

This establishes the FINITE lattice symmetry input for the replicated bounds. It does not establish a continuum $SO(4)$ tensor law; the counterexample in the tensor-covariance test note explains why that remains a distinct problem.

Section 36 / Fixed-degree curvature marks in the original mixed-history expansion

This note separates the quantitative part of the composite-source problem from its remaining interfaces. The degree of the local curvature polynomial and the number of its species are fixed. The number n of insertions is allowed to grow. The principal example is the 56 charge-conjugation-odd curvature cubics for $SU(N)$, $N \geq 3$, whose engineering dimension is six.

Section 36 / Fixed-degree curvature marks in the original mixed-history expansion

The source-free maps, real coupling run, characteristic functions and reference replacement laws are the ORIGINAL ones. A live marked island is always one unnormalized kernel. No estimate below multiplies integrated rarity masses that share hidden variables.

The algebraic cold-moment lemma below is unconditional. Its use in the physical construction imports the source-free preparation/domain and own-layer reserve statements identified in the stated results, sections 13–14. The all-order conclusion is stated with a precise clean incoming norm. It is not obtained merely from the previously proved bounds at each fixed order.

Section 36.2 / The actual Wilson cubic and the engineering obstruction

For the traceless Hermitian plaquette variable

$$X_p(U) = \frac{U_p - U_p^{-1}}{2i} - \frac{1}{N}\,\mathrm{tr}\left(\frac{U_p - U_p^{-1}}{2i}\right) I,$$

the real Wilson action obeys

$$\|X_p(U)\|^2 \le Ca(U_p).$$

Consequently a fixed local curvature polynomial P_p of degree p, using finitely many adjacent plaquettes, has the literal compact-group bound

$$|P_p(X)| \le C\left(\sum_{\text{its plaquettes } q} a(U_q)\right)^{p/2}. \tag{CH10}$$

For the 56 cubics this is an $A^{3/2}$ bound. A fixed linear combination costs the norm of its coefficient vector. In a complex prepared boundary chart, the original real-energy majorant acquires only the already allowed regular-background and record-count terms; these are included in $1+E$ in (CH4).

A dimension-six physical integrated insertion at mesh ℓ has the factor

$$\ell^{-2} Z\, P_3(X). \tag{CH11}$$

The matrix Z is the common one-insertion normalization, not a new matrix chosen for a source profile or tuple.

Cold moments alone do not remove (CH11). In particular, $\ell^{-2}\exp[-cP_0(x_\ell)]$ is not uniformly small as $\ell \to 0$, since x_ℓ grows only logarithmically in ℓ^{-1}. Likewise the optimization

$$\sup_{x \ge x_*} x^{Cn} e^{-c(\log x)^q} = \exp\bigl[O(n^{q/(q-1)})\bigr], \qquad q > 1, \tag{CH12}$$

does not give $C^n(n!)^s$ for any fixed s. Neither an order-dependent weak threshold nor a separate fixed-order early-stop argument establishes a uniform hierarchy bound.

The engineering factors must instead be assigned to their actual source roots before the history sum.

The clean local norm in the joint-curvature note, (JC41)–(JC43), is stronger than (CH27). Its input is a local source budget on an observation grid of spacing h:

$$\sup_{q} \sum_{i} r_i \Lambda_i e^{-\kappa d_h(q,q_i)} \le \eta. \tag{CH37}$$

There is at most a fixed number of separated source supports in each such cell. Here Λ_i is the fixed first-profile load, including its h^{-2} cubic factor and one fixed power of x_h. The number of occupied cells and source labels may be arbitrary.

The ordinary squarefree norm is submultiplicative with constant one. It is placed INSIDE the rooted local polymer norm, which retains every source origin. In particular (CH37) is not replaced by the global smallness condition $\sum_i r_i \Lambda_i \le \eta$.

36.8.1 8.1 An exhaustive representation of the source factors

The original R construction can retain the following three classes of source factors, without introducing a new physical observable:

1. The original compact-link polynomial $P_3(X_p(U))$, on its original physical links. Its real absolute value is bounded by a fixed constant times its coefficient.
2. A fixed gauge-covariant completion of a quadratic or cubic physical curvature kernel. Real parallel transport is isometric, so its real supremum is bounded by the kernel's absolute summable norm.

3. The compensating analytic remainder on its actual stored pair domain. It is read only at a prepared field/current pair covered by that domain, and is bounded by its stored analytic supremum.

The completion and remainder are the EXACT split (JC39). If a completion is extracted from a live kernel, its compensating exponential remains inside the SAME kernel as in (JC40). No arbitrary bounded extension of a regular function to a wild field is used. Terms involving the independent current stay inside class 3; they are not evaluated as new polynomials on wild link variables.

This list is closed under the actual state transitions. Coordinate changes, gauge substitutions, disintegration, dropout and swallowing are source-independent linear operations on the source-containing unnormalized kernel. They carry the original factors with them. The original preparation-domain inclusions used in history sections 13–14 justify every class-3 evaluation. A fresh completed analytic member is again class 3. An observable jet extraction adds the class-2 completion and its class-3 compensation by an identity. Scalar coefficients are recorded once. The source-free normalization denominator remains the original one.

Gaussian Taylor polynomials used to calculate a center or a finite jet are analysis representations. None of these transitions requires replacing the exact source factor by an unbounded polynomial surrogate on a live chart. The two- and three-curl kernel norms justify the completion bounds; higher field-jet completions are unnecessary for the separated mixed coefficients.

There is a necessary order of operations. First combine the exact completion/compensation identities and resolve the entire one-label center/stable transport, including the complete-excursion renewal, into its FIXED first normalization. Only then take the local absolute majorant. Bounding a fresh $e^{-zC_j}e^{zC_j}$ pair separately at every scale would create the false charge $\sum_j \|C_j\|$. These cancelling copies are not new source members in (CH38). The source ledger contains genuine births, genuine joinings, and the once-resolved one-label branch, with each remaining compensation owned by its actual birth record. Its scalar identities remain exact.

36.8.2 8.2 Single-label live epochs and first joining

Below h, an unconverted one-label branch is transported by the fixed first-insertion family. Its normalized regular part obeys the ordinary local first-profile bound, and its live part obeys the excursion estimate (CH18). That excursion is indexed at its actual new wild entry, not at the bare cutoff.

A complete live component containing two different source branches is ALREADY a joining, even if it has not healed. At that instant its full carrier and its actual own-layer volume supply (CH23)–(CH25). Every newly joined label's engineering and coupling factor is paid once, and the paid flag stays with its origin. Consequently a component never contains arbitrarily many unconverted labels while attempting to use only one fixed live-age reserve.

Before joining, distinct unconverted live branches belong to distinct complete live components and use their respective actual age records. After joining, the complete source coefficient stays inside one joint live kernel. A partial dropout follows the unique product-rule owner of each distinguished label; if the kernel has not actually factorized, it is retained as a joint state.

Any later bounded linear readout loss before scale h can be prepaid at the first joining from the same scale-labelled carrier. For a fixed readout exponent b, $(h/\ell)^b(1+\log(h/\ell))^q e^{-ch/\ell}$ is uniformly bounded. The available exponent b is independent of the source degree: the complete mixed function uses the ordinary zero-vacuum, zero-linear-jet readout, and is not repeatedly Taylor-expanded to an order

increasing with the number of labels. The one-label cubic uses its separate fixed third-order readout.

36.8.3 8.3 The pointwise local source charge

For the finite-prefix induction, assume the ALREADY resolved incoming regular members have ordinary local coefficient norm at most A. This is the quantity to be bounded by the recurrence below; it is not assumed to equal $C\eta$ at the start. The once-resolved original one-label forcing has norm at most $C\eta$. After engineering payment, the class-1 and class-2 factors are bounded pointwise; the class-3 factors are bounded on their original prepared domains. On any actual stored live component or wall carrier X, the incoming local rooted source norm gives

$$\sum_{\text{source members meeting } X} \|F_{\text{member}}(z)\|_{\boldsymbol{r}} \le C(\eta + A)N_X, \tag{CH38}$$

after weakening the spare carrier exponent. Here N_X is the number of actual unit cells in its OWN stored layers, including its determining collars. To see the bound, point every member at its first intersection with X, sum the remaining rooted carrier by its original exponential norm, and then use (CH37). Since the retained layer spacing is at most h, the h-cell source packing sum is uniform at each such intersection. Enlarging an intersection by a fixed collar changes only C. Long source-attaching paths retain their locality cost and are not counted as fresh dense sources. The members here are precisely the resolved ones specified at the end of subsection 8.1. No cancelled one-label center completion is charged again at an intermediate scale.

Let $\mathcal{M}_\omega^-$ be the ONE positive joint majorant of the original history kernel after its specified engineering, first-root, age and tree fractions have been spent. It still contains the retained real action and the remaining original record-volume reserve. It is not the original kernel with an unspent bank. The payment proves a bound for the source coefficient TIMES those allocated record factors; it need not bound each bare unweighted source function separately.

Triangle inequality and coefficient submultiplicativity, BEFORE any live integration, consequently give

$$\|\text{the paid source-containing kernel}\|_{\boldsymbol{r}} \le e^{C(\eta+A)N_\omega}\mathcal{M}_\omega^-. \tag{CH39}$$

If both sides of one genuinely new completion split remain after the symbolic cancellation, their two norms are charged ONCE to that birth, with one fixed larger constant. This is not a separate charge for every later rereading of that stored member. The measure under $\mathcal{M}_\omega^-$ is still the single original joint kernel's measure. This estimate has no product of conditional expectations, no product of integrated rarity factors, and no inverse of a source-containing island mass.

Choose the prospective bound on $\eta + A$ small compared with the original spare own-layer volume reserve. The extra factor in (CH39) is then paid by a fixed fraction of that reserve. All old denominator and chart factors are already accounted for by the original $C(\log x_j)N_j$ record cost in history section 13. No part of (CH39) requires a source-order-dependent weak floor or a new field cutoff.

36.8.4 8.4 The finite-prefix regular and live-exit recurrence

Here is the closure of the preceding incoming-norm estimate. Let A_n be the ordinary local norm of all PAID regular source members in a finite prefix ending at n, after the complete one-label and ordinary regular linear resolvents have been taken. Let L_n be the local norm of first-complete-healing and terminal-exit fluxes of the corresponding JOINT live components. The latter is an exit norm of fully unfolded excursions. It is not a norm of an arbitrary live kernel to be multiplied with another such norm.

Section 36.8 / The local coefficient norm through original R operations

Set $s = \eta + A_n$. Subtract the zero-source term in a complete excursion before estimating it. In the same positive kernel used in (CH39), its linear and at-least-two-input excesses are bounded respectively by

$$CsN_\omega\,\mathcal{M}_\omega^-, \qquad Cs^2N_\omega^2 e^{CsN_\omega}\mathcal{M}_\omega^-. \tag{CH39a}$$

The power two here is a product of POINTWISE incoming regular/compact source loads in one integrand. It is not L_n^2, and it grants no second rarity when two labels belong to the same live component.

The original record-volume reserve is sufficient to sum N_ω and N_ω^2 in these two estimates. Explicitly, spend one fixed remaining fraction τN_ω; for $Cs < \tau/2$, both $N_\omega e^{-\tau N_\omega}$ and $N_\omega^2 e^{-(\tau - Cs)N_\omega}$ are bounded by constants independent of the retained dimension and source count. The rest of the original completed-R majorant gives its ordinary rooted carrier and age sum. In its linear term, one actual new wild entry supplies the rare envelope $F_*(x_v)$; an already live ancestor keeps that original entry label and has only dwell propagation. Core Cauchy costs have fixed order two for a vacuum-subtracted mixed member, or order three for the one-label odd member. Thus $F_*(x)$ is a fixed polynomial times the original rare tail, independent of source count. The running tube gives

$$q_R \le C\sum_{v\le n} F_*(x_v) \longrightarrow 0 \quad \text{as the fixed weak floor increases.} \tag{CH39b}$$

Source-anchor first-root sums below h use (CH26); their order does not grow with the number of labels. The original local formation constant and all the weakened record constants are fixed.

Consequently there are constants C_0, C_1, C_2, independent of n, the coefficient algebra, the bare mesh and the torus, for which

$$\begin{aligned} A_n &\le C_0\eta + C_0L_n + C_1\frac{(\eta + A_n)^2}{1 - C_1(\eta + A_n)},\\ L_n &\le C_2\eta + q_RA_n + C_2\frac{(\eta + A_n)^2}{1 - C_2(\eta + A_n)}. \end{aligned} \tag{CH39c}$$

The forcing $C_2\eta$ includes a primitive source born inside a live component and its resolved one-label excursion. When several live components merge, their actual hidden integrals and original entry records remain inside the ONE unfolded component in (CH39a); no product of their integrated flux bounds is used. A fresh regular source entering such a component is another bounded input in its same exponential. This is why there is no term CL_n^2 in (CH39c).

Choose the weak floor once so that $C_0q_R < 1/4$. Substitution of the second inequality into the first gives

$$A_n \le C_3\eta + \tfrac{1}{4}A_n + C_4\frac{(\eta + A_n)^2}{1 - C_4(\eta + A_n)}. \tag{CH39d}$$

Take $K > 4C_3$, then choose η once small so that the last expression is less than $K\eta$ at $A_n = K\eta$. The positive finite-prefix majorant starts at zero with the source fugacities. The radial first-exit argument therefore proves $A_n \le K\eta$ and, by (CH39c), $L_n \le K'\eta$ for every prefix. The complete source-zero linear resolvents were taken BEFORE this argument; an unresummed factor greater than one at every ultraviolet step would not yield (CH39c).

This derives the uniform rows used here from the original positive completed-R majorant, the stored-domain bounds, and the explicit record-volume moments in (CH39a). It is stronger than merely specifying the three representation classes. The original full-core connected-formation theorem and the spare action/volume estimates remain the explicit source-free inputs.

36.8.5 8.5 Localization, scalar outputs, and terminal integration

The original wall Cauchy estimates and connected formation proof now act on this local coefficient norm. They use its triangle inequality and submultiplicativity, with the same common field domains. For ordinary completed regular members, their products are legitimate products of bounded functions. For a still-live component, (CH39) is used once on its joint kernel. Its replacement law remains source free and normalized. Thus the source extension does not introduce an L^1-product step which was absent from the original localization.

The actual full carrier retains all source anchors. Every mixed scalar output born below h therefore has at least one separation bridge. The first-root sum (CH26), together with the paid-label tree estimate, is bounded per h-cell. The single cubic scalar coefficient vanishes by charge conjugation. Scalar outputs are recorded once rather than transported through further source cumulants.

It follows that the clean local majorant (JC43), with a fixed smaller locality exponent, extends through the original R sum on this represented class. The source-excess scalar ledger and the surviving terminal source norm are at most $C\eta N_h$. The unmarked factor is the SAME complete vacuum V_N, including the regular R vacuum terms. Missing vacua in an individual live history are filed in its actual origin records as in history section 15.

This conclusion imports the original full-core local-potential/wall theorem used in (JC43), the original prepared-domain inclusions, and its exact completed-R action/record bounds. It does not follow from a numerical flat-background covariance bound or from the global Gevrey estimate alone. The additional source operations needed for the bounded-atom R extension have been checked above.

Section 36.10 / Primary source-free inputs

The preparation and localization facts used here are the ORIGINAL ones already identified in the history and cubic notes:

- T. Bałaban, *Renormalization group approach to lattice gauge field theories. I*, CMP 109 (1987), 249–301: the coarse linear response and Ward kernels, (4.2)–(4.5), together with the independent-current and inner-window domain results.

  II. Cluster expansions, CMP 116 (1988), 1–22: the frozen wall localization and full paired-domain function bounds.

  II. Localization, exponentiation, and bounds for the R operation, CMP 122 (1989), 355–392: (1.65), (1.70)–(1.72), and the completed-carrier construction (1.90)–(1.98).

These papers are not being cited as a published composite-source theorem. The new coefficient, cold-moment and source-tree estimates are the ones proved above; the source-free map and domain theorems remain explicit imports.

Section 37.4 / Live components require their actual stored records

The boundary norm in (1.67) of [8] uses a relative tree distance that omits interiors of live holes. Two separated marks can lie in one hole while that distance is zero. Therefore (164) cannot be applied to a live boundary norm alone.

Marked operator filing must keep each physical observable on its original stored variables until they are integrated. Its anchor stays in every bridge union. Full completed carriers include all completed components and source flags. Substitution of a later replacement field for the original observable is not an allowed shortcut.

Section 37.4 / Live components require their actual stored records

For a live state two payments are legitimate.

1. Keep the marked coefficient in its unnormalized operator until the full own-scale mass estimate is used. Only then use (164).
2. Charge normalization once at the actual first-readout record, and retain a paid tag. Use the original cold action and that record's own-scale budget. A reconstructed fine background is not a new record.

If a primitive stored cell meets one source support, squarefree algebra makes its multiplier exactly $1 + z_i A$. A polynomial in its stored energy is paid by a fixed small cold-action tilt and a finite Gaussian or Wilson moment. On scales much smaller than h, distinct supports occupy distinct marked cells; a fixed per-record charge pays their coupling factors separately. Nonlocal readouts retain their source bridges. If several labels enter one localized record, use its full marked carrier when its full mass is applied.

Waiting or merging never duplicates a source label. Actual waiting records pay live dwell. At a coarse merger only distance at that scale is available: no fine-scale distance may be assigned retroactively. Fine children have already paid their one-mark or full multi-mark costs before their scalar bounds are reused. Unintegrated children remain explicit operators with their own records. Each child kernel and each mark occurs once in swallowed-operator composition.

Section 39 / When a Gaussian coarse mean is independent of running weights

This section separates an exact cancellation from a tempting extension that fails. In an orthogonal Gaussian band decomposition, arbitrary positive running weights do not change the conditional mean at a retained level. A generic local quadratic counterterm at an intermediate level does change it. The distinction matters when trying to replace a completed composite coefficient by a derivative of a smooth classical lift.

The results here are finite-dimensional identities, followed by an explicit one-dimensional Gaussian counterexample. They do not supply the nonlinear Wilson comparison.

Section 40 / Exact completion identities for derivative insertions

This section proves an arbitrary-density Gaussian completion identity, identifies the finite triangular mixing that it supplies, and gives finite-lattice gauge-covariant point-splitting and commutator formulas on the unchanged Wilson configuration space. The ultraviolet-local Wilson interaction is not assumed to be a fixed smooth cylinder density. The near-mark attachment estimate remains explicit.

Differentiate (205) once in every leg variable. For products normal ordered at their individual vertices this gives

$$\mathbb{E}_\mu \prod_i :\prod_{a\in L_i} X_a:_C = \sum_{M\in\mathfrak{M}_{\mathrm{cross}}} \left(\prod_{(a,b)\in M} \ell_a C \ell_b^*\right) \mathcal{A}_{|U(M)|}((C\ell_a^*)_{a\in U(M)}). \tag{210}$$

Here M is a partial matching whose edges join distinct vertices, and $U(M)$ is its set of unmatched legs. Same-vertex Gaussian pairings were removed by normal ordering. Interaction attachments can still join several legs at one mark; these are precisely local composite-renormalization contributions.

Every propagator and attachment direction is the full C. If one first disintegrates using any sharp map Q, or nested maps, contractions acquire retained or hidden covariance colors. Summing the colors reconstructs C on every edge and $C\ell_a^*$ on every attachment. Multilinearity of (208), or of the justified derivatives in (209), proves this before absolute values are taken. Artificial block-face terms therefore cancel algebraically with the arbitrary density retained.

This identity does not bound its right side. A sharp coarse density can produce singular attachments and separated divergences, as the positive rank-one tilt of the exact-conditioning section shows.

Section 40.2 / Partial pairings and the attachment problem

Connected coefficients follow by the finite cumulant partition formula. Equivalently, finite coefficients of $\log \mathcal{A}$ are connected attachment hyperedges; together with covariance edges one retains graphs connected on the marks. For at most $K(D)m$ legs, matching and partition counts cost at most $C_D^m (m!)^{q_D}$. The missing estimate is analytic, not a demand for an infinite composite-source power series.

Section 40.4 / A far-attachment estimate

Let V_X be bounded by B_X on a fixed physical paired core of radius R_0. A derivative-of-curvature leg of order s at x produces the flat reference shift

$$v_{x,s} = C\ell^*_{x,s}. \tag{214}$$

At unit current scale, if $\operatorname{dist}(x, X) \geq d \geq 1$, its local $(A, \nabla A, \Delta^* A)$ norm is at most $C_s d^{-3-s}$. The first derivative has an extra inverse distance; the free Euler component vanishes away from the anchor. These lattice Green bounds are uniform in fine mesh. The bounded background-dependent first pair map changes only the fixed constant on a smaller core. Multidirectional Cauchy consequently gives

$$|D^r V_X[v_{x_1,s_1}, \ldots, v_{x_r,s_r}]| \leq r! B_X \left(\frac{C}{R_0}\right)^r \prod_{i=1}^{r} d_i^{-3-s_i}. \tag{215}$$

Section 40.4 / A far-attachment estimate

The original carrier exponential is retained. A large-field-origin member uses its own shrinking radius and rare factor. Fixed-order polynomial leg costs can use the existing carrier and cold-action reserves.

This proof fails when the anchor enters X or approaches it with the cutoff. The full Green shift is then singular in the local core norm. Substituting the microscopic spacing for d gives a divergent power, not renormalization. These near attachments, after their local subtractions and connected completion, require a new estimate.

Section 40.5 / Gauge-covariant point splitting on the same lattice

For spacing a, define a based plaquette field

$$X_{\mu\nu,a}(x) = \frac{1}{2ia^2}\left[U_{\mu\nu}(x) - U_{\mu\nu}(x)^{-1} - \frac{1}{N}\,\mathrm{tr}\{U_{\mu\nu}(x) - U_{\mu\nu}(x)^{-1}\}I\right]. \tag{216}$$

A reflected clover choice is also possible. It transforms in the adjoint at its basepoint. Put

$$T_\mu X(x) = U_\mu(x)X(x + a\hat{\mu})U_\mu(x)^{-1}, \qquad D_{\mu,a} = a^{-1}(T_\mu - I). \tag{217}$$

An invariant polynomial in finitely many such differences is an exact finite-cutoff gauge-invariant insertion. At engineering dimension D its compact-group bound is $C_D a^{-D}$ and its carrier has radius $R_D a$, with fixed R_D. Averaging the finite patterns can impose the required lattice reflections and tensor symmetries. Covariant Taylor expansion on a smooth classical connection recovers the intended curvature polynomial. These facts define candidates without altering the action; they do not prove a quantum limit.

Two exact identities expose the interacting corrections:

$$D_{\mu,a}(XY) = (D_{\mu,a}X)(T_\mu Y) + XD_{\mu,a}Y, \tag{218}$$

$$[D_{\mu,a}, D_{\nu,a}]X = a^{-2}(\mathrm{Ad}\,U_{\mu\nu} - I)T_\nu T_\mu X. \tag{219}$$

The second follows from $T_\mu T_\nu = \mathrm{Ad}(U_{\mu\nu})T_\nu T_\mu$. On the real compact group,

$$\|[D_{\mu,a}, D_{\nu,a}]X\| \le 2a^{-2}\|U_{\mu\nu} - I\|\,\|X\|. \tag{220}$$

In a small plaquette chart this is $C\|F\|\|X\|$. At fixed D the two identities generate finitely many types of dimension at most D. This proves algebraic finiteness of the mixing problem, not bounds on its coefficients. Fixed microscopic paths avoid a separate diverging physical-length line normalization; mesoscopic paths require additional estimates.

The argument **clarification.** Equations AI18–AI32d in the later contact-completion chapter make this reduction explicit for the full scalar derivative-curvature square, retaining the Wilson equation-of-motion completion and every lattice remainder. Its commutator cubic is charge conjugation even and is not covered by the protected 56-component charge conjugation odd cubic center. The corresponding coupled even-sector mixing estimate is still required.

Section 40.8 / Scope of existing perturbative theorems

Lüscher and Weisz [15] prove positive-flow-time finiteness to all loop orders after renormalization of the underlying four-dimensional theory. This supplies precise gauge-covariant smoothing identities in perturbation theory, not a nonperturbative Wilson limit or a uniform zero-flow-time composite bound. Fröb, Holland, and Hollands [16] prove perturbative bounds for arbitrary composite insertions,

Section 40.8 / Scope of existing perturbative theorems

organized by weighted spanning trees, with the required Ward identities. Those bounds are indexed by perturbative order. The present target concerns the actual finite-cutoff integral and its counterterms. Neither theorem is being imported as that estimate.

Section 40.9 / Finite Gaussian scheme comparison and periodic jets

For finitely many derivative-of-curvature variables of dimension at most D, suppose $C' - C$ has uniformly bounded diagonal derivative jets for all contractions that enter. Equation (211) is a finite triangular matrix with entries and inverse entries bounded by $C_D(1+M)^{q_D}$, where M bounds those jets. Its inverse changes the sign of $C' - C$. This is a polynomial observable identity under any fixed measure. It does not apply to sharp retained covariances whose derivative jets diverge. If only the conditioning grid changes and full C is kept, completed Wick observables agree exactly.

After completion, periodic coefficients have a precise finite-jet comparison. Let a be integrable and period one in four dimensions, with cell norm at most M, and put $a_{\ell,o}(x) = a((x-o)/\ell)$. For each cell $Q = o + \ell(k + [0,1]^4)$ use its center c_Q, and set

$$m_\beta = \int_{[-1/2,1/2]^4} a(y + \tfrac{1}{2}) \frac{y^\beta}{\beta!}\, dy. \tag{224}$$

Taylor's theorem for a smooth compactly supported completed test kernel K gives

$$\int a_{\ell,o}(x) K(x)\, dx = \sum_Q \sum_{|\beta| \le k} \ell^{4+|\beta|} m_\beta \partial^\beta K(c_Q) + \mathcal{R}_{k+1}, \tag{225}$$

$$|\mathcal{R}_{k+1}| \le C_k M \ell^{k+1} \sum_{Q \cap \operatorname{supp} K \neq \varnothing} \ell^4 \sup_Q |\nabla^{k+1} K|. \tag{226}$$

Include boundary cells by extending K as zero outside its support. Recentring finite jets is a finite triangular binomial transformation. A periodic phase translation preserves the zero cell moment; higher moments enter this finite transformation and the remainder.

For a dimension-$d \le D$ coefficient with factor ℓ^{d-D}, choose $k \ge D - d$. The remainder is then strictly summable if the completed kernel has the required derivative bounds. The jets differentiate the whole product of its source test and its separated correlation kernel. Leibniz's rule includes lower operator descendants; differentiating only the source test is incorrect.

For a mean-zero periodic distribution with Fourier growth $|\widehat{a}(n)| \le M(1+|n|)^{r_0}$, choose even $k > r_0 + 4$. The zero-mean solution of $(-\Delta)^{k/2} b = a$ has absolutely summable Fourier coefficients, hence $\|b\|_\infty \le C_k M$. Distributional integration by parts proves

$$\left| \int a_{\ell,o}(x) K(x)\, dx \right| \le C_k M \ell^k \|\nabla^k K\|_{L^1}. \tag{227}$$

The difference of translated phases has zero mean. Choosing also $k > D - d$ makes the power-counted remainder summable over fine geometric scales. These estimates are uniform in the phase.

Completed free-field separated kernels satisfy the derivative bounds by differentiating their full cross contractions. They therefore give a rigorous finite-dimensional Gaussian comparison. The pre-completion quantity $(h''_\sigma)^2$ has a cell norm of order σ^{-1} and violates the uniform M hypothesis. For Wilson's theory a bounded completed local coefficient and its derivative norm are the near-mark attachment input, not a consequence of periodic averaging.

Section 41 / A dimension-six averaged readout and the remaining current condition

This note proves three narrower results. The actual flat sharply conditioned curvature has a uniform first-derivative L^2 bound and an H^{1+s} bound for each fixed $0 < s < 1/2$. A symmetric quadratic readout therefore has a strict dimension-six Taylor remainder in an INTEGRATED norm, without a pointwise second derivative of curvature. Descendants and completed equation-of-motion insertions also have exact constructions on the same bare sequence. The complete nonlinear, wall-interpolated paired-current maps have not yet been proved to preserve the required fractional norm.

The original independent-current domain in Balaban I, (1.14), bounds the current in supremum norm. It does not impose a current Hölder norm. The perturbation directions in (3.14), (4.4) have first-order and covariant-Laplacian bounds. These are not automatically the fractional improvement proved below for an actual flat constrained lift.

41.4.1 4.1 Descendants

If O is an already constructed distribution-valued curvature quadratic field, define

$$(\partial^\alpha O)(f) = (-1)^{|\alpha|} O(\partial^\alpha f). \tag{D6.14}$$

This constructs $\Box\,\mathrm{tr}(F^2)$, for example, on the SAME bare sequence. At finite cutoff use finite differences and exact discrete summation by parts. These descendants need no new high-derivative conditional norm, and generally have nonzero separated correlations.

41.4.2 4.2 An exact contact equation-of-motion square

Let S_a be the actual Wilson action and X_b^A a left-invariant bond derivative in an orthonormal Lie-algebra basis. Haar measure is divergence-free for X_b^A. Put $E_b^A = X_b^A S_a$. For any smooth V_b^A, G,

$$\left\langle \{E_b^A V_b^A - X_b^A V_b^A\} G \right\rangle = \left\langle V_b^A X_b^A G \right\rangle. \tag{D6.15}$$

This is exact integration by parts in the physical measure.

Taking $V_b^A = E_b^A$ gives the gauge-invariant local representative

$$\mathcal{E}_a(f) = a^{-2}x_a^{-2}\sum_{b,A} f(b)\{(E_b^A)^2 - X_b^A E_b^A\}. \tag{D6.16}$$

Up to the fixed action convention, its leading classical density is $(D_\mu F_{\mu\nu})^2$. The second term is its required local completion. If all factors of G are supported away from the differentiated bonds,

$$\langle \mathcal{E}_a(f) G\rangle = 0. \tag{D6.17}$$

No sourced denominator or conditional estimate appears. An uncompleted force square does not have this property. Other equation-of-motion multiples use the appropriate V_b^A, including its derivative counterterm.

41.4.3 4.3 A continuum identity with fixed conventions

For anti-Hermitian connections use $\langle X, Y\rangle = -\mathrm{tr}(XY)$ and full ordered index sums. Define

$$E_\nu = D_\mu F_{\mu\nu}, \quad A_6 = \sum_{\rho,\mu,\nu} |D_\rho F_{\mu\nu}|^2, \quad B_6 = \sum_{\nu} |E_\nu|^2, \quad C_6 = \sum_{\rho,\mu,\nu} \langle F_{\mu\nu}, [F_{\rho\mu}, F_{\rho\nu}]\rangle.$$

Bianchi and $[D_\rho, D_\mu] = [F_{\rho\mu}, \cdot\,]$ give

$$D_\rho D_\rho F_{\mu\nu} = D_\mu E_\nu - D_\nu E_\mu + 2[F_{\rho\mu}, F_{\rho\nu}].$$

Taking the inner product with $F_{\mu\nu}$ proves

$$A_6 = 2B_6 - 2C_6 + \tfrac{1}{2}\Delta|F|^2 - 2\partial_\mu\langle F_{\mu\nu}, E_\nu\rangle. \tag{D6.18}$$

Consequently

$$\int f A_6 = 2\int f B_6 - 2\int f C_6 + \tfrac{1}{2}\int (\Delta f)|F|^2 + 2\int (\partial_\mu f)\langle F_{\mu\nu}, E_\nu\rangle. \tag{D6.19}$$

Quantum use needs the local completions in (D6.15). It cannot multiply an equation of motion by another singular field and discard the derivative counterterm.

The argument **clarification and extension.** The cubic C_6 here is charge-conjugation even, not a member of the protected 56-dimensional C-odd center. The transformation and its exact lattice counterpart are proved in the stated results, AI24 and AI30. That chapter also gives an exact smeared scalar identity, including the projected Wilson flux, cube defects, test-derivative terms, and the explicit Casimir counterterm. Its alternative compact completion AI32a–d gives an exact bare-basis reduction to a C-even cubic and quadratic descendant. The coupled even dimension-six mixing estimates remain separate; the candidate even-center qualification below is unchanged.

The full hypercubic list can be larger than the rotational scalar list. For example, squaring $D_\mu F_{\mu\nu}$ separately for each μ before summing is not the square of $\sum_\mu D_\mu F_{\mu\nu}$. Do not discard all derivative-curvature tensors by an $O(4)$-scalar classification before the continuum tensor law is proved. The full derivative tensor in (D6.12) is the safe finite center.

Section 41.6 / The precise remaining map lemma

A sufficient weaker replacement for an all-orders Schauder-germ theorem is the following.

For some fixed $0 < s < 1/2$, every actual clean inner readout and every field/wall interpolation used on an old member has local gauges with

$$\|\chi_X F\|_{H^{1+s}} \le C \operatorname{poly}(\operatorname{diam} X)\, \mathcal{N}(U, J), \tag{D6.21}$$

with matching localized derivative/tree bounds under coarse field variations and the ACTUAL old member's core normalization. Outer/removed readouts may instead have exponentially small response in this norm. R-origin members require it only at their admitted prepared evaluations and retain their shrinking-core losses.

One route is a weighted H^s, or BV-plus-sup, bound for the actual gauge-fixed Euler force. The flat box/tent argument proves this at zero background. Nonlinear parallel-transport weights can depend on remote fields, and intermediate interfaces can remain. A proof needs a fractional product/path estimate or summable interface jumps; counting only final block faces is insufficient.

A second route is to estimate each original localized block-resolvent parametrix in this fractional norm and then sum the actual typed-walk series. This need not preserve an exact Euler equation, but the endpoint estimate must survive every wall interpolation and nonlinear fixed-point substitution. The published low-regularity statement does not by itself include it.

If (D6.21), its physical own-current scaling, and its common analytic directional margins are proved, (D6.10)–(D6.13) supply the strict integrated quadratic derivative age gain. Add the finite even F^3 center and existing quadratic center, retain descendants, and pair each extracted counterterm with its origin. Relevant dimension-four mixing is triangular with its genuine engineering factor; the protected

Section 41.6 / The precise remaining map lemma

C-odd argument does not remove it. Higher ordinary field degree has a strict engineering gain. This is a candidate finite filtration, not yet an interacting first-insertion cocycle.

Physical completion would still require first-complete-healing/compatible-promotion, joint source-root bounds, and fixed normalization/near-attachment analysis. The present note closes the flat averaged-readout and exact descendant/EOM subproblems and identifies the additional regularity estimate.

Section 42 / Physical critical germs, nonlinear averages, and a covariant BV estimate

This note separates three questions that must not be conflated. A nonlinear constrained critical field need not have a scalar piecewise constant forcing. Literal criticality need not survive every auxiliary interpolation. Nevertheless, the **complete physical constraint current** admits a useful one-step covariant bounded-variation estimate. The latter estimate can be iterated through genuine averaging steps without propagating a nonphysical error term separately. It supplies a fractional-regularity route for low-dimensional derivative insertions; it is not an all-derivative invariant-domain theorem.

The actual averaging maps used below are the symmetric contour averages in Bałaban I, (0.4) or (0.12), and their actual composition in I (1.1). Their **outer** bond average is a reference-logarithm average. The symmetric group mean (0.5)–(0.10) is used for the **inner** endpoint contour variable (0.11) in the version (0.12). These are different operations. The independent current extension and its low-derivative domains are I (1.8)–(1.18), (3.11)–(3.17). The final marked-readout distinction is explained in Section 8. All estimates below use one fixed blocking factor L. Constants may depend on L and the compact matrix group, but not on the number of blocking steps.

Section 42.8 / The actual-map invariant-domain question

There are two different interpolation levels in the original construction. A literal nonlinear critical submanifold is not generally convex or closed under scalar interpolation. For example, let

$$S(x,y) = \tfrac{1}{2}(x^2+y^2) + \gamma x^2 y, \qquad Q(x,y) = x.$$

For small data the constrained minimizer is $H(b) = (b, -\gamma b^2)$. The hidden gradient at $tH(b)$ is $\gamma(t^2 - t)b^2$, which is nonzero for $0 < t < 1$ and nonzero b. Thus no multiplier makes $tH(b)$ critical. This elementary observation applies to an argument that asks every intermediate Taylor field to remain exactly stationary.

It is **not** enough to dismiss the original final readout. I (3.36), Lemma 4/(3.53), and the corresponding II formation formulas compose an inner U_j(box,exp(iB)) with its **own** current. That inner map is a constrained reconstruction even when its outer argument was assembled using an interpolated H_k(B'). The inner constraint scale is the old j scale. Being critical at that scale does not turn all its old-scale faces into smooth forcing on a current-k-scale ball.

The new analytic task for a high-derivative marked norm is therefore specific:

1. Identify the actual inner reconstruction and the current paired with it at every marked evaluation, including all localization parameters.
2. Prove either a uniform all-jet forcing bound on a protected own-scale neighbourhood, with an all-depth transport estimate for its marked coefficient, or a completed weak/finite-jet cancellation estimate that avoids those pointwise derivatives.
3. For every walk- or history-localized reconstruction, identify its exact equation and show that the added residual is supported outside the protected neighbourhood, or bound that residual in the same higher norm.
4. Close these estimates on a common analytic parameter domain. The original theorem's inclusion in a low-derivative independent-(U,J) domain is not this statement.

The real physical BV theorem above is a genuine part of item 2 for fractional regularity. It does not settle items 3–4 or the all-derivative version of item 2. A smooth analytic hull of critical germs may be the appropriate domain; literal stationarity at every intermediate Taylor point is unnecessarily strong.

42.8.1 8.1 Complexification

The algebraic identities, the Ward relation, and the local mean comparison are holomorphic. For complex variables the star in these identities means the transpose for the invariant bilinear Lie-algebra pairing, followed by estimates in a Hermitian absolute norm. It does not mean an anti-holomorphic Hermitian adjoint.

There is a concrete sufficient chart condition that avoids resetting a gauge-frame norm at every level. Suppose the initial fine field is in one actual physical chart

$$U_0(b) = e^{h_0 A'(b)} U_0^{\mathrm{real}}(b), \qquad \|A'\|_\infty \le M, \tag{PG24}$$

and its complex logarithmic plaquette curvature is bounded by α_0. Define directly, in this one common frame,

$$K_s = h_s^{-1} \sup_b \log \max\{\|U_s(b)\|, \|U_s(b)^{-1}\|\}. \tag{PG25}$$

Then $K_0 \le M$, up to the fixed matrix convention, and a path of physical length r in the scale-s connection has adjoint norm at most $\exp(CK_s r)$. The actual reference-log map, the endpoint mean, and non-Abelian plaquette telescoping give the coupled recurrences

$$\begin{aligned} K_{s+1} &\le K_s + C_L \ell_s \alpha_s e^{C_L K_s \ell_s}, \\ \alpha_{s+1} &\le e^{C_L K_s \ell_s}(\alpha_s + C_L \ell_s^2 \alpha_s^2), \end{aligned} \qquad \ell_s = L h_s. \tag{PG26}$$

For the first line, the literal reference bond R is a product of L old links, so $\log \|R^{\pm 1}\| \le \ell_s K_s$ with coefficient one. Every $C_i R^{-1}$ differs from the identity by $O_L(\ell_s^2 \alpha_s) e^{C_L K_s \ell_s}$, by filling its contractible loop. The outer exponential in (PG15a) therefore increases the logarithmic norm by at most this amount; division by ℓ_s proves the first line. The endpoint group mean is covered by the same relative-diagonal estimate.

For the second line, a local tree-gauge transformation has adjoint norm at most $\exp(C_L K_s \ell_s)$. Repeat the positive linear two-form comparison (PG19) in that frame, retain the quadratic Taylor error, and undo the same frame. The two transport factors are absorbed into the displayed exponential. Its principal coefficient tends to one as $K_s \ell_s$ tends to zero. There is no use of an associativity identity for either mean.

A joint bootstrap gives, for $M\ell_* + \alpha_0 \ell_*^2$ sufficiently small,

$$K_s \le C_L(M + \alpha_0 \ell_*), \qquad \alpha_s \le 2\alpha_0, \qquad \sum_s K_s \ell_s \le C_L(M\ell_* + \alpha_0 \ell_*^2). \tag{PG27}$$

Indeed, the first line of (PG26) adds at most $C_L \alpha_0 \sum_s \ell_s$ while $\alpha_s \le 2\alpha_0$ and $K_s \ell_s$ is small. The logarithm of the second line is bounded by $C_L \sum_s K_s \ell_s + C_L \sum_s \ell_s^2 \alpha_s$, and closes $\alpha_s \le 2\alpha_0$ after decreasing the common weak-field radius. Each one-step path has length $O_L(\ell_s)$, so the real contractions in (PG9) are replaced by factors $\exp(C_L K_s \ell_s)$. The coefficient of the inherited variation has the same factor. Their product is bounded by (PG27). Equations (PG15)–(PG18) thus extend to this one common complex physical chart, with bounded extra constants.

This proves a sufficient complex physical-family extension, not an assertion that an arbitrary independent-(U,J) pair is in that family. To apply it to a marked formation term one must identify a common fine physical chart (PG24) for the actual inner reconstruction and every allowed interpolation. The original bounded physical-potential response is relevant to that identification; no product of unrelated frame-equivalence constants may be charged once per depth. Bounded complex plaquette curvature alone, without a controlled comparison to a real field, is not the stated premise.

42.8.2 8.2 Fractional analytic hull under scalar interpolation

The failure of literal criticality under tH does not itself obstruct a fractional hull. In a fixed local continuum chart,

$$F(tA) = tF(A) + (t^2 - t)A \wedge A. \tag{PG28}$$

If A lies in $W^{1,\infty} \cap H^2$ and F(A) lies in H^{1+s}, $0 < s < 1/2$, then $A \wedge A$ lies in H^2 by the product rule. Thus F(tA) remains uniformly in H^{1+s} for bounded real or complex t.

There is also a direct lattice estimate, so no smooth continuum interpolation is needed. In a small physical potential chart, the logarithm of the four-link plaquette has the exact analytic form

$$F_h(A) = d_h A + [A_\mu, A_\nu] + h\,\mathcal{R}_h(A, \nabla_h A). \tag{PG28b}$$

The commutator is at one base site; changing a shifted quadratic argument to that site contributes h times a first difference and belongs to R_h. The BCH terms of degree at least three also have at least one factor h after division by h². On a fixed $W^{1,\infty}$ ball the convergent local analytic function R_h has

$$\|\mathcal{R}_h(A, \nabla_h A)\|_{H^1} \le C_M \|A\|_{H^2}. \tag{PG28c}$$

Differentiating its finite shifts uses only A, its bounded first differences, and its second differences in L^2. The mesh-frequency bound gives

$$h\|\mathcal{R}_h\|_{H^{1+s}} \le Ch^{1-s}\|\mathcal{R}_h\|_{H^1}.$$

Subtract $tF_h(A)$ from $F_h(tA)$ in (PG28b); the linear curl cancels, the remaining pointwise commutator lies in H^2, and the last displayed estimate bounds the remainder uniformly for $h \le 1$ and $s < 1$. This proves the same fractional-hull stability for scalar interpolation on the actual fine lattice. It does not identify the equation solved by an independently wall-modified inverse or by an inner re-minimization at another scale.

42.8.3 8.3 The old-scale normalization cannot be omitted

The terminal-variation bound after (PG18) has physical size $\ell_*^{-1}\|\lambda_k\|_1$. If the retained inner scale is an old scale r much smaller than the current macroscopic scale, replacing ell_* by r produces r^{-1}. It does not give a current-scale uniform H^s bound merely because the inner reconstruction is critical. In particular, an old dimensionless current of size $r^{2+\beta}$ corresponds, after the physical dimension-three current normalization, to size $r^{\beta-1}$. This is not uniformly bounded when $\beta < 1$.

There is an exact useful exception. Let U_k be the genuine minimizer of the original action subject to $M^k U = V$. For $j < k$, every field W with $M^j W = M^j U_k$ also satisfies $M^k W = V$. Hence U_k minimizes on that smaller constraint set. The uniqueness of the regular critical orbit gives

$$U_j(M^j U_k) = U_k \quad \text{up to the specified gauge.} \tag{PG28a}$$

For this actual nested readout one uses the k-scale multiplier and (PG18), so the spurious old-scale factor is absent. The conclusion does not cover an outer scalar-interpolated field, a separately wall-modified field, or a general outer datum. Those cases require uniform Sobolev stability of the **inner constrained reconstruction** on the chosen fractional analytic hull, or a direct compensated readout estimate. The flat Ritz projection has a promising H^{2+s} stability mechanism for s<1/2; its nonlinear and localized counterpart is a separate assertion, not a consequence of (PG28).

42.8.4 8.4 The protected-seed algebraic lemma

The companion local-residual note proves the following exact algebraic sufficient condition. Suppose local typed parametrices satisfy

$$\mathbb{L}P_z = I_z - T_z, \qquad \sum_z I_z = I, \qquad G = \sum_{n\ge 0} \sum_{z_0,\dots,z_n} P_{z_0} T_{z_1} \cdots T_{z_n}, \tag{PG29}$$

with absolute convergence. Attach each wall parameter once to the union of the complete typed carriers of a term. If every first carrier with a nonzero I_z output row in a protected region B lies in the seed where all wall parameters equal one, then

$$(\mathbb{L}G_s f)|_B = f|_B. \tag{PG30}$$

Applying L termwise yields positive I_z terms and negative T_z terms. On the protected rows, the first carrier adds no wall weight; summing I_z gives the identity, and each nonempty remaining string cancels the negative string with one fewer initial factor. This proves (PG30) without requiring each smooth partition cutoff to be locally constant. An augmented field/multiplier operator includes both Euler and constraint rows.

The actual carrier-once localization is not multiplicative: two separately localized factors meeting the same wall cube carry s^2, whereas their jointly localized product carries s. Consequently unlocalized identities between H, G, H_0 cannot simply be multiplied after independent localization. Matching (PG29) to every actual typed operator and to the successive mixed determining sets in R-II remains a separate estimate. The available II formulas display the initial-plus-typed-factor series and the carrier-once rule, but the full original parametrix identities referred to in the 1985 propagator paper require a separate proof The companion local-residual note records this distinction explicitly.

42.8.5 8.5 Source references and scope

The primary reference for the actual symmetric maps, their flat expansion, physical current, and paired domains is T. Bałaban, *Renormalization group approach to lattice gauge field theories. I*, Commun. Math. Phys. **109** (1987), 249–301, DOI 10.1007/BF01215223. Relevant locations are (0.4)–(0.12), (1.1), (1.8)–(1.18), (3.11)–(3.17), and Lemma 4/(3.53). Its printed pp. 285–286 explicitly use only Hölder control of first spatial derivatives in the old-readout renormalization argument; no all-order spatial regularity theorem is being imported from that discussion.

The one-step BV estimate is derived here from the displayed finite averaging maps, elementary tree-gauge/plaquette identities, and the actual Ward condition. Application to the full large-field and wall-localized formation machinery still requires the residual and analytic-domain checks stated above. These results narrow a concrete operator-regularity question; they do not complete the Yang–Mills construction.

Section 43 / A covariant variation bound and a protected-seed resolvent identity

This note records two different results. The first is a direct one-step estimate for the derivative of the actual contour/group averaging operation, with the physical gauge constraint retained. It gives a uniform bounded-variation bound for an iterated, unlocalized critical force. The second is an exact algebraic lemma for carrier-weighted parametrices. Matching the second lemma to every typed propagator in Bałaban's original localization remains a separate verification; the published low-regularity bounds do not establish that matching by themselves.

Section 43.1 / Physical normalization of the positive hold-and-line prolongation

Let the coarse spacing be ℓ, the fine spacing be $\eta = \ell/L$, with L fixed. Use the physical inner products

$$\langle A, J\rangle_\eta = \eta^d \sum_{z,\mu} \langle A_\mu(z), J_\mu(z)\rangle, \qquad \langle B, \lambda\rangle_\ell = \ell^d \sum_{y,\mu} \langle B_\mu(y), \lambda_\mu(y)\rangle. \tag{LR1}$$

Thus $U(z, z+\eta e_\mu)$ varies as $e^{i\eta t A_\mu(z)} U(z, z+\eta e_\mu)$, and a coarse link varies as $e^{i\ell t B_\mu(y)} V(y, y+\ell e_\mu)$. Write $B(y)$ for the L^d fine starting points assigned to the coarse cell centered at y. The flat box-and-line average is

$$(Q_{\mathrm{bl}} A)_\mu(y) = \frac{1}{L^{d+1}} \sum_{x \in B(y)} \sum_{i=0}^{L-1} A_\mu(x + i\eta e_\mu). \tag{LR2}$$

If $H\lambda$ is the block-constant hold, its adjoint is exactly

$$(T_0 \lambda)_\mu(z) = \frac{1}{L} \sum_{i=0}^{L-1} (H\lambda)_\mu(z - i\eta e_\mu). \tag{LR3}$$

The factor L^d from the two physical volume measures in (LR1) is essential for (LR3).

For the anisotropic variation

$$\mathrm{TV}_\ell(\lambda) = \ell^{d-1} \sum_{y,\mu,\nu} |\lambda_\mu(y + \ell e_\nu) - \lambda_\mu(y)|, \tag{LR4}$$

one has

$$\mathrm{TV}_\eta(H\lambda) = \mathrm{TV}_\ell(\lambda), \qquad \mathrm{TV}_\eta(T_0\lambda) \le \mathrm{TV}_\ell(\lambda). \tag{LR5}$$

Indeed one coarse interface is represented by exactly L^{d-1} fine interfaces, and $\eta^{d-1} L^{d-1} = \ell^{d-1}$. Each fine translation in (LR3) preserves variation, and their positive average contracts it. The same formulas give $\|T_0\lambda\|_\infty \le \|\lambda\|_\infty$ and $\|T_0\lambda\|_{1,\eta} \le \|\lambda\|_{1,\ell}$. These estimates have coefficient one, not a fixed constant raised to the number of scales.

Section 43.4 / Exact cancellation for a carrier-weighted parametrix

Here is the useful algebraic result, independent of positivity. Let $\mathcal{L} : E \to F$ be an operator on a finite lattice, possibly a block matrix on fields and constraint multipliers. Suppose local operators $P_z : F \to E$, $I_z, T_z : F \to F$ obey

$$\mathcal{L}P_z = I_z - T_z, \qquad \sum_z I_z = I_F. \tag{LR21}$$

Each triple is assigned one carrier C_z containing its kernels and background dependence. Assume the following series is absolutely convergent in the required low operator norms:

$$G = \sum_{n\geq 0} \sum_{z_0,\dots,z_n} P_{z_0} T_{z_1} \cdots T_{z_n}. \tag{LR22}$$

For parameters attached only to cubes outside a protected seed Ω, define

$$w(C) = \prod_{\Delta:\, \Delta\cap C\neq\varnothing} s_\Delta, \quad G_s = \sum_{n,\mathbf{z}} w(C_{z_0} \cup \cdots \cup C_{z_n}) P_{z_0} T_{z_1} \cdots T_{z_n}. \tag{LR23}$$

Each parameter occurs once per complete walk carrier, just as in II (1.6)–(1.10).

Let B be a set of output rows such that

$$I_z(x, y) \neq 0 \text{ for some } x \in B, y \quad \Longrightarrow \quad C_z \subset \Omega. \tag{LR24}$$

Then, on these rows,

$$(\mathcal{L}G_s f)|_B = f|_B. \tag{LR25}$$

To prove this, apply $\mathcal{L}$ termwise and use (LR21). The terms starting with $-T_{z_0}$ are indexed by all nonempty strings of T's. For a positive term starting with I_{z_0}, (LR24) implies

$$w(C_{z_0} \cup C_{z_1} \cup \cdots \cup C_{z_n}) = w(C_{z_1} \cup \cdots \cup C_{z_n})$$

on every nonzero row contribution. Summing I_{z_0} gives the identity. For $n \geq 1$ these terms cancel exactly the negative terms with a string one factor shorter. The $n = 0$ term is f. This proves (LR25), including for complex wall parameters and after any parameter differentiation permitted by absolute convergence.

This proof does **not** require every partition cutoff to be constant near B. For the usual parametrix $P_z = \chi_z G_{D_z} \eta_z$, the commutator $[\mathcal{L}, \chi_z] G_{D_z} \eta_z$ is included in T_z; overlapping cutoff derivatives are cancelled by the same telescoping argument. What matters is the complete first-carrier condition (LR24).

For a constrained background, the natural version of (LR21) is the augmented field/multiplier operator

$$\mathbb{L} = \begin{pmatrix} H & Q^* \\ Q & 0 \end{pmatrix}, \tag{LR26}$$

with its actual gauge normalization. If the local typed factors are entries of one expansion satisfying (LR21), the same proof gives the local Euler and constraint rows simultaneously. The multiplier forcing is then still present in the Euler row; it has not been converted into a constant or discarded.

Independent localization of already multiplied operators is not automatically this construction. For example, if two factors have carriers meeting the same external parameter cube,

$$w(C_1 \cup C_2) = s_\Delta, \qquad w(C_1)w(C_2) = s_\Delta^2. \tag{LR27}$$

Thus carrier-once localization is not an algebra homomorphism. An identity between unlocalized H, G, H_0 cannot be transferred by multiplying their separately localized versions without checking their shared typed factorization.

Section 43.5 / What the readable original construction verifies

Bałaban II (1.2)–(1.5) defines the nonlinear reconstruction through the three propagators H, G, H_0. Its (1.6) displays an initial local factor followed by typed local factors, and its wall multipliers are exactly carrier-once weights. The paper proves component separation and the original analytic/low regularity bounds. The algebraic identities (LR21) for those typed factors are referred back to the 1985 propagator paper, not displayed in II. A readable full copy of that older paper has not been obtained

R-II (1.61)–(1.65) performs three successive replacements of backgrounds and determining sets. The changed determining sets agree off own-scale boundary layers of width at most $2M$; the boundary gauge transform must be kept. Equation (1.65) establishes the original paired-domain inclusion. It is not a statement of high-jet or bounded-variation preservation.

There is also a distinction between intermediate arguments and the final old member. Scalar interpolation of a nonlinear critical perturbation need not itself be critical. However, I (3.25), (3.36) subsequently uses an inner reconstruction $U_j(\square_0, \exp(iB))$ with its own current. One cannot infer failure of this final critical readout merely from the nonstationarity of an earlier tH. Conversely, an old-scale critical reconstruction need not have a smooth forcing on one larger, current-scale ball: even flat sharp averaging introduces old-cell multiplier faces. This is why current-scale high spatial jets need a corrected-germ or integrated readout proof.

The precise remaining checks for the original wall construction are now:

1. identify the operators P_z, I_z, T_z, including all gauge and constraint blocks, in its actual typed expansion;
2. verify (LR24) for each of the successive R-II substitutions, at the stored scale used by that substitution;
3. keep the actual Ward-compatible multiplier in the Euler row and establish its local variation bound at mixed determining interfaces, or retain a compensating boundary residual with the required own-scale BV norm.

If these checks hold, (LR25) supplies a local equation and (LR19) supplies the nonlinear sharp-average force estimate on uniform pieces. Standard local fractional elliptic estimates would then give the H^{1+s} curvature hull needed by the dimension-six argument. No all-engineering-dimension invariant smooth-germ theorem follows from the present note.

Section 43.8 / Literal carrier weights and the projected-operator defect

The protected-seed lemma (LR21)–(LR27) assumes that the exact identity $\mathcal{L}P_z = I_z - T_z$ is filed on the **same complete carrier** for all three terms. This hypothesis is stronger than the statement that a propagator has a short-jump expansion. A nonlocal gauge projection creates a concrete reason to distinguish the two assertions.

43.8.1 8.1 An exact defect identity when the seed and residual carriers differ

Suppose

$$\mathcal{L}P_i = I_i - \sum_{\alpha} T_{i\alpha}, \qquad \sum_i I_i = I, \qquad T = \sum_{i,\alpha} T_{i\alpha}, \qquad \|T\| < 1. \tag{LR37}$$

Thus

$$G = \sum_{m\geq 0} \sum_{i,\omega} P_i T_\omega = P(I-T)^{-1}$$

is a right inverse of $\mathcal{L}$, and is its inverse when invertibility is assumed, as in the finite-dimensional example below. A word ω contains m residual factors, and the empty word is the identity. Let A_i be the seed carrier and $B_{i\alpha}$ the complete carrier of $T_{i\alpha}$. A residual may have $B_{i\alpha} \supsetneq A_i$, for example when a nonlocal projection in $\mathcal{L}$ reaches outside the local inverse domain.

For wall variables s_Δ, write

$$w_s(C) = \prod_{\Delta:\, \Delta\cap C\neq\varnothing} s_\Delta, \qquad G_s = \sum_{i,\omega} w_s(A_i \cup C_\omega) P_i T_\omega. \tag{LR38}$$

Each wall appears once in a complete carrier, irrespective of the number of visits. Assume absolute convergence of these weighted expansions. Let B be a protected output set, and suppose that whenever $1_B I_i \neq 0$, every wall meeting A_i is unweighted. Then the exact identity on protected rows is

$$\boxed{1_B(\mathcal{L}G_s - I) = \sum_{i,\alpha,\omega} \left[w_s(B_{i\alpha} \cup C_\omega) - w_s(A_i \cup C_\omega)\right] 1_B T_{i\alpha} T_\omega.} \tag{LR39}$$

To prove it, apply (LR37) to the first factor in every term of (LR38). The I_i terms sum, on B, to $\sum_\omega w_s(C_\omega)T_\omega$. Separate its empty word and reindex every nonempty word by its first residual factor. Subtract the $T_{i\alpha}$ terms obtained directly from (LR37). This gives (LR39), with no inequality or rearrangement beyond absolute convergence.

If $A_i = B_{i\alpha}$, the coefficient vanishes and this is exactly the previous telescope. It also vanishes when the difference of the two carriers meets only unweighted walls. In general, a long residual can add a weighted bridge between its seed and its protected output. Decomposing that bridge into short typed factors does not by itself remove the difference in (LR39).

For an unaugmented operator containing a genuinely nonlocal projection, a compact seed can have $\mathcal{L}P_i$ tails at arbitrarily distant outputs. In that case simply enlarging the seed to a common carrier would require an unbounded carrier. Resolving the tails into finite words instead produces the separate $B_{i\alpha}$ in (LR39). A local augmented system is a possible remedy, but its full block expansion must then be matched to the weakened operators actually used; the change is not justified by carrier enlargement alone.

43.8.2 8.2 A three-site calculation

This finite-dimensional example checks the distinction without making any assertion about the Wilson propagator. On three sites arranged as $1, 2, 3$, let

$$\mathcal{L} = I + \varepsilon(E_{13} + E_{31}), \qquad 0 < \varepsilon < 1, \qquad P_i = I_i = E_{ii}. \tag{LR40}$$

Then $T_1 = -\varepsilon E_{31}$, $T_3 = -\varepsilon E_{13}$, and $T_2 = 0$. Take the seed carriers $A_i = \{i\}$, and file every nonzero long residual on the complete connecting carrier $\{1, 2, 3\}$. Protect site 1; give sites $2, 3$ wall parameters s_2, s_3.

Every nonempty word has the same complete carrier, so the literal carrier-once weakening is

$$G_s = E_{11} + s_2 E_{22} + s_3 E_{33} + s_2 s_3 (G - I). \tag{LR41}$$

Since $\mathcal{L}(G - I) = I - \mathcal{L}$, direct multiplication gives

$$(\mathcal{L}G_s - I)_{13} = \varepsilon s_3(1 - s_2). \tag{LR42}$$

All seed carriers contributing to I_i on the protected row are unweighted. Nevertheless that row is not an exact inverse row. One may factor $E_{13} = E_{12}E_{23}$ and $E_{31} = E_{32}E_{21}$, thereby resolving every long residual into nearest-neighbor steps. The complete union carrier and (LR42) are unchanged.

If a convention leaves every zero-jump seed unweighted, the same example instead gives $G_s = I + s_2 s_3 (G - I)$ and $(\mathcal{L}G_s - I)_{13} = \varepsilon(1 - s_2 s_3)$. Thus the issue is not removed by that zero-jump convention.

The example does not say that the original non-Abelian construction has this residual. It says exactly that a short-jump expansion plus a protected local seed is insufficient to prove the same-carrier hypothesis. In a gauge problem, some resulting defects may be gauge directions or auxiliary scalar equations. Their scope can be decided only after the scalar companions and the physical constraint equations are retained.

43.8.3 8.3 What an exponential estimate would and would not provide

If a nonzero coefficient in (LR39) requires a complete residual carrier to join B to a weighted wall a distance R away, a tree-decay estimate can pay e^{-cR}. More precisely, suppose the same residual-word expansion has an absolutely summable majorant in an operator norm $\mathcal{X} \to \mathcal{Y}(B)$, retaining a spare factor $e^{-c\,\mathrm{length}(C)}$. Then

$$\|1_B(\mathcal{L}G_s - I)\|_{\mathcal{X}\to\mathcal{Y}(B)} \le Ce^{-c'R}. \tag{LR43}$$

For complex wall discs, their factors are paid from the same spare carrier exponent before the estimate is made. This is a consequence of (LR39) **in the norm actually supplied by the residual majorant**. It is not a way to upgrade that norm.

The distinction matters for the present fractional target. On a periodic chart, take one transverse component

$$A_N(x) = \varepsilon N^{-2}\sin(Nx_1).$$

For every fixed $\beta < 1$, its supremum, first-difference supremum, C^β seminorm of the first difference, and Laplacian supremum are all $O(\varepsilon)$, uniformly when the fine mesh resolves N. But

$$\|\Delta A_N\|_{H^s} \asymp \varepsilon N^s. \tag{LR44}$$

Choose, for example, a fine mesh much smaller than N^{-1} so the lattice symbols are comparable to their continuum values. A fixed $\varepsilon = e^{-cR}$ does not remove this divergence. Thus exponentially small errors in the old low norms, even supplemented by a Laplacian supremum, do not establish a cutoff-uniform H^s Euler residual. This is a norm counterexample, not an example of an actual localized Wilson field.

There is a positive flat comparison that identifies the extra smoothing one would need. At one homogeneous retained scale r, let

$$P_r = GS^*(SG^2S^*)^{-1}SG, \qquad G = \Delta^{-1},$$

be the smoothed scalar complement of the Landau projection. On mean-zero modes and for $0 \le s \le 1$,

$$\|dP_r\delta\|_{L^2\to H^s} \le C_s r^{-2-s}, \qquad 0 < r \le 1. \tag{LR45}$$

This estimate is uniform in the fine cutoff. On an alias fibre, put $C_2 = \sum_\nu |q_\nu|^2\lambda_\nu^{-2}$ and $v_\nu = d(p_\nu)\bar{q}_\nu\lambda_\nu^{-1}$. The operator is vv^*/C_2. The box identities give

$$\begin{aligned} \|v\|_2^2 &\le \lambda_0^{-1}, \\ \|v\|_{H^s}^2 &\le \lambda_0^{-1}(1+\lambda_0)^s, \\ C_2 &\ge c\lambda_0^{-2}. \end{aligned}$$

Hence its fibre norm is at most $C\lambda_0(1+\lambda_0)^{s/2} \le C_s r^{-2-s}$. The zero coarse mode contributes zero after the outside gradients. This proves (LR45).

Equation (LR45) is a real smoothing statement about the complete projection kernel. It is stronger than a low-core supremum estimate. It also shows the retained-scale price explicitly. To use it for an old layer in current units, either the complete layer/kernel sum must be performed, or an actual separation factor such as $r^{-2-s}e^{-cR/r}$ must be retained. The old spacing cannot be renamed as the current spacing. No non-Abelian or mixed-boundary version of (LR45) follows just from this flat calculation.

43.8.4 8.4 The literal the analysis

The following are the verifiable statements from the readable non-Abelian papers.

- Balaban II, CMP **116** (1988), (1.2)–(1.5), defines the nonlinear map using H, G, H_0 and permits replacing these operators by ones with the specified old bounds. This supplies the analytic weak-domain fixed point.
- Its (1.6)–(1.7) imports typed random-walk factors and their kernel/locality bounds from the background-propagator paper (3.107)–(3.108). Its wall rule weights each complete union carrier once.
- Equations (1.12)–(1.18) perform that replacement separately in the three operators and in the nonlinear linearizing change of variables. They establish the bounds and component localization used by the ordinary cluster construction.
- The physical equivalence of the two variational representations uses the gauge argument in the 1985 variational paper, (123)–(143), as recorded in Section 7 above.

None of these displayed statements identifies the localized triples as the blocks of one inverse for a physical augmented KKT operator. In particular, the fact that each factor vanishes near its own carrier boundary is not an identity equating the seed carrier with the carrier of its image under a nonlocal projected operator.

Dimock's primary Abelian paper, arXiv:1712.10029, provides a useful structural check. In Section 3.7.1, (277)–(280), its projected operator contains the nonlocal term $-dP\delta$; the commutator of the local parametrix has long tails from that term. Section 3.7.4, (310)–(314), inserts the preliminary projection expansion to produce short typed jumps and then applies complete-carrier wall weights, with its stated exception leaving zero-jump seeds unweighted. This confirms why the issue in (LR39) must be examined. It does not verify the non-Abelian typed identities or the nonlinear physical Ward equation, and it is not used here to assert that they fail.

The contents needed for the final identification are the actual background-propagator types in CMP **99** (1985), (3.107), including their scalar and constraint blocks.

The precise remaining alternatives are:

1. Prove that the published complete expansion is the block expansion of a **single** augmented operator with local protected rows and matching first-carrier identities. The previous protected-seed telescope would then give the local equations.
2. Compute its actual residual using an identity of the form (LR39), retain its scalar/constraint companions, and prove the required current-scale H^s bound using the actual kernel smoothing and complete layer geometry.

An arbitrary separately prescribed Ward defect can restore the endpoint terms, so the original Ward-restricted estimate cannot be applied to it. The later compatible-residual theorem CR supplies a stronger alternative for the actual raw hierarchy: when

$$E = J_{\mathrm{Wilson}}(u) - (D\mathcal{M}_r(U))^{\mathsf{t}}\lambda \quad \text{belongs to } H^s,$$

the exact gauge identity gives $d_U^{\mathsf{t}} E = -N^{\mathsf{t}} d_V^{\mathsf{t}} \lambda$. Its atomic point-embedding estimate controls the restored endpoint current. Thus an actual H^s Euler residual no longer needs a separate exact coarse Ward hypothesis. A low-norm error or an unrelated prescribed divergence error still does not supply that premise.

The later weak-chart theorem WC also constructs a small representative for the genuine real periodic

regular critical orbit. It does not construct a common complex wall chart. These improvements reduce the remaining localization obligation to the actual protected equations or their compatible H^s residual, together with the common auxiliary field domain. They do not identify the original separately weakened maps with that domain. The following augmented-system work investigates a replacement interpolation rather than asserting that identification.

Section 44 / A fractional interpolation hull and uniform flat reconstruction

This chapter proves two estimates relevant to the new dimension-six route. The first is that fractional curvature regularity survives the scalar and smooth-cutoff interpolations that destroy literal stationarity. The second is a scale-uniform Sobolev bound for the actual flat box/line constrained reconstruction. It also gives a valid telescoping comparison for nonlinear physical currents, using only flat propagation of the error terms. These facts do not establish the corresponding nonlinear, wall-localized reconstruction bound. The last section states that remaining task with its physical scale visible.

Section 45 / The centered reference-log mean has a quadratic Sobolev defect

This note proves an estimate for the ACTUAL centered one-step average, with one fixed odd blocking factor L. It includes the reference holonomy in Balaban I (0.4)/(0.12), and allows the contour means in (0.11). The outer operation is not replaced by a Karcher mean.

The result is a bound for coarsening a field in a specified small potential chart. A fixed explicit coarse gauge change removes the flat connector. In that analysis gauge, the nonlinear defect is $O(\ell^2)$ in a coarse fractional Sobolev norm and remains so after composition through arbitrarily many fixed-L steps. This does NOT yet prove stability of the nonlinear constrained minimizer: passing from the coarsening estimate to the inverse constraint problem, with its fine gauge and original target orbit, remains an additional step.

Section 46 / Exact gauge compatibility and the actual transverse averaging defect

The centered-mean chapter proves a uniform coarsening estimate after an explicit coarse gauge change. This chapter proves that the gauge change can be accounted for in the original averaging map itself. The key estimate concerns the node factors sampled on a much coarser lattice. In four dimensions, their zero constant moment supplies enough decay to sum all intermediate scales in a fractional Sobolev norm.

The resulting theorem is about the actual nonlinear averaging constraint, on a common small potential chart. It does not assume a smooth fine extension of a blockwise gauge transformation. It also does not establish the nonlinear constrained minimizer's regularity or identify separately wall-localized auxiliary maps with genuine minimizers.

Section 46.6 / Precise scope for the nonlinear reconstruction problem

The remaining reconstruction problem is not being hidden in (GQ26). Its right side contains the norm of the unknown minimizer u. To close that norm one still needs the actual Euler equation, its multiplier estimate, and the exact chosen fine gauge condition. In particular:

- A $W^{1,\infty}$ bound alone gives an H^1 bound on a fixed finite volume, not an H^{1+s} bound.
- A bound for the transverse part of u does not control an arbitrary longitudinal pure-gauge part. The original projected or background Landau condition must be used as stated, or a uniformly controlled gauge section must be constructed.
- The variational paper's adapted $Q_j(U_0, \eta A)$ includes gauge and coarse-coordinate transformations. Its physical orbit can be matched by the identities above, but its written gauge equation and analytic domains cannot be replaced silently by a raw formula.
- Mixed determining sets require the corresponding collars and extensions. Independently localized H, G, H_0 require the complete nonlinear compatibility checked in the local-residual chapter. The present proof does not identify those auxiliary maps with the genuine averaging chain.

For a common complex family satisfying the displayed strong norms, all finite maps in the proof are holomorphic and the resulting strong-norm estimates permit ordinary Cauchy estimates. Analyticity only in a weaker paired norm does not provide that common strong-norm premise.

Section 47 / An exact smooth right inverse of site block averaging

This note proves a mesh-uniform linear gauge-parameter lift. It replaces the sharp block hold by a smoothed extension while preserving the prescribed site average exactly. The construction works on compatible periodic lattices, commutes with coarse translations and the centered lattice symmetries, and has an exponentially decaying kernel in coarse units.

The theorem is about the flat site box average. It is not, by itself, a right inverse of the full nonlinear gauge cocycle of the actual reconstructed field. Section 7 states that remaining distinction explicitly.

Section 48 / A nonlinear periodic Wilson reconstruction estimate from the actual KKT equations

This chapter proves a current and curvature regularity estimate for the actual nonlinear, iterated Wilson constraint on one fixed physical torus. It uses the actual centered reference-log averages, their exact Ward equation, and the transverse constraint estimate proved in the gauge-compatibility chapter. It does not replace the constraint by a linear average or change the fine field by a discontinuous gauge transformation.

The admitted low-regularity premise is stated explicitly: the reconstructed field is already on the regular branch with a uniformly small physical $W^{1,\infty}$ norm and a bounded H^{1+s} norm. A current-scale $C^{1,\beta}$ bound would supply the latter for $s < \beta$ on a fixed physical volume. Its applicability to an old inner reconstruction is a separate original-domain check; an outer-field estimate does not supply that check by itself. The conclusion is the new derivative of curvature and the current's fractional regularity. No H^{1+s} bound for the reconstructed curvature is assumed.

The result concerns genuine minimizers, or genuine stationary points, of the constrained Wilson action. A separately wall-localized auxiliary background must first be shown to obey the corresponding equations, including its actual forcing and boundary terms. That interface remains separate.

Section 48.8 / Analytic parameters, scope, and the remaining original-map interface

At any finite cutoff, all identities above are analytic on the stated regular complex branch. Suppose a parameter family of actual constrained backgrounds $u(z)$ and inputs $A(z)$ is holomorphic there and satisfies the common bounds (NK4), (NK7), and an input curvature bound on one fixed parameter domain. Equation (NK6) gives a uniform strong-norm bound on that same domain. Ordinary Cauchy estimates then control parameter derivatives in the strong norms, with only the stated parameter-radius costs.

This does not deduce a common strong domain from analyticity in an unrelated weaker norm. The original regular-branch estimates and the actual equation must hold throughout the parameter domain.

The theorem supplies the nonlinear current/curvature estimate for the genuine periodic constrained reconstruction. Its limitations are specific:

- the weak radius can depend on the fixed physical torus size T;
- collars, mixed determining sets and boundary gauge data require their own actual equations and uniform extension estimates;
- a wall interpolation of separately localized H, G, H_0 is not automatically a genuine minimizer;
- the full dimension-six tensor center, its normalization and all operator mixing are not proved merely by this regularity gain.

The new estimate removes the negative-power retained-spacing obstruction for the genuine nonlinear periodic constraint, using its actual transverse defect and Ward equation. It leaves the original localized preparation/formation maps to be matched explicitly.

Section 48.9 / Dependencies

The actual finite mean, exact coarse gauge correction, and their uniform coarsening estimates are proved in the stated results and the stated results The actual physical-current comparison and the all-depth flat Schur bounds are proved in the stated results and the stated results The exact plaquette-variation incidence and flux conventions are written in the stated results D6.22–33.

The source-free low regularity and analyticity premises must be read in their original domains; this chapter does not enlarge those domains by assertion.

Section 50 / Finite-integrability potential charts for the nonlinear reconstruction estimate

The periodic KKT argument was first proved with a small $W^{1,\infty}$ potential. A norm-preserving nonlinear gauge choice is easier in $W^{2,p}$ for its gauge parameter, with finite $p > 4$, than in $W^{2,\infty}$. This chapter proves the corresponding finite-integrability versions of the actual averaging and KKT estimates. The physical curvature supremum is retained as a separate hypothesis. A $W^{1,p}$ potential does not by itself have bounded curvature.

All statements concern one fixed four-dimensional physical torus, nested periodic lattices, a fixed odd blocking factor, and the actual centered reference-log averages used in the preceding chapters.

Section 50 / Finite-integrability potential charts for the nonlinear reconstruction estimate

Constants may depend on the torus and on the displayed exponents, but not on either lattice spacing or the number of averages. No claim about mixed determining sets or wall-localized auxiliary fields is made here.

Section 50.2 / Mathematical scope

Let the current one-step spacings be a and $\ell = La$. The exact finite analytic expansion CM6, the quadratic constant-field cancellation CM7, and the algebraic subtraction CM8 are unchanged. Every quadratic term after subtraction contains a fine spatial difference of A over distance $O_L(\ell)$.

For $M = M_p(A)$ and $K = K_s(A)$, fixed-offset lattice sampling has its usual fixed-L bound in each of the norms used below. Telescoping the difference and applying (WP3)–(WP5) gives

$$\|N_\ell(A)\|_{L^p} \le C_{L,p,T}\ell^2(M^2 + M^3), \qquad \|N_\ell(A)\|_{H^s} \le C_{s,L,p,T}\ell^2(M + M^2)K. \tag{WP8}$$

For the second estimate the term in which the fractional difference falls on the undifferentiated factor is bounded by $\ell\|\nabla A\|_{L^p}\|A\|_{\mathcal{B}^s_{q,2}}$. The other term uses $\ell\|A\|_\infty\|\nabla A\|_{H^s}$. The extra ℓ in the quadratic coefficient gives the displayed ℓ^2. Terms of field degree at least three already carry ℓ^2; their convergent analytic series is bounded by the same estimates.

The explicit coarse straightening is an analytic local map with the same quadratic constant-field cancellation. Thus (WP8) holds for its defect as well. A single coarse inverse difference gives

$$\|N^{\mathrm{str}}_\ell(A)\|_{W^{1,p}} \le C\ell(M^2 + M^3), \qquad \|N^{\mathrm{str}}_\ell(A)\|_{H^{1+s}} \le C\ell(M + M^2)K. \tag{WP9}$$

The flat box-and-line average contracts L^p, its first-difference norm, and all the positive-order Bessel norms in question. Iterating (WP9) therefore gives, on one sufficiently small initial chart,

$$\sup_{j\le k} M_p(A_j) \le 2M_p(A_0), \qquad \sup_{j\le k} K_s(A_j) \le e^{C(M+M^2)r}K_s(A_0), \quad r = L^k a \le 1. \tag{WP10}$$

The argument sums $\sum_j \ell_j \le C_L r$; it does not multiply a fixed one-step operator loss over the microscopic depth. Unrolling only the flat tails gives the same r^2 weak defect and r strong defect as before.

Section 51 / Closing the low fractional norm on the identity-background projected Landau slice

This chapter removes the unknown H^{1+s} low norm from the nonlinear KKT estimate on one precisely specified gauge slice. It uses the proved finite-lattice mixed centered-sampling/biharmonic lift and the actual raw averaging constraint. The normalizing scalar operator is not replaced by a positive QGQ^* operator: its two outer maps are genuinely different.

The theorem assumes that the physical field already satisfies the displayed identity-background projected Landau condition and the physical KKT equation. It does not identify an arbitrary published background-dependent $R(U_0)$ with this projector, and it does not yet supply a norm-compatible nonlinear gauge transformation into the slice.

Section 51.2 / The mixed biharmonic lift already proved in LF

The exact finite-lattice heat-alias proof is in the stated results, LF6–18, and is not repeated here. With the notation of this chapter, that theorem applies to the actual distinct maps R and S in (PL5). On every nonzero coarse Fourier fibre,

$$q_0\lambda_0^{-2} \le c_{\mathrm{mix}}(\theta) \le 16q_0\lambda_0^{-2}, \qquad q_0 \ge (2/\pi)^4. \tag{PL14}$$

Here q_0 is the principal centered box alias and λ_0 the principal fine Laplacian symbol. The lower bound follows by pairing the signed aliases in each coordinate of the actual finite-lattice heat kernel. It is not an inference from a positive squared-alias Schur operator.

Consequently the exact lift

$$E_{\mathrm{mix}} = G^2R^*C_{\mathrm{mix}}^{-1}, \qquad SE_{\mathrm{mix}} = I \tag{PL15}$$

is well-defined on mean-zero coarse scalar fields. LF16 and LF18 give

$$\|E_{\mathrm{mix}}\psi\|_{H_a^t} \le C_t\|\psi\|_{H_r^t}, \qquad 0 \le t \le 4, \tag{PL16}$$

and

$$\boxed{\|d_aE_{\mathrm{mix}}\psi\|_{H_a^{1+s}} \le C_s\|d_r\psi\|_{H_r^{1+s}}, \qquad 0 < s < 2.} \tag{PL17}$$

These constants are independent of both lattice spacings, their ratio and the compatible torus volume. The gradient estimate does not use a Poincare constant. A constant scalar gauge parameter is fixed separately: G removes it, and it changes no flat one-form.

Equations (PL4)–(PL5) now imply the exact relation

$$\phi = E_{\mathrm{mix}}(S\phi). \tag{PL18}$$

This is the entire scalar input needed for the nonlinear slice closure below.

Section 51.5 / The finite-p slice corollary

The same closure works in the potential chart proved in the stated results, WP1–21. Fix

$$p > 8, \qquad 0 < s < \tfrac{1}{2}, \qquad s < 1 - \frac{4}{p}.$$

Replace the $W^{1,\infty}$ hypothesis of Theorem PL by

$$\|u\|_{W_a^{1,p}} + \|A\|_{W_a^{1,p}} + \|F_a(u)\|_\infty + \|F_a(A)\|_\infty \le M \le M_*(s,p,L,T). \tag{PL29}$$

The curvature supremum is a separate premise. In dimension four a $W^{1,p}$ potential, even for $p > 8$, does not imply bounded curvature. Keep the same genuine raw constraint, the same physical multiplier/constraint-qualification premise and the exact slice (PL2).

Corollary PL-p. Under these assumptions,

$$\boxed{\|u\|_{H^{1+s}_a} + \|\lambda\|_{H^s_r} + \|J_a(u)\|_{H^s_a} + \|F_a(u)\|_{H^{1+s}_a} \le C_{s,p,L,T}(\|A\|_{H^{1+s}_a} + \|F_a(A)\|_{H^{1+s}_a}).} \tag{PL30}$$

The constants are uniform in the fine cutoff and the retained spacing.

Proof. The flat projector and mixed biharmonic lift do not involve the low potential norm, so (PL17)–(PL19) remain exact. WP18 gives the full strong raw remainder

$$\|N_{\mathrm{raw}}(w)\|_{H^{1+s}_r} \le Cr(M + M^2)\|w\|_{H^{1+s}_a}, \tag{PL31}$$

and WP21 supplies (PL23) with $C_{s,p,L,T}$. Its mean-zero fine Euler equation also supplies (PL24) with this constant. No unknown high norm is inserted into the choice of the small radius.

The longitudinal argument (PL25)–(PL26) uses only the strong raw estimate (PL31), the uniform linear raw map and (PL17); it is unchanged. For the harmonic one-form, the exact equality of means (PL21) yields directly

$$\|u_H - A_H\|_2 \le Cr(M + M^2)(K_A + K_u). \tag{PL32}$$

This bound is sufficient and does not need the stronger r^2 weak remainder. Thus the complete low norm obeys

$$K_u \le C_{s,p,L,T}\mathcal{I}_s(A) + C_{s,p,L,T}(M + M^2)K_u. \tag{PL33}$$

Absorb the last term by the single fixed choice of M_*, then apply WP21. This proves (PL30).

This corollary does not change the gauge condition. Its use after a nonlinear $W^{2,p}$ site gauge requires that the gauge fix the actual retained centers, place the field in (PL2), and preserve the small $W^{1,p}$ potential and bounded-curvature chart. WP22–23 prove the conditional transfer of the current, curvature and raw target once such a gauge is available; constructing it is a separate theorem.

Section 51.6 / Exact scope of this result

The proof closes a genuine nonlinear retained-spacing estimate on the specified identity-background slice. It uses the original raw constraint and its true flat derivative, not a constraint with block means substituted for center samples.

The following tasks are still different mathematical statements:

- showing that an arbitrary regular minimizer can be put into this slice while preserving the needed physical low norms and the same raw target;
- replacing $\mathscr{R}_0$ by the precise background-dependent $R(U_0)$ on the full original prepared domain;
- extending the theorem to the actual mixed determining sets, wall-localized equations and boundary terms used in formation;
- extracting and normalizing the complete dimension-six operator family.

The finite-lattice formulas and the estimate are holomorphic on any common complex parameter domain where the stated low chart, actual KKT equations and projected slice hold. Thus strong-norm Cauchy estimates are available on such an admitted domain. That analytic premise is not inferred from gauge-orbit equivalence alone.

Section 51.7 / Relation to the surrounding notes

The nonlinear force and KKT bounds are NK1–34 of the stated results. The complete strong raw remainder is CM26–36 of the stated results. The exact gauge accounting and raw chart

Section 51.7 / Relation to the surrounding notes

bound are GQ1–26 of the stated results. The projector in (PL1) is stated here as an exact hypothesis; its appearance in the original background-gauge construction is being checked separately against the primary definitions.

Section 53 / The assembled periodic nonlinear curvature theorem

This chapter combines the actual averaging estimates, the Wilson KKT argument, the mixed biharmonic lift, and the controlled gauge section. The resulting a priori theorem no longer assumes a uniform high norm of the unknown constrained field. It is a theorem about genuine periodic constrained Wilson fields on a specified small physical chart. It is not an existence theorem for the continuum quantum theory.

The distinction from the earlier conditional KKT estimate is precise. Its term $(M + M^2)\|u\|_{H^{1+s}}$ is now absorbed after an exactly constraint-preserving gauge transformation. Curvature and the physical Euler current are then transferred back to the original representative. No high potential norm in that original representative is claimed.

Section 53.5 / What remains outside this theorem

The proof has removed the unknown-output-norm assumption from the genuine periodic small-chart reconstruction estimate. Three other requirements of the attempted Yang–Mills construction remain separate.

First, the original formation and large-field history maps use mixed determining sets, buffers, boundary gauge data, and independently weakened propagators. An auxiliary output can be treated by (PR4) only after its exact Euler equation, constraint and Ward identity have been matched to the stated ones, or after a proved local/residual extension replaces them. The carrier-weighting counterexample and transformed residual in the local-residual chapter remain relevant.

Second, a reconstruction estimate for a fixed family of curvature expressions does not define all renormalized local operators. The local interaction terms near a completed insertion, their subtractions, lower-dimensional mixing, and the compatibility of all tensor families are still construction estimates.

Third, the radii here may depend on the fixed physical torus. This chapter does not supply an infinite-volume limit, continuous tensor covariance, vacuum selection for the full operator algebra, or the identification of all interacting Yang–Mills relations.

These are not assumptions being silently added to Theorem PR. They are the reasons that the assembled periodic theorem is an auxiliary result within an incomplete continuum construction.

Section 54 / Stability of the periodic estimate under physical residuals

Theorem PR is an a priori statement for exact constrained critical fields. A localization argument has to identify those equations, or estimate its defects in the norms used by that theorem. This chapter proves a residual version under an exact coarse Ward identity. It also shows explicitly when a fine old-layer projection error can be summed without a microscopic cutoff loss. Neither statement identifies the actual independently weakened background maps with its hypotheses.

Section 54.5 / Scope of the extension

The residual estimate makes the next localization obligation explicit: a common small physical chart, a constraint defect measured by (RS2), and a physical Euler defect measured in H^s, with exact coarse Ward or a separately proved replacement. Exponentially small errors in a weaker norm do not satisfy that obligation. Conversely, a verified residual expansion with the stated smoothing and physical separation would suffice; exact coincidence with an unweakened constrained minimizer would not be necessary.

This chapter proves that implication and its scale bookkeeping. It does not claim that the published separately weakened H, G, H_0 already satisfy the hypotheses, and it does not settle the near-insertion operator estimates or the infinite-volume quantum construction.

Section 55 / Compatible Euler residuals control the raw endpoint current

The periodic reconstruction theorem originally used an exact coarse Ward equation. This chapter removes that separate hypothesis when the forcing error is an actual Euler residual. Gauge invariance then determines its divergence. The divergence is supported at the retained centers, with the physical point-restriction normalization. That atomic structure controls the potentially singular endpoint current.

The result concerns the actual composition of fixed-L centered raw averages. A single direct contour map with an arbitrarily increasing block factor has different thin-path estimates and is not silently substituted. The nonlinear proof below uses the exact full-space endpoint decomposition at each fixed-L step, not a Ward-restricted estimate on an inadmissible current.

Section 55.8 / Scope and use in localization

This theorem weakens a concrete requirement on a localization residual: exact coarse Ward is not needed separately if the residual is the actual difference between the Wilson Euler force and the genuine raw constraint-adjoint force. Its exact divergence compatibility then supplies the missing estimate. An arbitrary independently prescribed q cannot be inserted without paying the atomic norm (CR19).

The theorem does not establish the H^s norm of the residual for any particular wall-weakened propagator. It also does not infer current-scale weak chart bounds from old-layer bounds. Applying it to the original formation maps still requires their actual residual, target equation, determining regions, and boundary terms to be identified.

All arguments use physical currents and the actual raw hierarchy. Replacing the raw derivative by an unrelated adapted operator, or treating the nonlinear endpoint current as though it satisfied the exact Ward equation, would remove the identities that make the proof work.

The next lemma is also independent of the constraint and stationarity. It avoids assuming the derivative norm that it will prove.

Lemma WC-B. Fix $4 < p < \infty$ and T. Let $U = e^{au_0}$ have sufficiently small $\|u_0\|_{L^p_a}$, and suppose the plaquette logarithm $F_a(U)$ is defined and has small L^p_a norm. Then there is a periodic gauge e^{ϕ}, with scalar mean $\bar{\phi} = 0$, such that $U^{e^{\phi}} = e^{av}$ and

$$\delta_a v = 0, \qquad \|v\|_{W^{1,p}_a} \le C_{p,T}\big(\|u_0\|_{L^p_a} + \|F_a(U)\|_{L^p_a}\big). \tag{WC16}$$

Here smallness includes the common link logarithm branch. For $a \le 1$ that branch already follows from a sufficiently small L^p bound, since $p > 4$.

56.4.1 The gauge fixed point in $W^{1,p}$

Set $\Delta_a = \delta_a d_a$, the nonnegative scalar lattice Laplacian, and let $G_a = (\delta_a d_a)^{-1}$ on mean-zero scalar fields, with G_a zero on constants. The periodic discrete Calderón–Zygmund estimate gives

$$G_a \delta_a : L^p_a \longrightarrow W^{1,p}_a \tag{WC17}$$

with a mesh-independent constant. The actual gauge map is

$$a^{-1} \log\big(e^{\phi(x)} e^{au_{0,\mu}(x)} e^{-\phi(x+ae_\mu)}\big) = u_0 - d_a\phi + \mathcal{N}_a(u_0, \phi). \tag{WC18}$$

For $\phi \in W^{1,p}$, first insert $\phi(x + ae_\mu) - \phi(x) = aD_{a,\mu}\phi(x)$. Uniform discrete Sobolev embedding and the elementary point inverse bound give

$$\|\phi\|_\infty \le C_{p,T}\|\phi\|_{W^{1,p}}, \qquad a\|D_a\phi\|_\infty \le a^{1-4/p}\|D_a\phi\|_{L^p}. \tag{WC19}$$

Similarly $a\|u_0\|_\infty \le a^{1-4/p}\|u_0\|_{L^p}$. Thus all dimensionless BCH arguments are small on one $W^{1,p}$ ball, uniformly in $a \le 1$. Every remainder monomial contains a gauge factor and at least two factors in total. Any product with two physical difference factors and no undifferentiated ϕ carries an explicit factor a. Pointwise multiplication and (WC19) give

$$\|\mathcal{N}_a(u_0, \phi)\|_{L^p} \le C_{p,T}\big(\|u_0\|_{L^p} + \|\phi\|_{W^{1,p}}\big)\|\phi\|_{W^{1,p}}, \tag{WC20}$$

and its Lipschitz version with the same small coefficient. For example the potentially problematic quadratic term obeys

$$a\|(D_a\phi)^2\|_{L^p} \le a^{1-4/p}\|D_a\phi\|^2_{L^p}.$$

There is no $W^{1,p}$ norm of u_0 in this estimate.

Solve

$$\phi = G_a\delta_a\big(u_0 + \mathcal{N}_a(u_0, \phi)\big) \tag{WC21}$$

by contraction on the mean-zero $W^{1,p}$ ball. It gives $\|\phi\|_{W^{1,p}} + \|v\|_{L^p} \le C\|u_0\|_{L^p}$ and the exact equation $\delta_a v = 0$. The iteration stays in the real Lie algebra. At this stage no bound for ∇v has been used or obtained.

56.4.2 Absorbing the unknown derivative norm

At each finite cutoff every norm of v is finite. The inverse bound gives $a\|v\|_\infty \le a^{1-4/p}\|v\|_{L^p}$, so the same small BCH branch is valid. The exact plaquette expansion has the form

$$F_a(v) = d_a v + \mathcal{B}_a(v), \qquad |\mathcal{B}_a(v)(p)| \le C \sum_{b,b' \in \partial p} |v_b||v_{b'}|. \tag{WC22}$$

All degree-n terms, $n \ge 2$, have coefficient a^{n-2}, and the dimensionless link bound sums the higher terms into the displayed quadratic majorant. Finite shifts are retained. Thus

$$\|\mathcal{B}_a(v)\|_{L^p} \le C\|v\|_\infty \|v\|_{L^p} \le C_{p,T}\|v\|_{W^{1,p}}\|v\|_{L^p}. \tag{WC23}$$

Uniform flat Hodge estimates, $\delta_a v = 0$, and gauge covariance of the real curvature give

$$\|v\|_{W^{1,p}} \le C_{p,T}\big(\|F_a(U)\|_{L^p} + \|v\|_{L^p}\big) + C_{p,T}\|v\|_{L^p}\|v\|_{W^{1,p}}. \tag{WC24}$$

Choose the original L^p radius so the last coefficient is below one half. Absorption proves (WC16). Harmonic one-forms are covered by the retained L^p term, not inverted away. This proof establishes its own high norm after the low-norm gauge construction; it does not use an a priori $W^{1,p}$ bound for the output.

Section 56.7 / Limits of the conclusion

The three inputs performing different jobs should remain distinct.

1. The original **regular critical-orbit** theorem, rather than energy comparison alone, supplies the scale-improved curvature supremum (WC5).
2. The prescribed raw coarse field supplies the small global holonomies needed by the periodic patch construction.

3. The small-L^p Coulomb fixed point and finite-cutoff absorption create the derivative norm, rather than assume it.

No common complex gauge family is obtained merely from these real bounds. A complex extension would require a common physical-family domain and control of the relevant transition gauges; it cannot be inferred from compact-unitary telescoping after replacing it by unrelated complex frame norms.

Mixed determining sets, real boundary conditions, and independently weakened wall maps require their own local hypotheses. In particular, a wall-modified map is not covered by naming its output a minimizer. Its actual constraint, physical Euler equation, Ward identity and curvature improvement must first be established. The localization residual discussed in LR37–45 remains a separate issue.

Finally, the constants here depend on the fixed physical torus. This chapter does not prove an infinite-volume theorem or the completed multi-insertion bounds needed for a full Yang–Mills construction.

Section 57.7 / A direct endpoint-mean lemma that would verify the anchor

Here is a precise finite algebraic condition sufficient for (CX22). It must be matched to the source's actual gauge definition, not inferred from analyticity alone.

Suppose an axial reference transport $P_0(y,x)$ and a Landau transport $P(y,x)$, with the same endpoints, obey the gauge relation

$$P(y,x) = u(y)P_0(y,x)u(x)^{-1}. \tag{CX25}$$

Suppose the source's normalization at the retained site is the covariant group-mean condition

$$\mathsf{M}_{x\in B(y)}\left(P_0(y,x)u(x)P_0(y,x)^{-1}\right) = 1, \tag{CX26}$$

where M is right equivariant. Substituting $u(x) = P(y,x)^{-1}u(y)P_0(y,x)$ into (CX26) gives the exact formula

$$u(y) = \left[\mathsf{M}_{x\in B(y)}\left(P_0(y,x)P(y,x)^{-1}\right)\right]^{-1}. \tag{CX27}$$

If the paths have total physical length at most $C_L r$, and the Landau field differs from its real reference by a physical potential b, product telescoping gives

$$\|P_0P^{-1} - 1\| \le C_L r\|b\|_\infty \exp(C_L r\|b\|_\infty).$$

The same bound holds for finite admitted endpoint means by their analytic relative-diagonal estimate. Consequently

$$\|u(y) - 1\| \le C_L r\|b\|_\infty \tag{CX28}$$

on the common branch. A hierarchical version has the same estimate when its total path length is bounded by $\sum_j C_L\ell_j \le C_L r$. This supplies a bounded anchor on the whole source ball and a small anchor on the fixed physical ball.

The readable source passage is precise but not the formula itself: CMP102 p281 imposes the group-averaged normalization $\overline{R_0u}^{\,j} = 1$; Bałaban I pp265–266 says the inverse axial/Landau gauge is expressed by formula (106) of *Averaging operations for lattice gauge theories*, CMP **98** (1985), and is

Section 57.7 / A direct endpoint-mean lemma that would verify the anchor

analytic on an absolute small $\|\mathscr{H}\|_{\infty}$ ball. The literal definition of that group average and its relation to the endpoint transports must confirm (CX25)–(CX26), or provide an equivalent uniformly bounded formula, before (CX28) is assigned to that source bridge.

An analytic function on a fixed complex domain that is unitary for real arguments need not have a uniform complex bound: $z \mapsto e^{iNzX}$ is a counterexample as N grows. Therefore (CX22) is not being silently inferred from the source's analyticity statement.

Section 59.3 / The coarse weighted Hodge projection

The matrix C in (FW6) is diagonal in spatial components, with entries c_μ. On a nonzero coarse mode define

$$P_C = I - d_r(\delta_r C^{-1} d_r)^{-1}\delta_r C^{-1}. \tag{FW16}$$

On the harmonic mode set $P_C = I$. This is the projection with gradient kernel and range

$$\{B : \delta_r C^{-1} B = 0\}.$$

It obeys the uniform finite-p estimate

$$\|P_C B\|_{W_r^{1,p}} \le C_p \|B\|_{W_r^{1,p}}, \qquad 1 < p < \infty. \tag{FW17}$$

Here is the multiplier justification. In retained units $C^{-1} = \lambda_0 A^{-1}$, where A is the positive diagonal matrix from (FW12)–(FW13). The scalar λ_0 cancels in (FW16). The remaining scalar denominator is $D(z)^* A(z)^{-1} D(z)$, bounded below by $c|D(z)|^2$. The derivatives of A are uniformly bounded. Thus the symbol of (FW16) has the uniform bounds

$$|\partial_z^\beta P_C(z)| \le C_\beta |z|^{-|\beta|}$$

near zero and bounded derivatives elsewhere. The usual dyadic kernel proof of the discrete Calderón–Zygmund estimate applies: each annular piece has a rescaled integrable kernel, and summing gives

Section 59.3 / The coarse weighted Hodge projection

the uniform size, cancellation and first-difference bounds. Periodization gives the same bounds on compatible finite tori. This proves its L^p estimate; coarse physical differences commute with this multiplier and give (FW17). The assigned harmonic value adds only the bounded constant projection.

For B set

$$B' = P_C B, \qquad c = \Delta_r^{-1}\delta_r(I - P_C)B, \qquad d_r c = (I - P_C)B. \tag{FW18}$$

The scalar c has zero mean. The ordinary uniform flat elliptic estimate on the fixed physical torus gives

$$\|c\|_{W_r^{2,p}} \le C_{p,T}\|B\|_{W_r^{1,p}}. \tag{FW19}$$

Section 60.4 / One common carrier expansion for every block

Choose overlapping periodic localization boxes X_z of side cM_1, with a fixed overlap number. Let χ_z be smooth physical cutoffs supported a fixed fraction of M_1 inside X_z, with

$$\sum_z \chi_z^2 = 1, \qquad \|D^j \chi_z\|_\infty \le C_j M_1^{-j}. \tag{AL20}$$

Use the same continuous partition sampled on each field type. The identity in (AL20) then holds exactly on every type. Let Γ_z be the inverse of the same flat augmented matrix on the periodic box X_z.

Write $\mathbb{D}$ for either $\mathbb{A}$ or $\mathbb{K}$. Set

$$P_z = \chi_z \Gamma_z \chi_z, \qquad I_z = \chi_z^2, \qquad T_z = -[\mathbb{D}, \chi_z]\Gamma_z \chi_z.$$

Because the support collars exceed the finite range of every entry,

$$\mathbb{D}P_z = I_z - T_z, \qquad \sum_z I_z = I. \tag{AL21}$$

All three terms are filed on the same complete box X_z. The scalar and constraint components are included in that box; they are not assigned a second independent weakening.

Every commutator is bounded by C_β/M_1 from $\mathcal{X}_\beta$ to $\mathcal{Y}_\beta$. In particular:

- The second-order commutators in the first two rows use first derivatives of A, ψ.
- The third row uses $D\chi\, A$ and $D\chi\, D\lambda$, controlled in C^β.
- The averaging commutators contain $\chi(x) - \chi(y)$ with $|x - y| \le C$, hence cost M_1^{-1}.

- The same estimates hold for the added coarse-constraint row and multiplier column.

Together with (AL18) this gives a small residual. Nonzero products of residual factors must follow overlapping boxes, so their number is at most D_0^m at length m, with D_0 fixed. For M_1 sufficiently large,

$$\Gamma = \sum_{m\ge 0} \sum_{z_0,\dots,z_m} P_{z_0} T_{z_1} \cdots T_{z_m} \tag{AL22}$$

converges absolutely in the displayed localized operator norms and equals the augmented inverse.

Now choose the wall cells to have side at least a fixed multiple of M_1. Every localization box then meets at most c_0 wall cells. For a complete walk carrier $X = X_{z_0} \cup \cdots \cup X_{z_m}$, set

$$w_s(X) = \prod_{\Delta:\Delta\cap X\neq\varnothing} s_\Delta, \qquad \Gamma_s = \sum_\omega w_s(X_\omega)\Gamma_\omega. \tag{AL23}$$

Each wall occurs once. For a prescribed finite complex radius $R_w > 1$, the weighted length-m majorant is at most

$$C_\beta R_w^{c_0} \left(\frac{D_0 C_\beta R_w^{c_0}}{M_1}\right)^m. \tag{AL24}$$

This parameter choice is noncircular: first fix R_w and the required spare exponential, then choose M_1 large enough for (AL24), then take the wall cells larger than the localization boxes. The weak field ceiling in a later perturbative application is chosen afterward. Since these are fixed retained-scale constants, no fine-cutoff dependence is introduced. One can reserve an additional exponential in m, which becomes exponential spatial locality because a single step has bounded diameter $O(M_1)$.

Let B_+ be protected. Declare unweighted every wall meeting X_z whenever I_z has a nonzero row in B_+, for any of the field types. The same-carrier reindexing from LR then gives the exact identity

$$1_{B_+}\mathbb{D}\Gamma_s = 1_{B_+}. \tag{AL25}$$

There is no long projection tail in this calculation. The projected term only appears after eliminating scalar blocks, and those blocks have already received the same carrier weight.

Setting wall parameters to zero removes all walks meeting those cells. The surviving operators depend only on their complete connected carriers and split between the corresponding components. The local nonlinear fixed-point iteration has the same property if every nonlinear vertex is filed on its actual finite carrier. This last sentence is a property of such a local iteration; it does not assert that an arbitrary nonlinear raw chart already has its required bounds.

For the constrained system, extract

$$C_s = \pi_A \Gamma_s \iota_f, \qquad H_s = \pi_A \Gamma_s \iota_B, \tag{AL26}$$

and retain all accompanying scalar and coarse multiplier blocks. At $s = 1$, these are $C, H_0 = H$ from (AL16). On protected coarse rows,

$$QC_s = 0, \qquad QH_s = I. \tag{AL27}$$

The protected first, second, and third rows hold at the same time. Thus (AL26) is a genuinely compatible flat triple, with $H = H_0$, rather than three independently chosen kernel families.

One should not infer that Γ_s is a global inverse of a new physical operator or that it is symmetric for every complex s. Neither claim is needed for (AL25). A use requiring global transpose identities must prove them, or carry their actual defects. The local augmented identities are exactly what has been established.

Section 61 / Gauge-invariant readouts on the controlled complex periodic orbit

This note records a limited consequence of WC and CX5–20. The unresolved anchor bound CX22 is unnecessary for a holomorphic gauge-invariant **scalar value** evaluated on the genuine periodic critical orbit. It remains relevant to the route that controls both the potential and curvature while fixing

Section 61 / Gauge-invariant readouts on the controlled complex periodic orbit

the exact raw coarse field. The later CK theorem constructs a same-target holomorphic $W^{1,p}$ branch without an anchor bound, but does not supply its complex curvature supremum. The flat augmented wall theorem AL supplies a separate family; its strong bounds cannot be assigned to the CX family without an identification.

Section 61.3 / The exact relation to the flat wall theorem

AL constructs a jointly weakened **flat boxed** augmented system. On its protected rows it proves the required old weak bounds and an H^{1+s} curvature estimate. CX constructs a controlled complex representative of the genuine nonlinear periodic critical orbit.

These are not presently identified as the same family. In particular, one cannot combine the supremum bound from CX with the strong bound from AL and attribute both to an unverified wall-deformed Wilson background.

A limited composition is legitimate. Suppose A_s is any holomorphic family that is actually bounded in the CX input ball, and its value at $s = 1$ is the intended physical input. Then

$$\widehat{\mathcal{O}}_s = \mathcal{O}_P(A_s) \tag{IR10}$$

Section 61.3 / The exact relation to the flat wall theorem

has the periodic value bound (IR5), is holomorphic, and has the correct scalar endpoint. If A_s is supplied by AL with sufficiently small data on the fixed torus, its weak bound can verify the first premise.

But component dependence of A_s does not imply component dependence of (IR10). The full periodic reconstruction applied after A_s can reintroduce dependence on other components. Thus (IR10) alone does not supply the factorization or carrier estimates required by the localization theorem.

Alternatively, one may evaluate a gauge-invariant trace directly on an AL output. That uses AL's proved bounds, but identifying its endpoint with the intended nonlinear physical readout is a separate obligation. Gauge invariance removes a difference that is known to be a gauge transformation; it does not remove an arbitrary difference between two background maps.

There is also a precise local version: if an actual auxiliary field is proved gauge-equivalent to the controlled periodic representative on every bond and site of the readout carrier, the same invariant value bound holds on that protected carrier, without a global anchor bound. The carrier must include every transport path used by the readout. Neither protected augmented equations alone nor agreement of a few low field jets proves this local orbit comparison. AL does not currently provide it for the nonlinear wall field.

Section 63 / A test that a tensor rotation argument must pass

The scalar angular theorem uses the scalar transformation law in both its reflection form and its angular periodicity. Replacing its labels by tensor indices is not a formal extension. Here is an explicit positive local example with all lattice mirror symmetries and a subleading short-distance covariance, but without the proposed continuum tensor transformation law.

Section 64 / The exact Ward identity and the required cutoff estimate

The tensor covariance test suggests using the physical Ward identities to determine the transformation law of the larger field family. This section records the exact finite-cutoff identity and isolates the additional estimate needed to pass it to the continuum. It does not infer that estimate from a classical smooth-field expansion.

Section 64.3 / What the existing primary results provide

Del Debbio, Patella and Rago, *Space-time symmetries and the Yang–Mills gradient flow*,

Their equations (6.14)–(6.15) explicitly distinguish finiteness of the stress insertion from vanishing of the residual; the text at that point invokes recovery of the continuum symmetry. It therefore does not supply the needed construction estimate for the present Wilson sequence merely by naming a nonperturbative normalization prescription.

Positive flow-time probes are useful for separating contact singularities, but their continuum finiteness cannot be imported as a nonperturbative existence theorem here. Lüscher and Weisz, *Perturbative analysis of the gradient flow in non-abelian gauge theories*,

Likewise, the arbitrary-composite bounds and gauge Ward identities of Fröb, Holland and Hollands, *All-order bounds for correlation functions of gauge-invariant operators in Yang–Mills theory*,

Their weighted-tree estimates give a useful model for the desired attachment bounds; summability at fixed nonzero coupling is an additional issue.

The Ward route is therefore mathematically concrete: establish (WT4), (WT5), and the stress insertion limit on the same physical operator hierarchy. They are still unproved inputs in this attempt.

Section 65 / From separated product tests to the full flat-test growth condition

This is an auxiliary implication. Its input is a uniform separated-cube bound for the actual cutoff correlations. It does not assert that the Yang–Mills correlations already satisfy that input. The point

Section 65 / From separated product tests to the full flat-test growth condition

is to avoid an extra, unjustified assumption that the correlations have uniformly bounded pointwise kernels.

Section 66 / A scalar insertion density from a finite-order source norm

This note isolates a reconstruction consequence of the scalar source construction. Its hypotheses include the actual ordinary-source bound, the strictly local contact normalizer, and the physical all-mirror reflection-positive state. It does not replace any of those inputs by a claim about a reference replacement measure. The raw reflection argument obtaining the uniform insertion form is proved in the history section; the passage from that form to a point insertion is proved here.

Section 66.5 / Scope

The deduction is an explicit theorem from the physical source and mirror inputs. It shows that replacing a C^1 source norm by the proved C^5 norm costs four extra powers of the time margin, and does not prevent the analytic-density construction. It does not establish the ultraviolet two-point law, the mirror inputs for a different observable family, or the dimension-filtered correspondence with every gauge-invariant curvature polynomial. Those remain separate parts of a Yang–Mills existence argument.

Section 67.1 / Conditional statement and exact scope

This note proves a functional-analytic implication for a centered scalar two-point function. Its ultraviolet premise is a uniform bound on a fixed class of nonnegative C^5 shapes. No C^1 source theorem is used. The reflection and spectral premises are those of the actual physical functional, not of an auxiliary Gaussian or adaptive-history reference measure. The interacting ultraviolet comparison and the source-free preparation estimates needed to obtain that premise remain inputs.

Write a point of $\mathbb{R}^4$ as (y, z), where $y = (x_4, x_1) \in \mathbb{R}^2$ and $z = (x_2, x_3) \in \mathbb{R}^2$. Let

$$R_\theta(t, x) = (t\cos\theta - x\sin\theta,\, t\sin\theta + x\cos\theta), \qquad \Theta(t, x) = (-t, x). \tag{A1}$$

These transformations leave z fixed. Put $n = R_{\pi/4}(1, 0)$ and $\Theta_n = R_{\pi/4}\Theta R_{-\pi/4}$.

The assumptions are as follows.

1. S_2 is a real, symmetric bilinear functional on disjointly supported $C_c^5(\mathbb{R}^4)$ tests, complexified bilinearly. It is translation invariant and invariant under signed coordinate permutations. For fixed disjoint compact supports K, L, it obeys

$$|S_2(f, h)| \le C_{K,L}\|f\|_{C^5}\|h\|_{C^5}. \tag{A2}$$

2. The coordinate reflection form is positive. Its Hilbert quotient has the actual translation action

$$T(s, b) = e^{-sH+ibP}, \qquad s \ge 0, \quad b \in \mathbb{R}, \qquad H \ge |P|, \tag{A3}$$

with strongly commuting self-adjoint generators. On test classes, $T(s,b)[f] = [\tau_{(s,b)}f]$, whenever f is supported at positive time. Here $(\tau_v f)(x) = f(x-v)$. The inequality in (A3) is the exact coordinate-pair consequence of the spectral cube cone. Coordinate reflection positivity by itself is not being substituted for that spectral assertion.

3. The physical diagonal reflection form is also positive:

$$S_2(\Theta_n \overline{f}, f) \geq 0 \quad \text{if } \operatorname{supp} f \subset \{n \cdot y > 0\}. \tag{A4}$$

4. For the fixed constant $\vartheta_* > 0$ constructed in Section 3, there are $d_0 > 0$ and a finite nonnegative function $u(d)$, with $u(d) \to 0$ as $d \downarrow 0$, such that

$$|S_2(f,h)| \leq u(d)\|f\|_\infty \|h\|_\infty \tag{A5}$$

for every pair in the following shape class. Both tests are nonzero, nonnegative C_c^5 functions, each supported in a ball of radius d, with

$$\begin{aligned} d \leq \operatorname{dist}(\operatorname{supp} f, \operatorname{supp} h) \leq 6d, \\ \int f \geq \vartheta_* \|f\|_\infty d^4, \\ \max_{1 \leq |\gamma| \leq 5} d^{|\gamma|} \|\partial^\gamma f\|_\infty \leq \vartheta_*^{-1} \|f\|_\infty, \end{aligned} \tag{A6}$$

and the same conditions hold for h.

Only the reflected pairs constructed below are needed in (A5). Their two-point values are nonnegative by coordinate reflection positivity, so an absolute-value upper bound is sufficient. The constant ϑ_* is fixed by explicit shapes, independently of S_2. A premise for any smaller shape parameter is stronger and also suffices.

Theorem. Under (A2)–(A6), every equal-real-transverse-label two-point kernel is radial in the y-plane. The same conclusion holds for mixed and complex transverse labels. Thus every nonzero two-point angular harmonic vanishes.

The kernel meant here is constructed in Section 2. If the physical functional has the continuous off-diagonal four-dimensional density supplied by the companion scalar insertion theorem, this also gives rotation invariance of the complete two-point distribution off the coincidence diagonal. No conclusion about all higher moments, the mass gap, or a full gauge-invariant observable algebra is included.

For an initially uncentered physical functional with a unit invariant vacuum vector, one may first subtract the vacuum projection. This is an orthogonal projection commuting with the translation generators, so it preserves the reflection forms and the cone. Vacuum uniqueness is not required. In this note S_2 already denotes that centered functional whenever the ultraviolet estimate concerns connected correlations.

Section 67.3 / Fixed C5-admissible shapes and an explicit shape constant

All functions in this section are fixed independently of the physical functional. Define the C^6 smooth step

$$F(s)=\begin{cases}0, & s\le 0,\\ 12012\displaystyle\int_0^s v^6(1-v)^6\,dv, & 0<s<1,\\ 1, & s\ge 1.\end{cases} \tag{A19}$$

The normalization is exact, since $\int_0^1 v^6(1-v)^6\,dv=1/12012$. Also $F(1-s)=1-F(s)$ on $[0,1]$. Put

$$a(t)=F(8t-17)F(23-8t). \tag{A20}$$

Then $0\le a\le 1$, its support is $[17/8,23/8]$, it is one on $[9/4,11/4]$, and $\int a=5/8$. Its two transition intervals do not overlap, so $\|a^{(k)}\|_\infty\le 8^k\|F^{(k)}\|_\infty$ for $1\le k\le 5$. Every derivative constant here can be read from the degree-thirteen polynomial

$$F(s)=12012\sum_{j=0}^{6}\frac{(-1)^j\binom{6}{j}}{7+j}s^{7+j}\quad(0\le s\le 1). \tag{A21}$$

For the spatial factor use EIGHT interval convolutions. Let

$$v=\mathbf{1}_{[-1/8,1/8]},\qquad w=v^{*8},\qquad b=w/w(0). \tag{A22}$$

This spline belongs to $C_c^6(\mathbb{R})$, is supported on $[-1,1]$, is positive on $(-1,1)$, and has maximum one. For the last assertion write $w=v^{*4}*v^{*4}$ and use Cauchy–Schwarz for its autocorrelation. The explicit spline formula is

$$w(s)=\frac{1}{7!}\sum_{j=0}^{8}(-1)^j\binom{8}{j}(s+1-j/4)_+^7. \tag{A23}$$

It gives

$$w(0)=\frac{151}{5160960},\qquad c_b:=\int b=\frac{315}{604},\qquad \widehat{b}(q)=c_b\left(\frac{\sin(q/8)}{q/8}\right)^8. \tag{A24}$$

Section 67.3 / Fixed C5-admissible shapes and an explicit shape constant

The Fourier convention is $\widehat{b}(q) = \int b(s)e^{isq}ds$. The value of the quotient at zero is one.

Here is a C^6 transverse partition. Set

$$q(s) = b(4s/3), \qquad D(s) = \sum_{j\in\mathbb{Z}} q(s-j), \qquad \chi(s) = q(s)/D(s), \qquad \psi(z) = \chi(z_1)\chi(z_2). \tag{A25}$$

There are at most two nonzero summands in D. Let $\beta_* = \min_{|s|\le 2/3} b(s) > 0$. A nearest integer gives $D \ge \beta_*$. Thus $0 \le \chi \le 1$, its support is $[-3/4, 3/4]$, and the integer translates of ψ sum to one on $\mathbb{R}^2$.

Take the C^6 pedestal

$$\rho(s) = \begin{cases} 1, & |s| \le 1, \\ F(3-2|s|), & 1 < |s| < 3/2, \\ 0, & |s| \ge 3/2, \end{cases} \qquad \widetilde{\psi}(z) = \rho(z_1)\rho(z_2). \tag{A26}$$

Then $\psi \le \widetilde{\psi}$, and $\int \widetilde{\psi} = 25/4$.

For a completely specified finite derivative constant, put

$$\begin{aligned} C_a &= \max_{0\le k\le 5} \|a^{(k)}\|_\infty, \\ C_b &= \max_{0\le k\le 5} 4^k \|b^{(k)}\|_\infty, \\ C_{\mathrm{ped}} &= \max_{|\gamma|\le 5} \left(2\|\partial^\gamma \widetilde{\psi}\|_\infty + \sum_{\beta\le\gamma} \binom{\gamma}{\beta} \|\partial^\beta \psi\|_\infty \right), \\ C_{\mathrm{sh}} &= \max(1, C_a C_b C_{\mathrm{ped}}), \\ \vartheta_* &= \frac{1}{2} \min\left\{ \frac{c_b}{120}, \frac{1}{(5/2)^5 C_{\mathrm{sh}}} \right\}. \end{aligned} \tag{A27}$$

These definitions contain no unknown physical constant. For instance, if a purely arithmetic upper bound is desired, (A23) gives

$$\|b^{(k)}\|_\infty \le \frac{2^{15-k}}{w(0)(7-k)!} \quad (0 \le k \le 5). \tag{A28}$$

If $q_k = (4/3)^k \|b^{(k)}\|_\infty$, bounds for the derivatives of χ follow recursively from

$$M_0 = 1, \qquad M_k = \beta_*^{-1} \left(q_k + 2\sum_{\ell=0}^{k-1} \binom{k}{\ell} M_\ell q_{k-\ell} \right). \tag{A29}$$

The pedestal derivatives follow directly from (A19), (A26). Thus (A27) is an scope positive universal number; optimizing it has no role in the argument.

Section 68 / Analytic scalar kernels and all-order angular closure

This note proves a conditional implication from the scalar source, reflection, and two-point inputs. It does not import the unpublished construction as a theorem. In particular, positivity below comes from actual Hilbert vectors, and the final distributional step does not identify a distribution merely from its values on a dense open set.

Section 68.8 / All angles, all planes, and the scalar conclusion

We have local invariance for every sufficiently small angle near each point of $\mathcal{C}_n$. For a compact off-diagonal test and a fixed finite angle, the union of its rotated supports is compact in $\mathcal{C}_n$. Cover

Section 68.8 / All angles, all planes, and the scalar conclusion

it by finitely many preceding neighborhoods, take the minimum of their positive angle bounds, and divide the angle into finitely many smaller steps. A partition of unity allows the local distributional identities to be applied on each step. Composition proves invariance for the prescribed angle.

Hypercubic conjugations give this circle of rotations in every coordinate plane. Plane rotations generate $\mathrm{SO}(4)$, as seen by successively zeroing the entries below the diagonal of an orthogonal matrix. An available coordinate reflection supplies the other component of $\mathrm{O}(4)$. The centered one-point functional is zero; restoring a translation-invariant mean $c \int f$ commutes with these operations.

Finally rotations act continuously on Schwartz space, preserve all coincidence diagonals, and preserve flatness there. The unique diagonal-flat extension and its constrained tensor density therefore inherit the same covariance. The C5 all-cutoff bounds are sufficient; no C1 source theorem has been substituted.

The result is the following conditional scalar implication: the actual operator-strength plate input, coordinate cube cones and reflection forms, hypercubic symmetry, all-cutoff flat-test construction, and the separate two-point angular elimination imply $\mathrm{O}(4)$ covariance of the entire scalar hierarchy. The analytic kernels, boosted Gram identity, polygon divisibility, integer convexity, and exceptional-support arguments have all been proved above. This conclusion does not supply those upstream physical inputs or identify the scalar cyclic theory with the full Yang–Mills observable family.

Section 69 / Energy bounds and the scalar pure-phase question

The purpose of this section is to isolate a route to a unique-vacuum scalar component after the scalar reconstruction inputs have been established. It neither assumes that ergodicity survives a weak limit nor identifies a scalar component with the complete Yang–Mills field family. The essential new estimate is a one-sided energy bound derived from the time-regularized insertion density.

Section 70 / A four-shape test that survives pure-phase decomposition

A nonzero fourth connected function of a mixed state does not by itself prove that any of its pure components is non-Gaussian: mixing Gaussian covariances can produce such a function. The following

Section 70 / A four-shape test that survives pure-phase decomposition

finite shape test repairs that inference. It applies once the physical scalar four-point ultraviolet comparison and the scalar pure-phase theorem have both been established.

Section 71 / An explicit test of what the scalar ultraviolet law identifies

The purpose of this calculation is to check a logical implication used in the proposed construction. A scalar two-point law proportional to $|x|^{-8}\log^{-2}(1/|x|)$, even together with the scalar Wightman axioms and a unique vacuum, does not identify Yang–Mills dynamics. Here is a positive generalized free field with exactly that leading law. The calculation fixes the normalization and the separated-test limit, rather than relying only on the general observation that two-point functions leave higher correlations undetermined.

Section 72.1 / Two-point normalization, separated tests and phase hypotheses

The following implications are mathematically consistent under their stated hypotheses.

- **Raw cell bound to SP2.** In marked-history Section 21, the formal adjoint of `T_D A = D tau_ell A` is the stated reflected-conjugate insertion. Both parity cases of log convexity follow from raw reflection Cauchy–Schwarz. The coarser endpoint cells have a number independent of the long column length, so their supremum bounds disappear after taking the column root. This genuinely removes dependence on the degree and coefficients of the outside polynomial. Fixing the normalizer at fixed mesh before taking the long-column limit is necessary and is done in the stated argument.
- **SP2 to analytic bounded point insertions.** The time shifts in SP4 agree exactly with the defining form. The compact four-dimensional fifth-Laplacian parametrix is C^5, and its scaled C^5 norm is O(r), while the smooth remainder costs O(rˆ-9). Ten analytic derivatives therefore give the claimed margin exponent p+9. This does not use a bound on the diverging C^5 norms of point mollifiers: the resulting norm-continuous analytic density supplies the norm limit afterwards.
- **Two-point angular elimination.** Dilation averaging correctly avoids the Fourier zeros of the spline. The spectral cone pays the polynomial Fourier decay, leaving hˆ4 Gamma(7h). Transverse patch counting then yields Gamma(t)=o(tˆ-8). The three-mode positive-measure identity uses positive coefficients and really does force the diagonal measure onto the coordinate cone, after which both signs of each remaining nonconstant mode are obtained.
- **Higher angular closure.** The boosted positivity is an actual vector identity; the involution is `(v0,v1)*=(conj(v0),-conj(v1))`, which commutes with the imaginary rotation. The top coefficient is positive at an arbitrarily oriented mirror only after inserting its fixed unit phase. The polygon argument releases the other labels by a zero-row argument followed by real-analytic continuation. Divisibility plus integer convexity and linear growth then forces all angular indices to be controlled by the two-point index. The common-cut argument deals with distributions on exceptional projection sets.

Section 72.1 / Two-point normalization, separated tests and phase hypotheses

- **Finite energy transfer.** EP2–EP5 is a sound finite-band argument. The graph-norm approximation in EP5a is important: it identifies the constructed bounded extension with the closure of the Wightman field, rather than merely identifying sesquilinear forms on a dense set.

One minor presentational correction: D1 uses scaled C^5 norms, while history (20.14) is displayed with unscaled C^5 norms. The patch proof gives the stronger scaled version directly. For a patch cutoff of width h/n,

$$\|\chi_{h/n} f\|_{C^5} \le C(n/h)^5 \|f(c + h\,\cdot)\|_{C^5}.$$

because a derivative falling on f contributes hˆ-1 and a derivative falling on the cutoff contributes n/h. Thus b=5 and p=9 remain available. The displayed unscaled inequality alone would not imply that precise D1 statement.

Section 72.2 / Exact upstream dependencies of the source-to-SP2 route

The functional-analytic proof starts from the following **physical** estimate, with constants uniform in the mesh and compatible torus:

$$\left| e^{-L_{a,T}[J]} \left\langle e^{\mathcal{O}_a(J)} \right\rangle_{a,T} \right| \le e^{C\,\mathrm{Vol}(T)}, \qquad \|J\|_{C^5} < r_0. \tag{FA1}$$

Here L is a genuinely local finite-jet functional, including both clean E and regular R vacuum births. It must have the stated reality, translation, and reflection identities. The raw SP2 argument needs the whole ordinary complex disc in (FA1), not just every fixed mixed derivative at J=0.

The precise model-specific proof obligations immediately upstream of (FA1) are:

1. **Actual ordinary source formation.** The actual frozen-background local formation maps, including all terms of R origin on their original shrinking cores, must be bounded and holomorphic in one common ordinary source domain throughout arbitrarily many ultraviolet steps. A clean Gaussian comparison, a source-free stability theorem, or individually bounded finite source jets does not imply this. Local-map (14.10)–(14.15) and history Sections 15 and 19 are the proposed new induction; their full actual-map hypotheses remain the dependency that must be checked against the primary preparation and formation maps.
2. **Universal regular source-vacuum locality.** The family nu_j(X;J) must be the local scalar coefficient of the exact physical expansion, with a completed carrier that contains all determining source and history data. The key quantitative statement is the rooted ordinary-source bound (local-map 15.1 / history 15.4). Its strict carrier dependence is what makes a source Taylor jet a valid local counterterm and what makes a separated-source derivative gain a bridge. Bounds on a final field argument alone do not prove this determining-data locality.
3. **One common vacuum and actual terminal stability.** After factoring the same V+E for every history, the integrated original partition function must satisfy both sides of history (15.6), equivalently local-map (15.10), with constants per terminal cell. In particular the lower estimate must use the completed branch with no surviving terminal island and must include healed R vacuum births. An all-clean event, an auxiliary normalized reference measure, or a bound for individual unnormalized kernels does not give that lower estimate with the same V.
4. **Compatible periodic comparison.** Cell dissemination and the subsequent thermodynamic passage require the same source normalizer on the compared periodic volumes, or the proved

scalar discrepancy tending to zero. Nonwrapping equality has to be established term by term for the complete determining carrier before the ultraviolet root count is summed. Two separate local coefficient bounds leave a divergent bulk count and do not suffice.

Once these four items are established for the original Wilson observable, the derivation of SP2 does not require a further estimate uniform in arbitrary outside polynomials: the raw long-column argument supplies that uniformity. Conversely, SP2 should not be presented as an stated physical input merely because the long-column implication has been checked.

The independent noncollapse/rotation input is the **full physical ultraviolet comparison** on the same bare sequence. Its upper bound must be uniform over the fixed C^5-positive shape class used in A5–A6; its positive coefficient must survive the retained-field terms, complete history complement, and periodic-volume transfer. A pointwise clean-window Gaussian calculation or a calculation for one pair of profiles does not supply the shape-uniform upper bound. The four-shape conclusion similarly needs the four physical cumulant comparisons with one common running normalization.

I have not disproved the proposed new actual-map inductions in the shared notes. The distinction above is between a checked downstream theorem and the exact model-specific claims that theorem still requires; the word “conditional” cannot itself discharge those claims.

Section 72.3 / A precise padding lemma for the Euclidean phase integral

The terse norm-limit assertion at the end of the non-Gaussian-phase note can be replaced by this lemma. It introduces no new physical estimate once the reconstructed field has been identified with its analytic insertion density.

Assume the reconstructed scalar theory has bounded point insertions $B_s(x)$, $s > 0$, satisfying

$$\|B_s(x)\| \le C s^{-q},$$
$$B_s(x) = e^{-(s-r)H} B_r(x) e^{-(s-r)H}, \qquad 0 < r < s.$$

with x a real Minkowski translation of the insertion and $H \ge 0$. Let M be the algebra in the Driessler–Summers generalized-energy-bound decomposition. Its Poincare representation, H spectral projections, and bounded compact-frequency field regularizations belong to M.

72.3.1 3.1 Finite energy cutoffs converge in operator norm

Set $Q_R = \mathbf{1}_{[0,R]}(H)$. Taking r=s/2 in the padding identity gives

$$\|B_s(x) - Q_R B_s(x) Q_R\| \le 2e^{-sR/2} \|B_{s/2}(x)\|. \tag{FA2}$$

This estimate is the missing reason finite-energy approximants converge in norm. Strong convergence of Q_R to 1 alone would not have sufficed for an arbitrary bounded operator.

72.3.2 3.2 Compact-frequency approximate identities also converge in norm

Padding bounds H B_s and B_s H. The spectral cone bounds the corresponding spatial momentum products. Consequently conjugation of B_s(0) by real spacetime translations is norm continuous, indeed norm differentiable with finite s-dependent bounds.

Choose a Schwartz approximate identity g_epsilon with compact Fourier support and integral one. Then

$$B_s(g_\epsilon, x) = \int g_\epsilon(y - x) B_s(y)\, dy \;\longrightarrow\; B_s(x). \tag{FA3}$$

in operator norm. Compact Fourier support is preserved by scaling and translation. The identity with the actual smeared field is

$$B_s(g_\epsilon, x) = e^{-sH} \overline{\phi(g_\epsilon, x)} e^{-sH}.$$

For these tests EP5a gives the domain inclusion. Bounded compact-energy approximants belong to M; (FA2) or the corresponding energy-bound cutoff argument puts the smeared sandwich in M. Then the norm closure of M and (FA3) put B_s(x) in M.

This use of the DS algebra is justified by its strong-commutation result for compact-frequency fields (Proposition 3.9), its inclusion of the Poincare representation (Proposition 3.10), and the direct-integral proof of Theorem 2.3. It does not require assuming that every unsmeared field closure is affiliated before the bounded approximation is constructed.

72.3.3 3.3 Fiber products and domination

First choose countably many s, x, and approximate identities. Remove one null set on which any of their decomposition identities or bounds fail. Norm continuity and padding extend to the remaining parameters on that common full-measure set.

For finitely many strictly ordered Euclidean insertions choose a fixed a>0 smaller than half every intervening time gap. Their kernel is an expectation of a product of B_a factors and bounded positive-time semigroups. Every factor decomposes over the same phase space. Its fiber expectation is bounded by

$$C^n a^{-qn} \|\Omega_\xi\|^2. \tag{FA4}$$

times the fixed L1 norms of transverse smears when present. The right side is integrable because integral ||Omega_xi||^2 dnu(xi)=1. Thus the finite product and its compact-support integrals pass through the phase integral by dominated convergence. Agreement with the continued phase Wightman moments follows first for the compact-frequency approximants and then by analytic uniqueness on a common ordered tube.

For the four small radial-bump configurations one can choose the bump radius sufficiently small that each configuration admits a strictly ordered time projection. Component proper Poincare covariance supplies connected Euclidean covariance after continuation, so each configuration can be rotated into such a projection. The same finite phase integral then applies to all four terms in NG3.

The nonconstant-component conclusion actually needs less: if almost every Wightman component is a constant field, its Wightman mixture already has constant connected two-point distribution, whose Euclidean continuation has the d^8 scaling. Thus EP8 alone contradicts that possibility without needing the full Euclidean phase-integral lemma.

Primary source checked: Driessler–Summers, *On the decomposition of relativistic quantum field theories into pure phases*, HPA 59 (1986), Theorem 2.3 and Section III, pp. 335–343,

Section 72.3 / A precise padding lemma for the Euclidean phase integral

periodica.ch/cntmng?pid=hpa-001%3A1986%3A59%3A%3A1439 . The theorem really allows a degenerate vacuum and produces proper Poincare-covariant clustering components under the compact-frequency generalized H-bound. The formulae and estimates (FA2)–(FA4) above are the analysis additional argument.

This section records two issues found Section 7 below states the outcome after both were corrected in the integrated text.

72.5.1 5.1 Valid algebraic deductions

The weighted squarefree norm and the one-joint-energy estimate in JC29–JC32 / CH4–CH8 are correct. Their constants are independent of the number of labels. They give a global Gevrey coefficient estimate under a small sum of source loads, not a local replicated-cell estimate under a bounded load per cell.

The separate Gaussian local-slot lemma CH32–CH36 is also correct. Its essential hypotheses are a bounded covariance operator in the ACTUAL cold-energy/readout coordinates, bounded column overlap of the local quadratic energies, and a trace cost proportional to the number of slots. One common Gaussian variable repeated in every slot violates the column-overlap hypothesis and recovers the factorial growth; therefore that hypothesis cannot be omitted. This is a useful explicit sufficient condition if Gaussian polynomial source slots are retained.

The exact compact split JC39 and compensating-factor identity JC40 preserve every finite-cutoff physical source derivative. This remains true after every source-independent linear transfer, provided both the completion and compensation retain all their original source labels and carriers. The bounded compact completion is not, by itself, a bounded extension of the analytic remainder onto a wild field. CH Section 8 correctly keeps that distinction.

The geometric once-per-label payment has a coherent justification: selected path portions lie in disjoint h/4 source neighborhoods, their creation scales do not exceed the first-joining scale, and the original source anchors are retained. The live version needs the complete unit-layer component cost, not a relative tree crossing a hole for free. See the appropriate original M-cube convention and the explicit payment of the growing-component geometry in R-II (1.67) and (1.93)–(1.97). That is the right geometry for CH23 and CH38.

72.5.2 5.2 A concrete reserve-accounting correction

As first written, CH39 identifies its right-hand K0 as the original source-free kernel, although engineering and live-age reserves have already been spent. The majorant on the right must instead be the **same one joint cold majorant with exactly those spent reserves removed**. It cannot remain the unchanged numerical original kernel.

A one-line example explains the distinction. Suppose one live epoch of age n contributes a source coefficient L^(2n) and has original bank weight exp(-Bn). Engineering payment gives

$$L^{2n}e^{-Bn} = e^{-(B-2\log L)n}.$$

It does not give a constant multiple of exp(-Bn). At source radius r, an unchanged-K0 version of CH39 would require

$$\log(1 + rL^{2n}) \le C\eta n.$$

For fixed r>0 the left side divided by n tends to 2 log L, so the assertion fails uniformly when eta is made small with C fixed. Removing the declared fixed engineering reserve from the joint bank repairs the issue. This repair needs no product of independent island integrals and changes no physical source identity; it is a distinction between the exact source-free integrand and the enlarged positive majorant used for its bound.

The same point applies to the source-tree reserve used in JC25b. Payment establishes a weighted coefficient inequality. Any subsequent assertion that the resulting atoms are pointwise bounded must say which corresponding part of the common positive majorant has been enlarged.

72.5.3 5.3 The strongest conclusion justified without assuming the desired local norm

JC41–JC43 are a valid **conditional local-majorant implication** if the actual full-core formation map has an operator-valued positive connected-kernel majorant in that local coefficient norm, and if the geometric payment is implemented on its scale-labelled source forest before taking absolute values. Analyticity at a flat numerical background or separate finite-order estimates do not supply this input.

CH38 is an elementary pointing estimate **for incoming source members already obeying the ordinary local rooted norm**, including summability of their resolved ages. The three represented-state classes alone do not imply that norm: a bounded analytic function in each class can have arbitrarily large numerical coefficient, and primitive CH37 bounds only the first input loads. Thus the genuine closing step is an induction for the norms of newly healed multi-label R members.

A sufficiently explicit proof would:

1. State a finite-prefix bound for the ordinary LOCAL norm of all regular incoming source packets and a separate one-JOINT-kernel majorant for each still-live component.
2. Apply the pointing estimate only to those incoming packets; take the compact completion kernel costs at their actual birth/healing/promotion, with every compensation included.
3. Use the remaining original deterministic volume and tree reserves to bound the zero-source difference of a completed R formation, and its proper-joining remainder, in that SAME local norm. For regular source insertions this follows from a positive estimate of the form `||exp(F)-1|| <= ||F|| exp(||F||)` and its quadratic remainder, before the one common live integration. Unresolved live packets must remain joint; they cannot enter a product of their integrated norms.
4. Display the resulting finite-prefix scalar inequality, with the one-label excursion contribution resolved linearly and a genuine regular joining term of order J^2 or its analytic majorant. Its linear/resolvent and proper-joining constants must be independent of the number of labels, the cutoff depth, and the number of occupied h-cells.
5. Close that inequality at one small eta. Only after this step is CH38 valid for all output members and can CH39–CH43 be asserted without making the desired extended norm a premise.

This is the exact dependency in the replicated-cell claim. The source-free full-core local-potential/wall theorem, preparation-domain inclusions, and completed-R reserve bounds are the relevant primary

Section 72.5 / The joint local estimates

inputs. The new ordinary-local coefficient induction is not a consequence of the global Gevrey theorem or of the finite-torus factorial bound. Once that induction is proved, the terminal lower bound using a radius x_hˆ(-1/2) compact-link core gives the stated polynomial-per-cell dependence; this part does not introduce a new ultraviolet power.

The first cubic center should enter its fixed 56-dimensional normalization only at the original covariance promotion. During the delay, the complete tagged local cubic kernel stays in the state.

The replicated estimate further requires exact lattice-reflection covariance of the fixed first normalization; bounding a conjugated reflection by a cutoff-dependent condition number is not a replacement. Neither the local coefficient estimate nor these lattice reflections supplies the missing continuous tensor rotation law.

No counterexample to a properly stated full-core, reserve-aware local-majorant theorem was found. The established new mathematical content is the Gevrey cold estimate, the independent local Gaussian-slot lemma, exact completion identities, and the geometric once-per-label payment. The unconditional physical replicated-cell theorem requires the explicit local R induction just identified.

Section 72.6 / Independent check of the finite Gaussian buffer lemma

See BA1–BA14 in the stated results directly. The conditional covariance and its nondegeneracy criterion are correct. In particular `ker Gamma = ran Q*` follows from `Gamma=C^(1/2)(I-P)C^(1/2)`, and injectivity of T* is exactly the additional fact needed for the range-intersection formulation of strict positivity.

Continuing the Gaussian density while evaluating A only on real variables proves the entire continuation and the displayed Cameron–Martin bound. The directional derivative is the stated Wick polynomial; its L^2 norm squared is the permanent of the directional Gram matrix. This gives the square-root-factorial derivative bound and the reciprocal-square-root-factorial Taylor remainder. The approximation map has rank at most the number of monomials of total degree at most p in rank(S) variables.

The chapter correctly requires fixed p as the region shrinks and separately requires a common finite-dimensional range. Uniformly bounded ranks of different subspaces do not by themselves produce one finite-rank phase-space approximant. The projection inequality BA14 is correct in one fixed ambient normed space. The additional requirement to realize the polynomial coordinates and their coefficient functionals in the reconstructed energy-damped framework is explicitly retained. I found no hidden claim of interacting Wilson smoothing in this finite-dimensional lemma.

Section 72.7 / Hypotheses and conclusions after the replicated-cell corrections

I reread the revised CH Sections 8.3–8.4. They now contain both repairs identified in Section 5

- CH38 is explicitly an **incoming** norm estimate with size C(eta+A), where A is the regular norm being bootstrapped. It no longer assumes that all newly formed outputs already have the desired bound.
- CH39 now uses the one remaining joint positive majorant after the declared engineering, first-root, age, and tree reserves have been spent. It no longer makes the incorrect comparison with the unmodified source-free kernel.
- CH39a bounds the linear and proper-joining excesses inside that same majorant by `C s N` and `C s^2 N^2 exp(C s N)`. Fixed remaining volume reserve pays the first two record moments.

There is only one actual rare entry per lineage/component; merging unresolved components does not produce a product of integrated exit norms.

- The resulting regular/first-exit inequalities CH39c imply CH39d. The positive finite-prefix majorant is continuous under radial scaling of the finite source fugacities, starts at zero, and cannot first reach K eta because the upper bound there is strictly smaller. This justifies the stated first-exit closure independently of the cutoff depth and source count.

The new recurrence addresses the missing logical step. Under the explicitly stated original-map hypotheses, I find no remaining algebraic gap in the deduction of the ordinary local replicated-cell bound. In particular, the conclusion is no longer being inferred from the global factorial estimate.

The nonalgebraic hypotheses used here are still precise: the original full-core operator-valued connected-formation bounds; the prepared-domain inclusion for every stored regular readout; and the positive completed-R majorant with sufficient **simultaneously unspent** deterministic tree, unit-layer volume, and age reserve. Those hypotheses must hold for the exact frozen source-zero maps and domains being used, rather than for a replacement measure or a numerical flat-background covariance. Within that stated source-free framework the revised proof supplies the new coefficient extension; it does not cite a pre-existing published composite-source theorem.

The corrected handling of cubic R origins is also necessary: local third kernels are retained at healing, while the universal 56-dimensional center receives them only at compatible covariance promotion. The finite delay is absorbed in the stated rare envelope; it does not become an extra center at every fine birth scale.

The strongest conclusion of this follow-up $_{\text{the analysis}}$ is therefore a reserve-aware, conditional **local replicated coefficient theorem for the fixed protected quadratic/cubic family** on one Wilson sequence. Passing it through physical chessboard still requires exact lattice-reflection covariance of that fixed first family. Continuous tensor rotation covariance, a complete hierarchy for arbitrary curvature derivatives, and a microscopic phase-space estimate remain distinct tasks. Neither the revised local theorem nor the finite Gaussian buffer lemma proves those tasks.

Subsequent lattice-symmetry update. The joint-curvature chapter, Section 11, fixes clover representatives once and proves finite lattice-reflection intertwining from the original symmetric blocking map. It includes the quadratic large-field first-center correction, retains partition tags until compatible promotion, and uses equivariant pointed extraction before the translated-root sum. That supplies the finite lattice-symmetry input under the same original-map framework. It does not supply continuous tensor rotation covariance.

Section 73 / A possible route through bounded local observables: an exact Gaussian lemma

The completed-insertion calculation exposes an interacting near-mark estimate that has not been proved. There is a potentially different route: first approximate every bounded local observable by a finite-dimensional space of energy-bounded point fields, and then construct their products. Bostelmann's microscopic phase-space condition makes that route precise. It is an additional short-distance condition, not a consequence of the scalar reconstruction theorem. The analysis gives an explicit generalized-free counterexample to that implication.

This chapter proves the finite-dimensional Gaussian smoothing statement needed at the beginning of such a route. It also identifies the extra uniformity that an actual Wilson argument would

Section 73 / A possible route through bounded local observables: an exact Gaussian lemma

have to establish. No infinite-dimensional or interacting smoothing theorem is inferred from the finite-dimensional identity.

Section 73.5 / What this route would require for Wilson fields

Bostelmann's microscopic condition asks for finite-rank approximation of the energy-damped restriction of normal functionals to every sufficiently small local algebra, with an error smaller than an assigned power of its radius. His results then give finite-dimensional point-field spaces and operator product expansions. The condition quantifies over the local algebra's entire unit ball. Bounds for a single chosen scalar and its moments do not have that quantifier.

To turn the finite-Gaussian lemma into a Wilson proof, one would need the actual renormalization kernels to provide a counterpart of (BA3)–(BA4) for bounded gauge-invariant local observables, uniformly over complete large-field histories, together with (BA10)–(BA13) in the reconstructed energy-damped norm. Gauge constraints and zero modes require their real coarea treatment. The local approximation spaces would have to be compatible across scales and contain the required curvature and derivative types. Their covariance under spacetime transformations would also have to be proved.

The ordinary source theorem for the action density does not quantify over arbitrary bounded observables, and a bound on a conditional mean does not control Σ^{-1} in (BA4). The sharp-conditioning examples show why those distinctions are substantive. Conversely, (BA3)–(BA12) show that nonanalytic microscopic observables need not obstruct a bounded-observable route when a genuine observation buffer and its Cameron–Martin estimates are available.

Primary references for the alternative framework: H. Bostelmann, *Phase space properties and the short distance structure in quantum field theory*, arXiv:math-ph/0409070, Section II; and *Operator product expansions as a consequence of phase space properties*, arXiv:math-ph/0502004, equations (2.5)–(2.7). The finite-Gaussian estimates (BA1)–(BA14) above are proved here.

Section 74 / The exact scalar conclusion and the remaining Yang-Mills task

This section puts the scalar implications in their logical order. It is useful to state the combined result separately from the much larger operator-family target. The hypotheses below are physical cutoff statements; an auxiliary reference replacement law cannot be substituted for them.

Section 76 / Continuation

This chapter revisits the curvature-supremum premise of PR. It is not legitimate to deduce bounded curvature directly from a $W^{1,p}$ potential. There are, however, two additional consequences of the actual critical equation.

First, CK supplies a curvature bound independent of the microscopic cutoff, with an explicit retained-scale loss $r^{-4/p}$. Second, the fractional PR argument can be run without assuming any curvature supremum: the inverse lattice bound at each intermediate spacing is multiplied by a summable scale factor. The latter statement gives a common complex fractional chart after CK and the retained-node gauge section.

These conclusions concern the actual periodic critical branch. They do not identify independently wall-modified formations with that branch, replace the original off-shell paired maps, or provide the full quantum observable hierarchy. A bound uniform also as $r \downarrow 0$ for the matrix curvature supremum is not asserted.

Use the endpoint-removed operators from CK:

$$\mathcal{A}_j = DM_j(U_j) + d_{U_{j+1}}\rho_j.$$

They have the exact gauge chain

$$\mathcal{A}_j d_{U_j} = d_{U_{j+1}} S_j. \tag{WB5}$$

The full-space decomposition PG15d–e, in the common complex physical frame, compares $\mathcal{A}_j^{\mathsf{t}}$ with the flat normalized box/line prolongation T_j^0. Its row and column errors obey

$$\|\mathcal{A}_j^{\mathsf{t}} - T_j^0\|_{L^q\to L^q} \le C_{L,q}(\ell_j M + \ell_j^2\alpha_j) \le C_{L,q}(\ell_j M + \ell_j^{2-4/p} m), \quad 1 \le q \le \infty. \tag{WB6}$$

Here q denotes the norm exponent, independently of the p defining the weak chart. At the endpoints $q = 1, \infty$, the estimate follows from the literal physical-volume row and column sums. Interpolation is performed on this full-space operator, not on a Ward-constrained domain.

For detail, the transported principal part has the same normalized positive scalar weights as the flat filter. Each path coefficient differs from the identity by $C\ell_j M$. The relative-contour, group-mean and reference-bond errors have row and column size $C\ell_j^2\alpha_j$, by the fixed-L analytic expansion. The endpoint gradient has been removed exactly. These are precisely the terms in PG15d–e; (WB3) verifies its small-loop premise at each step.

The flat prolongation is an L^q contraction with coefficient one. Therefore

$$\prod_{j<k}\left[1 + C(\ell_j M + \ell_j^{2-4/p} m)\right] \le C, \tag{WB7}$$

because both geometric sums are bounded independently of a, r, k.

If the terminal multiplier satisfies $d_{U_k}^{\mathsf{t}}\lambda_k = 0$, propagate the complete currents by the actual differential:

$$\lambda_j = DM_j(U_j)^{\mathsf{t}}\lambda_{j+1}.$$

The actual raw gauge chain preserves the Ward equation. CK7 then identifies each propagation with $\mathcal{A}_j^{\mathsf{t}}$, and (WB7) proves

$$\|\lambda_j\|_{L^q} \le C_q\|\lambda_k\|_{L^q}. \tag{WB8}$$

The same conclusion applies to the endpoint-removed chain on arbitrary terminal currents, but only the Ward-compatible chain is identified here with the physical current.

For the fractional comparison set

$$e_j = \lambda_j - T_j^0\lambda_{j+1}.$$

Section 76.2 / Full-space adjoint estimates and physical-current propagation

At its own mesh, the elementary inverse estimate and (WB6) give

$$\|e_j\|_{H^s_{h_j}} \le C\left(M\ell_j^{1-s} + m\ell_j^{2-s-4/p}\right)\|\lambda_k\|_2. \tag{WB9}$$

Unroll this identity exactly. Every tail applied to an error is a flat direct prolongation, bounded on H^s by FH22–23 for $s < 1/2$. Thus

$$\boxed{\|\lambda_0 - Q_{\mathrm{bl}}^{\mathsf{t}}\lambda_k\|_{H^s_a} \le C\left(Mr^{\,1-s} + mr^{\,2-s-4/p}\right)\|\lambda_k\|_2 \le Cm\|\lambda_k\|_{H^s_r}.} \tag{WB10}$$

Both exponents in the sum are positive under (WB1). No separate nonlinear propagation of an error field is used. In particular, no Ward identity is assigned to e_j.

This replaces the bounded-curvature comparison FH32 wherever it was used in the periodic KKT proof.

Section 77 / Curvature estimates from the weak critical branch

This note proves the flat endpoint estimate requested complex curvature argument. It uses the uniform analytic offset kernels already proved in FW8–15 of the stated results, and the exact coarse weighted Hodge projection FW16. It does not assert a generic Riesz projection bound on L infinity.

Section 77.4 / The positive order makes the kernel absolutely summable

Here is the elementary endpoint multiplier fact used on FE13. A periodic matrix multiplier t smooth away from zero, with t(0)=0 and |partialˆalpha t(z)|<=C_alpha |z|ˆ{sigma-|alpha|} for some sigma>0, has an absolutely summable lattice convolution kernel, provided those bounds hold through a fixed order greater than four. Its l1 norm is bounded by the indicated constants.

To prove this, split into one smooth part away from zero and dyadic annuli |z| asymptotic to epsilon=2ˆ{-j}. On an annulus the Fourier kernel obeys, by rescaling and integration by parts,

$$|k_j(y)| \le C_N \epsilon^{4+\sigma} (1+\epsilon|y|)^{-N}, \qquad N > 4. \tag{FE14}$$

Its l1 norm is at most C epsilonˆsigma. The sum over j converges. The away-from-zero part is a smooth periodic multiplier with a summable kernel as well. The absolutely convergent resulting Fourier series has the assigned value t(0)=0. Periodizing its kernel therefore proves the same bound on every compatible finite torus, including its zero mode.

Applying this with sigma=1 to FE13 gives

$$\sup_{\xi} \sum_{y} |t_\xi(y)| \le C, \tag{FE15}$$

uniformly in n. Combining FE10–13 now gives the exact representation

$$(d_a H_{\mathrm{bl}} B)(y+\xi) = (d_r B)(y) + (t_\xi * d_r B)(y) \quad \text{in retained units.} \tag{FE16}$$

The first term is simply a fixed coarse two-form hold at the chosen offset. It is not required to be a cochain interpolation: FE12 already uses only the exact closed two-form d_rB. The hold has norm one. Young's inequality, followed by the physical average over the nˆ4 fine offsets, proves FE1 for every p from one to infinity. Only the l1 kernel estimate is used, so the two endpoints are included.

Section 78 / Uniform endpoint stability of the actual flat curvature projection

This chapter closes a specific periodic gap left in CX and CK. The same-target holomorphic critical branch constructed in CK has a matrix curvature supremum bounded independently of both lattice spacings, when the input has the corresponding bounded curvature. The proof uses the flat endpoint theorem FE and the weak estimates WB. It does not use a holomorphic gauge bridge to an independently chosen representative, or an irreducibility assumption. It does not identify the original off-shell paired maps or the mixed-boundary renormalization operations.

Section 79 / A common complex curvature bound for the actual critical branch

The common complex bound CC extends to a genuine residual equation. This is an a priori statement about the actual Wilson current and actual raw mean. It does not assert that an independently weakened operator has a small residual. Its useful feature is that no separate small Ward defect has to be postulated: the physical Noether identity controls that defect, including its atomic endpoint contribution.

Section 79.5 / Scope of the residual statement

RC3 can be used for an actual localized configuration once its physical Euler residual E is proved to have the indicated Lp bound, and once its constraint error and weak multiplier bound are proved. It does not prove those premises by giving an arbitrary weakened matrix the same symbols. It also gives no derivative of F beyond the stated supremum. The high-jet example HJ remains compatible with RC3 even when E=b=0.

The older unrestricted negative-Sobolev inverse counterexample FI is not contradicted. RC3 controls curvature, uses an Lp residual, assumes the stated actual weak solution and multiplier, and exploits its exact Noether compatibility. It is not an inverse on arbitrary off-shell negative-Sobolev forcing.

Section 80 / Curvature stability for an actual Euler residual

See CC1–34 against the actual operators in FW, CK, WB, and the new flat endpoint proof FE. No defect was found in the stated periodic theorem. The analysis concerns that theorem's explicit input class and does not supply the separate original-source or quantum-insertion interfaces.

Section 80.1 / Negative-norm Ward control has no hidden retained-scale inverse

In physical tangent coordinates,

$$d_V - d_r = \sum_{\ell} M_{c_\ell(B)} \tau_\ell, \qquad \|c_\ell(B)\|_{W_r^{1,p}} \le CM, \tag{CA1}$$

where each tau is one fixed lattice shift. The exact coefficient has the form (Ad exp(rB)-I)/r, up to the fixed sign and endpoint convention. This is zero order, rather than a first derivative of the gauge parameter.

For p>4, multiplication by c in W^{1,p} is bounded on W^{1,p'}: the derivative of c is paired with an undifferentiated test in L^{p/(p-2)}. The uniform four-dimensional embedding W^{1,p'} into L^{4p/(3p-4)}, followed by the fixed-volume inclusion, supplies this exponent. The undifferentiated c is bounded by Morrey. Duality gives

$$\|R_B \lambda\|_{W_r^{-1,p}} \le CM \|\lambda\|_{W_r^{-1,p}}. \tag{CA2}$$

The same argument controls conversion from the logarithmic multiplier to the physical one, if that conversion is needed at the CK interface. Exact Ward makes R_B lambda a mean-zero flat divergence. The fixed-torus operator d_rG_r:W_r^{-1,p}_0 to L_r^p is uniformly bounded. Thus CC12 indeed has size CM squared with no r^{-1}; using CC7 on this term first would unnecessarily lose that conclusion.

The separate endpoint-current comparison CC6 is allowed to use CC7: its geometric sums contain r or r^{2-4/p}, which pay the single r^{-1}. Every error is propagated through a whole flat tail, so no Ward identity is incorrectly assigned to an isolated error.

Section 80.5 / Result and exact scope

The checked chain proves

$$\|F_a(w(q_{r,\mathrm{raw}}(A)))\|_\infty \le C(\|F_a(A)\|_\infty + \|A\|_{W_a^{1,p}}^2) \tag{CA6}$$

Section 80.5 / Result and exact scope

on a common small periodic CK weak chart, uniformly in a, r, and the blocking depth, with the physical torus and finite p>8 fixed. The common complex input ball with norm $W^{1,p} + \|dA\|_\infty$ therefore gives the claimed uniform holomorphic matrix-curvature bound and its parameter Cauchy estimates. No real-axis maximum principle or complex gauge bridge is used.

This closes the periodic same-target complex-curvature question for the specified branch. It does not prove an identification of the old off-shell paired coordinate maps, a common localized coarea transition, or the arbitrary-insertion mixing estimate AI45. The optional strong fractional consequence still uses the input strong norms and the earlier GS/WB theorems exactly as CC34 states.

Section 80.6 / Independent check of the physical residual extension RC

RC1–20 also pass. The sparse-site dual test has the exact physical factor a^{-1}(a/r){4/p'}, so its reciprocal is r^{4-4/p}a{-3+4/p}, as in RC5–7. This is an upper bound on the Ward residual obtained from the actual Noether identity, not an unjustified norm estimate for an arbitrary endpoint lift. The explicit multilinear coarse interpolant in RC8–9 separately gives the required Wˆ{-1,p} bound and is valid for every finite p'>1; it does not assert a point trace theorem for an arbitrary Sobolev function.

The exact raw gauge chain yields $\|q_j\|_p = (r/h_j)^{4-4/p}\|q\|_p$. Substitution into the full-space endpoint recursion gives its three q-dependent exponents exactly:

$$-2 + 4/p, \qquad -1, \qquad -3 + 4/p. \tag{CA7}$$

For p>8 all geometric sums are dominated by the final, finest-scale power aˆ{-3+4/p}. The sparse-site Noether estimate pays that power. Consequently RC16 is a valid unrestricted-current comparison **for this actual compatible residual**, with size CM squared plus $C\|E\|_p$. It is not the false unrestricted negative-Sobolev inverse disproved in FI.

The coarse longitudinal term uses the separate negative-norm estimate. Only q+R_B lambda is necessarily mean zero, and RC17 correctly lets G remove their individual constants. The fine particular solution then has curvature bounded by CM squared plus $C\|E\|_p$. Adding the raw constraint residual changes the coarse curl by exactly d_r b; no uncontrolled longitudinal part is measured. The transverse identity and FE endpoint estimates therefore prove the displayed RC3, uniformly in both lattice spacings on the fixed torus. Its hypotheses still include the actual physical residual equation and the weak multiplier bound; an arbitrary weakened localization has not been shown to satisfy them

Section 81.9 / Scope and the remaining gluing condition

BI3–4 prove the requested flat boxed Banach inverse. BI28 quantifies the perturbation for the stated actual Wilson Hessian, actual endpoint-removed A_U, and covariant site operator S_U. It is valid for small W1p reference charts with p>4, or for the stronger small C1,beta charts, with every scalar and coarse type retained.

It does not replace A_U by DM_raw on arbitrary currents. It does not say that A_U is the derivative of one globally integrated gauge connection; a separate stepwise nonlinear constraint can have exactly this derivative. It also makes no claim that the reference Hessian kills gauge directions off shell. Endpoint gauge cancellation remains an identity to prove for the completed nonlinear equations.

Finally, periodicizing a potential by a scalar cutoff is not equivariant under an arbitrary nonconstant change of background gauge. Consequently the periodic tile inverse just constructed cannot by itself be glued in different local gauges and called an exact common covariant parametrix. A gauge-equivariant extension convention or a proved local boundary inverse is required. Those are concrete additional tasks, distinct from the Banach inverse and smallness estimates established here. No original off-shell source map or coarea formation has been identified by this theorem.

Section 82 / A uniform Banach inverse for the boxed augmented system

This chapter supplies an explicit local extension for the covariant augmented inverse. It avoids choosing one global axial gauge on the physical volume. Each extension depends only on its complete local box, and its restricted inverse kernels transform covariantly under changes of the input gauge. The weak gauge used for estimates is not required to preserve an independently fixed coarse target.

Section 83.1 / Result and precise scope

There are two distinct conclusions.

First, the physical critical branch in CK cannot literally replace the original background response on its entire independent-(U, J) domain. The original response vanishes at zero fluctuation for every admitted background, including noncritical backgrounds. Section 2 gives a concrete counterexample to that identification.

Second, there is a constructive periodic alternative. The explicit local straightening in CM17 is a genuine nonlinear map, gauge equivalent to the actual raw average, and has the box-and-line derivative at the identity. Its full adjoint has a uniform fractional bound without a Ward restriction. Using this map in the flat augmented system AL gives a nonlinear Wilson wall family with:

- one common analytic domain and one common wall expansion for every type;
- the actual nonlinear Wilson action in the protected equations;
- cutoff-uniform weak bounds and protected fractional curvature/current bounds;
- an endpoint which is a genuine critical configuration for the raw constraint, after a controlled fine gauge correction.

The construction below is on a **single small periodic potential chart**, in retained units. It does not include wild regions, mixed determining boundaries, or the original independent-current extension. A gauge-invariant readout of the physical periodic endpoint can be identified using the published small-orbit uniqueness theorem. An entire already gauge-fixed fluctuation integrand requires more: its canonical coordinates, tests, and coarea density must be transported together. Section 9 states the exact finite-cutoff identity and the remaining quantitative obligation.

The flat inverse and common-carrier bounds used below are the proved AL1–AL27, not an assumed nonlinear extension of the background-propagator paper.

Section 83.3 / A true nonlinear constraint with a controlled full adjoint

Use centered odd blocking, fine spacings a_j, next spacings $\ell_j = La_j$, and final retained spacing $r \le 1$. Let q_j be the physical logarithm of the actual one-step reference-log mean. CM17 defines

$$g_j(A) = \exp[-r_j(A)], \qquad q_j^{\#}(A) = \ell_j^{-1} \log\Big(\mathcal{M}_j(e^{a_j A})\, g_j(A)\Big). \tag{NL7}$$

Here r_j is the actual flat endpoint connector, including its centered endpoint/path convention. It is a dimensionless linear local function, and

$$Dq_j^{\#}(0) = Q_{\mathrm{bl},j}. \tag{NL8}$$

This is an actual nonlinear map, not the assertion that the endpoint-removed connection $\mathcal{A}_j$ of CK is an integrable Jacobian.

The distinction matters. In general the averaged endpoint one-form $\rho = \sum w_i\, dP_i P_i^{-1}$ is not a Maurer–Cartan form. For two independent transports $P_1 = e^{xX}$, $P_2 = e^{yY}$, with nonzero weights and $[X, Y] \neq 0$, the mixed part of $d\rho - \rho \wedge \rho$ is nonzero. Thus one cannot integrate the removal of all endpoints merely by claiming $dh\, h^{-1} = \rho$. The explicit map (NL7) avoids that assertion.

83.3.1 3.1 Uniform row and column bounds

Suppose every intermediate analyzed potential has supremum at most M. This condition follows from a sufficiently small initial supremum: the finite analytic expansion and the literal flat contraction give

$$\|q_j^{\#}(A) - Q_{\mathrm{bl},j}A\|_\infty \le C_L \ell_j \|A\|_\infty^2, \qquad \sup_j \|A_j\|_\infty \le 2\|A_0\|_\infty, \tag{NL9}$$

since $\sum_j \ell_j \le C_L r$. For stronger $C^{1,\beta}$ intermediate bounds one may use CM/WP, but they are not needed in the following adjoint estimate.

Let $\mathcal{E}_a(A)$ convert a logarithmic tangent to the physical left-trivialized tangent. The physical differential of (NL7) is

$$\mathcal{Q}_j^{\#} = \mathcal{E}_{\ell_j}(q_j^{\#}(A_j)) Dq_j^{\#}(A_j) \mathcal{E}_{a_j}(A_j)^{-1}. \tag{NL10}$$

At fixed L, every row is a fixed finite analytic expression in the dimensionless potentials $\ell_j A_j$. At the origin it is exactly the normalized positive box-and-line row. Each input bond belongs to only a fixed number of determining patches. Therefore, with the physical volume normalization of the transpose,

$$\begin{aligned} &\|\mathcal{Q}_j^{\#} - Q_{\mathrm{bl},j}\|_{L^p\to L^p} + \|(\mathcal{Q}_j^{\#})^{\mathsf{t}} - Q_{\mathrm{bl},j}^{*}\|_{L^p\to L^p} \le C_{L,p}\ell_j M, \\ &\|(\mathcal{Q}_j^{\#})^{\mathsf{t}}\|_{L^p\to L^p} \le 1 + C_{L,p}\ell_j M. \end{aligned} \tag{NL11}$$

One can first prove the absolute row and column sums, then interpolate. The factor L^4 in the physical adjoint is canceled by the normalized one-step weights, exactly as in LR1–LR3. No derivative of curvature, or curvature supremum, occurs here.

The tangent chart factors at intermediate endpoints telescope when these physical differentials are composed. They must not be inserted anew at every step with an independent fixed norm loss.

83.3.2 3.2 A flat-tail lemma in $W^{s,p}$

For

$$1 < p < \infty, \qquad 0 < s < 1/p, \tag{NL12}$$

the entire flat adjoint tail from one intermediate mesh to a finer mesh has a uniform $W^{s,p}$ bound. It is the direct block hold followed by normalized line averages. Line averages contract translation-difference norms.

For block hold, a translation by h, $|h| \le b$, changes only a layer of relative width $C|h|/b$ at a mesh-b interface. Its contribution is bounded by

$$\|\tau_h H_b f - H_b f\|_p^p \le C\frac{|h|}{b}\sum_\nu \|\tau_{be_\nu} f - f\|_{p,b}^p. \tag{NL13}$$

Integrating against the fractional weight gives a finite factor $\int_0^b t^{-sp}\,dt$, precisely because $sp < 1$. The part $|h| \ge b$ is compared to coarse translations, with the leftover fraction of a cell estimated by (NL13). This proves the assertion for the discrete Slobodeckij norm, uniformly in the ratio of meshes and under periodization. Equivalent discrete fractional norms may be used with one fixed equivalence constant at the end.

83.3.3 3.3 Full-adjoint fractional bound

A fractional fine inverse inequality applied to (NL11) gives

$$\|[(\mathcal{Q}_j^{\#})^{\mathsf{t}} - Q_{\mathrm{bl},j}^{*}]v\|_{W_{a_j}^{s,p}} \le C_{L,s,p}\ell_j^{1-s}M\|v\|_{L^p_{\ell_j}}. \tag{NL14}$$

This does not differentiate the coefficient fields. It uses $a_j^{-s} = L^s\ell_j^{-s}$.

In the exact product difference, put the **whole flat fine-prefix** to the left of the distinguished error and the nonlinear coarse-suffix to its right. The left prefix is controlled by (NL12)–(NL13); the right suffix has norm at most $\exp(CM\sum \ell_j)$ by (NL11). Summing $\sum \ell_j^{1-s} \le C_{L,s}r^{1-s}$ proves

$$\boxed{\|(\mathcal{Q}^{\#}(u))^{\mathsf{t}}\lambda\|_{W_a^{s,p}} \le C_{s,p,L}\left(\|\lambda\|_{W_r^{s,p}} + Mr^{1-s}\|\lambda\|_{L_r^p}\right).} \tag{NL15}$$

The weaker L^p and supremum bounds follow from (NL11). The logarithmic derivative $Dq^{\#}(u)^{\mathsf{t}}$ obeys the corresponding bound whenever its fine endpoint chart factor is a bounded $W^{s,p}$ multiplier; in the $C^{1,\beta}$ ball used below this follows directly. At retained spacing one the coarse chart factors have uniformly bounded finite differences.

This lemma is for arbitrary coarse λ. There is no Ward assumption.

Section 83.10 / A separate source-transport caution discovered in this pass

The generic formula in the argument (6.2)/(13.18),

$$r_0(y,b)\,D_0(b)^{-1}\int q_0(\xi,b)M(\xi,b)\,d\xi \tag{NL43}$$

is mass preserving when $D_0(b)=\int r_0(y,b)\,dy$ and depends only on the retained exterior datum. A correction to the initial reading is essential: the legible R-II (1.10) contains $U_{k,Z}(V'V_\Lambda)$, with the relaxed reference V_Λ, rather than an arbitrary untranslated interior V_k. Equation (1.3) fixes the constraints only on $Z\setminus\Lambda$; the paragraph following (1.34) says that V_Λ equals the original field outside Λ. Equations (1.38)–(1.43) construct its analytic critical continuation. These are substantial evidence for an exterior-determined reference orbit. They must be used instead of treating the denominator as a generic convolution at the new field.

Here is the exact gauge part of that repair. If two reference representatives with the same exterior obey $K_2=K_1^h$, $h=1$ on the retained boundary, then the relative variable changes by

$$\eta_b\longmapsto h(x_b)\eta_b h(x_b)^{-1},\qquad \eta_b K_{1,b}\longmapsto(\eta K_1)_b^h. \tag{NL43a}$$

Product Haar measure, a class-norm cutoff about 1, and a relative tree delta $\prod_{b\in G_0}\delta_1(\eta_b)$ are unchanged. The Wilson reference action is unchanged by gauge covariance of its reconstruction. Its denominator is consequently independent of this gauge choice.

Section 83.10 / A separate source-transport caution discovered in this pass

To identify (NL43) literally, one still has to match the original old/new integration coordinates: the output reference factor must be integrated with this same relative Haar/cutoff/tree measure. In particular, a gauge choice depending on the variable currently being integrated cannot be treated as a fixed Haar substitution without the original simultaneous gauge or Faddeev–Popov identity. Printed p.378 does retain χ and $\delta_{G_0}(V_k')$, $V_k' = V_k V_\Lambda^{-1}$, until the completed small-component split; this is the correct place to check that match. Here b includes every frozen shell/determining variable and all the omitted $\widetilde{T}_k, T_h, V_h$ variables, not just the geometric exterior of Z. The proof of Proposition 1 on printed p.359 explicitly fixes a regular extension from $Z \cap \Lambda^c$ before solving its relative critical equation. That reference may therefore be fixed once for b, independently of the later current output variable.

In general $r(y)(Kq)(y)/(Kr)(y)$ does not preserve the integral of q. The later normalization $T/T1$ in R-II (1.103) does not establish the missing physical-source identity. This question is being the analysis separately source/large-field module. No result in Sections 3–8 uses the generic mass-preserving replacement as a premise.

Section 84 / An actual nonlinear constrained wall family and the integration identity it does not yet replace

This chapter combines the actual one-step endpoint transports with the covariant periodic extension CE and the five-type inverse BI. It constructs a nonlinear wall family whose individual values are gauge covariant. Its physical endpoint agrees with a genuine regular critical input. It does not identify the original independent-current response away from that physical graph, and it does not replace mixed determining constraints by a single uniform blocking constraint.

Section 84 / An actual nonlinear constrained wall family and the integration identity it does not yet replace

All spacings and norms below are physical. Rescale the retained spacing to one; the fine spacing is $a = L^{-k}$, with fixed odd L. Fix $p > 8$ and $0 < s < 1/p$. Constants may depend on these choices and the fixed localization geometry, but not on a, k, or the number of regular boxes. Transposes in identities are complex-bilinear invariant transposes. Norms are absolute norms.

Use the complete products A_U, S_U above and the actual relative-chart Wilson Hessian $H_{\log,W}(U)$. The linear matrix is exactly BI26:

$$\mathbb{K}_U \begin{pmatrix} b \\ \psi \\ \phi \\ \nu \\ \mu \end{pmatrix} = \begin{pmatrix} [H_{\log,W}(U) + A_U^{\mathsf{t}} \kappa A_U] b + d_U \Delta_U \phi + A_U^{\mathsf{t}} \mu \\ \Delta_U \psi + S_U^{\mathsf{t}} \nu \\ -\delta_U b + \psi + \Delta_U \phi \\ S_U \phi \\ A_U b \end{pmatrix}. \tag{CL19}$$

The coarse positive weight κ is fixed and bounded above and below. The Hessian includes its chart-current term; it is not assumed to annihilate gauge directions at a noncritical reference.

BI proves its inverse, in a small $W^{1,p}$ reference chart, on

$$\begin{aligned} X_p &= W^{1,p}(b) \oplus L^p(\psi) \oplus W^{2,p}(\phi) \oplus \ell^p(\nu, \mu), \\ Y_p &= W^{-1,p} \oplus W^{-2,p} \oplus L^p \oplus \ell^p \oplus \ell^p, \end{aligned} \tag{CL20}$$

with exponentially summable cell majorants. It also proves the stronger $\psi \in W^{1,p}$ variant when the second source is $W^{-1,p}$.

On a regular carrier the local inverse is defined using CE's canonical tree, cutoff, periodization, and undoing of that same tree gauge. This is an intrinsic connection extension, not a periodization in an arbitrary unrecorded potential chart. CE proves exact covariance of its restricted kernels, including the artificial periodic boundary. Each extension and all its tree paths are filed on its whole fixed box.

The norms transported back from a weak analysis chart are covariant Sobolev norms. Bounded adjoint transports alone do not control ordinary derivatives through a rough tree gauge. In the weak chart, $\nabla_U - D$ is a zero-order $O(w)$ coefficient, and $\nabla_U^2 - D^2$ consists of wD, Dw, w^2. For $w \in W^{1,p}$, $p > 4$, the covariant $W^{1,p}$ and $W^{2,p}$ type norms are uniformly equivalent to the ordinary ones; negative norms follow by duality. Exact covariance then transports those covariant norms with the bounded pointwise adjoint factors supplied by CE.

With the scalar partition used in AL, the local inverse obeys

$$\mathbb{K}_U(\chi_z \Gamma_z(U) \chi_z) = \chi_z^2 + [\mathbb{K}_U, \chi_z] \Gamma_z(U) \chi_z. \tag{CL21}$$

The identity uses an inner collar containing every complete stencil of every field type. The three terms are assigned one and the same complete carrier. BI's estimates and covariant product rules bound the commutators by C_p/M_1. Choose M_1 first, and then the field ceiling so CE's weak chart lies inside the fixed BI ball. The complete-carrier series of AL gives $\Gamma_{U,\mathbf{s}}$ with a common complex wall-polydisc and exact gauge covariance.

Section 84.4 / The covariant Banach inverse and its local definition

This choice is noncircular: the extension constant may grow with the already fixed box size, but the weak-reference perturbation radius is fixed before that field ceiling is decreased. At zero walls the maps depend on their actual connected components. It is the subsequent connected expansion, not a false assertion about arbitrary disconnected mixed derivatives of a product, that organizes connected source activities.

Section 84.10 / Scope after the uniform-scale construction

For regular single-scale carriers, the construction now provides the following concrete items in one common analytic domain:

- a genuine nonlinear constraint built from the actual endpoint transports and actual raw mean;
- exact joint gauge covariance of every wall value, through CE's intrinsic local extension and BI's complete typed inverse;
- the physical protected Euler equation and the compatible scalar equation (CL27), with the fractional and low bounds (CL28), (CL33);
- exact matching (CL38) on the original physical critical graph.

The remaining model-specific step is to identify the readouts used at every original formation with that graph, or to prove the analogous theorem for their actual mixed determining constraints. That requires a single compatible augmented inverse for the layered constraint rows and their scalar gauge map, with the own-scale norms and all collar carriers retained. It is not supplied by the single-scale inverse above.

The separate conditional replacement/source-mass identity in R must also retain the actual relative reference fiber measure. Neither a new classical saddle nor equality (CL38) changes an old normalizing denominator or proves a statement about a different source-bearing history integral. The next section closes the actual nonlinear endpoint algebra for the two-scale periodic model MB; it does not yet assemble all original determining layers or action boundaries.

Section 84.11 / A genuine two-scale nonlinear seam, with assigned node depths

Use exactly the mixed geometry in MB1: refined blocks have retained spacing one, unrefined blocks have retained spacing L, and the optional inward unit rows retain their junction nodes. Let I be the unit and junction nodes and C the unrefined block centers. The common microscopic spacing is $a = L^{-k}$. Every actual mixed raw row is the corresponding row of the prescribed k- or $(k+1)$-step composition. Denote that mixed connection by $M_m(V)$; no direct all-depth mean is substituted.

Let $G_j(U,V)$ be the exactly composed relative coarse gauge in (CL15), so that the uniform relative output at spacing $L^j a$ is

$$V_j = M_j(V)^{G_j(U,V)}. \tag{CL39}$$

Assign a depth to every actual mixed node, including junctions:

$$G_m(U,V)(y) = \begin{cases} G_k(U,V)(y), & y \in I, \\ G_{k+1}(U,V)(y), & y \in C. \end{cases}$$

Define the mixed relative connection by the literal nodal gauge transformation

$$\mathcal{B}_{m,U}(V)_e = G_m(U,V)(e_-)\, M_m(V)_e\, G_m(U,V)(e_+)^{-1}. \tag{CL40}$$

The same node factor is used on every edge incident to a node. Thus (CL40) is a well-defined gauge transformation of the actual mixed raw connection, not independent changes of its row targets. It is analytic, local on complete retained carriers, jointly gauge covariant, and equals $M_m(U)$ when $V = U$.

84.11.1 11.1 The outer-endpoint factor has an exact order

Write $U_1 = M_k(U)$, $V_1 = M_k(V)^{G_k(U,V)}$, where the subscript one now denotes retained spacing one. Set

$$t(z) = h(U_1, V_1)(z).$$

The next uniform relative output is $V_L = \mathcal{M}(V_1)^t$, and the exact composition gives

$$G_{k+1} = t\, N_{L,1} G_k. \tag{CL41}$$

For a long mixed row e ending at an outside unit node $z = e_+$, substitution into (CL40) gives

$$\mathcal{B}_{m,U}(V)_e = V_L(e)\, t(z). \tag{CL42}$$

If instead $z = e_-$ is the outside tail, the formula is $t(z)^{-1} V_L(e)$. At two coarse endpoints there is no seam factor; on a unit or retained inward row the output is $V_1(e)$. These noncommutative orders follow by cancellation of the adjacent $G_k(z)$ factors and must not be reversed.

The factor $t(z)$ reads only the internal unit links of the refined L-block centered at z, through the actual endpoint transport. All those links are retained unit rows in MB's geometry. Consequently the map from the uniform product rows to (CL40) is triangular: it leaves the unit rows unchanged, and changes a coarse outward row only by a function of those unchanged internal rows. Its inverse uses the opposite side factor computed from the same internal rows. In particular a prescribed mixed target determines its seam correction without inspecting an unprescribed field. When the internal target equals the reference, $t(z) = 1$.

At the identity the differential of this triangular map is $T_m = I - C$ in MB22–24, with $C^2 = 0$. The sign difference between the connector called K there and ρ in (CL7) is the fixed gauge-generator convention; the resulting physical derivative is

$$D\mathcal{B}_{m,1}(1) = Q_{\mathrm{red},m} = T_m Q_m. \tag{CL43}$$

Thus the relevant flat comparison is $Q_{\mathrm{red},m}$, not Q_m with its different outside-endpoint incidence.

84.11.2 11.2 The mixed full scalar cocycle is still a site map

With U fixed, a fine gauge g changes (CL40) by the actual mixed nodal gauge

$$\gamma_m(U,V;g)(y) = G_m(U,V^g)(y)\, g(y)\, G_m(U,V)(y)^{-1}. \tag{CL44}$$

At a node of depth j this is the composed scalar cocycle of that very prefix. The cancellation (CL11)–(CL12) therefore holds at each step. Its differential $\mathcal{S}_{m,U,V}$ is the actual prefix site map at that node's assigned depth. There is no new point-supported term at a junction and no raw fine derivative of the gauge parameter.

If $\mathcal{Q}_{m,U,V}$ is the physical differential of (CL40), then

$$\mathcal{Q}_{m,U,V} d_V = d_{\mathcal{B}_{m,U}(V),\,\mathrm{node}}\, \mathcal{S}_{m,U,V}. \tag{CL45}$$

Here the derivative on the right is incidence on the actual mixed node graph, divided by the physical length of each row. At the identity the two operators are $Q_{\mathrm{red},m}$ and the scalar S_m of MB. Retained inward rows use the depth-k factor at both endpoints; their junction scalar averages are essential to this identity.

The full-space bounds also persist. Split the construction into the common fine-to-unit prefix followed by this one mixed top step. The prefix has (CL17)–(CL18). The top step has finitely many unit coordinates per L-block, uniformly bounded overlap, and uniformly bounded physical analytic kernels. Its perturbation from (CL43) and from the top scalar map is $O_L(M)$ on a common small supremum chart. In particular, for $p > 8$ and $0 < s < 1/p$,

$$\begin{aligned} \|\mathcal{Q}^{\mathsf{t}}_{m,U,V}\lambda\|_{W^{s,p}_a} &\le C_{L,p,s}\|\lambda\|_{\ell^p_m}, \\ \|\mathcal{S}^{\mathsf{t}}_{m,U,V}c\|_{W^{s,p}_a} &\le C_{L,p,s}\|c\|_{\ell^p_m}. \end{aligned} \tag{CL46}$$

The same statements hold in the uniformly local versions and at the L^p, L^∞ endpoints without the fractional derivative. One can prove (CL46) directly by a whole flat fine-prefix Duhamel expansion: each error costs $\ell_j^{1-s}M$, these scales sum geometrically, and the single mixed top map costs only C_L. No C_L factor is iterated over the microscopic depth. No Ward restriction on λ or c is used.

84.11.3 11.3 The nonlinear periodic two-scale theorem

Use the genuine mixed logarithmic constraint

$$q_{m,U}(b)_e = r_e^{-1}\log\Big[\mathcal{B}_{m,U}(e^{ab}U)_e\, M_m(U)_e^{-1}\Big], \qquad r_e \in \{1, L\}. \tag{CL47}$$

Its derivative at zero is the mixed compatible $A_{m,U}$, whose flat value is (CL43). Let $S_{m,U}$ be the corresponding scalar map at zero. The covariant five-type matrix formed with these operators is an $O_{L,p}(M)$ perturbation of the MB20 matrix with $Q_{\mathrm{red},m}$, by (CL46), BI's relative-chart Wilson Hessian estimate, and the same weak covariant product bounds. MB24 supplies the exact flat kernel/coercivity equivalence, so its inverse is uniform for a fixed small M.

Consequently the construction in Sections 4–8 applies to this periodic two-scale constraint, with the row weights r_e^4, row masses comparable to r_e^{-2}, and junction scalar rows kept. The CE artificial box also carries a fixed periodic extension of the two-scale pattern; it agrees with the actual pattern on the entire inner carrier. The complete local carrier contains the seam factors, endpoint trees and the artificial pattern. Every reference and nonlinear vertex uses the weak filing convention preceding (CL25).

The protected physical equation has the same form (CL26), with the mixed operators. Its Noether consequence is

$$\Delta_W \omega = -\mathcal{S}^{\mathsf{t}}_{m,U,W} \delta_{\mathcal{B}_{m,U}(W),\,\mathrm{node}} \lambda. \tag{CL48}$$

The mixed coarse incidence has bounded ℓ^p_m norm at these two retained lengths. Thus the entire bootstrap (CL28)–(CL33) holds in the smaller retained units, with constants depending on L, p, s, and the fixed carrier geometry only. The same complex-bilinear scalar onto argument eliminates ω, ϕ at the unweakened endpoint. For a genuine small periodic mixed critical input the actual nodal Ward identity kills the extra vertical differential in (CL40), so $b = 0$ with its original multiplier solves the system. Uniqueness again proves $W_{\mathbf{1}}(U) = U$ exactly in the input gauge.

This is a theorem for the specified periodic two-scale geometry. It now includes the nonlinear seam and compatible Ward identity; those are no longer merely flat target-recovery hypotheses. Global many-scale localization still requires a cover whose full row carriers meet only adjacent layers, the scaled commutator estimates on overlaps, and the actual treatment of action or hole boundaries. CMP 102's adjacent-layer cube geometry is a concrete starting point, not by itself a replacement for those checks.

84.11.4 Inputs and primary identification

The actual endpoint transports and raw averaging conventions are those of Bałaban I, CMP 109 (1987), (0.4)–(0.12), including its inner symmetric group mean. The finite one-step differential identities and norms were proved in CK and PG, with the actual reference-log outer mean retained. The present relative gauge (CL3) is a newly defined map using those actual transports; it is not attributed to the original paper.

The local periodic extension CE and typed inverse BI are proved in the argument notes. AL supplies the common-carrier random-walk algebra. WC/GS supply the weak local Coulomb analysis gauge, and WB supplies the exact weak Wilson error identity used in (CL29). No unavailable source formula is invoked to identify the new off-shell maps with the old ones.

Section 85 / A covariant nonlinear wall reconstruction on regular carriers

This note proves a local mixed-scale replacement for the periodic single-scale inverse BI. The local model has retained lengths 1 and L, with arbitrary unions of aligned L-blocks assigned to the coarser scale. The constants are uniform in the fine mesh, volume, and shape of that union. An arbitrary subset of inward unit bonds may be retained as well; their junction nodes are kept explicitly.

The proof does not infer a mixed theorem from the single-scale labels. It verifies the row factorization, scalar kernel inclusion, field-row inf-sup estimate, and mixed gauge-slice coercivity. A final exact linear identity isolates the outer-endpoint correction needed for the raw constraint. Global many-scale gluing and the nonlinear seam factors are separate statements unless explicitly established below.

Section 85.3 / A field-row right inverse with actual interface support

Here is the row pivot calculation. For a retained length r=Na, N=2m+1, and a field supported strictly inside one endpoint pillar, take only component mu nonzero and set

$$A_\mu(y+au) = b\,\varphi(u_\mu)\prod_{\nu\neq\mu}\chi(u_\nu). \tag{MB7}$$

The right and left full Q_r rows touching that pillar count the longitudinal fine bond respectively u+m+1 and m-u times. Put

$$\beta = N^{-1}\sum_u \chi(u), \qquad \sigma = N^{-1}\sum_u (u/N)^2\chi(u).$$

Their response functionals are

$$\ell_+(\varphi) = \beta^3 N^{-2}\sum_u (u+m+1)\varphi(u), \qquad \ell_-(\varphi) = \beta^3 N^{-2}\sum_u (m-u)\varphi(u). \tag{MB8}$$

Section 85.3 / A field-row right inverse with actual interface support

Choose an even nonnegative profile chi with positive beta and sigma. The two exact dual profiles are

$$\begin{aligned}\varphi_+(u) &= \beta^{-4}\chi(u) + \frac{m}{N}\frac{\beta^{-3}}{\sigma}\frac{u}{N}\chi(u),\\ \varphi_-(u) &= \beta^{-4}\chi(u) - \frac{m+1}{N}\frac{\beta^{-3}}{\sigma}\frac{u}{N}\chi(u).\end{aligned} \tag{MB9}$$

Evenness gives sum u chi=0. Substitution proves

$$(\ell_+(\varphi_+), \ell_-(\varphi_+)) = (1,0), \qquad (\ell_+(\varphi_-), \ell_-(\varphi_-)) = (0,1). \tag{MB10}$$

There are no other same-level responses: the transverse support is in one centered block, and the other vector components vanish.

For a unit mixed row assign an endpoint outside Omega and use these profiles inside its unit pillar. This also works for an inward row. For an L-row choose an endpoint in C and place the profile in its central UNIT child, not merely in an unspecified part of the L-pillar. Use chi supported in a fixed middle part of that unit child. In the variables of MB8 its beta and sigma have positive lower bounds depending on L; its first physical differences are bounded by C_L. It is consequently invisible to every retained unit row, including inward rows, whose carriers are unions of outside or junction unit pillars.

The small fine meshes require no limiting argument. For n>=3 below the smooth-profile range use the two longitudinal positions -1,0 inside the central unit child and the center in each transverse direction. Their two response columns have determinant -N^{-9}; hence they are independent. There are only finitely many such n. When n=1, unit rows are literal fine bonds. A unit seed is that bond itself. For an L-row the fine bonds at positions -1 and 0 around its coarse center are interior to the unrefined L-block, so they are not any retained inward unit row. Their same two moments are independent for L>=3. These cases have fixed L-dependent bounds as well.

Let E^0 be the sum of these seeds. At most eight seeds use one assigned endpoint. Their supports and first-difference collars have bounded overlap. Thus

$$\|E^0 b\|_{W_a^{1,p}} \le C_{L,p}\|b\|_{\ell_m^p}, \qquad Q_m E^0 = I + T, \tag{MB11}$$

where T maps unit-row data to L-row data and all other blocks vanish. In particular T^2=0. Its norm is bounded by C_L using the normalized averaging kernels and the displayed seed bound. It follows exactly that

$$E_{\mathrm{row}} = E^0(I-T), \qquad Q_m E_{\mathrm{row}} = I, \qquad \|E_{\mathrm{row}} b\|_{W_a^{1,p}} \le C_{L,p}\|b\|_{\ell_m^p}. \tag{MB12}$$

This proves the inf-sup estimate. It does not iterate an uncontrolled constant over the microscopic depth. All determining supports lie in a fixed L-dependent enlargement of their original row carriers.

Section 85.5 / The full five-type mixed Banach inverse

Use Q_m,S_m in the complete five-type system:

Section 85.5 / The full five-type mixed Banach inverse

$$\begin{aligned}(\Delta_1 + Q_m^*\kappa Q_m)A - d\psi + Q_m^*\mu &= f,\\ \Delta\psi + S_m^*\nu &= g,\\ -\delta A + \psi + \Delta\phi &= h,\\ S_m\phi &= k,\\ Q_m A &= B.\end{aligned} \tag{MB19}$$

The scalar saddle inverse MB4–6, projected coercivity MB18, and field inf-sup MB12 prove its energy inverse exactly as in AL14–16. Equivalently eliminate the scalar block: its homogeneous contribution to the first row is d R_m delta, and its prescribed sources give a bounded Hˆ{-1} forcing. The remaining constrained positive operator is delta d+d R_m delta+Q_mˆ* kappa Q_m. MB18 and MB12 give its saddle inverse. Restoring the scalar variables proves the full typed result, including all harmonic and scalar constant modes.

Conjugate the matrix by smooth physical exponential weights. Every average has range O(L); commutators of differential entries have the same lower orders as in AL. Thus the energy inverse has exponential unit-cell bounds, with rates and constants depending on L only. Its mixed adjoints send bounded coarse values to bounded fine values: all site and row averaging kernels have bounded physical L2 norms, bounded normalized row/column sums, and only a fixed overlap. The scalar junction overlaps do not introduce a fine-mesh factor.

The coefficient-localization and literal Poisson proof BI5–13 now applies without a change of principle. For every 2<=p<infinity,

$$\begin{aligned}X_{m,p} &= W^{1,p}(A) \oplus L^p(\psi) \oplus W^{2,p}(\phi) \oplus \ell^p_m(\nu,\mu),\\ Y_{m,p} &= W^{-1,p}(f) \oplus W^{-2,p}(g) \oplus L^p(h) \oplus \ell^p_m(k,B),\\ \|K_m^{-1}\|_{Y_{m,p}\to X_{m,p}} &\le C_{L,p}.\end{aligned} \tag{MB20}$$

There is also the stronger psi W1p/g Wˆ{-1,p} variant. Both estimates have exponentially summable unit-cell majorants. Uniformity includes the number and shape of refined/coarse components. The proof is on the periodic ambient graph; it has not replaced the actual action boundary by a periodic or Dirichlet boundary without comparison.

For the statements about the actual fixed-L hierarchy, take n=Lˆk. For a reference in one small uniformly local W1p chart, p>4, the endpoint-removed PRODUCT rows at the assigned retained lengths, and their prefix site rows at the junctions, have full-space row/column error O_L(M) from Q_m,S_m. BI's geometric sums stop at length at most L, and the retained carriers have bounded overlap. The Wilson Hessian and covariant differential errors are exactly those in BI17, BI23. Thus this product-row five-type matrix is an O_{L,p}(M) perturbation in MB20 and has the same uniform inverse. The actual NODE-compatible relative map instead has the flat derivative Q_red,m of Section 6, not Q_m. For that map the comparison must be made to the Q_red,m version of MB20 proved there. This distinction retains its order-one seam correction. Both perturbation statements include the current-dependent relative-chart Hessian term and do not require the reference to be stationary.

Section 85.6 / The flat outer-endpoint correction is triangular

The preceding mixed Q_m uses the direct box-and-line rows. To relate it to actual raw constraints, retain the true composed connectors. In this section and the raw-recovery section, n=Lˆk so the unit

Section 85.6 / The flat outer-endpoint correction is triangular

prefix is precisely the actual fixed-L composition. Write Q_raw,1=Q_1-d_1K_1 and Q_raw,L=Q_L-d_LK_L. The exact one-step composition is

$$K_L = N_{L,1}K_1 + \rho_{L,1}Q_1. \tag{MB21}$$

Here rho is the actual centered endpoint connector, acting on unit one-forms. It reads only internal unit rows in its L-block. At every actual mixed node use the connector at that node's assigned level: K_node=K_1 on I and K_node=K_L on C. Let D_node be ordinary oriented incidence on the actual mixed nodes, divided by the length of its row.

For a coarse outward row with outer endpoint z define

$$(Cb)(c) = \frac{\varepsilon(c,z)}{L}\rho_{L,1}(b_{\mathrm{int}})(z), \qquad \varepsilon(c,z) = \begin{cases} +1, & z = c_+, \\ -1, & z = c_-. \end{cases} \tag{MB22}$$

Set C=0 on all other rows. The internal unit rows in b_int lie in the outside L-block at z and are all retained. This map reads only unit internal rows and outputs only coarse outward rows. Hence

$$C^2 = 0, \qquad T_m = I - C, \qquad T_m^{-1} = I + C, \qquad \|T_m\|_{p\to p} + \|T_m^{-1}\|_{p\to p} \le C_{L,p}. \tag{MB23}$$

The physical Lˆ{-1} in MB22 compensates the connector length O(L). Its bounded finite-stencil coefficients also prove the displayed norm bound, with no microscopic trace estimate. Substitution of MB21 at the outer endpoint gives the exact mixed identity

$$Q_{\mathrm{raw},m} = Q_{\mathrm{red},m} - D_{\mathrm{node}}K_{\mathrm{node}}, \qquad Q_{\mathrm{red},m} = T_m Q_m, \qquad Q_{\mathrm{red},m}d = D_{\mathrm{node}}S_m. \tag{MB24}$$

For the last identity use K_node d=S_m-N_node together with the exact raw gauge identity Q_raw,m d=D_node N_node. It also verifies the orientation signs in MB22. The same formula covers retained inward rows: their two endpoints both use level-one connectors, and their junction scalar row was included in S_m from the start.

Because T_m is boundedly invertible, Q_red,m has the same kernel, coercivity, and inf-sup bounds as Q_m. Its mass form Q_red,mˆ* kappa Q_red,m is positive and equivalent to the previous row mass. It still has bounded retained range and bounded physical kernel sums. Therefore the complete theorem MB20 also holds with Q_red,m replacing Q_m. In this version the gauge chain uses actual node incidence, rather than the outside block average in MB3.

Section 85.10 / Every component has its own scale factor

The physical row mass is kappa(c)=kappa_0(c) r_cˆ(-2), with fixed positive bounds on kappa_0. In a tile of base length ell use x=ell x' and the following CONSTANT local changes of variables:

$$\widetilde{X} = (\ell A,\ \ell^2\psi,\ \phi,\ \ell^4\nu,\ \ell^3\mu),$$
$$\widetilde{Y} = (\ell^3 f,\ \ell^4 g,\ \ell^2 h,\ k,\ \ell B). \tag{MB30}$$

Section 85.10 / Every component has its own scale factor

All fields on the right are evaluated at ell x'. Fine physical volume becomes (a/ell)^4; a row or node of physical length r has rescaled volume (r/ell)^4. Its new mass coefficient is ell^2 kappa(c), bounded above and below because r/ell is 1 or L. Thus the rescaled matrix is literally the MB/CL matrix, not one whose different field types have been assigned the same dimension. For example d Delta phi and the Wilson Hessian acting on A both acquire ell^(-3), while Delta psi and S^*nu acquire ell^(-4). The third row acquires ell^(-2).

For definiteness one may define the global adapted X norm by taking the local rescaled positive Sobolev norms on fixed enlargements of the primary pillars, followed by an ell^p sum (or a supremum for the uniformly local norm). Coarse variables use the individual rescaled node/row values in MB30, with their physical volume convention. For the negative rows use the corresponding quotient norm of sums of locally supported distributions: first represent a distribution as derivatives of coefficient fields, as in BI5, partition those coefficients on the primary pillars, and then use each pillar's constant rescaling MB30. Smooth fixed-size local cutoffs give the equivalent local negative-Sobolev description. This definition does not presume that a global negative-norm representation has local support. Bounded overlap and the fixed adjacent scale ratio give equivalence between the primary-cell norm and the cover norm.

At p=2 the positive fine terms are the familiar adapted energy norms:

$$
\begin{aligned}
\|A\|_X^2 &\asymp \|\sigma^{-1}A\|_2^2 + \|DA\|_2^2,\\
\|\psi\|_X^2 &= \|\psi\|_2^2,\\
\|\phi\|_X^2 &\asymp \|\sigma^{-2}\phi\|_2^2 + \|\sigma^{-1}D\phi\|_2^2 + \|D^2\phi\|_2^2,\\
\|\nu\|_X^2 &= \sum_v r_v^8 |\nu(v)|^2, \qquad \|\mu\|_X^2 = \sum_c r_c^6 |\mu(c)|^2,\\
\|k\|_Y^2 &= \sum_v |k(v)|^2, \qquad \|B\|_Y^2 = \sum_c r_c^2 |B(c)|^2.
\end{aligned}
\tag{MB31}
$$

The source negative norms are their dual energy types. These powers also show the physical pairing consistency of MB30. For general p the displayed local rescaling is the definition; no derivative is taken of a discontinuous field sigma(x). On overlaps the CONSTANT tile rescalings differ only by fixed adjacent-scale factors.

For a covariant reference use covariant differences in these norms. The CE weak analysis charts identify them with the ordinary MB norms on a tile. This retains covariance under a rough tree gauge without ever treating its ordinary derivatives as bounded.

Section 85.12 / A global mixed parametrix without a depth product

Let K_U be the actual mixed five-type matrix with the derivative of the nodal relative constraint CL40, its prefix site map, physical row masses r_cˆ(-2), and the actual Wilson relative-chart Hessian. On every interior tile its complete entries coincide with the two-scale model on the inner carrier. Use CE for its reference extension and MB20 with Q_red,m for its periodic inverse Gamma_z. On a bare boundary tile use MB32 with the literal action graph.

This entry equality follows from the definitions, not just from a kernel-diameter estimate. Keep the actual grid phase, assigned node depth, endpoint path family and logarithm branch on the local tile. The CE extension equals U on the union of every microscopic prefix read set for an inner row. Induction through the actual raw averages therefore makes all intermediate reference links equal there; the relative endpoint gauges and their derivatives agree by the same induction. Hence the local A_U and S_U coefficients, including the outer-node factor and every junction prefix, are literally the global coefficients. In the rescaled tile a/ell=Lˆ{-j}, so its fine-to-unit prefix has exactly the original j steps. The actual Wilson Hessian reads the same plaquettes. No independently chosen mean or shorter prefix is substituted.

Assume the reference has the own-scale regular bounds

$$\sup_p \sigma(p)^2 |F_a(U)_p| \le \epsilon, \qquad U_b = e^{a z_b} \bar{U}_b, \quad \bar{U}_b \in G, \qquad \sup_b \sigma(b) |z_b| \le \epsilon. \tag{MB33}$$

At a two-layer tile the factors converting these to the smaller local units are fixed powers of L. CE, rescaled by ell_z, gives one weak analysis chart with norm C_{R_loc,L,p} epsilon. Choose the field ceiling AFTER the localization radius so this lies in the fixed MB/BI perturbation ball. The inverse bounds are then the same volume-uniform C_{L,p} bounds as in MB20; growth of CE's extension constant changes the ceiling, rather than multiplying the inverse constant at each scale. The real case uses z=0.

In the locally rescaled types, the commutators obey

Section 85.12 / A global mixed parametrix without a depth product

$$\|[K_U,\chi_z]\|_{X\to Y}\le \frac{C_{L,p}}{R_{\mathrm{loc}}}. \tag{MB34}$$

Here are the entry checks. The order-three entry d_U Delta_U has an order-two commutator, bounded W2p to W^{-1,p}; every term has a cutoff difference. The Delta_U psi commutator maps Lp to W^{-2,p}; its differences remain in that weak type. The third row uses the usual first/second-order product rules. For Q,S and their transposes, each kernel entry has the factor chi_z(input)-chi_z(output). Its complete carrier has diameter O_L(ell_z), so this difference is O_L(R_loc^{-1}); the physical row/column bounds give the operator estimate. The same holds for the node-compatible seam factors and the row mass. Finally the Wilson Hessian is estimated in weak divergence/product form in the CE chart, retaining its relative-chart current term. Its commutator loses one differential order and has the same cutoff factor. No sharp wall multiplier acts on a preformed negative-Sobolev force. The analogous scalar biharmonic block uses its weak Delta_U–Delta_U form and has the same gain.

Define P= sum_z chi_z Gamma_z chi_z. All type restrictions and extensions are made on the same complete carriers. Then exactly

$$K_UP=I+E,\qquad E=\sum_z[K_U,\chi_z]\Gamma_z\chi_z,\qquad PK_U=I+E',\qquad \|E\|_{Y\to Y}+\|E'\|_{X\to X}\le\frac{C_{L,p}}{R_{\mathrm{loc}}}. \tag{MB35}$$

The left identity is obtained in the same way, commuting the input cutoff through K_U. Any global/local entry discrepancy vanishes because the support and every involved stencil lie in the matched inner carrier. Bounded cover overlap and the local inverse majorants prove the norm bounds in both the summed and uniformly local types. For negative sources the coefficient localization is performed before applying the cutoffs, as specified after MB30.

Choose R_loc large enough that the last bound is below one half. Then P(I+E)^(-1) is a right inverse and (I+E')^(-1)P is a left inverse. They coincide, and

$$\|K_U^{-1}\|_{Y\to X}\le C_{L,p}. \tag{MB36}$$

The construction has no product indexed by the number of determining levels. Each local comparison sees at most two lengths, and the remaining product is a convergent spatial parametrix series with ratio C/R_loc. Local exponential majorants, finite overlap, and the complete-carrier radius bound give an exponential majorant in the metric with line element ds/sigma(x), with a fixed possibly smaller rate. Equivalently, each step in the cover graph has bounded metric diameter O_L(R_loc), and its positive block majorant has two-sided sum below one. A sufficiently small fixed exponential weight leaves that sum below one.

For the common complex wall domain, attach wall variables to the ADAPTIVE LOCALIZATION cells of diameter R_loc sigma, not separately to every primary unit pillar used to define the norm. A complete local inverse box, including its CE extension and stencil collar, then meets at most C_L such wall cells, by the same bounded-overlap packing. File its P/I/T factors on that SAME complete box. File nonlinear force vertices in the weak divergence/product convention of CL25, before variation. For any fixed wall radius R_w>1 the weighted residual majorant is at most R_w^(C_L) C_{L,p}/R_loc. Choose R_loc after R_w so this is below one half, then choose the separation H and finally the regular-field ceiling. This gives one common polydisc without a hidden R_w^(c R_loc^4) factor from counting primary pillars inside a box. The reference extension and every wall coefficient remain gauge covariant by CE and CL's nodal construction.

Section 85.12 / A global mixed parametrix without a depth product

The scalar constrained biharmonic block can be glued with the same cover and weak commutators. At p=2 it gives, on ker S_U, $\|\phi\|_{H^2_{\mathrm{ad}}} \le C\|\Delta_U\phi\|_2$: apply its inverse to (Delta_Uˆ2 phi,0) and use Delta_U:H2 to L2 and its transpose. Its bilinear inverse also gives a bounded symmetric projection onto Delta_U ker S_U, including on the admitted complex domain. Thus the scalar onto step used in CL34–35 is not left as an unproved consequence of the one-form inverse.

Section 85.13 / The precise strengthened scope

MB36 is a global adapted weak inverse for the specified separated mixed hierarchy, with the actual compatible nodal relative derivative, actual inward-node convention, and pinned bare action boundary. Its assumptions include the quantitative separation, complete-carrier geometry, and own-scale regular reference domain stated above. It does not prove those regular reference properties for an arbitrary Wilson configuration. It also does not replace the relative nodal derivative by the unrestricted raw derivative.

The companion nonlinear construction CL39–48 proves the two-scale noncommutative seam identity that Section 8 left open. With MB36, its weak vertex convention and local protected bootstrap can now be applied in the own-scale types on this hierarchy. Any claimed identification with a particular original formation still has to use that formation's actual nested constraint fiber and boundary data. Neither this inverse nor equality of a classical critical graph changes a conditional reference measure, a normalizing denominator, or an operator mixing estimate.

Section 86 / A two-scale mixed inverse and its exact flat seam correction

This note combines the global mixed inverse MB27–36 with the genuine nodal relative constraint and physical-gradient augmentation CL. It proves a classical wall reconstruction on the stated regular mixed hierarchy, including inward junctions and its pinned bare boundary. The physical endpoint is exactly the input whenever that input is an actual constrained critical configuration.

The theorem is not an estimate for the original independent-current extension away from its physical graph. Nor does it prove that every inherited marked source stays in an invariant analytic core, or that a conditional normalizing denominator can be replaced. These distinctions remain after the classical mixed reconstruction has been constructed.

Section 86.1 / Geometry, a fixed analytic tube, and the theorem

Fix odd L>=3, p>8, and 0<s<1/p. Use the actual separated determining hierarchy of MB27: xi_j=Lˆj a, aligned nested regions, primary pillars, interior/outward rows, and the chosen subset of inward rows. Every retained inward endpoint has its finer-level junction scalar row. The physical row length is r_c and the assigned node length is r_y. The outer action boundary has the literal dynamic bond/site graph of the action. Fixed exterior or ghost variables are excluded from the unknowns; gauge variations there vanish. Every unknown in a boundary tile is pinned by Q_0=S_0=I, as supplied by the j=0 bare collar.

Fix a wall radius R_w>1. Choose the adaptive localization radius R_loc large enough for the weighted parametrix, then a separation H>=C_L R_loc, and finally the field ceiling. Wall variables belong to adaptive cells of size R_loc sigma, where sigma is the primary pillar scale. A complete inverse

box meets at most C_L wall cells. All P/I/T factors use the same complete box, with its full prefix carriers, and all nonlinear vertices are filed in weak form before variation. These are the conventions proved in MB and corrected in CL25; no sharp hold multiplies an already formed negative-norm force.

Here is a fixed Banach-coordinate domain for holomorphy. Fix a real compact reference bar U with

$$\sup_p \sigma(p)^2 |F_a(\bar{U})_p| \le \epsilon. \tag{MC1}$$

For U=exp(aZ) bar U impose

$$\|Z\|_{\mathcal{A}^{1,p}_{\bar{U}}} \le \epsilon, \qquad \sup_p \sigma(p)^2 |F_a(U)_p| \le \epsilon. \tag{MC2}$$

The first norm is the supremum over primary cells of the W1p norm of ell Z(ell x') in the fixed local analysis frame of bar U, or its equivalent covariant norm. The cell length is ell. On boundary cells ell=a and the scaled lattice has spacing one. The complex curvature bound is a separate condition; it does not follow from W1p alone. The relevant logarithm branches are those of the actual raw mean and relative endpoint transports. Their common validity follows from the small regular charts on the complete carriers.

Use fixed bar-U covariant adapted spaces. Their local scale factors are

$$\begin{aligned} X: &\quad (\ell b, \ell^2\psi, \phi, \ell^4\nu, \ell^3\mu), \\ Y: &\quad (\ell^3 f, \ell^4 g, \ell^2 h, k, \ell B), \end{aligned} \tag{MC3}$$

with local positive types (W1p,Lp,W2p,ellp,ellp) and negative types (W^{-1,p},W{-2,p},Lp,ellp,ellp). Take the uniformly local norm, defined by MB30 and its coefficient localization of negative rows. The W1p condition in MC2 makes these fixed spaces uniformly equivalent to the U-covariant types. In particular the differentiated connection coefficient in the scalar W2p type is controlled. MB's fiberwise inverse requires only the weaker supremum link-perturbation condition; that weaker condition is not silently called a fixed-norm analytic Banach tube here.

Mixed classical reconstruction theorem. There is a fixed epsilon_*>0, depending on the displayed fixed geometry and exponents but not on the fine cutoff, volume, or number of determining levels, with the following properties. On a smaller open tube MC1–2 and the common wall polydisc |s_Delta|<R_w, there is a unique small holomorphic five-type solution x_s=(b_s,psi_s,phi_s,nu_s,mu_s) of the localized augmented equations below, and

$$\|x_{\mathbf{s}}\|_{X,\mathrm{uloc}} \le C\epsilon, \qquad W_{\mathbf{s}}(U) = e^{ab_{\mathbf{s}}} U. \tag{MC4}$$

Every wall value is jointly gauge covariant. On an actual protected carrier, in its own units ell and with sufficiently many complete collars, it has

$$\begin{aligned} \|\ell^2 \omega_{\mathbf{s}}\|_{W^{2+s,p}} + \|\ell^3 J_{\Omega_0}(W_{\mathbf{s}})\|_{W^{s,p}} + \|\ell^2 F_a(W_{\mathbf{s}})\|_{W^{1+s,p}} \le C\epsilon, \\ \omega_{\mathbf{s}} = \Delta_U \phi_{\mathbf{s}}. \end{aligned} \tag{MC5}$$

The rescaled fields are evaluated at ell x'. The curvature/current norms are taken in the controlled final analysis frame, equivalently with covariant differences and transports of W_s. The auxiliary

omega norm refers to that same final frame. It is not an unproved high-derivative norm in the original rough reference frame.

At all walls equal to one, omega=phi=0 and W_1 is a true physical critical orbit for the actual mixed raw constraint, with the same prescribed coarse orbit as U. It has a bounded small fine gauge representative with exactly the original raw target. If U itself is an actual regular constrained critical input, then in its original input gauge

$$W_{\mathbf{1}}(U) = U. \tag{MC6}$$

The theorem makes no current stationarity assumption for an arbitrary reference U in MC2. The additional critical-input hypothesis is used only in the exact graph identity MC6.

Section 86.2 / The actual mixed relative constraint

For a node y of assigned depth j(y), use CL's composed relative gauge G_(j(y))(U,V)(y), with the exact actual prefix of that depth. Set G_m(U,V)(y) equal to that value. On an actual mixed row e define

$$\mathcal{B}_{m,U}(V)_e = G_m(U,V)(e_-)\, M_m(V)_e\, G_m(U,V)(e_+)^{-1}, \qquad q_U(b)_e = r_e^{-1} \log\left[\mathcal{B}_{m,U}(e^{ab}U)_e M_m(U)_e^{-1}\right]. \tag{MC7}$$

M_m is the actual prefix raw average on each determining row. The same G_m(y) is used on every row incident to a node, including inward junctions. At an outward seam this is the exact triangular relative map CL41–43: an outside head contributes right multiplication by t(z), and an outside tail contributes left multiplication by t(z)^(-1). That t reads only the finer internal rows of its outside block. This specifies its noncommutative order and its dependence on prescribed data. It is not a choice of an independent gauge for each row.

Let A_U=Dq_U(0), and let S_U be the fine-gauge cocycle derivative at b=0. At the identity these are Q_red,m and S_m, respectively. The full physical differential Q_(U,W) and full site cocycle S_(U,W) obey exactly

$$\mathcal{Q}_{U,W} d_W = d_{\mathcal{B}_{m,U}(W),\mathrm{node}} \mathcal{S}_{U,W}. \tag{MC8}$$

The cancellation of the retained node term is CL11–12 at each actual prefix. Thus S_(U,W) is a site average with analytic transport coefficients, not a derivative of a thin path force. Its assigned-depth definition is why MC8 survives at a junction.

On every enlarged adaptive tile, all microscopic prefix read sets lie inside its complete carrier. The periodic extension agrees with U there. Induction through the actual raw means and actual relative endpoint means makes the local and global intermediate fields and their derivatives identical. The rescaled fine mesh a/ell=L^{-j} has exactly the original j prefix steps. Consequently the genuine global coefficients in MC7–8 coincide with the two-scale CL/MB coefficients on the inner carrier. Diameter bounds alone would not prove this equality.

Section 86.7 / The global scalar onto estimate

At walls one, all equations hold globally. Use the actual scalar biharmonic inverse, glued separately in MB36, on the allowed fine scalar space with zero fixed exterior values. Let R_U be its bounded symmetric complex-bilinear projection onto Delta_U ker S_U. Its finite-cutoff formula is the corresponding constrained Delta_U–Delta_U Gram inverse. The scalar inverse and its type bounds prove

$$\zeta \longmapsto (R_U \Delta_U \zeta, S_U \zeta) : H^2_{\mathrm{ad}} \longrightarrow \operatorname{Ran} R_U \oplus \ell^2_{\mathrm{node,ad}} \quad \text{is uniformly onto.} \tag{MC19}$$

Here H2_ad is exactly MB31's adapted energy norm; the scalar range has ordinary fine L2 norm, and the target node norm is the count norm in MB31. Constants are fixed by the actual scalar averages, not dropped. Complex symmetry is obtained from the bilinear saddle inverse, not from a supposed Hermitian positivity of a complex projection.

The onto map with W instead of U in its first differential and with the full scalar cocycle in its second row,

$$\zeta \longmapsto (R_U \Delta_W \zeta, \mathcal{S}_{U,W} \zeta), \tag{MC20}$$

is a uniformly small perturbation. On each rescaled cell, Delta_W-Delta_U:H2 to L2 is $O(\|b\|_{W^{1,p}})$; the coefficient Db in Lp pairs with the H2 embedding into L^{2p/(p-2)}. The site-map difference is $O(\|b\|_{W^{1,p}})$ in the same assigned-depth row norm. Sum these energy bounds over primary cells using bounded carrier overlap. Their scale factors are exactly MC3/MB31, so no new local-length power or level count appears. MC4 makes the perturbation smaller than half the reference inverse bound. Therefore MC20 is onto on the whole hierarchy, with its actual boundary gauge restrictions.

Section 86.10 / What has now closed, and what remains independent

The proved classical package now has one compatible construction for the separated regular mixed hierarchy: its actual nodal seam, all junction scalar rows, its uniform adapted weak inverse, its literal bare boundary, one common gauge-covariant wall family, the protected physical Noether equation, the own-scale fractional curvature/current estimates, and exact physical-graph identity.

Three distinct domain statements should not be conflated. MB has a fiberwise coefficient/inverse bound under a supremum complex link perturbation and curvature bound. MC2 supplies a fixed first-order Banach tube for holomorphic nonlinear reconstruction. An inherited marked source needs, in addition, an all-depth invariant class of its actual completed regular and prepared rough-boundary readouts. The first two statements do not establish the third.

In particular the original independent-current off-graph response need not equal this physical-gradient extension. A source-only use of MC6 must identify the actual completed critical graph to which that source is applied, and must verify its locality and analytic range in the same marked-source norm. The strict old-scale/age remainder after dimension-filtered mixing remains an estimate to prove for arbitrary operators. Finite-dimensional counterterm spaces and finite-source jets do not supply that estimate alone.

Finally neither the new classical saddle nor the graph identity alters a conditional reference fiber measure, a coarea factor, an old numerator cutoff, or a normalizing denominator. The scalar source/naturality comparison must retain their actual placement. No quantum continuum existence or mass-gap conclusion follows from this classical closure by itself.

Section 87 / Nonlinear closure on the separated mixed regular hierarchy

This note separates a proved geometric scaling lemma from the still required norm of an actual marked birth. It does not assert that a new derivative-source class is invariant under the complete formation.

Section 87.1 / Mathematical scope

Normalize the current retained scale to one, with microscopic mesh a. Let X be a birth-unit carrier contained in a contractible cube of diameter at most D, and let its current physical image be $x_0 + rX$, $a \le r \le 1$. Assume a larger cube of radius CDr lies in one protected current regular window. On a fixed controlled analysis chart suppose

$$U_i(x) = e^{aA_i(x)}, \qquad \|A\|_{C^{1,\beta}} + \|F_a(U)\|_\infty + \|J_a(U)\|_\infty \le \epsilon, \qquad Dr \le c. \tag{DA1}$$

The field and the chart can be complex. The constants in this note may depend on fixed D and the admitted current chart, but not on a/r. CL33 or the protected own-unit MC estimate supplies this input on its proved windows.

Choose x_0 to be a microscopic site and put

$$S(x) = \sum_i (x_i - x_{0,i}) A_i(x_0), \qquad g(x) = e^{S(x)}. \tag{DA2}$$

In the convention $U^g(x, x + ae_i) = g(x)U_i(x)g(x + ae_i)^{-1}$, the logarithmic potential of U^g vanishes on the four forward bonds at x_0. More generally,

$$\begin{aligned} \|A^g\|_{\infty, x_0 + rX} &\le C_D r(\epsilon + \epsilon^2), \\ \|D_a A^g\|_\infty + [D_a A^g]_\beta &\le C_D(\epsilon + \epsilon^2), \\ \|g^{\pm 1}\|_\infty &\le e^{C_D r\epsilon}. \end{aligned} \tag{DA3}$$

There is no derivative of curvature in this assertion.

For an explicit verification, use

$$e^{aA_i^g(x)} = e^{S(x)} e^{aA_i(x)} e^{-S(x) - aA_i(x_0)}. \tag{DA4}$$

At $x = x_0$ this is exactly the identity. Subtract $A_i(x) - A_i(x_0)$ from a^{-1} log of (DA4). The remaining analytic expression is $O(|S(x)|\{|A(x)| + |A(x_0)|\}) = O_D(r\epsilon^2)$, with uniform first-difference and Hölder bounds on the fixed logarithm ball. This proves (DA3); the divisibility by a is kept before taking a fine difference. Equivalently one can differentiate the analytic finite-dimensional function in (DA4) with respect to its two physical endpoint variables. The coefficients and their differences are bounded uniformly as a tends to zero.

Use old coordinates $y = (x - x_0)/r$ and fine mesh $\xi = a/r$. Define $\widetilde{A}(y) = rA^g(x_0 + ry)$. Then

$$\|\widetilde{A}\|_\infty + \|D_\xi \widetilde{A}\|_\infty + [D_\xi \widetilde{A}]_\beta \le C_D r^2 \epsilon \tag{DA5}$$

after fixing the small ϵ ceiling. The C^β seminorm of the first derivative actually acquires the extra factor r^β; we have weakened it in (DA5).

This is a gauge on a fixed contractible carrier. It is not asserted on a wrapping carrier or a domain surrounding a wild hole. Nor is an uncontrolled large constant D absorbed into C_D in an infinite carrier sum.

Section 87.4 / A fixed finite age threshold, with explicit scope

Let $r = L^{-m}$, $m = k - j$. Suppose the actual running radii obey

$$\frac{\epsilon_k}{\min(\alpha_{0,j}, \alpha_{1,j}, \gamma_j)} \le C(1+m)^q \tag{DA9}$$

on the trajectory being used. For fixed L, fixed carrier size D, and fixed original admission constants, there is a finite m_0 such that (DA8) has a uniform strict margin for every $m \ge m_0$. Indeed $L^{-2m}(1+m)^q$ tends to zero, and the current has the additional L^{-m} gain. This argument does not presume that the couplings are pointwise monotone.

The constants in CL/MB depend on the fixed blocking parameter L. Consequently one cannot instead infer $C_L L^{-2} < 1$ merely by saying that L will be taken large. A uniform-in-L estimate would be needed for that inference. The finite-age conclusion above needs no such estimate.

There is a legitimate conditional hybrid, but it has real input requirements:

1. For $m \ge m_0$ and bounded contractible old carriers contained in a protected regular window, use the direct readout and (DA8).
2. For the finitely many recent ages, retain an original admitted low-core representation only if the **same marked member** has a proved norm on that original paired core.
3. For large, wrapping, or boundary carriers, retain their actual admitted tail or joint operator representation with its known norm. An exponential coefficient does not authorize evaluating an analytic function outside its domain.

The finite number of recent ages may change a contraction constant but does not itself cause a scale divergence. It still must be controlled by the actual formation estimate. Similarly a long carrier's locality exponent can pay its age only after a valid readout norm has been retained.

For a new MC-based derivative-source birth, items 2 and 3 are not automatic consequences of the source-free theorem. One must prove its ordinary paired-core norm, or preserve another admitted representation through the formation. The direct local scaling lemma therefore removes repeated age-composition losses where its hypotheses apply; it does not establish a global invariant new-source class by itself.

Section 88 / Direct old-scale admission on a protected regular window

This chapter records the exact interface between a new analytic extension of an observable and the original integration variables. Its purpose is to avoid two unnecessary, and generally false, identifications: equality of two extensions on every independent-current input, and equality of two gauge-fixed coordinate densities without their Jacobians. The physical identity proved here needs neither. Its quantitative use still requires the stated domain and carrier estimates.

Section 89 / Physical source identities and exact locality of completed carriers

This note isolates the actual source interface It does not assert a construction of four-dimensional Yang–Mills theory. In particular, the original paired-current extension, the physical critical family, and a newly constructed periodic weak critical family are different maps away from the physical graph.

The bounded first difference of the local formation map follows from the explicit small-potential calculation; it need not be assumed as an additional source theorem. The original large-field operator also accepts an arbitrary carrying function, and a retained positive-energy reserve supplies ordinary complex-source, linear, and proper-joining bounds before the one joint integration. Sections 13.5–13.7 identify the actual reference fiber, and Sections 18–19 close the original scalar source and vacuum bookkeeping. Section 20 gives the final scope. Sections 7–10 retain the earlier normalization tests and a bounded-component alternative; their intermediate unresolved-reference wording is superseded by those later identifications.

The original-map matching is substantive: the actual stored source functions and every preparation operation must obey the displayed input norms and integral identities. Equality only at source zero would not supply that matching. The deductions here use the original source-free domains and estimates; they do not identify the new periodic critical extension with every original off-shell readout.

89.2.1 2.1 Explicit input, not a hypothesis about the output logarithm

At one RG step let Y be a fine localization domain, $d_f(Y)$ its tree size, and let $\mathcal{U}(Y)$ be the original admitted paired field/current domain. Let the potentials be analytic $\mathcal{A}$-valued functions satisfying

$$\|V_Y(B)\|_{\mathcal{A}} \le a_Y \left(1 + M^{-4} \sum_{b \subset Y} |B_b|^2\right), \qquad \sup_q \sum_{Y \ni q} e^{\mu d_f(Y)} a_Y \le a_*. \tag{S4}$$

The norm includes the supremum over the actual paired domain and its assigned fluctuation box. It is not a norm just at the trivial field. Original scalar potentials and inherited source potentials are added inside this same norm.

The original conditioned Gaussian and its wall parameters must have the following source-free properties: a positive real reference integral; uniformly bounded local Gaussian covariance in the physical fluctuation coordinates; the conditioning cancellation; and fixed wall-parameter discs on which the complex Gaussian errors consume only a small quadratic tilt. The original clean proof supplies these properties in its equations (2.15)–(2.25). The complete determining carriers of each potential must be retained by the wall construction.

For clarity, write p_* for the small failed-test activity, after a fixed Gaussian tilt has been reserved. This is a scalar number, not a source-dependent probability. The exact Mayer terms contain the required real cutoff restrictions. No contour is shifted through those restrictions.

89.2.2 2.2 Proposition: ordinary formation and a rooted analytic derivative

Fix sufficient spare locality $\mu - \kappa$ to pay the carrier-entropy and covering constants, and fix the original geometric parameters. The map $\mathcal{F}$ below is the connected remainder after the explicit Gaussian and coordinate normalizations have been extracted. There are fixed $a_* > 0$, $p_* > 0$, and C such that this local formation map is analytic on the potential ball in (S4), with

$$\|\mathcal{F}(V)\|_\kappa \le C(\|V\|_\mu + p_*). \tag{S5}$$

Here the output norm is the rooted polymer norm

$$\|E\|_\kappa = \sup_q \sum_{X \ni q} e^{\kappa d_c(X)} \sup_{\mathcal{U}(X)} \|E_X\|_{\mathcal{A}}. \tag{S6}$$

For $\|V\|_\mu \le a_*/4$, $\|H\|_\mu \le a_*/4$, after reducing the fixed failed-test ceiling so that $p_* \le a_*$,

Section 89.2 / A local formation theorem with a proved first-difference bound

$$\begin{aligned} \|D\mathcal{F}(V)[H]\|_\kappa &\le C\|H\|_\mu, \\ \|\mathcal{F}(V+H)-\mathcal{F}(V)\|_\kappa &\le C\|H\|_\mu, \\ \|\mathcal{F}(V+H)-\mathcal{F}(V)-D\mathcal{F}(V)[H]\|_\kappa &\le Ca_*^{-1}\|H\|_\mu^2. \end{aligned} \tag{S7}$$

The same conclusions hold with one distinguished incoming carrier. Every derivative term has a connected output carrier containing that carrier and all source anchors that actually occur in the term. This is a statement about the chosen connected expansion, not an assertion that an arbitrary representation of the final logarithm has the same support.

Proof. Select a finite family $\mathcal{D}$ of potential factors, a set P of failed coordinate tests, and a final coarse domain X. Give each selected potential an auxiliary multiplier ζ_Y. On the discs

$$|\zeta_Y| \le (W/a_Y)e^{-\mu_0 d_f(Y)}, \tag{S8}$$

where μ_0 is below the input locality exponent and above the fine-domain entropy exponent, the total potential is bounded by a small local quadratic tilt plus a constant times the number of coarse cells in X. The real conditioned Gaussian must be integrated jointly with its auxiliary exterior Gaussian: at zero error the two induced exterior quadratic forms cancel. The residual tilt costs $e^{C|X|}$, counting coordinates in X, not the whole exterior field space.

The expression vanishes whenever one selected ζ_Y vanishes. Banach-valued Schwarz on those discs and Cauchy on the wall discs therefore give a bound of the form

$$\|T(X,\mathcal{D},P)\|_{\mathcal{A}} \le e^{C_0|X|}\prod_{Y\in\mathcal{D}}(Ca_Y e^{\mu_0 d_f(Y)})p_*^{|P|}(R_c-1)^{-h}, \tag{S9}$$

where h is the number of differentiated coarse walls outside the root collar. The standard carrier covering inequality

$$|X| \le C_h\left(|\mathcal{D}| + L^{-1}\sum_{Y\in\mathcal{D}} d_f(Y) + |P| + h\right) \tag{S10}$$

converts the spare decay, the small potential factors, failed-test factors, and large wall radii into exponential output decay. Summing these positive bounds gives local activities of rooted norm $C(\|V\|_\mu + p_*)$. The convergent hard-core connected series gives (S5). These are precisely the estimates needed from the ordinary Gaussian/Mayer proof; taking a Banach norm before every positive integral and absolute sum changes none of them. The explicit compatible order of the geometric and radius choices is given marked-bridges Section 5.3.

For (S7), consider the analytic function $z \mapsto \mathcal{F}(V+zH)$. On $|z| \le a_*/(2\|H\|_\mu)$ its output norm is at most Ca_*. Cauchy bounds its Taylor coefficients by

$$\frac{1}{m!}\|D^m\mathcal{F}(V)[H,\dots,H]\|_\kappa \le Ca_*\left(\frac{2\|H\|_\mu}{a_*}\right)^m. \tag{S11}$$

The linear bound and the geometric sum of the terms of order at least two prove (S7). A distinguished direction (H) supplies a distinguished selected vertex in the same connected series. Connectivity and the original carrier union then give the pointed statement, with any fixed pointing polynomial paid from spare locality. No global source zero-free theorem was used: the logarithm here is the locally convergent connected series. □

89.2.3 2.3 Consequence for the proposed scalar induction

The “bounded first difference of the generic formation map” in local-map Section 14, and in the tagged recursion of marked-bridges Section 19, is therefore a consequence of the preceding local calculation. It is not necessary to assume a separate published source theorem.

What still must be supplied at each application is (S4) for the actual incoming readouts. A small bound on $\mathcal{F}(V)(1)$, or on one Gaussian quadratic form, does not supply (S4). Pure-clean members use their fixed original core; R-origin members use their original shrinking core and the explicit radius factors in their paired marginal extraction. They cannot be relabelled into a fixed-core class merely because the original notation later calls them (E).

89.4.1 4.1 The precise positive input

For a completed or still-live history ω, let N_ω be its actual own-layer record count and let $\mathscr{E}_\omega(\xi) \geq 0$ be the still available positive action/Gaussian energy. Every energy contribution is owned once. A later merger does not create a second copy of an earlier energy reserve.

Suppose the source-free positive comparison obeys

$$\int \exp\{\tau N_\omega + \sigma \mathscr{E}_\omega(\xi)\}\, d\nu_\omega(\xi) \leq B_\omega, \qquad \tau, \sigma > 0. \tag{S13}$$

Here $d\nu_\omega$ already includes the complex-background comparison cost from (S12), together with all engineering, tree, and age reserves declared spent before this estimate. Thus B_ω is the enlarged

remaining positive majorant, not the original numerical unsourced kernel. Its bound may contain rarity, live dwell, and tree factors.

Suppose the total represented source exponent inside this same integral satisfies

$$\|F_\omega(\xi,z)\|_{\mathcal{A}} \le s\big(N_\omega + \mathscr{E}_\omega(\xi)\big). \tag{S14}$$

This is an incoming norm condition. For a primitive scalar Wilson source it follows from (S3) on the action weight that actually carries the original plaquette. For prepared analytic members it follows from their original-domain sup norms and a rooted pointing sum. A completion cannot replace an unbounded wild-field remainder unless its compensating factor remains inside this same integral.

89.4.2 4.2 Proposition: whole-kernel, linear, and joining bounds

If $s \le \frac{1}{4}\min(\tau,\sigma)$, then

$$\begin{aligned} \Big\| \int e^{F_\omega}\, d\nu_\omega \Big\|_{\mathcal{A}} &\le B_\omega, \\ \Big\| \int (e^{F_\omega} - 1)\, d\nu_\omega \Big\|_{\mathcal{A}} &\le C_{\tau,\sigma} s B_\omega, \\ \Big\| \int (e^{F_\omega} - 1 - F_\omega)\, d\nu_\omega \Big\|_{\mathcal{A}} &\le C_{\tau,\sigma} s^2 B_\omega. \end{aligned} \tag{S15}$$

If $F = F_1 + F_2$ with $\|F_i\| \le s_i(N+\mathscr{E})$, the mixed proper-joining part is bounded by $C s_1 s_2 B_\omega$, provided $s_1 + s_2$ satisfies the same ceiling.

Proof. In any submultiplicative Banach algebra,

$$\|e^F\| \le e^{\|F\|}, \quad \|e^F - 1\| \le \|F\| e^{\|F\|}, \quad \|e^F - 1 - F\| \le \tfrac{1}{2}\|F\|^2 e^{\|F\|}. \tag{S16}$$

Put $t = N + \mathscr{E}$ and $c = \min(\tau,\sigma)$. The factors t and t^2 are bounded by constants times $e^{ct/2}$, and $s \le c/4$. Integrating (S16) against the one positive measure and using (S13) proves (S15). The mixed identity

$$e^{F_1+F_2} - e^{F_1} - e^{F_2} + 1 = (e^{F_1} - 1)(e^{F_2} - 1) \tag{S17}$$

proves the last statement for commuting source values. Equivalently, one can use its two-parameter integral remainder. There is no product of separately integrated island masses. □

89.4.3 4.3 What the original large-field proof does and does not supply

The source-free paper obtains the small mass by reserving positive portions of the Wilson and Gaussian actions for failed tests, then paying normalization, geometry, and renewal costs. Its (1.73) permits arbitrary carrying functions; (1.76)–(1.89) are the positive large-field mass calculation; (1.92)–(1.100) perform the final connected resummation. Fixed reductions of those positive action fractions retain the large-field estimate after the weak ceiling is adjusted.

Accordingly, a fixed additional energy tilt in (S13) is legitimate when it is charged to an explicitly unspent action fraction, and a fixed record-volume tilt is paid by the existing growing creation/renewal

Section 89.4 / A reserve lemma that proves the ordinary-source excess estimates

credit. The proof cannot be shortened to "there was a small scalar mass, therefore all exponential moments are bounded." A positive measure can have tiny mass and arbitrarily large exponential moments. The once-owned action ledger is the needed additional fact.

Equations (S12)–(S17) supply the ordinary Banach-valued estimate once that ledger has been matched. They also justify the linear and proper-joining powers used in CH39a. An unresolved live component stays one atom: if several flags occur on the same component, their masks are multiplied before (S15) is applied. One cannot obtain a square of the rarity merely from two flags on the same component.

Section 89.5 / Exact observable transport is move-by-move

The required exactness is an integrated identity with the physical observable on its original stored variables. It is not equality of the new reference field with a deterministic block average of the original Wilson field.

The following elementary lemma makes the sufficient condition precise. Suppose a finite sequence of operations consists of:

1. a decomposition of unity independent of the source;
2. an invertible change of variables, with its exact Jacobian;
3. Fubini integration or a regrouping justified by an absolute majorant;
4. gauge integration/removal valid for gauge-invariant carrying functions;
5. an actual normalized replacement with density $r_0(y, b)$, normalization $D_0(b) = \int r_0(y, b)dy$, and old numerator integrated as

$$\frac{r_0(y, b)}{D_0(b)} \int q_0(\xi, b)M(\xi, b; z)\, d\xi. \tag{S18}$$

Then the complete integrated output equals the original integral with the same multiplier (M), coefficientwise and on any ordinary source domain on which the majorants above apply.

Proof. The first four operations preserve an arbitrary admissible carrying function by their defining identities. For the fifth, integrate (y); its factor integrates to one. Iterate these equalities. In a gauge-removal step the transformation acts on every stored old variable that appears in (M); gauge invariance must be checked for that joint action. Dominated integration gives the ordinary analytic statement. □

The readable R-II formulas support this formulation: (1.71) is an ordered stored integration, (1.70) is an exact action completion, and printed p.378 expressly allows gauge-delta removal only after the corresponding gauge fields are integrated. They do not say that each elementary operation is itself the conditional kernel (S18). Nor does equality of the scalar partition at (M=1) prove the lemma for an arbitrary (M).

Thus any use of an unexposed R-I preparation step must import its actual change-of-variables or normalized-replacement identity on carrying functions. It must not infer that identity from source-free normalization alone. The source formulas require this matching estimate.

Section 89.7 / Precise remaining interface and a test for circularity

The generic connected formation and one-joint-kernel estimates above are proved implications with explicit inputs. The first original-map statement still to be verified in a complete scalar construction can be written as follows.

Actual source-carrying preparation statement. On the frozen original Wilson trajectory, every incoming scalar source state produced by the homogeneous/heterogeneous induction has all of the following properties, simultaneously:

- every regular stored readout obeys (S4) on its original paired domain, with its full source/determining carrier and the fixed clean versus shrinking R-origin distinction;
- every live component has a single representation satisfying (S12)–(S14), after exactly the declared action, volume, tree, and age reserves have been spent;
- every preparation, dropout, swallowing, reference replacement, and gauge-removal step satisfies the carrying-function identity of Section 5;
- the universal vacuum births generated by this same construction have the nonwrapping determining-data equality needed to compare periodic volumes, and their source-zero convention agrees with the one used in the lower estimate.

The bound in the first bullet is an incoming assertion, not the desired output bound. The source-free paired-domain theorem justifies composing an incoming analytic function with the original readout; it does not bound an arbitrary previously unbounded source function. Conversely, once the incoming norm is known, no new numerical flat-background or sourced-minimizer theorem is required for (S7).

The existing clean spatial and permanently tagged recursions are usable only in this order: prove their incoming readout estimate; apply (S7) or (S15); extract the actual paired marginal; then close the scalar majorant. Saying “the source class is stable” in order to establish the input to its own formation theorem would be circular. Likewise, CH38 is a pointing inequality for an already bounded incoming local state, not a proof that all newly healed states have that bound.

Two distinctions remain particularly important:

1. The original off-shell paired extension has zero fluctuation response at zero fluctuation for every admitted pair. A new physical stationary map generally does not. Replacing the former by the latter would require a new equality or a new extension argument.

Section 89.7 / Precise remaining interface and a test for circularity

2. The R-II carrying-function inequality is genuinely stronger than a scalar mass bound, but it does not by itself identify every earlier R-I preparation with a normalized replacement. The exact integrated transport must be matched move by move.

These are the remaining model-specific interfaces of the ordinary-source route as the analysis here. They are separate from the higher derivative insertion and common complex curvature questions addressed by the argument 5 modules.

Section 89.11 / The incoming regular source norm can be specified formation

This section addresses a different potential circularity. It derives the quadratic-growth input from the old member's analytic norm and the original readout bounds. It does not assume that the new source member already belongs to the class that is being proved invariant.

89.11.1 11.1 A quantitative joint-jet lemma

Let E be analytic on the radius-R ball of a complex normed field/current chart, with values in $\mathcal{A}$, and suppose

$$E(0) = 0, \qquad DE(0) = 0, \qquad \sup_{\|z\|<R} \|E(z)\| \le b. \tag{S37}$$

On a fixed smaller ball, Cauchy's estimate and the fundamental theorem of calculus give

$$\|E(z+v) - E(z)\| \le \frac{Cb}{R^2}(\|z\| + \|v\|)\|v\|, \qquad \|E(z)\| \le \frac{Cb}{R^2}\|z\|^2. \tag{S38}$$

The constants are independent of the dimension of the chart: restrict the Cauchy argument to the two-dimensional span of the directions concerned. Carrier-size factors in the actual local chart can be retained explicitly and paid from spare locality.

For a completed regular gauge-invariant member, the first joint jet vanishes by simultaneous constant color conjugation. Its differential at the trivial field/current pair is an invariant linear functional on a

sum of adjoint representations, and the semisimple Lie algebra has no such nonzero functional. This applies coefficientwise to ordinary source values. It does **not** apply to the outer-field differential of a still-live kernel when its stored colored variables are held fixed: that kernel is only jointly invariant. Such kernels remain in Section 4 until they have genuinely healed into a regular function.

89.11.2 11.2 Apply the actual readout before taking its norm

Consider an old regular member born at scale j, read at scale $k \geq j$, and write $r = L^{j-k}$. First retain the original small-carrier versus remote/large-carrier split of I (3.5)–(3.17). The displayed estimate below is for the local part whose two readouts admit a common smaller birth chart. Let P_X be a fixed polynomial in the old carrier diameter, including its determining collars. The source-free readout estimate used here is the following statement about the original maps, with their original paired domains and spare radii:

In this subsection α_k is the **current analytic ceiling for the birth-core class being estimated**, not necessarily a physical fluctuation amplitude of the form $g_k \log^q x_k$. For the pure-clean fixed-core calculation it is the fixed clean ceiling. An R-origin member retains its original shrinking birth radius and the actual delayed preparation/admission rule, including after the source-free notation relabels it as E. Its mere appearance in a clean output does not put it on a fixed clean field tube. The estimates below are applied on these admitted class-specific windows; a transition to a larger current chart retains the original delay and radius conversion in the tagged kernel of (S44b).

$$\begin{aligned} \|z_{j,k}(b,0)\| &\leq C P_X r^2 \alpha_k, \\ \|z_{j,k}(b,B) - z_{j,k}(b,0)\| &\leq C P_X r^2 g_k \, \mathfrak{b}_X(B), \\ \mathfrak{b}_X(B)^2 &\leq C \sum_{b'} e^{-c\, d(b',X)} |B_{b'}|^2. \end{aligned} \tag{S39}$$

The two readouts and the connecting analytic segment must lie in the fixed smaller radius-R_j birth chart. On this local part, the margin $\delta_k = g_k P_k \ll \alpha_k$, the scale factor r^2, and the original near-age margin give that inclusion. If the carrier polynomial prevents this smaller-chart inclusion, retain the term in the admitted-domain remainder described below; do not extend (S39) to it by fiat. In particular, a single global tree chart on a carrier of old-scale diameter comparable to r^{-1} need not satisfy a diameter-independent $Cr^2\alpha_k$ bound. A constant-curvature field already shows the need for its carrier-size factor. A crossing of a genuinely wild component is not a regular window and is not inserted into either regular estimate.

Equation (S39) uses the original nested paired-domain condition I (1.15)–(1.16), the actual off-shell substitutions and simultaneous field/current/fluctuation gauges I (3.24)–(3.32), their range verification I (3.36)–(3.42), and the localized response II (1.21)–(1.36). The derivative bounds I (4.17)–(4.18) and the prepared-domain inclusion R-II (1.65) are used on their stated domains. In particular, this concerns the original joint field/current readout; a different physical stationary map cannot be substituted for it. The argument below needs these source-free range and response statements, not a new source-dependent version: all maps are fixed while the analytic function E being composed varies.

Put $u_j = b_j / R_j^2$. Applying (S38) to (S39) yields the explicit incoming fluctuation potential bound

$$\|E_j(z_{j,k}(b,B)) - E_j(z_{j,k}(b,0))\| \leq C P_X^2 u_j r^4 \left(\alpha_k g_k \mathfrak{b}_X(B) + g_k^2 \mathfrak{b}_X(B)^2 \right). \tag{S40}$$

Since $g_k \le \delta_k \ll \alpha_k$, the right-hand side is bounded by $CP_X^2 u_j r^4 \alpha_k \delta_k (1 + \mathfrak{b}_X(B)^2)$. Spend a fixed part of the old locality exponent to sum P_X^2.

For the complementary pieces keep the original admitted representation and its actual source-free localization tail. I (3.7)–(3.17) differentiates the member on its admitted pair domain: either the response has an exponential distance tail, or a large carrier supplies that tail from its coefficient. Its Cauchy cost is a first derivative on the old spare core, not a Taylor expansion about an inadmissible flat chart. With the birth radius retained explicitly, this gives a bound of the form

$$\|\Delta E^{\mathrm{rem}}_{j,k,X}(B)\| \le C \frac{b_{j,X}}{R_j}\, g_k\, P_X\, \mathcal{T}_{j,k,X}\, \mathfrak{b}_X(B), \tag{S40a}$$

where $b_{j,X}$ are the individual coefficient majorants whose old rooted sum is b_j, and a fixed part of the old carrier decay is left unspent. For remote or physically large carriers the original split gives $e^{-c/r}$; every fixed inverse power of r in its covering or response estimate can be paid from that tail. On a class-preserving admitted window, a local carrier excluded only because its polynomial chart bound is too large satisfies $P_X r^2 \alpha_k \gtrsim R_j$, which forces a large old diameter. The same conclusion follows from its unused carrier decay. In the fixed-clean class the radius ratios are fixed. In the still-shrinking R class the running tube gives $R_j \le C\alpha_k$ and $\alpha_k/R_j \le C(1 + k - j)^p$, with fixed p. Thus, on those windows and after the spare-decay payment, $P_X \mathcal{T}_{j,k,X} \le C_N r^N$ for any fixed N. The finitely many near ages use the original spare-core bound with fixed class constants.

That polynomial-age comparison is not asserted for a shrinking R birth against an arbitrarily chosen fixed clean ceiling: $\alpha_{\mathrm{clean}}/R_j$ can depend on x_j, even at a short age. If the original preparation assesses such a transition, retain its actual admission delay and radius-explicit bound in $W_{k,j}$ of (S44b). Its fixed polynomial radius conversions belong in the tagged rare envelope at birth, as in marked-bridges Section 19.4. Neither a small rarity coefficient nor an administrative promotion supplies a missing analytic range inclusion.

On the admitted class-preserving windows, choosing $N \ge 4$, using $R_j \le C\alpha_k$ in that class and $g_k \le \delta_k$, bounds (S40a) by $Cu_j r^4 \alpha_k \delta_k (1 + \mathfrak{b}_X(B)^2)$, in the remaining rooted locality norm. No polynomial or exponentially small coefficient licenses evaluation outside the old domain; it pays the numerical estimate only after the original range theorem has supplied a valid evaluation and Cauchy neighborhood.

The number of old roots of scale j in one current cell is at most Cr^{-4}. Summing the local part and the admitted-domain remainder, the exponentially localized coefficients in the last line of (S39), and then the old root locations, gives the classwise estimate

$$\|V_k^{\mathrm{old,fluct}}\|_\mu \le C\alpha_k \delta_k \sum_{j \le k} u_j = C\alpha_k^2 \theta_k \sum_{j \le k} u_j, \qquad \theta_k = \delta_k/\alpha_k. \tag{S41}$$

This is (S4), up to the fixed geometric normalization of the local quadratic sum. The formation input is the sum of the classwise contributions, together with any admitted delayed transitions in their radius-explicit tagged estimate. It does not identify the fixed clean core with a shrinking R core. The constant term $E_j(z_{j,k}(b, 0))$ is transported separately as the old state; it is not falsely declared a small new fluctuation birth. There is no factor C^{k-j}, and no strict age decay has been claimed for the unsubtracted quadratic member.

89.11.3 11.3 The primitive scalar starts in this class

On complex links use the holomorphic extension

$$\psi(U_p) = 1 - \frac{1}{2N}\left(\operatorname{Tr} U_p + \operatorname{Tr} U_p^{-1}\right). \tag{S42}$$

It agrees with a_p on compact real links and has zero constant and linear jets at the identity. If $U_p = e^{\eta^2 F_p}$ on a regular unit cell, then $|\psi(U_p)| \le C\eta^4 |F_p|^2$. The $O(\eta^{-4})$ plaquettes in a fixed physical cell cancel that factor exactly. Thus a scalar profile of sup norm ρ gives a local analytic birth of size $C\rho\alpha^2$, with no microscopic volume loss. Applying the same Taylor estimate to the original regular response gives (S40), including its local quadratic fluctuation growth. This verifies the initial incoming bound directly; it does not use a presumed source theorem.

If the source has been multiplied by a bare or running normalization factor, that factor belongs in ρ. The above statement does not silently replace it by a bounded number.

The heterogeneous initialization needs the additional spatial oscillation factor. Let x_p be a primitive plaquette's sampling point and let y_p be the prescribed anchor of its bare local homogeneous packet, with $|x_p - y_p| \le Ca$. Split its source exactly into the frozen part $w(y_p)\psi(U_p)$ and $[w(x_p) - w(y_p)]\psi(U_p)$. The profile budget (S2a) gives

$$\sup_{z \in D_{\boldsymbol{r}}} |w(z, x_p) - w(z, y_p)| \le C\rho a^{\alpha}, \qquad M_0^{\mathrm{het}} \le C_0 \rho \ell_0^{\alpha}, \quad \ell_0 = a. \tag{S42a}$$

The second estimate is the same quadratic primitive calculation above, now with the smaller coefficient; the number of primitives in one bare rooted stencil is fixed. If the prescribed anchor is the sampling point, that primitive defect is zero. The weaker bound $C\rho$ for the unsplit primitive would not suffice to initialize (S44).

89.11.4 11.4 Finite-step induction and the heterogeneous closing inequality

At a finite RG step, assume only that the already constructed old regular coefficients have their stated analytic birth norms and that the live kernels have their single-integral representation from Section 4. Compose the old coefficients with the unchanged original readouts. Formula (S41) verifies the formation input. Equations (S5)–(S7) then construct the new regular coefficients and bound their first difference. The completed R output is formed from the actual joint kernel and the connected resummation; it is assigned its own shrinking birth radius. A sourced regular reference denominator, when required by an exact replacement identity, contributes the analytic logarithm (S33), hence another birth with the same bounded-first-difference property. Constants depending on M are fixed before the weak floor is chosen.

For a smooth heterogeneous source, first form the locally frozen **homogeneous reference packet**: evaluate each complete pointed homogeneous member and its own counterterm at its anchor value $c = w(z)$. For a prospective output carrier anchored at y, freeze those scalar arguments to $w(y)$. Their differences are bounded by $\rho \ell_k^{\alpha} d(X)^{\alpha}$. Cauchy is applied only to the actual scalar parameter c of these homogeneous members. Differentiate the member together with its counterterm before using its strict-age estimate, and use the summable homogeneous coupling-increment derivative for the current action coefficients. The bounded formation derivative then propagates this input difference. Spare carrier decay pays $d(X)^{\alpha}$, giving the direct heterogeneous birth size $C\rho\ell_k^{\alpha}$.

The already constructed heterogeneous defects are **not** differentiated along an arbitrary pointwise profile direction; their entire fluctuation input is bounded by (S41). This separation avoids presuming an all-depth fine-plaquette product disc, which would not follow from a smooth-profile theorem. It is the scalar homogeneous disc, paired old-age estimate, and bounded derivative of the formation map that justify the freezing step.

If $M_k = \sum_{j\le k} u_j^{\mathrm{het}}$, this gives the actual finite-step majorant

$$M_{k+1} \le (1 + C\theta_k)M_k + C\rho\ell_{k+1}^{\alpha}. \tag{S43}$$

Choose the source-free fluctuation/analytic margin so that $q = 1 + C\sup_k \theta_k < L^{\alpha}$. Starting with (S42a), iterating (S43) and dividing by $\rho\ell_k^{\alpha}$ bounds its right side by $C_0(q/L^{\alpha})^k + C\sum_{h=0}^{k-1}(q/L^{\alpha})^h$. The geometric sum is bounded independently of the bare depth, giving

$$M_k \le C'\rho\ell_k^{\alpha}. \tag{S44}$$

For R-origin births, replace the last term of (S43) by $C\rho\ell_{k+1}^{\alpha}\widehat{F}_{k+1}$, where $\widehat{F}_k$ is the nondecreasing envelope of the once-paid rare birth bounds. The same induction gives $M_k \le C'\rho\ell_k^{\alpha}\widehat{F}_k$. All fixed inverse powers of the R birth radius and all fixed M-dependent denominator constants are included in that envelope at birth. An already-live root does not receive an extra rarity at healing.

The homogeneous family is constructed in the following subsection by the actual triangular scalar coupling recursion. Its renormalized old members use the original paired strict-age Ward estimate. The argument does not replace that scalar recursion by a claimed contraction for an arbitrary local coupling function. With the scalar family in place, (S43)–(S44) prove the heterogeneous incoming norm without assuming its invariance.

This closes the regular scalar **analytic input loop** under the listed original readout estimates. The physical partition identity requires the separate same-fiber argument of Section 13. It does not prove a high-derivative source readout or replace a still-live joint function by a regular member with zero first joint jet.

89.11.5 11.5 The homogeneous disc is constructed before differentiating it

For completeness, spell out the finite-step bootstrap used from the earlier marked-bridges Sections 11–12 and 19.4–19.6. This avoids treating the homogeneous family in Section 11.4 as an unexplained source-stability assumption.

First retain the pure-clean scalar history estimate on its independent past-coupling half-discs. With the original real tests fixed, its fresh non-Gaussian remainder is bounded there by $Cg_j(\log x_j)^q$, plus its Gaussian tail; the field-dependent Gaussian coefficient is independent of the scalar current coupling. Scalar-output Cauchy with derivative phases gives a weighted sum of the historical derivatives. The half-disc radii are bounded below by $cx_j/(\log x_j)^M$ along the actual running tube. Consequently the clean coefficient satisfies

$$\begin{aligned} |\beta_{j+1}^{\mathrm{cl}}(x+w) - \beta_{j+1}^{\mathrm{cl}}(x)| &\le a_j \max_{t\le j}|w_t|, \\ a_j &\le Cx_j^{-3/2}(\log x_j)^Q, \\ \sum_j a_j = \epsilon_{\mathrm{cl}}(h) &\longrightarrow 0 \quad (h \longrightarrow \infty). \end{aligned} \tag{S44a}$$

The fixed powers in this formula are harmless. The actual tube bounds past x_t below by the current value up to a fixed additive constant and bounds the sum by a convergent integral. This estimate is the previously proved clean constant-parameter theorem, with no sourced live component in its proof. It is a necessary quantitative input from that earlier calculation, not a conclusion of a bound only at real source zero.

Section 89.11 / The incoming regular source norm can be specified formation

Keep permanent R-origin labels, including the cutoff-comparison tails, through all later formations. Any connected term containing such a label remains in the tagged part. On the provisional bound $\max_t |w_t| < 2r$, Sections 2, 11.1–11.2, 14, and 19 apply to the already constructed finite prefix. The radius-explicit paired Ward estimate gives the tagged birth recurrence

$$u_{j+1} \le d_{j+1} + C\theta_j \sum_{t\le j} W_{j,t} u_t, \qquad \sup_j C\theta_j \sum_{t\le j} W_{j,t} < \tfrac{1}{2}, \qquad d_{j+1} \le F(x_{j+1}), \tag{S44b}$$

where the finite promotion delay, original shrinking radii, and their fixed polynomial conversion factors are included in W and the superpolynomial rare envelope F. The explicit kernels and delay count are those proved in marked-bridges Section 19.4. This recurrence gives $u_j \le CF(x_j)$. After including the finite promotion-batch multiplicity in F, the tagged scalar marginal obeys $|\beta^{\mathrm{tag}}_{j+1}(c)| \le F_j$ and $\sum_j F_j = \epsilon_R(h) \to 0$, uniformly on the provisional disc. Its estimate follows from the one-joint-integral reserve before any source derivative is taken.

Choose a fixed positive r smaller than one sixteenth of the available additive coefficient allowance and the fixed cold-source allowance. Increase the weak floor so that $\epsilon_{\mathrm{cl}} < 1/8$ and $\epsilon_R < r/8$. For the bare exponent in (S2), a constant source c changes the leading bare coefficient by $-c$. Write $w_j(c) = x_j(c) - x_j(0)$, so $w_0(c) = -c$. The exact triangular recursion on a finite prefix yields

$$\max_{t\le j+1} |w_t(c)| \le r + 2r\epsilon_{\mathrm{cl}} + 2\epsilon_R < 2r, \qquad |c| < r. \tag{S44c}$$

Each next local integral and connected series is holomorphic on this same disc by the proved formation and live estimates. Induction therefore constructs the family on $|c| < r$ at every ultraviolet depth and closes the provisional inequality. No derivative of a yet-unconstructed R family is used. Set $r_{\mathrm{hom}} = r$; ordinary Cauchy on its smaller concentric disc now justifies the scalar packet derivatives used in Section 11.4. The same argument with the source sign reversed is equivalent. It proves no all-depth analytic product domain for independent rough spatial couplings.

Section 89.12 / A primitive complex Wilson tilt with exact own-scale counting

The positivity used in (S3) belongs to real compact fields. The following lemma gives a precise replacement when an actual stored field has a complex deformation. It requires covariant first differences against that actual real field, not merely ordinary first differences in an unrelated reference gauge.

89.12.1 12.1 Covariant plaquette displacement

Let U be any real $SU(N)$ lattice field at spacing η, with no small-curvature assumption. Write

$$W_\mu(x) = e^{\eta A_\mu(x)} U_\mu(x), \qquad \|A\|_\infty + \|D_U A\|_\infty \le \alpha, \tag{S45}$$

where $A_\mu(x) \in \mathfrak{sl}(N, \mathbb{C})$, every component is compared in every coordinate direction, and

$$(D_{U,\nu} A_\mu)(x) = \eta^{-1} \left[\mathrm{Ad}_{U_\nu(x)}\, A_\mu(x + \eta e_\nu) - A_\mu(x) \right]. \tag{S46}$$

Assume $\eta\alpha$ lies in a fixed small ball. For every elementary plaquette,

$$W_p = e^{K_p} U_p, \qquad \|K_p\| \le C\eta^2(\alpha + \alpha^2). \tag{S47}$$

Proof. Move the four small exponentials in the oriented plaquette product to its base point. Their linear terms are the two positive transported components and the two negative transported components. Parallel transport along a path of at most four edges changes a component from its value at the base by at most $C\eta\alpha$, by telescoping (S46). Real transports are isometries in an invariant Hermitian norm. This applies also to the closed path: $(\mathrm{Ad}_{U_p}-1)A_\mu$ is $O(\eta\alpha)$, even when U_p is large. The constant-component terms cancel in the signed sum. Multiplication by η therefore leaves $O(\eta^2\alpha)$. The analytic BCH remainder of the four exponentials is $O(\eta^2\alpha^2)$. Their product is in a fixed identity neighborhood, proving (S47) with the principal local logarithm. □

It would be incorrect to omit the closed-path comparison and bound it merely by $C\alpha$. That would lose one power of η and make the four-dimensional volume count diverge.

89.12.2 12.2 The source is bounded by real action plus a record charge

For real U_p, the holomorphic plaquette function (S42) obeys

$$\left|D\psi_{U_p}[KU_p]\right| \le C\|K\|\,\|U_p-1\|, \qquad \|U_p-1\|_{\mathrm{HS}}^2 = 2Na_p(U). \tag{S48}$$

The first inequality follows directly by differentiating the two trace terms; their linear coefficient contains $U_p - U_p^{-1}$. The second-order Taylor remainder on a fixed K-ball is at most $C\|K\|^2$. Young's inequality and (S47) imply, for every fixed $0<\varepsilon\le 1$,

$$|\psi(W_p)| \le (1+\varepsilon)a_p(U) + C\varepsilon^{-1}\eta^4(\alpha+\alpha^2)^2. \tag{S49}$$

Suppose a stored own-scale record consists of unit cells, with at most $C\eta^{-4}$ plaquettes per cell. Then, for arbitrary complex scalar coefficients $|w_p|\le d$,

$$\left|\sum_{p\ \text{in record}} w_p\psi(W_p)\right| \le (1+\varepsilon)d\sum_p a_p(U) + C\varepsilon^{-1}d(\alpha+\alpha^2)^2 N_{\mathrm{record}}. \tag{S50}$$

The cancellation of η^4 and η^{-4} is exact. No supremum over a fine-cell derivative and no sum over hypothetical reconstructed copies of a record is introduced.

For an own-scale coupling x_j, let a fixed unused fraction ϑx_j of the real Wilson action belong to that record. If

$$d_j \le c\frac{x_j}{(\log x_j)^{M_0}}, \qquad \alpha_j \le Cx_j^{-1/2}(\log x_j)^q, \qquad M_0 > 2q, \tag{S51}$$

then $(1+\varepsilon)d_j \le s\vartheta x_j$ and $C\varepsilon^{-1}d_j(\alpha_j+\alpha_j^2)^2 \le s$, after one fixed reduction of c and increase of the weak floor. Summing over actual records, each with its action counted once, gives precisely the primitive contribution

$$\|F_{\mathrm{primitive}}\|_{\mathcal{A}} \le s(\mathscr{E}_\omega + N_\omega) \tag{S52}$$

needed in (S14). The statement is pointwise in the still-unintegrated real variables and therefore is compatible with a single joint cold integral.

89.12.3 12.3 Exact limitation of the current application

For regular prepared fields the original complex paired domain supplies covariant first-difference control, so (S45) is an actual low-regularity condition. For a deformed field inside a wild live component, it is essential to check that the covariant derivative in that condition is taken against the same real field whose Wilson action supplies the reserve. A bound against a smooth reference field alone does not establish (S45) against an arbitrary cold field. Alternatively one may use the original pointwise action-comparison identities (1.38)–(1.48), with their assigned couplings and their once-owned remainder costs, to establish (S52) directly.

The distinction is sharp even on a flat real field. On a plaquette take $U_\mu = H$ on both μ-edges and $U_\nu = 1$, where H is fixed and noncentral. Take the ordinary constant perturbation $A_\mu = 0$, $A_\nu = C$, with $[H, C] \neq 0$. Its ordinary first differences vanish, but

$$U_p = 1, \qquad W_p = He^{\eta C}H^{-1}e^{-\eta C} = \exp\{\eta(\mathrm{Ad}_H - 1)C + O(\eta^2)\}. \tag{S52a}$$

The plaquette action is then of order η^2, not η^4, for a nonzero anti-Hermitian choice of C. The missing hypothesis is exactly the covariant difference bound: here $D_{U,\mu}A_\nu = \eta^{-1}(\mathrm{Ad}_H - 1)C$. This is a counterexample to replacing covariant control by ordinary control, not a counterexample to the actual prepared R maps.

Thus (S45)–(S52) are a proved primitive-source domination theorem and an explicit test for the original stored representation. They do not replace that representation by a source-dependent stationary map, and they do not assume positivity for a complex Wilson action.

There is a simpler complementary case. If the plaquettes are made from the **actual stored group coordinates at their own lattice spacing**, rather than a reconstructed finer field, a link-log deformation of size α_j gives the crude estimate $\|K_p\| \le C\alpha_j$. Then

$$\sum_p |w_p \psi(W_p)| \le (1+\varepsilon) d_j \sum_p a_p(U) + C\varepsilon^{-1} d_j \alpha_j^2 N_j, \tag{S52b}$$

because their number is $O(N_j)$. No covariant derivative is needed in this case. The bounds in (S51) again pay both terms. This counting argument must not be applied to a reconstructed fine field with $O(\eta^{-4})$ plaquettes per actual coarse integration coordinate; that is exactly the case in which (S47), or an equivalent original action comparison, is needed.

Section 89.14 / A once-owned live-source ledger, derived from the incoming atoms

The norm condition (S14) should not be restated as a theorem about its own output. Here is a sufficient incoming classification and its pointwise proof. The classification is imposed on the **unexpanded joint integral** such as R-II (1.70)–(1.71), before replacing its boundary exponentials by selected Mayer factors. No ratio of a sourced Mayer activity to a possibly vanishing source-zero activity is used.

89.14.1 14.1 The actual records and the admissible atom types

Let $\mathcal{R}_\omega$ be the integration/retention records of a fixed history. A record includes its birth or operation scale, its component, and its actual own-scale coordinates. Distinct stored layers at the same physical location are distinct records. Looking at an already stored variable through another reconstruction does not make a new record. Put

$$N_\omega = \sum_{R\in\mathcal{R}_\omega} |R|, \qquad \mathscr{E}_\omega = \sum_{R\in\mathcal{R}_\omega} \mathscr{E}_R, \quad \mathscr{E}_R \geq 0, \tag{S59}$$

where each retained positive energy belongs to exactly one record. The required source atoms are the following.

- **Explicit Wilson or Gaussian atoms.** Their coefficients are bounded by the allotted additive source allowance. Real Wilson atoms use (S3); genuinely deformed own-coordinate atoms use

(S52b); reconstructed regular Wilson atoms use (S45)–(S50). A canonical Gaussian atom is bounded by a small fraction of its retained quadratic energy plus a fixed per-coordinate constant. These bounds concern the actual stored variables, before integration.

- **Admitted regular atoms.** A completed regular member or a newly formed compensation is evaluated on the original prepared readout in its admitted pair domain. Its supremum is bounded by its known analytic member norm. At each stored scale the rooted sum of these incoming coefficients, with their actual localization carriers, is at most $C\rho$. Section 11 derives this input for the heterogeneous regular state; the homogeneous paired remainder uses its original strict-age estimate. A live kernel at fixed colored old variables is not put into this category by asserting a zero first jet.
- **Missing vacuum atoms.** These are the scalar terms from the exact identity (S20). Each has the same rooted bound as the newly formed regular member evaluated at the trivial pair. The atom is attached to the first record that excludes its universal carrier from the displayed regular action.
- **Reference-normalization atoms.** For a matched sourced reference factor, use the actual logarithmic ratio (S33), including its constant part (S35a). It is attached to that reference insertion, on its bounded component. A later rereading of the same normalizer does not charge it again.

The scalar coefficient ρ here is an incoming source norm on a common fixed ordinary disc. If a leading homogeneous Wilson coefficient is placed in the first category, it is not also included as an unbounded analytic regular atom. The same rule applies to every paired marginal counterterm. Its analytic remainder and its explicit compact/Wilson completion are kept together algebraically but are charged in their corresponding categories once.

89.14.2 14.2 Pointing proves the record charge

For clarity, consider all regular atoms at one scale, with complete carriers X and coefficients a_X satisfying

$$\sup_q \sum_{X \ni q} a_X \le C\rho. \tag{S60}$$

Every atom moved into the live exponent has a carrier meeting an actual record at that scale; otherwise it remains in the external regular action. Assign it to the first such record and choose one intersected cell. Then

$$\sum_{X \ mmoved \ in} a_X \le \sum_{R \in \mathcal{R}_\omega} \sum_{q \in R} \sum_{X \ni q} a_X \le C\rho N_\omega. \tag{S61}$$

This argument also covers the missing vacuum atoms. A localization or pointing polynomial is paid from the spare exponential carrier norm before (S60). The reference atoms contribute at most $C_M\rho$ per actual bounded reference insertion, hence at most $C_M\rho N_\omega$. The explicit Wilson/Gaussian atoms contribute at most $C\rho\mathscr{E}_\omega + C_M\rho N_\omega$ by Section 12 and the elementary quadratic-energy estimate. Thus the **incoming classification**, rather than a presumed live-source estimate, gives

$$\|F_\omega\|_{\mathcal{A}} \le C_M\rho(\mathscr{E}_\omega + N_\omega). \tag{S62}$$

Choose ρ so that $C_M\rho$ fits the unused fractions in (S13). Equations (S15)–(S17) now prove the whole-kernel, linear-excess, and proper-joining bounds in one integral. All fixed geometric constants are chosen before this source radius. The radius does not shrink with the number of stored layers.

89.14.3 14.3 Why the owner rule survives the exact moves

The following statements are algebraic and do not require a bound on a normalized live conditional expectation.

1. A source-free test splits a history into branches; within one branch it does not copy any source atom or its energy reserve.
2. A fixed change of variables carries the same atom and the same owner. Its representation changes, not its multiplicity.
3. When an old regular readout is rewritten, the exact identity $E(U_{\mathrm{old}}) = E(U_{\mathrm{new}}) + [E(U_{\mathrm{old}}) - E(U_{\mathrm{new}})]$ transports the first term as the old state. Only the displayed difference is a new formation input. Its norm is (S40)–(S41), or the original localized boundary difference bound on a prepared R domain. Treating both terms as independent unbounded copies of the old state would be an incorrect ledger.
4. Completing the regular action adds and subtracts the same source member. The outside copy belongs to the regular state; the compensating inside copy gets its first missing-carrier owner. Identical completion/inverse pairs are canceled before absolute norms are taken.
5. A merger unions the record sets and the atom lists before integration. An atom that was already live keeps its owner. Only genuinely new readout or completion differences receive new operation records. There is no product of independently estimated masses of overlapping live descendants.
6. At first complete healing, the joint integral is evaluated with (S15) and then locally exponentiated. The result is a new bounded regular birth, with its rare factor and its own birth radius. Its old internal atom list is no longer charged as a second external copy. Partial dropout of unrelated variables is not a second healing of the same source root.
7. Evaluating the new birth at the trivial pair creates its universal vacuum value. Its norm is at most the already established birth norm. Subtracting this value and filing any later missing copy therefore uses (S61); it does not assume a separate bound for an otherwise unconstructed healed vacuum.

These rules are the pointwise versions of the ordered stored integrations and action completion in III (2.20), R-II (1.70)–(1.72). The original prepared-domain theorem is used only to place each regular readout in the domain on which its **already known** member norm applies. In particular, it is not used to declare a wild field itself regular.

89.14.4 14.4 What is now reduced to an actual-coordinate check

For real compact stored fields, the primitive part of (S62) follows immediately from the actual positive action allocation. For complex external fields, it follows if every affected primitive is identified either as an actual own-coordinate atom with the link displacement bound in (S52b), or as a reconstructed regular atom with the covariant estimate in (S45). A direct original action-comparison ledger can replace either verification. Ordinary first-derivative control against an unrelated real field is not enough, as (S52a) shows.

Consequently, the live incoming estimate has been reduced to a concrete field/coordinate classification, a pointwise source budget, and a once-owned record assignment. The analytic regular part and the

Section 89.14 / A once-owned live-source ledger, derived from the incoming atoms

healed vacuum part no longer require a hypothesis asserting their own source stability. The remaining literal identification concerns (a) the matched reference fiber of Section 13, and (b) the primitive complex deformation on each original stored cold piece. No assertion of physical Wilson-source matching is made while either identification is outstanding.

Section 89.15 / Current dependency ledger after the component and atom proofs

This section supersedes a broader reading of the interface list in Section 7. In particular, the elementary conditional-probability identity is proved; the outstanding question is its literal coordinate match, or the source-free estimates for the explicitly defined alternative replacement (S58).

89.15.1 15.1 Universally proved mathematical statements

Statement	What is actually proved	What it does not assume
Connected formation, S5–S11	The original positive Gaussian/wall estimates extend to a commutative source Banach algebra; the formation map has a bounded first difference and proper quadratic remainder.	An a priori source zero-free theorem or a bound for an already constructed sourced output.
Bounded reference normalization, S30–S35c	A fixed-dimensional regular reference integral has a common ordinary source disc, analytic log, and the growing additive coupling allowance after constant extraction.	A product factorization of an arbitrary full live component or a sourced island-mass inverse.
Regular incoming norm, S37–S44	The original low-regularity readout bounds, the zero first joint jet, and the actual root count give the quadratic-growth fluctuation input and close the heterogeneous recurrence.	A strict age gain for an unsubtracted quadratic member or zero outer jets of a live kernel with colored stored variables fixed.
Primitive complex tilt, S45–S52b	Real Wilson energy plus a once-per-record charge dominates a complex source under the stated covariant deformation, or under the own-coordinate link bound.	Positivity of the complex action, or ordinary-gradient control in place of covariant control.
Gauge fiber and exact source transport, S53–S58	Tree-Haar disintegration, its Jacobian, gauge independence of a relaxed reference, and exact marked normalization on the matched conditional fiber.	A constant gauge substitution when the gauge actually depends on integration variables.

Statement	What is actually proved	What it does not assume
Live excess and vacuum algebra, S59–S62 and S15–S21	The listed incoming atoms and owner rule give the pointwise source tilt, one-joint-integral bounds, and exact universal-vacuum completion.	Products of overlapping live masses, a second rarity at healing, or a global complex logarithm of the physical partition.

89.15.2 15.2 Literal source-free facts verified in the primary formulas

The R-II reference is the relaxed V_Λ, not an arbitrary copy of the current interior. Its variational problem fixes the retained constraints; the paper proves local uniqueness and analytic continuation in the specified axial gauge. The denominator analysis is on a component of size at most $100M$, with a positive gauge-fixed quadratic form and an analytic logarithmic normalization. The older stored operations suppressed before (1.1) remain unchanged and can be held in the conditional base b.

The subsequent formulas give exact Wilson/action completions, original paired-domain inclusion for the prepared old members, factorization of the final stored integrations in components, and a positive-real/complex-comparison estimate for an arbitrary carrying function. The positive cold calculation allocates actual Wilson and Gaussian energy and counts actual stored coordinates. These are stronger facts than a mere small bound for $T1$, but they must be applied in their actual order and to their actual variables.

The source-free full-density description includes both clean and R vacuum subtractions and gives conditional stability bounds with its stated normalization. These facts justify the unmarked reference bounds used above. They do not, without the following identifications, certify every physical source comparison in the earlier scalar synthesis.

89.15.3 15.3 Remaining model-specific identifications

Current reference fiber. The fixed-extension construction on p.359 now resolves the reference's independence from the later free output, as explained in Section 13.5. Match the CURRENT $V_k|_\Lambda$ old/new coordinates and the output relative $\chi\delta_{G_0}$ measure to (S53)–(S58). Older omitted variables belong to b, even if physically located inside a larger component. The remaining question is not whether a true conditional normalization preserves sources; that theorem is proved. It is the exact placement of the output measure and old numerator in the original preparation. Alternatively define (S58) as a new replacement and verify its old numerator/new determining maps satisfy the same actual hypotheses of the R-II cancellation and prepared-range proofs. Its exact preservation of the Wilson integral is then automatic by construction.

Complex live primitives. For every primitive remaining in the unfolded original cold integral, identify it as an actual own-coordinate atom, a reconstructed regular atom, or a term covered by the original pointwise Wilson comparison. Check its deformation against the same real field whose action reserve is spent. Sections 12–14 prove the bounds after this classification, and prove the regular, compensation, and healed-vacuum charges from already known norms. They do not certify an uninspected off-shell deformation by calling its source class stable.

Common vacuum and the terminal comparison. The exact algebra (S20) is established, as is its pointing estimate once the constructed source births have their rooted norm. To conclude a physical normalized bound, the source-zero terminal lower estimate must use exactly the same

Section 89.15 / Current dependency ledger after the component and atom proofs

deterministic V_a, including R, Gaussian, Haar, and gauge-volume constants. Either check equality with the published vacuum convention or bound their difference by terminal volume as in (S22). A positive terminal regular sector must be shown admissible by its actual tests; a selected signed Mayer term is not such a sector. The exact nonwrapping equality of the resulting **sourced** local coefficients across compatible tori likewise needs their completed determining carriers and the literal same-kernel construction. Neither follows just by writing a common formal symbol V_a.

Relation to the new physical periodic estimates. The CC/FE/WB periodic curvature theorems concern actual critical maps on explicit common complex input balls. They do not identify the original off-shell paired maps or every mixed/wall-dependent readout appearing above. They can supply a new source-only extension on a physical graph, but that route still needs its local covariant comparison. Conversely, the conditional source estimates in this note do not construct the higher-jet quantum insertion hierarchy.

These remaining statements are separate. A successful reference-fiber identification would remove the first one without automatically establishing the complex primitive classification, the terminal-vacuum comparison, or the arbitrary-operator construction.

Section 89.17 / Reconstructed regular primitives can be paid by volume alone

The primitive comparison with a cold real field is unnecessary on a part that already belongs to its own-scale complex regular domain. The following exact counting lemma separates that case from genuinely unresolved group variables.

Let a be the fine spacing. A determining layer consists of N_j blocks of side ℓ_j, and its assigned fine plaquettes are denoted by $\mathcal{P}_j$. Assume bounded overlap and the actual complex plaquette estimate

$$\#\mathcal{P}_j \le C(\ell_j/a)^4 N_j, \qquad \|U_p - 1\| \le C(a/\ell_j)^2\alpha_j \quad (p \in \mathcal{P}_j), \qquad \alpha_j \le \alpha_*. \tag{S71}$$

Here $\mathcal{P}_j$ is an assignment of actual plaquette terms, not a collection of every reconstructed copy of the same term. Take α_* sufficiently small that $\|U_p^{-1}\| \le 2$. The holomorphic Wilson function obeys the exact matrix identity

$$\psi(U_p) = -\frac{1}{2N}\operatorname{Tr}\big[(U_p - 1)^2 U_p^{-1}\big]. \tag{S72}$$

It follows without a choice of logarithm, a real comparison field, or a covariant derivative estimate that

$$\sum_{p\in\mathcal{P}_j} |w_p \psi(U_p)| \le C d_j \alpha_j^2 N_j, \qquad d_j = \sup_{p\in\mathcal{P}_j} |w_p|. \tag{S73}$$

Indeed each summand is bounded by $C d_j (a/\ell_j)^4 \alpha_j^2$, and the two lattice powers cancel exactly. This argument also holds in the supremum norm on a common complex source domain.

For an ordinary fixed source disc, $d_j \le C\rho$ and $\alpha_j \le \alpha_*$ give a charge $C\rho\alpha_*^2 N_j$. For the larger homogeneous coupling discs in (S51), the bound $d_j\alpha_j^2 \le Cc(\log x_j)^{2q-M_0}$ gives the same conclusion after choosing the fixed constant c small. Thus the reconstructed regular primitive consumes only the per-record volume reserve; its real Wilson energy remains available for the original rarity estimate.

This proves the following exhaustive-classification criterion. Suppose each explicit primitive in the unexpanded live integrand is assigned, with bounded multiplicity and its existing owner, either:

1. to an actual own-spacing stored group plaquette with the deformation bound in (S52b); or
2. to an actual determining-layer plaquette set satisfying (S71).

Then the primitive part of (S62) follows on the whole complex input domain. There is no remaining requirement to estimate its covariant derivative against an arbitrary cold field. Unresolved group variables in the first class still use their retained positive energy; the second class uses (S73). The regular, compensation, reference, and vacuum atoms are already treated in Section 14.

The actual-map issue has therefore become a coverage check: the full original determining-layer regularity must cover every reconstructed primitive that is not an own-spacing group variable, on the same admitted complex input domain used by the carrying operator. A range theorem for completed analytic members alone does not cover a distinct cold reconstruction automatically. Conversely,

Section 89.17 / Reconstructed regular primitives can be paid by volume alone

whenever the original mixed-field regularity theorem supplies (S71) for that reconstruction, no stronger smoothness or same-real-field comparison is needed. This lemma does not label an arbitrary wild fine configuration as regular.

Section 89.19 / Primitive coverage and the unused positive reserve in the original scalar

This section concerns ordinary scalar plaquette sources on a fixed small source disc. It does not assert the growing source discs needed for arbitrary higher-degree composite operators. The maps and selectors remain those of the real source-zero construction.

89.19.1 19.1 The two primitive estimates require only the original low regularity

An explicit plaquette made from actual own-spacing stored compact links has $0 \le a_p \le 2$. A plaquette made from four links in a fixed complex tube of those compact links has $|\psi(U_p)| \le C$. The same bound holds if the whole plaquette is conjugate to such a representative by any invertible complex matrix, because its trace and inverse trace do not change. Therefore

$$\sum_{p \text{ on own records}} |w_p \psi(U_p)| \le C\rho N_{\text{own}}, \qquad \sup_p |w_p| \le \rho. \tag{S82}$$

Only actual own-spacing plaquettes are counted here. Their number is bounded by a fixed multiple of the number of stored own-spacing links/cells. A long fine reconstruction is not a single such plaquette.

For that reconstruction, the required estimate is supplied by the original weighted regularity. Write $\ell_j = L^j a$. The real reference satisfies the plaquette bound in CMP102 (2), and its analytic relative field, in the controlled Landau representative, satisfies (19):

$$\|U_{0,p} - 1\| \le C\epsilon_0 (a/\ell_j)^2, \qquad |A| \le \epsilon_2 \ell_j^{-1}, \qquad |\nabla_{U_0} A| \le \epsilon_2 \ell_j^{-2}. \tag{S83}$$

Proposition 9 explicitly carries these conditions to the analytic response, with $\epsilon_2 = B_5 \epsilon_1$ as in (173). The same low-norm estimates are used in the original full paired-domain extension and the prepared readout inclusion, rather than supplied by the new periodic critical map.

Transporting the four relative link increments around a plaquette gives a relative loop logarithm bounded by $C(a/\ell_j)^2(\epsilon_2 + \epsilon_2^2)$. The proof is the covariant telescoping argument of Section 12, now in the actual regular reference U_0. Combining this with the first bound in (S83), and then using (S72), gives

$$|\psi((e^{aA} U_0)_p)| \le C(a/\ell_j)^4 (\epsilon_0 + \epsilon_2 + \epsilon_2^2)^2. \tag{S84}$$

An axial-to-Landau gauge bridge need not have a uniformly bounded complex matrix norm for this estimate: the scalar trace is invariant under that bridge. Analyticity remains the analyticity of the original readout; choosing a controlled representative to establish its scalar supremum bound does not change the function.

The reference and fluctuation ceilings in the original admissible domains can be chosen once below a fixed ϵ_*. For a fixed ordinary source disc, (S84) and the determining-layer count prove (S73) with $\alpha_j \le C\epsilon_*$. No estimate $\alpha_j \asymp g_j \log^q x_j$, and no unread precise exponent in R-II (1.45), is needed for this conclusion.

89.19.2 19.2 Why the primitive list is exhaustive

Keep an ancestry tag on each original plaquette source until it is integrated into a regular connected member. Use the following rule through the actual stored-variable representation.

- A primitive expressed in still-stored group variables is estimated at the scale of those variables. Original real Haar factors remain real. In R-II (1.71), the paper states immediately after the formula that only the quadratic form in B and $V(X^\sim)$ depend on the external pair; the remaining cold density is real and positive. One does not complexify an untouched old Haar field merely because the outer pair is complex. If a relative chart is actually used on an own-spacing field, its prescribed small relative links give the fixed tube in (S82).
- A primitive still expressed through a regular small-field reconstruction uses that reconstruction's actual determining layers. The old fields in (1.19) and (1.50), the relaxed background in (1.43)–(1.45), and the new background in (1.33)–(1.35) are precisely such readouts on their prescribed

domains. The original real regularity theorem and analytic response bounds give (S83) there; the original prepared-domain inclusion (1.65) is retained when old regular arguments are subsequently localized. Formula (S84) is applied separately in each assigned layer. At an interface use the original determining collars and a fixed enlargement of the finer layer; their bounded overlap changes only the fixed counting constant. A part outside the reconstruction domain is still an own-spacing retained variable, so it goes into the previous case.

- When the clean integration of a source carrier is performed, its source contribution becomes a regular connected member. Its complete source coefficient, rather than the old primitive again, is the incoming object for (S37)–(S44). Such a member is covered by its already established analytic norm on the original paired domain.
- Adding and subtracting the leading Wilson completion produces an explicit Wilson atom on the new determining field and an opposite compensation on its actual carrier. The first uses (S82) or (S84); the analytic remainder uses the regular-member norm. Equal completion/inverse terms are canceled before absolute bounds. These are the only new explicit Wilson atoms introduced by the action bookkeeping.
- At partial dropout or swallowing, keep an unresolved primitive under the same joint stored integral. Do not replace it by a product of separately normalized child expectations. Its ancestry tag, old spacing, and owner persist. At first complete healing the joint source function is integrated and becomes a new bounded regular birth; the old primitive is no longer a second external atom.

This is exhaustive because the operations are tests, coordinate/gauge substitutions, the displayed old-readout rewritings, integration, and completion of the regular action. A change of variables changes the representation of an existing primitive; it does not create a new physical plaquette source. The nonlinear source-free reconstruction can create a complicated function of its arguments, but its value is still covered by (S84) on that map's admitted domain. New functions produced by clean integration belong to the regular-member case. No conclusion about high derivatives of those reconstructed functions is required.

The determining layers satisfy the original disjoint shell decomposition $B^j(\Gamma_j) = \Omega_j \setminus \Omega_{j+1}$, with the prescribed collars. Thus their physical volumes give exactly the fine count in (S71), up to that bounded overlap. A primitive on an old finer layer is charged to $|\Gamma_j|$ at that layer, never to one cell at the newest scale. Keeping these records is essential: it is what prevents a false fine-volume estimate in (S82).

It follows that the scalar primitive part of the live exponent is bounded by

$$\|F_{\text{primitive}}\|_{\mathcal{A}} \le C\rho N_\omega \tag{S85}$$

on the original admitted complex domain. Together with the regular, reference, and vacuum atoms already proved in Section 14, this gives (S62). The model-specific input in this proof is the published original low-regularity/range theorem for these very readouts. It is not an assumption that their sourced output is small, and it does not identify a new periodic critical map with an off-shell original map.

89.19.3 19.3 A genuine reserve follows from the positive cold proof

The positive reserve used in (S13) must be retained before applying the source bound. The required fact is stronger than a scalar bound on $T1$, but it follows from the same positive calculation in R-II (1.76)–(1.89). Here is the quantitative reason.

Section 89.19 / Primitive coverage and the unused positive reserve in the original scalar

After the original action cancellation and real comparison, write $\mathscr{E}_\omega$ for the sum of its positive Wilson and Gaussian energies, each assigned once to its actual record. Fix a small number $\theta > 0$ before choosing the weak-coupling floor and the large-field geometry. Run the positive estimates with coefficients $1 - 2\theta$ multiplying those energies. The Gaussian integral bounds change by at most $e^{C\theta n}$ on a record with n actual coordinates. The original powers of g_j, Jacobians, and reference normalizers remain those of the unchanged integral; the reduction of a Gaussian energy merely changes a fixed per-coordinate constant in its numerical upper bound. A Wilson failed-test lower bound $\mathscr{E}_R \geq cP_R$ gives the same rare factor with c replaced by $(1 - 2\theta)c$. Spatial and age payments likewise retain positive exponents after their assigned energy fractions are reduced.

The source-free cold proof already pays its coordinate entropy and normalizer costs by the much larger failed-test/age bank. Its parameters are chosen with strict inequalities. Replacing their constants by the preceding fixed factors and increasing the weak floor leaves those inequalities valid. In particular there are fixed positive θ, τ and a slightly weakened original rare majorant B'_ω such that

$$\int \exp\{\theta\mathscr{E}_\omega + \tau N_\omega\}\, d\nu_\omega \leq B'_\omega. \tag{S86}$$

This does not alter a test, reference denominator, reconstruction, or physical coupling. It only retains unused fractions in the proof of the same positive integral. The source-free comparison errors have already consumed their assigned fractions before θ is chosen. If a record is merged or partially integrated, its energy is not reassigned to a second descendant.

Choose ρ with $C_M\rho < \min(\theta, \tau)/4$. Equations (S62), (S85), and (S86) give the ordinary complex-source estimate, the linear excess estimate, and the same-kernel proper joining estimate (S15)–(S17). The reduction of the rare exponent is harmless: it remains a positive coefficient multiplying the original large-field test function, and dominates every fixed polynomial normalization cost after the weak floor is chosen.

After first complete healing and the original connected exponentiation, the new source-excess regular R member therefore has the same carrier bound as the unmarked member, multiplied by $C\rho$. Its flat value has the same bound, so the universal source-excess vacuum charge used in (S75) follows rather than being imposed. Two source packets inside the same live component share the one rare majorant in (S86); this statement never multiplies two masses of the same component.

The remaining physical identity is the literal matched output fiber of Section 13. Once that identity is fixed, the ordinary scalar-source estimate and the common-vacuum bound (S81) follow using the original readout domains and positive estimates just specified. This conclusion does not establish higher composite-operator mixing or the separate local common-chart construction for a new nonlinear wall family.

Section 89.20 / Final source-interface ledger

This section supersedes the unresolved-reference wording in Sections 7, 8, 13.4–13.5, 15, and 16.3. Those sections retain the intermediate tests and counterexamples because they explain why the matching had to be done. Their conditional algebra is not a claim that the final original-map identification remains missing.

89.20.1 20.1 What is proved for ordinary scalar sources

On a weak source-zero Wilson trajectory to which the published original clean and large-field induction applies, keep its maps, tests, gauge conventions and reference densities fixed. Ordinary scalar plaquette

sources on the finite-profile polydisc (S2a), with the fixed small radius choices (S2b), have the following extension. In particular this statement does not grant a cutoff-independent arbitrary fine-plaquette product disc in an unweighted ℓ^∞ norm.

- The local Gaussian/Mayer formation estimate is Banach-valued and has the bounded first derivative and proper remainder in (S5)–(S11). Its constants do not depend on the number of source parameters.
- Primitive, homogeneous paired, and heterogeneous regular source inputs satisfy the quadratic-growth norm by Section 11. The freezing step differentiates only actual scalar homogeneous packets; it does not presume an all-depth arbitrary fine-source product disc.
- The reference operation carries the old marked numerator by (S58b). Its conditional normalization, tree measure, old field slots, and output selector sum agree with the collective original identities listed in Section 13.7. It preserves the original marked Wilson integral.
- The actual low-regularity and prepared-range theorems supply the regular reconstructed primitive estimate. Own-spacing cold variables and reconstructed layers exhaust the primitive list by Section 19.2. The primitive, regular, compensation and healed-vacuum terms therefore obey the once-owned pointwise live bound (S62).
- Retaining fixed unused fractions in the original positive cold proof gives (S86), so the whole-kernel, linear, and proper-joining source estimates follow before the single joint integration. Completed source-excess R births and their vacuum values inherit their rare carrier norms. No product of overlapping live masses and no sourced island inverse is used.
- With $V_a = V_a^{\mathrm{pub}}$ and only source-excess vacuum births extracted, the source-zero normalization is identical by (S74)–(S76). The full positive terminal density lower estimate and the retained regular Haar core give (S79), hence the common-vacuum normalized bound (S81).

The genuinely model-specific imports are the **published source-free** Gaussian/wall estimates, the actual low-regularity/range theorem on its localized paired domains, the exact original preparation identities, and the positive cold estimate with its strict parameter margins. The other quantitative dependency is the earlier proved pure-clean scalar history estimate (S44a); Section 11.5 explicitly combines it with the newly bounded tagged births to construct the homogeneous disc before differentiating it. These are not an assumed full ordinary-source stability conclusion. The heterogeneous incoming recurrence, preservation of the old marked integral, one-kernel live estimates, and source-vacuum bookkeeping are the deductions made here.

89.20.2 20.2 What this does not finish

This local and history source extension does not by itself reprove every global implication in the scalar synthesis. Applying that synthesis still requires its quantitatively normalized classical/Gaussian source transport, physical ultraviolet comparison, and the stated reconstruction hypotheses on the same sequence. The vacuum $\mathcal{E}_a$ in (S81) is initially the complete localized analytic source functional. Section 21 supplies the additional source-Taylor remainder argument and replaces it by the strictly local finite-jet counterterm required for the within-torus reflection argument; (S81) alone would not perform that replacement. Exact intertorus nonwrapping equality is addressed separately in Section 21.4 and requires the complete local constructions to agree. No ambient-torus dependence may be suppressed inside a stored operator.

Nor does this result identify the new periodic CK/CL critical family with the original off-shell paired family. A new source-only extension on the physical graph has its separate PS interface and mixed-domain obligations. The present scalar argument instead uses the original low-regularity maps, for

Section 89.20 / Final source-interface ledger

which plaquette traces require no higher derivative bound.

Finally, general covariant-derivative composites require completed quantum insertion estimates and dimension-filtered mixing beyond the scalar source class. The actual-minimizer higher-jet counterexample HJ shows that the missing estimate cannot simply be obtained by assuming arbitrary conditional-background smoothness. Nothing in the ordinary scalar-source proof removes that obstruction or proves the complete Yang–Mills field family, a mass gap, or the requested full existence theorem.

The free normalization calculation is sound. For the Euclidean null momentum and transverse polarization in DN5, the two nonzero directional-current entries are -1 and $+1$; their sum vanishes and their squared sum is 2. Thus the quadratic field is not eliminated by the physical free equations. Independent exact integer matrix multiplication for DN7 gives

$$\operatorname{tr}(F_1F_2F_3) = -1, \qquad \operatorname{diag}(F_1F_2F_3) = (0,-1,0,0).$$

For three metric-skew matrices the Lorentz trace is alternating. The antisymmetric color tensor therefore reinforces, rather than cancels, the permutation sum. Both form factors are nonzero on open sets of physical momenta. A small spatial test bump and a nonnegative positive-time bump make the common smearing transform nonzero there. The Wick quadratic and cubic states belong to different particle-number sectors, giving the positive diagonal free Gram matrix.

The ordered-product estimate DN13 needs no simultaneous diagonalization or commutation. Its integral comparison has the correct direction. DN14 also has the correct factor in the exponent. These are conditional transfer lemmas: the actual interacting transfer and Gram comparison remain unproved in DN, as they should. The monotone-decrement hypothesis can be replaced by the actual lower running tube: if $x_j \geq x_n + c(n-j) - C_0$ and $x_n > C_0$, then

$$\sum_{j=m}^{n-1} x_j^{-1} \leq c^{-1} \log \frac{x_n - C_0 + c(n-m)}{x_n - C_0} \leq c^{-1} \log \frac{x_m}{x_n - C_0}.$$

This gives the same polynomial matrix-product control up to a fixed weak-floor constant.

Section 90.2 / Physical-source naturality PS1-9

The nested variational identity is valid with the same action, boundary conditions, and actual ordered constraint composition. It does not imply an equality for unrelated mixed boundaries. The identity theorem extends an already proved physical equality but does not supply an analytic radius.

The source-only replacement in PS6–7 is valid when its actual insertion argument is the verified constrained minimum. An arbitrary fluctuating connection is not such an argument. PS8–9 correctly keeps the source-free kernel and the original integration support unchanged. The physical state must include all still-stored old variables and their old field slots.

The confirmed S53–58b reference carrying operation fits this framework: its denominator and conditional reference law are source independent, the mark stays in the old numerator, and normalization occurs after summing new selector labels. No old source may be silently evaluated on the new output coordinate. Completion adds and subtracts identical terms, so it is a natural exact algebraic operation on the carrying function. Logarithms are fixed at the common source-zero branch.

PS's formerly outstanding matched R-fiber item is therefore resolved for this frozen original construction. Its separate mixed-domain and complete intertorus carrier requirements are not consequences of PS's algebra alone.

Section 90.3 / Source radius and the incoming-state loop

The final source statement now has an explicit finite-profile domain S2a:

$$w(z)=\sum_i z_i f_i, \qquad \sum_i r_i\|f_i\|_{C^\alpha}\le \rho, \qquad |z_i|<r_i, \qquad 0<\alpha<1.$$

Section 90.3 / Source radius and the incoming-state loop

The fixed ρ is chosen below the formation allowance, the homogeneous scalar half-disc, and the unused live energy/volume reserves. This does not assert an all-depth arbitrary fine-plaquette ℓ^∞ product disc. Every insertion normalization belongs in the displayed budget.

The original S11 wording left the homogeneous family as a reference to another chapter. New S44a–c makes the dependency explicit. The clean historical row-Cauchy estimate is a separately proved constant-parameter input. The provisional homogeneous leading-coefficient bound, the tagged birth Volterra estimate, and the one-joint-integral rare reserve close a finite-step holomorphic induction before any tagged source derivative is taken. The scalar packet derivatives used for spatial freezing therefore do not assume the desired full source theorem.

For heterogeneous members, the entire old fluctuation difference is bounded by the joint zero-jet/Cauchy estimate S38 and the original low-regularity readout S39. The r^4 field factor cancels the old-root count; the additional fluctuation factor gives S41. Only the already constructed homogeneous scalar packet is differentiated in the freezing step. The unextracted heterogeneous quadratic tensor is retained, and its source oscillation supplies the geometric factor in S43–44. There is no claim of age decay for an unsubtracted homogeneous marginal.

The fixed clean core and the shrinking R-origin core remain distinct. Each R radius conversion is paid at birth and retained in its rare envelope; it is not silently replaced by fixed-core analyticity after promotion.

Section 90.4 / Primitive coverage, the reference fiber, and the vacuum

The scalar primitive classification is exhaustive for the listed original operations. Untouched old Haar variables remain real; own-spacing chart atoms use their compact tube. Finer regular reconstructions use the original covariant relative-field bounds and their actual determining-layer counts. At a layer interface the finer valid collar owns the plaquette. Once a primitive has been integrated into a regular member, its old copy is not charged again. A live component keeps one joint kernel through mergers and partial dropouts.

The reconstructed trace estimate requires the original covariant first-difference bound. An ordinary first derivative in an unrelated gauge is insufficient, as S52a demonstrates. The current proof uses the controlled representative only to bound a gauge-invariant trace and makes no unjustified bound on a complex gauge-conjugation matrix.

The unused reserve S86 is a strengthened estimate on the same original positive integral, obtained by retaining fixed fractions of its Wilson/Gaussian energies. The maps, couplings, and selectors do not change. Actual record energies and volumes are owned once. This is stronger than a bound on an unmarked mass and supplies the pointwise source tilt, linear excess, and proper joining before integration.

The published source-zero vacuum is fixed at the start. Only source-excess scalar births are extracted. At zero source the kernel, normalizers, and scalar branches are the original ones, so S76 uses the same vacuum in numerator and denominator. The terminal lower bound comes from the full positive source-zero density and its genuine regular Haar core, not a signed cluster term or the event that no fine-scale test ever failed.

Section 90.5 / The strict-local scalar normalizer is now a consequence

New S87–93 applies the earlier fourth-order Taylor argument to the source coefficients established in this note. It gives the complete-carrier ordinary source bound, uses a genuine globally admissible C^α direction to compare a source with its fourth spatial jet, and gains $\ell^{5-\alpha}$ before the four-dimensional root count. The remaining scale factor $\ell^{1-\alpha}$ is summable. All E and healed R vacuum types are included.

Consequently, for a fixed C^5 radius independent of cutoff and compatible volume,

$$\left| e^{-\mathcal{L}_{a,T}(J)} Z_{a,T}[J]/Z_{a,T}[0] \right| \leq e^{C\,\mathrm{Vol}(T)}.$$

The counterterm is a sum of holomorphic functions of the jet through order four at a single point. Finite symmetry averaging preserves this bound and strict locality. Positive-side reflection is an exact diagonal congruence of the original Wilson reflection form. This closes the within-torus strict-local source input; cross-volume equality is unnecessary for this conclusion.

Section 90.6 / Exact intertorus matching has a narrower source-definition requirement

The published source give exact component support, interior whole-lattice regular coefficients, and complete R component/bridge unions. They justify source support and the universal local regular family used in the preceding argument. They are not merely norm estimates.

Literal equality of a completed nonwrapping term on two tori additionally uses the same elementary typed local-factor construction on isomorphic finite domains and the same volume-independent marginal extraction. With that original local-choice convention, the naturality proof is termwise: the kernels, fixed-point equations, relative Haar/tree measures, old slots, selectors, and unmarked denominators coincide. The marked extension introduces no new ambient dependence.

The logical caution is specific. Dependence on the external field restricted to a set does not alone exclude an explicit ambient-volume constant. The local-factor definition of the original propagator construction is the required source-definition input for exact intertorus equality. No actual counterdependence has been found. The within-torus theorem above does not require this additional identification.

Also, a C^5 bound alone does not recover the sharper $d^{4-2\alpha}$ volume-transfer error used in the ultraviolet comparison. On scale-d profiles it costs d^{-5} per label; packing two labels supplies only d^4. The earlier positive-power transfer needs a common C^α cell normalizer or a comparably strong physical comparison. Vanishing separated contact derivatives does not enlarge the admissible source radius.

The final reread substantially narrows the needed equality. New source Section 22, S94–99, shows that the common C^α normalizer is constructed entirely from source births on ONE reference torus. The exact reflected transfer inequality transports its bound to larger volumes. No sourced coefficient on two tori is identified. The sole cross-volume renormalization input is the source-zero vacuum-density lower comparison S94. Consequently the outstanding literal typed-factor check is confined to the universal source-zero vacuum family.

For the flat/Abelian local construction there is an explicit primary definition: Dimock, arXiv:1712.10029, (278), uses the enlarged cube's own periodic inverse and local projector; (310)–(312) gives its short typed factorization. This verifies autonomous local boundary data in that case. The non-Abelian background version of the elementary factors in CMP99 (3.107) is still the exact construction convention to verify. Inherited members inside a source-zero fluctuation integral are evaluated at nonflat fields, so finite jets at the identity are insufficient. The diagnostic S100–101 shows that an unmatched g^{2m} error

Section 90.6 / Exact intertorus matching has a narrower source-definition requirement

per cell, however high its fixed jet order, does not survive the four-dimensional ultraviolet root count. This is a limitation of the proposed inference, not an actual counterexample to the original factors.

S102–104 also proves an alternative comparison lemma. Two source-zero vacua which both give $\pm C\,\mathrm{Vol}(T)$ stability for the SAME positive Wilson partition function differ by at most that order. Therefore a second canonical decomposition with nonwrapping matching would transfer the required vacuum-density bound to the original decomposition. No second such complete source-zero construction is claimed: agreement of new augmented maps on the physical critical graph alone does not imply the requisite stability premise.

Section 90.7 / Hypotheses and conclusions

No remaining circular ordinary-source stability assumption was found after making the homogeneous bootstrap and source radius explicit. The original source-free range, Gaussian, positive reserve, and preparation theorems remain imports. Their reuse is restricted to the actual maps and domains stated in S; a new periodic critical family is not substituted for the original independent-current extension.

The physical ultraviolet coefficient comparison, the complete operator-family mixing problem, and any claimed full Yang–Mills or mass-gap conclusion remain separate. The dimension-six free normalization proves nondegeneracy of its free reference, not the interacting insertion theorem.

Section 91.1 / Hypotheses and conclusions

I found no counterexample to the conditional scalar closure after the repairs below. I did find two places where a short formulation could support a false stronger inference.

First, the initial heterogeneous norm must be

$$M_0^{\mathrm{het}} \le C\rho a^{\alpha}. \tag{SA1}$$

The printed bound $C\rho$ for an unsplit primitive is insufficient: the recurrence $M_{k+1} \le qM_k + C\rho\ell_{k+1}^{\alpha}$, with $q > 1$, allows a contribution $q^k M_0$. Root added S42a. It splits each primitive into its homogeneous packet at an anchor and the exact oscillation remainder. For an anchor at distance at most Ca, the profile budget gives the missing factor a^{α}; an anchor at the sampling point gives zero initial defect.

that repair. The solution of the recurrence is

$$\frac{M_k}{\rho\ell_k^{\alpha}} \le C_0(q/L^{\alpha})^k + C\sum_{h=0}^{k-1}(q/L^{\alpha})^h, \tag{SA2}$$

which is uniform when $q < L^{\alpha}$.

Second, the pointwise readout bound S39 cannot mean that one diameter-independent flat tree chart works for every carrier. A constant-curvature connection on a carrier of old-scale diameter comparable

to r^{-1}, where $r = L^{j-k}$, already gives tree coordinates of order $r\alpha_k$, rather than $r^2\alpha_k$, at distant edges. This is a counterexample to that stronger chart claim, not to the rooted formation estimate.

At root's request, I revised Section 11.2 to retain a carrier polynomial P_X in S39–S40 and to treat complementary pieces using the original admitted-domain response and localization tail. New S40a makes this second estimate explicit. The original evaluation must remain valid before a small coefficient or exponential tail is used. No decay factor authorizes evaluation outside an inherited member's analytic domain.

Section 91.2 / What was specified in the primary source-free texts

The following locators were read in the actual primary papers.

- CMP109 I defines the original paired space as a union of gauge orbits, with nested readout/current conditions (1.15)–(1.16). Its remote/large-carrier split and admitted-domain Cauchy argument are (3.5)–(3.17). Its actual off-shell substitutions, simultaneous gauge transformations, and range verification are (3.24)–(3.42). The derivative bounds used in the local analysis are (4.17)–(4.18).
- CMP116 II, (1.21)–(1.36), supplies the localized response and old fluctuation-potential construction. In particular the old-age summation in the discussion leading to Lemma 1 is already part of that construction.
- R-II fixes an exterior extension before its relative variational equation on printed p.359. Its new output and old reconstruction are separately displayed in (1.34) and (1.50). The prepared range statement (1.65) explicitly places the old compositions in their analytic domains. Printed p.378 distinguishes equality of densities from equivalence under the relevant gauge-field integral.
- CMP102 Proposition 9 explicitly retains the relative regularity conditions (19)–(21) for its analytic response.

These are useful source-free inputs. They must not be replaced by the claim that every field with a small generic first-order Sobolev norm belongs to the original paired domain. The additional nested conditions are substantive. Nor may the new periodic classical maps be silently substituted for the independent-current extension in I.

Section 91.3 / Rooted quadratic growth after the carrier split

For a local piece whose readouts and connecting analytic segment lie in the smaller birth chart, the joint zero-jet argument gives

$$|\Delta E_{j,X}(B)| \le C P_X^2 \frac{b_{j,X}}{R_j^2} r^4 (\alpha_k g_k \mathfrak{b}_X(B) + g_k^2 \mathfrak{b}_X(B)^2). \tag{SA3}$$

The zero linear jet follows from color invariance of a completed regular member for a semisimple gauge algebra. It does not apply to an outer derivative of a live function with its old colored variables held fixed.

Spend fixed spare carrier decay to sum P_X^2. The response weight $\mathfrak{b}_X^2$ is controlled by an exponentially weighted sum of actual current fluctuation coordinates. There are $O(r^{-4})$ old roots per current cell. Thus the r^4 in (SA3) is precisely the factor needed for the rooted sum; it does not supply a further strict age gain for a general quadratic member.

For a complementary regular piece, use first-derivative Cauchy on its original admitted core, together

with the original response or large-carrier tail:

$$|\Delta E^{\mathrm{rem}}_{j,k,X}(B)| \le C\,\frac{b_{j,X}}{R_j}\,g_k P_X \mathcal{T}_{j,k,X}\mathfrak{b}_X(B). \tag{SA4}$$

This does not use a flat Taylor chart. For the original remote or physically large split, part of the tail is $e^{-c/r}$. If the only reason for rejecting the local flat chart is $P_X r^2 \alpha_k \gtrsim R_j$, that condition itself forces a large old carrier. Because P_X is a fixed polynomial and α_k/R_j grows at most polynomially in $k-j$ within the shrinking-core class, its unused exponential carrier decay likewise dominates every prescribed power of r. The finitely many near ages use the original spare-core estimate.

The radius comparisons here are made within admitted birth-core windows. In Section 11.2, α_k denotes the current analytic class ceiling, not a mandatory physical fluctuation amplitude. Clean cores are fixed, while still-shrinking R cores use the actual running tube. On those windows $R_j \le C\alpha_k$, and the opposite ratio is fixed or at most polynomial in the age. One must not give a fixed clean member a new shrinking radius and then pretend these constants are unchanged. Equally, an R birth evaluated against a fixed clean ceiling can have $\alpha_{\mathrm{clean}}/R_j$ growing with x_j at short age. Its actual promotion/preparation delay and radius conversion must remain in the tagged kernel S44b, with fixed polynomial conversions in its rare envelope. No administrative relabeling proves admission to a larger field tube.

On a class-preserving admitted window, after reserving enough of the tail, (SA4) is bounded by

$$C\frac{b_{j,X}}{R_j^2} r^4 \alpha_k \delta_k (1 + \mathfrak{b}_X(B)^2), \tag{SA5}$$

since $g_k \le \delta_k$. Combining the two parts and their rooted sums gives the classwise S41. The input is the sum of those contributions and any actual admitted delayed transitions with their radius-explicit tagged bounds. This checks the rooted estimate under the original domain/range inputs; it is not a claim that S39 holds as one global chart estimate on every carrier or after any chosen change of field radius.

Section 91.4 / The homogeneous disc does not move a sharp physical cutoff

See the fixed-coordinate construction in marked-bridges Sections 11.1–11.2 and 19.2/19.6, on which S44a–c relies. The coordinate parametrization uses the real reference g_s, including in the nonlinear constraint correction. It does not replace that reference by a complex $g_s(c)$ inside a change of integration variables.

The elementary Gaussian identity making this precise is as follows. On the actual n-dimensional independent real fluctuation space, let $Q = C^{-1}$, and take t in a fixed neighborhood of one with $\operatorname{Re} t > 0$. On the same real coordinate space,

$$d\mu_{C/t}(B) = t^{n/2} \exp\left[-\frac{t-1}{2}\langle B, QB\rangle\right] d\mu_C(B), \tag{SA6}$$

with the holomorphic branch normalized at $t = 1$. Consequently, for the unchanged real cutoff set D_0,

$$\int_{D_0} e^{-\frac{1}{2}t\langle B,QB\rangle + V_t(B)}\, dB = Z_1(Q) t^{-n/2} \int_{D_0} e^{V_t(B)}\, d\mu_{C/t}(B). \tag{SA7}$$

There is no complex rescaling of B, no moved cutoff, and no unwritten boundary term in this identity. A separately normalized gauge Gaussian requires its compensating scalar normalization, as the earlier text expressly says.

Section 91.4 / The homogeneous disc does not move a sharp physical cutoff

The background-dependent full Gaussian determinant coefficient is unchanged by multiplication of the precision by a scalar t. The scalar factor $t^{-n/2}$ belongs in the vacuum/count ledger. The cutoff-dependent Gaussian integral in (SA7) still depends on t; it cannot be declared constant. The existing construction retains that dependence among the cutoff activities.

When a fixed physical outer cutoff is compared with the actual smaller III cutoff, the text uses the exact tail-flag interpolation and retains its local logarithmic difference, vacuum value, and marginal extraction. Its smallness follows from a common positive Gaussian reserve. It is not an identity between two unequal cutoff integrals.

With these points retained, S44a is a clean historical Cauchy input. The tagged Volterra estimate is applied on a provisional finite prefix, and S44c closes that prefix before a tagged source derivative is taken. I found no circular appeal to an already existing sourced island logarithm in this bootstrap.

Section 91.5 / Exact source transport and its limits

The finite-graph Haar identity S53–S54 is correct for a forest with the stipulated fixed roots and permitted gauge vertices. For fixed retained datum b, each tree variable is a left/right Haar translate of the newly solved gauge variable. Each remaining link is a left/right translate of its relative coordinate. The Jacobian is one.

The decisive normalization calculation is

$$\sum_{\alpha} \int \frac{\chi_{\mathrm{ref}}(C) r_b(CR(b))}{D(b)} \left[\int q_0(\xi, b) M(\xi, b) \pi_\alpha(C, \xi, b)\, d\xi \right] dC = \int q_0(\xi, b) M(\xi, b)\, d\xi. \tag{SA8}$$

Its two requirements are that the reference factor is normalized at fixed full b, and that the selector sum is one before normalization is used. The observable remains a function of the old variables ξ. A selected history need not have unit mass.

The separate old and new slots and the fixed reference construction in the scope R-II formulas support this interpretation. An old variable on which the reference depends must remain in b until its own integration occurs; its denominator need not factor out of that later integral. Gauge removal is an integrated operation and requires joint invariance of every retained old slot and the observable. These restrictions are present in the current text.

No false density equality or hidden evaluation of the old observable on the replacement coordinate was found. This conclusion uses the published unmarked preparation identity with its original slots and supports. It would not follow from the mere existence of a quotient or from equality of two unmarked final partition functions.

Section 91.6 / Primitive coverage, reserve, and formation

The primitive trace estimate has the correct engineering count. A controlled covariant first difference bounds a reconstructed plaquette displacement by $C(a/\ell)^2\epsilon_*$. The symmetric trace observable has a quadratic zero, hence its size is $C(a/\ell)^4\epsilon_*^2$. The number of original plaquettes in the assigned own-scale layer is $C(\ell/a)^4 N_\ell$. These factors cancel. A genuinely own-spacing retained compact plaquette instead uses its compact bound and its actual own-spacing record count.

This estimate is gauge invariant. It neither requires nor implies a bound for an unrelated complex conjugating matrix. However, the mixed old/new readouts must still be the ones covered by the original range theorem. An arbitrary product of an uncontrolled complex background with a wild

stored field is not covered by the two primitive estimates merely because its factors have familiar names. The actual domain statements, determining shells, and collars are essential to Section 19.2.

The reserve implication S13–S17 is an elementary correct one-measure argument. S86 is stronger than an unmarked mass bound. Its justification is the positive cold calculation with fixed unspent energy fractions and per-record volume allowance, not the small numerical value of $T1$. Changing a positive Gaussian coefficient by a fixed factor changes its integral by an exponential per-coordinate cost. A fixed fractional reduction of a failed-test energy changes its rarity exponent by the same fraction. The source-free surplus must pay those changes once in its actual record ledger. The current proof states that strengthening as a reuse of the source-free positive calculation; it does not infer exponential moments from mass alone.

No multiplication of two independently estimated masses of a shared component occurs in the corrected joining bound. The masks are multiplied and estimated in the same joint integral. The first and second source excess powers then follow from the exponential remainder formulas.

For local formation, the derivative bound S7 follows from S5 by one-dimensional Cauchy on $V + zH$, with a radius proportional to $a_*/\|H\|$. This does not require a zero-free disc for the full partition function. The connected local expansion must first be constructed with the stated Gaussian/wall and quadratic-growth inputs. The analysis does not replace that source-free Gaussian construction by a scalar partition bound.

Section 91.7 / Later consequences and the remaining scope

Given S87 and exact completed-carrier source support, the local Taylor argument S88–S91 has the correct powers: $r_X^{5-\alpha}$, followed by a four-dimensional root count, leaves the summable factor $\ell^{1-\alpha}$. The cutoff is needed to make the Taylor remainder a globally admissible C^α direction. Disjoint positive-side jets give the diagonal congruence in S93.

The common-normalizer argument S94–S99 still requires S94. Its construction uses sourced coefficients only on the reference torus, so it does not require matching sourced coefficients between two tori. Nothing checked here proves the missing full non-Abelian typed local-factor naturality in CMP99.

Likewise, the fixed-jet diagnostic S100–S101 tests an invalid inference; it is not an example of actual ambient-volume dependence in Bałaban's construction.

The phrase "unconditional within-torus consequence" should mean independence from the extra cross-volume premise S94, while retaining the source-free inputs of the scalar theorem. It cannot mean an unconditional interacting continuum existence theorem. The physical ultraviolet identification, arbitrary insertion mixing, and common continuum hierarchy remain separate obligations.

Section 92.1 / Result and scope

The elementary fixed-measure route does **not** establish S94. It gives an exact gluing identity and a boundary estimate, but the latter costs $O(\beta_a a^{-3}/\ell_*)$ per physical volume. Along the prescribed four-dimensional trajectory, $\beta_a \asymp \log(1/a)$, this is not uniform. Reflection positivity supplies the opposite side of the pressure comparison. Conditional factorization does not remove the interface free energy.

This is a missing estimate, not a counterexample to S94 or to the original Wilson measure. No inaccessible typed-factor definition is assumed below. No source norm is substituted for a source-zero partition-function comparison, and no coupling is retuned.

The fixed inputs are S1–S2, S74–S81, S94–S95, and the exact projected transfer identity in marked-bridges Section 18.1. The original vacuum stability used below is also the FIRST inequality in S102. No second vacuum $\widetilde{V}$, proposed in the other inequality of S102, is used or asserted to have been constructed. The discussion does not edit or change those notes.

Write

$$T_* = (\mathbb{R}/\ell_*\mathbb{Z})^4, \qquad T = qT_*, \qquad n = \ell_*/a, \qquad M = q^4. \tag{SV1}$$

All lattices and reflection planes are compatible with the construction. Use normalized Haar measure and exactly the same homogeneous bare coefficient $\beta_a = x_a$, chosen by the existing tuned trajectory:

$$s(U) = 1 - \frac{1}{N}\operatorname{Re}\operatorname{Tr} U, \qquad 0 \le s \le s_{\max} \le 2, \qquad Z_{a,R} = \int e^{-\beta_a \sum_{p\subset R} s(U_p)} \prod_{b\subset R} dU_b. \tag{SV2}$$

Any fixed group-dependent convention in the inverse coupling is included in β_a. There are $6\,\mathrm{Vol}(R)/a^4$ plaquettes.

Set $p_{a,R} = \mathrm{Vol}(R)^{-1} \log Z_{a,R}$. The original source-zero stability bound follows from S74–S81 (the upper bound at $z = 0$, together with S79), equivalently the first inequality in S102:

$$|\log Z_{a,R} - V_{a,R}| \le C_{\mathrm{st}}\,\mathrm{Vol}(R),$$

then

$$\left| \left(\frac{V_{a,T_*}}{\ell_*^4} - \frac{V_{a,T}}{M\ell_*^4} \right) - (p_{a,T_*} - p_{a,T}) \right| \le 2C_{\mathrm{st}}. \tag{SV3}$$

Thus the missing one-sided vacuum comparison is equivalent, up to this already available constant, to

$$p_{a,T_*} - p_{a,T} \le C. \tag{SV4}$$

A common bulk normalization $Z_{a,R} \mapsto e^{c_a\,\mathrm{Vol}(R)} Z_{a,R}$ cancels exactly in (SV4). Independent volume-dependent normalizations cannot be inserted.

Section 92.8 / Hypotheses and conclusions

The following facts are now established directly for the exact finite-cutoff Wilson integral at the existing tuned coupling:

1. The vacuum comparison reduces to a physical pressure comparison by (SV3).
2. The same-variable gluing identity is (SV5).
3. The elementary cut-boundary bound is (SV7)–(SV8), with its ultraviolet divergence explicit.
4. Independent block gauge symmetry explains why a local small plaquette estimate does not improve the Jensen comparison (SV9).

Section 92.8 / Hypotheses and conclusions

5. The exact missing signed boundary estimate is (SV11).
6. The uniform normalized transfer trace estimate (SV16) is an alternative sufficient condition.
7. Any change of bare trajectory has the explicit additional cost (SV19), or (SV20) under its stated extra premise.

Neither (SV11) nor (SV16) has been proved by this continuation. Accordingly S94 remains open at the same narrow source-zero intertorus interface. Original typed-factor naturality, a second source-zero construction satisfying S102, or a genuine uniform boundary/spectral estimate for the same Wilson measure would close it. Finite range, reflection positivity, and elementary coupling monotonicity alone have not done so.

Section 93 / Exact contact completions and a local finite-jet Wilson

This note extends the exact finite-cutoff sector of the earlier dimension-six and completed-insertion chapters. It does **not** establish the arbitrary composite-operator theorem. Its new results are:

1. an ordered, all-degree Haar equation-of-motion completion, including its contact multiplication rule;
2. an exact, physically smeared lattice reduction of the scalar first derivative-curvature square, with the Wilson force, its counterterm, and every cube and transport remainder explicit;
3. a finite-dimensional, normalized conditional Wilson Taylor theorem on a fixed microscopic block, requiring no exponential source for the inserted observable;
4. a fixed-degree moment estimate using one common positive integration majorant. Its remaining hypothesis is stated, not borrowed from a conditional source theorem.

The second result removes the current attachment exactly. It leaves an explicit finite-range cubic, a quadratic descendant, and terms which are classically small but still require quantum local mixing. The third result provides an actual local jet estimate, but not field-independent continuum normalization of those jets.

Section 93.2 / The exact EOM module and all ordered products

For a globally smooth Lie-algebra-valued bond coefficient V, define

$$\mathscr{E}_a[V] = a^4 \sum_{b,A} \{J_b^A V_b^A - \kappa_a X_b^A V_b^A\}. \tag{AI7}$$

Haar integration by parts gives, for every globally smooth G,

$$\langle \mathscr{E}_a[V] G \rangle_{\mu_a} = \kappa_a a^4 \sum_{b,A} \langle V_b^A X_b^A G \rangle_{\mu_a}. \tag{AI8}$$

The proof is the integral of X_b^{A(V_bA G e^{-beta_a S_0}); the Haar integral of an invariant derivative is zero. This also fixes all factors in AI8. If V transforms as a tangent vector under lattice gauge transformations, AI7 is gauge invariant: it is the weighted divergence of that vector field.

In particular AI7 has zero correlation with any product of observables whose determining bonds are disjoint from the differentiated bonds b. This is exact for every a, beta_a, and finite volume. It is not merely an asymptotic Schwinger-Dyson assertion.

The multiplication rule is

$$P\mathscr{E}_a[V] = \mathscr{E}_a[PV] + \kappa_a a^4 \sum_{b,A} V_b^A X_b^A P. \tag{AI9}$$

Thus the vanishing sector is a contact-completed EOM module. Multiplying an EOM representative by another field produces the displayed contact term; one cannot set an uncompleted EOM factor to zero inside arbitrary products. Covariant spacetime derivatives of J can first be moved onto V and the test function by exact lattice summation by parts, and then completed by AI7. The resulting coefficient V and every link derivative of V remain present.

There is an all-degree form. Combine b,A into i and put

$$K_i^\dagger = J_i - \kappa_a X_i, \qquad \mathcal{H}_{i_1\dots i_m}[V] = K_{i_1}^\dagger \cdots K_{i_m}^\dagger V. \tag{AI10}$$

The dagger is a weighted adjoint, not complex conjugation of coefficients. Repeated integration by parts proves

$$\langle \mathcal{H}_{i_1\dots i_m}[V] G\rangle = \kappa_a^m \langle V X_{i_m} \cdots X_{i_1} G\rangle . \tag{AI11}$$

Its leading term is J_i1 … J_im V. At degree two the exact expression is

$$\mathcal{H}_{ij}[V] = J_i J_j V - \kappa_a\{(X_i J_j)V + J_j X_i V + J_i X_j V\} + \kappa_a^2 X_i X_j V. \tag{AI12}$$

If [X_i,X_j]=c_ijˆk X_k in the convention AI3, then [K_i$^{\text{dagger,K_j}}$dagger]=-kappa_a c_ijˆk K_kˆdagger. This follows by taking weighted adjoints of the commutator; it does not assume a potentially opposite sign convention for left versus right invariant vector fields. Symmetrizing AI10 is possible, but its lower-degree commutators must be kept.

The support of AI10 lies in the support of V and the plaquette stars of its differentiated bonds. If any differentiated bond is absent from the support of G, the right side of AI11 is zero. In particular, arbitrary fixed-degree completed EOM insertions have identically zero separated correlations and connected correlations. The constants in this separated statement do not grow with the cutoff because the correlation is zero. This proves no bound for coincident contact distributions, and it gives no reduction of arbitrary curvature words to the EOM module.

For completeness, exact coarse pushforward preserves this divergence interpretation. For a globally smooth coarse observation M and test g,

$$\langle \mathscr{E}_a[V] g(M)\rangle = \kappa_a a^4 \langle (DM)V \cdot dg(M)\rangle . \tag{AI13}$$

In measure language its pushforward is minus the divergence of the vector-valued measure kappa_a aˆ4 M_*((DM)V mu_a). This identity by itself does not bound that conditional vector measure or make it local. It cannot be substituted for a missing RG attachment estimate.

93.2.1 2.1 An explicit Casimir counterterm

Each plaquette meeting b contains that link exactly once. Consequently

Section 93.2 / The exact EOM module and all ordered products

$$\sum_A (X_b^A)^2 S_0 = C_F \sum_{p \ni b} \mathrm{Re}\,\mathrm{tr}\, P_p, \qquad \sum_A X_b^A J_b^A = C_F a^{-3} \sum_{p \ni b} \mathrm{Re}\,\mathrm{tr}\, P_p. \tag{AI14}$$

The exact completed current square is therefore

$$|J_b|^2 - C_F \beta_a^{-1} a^{-6} \sum_{p \ni b} \mathrm{Re}\,\mathrm{tr}\, P_p. \tag{AI15}$$

Since Re tr P=N-s(P), this is a fixed identity-plus-Wilson-action mixing, with known coefficients. It is an exact finite-cutoff computation, not merely the classical quadratic jet of a proposed counterterm.

Section 93.3 / Exact scalar dimension-six reduction on the Wilson lattice

All sums over curvature indices below are full ordered sums. Write

$$B_{\rho\mu\nu} = D_\rho X_{\mu\nu} + D_\mu X_{\nu\rho} + D_\nu X_{\rho\mu}. \tag{AI16}$$

This is the additive flux Bianchi defect; it is not set to zero. Define

$$\begin{aligned} A_a^{\mathrm{sym}}(f) &= \frac{a^4}{2} \sum_{x,\rho,\mu,\nu} f(x)\{|D_\rho X_{\mu\nu}(x)|^2 + |D_\rho^* X_{\mu\nu}(x)|^2\}, \\ R_a(f) &= a^4 \sum_{x,\rho,\mu,\nu} f(x)\langle X_{\mu\nu}, [D_\rho^*, D_\mu] X_{\rho\nu}\rangle, \\ V_{f,\nu}(x) &= \sum_\mu (D_\mu^* f)(x) T_\mu^{-1} X_{\mu\nu}(x), \qquad \Delta_a = -\sum_\rho D_\rho^* D_\rho \quad \text{on scalar functions.} \end{aligned} \tag{AI17}$$

The reflection-symmetric density is useful here: its physical test weight on the forward energy is (f(x)+f(x+a rho))/2. No discarded O(a DX squared) term is hidden in that choice.

The exact smeared identity is

$$\begin{aligned} A_a^{\mathrm{sym}}(f) = 2a^4 \sum_x f|J|^2 + 2R_a(f) + \frac{a^4}{3} \sum_{x,\rho,\mu,\nu} f|B_{\rho\mu\nu}|^2 + 2a^4 \sum_{x,\nu} \langle V_{f,\nu}, J_\nu\rangle \\ + \frac{a^4}{2} \sum_{x,\mu,\nu} (\Delta_a f)|X_{\mu\nu}|^2 + a^4 \sum_{x,\rho,\mu,\nu} (D_\rho f)\langle T_\rho X_{\mu\nu}, B_{\rho\mu\nu}\rangle. \end{aligned} \tag{AI18}$$

Derivation. At each site antisymmetry gives sum|B|^2=3 sum|DX|^2-6 sum<D_rho X_mu,nu,D_mu X_rho,nu>. Move D_rho to its adjoint in the cross term, commute D_rho* past D_mu, and move D_mu to its adjoint. The commuting part is f|J|^2. Commuting past the scalar test produces

$$\sum_{\mu,\nu} \langle [D_\mu^*, f] X_{\mu\nu}, J_\nu\rangle + \sum_{\rho,\mu,\nu} \langle X_{\mu\nu}, [D_\rho^*, f] D_\mu X_{\rho\nu}\rangle. \tag{AI19}$$

The first term is <V_f,J>. Changing variables by one rho step makes the second term -sum(D_rho f)<T_rho X_mu,nu,D_mu X_rho,nu>. Substitute AI16 and interchange mu,nu in one term. This equals

$$-\frac{1}{2}\sum (D_\rho f)\langle T_\rho X_{\mu\nu}, D_\rho X_{\mu\nu}\rangle + \frac{1}{2}\sum (D_\rho f)\langle T_\rho X_{\mu\nu}, B_{\rho\mu\nu}\rangle. \tag{AI20}$$

Finally <T_rho X,D_rho X>=D_rho|X|^2/2+a|D_rho X|^2/2. Scalar summation by parts gives Delta_a f. Moving the remaining forward energy term to the left produces exactly the symmetric weight in AI17. This proves AI18 with no continuum interpolation or approximate Leibniz rule.

93.3.1 3.1 The exact point-split cubic and transport remainder

Put

$$W_{\rho\mu}(x) = T_\rho^{-1} P_{\mu\rho}(x), \qquad Y_{\rho\mu\nu} = T_\mu T_\rho^{-1} X_{\rho\nu}. \tag{AI21}$$

The first formula means the transported plaquette at x-a rho. The exact shift commutator is

$$[D_\rho^*, D_\mu] X_{\rho\nu} = a^{-2} (\mathrm{Ad}\, W_{\rho\mu} - I) Y_{\rho\mu\nu}. \tag{AI22}$$

For any unitary W set A=W-I, C=W^{-1}-I. Direct multiplication gives

$$\begin{aligned} (\mathrm{Ad}\, W - I)Y &= \left[\frac{W - W^{-1}}{2}, Y\right] + \mathcal{R}_W(Y), \\ \mathcal{R}_W(Y) &= AYC - \tfrac{1}{2}(ACY + YAC), \qquad |\mathcal{R}_W(Y)| \le C\|W-I\|^2 |Y|. \end{aligned} \tag{AI23}$$

The traceless projection in AI4 changes (W-W^{-1})/2 by a scalar matrix, which has zero commutator. Thus no central remainder is being omitted. Moreover a^{-2} pi(W-W^{-1})/2=-T_rho{-1}X_rho,mu exactly. It follows that

$$\begin{aligned} R_a(f) &= -C_a^{\mathrm{split}}(f) + R_{4,a}(f), \\ C_a^{\mathrm{split}}(f) &= a^4 \sum f\, \langle X_{\mu\nu}, [T_\rho^{-1} X_{\rho\mu}, T_\mu T_\rho^{-1} X_{\rho\nu}]\rangle, \\ R_{4,a}(f) &= a^4 \sum f\, a^{-2} \langle X_{\mu\nu}, \mathcal{R}_{W_{\rho\mu}}(Y_{\rho\mu\nu})\rangle. \end{aligned} \tag{AI24}$$

This is a finite microscopic point-split cubic. Replacing it by a chosen onsite cubic is an additional comparison, not an equality already proved by AI24.

It is charge-conjugation **even**. In the anti-Hermitian convention charge conjugation sends F to -F^T and sends a transported F in the same way. For three such fields, [(-Y^T),(-Z^T)]=-[Y,Z]^T, and invariance of the trace gives < -X^{T,-[Y,Z]}T>=<X,[Y,Z]>. Thus the cubic in AI24 is not a member of CH's 56 charge-conjugation-odd symmetric-color cubic center. The latter's absence of a degree-two field jet after formation follows from its odd parity; that argument is unavailable here. A C-even cubic can mix with quadratic derivative-curvature operators of the same engineering dimension. D6 already identifies this as a separate even center.

93.3.2 3.2 The cube remainder is an explicit quadratic plaquette defect

An exact six-face identity, obtained by sorting the three covariant fundamental shifts in two orders, is

$$P_{\rho\mu}(T_\mu P_{\rho\nu})P_{\mu\nu} = (T_\rho P_{\mu\nu})P_{\rho\nu}(T_\nu P_{\rho\mu}). \tag{AI25}$$

Let A_1,A_2,A_3 and C_1,C_2,C_3 denote its two triples of factors, in the written order, and set alpha_i=A_i-I, gamma_i=C_i-I. Define Q(alpha)=alpha_1 alpha_2+alpha_1 alpha_3+alpha_2 alpha_3 +alpha_1 alpha_2 alpha_3. Expanding the two products gives

$$a^3 B_{\rho\mu\nu} = \pi_{\mathfrak{su}(N)}\{Q(\alpha) - Q(\gamma)\}. \tag{AI26}$$

The sign follows from sum pi(C_i)-sum pi(A_i)=a^3B. Since all factors are unitary, AI26 proves

$$|B_{\rho\mu\nu}| \le C a^{-3} \sum_{p \text{ on this cube}} \|P_p - I\|^2. \tag{AI27}$$

No logarithmic Bianchi identity and no small-plaquette assumption has been used. Projection commutes with the unitary transports, so the same argument applies to the traceless flux actually entering the Wilson force.

93.3.3 3.3 Complete removal of both current terms

AI7 applied to V=fJ+V_f removes the two current terms in AI18. The derivative counterterm is completely explicit. In addition to AI14, direct differentiation of the incoming plaquette yields

$$\sum_A X^A_{(x,\nu)} V^A_{f,\nu} = C_F a^{-2} \sum_{\mu\neq\nu} (D^*_\mu f)(x)\, \mathrm{Re}\,\mathrm{tr}\, P_{\mu\nu}(x - a\hat{\mu}). \tag{AI28}$$

Combining this with f times AI14, shifting the lattice sum, and counting the four bonds of each unoriented plaquette gives

$$\mathcal{C}_a(f) := 2\kappa_a a^4 \sum_{b,A} X^A_b (f J^A_b + V^A_{f,b}) = 8 C_F \beta_a^{-1} a^{-6} a^4 \sum_{x,\mu<\nu} f(x)\, \mathrm{Re}\,\mathrm{tr}\, P_{\mu\nu}(x). \tag{AI29}$$

Thus the exact same-measure identity is

$$\begin{aligned} A^{\mathrm{sym}}_a(f) - \mathcal{C}_a(f) &= 2\mathscr{E}_a[fJ + V_f] - 2C^{\mathrm{split}}_a(f) + \frac{a^4}{2}\sum (\Delta_a f)|X_{\mu\nu}|^2 + \mathcal{R}_a(f), \\ \mathcal{R}_a(f) &= 2R_{4,a}(f) + \frac{a^4}{3}\sum f|B_{\rho\mu\nu}|^2 + a^4 \sum (D_\rho f)\langle T_\rho X_{\mu\nu}, B_{\rho\mu\nu}\rangle. \end{aligned} \tag{AI30}$$

The EOM term has exactly zero correlation with observables on disjoint determining bonds. AI30 is therefore a genuine removal of the troublesome near-current attachment for this scalar and this lattice representative, including variable physical tests. It does not identify the remaining point-split cubic with every previously normalized cubic representative.

93.3.4 3.4 What still prevents the proposed scalar shortcut

Let e_C=sum_{p in C}s(P_p) on a fixed enlarged cube carrier. Unitarity gives $\|P - I\|_F^2 = 2s(P)$. From AI4, AI23, and AI27, pointwise bounds for the local summands in AI30 are

$$|r_{4,a}| + |B|^2 \le Ca^{-6}e_C^2, \qquad |(Df)\langle TX, B\rangle| \le C\|D_a f\|_\infty a^{-5}e_C^{3/2}. \tag{AI31}$$

On a fixed smooth classical connection these become O(a^2 F^4) and O(a |Df| F^3), respectively. On the Wilson measure the cold moment e_C^{2=O(beta_a{-2})}, even if separately available, would give an a^{-6} beta_a^{-2} bound rather than a vanishing one. No power of a can be discarded before the local lower-dimensional pieces of this remainder are extracted.

A precise elementary warning is furnished already by a scalar Gaussian curvature component Z_a with variance c_a asymptotic to a^{-4}. Wick's identity reads

$$a^2 Z_a^4 = a^2 : Z_a^4 : +6a^2 c_a : Z_a^2 : +3a^2 c_a^2. \tag{AI32}$$

After removing the vacuum term, the quadratic mixing still has size a^{-2}. This is not a counterexample to a cancellation special to AI30; it disproves the general inference that its classical a^2 F^4 bound already makes the quantum remainder negligible. A cancellation particular to the sum in AI30, or a fixed physical mixing subtraction with a strict completed remainder bound, must actually be proved.

Thus an exact remaining target for this scalar is a finite local subtraction of R_a(f), fixed independently of the other source locations, such that its completed separated correlations are O(a^epsilon), together with a comparison of C_a^split to the chosen normalized cubic family. The current term itself no longer forms part of that target.

93.3.5 3.5 An exact alternative choice of the cubic completion

There is a useful algebraic alternative to first proving that R_a vanishes. Define a new **bare observable**, without changing the Wilson measure, by

$$C_a^\sharp(f) := C_a^{\mathrm{split}}(f) - \tfrac{1}{2}\mathcal{R}_a(f). \tag{AI32a}$$

This is a genuine local density tested against f: the term containing D_rho f in R_a can be moved onto its scalar coefficient by exact summation by parts. More explicitly its density is

$$c_a^\sharp = c_a^{\mathrm{split}} - r_{4,a} - \tfrac{1}{6}\sum_{\rho\mu\nu}|B_{\rho\mu\nu}|^2 - \tfrac{1}{2}\sum_\rho D_\rho^* \sum_{\mu\nu}\langle T_\rho X_{\mu\nu}, B_{\rho\mu\nu}\rangle. \tag{AI32b}$$

It is globally smooth, gauge invariant, C even, and supported in a fixed enlarged microscopic carrier. Its constant, linear, and quadratic ordinary field jets at the trivial connection vanish. Indeed X starts linearly, the linear part of B is zero by the exact flat lattice Bianchi identity, R_4 starts in fourth field degree, and X B starts in third degree. On a smooth classical connection its leading symbol is the same continuum scalar cubic as C_split; the added terms are its explicit lattice corrections, not discarded errors.

The same fixed carrier has the unconditional compact bound

$$|c_a^\sharp(x)| \le Ca^{-6}e_C(x)^{3/2}. \tag{AI32c}$$

For its cubic part this is immediate from |X|<=Ca^{-2}e_C{1/2}. The fourth-degree terms in AI31 satisfy e_Cˆ2<=C e_Cˆ{3/2}, because e_C is bounded on this fixed compact carrier. The last difference in AI32b costs aˆ{-1}, while |X B|<=Ca^{-5}e_C{3/2}. Enlarging the carrier by one site proves the displayed bound. Thus this completion has the same cubic cold-growth estimate as a primitive cubic, despite containing higher field-degree words.

Equation AI30 becomes the exact triangular identity

$$A_a^{\mathrm{sym}}(f) - \mathcal{C}_a(f) = 2\mathscr{E}_a[fJ + V_f] - 2C_a^{\sharp}(f) + \tfrac{1}{2}Q_a(\Delta_a f), \qquad Q_a(g) = a^4 \sum_{x,\mu,\nu} g(x)|X_{\mu\nu}(x)|^2. \quad \text{(AI32d)}$$

This gives an exact bare-basis reduction of the full scalar derivative operator to a selected C-even cubic completion, a quadratic descendant, and known EOM/lower-dimensional terms. It uses no previously normalized cubic. If C_sharp and Q are constructed and normalized jointly, AI32d then defines the corresponding derivative-scalar renormalization on the same Wilson sequence, with contact multiplication governed by AI9. It does not prove that the naive C_split regulator and C_sharp have the same limit before their mixing is fixed. Nor does the vanishing bare quadratic jet prevent quadratic terms from being generated by integration.

Section 95 / A nondegenerate free normalization for the coupled

AI reduces the reflection-even hypercubic scalar problem at dimension six to two genuine directions after equations of motion, descendants and lower dimensions are accounted for: the completed cubic C-sharp and the residual directional-current quadratic H. This chapter checks that the two directions have a nondegenerate free physical normalization. It also proves the elementary matrix-transport estimate needed once an actual interacting transfer bound is available. The latter bound is stated explicitly and is not proved here.

Section 96 / A covariant smeared dimension-six remainder on the actual lattice

This note proves a regular readout estimate. It uses the protected fractional core proved in CL28–33 (and the mixed own-scale version MC); it does not prove a quantum first-insertion transfer or the invertibility of the candidate even dimension-six center. The estimate is an identity and inequality of actual lattice sums. No product of lattice fields is identified with a product of continuum interpolants.

The general path estimate below has a remainder of order r^{2+s}. That already gives the strictly summable r^s gain after dimension-six engineering rescaling. Section 5 proves the sharper r^{2+2s} result for a single coordinate path. We do not infer that stronger exponent for an arbitrary ordered path merely from its length bound.

Section 96.7 / Descendants, contacts, and the actual remaining insertion problem

The quadratic descendant in (CD17) is part of the pointed local projection. It does not vanish when f is not constant. It can be defined as a distributional derivative of the already retained quadratic family, using exact lattice summation by parts at finite cutoff. There is no new conditional high-jet estimate in this step.

The exact identities AI30 and AI32a–d in the stated results supply a fixed compact completion $C_a^\sharp$, with the same cubic cold growth, and reduce one derivative scalar to that completion, a quadratic descendant, an EOM/contact representative, and literal lower-dimensional terms. The complementary hypercubic quadratic direction H_a in AI33–35 remains. The projection in (CD17) retains the entire derivative-curvature tensor first, so it is compatible with that distinction and does not incorrectly remove H_a by an $O(4)$ scalar identity.

There is also an exact limitation. A Wilson equation-of-motion identity inside the full measure is not a pointwise identity for an effective background field. Its pushforward includes the derivative of the remaining effective interaction and the boundary/contact terms. Thus (CD17) does not authorize deleting a current-dependent effective vertex. The known compact EOM completions must be carried through the actual integration as in AI9–13.

What is proved here is the following conditional regular-age corollary:

- if a completed marked birth has its quadratic part represented by the actual covariant carrier kernels in (CD22), including the exact translation/phase pairing needed in (CD12), or a separately proved coefficient-difference norm;
- if its cubic and higher parts obey the common incoming analytic coefficient norm, with all original roots, source tests, and current/EOM completions retained; and
- if the readout lies on the protected own-scale CL/MC fractional core, with its original analytic admission margin,

then subtracting the full local dimension-four and dimension-six moments leaves the strict integrated age gain (CD25), with the shrinking-core factors shown in (CD26).

The missing interaction step is concrete: the actual Wilson first-insertion formation for the coupled even family $(C_a^\sharp, H_a)$, together with its lower quadratic/vacuum sectors, descendants, and EOM/contact module, must be shown to map this coefficient-kernel class into itself. Its finite center matrix must be estimated and shown uniformly invertible after engineering rescaling; its fluctuation differences

must carry the same rooted norm and the strict remainder just proved. Charge conjugation does not remove this family's lower-degree mixing. Neither finite dimensionality nor a regular classical readout estimate supplies those operator estimates.

The result therefore closes a previously missing **regular fractional quadratic remainder**, not the full quantum dimension-six construction and not arbitrary higher-derivative insertion families.

Section 97 / The actual coarse mean preserves the regular dimensionsix readout

This note proves a regular coarse-to-fine comparison for the ACTUAL centered mean of CM/GQ. No new averaging constraint is introduced. The proof uses a small potential chart with H^{2+s} and $C^{1,\beta}$ control. It compares the gauge-invariant tested observables H, $C^\sharp$, and the explicitly retained quadratic descendants. It does not identify a wall-deformed map with a genuine mean, replace an unmarked integration kernel, or prove an arbitrary history-dependent kernel has translation symmetry.

Section 97.7 / Raw gauge, regular age, and exact limits of the conclusion

Let V_r be the ORIGINAL raw iterated mean of U. GQ4 gives $e^{rA_k} = V_r^{h_k}$ exactly. Both sides of (CB30), (CB34), and (CB35) are gauge-invariant trace values, including their covariant differences and path transports. They may therefore be evaluated in the controlled straightened representative e^{rA_k}. No new estimate for derivatives of the accumulated raw gauge is required for these scalar VALUE identities. For complex links, the polynomial flux formulas and the invariant bilinear trace preserve the same identity. An estimate for a gauge-dependent raw potential, or for an untraced tensor in its original frame, would be a different assertion.

Thus CD can be applied directly to a regular universal coarse-curvature member on its OWN r-lattice, using (CB19) and the full coarse root sum. After the complete dimension-six moment projection, (CB30), (CB34), and (CB35) identify its retained H, $C^\sharp$, and descendant values with the corresponding fine protected readouts up to a strict r^s error.

Three restrictions are essential.

1. The dimension-FOUR quadratic family is kept exactly as its actual lower-dimensional state. A mere $O(r^s)$ approximation of that family cannot be multiplied by a dimension-six normalization r^{-2}. No such replacement is made here.
2. CD still requires the actual completed coarse translation/reflection pairing, including phases, or a proved norm for coefficient differences. An arbitrary root- or history-dependent coefficient may oscillate at frequency $1/r$. Coarsening the field does not repair that defect in the coefficient kernel.
3. Every path and determining prefix is inside the admitted regular chart. A carrier meeting a rough hole remains in its actual boundary/live operator. Exponential coefficient decay does not authorize evaluating a member outside its analytic domain.

For a graded stable member, the high-field tail has a useful separate algebraic count. A full-core bound $C\alpha_j^d g_j^{-d}$, $d = 2$ or 3, gives a FOURTH-ORDER TAYLOR REMAINDER coefficient bound $Cg_j^{-d}\alpha_j^{d-4}$ on a fixed smaller core. On a current canonical regular readout its four-field part therefore has size

$$g_j^{-d}\alpha_j^{d-4}(g_k r^2)^4. \tag{CB36}$$

After the r^{-4} old-root count and the r^{-2} dimension-six engineering factor, this is

$$r^2\alpha_j^{d-4}g_k^{4-d}(g_k/g_j)^d. \tag{CB37}$$

The low quadratic and cubic kernels instead retain their own canonical grades g_j^{-2} and g_j^{-3}. This calculation does not prove their incoming kernel norms or the actual marked formation row. It shows precisely why the high analytic tail needs no extra absolute g_j^{-d} loss after a correct graded readout.

Section 97.8 / Dependencies

The nonlinear map is exactly CM1–19, with its gauge-orbit identity GQ4. The strengthened estimates (CB5)–(CB8) are proved here from the same fixed-L kernel and its zero constant quadratic moment. The positive cochain identities are derived in (CB9)–(CB12). The observable definitions and exact cube/SBP identities are AI4–6, AI24–27, AI32a–d, and AI33–35 in the stated results. The regular moment subtraction referred to in Section 7 is CD12–26, with its completed translation/pairing hypothesis retained.

For the actual original mean formula, see T. Bałaban, *Renormalization Group Approach to Lattice Gauge Field Theories. I*, CMP 109 (1987), (0.4), (0.11), and (0.12):

The conclusion is a regular classical readout bridge. It does not assert an off-shell equality of new and old localization maps, or a complete quantum dimension-six normalization theorem.

Section 98 / Primitive source repair while inheriting the original

This note identifies a possible source-extension architecture precisely. It proves the physical pointwise identity and the low-core admission needed by that architecture. Section 5 also proves the canonical Gaussian polynomial-growth and $O(g)$ primitive comparison on a complete flat identity-chart determining family, using the original response bounds and the protected MC output. The full fixed physical core has an unavoidable polynomial g^{-d} cost. Its weighted inheritance, the marked rare-field transfer, the complete moment projection, and the actual dimension-six formation/center estimate remain separate obligations; they do not follow from MC alone.

Section 98.4 / Ordinary-core inheritance after formation: what is and is not needed

Keep the original unmarked formation and all its prepared readout maps. Insert (PA4) only for the selected primitive marks. A new marked birth is then an analytic function

$$E_{j+1}^{\mathrm{mark}}(X;U,J) \tag{PA8}$$

Section 98.4 / Ordinary-core inheritance after formation: what is and is not needed

on the original paired input core, if the marked Gaussian/cluster construction proves its norm there. Later ages evaluate the function (PA8) using the original admitted prepared maps. They do not apply MC anew to the argument of the whole activity.

This makes the domain issue precise. The required inheritance theorem is the ORIGINAL prepared-map admission theorem acting on an analytic function already bounded on its input core. There is no additional demand that an arbitrary MC output preserve the same radius under repeated compositions. The direct old-scale lemma DA is still useful for a sharper regular age estimate, but it is not needed to justify every basic inherited evaluation.

Two norms must be distinguished in the formation and inheritance steps. Write $\mathcal{O}_{d,g} = g^{-d}\mathcal{O}_d$, with $d = 2$ for H and $d = 3$ for $C^\sharp$. On a full fixed PHYSICAL background core of radius α, a usable single-source estimate has the form

$$\begin{aligned} |\mathcal{O}^\sharp_{d,g}(U, J, gB)| &\le C_{\mathcal{O}} g^{-d}\{\alpha + g\sqrt{\mathcal{E}_X(B)}\}^d \\ &\le C_{\mathcal{O}}(1 + \alpha/g)^d(1 + \mathcal{E}_X(B))^{d/2}, \qquad \text{(PA9)} \\ \mathcal{E}_X(B) = \sum_y w_X(y)|B(y)|^2, &\quad \sum_y w_X(y) \le C. \end{aligned}$$

The physical normalization and correct lower-dimensional subtractions are retained. An $O(1)$, B-only bound on this whole fixed physical core is FALSE: a smooth actual minimum of fixed small amplitude α, with $H \neq 0$, already has $g^{-2}H = O(\alpha^2/g^2)$ at $B = 0$. MC is the identity on that minimum and cannot remove this growth.

On a finite completed carrier, (PA7) and the linear small-input protected MC bound give the displayed estimate with $\mathcal{E}_X = \sum_{y\in X^+} |B_y|^2$: first use $\|B\|_\infty$, then bound it by that energy. Its Gaussian moment costs a fixed polynomial in the carrier volume. The stronger normalized local-energy version displayed in (PA9), with $\sum_y w_X(y) \le C$, requires the corresponding localized response/energy envelope. It is not inferred from the finite-carrier supremum estimate alone.

On a fixed CANONICAL external ball, where the physical background coordinates are gz with $\|z\| \le C$, the corresponding bound is $C(1 + \|z\| + \sqrt{\mathcal{E}_X})^d$. The genuinely $O(g)$ nonlinear comparison with the leading free mark is proved in Section 5 on its complete flat-data canonical class. Both estimates must survive the indicated wall expansions. A global cutoff supremum for B cannot replace a local energy envelope if doing so removes the desired coupling gain.

With these primitive estimates, the ordinary finite source-algebra Gaussian/Mayer majorant can be applied with the unchanged original reference measure and admitted activities. This supplies the norm of (PA8), provided the actual formation bound is checked; it is not inferred from pointwise analyticity alone. Only then is the original prepared-map theorem sufficient for the future low-core inheritance of this SAME marked member. Its full-core norm can carry the explicit polynomial g^{-d}. A weighted inherited-source ledger must preserve those factors; one cannot relabel the full-core norm as an $O(1)$ canonical norm when applying that theorem.

Finally a dimension-six center estimate and a strict remainder norm are separate from the mere bound of (PA8). CD supplies the regular fractional quadratic remainder under its completed translation/pairing hypothesis. The actual coupled $(C^\sharp, H)$ center, the lower-degree mixing, and all current/contact pushforwards still have to be estimated in the one-step marked formation.

Section 98.6 / The inherited unmarked interaction has its own coupling gain

The separate note the stated results, IV1–12, verifies the inherited part of MT12 directly for the ACTUAL original localized fluctuation interaction. II (1.14), (1.16), and (1.17) give $D_s(0) = A_{0,s}(0) = \mathcal{H}_{k,s}(0) = 0$ for every complex wall parameter. The original distinguished fluctuation variation therefore vanishes before wall differentiation. The Ward/counterterm pairing and absolutely convergent age summation preserve this identity in each localized term:

$$V_k'(Y;U,J,0) = 0, \qquad |V_k'(Y;U,J,B)| \le CE_0 g_k \|B\|_{\infty,Y} e^{-\kappa_* d_k(Y)}. \tag{PA14}$$

The second estimate follows by radial Schwarz from the literal ϵ_1/g_k product ball and $CE_0\epsilon_1 e^{-\kappa_* d_k(Y)}$ bound of II Lemma 1, (1.33)–(1.36). The remark after II (1.29) explicitly permits this $g_k|B|$ improvement.

Here the external pair is held fixed and may be any admitted original pair. Unlike the canonical primitive argument of Section 5, no claim that its determining exterior is flat is needed. A finite-carrier energy is introduced only after the Schwarz step. This settles the inherited UNMARKED interaction input, not the weighted induction for a new normalized marked dimension-six family. The fresh action/Jacobian/comparison vertices retain their own MT12 verification.

Section 99 / Polynomial growth of the actual inherited fluctuation

This note verifies the inherited part of the interaction estimate MT12. It uses the original localized, marginal-subtracted fluctuation action of Bałaban I–II, rather than interpolating estimates for two different representations of an old activity. The result is uniform on the original admitted paired field/current core. It does not prove the fresh Wilson/Jacobian part of MT12, or a marked dimension-six mixing theorem.

Section 99.4 / Scope of the conclusion

The proof uses the original uniform clean incoming E_0 class and its original marginal subtraction. It closes the inherited contribution to the interaction comparison $L_V(O) - L_0(O)$ once the finite-degree marked formation estimate is invoked. The fresh Wilson, coordinate-Jacobian, gauge-fixing and Gaussian-comparison vertices still require their own polynomial estimates. This note does not replace them by (IV10).

Nor may an R-origin activity on a shrinking core, or a normalized dimension-six marked activity with an explicit g_j^{-d} full-core norm, be silently placed in the uniform clean E_0 class. Their actual core, counterterm, and weighted source norm remain part of their induction. Likewise the off-shell high-jet obstruction for an arbitrary original (U, J) does not disappear: (IV10) bounds the UNMARKED original interaction, not a primitive DF-insertion.

For canonical primitive comparisons the separate PA10 flat-data restriction is still needed. It is unnecessary for the B-only Schwarz argument here because (IV4) holds for every admitted fixed external pair.

Section 100 / A finite marked transfer test for the even dimension-six

This note separates three questions that must not be merged. The actual flat Gaussian transfer has the expected principal two-dimensional symbol. A fixed number of polynomial marks can be passed through the same connected Gaussian and rare-field estimates as the ordinary formation map. Neither assertion makes the unmodified higher-derivative observable bounded on an independent-current background core: an explicit Cartan example disproves that inference.

The last sections give a precise primitive-slot repair using MC and prove its canonical Gaussian moment estimate, with a separate polynomial-in-g^{-1} bound on the original physical core. The actual clean unmarked vertices then give a clean one-step primitive-column matrix I+O(g). The full marked stable/live feedback row remains explicit in MT36. All source insertions remain on the original Wilson measure. No ordinary complex source disc for a cubic polynomial is postulated.

The closing sections prove the compatible intersection of the physical-core and canonical jet norms, a same-wall proper fluctuation-difference estimate, and the actual running Gaussian packet semigroup. They retain the global hybrid source read-set and admitted-completion identity, and the lower/contact and live feedback requirements, as explicit remaining interfaces.

Section 100.9 / What strict age pays, and the remaining marked attachment row

Suppose the actual running tube satisfies x_j<=x_k+b_+(k-j)+C_drift for j<=k, and x_k>=h. A uniform upper step bound gives this with C_drift=0; pointwise monotonicity is not needed. For any fixed q>=0,

Section 100.9 / What strict age pays, and the remaining marked attachment row

$$\left(\frac{g_k}{g_j}\right)^q \le \left(1+\frac{C_{\mathrm{drift}}}{h}\right)^{q/2} \exp\!\left(\frac{qb_+}{2h}(k-j)\right). \tag{MT34}$$

Consequently, choosing the weak floor with qb_+/(2h)<s log(L)/2 gives

$$\sum_{j\le k} L^{-s(k-j)}(g_k/g_j)^q \le \frac{(1+C_{\mathrm{drift}}/h)^{q/2}}{1-L^{-s/2}}. \tag{MT35}$$

The same argument applies to any fixed polynomial ratio of running radii and couplings. Thus CD's strict r^s gain really does dominate such RELATIVE costs. It does not absorb an arbitrary absolute g_j^{-q} loss at a fixed age: with k-j=1 that factor can diverge as the ultraviolet endpoint recedes.

This is why the next attachment row must retain a graded marked state, rather than just the large full-core supremum. For a completed stable marked birth R_j with its full dimension-six projection removed, the needed estimate is for the actual connected coefficient obtained by attaching R_j to one distinguished interaction vertex, on their one joint Gaussian or cold kernel. In schematic operator notation its target form is

$$\left\| \mathsf{P}_6 \mathcal{L}^{(2)}_{V_k}(R_j \circ \Phi_{j,k},\, V_k) \right\| \le C g_k L^{-s(k-j)} (g_k/g_j)^q \, \|R_j\|_{\mathrm{graded}}. \tag{MT36}$$

All quadratic and cubic coefficient kernels, their anchor moments, the quartic analytic tail, and the full lower/contact filing are parts of this norm. R-origin births require their original shrinking radii, finite promotion wait, and one once-owned rare factor; an unhealed component is not assigned a regular Taylor center.

The finite-mark formation lemma controls the attachment once its incoming polynomial envelope has this graded norm. CD controls its completed regular readout once that kernel class and protected-domain admission have been proved for the actual marked coefficient. MT25a alone does not prove the graded incoming bound in MT36, and neither does the primitive column estimate MT33. In particular one must not apply CD termwise to an original off-shell wall readout whose independent current hides a large actual D F.

The work above therefore reduces the remaining scalar dimension-six task to a concrete stable/live marked row. The primitive clean columns, their true Gaussian principal matrix, the O(g) fresh and inherited unmarked attachment loads, and the finite-source moment mechanism have been checked. A full all-depth center cocycle and arbitrary composite-operator theorem are not claimed.

Section 100.10 / A useful graded regular-window estimate for that row

The full-core loss can be handled more precisely than by calling every marked norm polynomially large. Here is a rigorous local estimate, with its incoming kernel assumptions visible. Let a completed birth E_j carry one physical label of canonical degree d=2 or 3. Assume

$$\|E_j\|_{\alpha_j} \le A_j(\alpha_j/g_j)^d, \qquad \|K_{2,j}\|_{2+s} \le A_j g_j^{-2}, \qquad \|K_{3,j}\|_{\beta} \le A_j g_j^{-3}. \tag{MT37}$$

The latter are the actual covariant quadratic and cubic kernel/moment norms after the physical own-current substitution and the quadratic covariant completion. They are stronger than an ungraded supremum alone. The exact dimension-four family is kept separately. The TOTAL dimension-six projection of the stable member vanishes, with its compact completions, contacts, and pointed

descendants retained. In particular the area-commutator cubic supplied by the quadratic kernel in CD17 can cancel the local cubic moment. Those two terms are combined before taking absolute values; neither is assumed separately zero. All paths in this paragraph stay in one admitted protected current window.

First restrict to the physical coarse graph, with its own current substituted. Subtract the WHOLE declared covariant quadratic completion and then the whole cubic completion, as well as the lower/-contact packet. Denote the remaining physical analytic function by $E_{j,>=4}$. It has vanishing ordinary physical field jets through degree three. This is not the Taylor tail in arbitrary independent (U,J) variables. The completions are bounded on the same admitted physical core, so radial Schwarz gives

$$\|E_{j,\geq 4}(z)\| \leq CA_j(\alpha_j/g_j)^d(\|z\|/\alpha_j)^4, \qquad \|z\| \leq \alpha_j/2. \tag{MT38}$$

This is a Banach-valued radial Cauchy estimate for that physical pullback and does not differentiate the background in space. In particular the quartic tail is not bounded by the entire g_j^{-d} supremum after composition.

On a fixed canonical current family, MT23 gives a protected field/current with norm O(g_k q_∞). For an old contractible carrier of relative size r=L^{j-k}, DA5–8 gives its actual direct old-core argument of size C r^2 g_k q_∞ (and current size C r^3 g_k q_∞). Use the direct readout only when its explicit admission margin holds. The kernel moment calculation CD24–26, the cubic root comparison, and MT38 then give, INCLUDING the r^{-4} root count and r^{-2} dimension-six normalization,

$$\begin{aligned} |R_{j\to k}| \leq CA_j\Big\{ & r^s(g_k/g_j)^2 q_\infty^2 \\ & + r^\beta(g_k/g_j)^3 q_\infty^3 \\ & + r^2(g_j/\alpha_j)^{4-d}(g_k/g_j)^4 q_\infty^4 \Big\}. \end{aligned} \tag{MT39}$$

For the last line, the r^8 from four fields is reduced to r^2 by the six powers in the integrated count. The coefficient is exactly $\alpha_j{}^{\{d-4\}g_j}\{-d\}g_k^4$ =(g_j/alpha_j)$^{\{4-d\}(g_k/g_j)}$4. This explicit cancellation explains why the large full-core bound need not itself destroy a canonical estimate. The sup-disc admission is checked before replacing powers of q_∞ by the one local energy polynomial.

More usefully, the quartic coefficient is exactly $\alpha_j{}^{\{d-4\}g_k^{4-d}(g_k/g_j)}$d. For a fixed clean α_j=alpha_*>0 it therefore has an additional O(g_k) factor when d=3 and O(g_k^2) when d=2, with only a relative coupling ratio. No monotonicity of the running trajectory is needed for this algebraic fact. For shrinking R cores the radius factors are not silently constant.

If the SAME localized marked readout entering formation has the representation and protected domain just used, MT11–13 now prove its interaction attachment estimate MT36 directly, with the three explicit weights in MT39. MT34–35 sum the relative coupling ratios. This is a proved regular-window row under MT37 and the stated actual-readout condition.

There is a possible source-only hybrid that identifies the remaining domain question. For sufficiently old bounded carriers, DA permits reading the original admitted E_j on ONE shared current protected MC field/current, in old units. At the nested critical endpoint it has its original physical value. For the finitely many recent ages keep the original prepared representation on the now established paired core. This does not require a nested composition of one MC map per past age. Nevertheless the full construction must show that these choices produce one completed carrier class, preserve the

Section 100.10 / A useful graded regular-window estimate for that row

translation/phase pairing needed by CD, and retain MT37 under the actual connected formation and live joins. Long, wrapping, and wild-crossing carriers must keep their admitted original norm until their separate estimate has been paid.

The current proof does not identify the old marked coefficient at an arbitrary off-shell wall value with that shared protected field. Such an identification is a source extension to be constructed and checked, not a consequence of equality at all walls one. This is the precise remaining interface after the primitive-column theorem, rather than an appeal to arbitrary higher spatial background jets.

Section 100.11 / The two norms form a usable graded intersection

The preceding estimate can be strengthened so that the physical-core norm and the canonical low-jet norm propagate together. This is an algebraic and analytic lemma about a COMPLETED regular family; the actual physical restriction, coarse kernel representation, and complete root/phase pairing remain hypotheses of its application.

Fix a clean physical radius $\alpha>0$ and a source label d in {2,3}. Let E_j be analytic on the admitted physical coarse-coordinate core, after its own-current substitution. Keep an exponential carrier norm with sufficient reserve for a fixed power of carrier size and path diameter. Suppose

$$\|E_j\|_\alpha \le A_j \alpha^d g_j^{-d}, \qquad g_j^m \|K_{m,j}\|_{\mathrm{mom}} \le A_j, \quad m=2,3. \tag{MT40}$$

The kernel K_3 is taken after subtracting the whole chosen quadratic covariant completion. The norm includes its anchor and path moments. Lower fields and EOM/contact packets have their separate triangular rows. In particular a contact coefficient is not set to zero merely because its support misses a test currently under discussion.

There is also a physical Cauchy bound for these same kernels. A carrier with N actual coarse coordinates has at most N^m ordered field-derivative entries. Cauchy on a fixed smaller physical polydisc bounds each entry by $C_m \alpha^{-m} \|E_j\|_\alpha$. The finite coarse curl homotopies and the stated path/anchor moments cost a fixed polynomial in N and in the carrier diameter. This polynomial is paid by the spare exponential carrier weight. Subtracting the quadratic completion before the cubic homotopy only changes that fixed constant. Consequently

$$\|K_{m,j}\|_{\mathrm{mom}} \le C A_j a_m^{(d)}(j), \qquad a_m^{(d)}(j) := \min\{g_j^{-m},\, \alpha^{d-m} g_j^{-d}\}, \quad m=2,3. \tag{MT41}$$

The same conclusion follows by retaining separately the two estimates rather than literally using the minimum. The canonical estimate uses Cauchy in v=g_j z on a FIXED z ball; the physical estimate uses a fixed v ball. The microscopic prefix links are not additional coarse Cauchy variables.

This intersection is stable under the finite local projection. Indeed the declared compact completion of an order-m kernel has physical norm at most $C\alpha^m\|K_m\|$ and begins in ordinary field degree m. Thus its physical bound follows from the second term in MT41, while its canonical order-m jet follows from the first. Higher ordinary jets of the quadratic completion acquire the explicit extra powers of g_j; its canonical third jet is at most C g_j times its canonical quadratic norm. The cubic kernel remaining after that subtraction satisfies the same estimates.

The completion in this argument must be the admitted PA/MC source extension, or a specified own-coarse-lattice completion with its declared pullback. It is NOT the unmodified fine H(U,J) on the independent-current core; MT2–5 disproves the required boundedness for that choice. With the declared completion, the physical analytic remainder after both subtractions has norm at most C A_j

Section 100.11 / The two norms form a usable graded intersection

$\alpha^d\, g_j^{-d}$ and vanishing jets through degree three. Therefore its radial fourth-order estimate is exactly MT38.

Let r=L^{j-k}, and suppose the actual direct regular readout has been admitted on the protected current window. Apply CD to the whole covariant quadratic/cubic sum, including the joint cancellation of the cubic center, and the radial estimate to the remaining physical pullback. At a physical field/current amplitude t this yields the envelope

$$|R_{j\to k}| \le CA_j \left\{ r^s a_2^{(d)}(j) t^2 + r^\beta a_3^{(d)}(j) t^3 + r^2 \alpha^{d-4} g_j^{-d} t^4 \right\}. \tag{MT42}$$

Here the exact dimension-four family remains separate. The r^{-4} root count and r^{-2} dimension-six normalization are included. If the quadratic path-area cubic and the cubic kernel moment cancel, that cancellation is used before estimating the three remainders in MT42.

On a full physical current core use $t \le C(\alpha + g_k\sqrt{E})$, and take $g_k \le c\alpha$. After normalizing by the CURRENT physical-core factor (g_k/alpha)^d, MT41 gives

$$(g_k/\alpha)^d |R_{j\to k}| \le CA_j (g_k/g_j)^d (r^s + r^\beta + r^2)(1+E)^2. \tag{MT43}$$

This is a relative coupling weight in the physical norm as well as in the canonical norm. There is no remaining absolute g_j^{-d} loss.

On the fixed canonical current family take $t = g_k q_\infty$. The exact canonical weights of the three terms are

	$d=2$	$d=3$
quadratic	$r^s (g_k/g_j)^2$	$r^s (g_k/g_j)^2$
cubic	$r^\beta (g_k/\alpha)(g_k/g_j)^2$	$r^\beta (g_k/g_j)^3$
quartic tail	$r^2 (g_k/\alpha)^2 (g_k/g_j)^2$	$r^2 (g_k/\alpha)(g_k/g_j)^3$.

(MT44)

The table supplies upper bounds; for d=3 the physical cap can further improve the quadratic entry. Powers of q_∞ are respectively two, three, and four. The actual sup-disc admission is checked first and those powers are then put into ONE joint quadratic energy reserve. The same finite formation estimate MT9 and the same canonical coarse-coordinate Cauchy estimate recover the two output norms. The physical Cauchy projection preserves the d=2 full-core degree because its normalized cubic coefficient is O(g_k/alpha); no extra inverse power appears when that center is subtracted.

Attaching one distinguished actual clean interaction vertex multiplies these envelopes by Cg_k via MT11–13. The Gaussian-density comparison on the canonical external ball has the same factor by MT31. Its normalization trace remains a vacuum/contact entry. All relative weights in MT43–44 are age summable by MT34–35. This proves the bounded graded formation step once the incoming completed physical kernel class has the declared actual-readout representation.

One remaining bookkeeping point is important even at the Gaussian reference. A stable total C-center does not mean that the cubic kernel moment alone vanishes: it can be the negative of the quadratic path-area contribution. That contribution has canonical size

$$g_k^3 \|K_{2,j}\|_{\mathrm{mom}} \le CA_j g_k (g_k/g_j)^2. \tag{MT45}$$

It therefore belongs among the explicit $O(g_k)$ graded corrections. Exact annihilation of the Gaussian stable-to-center block concerns the principal graded free part with completed root phases, not a separate zero assertion for every compact-completion summand.

Section 100.11 / The two norms form a usable graded intersection

These formulas remove the coupling-grade obstacle in the proposed regular kernel bridge. They do not assign a regular core to a live/dropout coefficient, justify a path crossing a wild component, or establish translation pairing for a history-dependent root coefficient. Those are separate actual-map statements.

Section 101 / The dimension-six marked transfer

The argument concerns MT8–26 in the stated results and the primitive architecture in the stated results. The target notes changed

The two-norm correction, the supremum-based radial disc, and the two-momentum alias argument described below have now been incorporated into MT. No full marked induction or continuum construction is established by these corrections.

Section 101.1 / Verdict and the distinction that matters

The canonical polynomial estimate is valid on a fixed canonical external-field ball, conditional on the literal primitive map having the stated analytic radial family. It is not a coupling-uniform bound on the whole fixed original physical core. The latter assertion is false even on genuine smooth constrained minima, where MC is exactly the identity.

The corrected MT25a/PA9 permits the unavoidable explicit g^{-d} cost in the full physical-core norm. Finite nilpotent mark formation permits such a cost: a mark is linear data and need not be a small ordinary potential. This makes the two bounds compatible. It does not prove that the resulting two-norm marked state has a closed, uniform multiscale recurrence.

The free H alias calculation holds. Its pointed extension needs two independent momenta; the corrected MT19a now provides that argument. The free calculation and the conditional formation lemma do not establish the interacting MT26 without the actual primitive identifications, the actual polynomial unmarked-vertex bounds, and the completed remainder-kernel norm.

Section 101.2 / A genuine counterexample to a uniform full-core mark norm

Let the original physical birth core have a fixed small radius $\alpha > 0$, independent of g. Work on a fixed regular periodic geometry with a nonzero sufficiently small coarse momentum p. In a Cartan direction take coarse potential data

$$A_2(y) = \epsilon \cos(py_1)T, \qquad A_\mu = 0 \quad (\mu \neq 2), \qquad 0 < \epsilon \ll \alpha. \tag{MA1}$$

Use the actual analytic constrained minimum for these determining data, denoted $U(\epsilon)$. Its linear curvature reconstruction has a nonzero directional current in this mode. Hence, at this fixed lattice and geometry,

$$H_a(1; U(\epsilon)) = \epsilon^2 h_a + O(\epsilon^3), \qquad h_a > 0. \tag{MA2}$$

The positivity follows directly from the squared complementary directional-current form; the leading current is nonzero for the chosen transverse mode. Choose one sufficiently small fixed ϵ, after the geometry is fixed.

An overall constant coefficient of the Wilson action does not change this constrained minimum. At zero fluctuation, all repairs which satisfy the claimed physical identity leave it unchanged. Consequently

$$g^{-2} H_a(1; U(\epsilon)) = g^{-2}\{\epsilon^2 h_a + O(\epsilon^3)\} \longrightarrow +\infty \quad (g \downarrow 0). \tag{MA3}$$

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Thus a bound $C(1+\mathcal{E}(B))$, with C independent of g and of the external point in that fixed physical core, fails at $B=0$. This example uses actual minima and is independent of the rough, noncritical example MT2–5.

The normalization issue is also visible without an example. If physical external coordinates are $A=gz$, then the original core $\|A\|\le\alpha$ allows $\|z\|\le\alpha/g$. The canonical estimate gives

$$|g^{-d}\mathcal{O}_d|\le C(1+\alpha/g+\sqrt{\mathcal{E}})^d, \tag{MA4}$$

and its canonical nonlinear error Cgq^{d+1} can be as large as $C\alpha^{d+1}g^{-d}$ on that core. A physical radius $g(\log g^{-2})^q$ instead produces logarithmic losses; it still does not give a uniform canonical unit ball.

There are legitimate repairs:

- retain both the full-core bound, with its explicit g^{-d} coefficient, and the smaller canonical bound used for center extraction;
- formulate and prove a weighted full-core norm, for example with weight $(1+\|A\|/g)^d$;
- establish a genuinely new $O(g)$ physical-radius domain and its exact prepared-map admission theorem.

The first repair is the one now stated in MT25a/PA9. It must remain visible in inherited norm estimates. A source-independent original domain theorem justifies evaluation there; it does not remove the coefficient g^{-d}.

Section 101.3 / The analytic radial disc must precede Gaussian moments

The earlier choice

$$q_2 = 1 + \|z\| + \left(\sum_{b \in Y^+} |B_b|^2 \right)^{1/2} \tag{MA5}$$

did not justify the statement that the pointwise physical cutoff puts $t = g$ inside $|t|q_2 < c_0$. For N admitted coordinates, take $B_b = \delta/(2g)$. Every pointwise cutoff holds, whereas gq_2 grows like $\delta\sqrt{N}/2$. A polynomial carrier factor after Gaussian integration cannot repair failure of an analytic-domain inclusion before the integration.

The corrected choice

$$q_\infty = 1 + \|z\| + \max_{b \in Y^+} |B_b| \tag{MA6}$$

repairs this particular defect. A uniform physical supremum response bound can establish the disc $|t|q_\infty < c_0$. Only after proving Cauchy estimates on this disc may one use $\max|B| \le (\sum |B|^2)^{1/2}$ to get a polynomial moment.

If an energy with total weight at most one is desired, on a finite carrier with N current coordinates put $\mathcal{E}_Y = N^{-1} \sum_{b \in Y} |B_b|^2$. Then the polynomial moment coefficient can acquire $N^{m/2}$. Fixed powers of N can be paid from spare carrier decay. This is different from increasing the radius of the analytic disc. On an infinite Gaussian field the global supremum is unsuitable; first use an actual finite-carrier localization or a proved local weighted analytic estimate.

Once the literal field family is analytic on this disc, vanishes at zero, and has the uniform MC output bound there, the one-variable argument is valid. If

$$|\mathcal{O}_d(W(t))| \le C|t|^d q_\infty^d \quad (|t|q_\infty < c_0), \tag{MA7}$$

Section 101.3 / The analytic radial disc must precede Gaussian moments

then $t^{-d}\mathcal{O}_d(W(t))$ has a removable singularity. Cauchy on a fixed smaller disc gives

$$\left|g^{-d}\mathcal{O}_d(W(g)) - P_d\right| \le C g q_\infty^{d+1}. \tag{MA8}$$

Here P_d is the leading homogeneous canonical polynomial of the chosen map. For $d = 2, 3$, this is exactly the useful MT25 estimate. Lower/Wick corrections which do not vanish at $t = 0$ must be carried separately, as MT explicitly requires.

For H, the protected H^{1+s} and supremum curvature bounds control the actual flux derivative through the analytic equivariant divided-difference formula. For C-sharp, AI32b and the exact cube product give the cubic small-curvature bound. These estimates avoid the false high-derivative assertion on an arbitrary low regularity off-graph background.

Section 101.4 / The literal map, its boundary data, and its zero base

The original MT21 used a post-repair $\mathcal{R}_j(\Psi(U, J, B))$, whereas PA4 repaired the outer background and replaced the independent current by its physical current before the original response. These are distinct off-graph analytic extensions. Agreement on physical input does not identify their analytic domains or prove the radial estimate for one from the estimate for the other.

The revised MT21 chooses the two-repair PA4 architecture. That is an admissible choice, provided its composition is estimated on one common smaller domain with the same determining hierarchy. The protected MC current bound is available for the pre-repaired background, so no objection based merely on fractional Sobolev control of the current is warranted.

PA10–13 gives another concrete way to obtain a radial family: choose the identity chart, scale the first-order potential, independent current, and physical fluctuation, and invoke the smaller first-three-clause inclusion for the original paired domain. This does not assume that the whole original domain is star-shaped. It supplies a valid route for the one-post-repair version whenever its literal original response has the stated range. For the chosen two-repair version, include both repairs and the physical-current substitution in the same radial family.

There is one essential base-point qualification. Every determining datum that affects the value must be accounted for, including retained boundary data, reference configurations, and near-window data. The assertion

$$\Psi_0 = 1 \tag{MA9}$$

is false if the construction still sees an unscaled nonflat exterior or nontrivial fixed boundary at $t = 0$. An arbitrary response around such an exterior can have a nonzero constant term. Holomorphy alone then gives neither the factor t^d nor the removable singularity in MA8.

A canonical center computation can use an all-data flat base on the actual finite determining problem. A general nonflat boundary/live slot must retain its boundary state, its actual nonzero base, or a separately proved localized comparison. Exponential closeness to another saddle does not establish the exact physical source identity.

No change of real integration variables is implicit in this pointwise t-family. The real cutoff remains the original reference cutoff. Merely using complex t for Cauchy is not a same-measure defect.

Section 101.5 / What finite-mark formation does and does not prove

MT9 and MT11 are sound conditional finite-mark estimates when the stated source-free joint comparison leaves one positive quadratic reserve and the actual mark bounds hold on every required

Section 101.5 / What finite-mark formation does and does not prove

wall/background domain. The elementary estimate

$$(1+E)^{d/2}e^{-\delta E} \le C_{d,\delta} \tag{MA10}$$

uses that reserve once. The auxiliary exterior Gaussian must remain in the same integration so that its source-free cancellation is preserved. For two marks, bound their product using the union energy, not a product of two overlapping masses. The finite partition formula is compatible with the stated Gevrey factor $C_D^k(k!)^{\max(1,D/2)}$.

The interpolation proof of MT13 is also valid if the actual unmarked vertices obey MT12. A small quadratic-growth norm with a cutoff factor gP^m is not, by itself, an $O(g)$ polynomial moment bound. The cutoff factor must be removed at the incoming vertex, before the marked estimate is applied.

There are concrete ways to verify parts of MT12:

1. For an analytic inherited fluctuation-difference coefficient on a fixed physical supremum disc, with zero value at zero fluctuation, radial Schwarz gives a factor $g\|B\|_\infty$. The original age summation must already have been performed in the coefficient norm. Re-summing independent old-birth supremum bounds can lose the required uniformity.
2. A cubic Wilson remainder in physical fluctuation $b = gB$, after its g^{-2} action normalization, is bounded by $Cg\|B\|_\infty^3$ on the fixed physical chart.
3. A Gaussian precision perturbation admits the exact same-coordinate treatment in Section 7 below.

These mechanisms are verification routes, not permission to declare every coordinate, coarea, inherited, boundary, and reference term covered without identifying it.

The currently scope primary clean input is explicit about analyticity: II Lemma 1, (1.33)–(1.36), supplies a localized analytic fluctuation family on the complex physical supremum ball; its construction (1.2)–(1.21) preserves the zero-fluctuation base. II (1.37)–(1.43) treats the normalized Wilson remainder. These are useful primary inputs for the above arguments.

Section 101.6 / Free H aliases and the pointed bilinear extension

For the actual centered box-and-line filter, one convenient one-dimensional factor is

$$b_m(u) = \frac{\sin((u + 2\pi m)/2)}{n \sin((u + 2\pi m)/(2n))}. \tag{MA11}$$

The component-μ filter is a product of one such factor in every coordinate and one additional factor in direction μ. The principal factor is $1 + O(|\theta|^2)$, and every nonprincipal alias vanishes at zero. The actual FW9 derivative bound gives

$$|\partial_\theta q_{\mu,\nu}(\theta)| \le C \prod_i (1 + |\nu_i|)^{-1}. \tag{MA12}$$

Its square is summable uniformly in the depth n. Integrating the derivative along the real small-momentum segment proves

$$\sum_{\nu \neq 0} |q_{\mu,\nu}(\theta)|^2 \le C|\theta|^2. \tag{MA13}$$

For example,

$$\sum_{m=1}^{n-1} |b_m'(0)|^2 = \frac{1}{4n^2} \sum_{m=1}^{n-1} \csc^2(\pi m/n) = \frac{n^2 - 1}{12n^2}. \tag{MA14}$$

This illustrates the required uniformity. A bounded averaging row sum alone would not establish it for an arbitrary filter.

The nonprincipal reconstruction alias has the exact form

$$h_{\mu,\nu} = \frac{\lambda_0}{D_\mu}\frac{\overline{q}_{\mu,\nu}}{\lambda_\nu}. \tag{MA15}$$

Two fine difference multipliers have product bounded by $C\lambda_\nu$. Since P_C is bounded, MA13 proves an offset L^2 bound $C|\theta|^3|B|$ for the nonprincipal directional-current leg. The principal leg has leading term

$$M_2(\theta)B = (-i\theta_\mu)(i\theta_\mu B_\nu - i\theta_\nu B_\mu). \tag{MA16}$$

The identity $d_r P_C = d_r$ ensures that no singular longitudinal contribution was discarded.

For a pointed test, write the offset leg, including its phase, as $J_\xi(\theta) = M_2(\theta) + R_\xi(\theta)$ with $\|R(\theta)\|_{\ell^2_\xi} \le C|\theta|^3$. For two independent external momenta θ, φ, cell Cauchy–Schwarz gives

$$\begin{aligned}\left|\langle J(\theta), J(\varphi)\rangle_\xi - \langle M_2(\theta), M_2(\varphi)\rangle_\xi\right| \le C\{&|\theta|^3|\varphi|^2 + |\theta|^2|\varphi|^3 \\ &+ |\theta|^3|\varphi|^3\}|B|\,|B'|.\end{aligned} \tag{MA17}$$

The estimate also holds for the holomorphic bilinear pairing: absolute Cauchy–Schwarz bounds the trace products. For a bounded offset test, include its multiplication operator; the leading coefficient uses the actual cell average and shifted-test convention. Offset phases can add only higher total degree.

Every error has total potential degree at least five. Thus the pointed fourth potential jet is the asserted H jet. A constant-test diagonal energy alone would not prove this: total derivatives are invisible on that diagonal. The two-leg calculation does prove it.

The C-sharp leading cubic uses the unchanged first curvature jet, with its specified compact completion and descendants. Gaussian contraction of a cubic cannot produce a quadratic; color invariance removes a scalar linear contraction. With vacuum and lower/contact data kept, this verifies the free dimension-six scalar quotient $T_0 = L^{-2}I_2$. It is a leading free-symbol statement, not a bound on the full interacting transfer.

Section 101.7 / A precise Gaussian precision lemma that can supply one MT12 input

Use the original fixed current coordinates B, on their positive gauge-fixed fluctuation space. Suppose the actual precision $K(A)$ is holomorphic on a fixed physical background ball, $K(0) = K_0$, has a uniform positive gap at the real base, and has a uniform exponentially localized kernel norm there. Cauchy on a smaller ball gives, for a fixed canonical external z,

$$\|K(gz) - K_0\|_{\mathrm{loc}} \le Cg. \tag{MA18}$$

With normalized Gaussians, the identity in the same integration coordinates is

$$d\mu_{K(gz)}(B) = \det(K(gz)K_0^{-1})^{1/2} \exp\Big[-\tfrac{1}{2} B \cdot (K(gz) - K_0)B\Big]\, d\mu_{K_0}(B). \tag{MA19}$$

The determinant branch is the one equal to one at the flat base. No cutoff is moved. A row/column Schur bound turns the quadratic exponent into $Cg \sum |B|^2$. Localizing its kernel with the actual carrier convention gives MT12 with degree two. The localized log determinant is retained as its vacuum/normalizing contribution. The positive gap and smallness of MA18 keep the interpolation in the same quadratic comparison allowance.

Section 101.7 / A precise Gaussian precision lemma that can supply one MT12 input

Uniform positivity alone does not imply MA18. Nor does analyticity on a shrinking physical radius automatically give an $O(g)$ derivative bound: Cauchy can cost the inverse radius. The needed fixed-core local kernel analyticity must be identified for the actual precision. II (2.15)–(2.17) provides the relevant original matrix-kernel and determinant comparison framework; (2.23)–(2.25) keeps the joint exterior cancellation. These facts support this route without authorizing an arbitrary substitute Gaussian.

The argument gives a real counterexample to the former coupling-uniform full-core estimate and a real failure of the former large-carrier radial-disc implication. Both have explicit repairs, now reflected in MT/PA.

The free aliases and finite-mark positive-reserve argument survive. The radial Cauchy calculation survives on its explicit canonical class, with all determining data and the zero base specified. The remaining questions concern actual slot coverage, literal source-wall compositions, the complete unmarked vertex decomposition, inherited two-norm estimates, and the completed dimension-six kernel/contact remainder. Those are mathematical proof obligations, not consequences of a positive free Gram matrix or of pointwise equality alone.

101.8.1 Subsequent clean-input verification

The later MT §8 supplies the clean verification outlined in Section 5 The order of subtraction is correct: I (2.10)–(2.13) removes the constrained linear current term before I (3.1) defines its independent-current extension. Thus the argument does not falsely require an arbitrary independent current to annihilate a fluctuation. The original complex fluctuation ball, coefficientwise zero-fluctuation base, and already aggregated old-age estimate support MT28–30. The same-coordinate score MT31 is the derivative of MA19, with its trace/vacuum part retained.

With the stated original fixed-core local precision bounds and the actual admitted primitive slots, these steps justify the clean primitive-column conclusion MT32–33. The later MT36 attachment row for inherited stable/live marked data remains a different statement. The argument therefore no longer lists every clean MT12 input as open.

Section 101.9 / A precise partial closure through finite coarse coordinates

The following route genuinely closes the coefficient-counting obstruction for a scoped regular family. It is stronger than deducing a fine-coordinate coefficient norm from a dual C^1 bound, which would be invalid.

101.9.1 9.1 The finite-coarse coefficient lemma

Assume a family of physical pullbacks

$$\mathscr{F}_X(V;f) = E_X(U_j(V), J_{\mathrm{phys}}(U_j(V)); f) \tag{MA20}$$

has already been localized with its actual original response and complete carriers. On a completed coarse carrier X, assume:

1. The independent coarse logarithmic coordinates have a common polydisc of radius R_j, and number $N_X \le C|X|$, in current birth units. The same physical minimum and its actual current lie in the original admitted paired domain throughout this polydisc.
2. The function is gauge invariant, linear in the discrete test f, and admits a carrierwise bound

$$\|\mathscr{F}_X(V;\cdot)\|_{(\ell^\infty_{\mathrm{anchors}})^*} \le b_X. \tag{MA21}$$

   The rooted sum of $e^{\mu d(X)} b_X$ is bounded.
3. The regular whole-root family is translation/reflection covariant on the old coarse lattice, with actual source offsets, paths, and phases retained. Boundary/history coefficients without this property are excluded from this regular family.

The first two assumptions can be verified with source-valued formation in the nilpotent Banach algebra $\mathbb{C} \oplus \ell^1(\text{anchors})$. The primitive estimates bound the sum of absolute microscopic anchor coefficients using the physical integration weights. The one-reserve proof then retains this norm. This formulation prevents an unjustified exchange of the supremum over f with the output-polymer sum.

Finite-dimensional duality identifies MA21 with an ℓ^1 anchor norm. Banach-valued multivariable Cauchy therefore bounds each coefficient of degree $m \le 3$ by

$$\|c_{X,i_1,\dots,i_m}\|_{\ell^1_{\mathrm{anchors}}} \le C_m b_X R_j^{-m}. \tag{MA22}$$

Summing the N_X^m coarse-coordinate choices gives

$$\sum_{i_1,\dots,i_m,\ \mathrm{anchors}} |c_{X,i_1,\dots,i_m}| \le C_m b_X R_j^{-m} N_X^m. \tag{MA23}$$

There is no factor counting all microscopic anchors inside a coarse cell. Their absolute sum was taken in MA21. Localizing the pullback before this count is essential: using all variables of a current-size box containing r^{-4} old cells would introduce spurious inverse powers of the relative scale.

On a contractible completed coarse carrier, a fixed finite cochain homotopy writes every gauge-closed potential leg as a sum of plaquette-curvature legs. The number and length of these coarse paths are bounded by fixed powers of the carrier diameter. Apply it twice to the quadratic Ward kernel. Choose an actual gauge-invariant compact quadratic completion with those paths, and subtract that entire completion. Its cubic terms must also be subtracted. The remaining degree-three kernel has the homogeneous linear Ward identity in each leg, so the same homotopy applies three times.

Every fixed-degree coefficient sum and every fixed path moment therefore satisfies a bound

$$C_m b_X R_j^{-m} (1 + \operatorname{diam} X)^{p_m}. \tag{MA24}$$

Spare exponential locality pays this polynomial. One can choose the homotopy/completion reflection covariantly by including its finite reflected tensor/path partners. This changes only fixed constants and preserves the prescribed jets.

This proves a coarse quadratic and cubic curvature-kernel norm. The argument must retain R_j^{-m}: if the useful canonical ball has physical radius $O(g_j)$, these are real normalization factors. A full physical-core bound carrying g_j^{-d} is not silently substituted for a sharp canonical coefficient bound.

For the order-at-least-four tail, expand this SAME physical pullback in the coarse variables. Its physical quadratic and cubic kernels do not imply vanishing independent-(U, J) jets of the original extension. The nonlinear compact completions also contribute higher ordinary field degrees; retain those contributions in the remainder. Their operator bounds on a smaller physical coarse ball follow

by bounding the path transports by $e^{C\alpha_j \operatorname{diam} X}$ and reserving locality. Use the physical-core Cauchy bound as well as the canonical one for each coefficient. For example, for an H label a cubic coefficient has the stronger physical-core bound $CA_j\alpha_j^{-1}g_j^{-2}$, even though the separate canonical estimate only says $CA_jg_j^{-3}$. The stronger bound prevents an artificial g_j^{-3} full-core completion cost for an H input. After these bounded completions, radial Cauchy gives the quartic remainder with its original full-core normalization.

101.9.2 9.2 Why the source phases can be completed in this subclass

In MA20 all field endpoints lie on the old coarse lattice. After re-rooting a curvature leg, its displacement h is therefore a coarse-lattice vector. The shift of the complete coarse root sum by h is an exact reindexing. A microscopic source anchor may still have an arbitrary subcell offset z; its absolute coefficient sum is already controlled by MA21. The pointed CD17a identity retains precisely such offsets, including its derivative-of-test term.

Reflection must act on the complete carrier, field labels, path, and source offset. It is not legitimate to average an isolated coefficient while leaving these data fixed. With the completed reflection covariance in assumption 3, the required partner identities hold. This supplies the algebraic premise of CD22 for this universal regular family.

For filing a current local contribution, use a smooth current-scale partition $\sum_Q \chi_Q = 1$ on the source root and the test $f\chi_Q$. Its $C^{2,s}$ norm is controlled independently of the old relative scale. Require the support and every short carrier/path to lie inside the protected regular window. Long and collar-crossing carriers remain in their original admitted tail or boundary/live representation. A sharp indicator of a partial root set would invalidate the reindexing unless its difference were explicitly retained.

101.9.3 9.3 A counterexample showing why assumption 3 is substantive

Bounded local kernels are not enough. In one coordinate, take X constant, $Y = N^{-1-s}\cos(Nx)$, a coefficient $c(x) = \cos(Nx)$, and $h \asymp N^{-1}$ with Nh fixed away from $2\pi\mathbb{Z}$. Embed this example in a fixed box and use cutoffs away from the collar if necessary. The H^{1+s} norms of X, Y stay bounded. For a constant scalar test, the reflected difference satisfies

$$\int c(x)X\left\{\frac{Y(x+h)+Y(x-h)}{2} - Y(x)\right\}dx \asymp h^{1+s}. \tag{MA25}$$

The displayed derivative-curvature correction in CD13 vanishes where X is constant, and the test-derivative correction is zero. An $O(h^{2+s})$ remainder is therefore false for this arbitrary coefficient family. The path range is one old-scale step and its coefficient is bounded.

This is not a counterexample to CD, which explicitly assumes completed translation pairing. It proves that analyticity, gauge Ward identities, and exponential locality cannot manufacture that pairing for arbitrary history-dependent coefficients. The scoped lemma above supplies it through a separate symmetry hypothesis and an exact full-root sum.

101.9.4 9.4 The remaining coarse-to-fine bridge has a viable proof

The CM/GQ centered actual averaging route can bridge the coarse kernel norm to the protected field. The necessary stronger estimate follows by upgrading CM14 with the tame product inequality at order

Section 101.9 / A precise partial closure through finite coarse coordinates

$1+s$:

$$\|N_\ell(A)\|_{H^{1+s}_\ell} \le C_L\ell^2\{M_0\|A\|_{H^{2+s}_a} + (M_1+M_0^2)\|A\|_{H^{1+s}_a}\}. \tag{MA26}$$

The zero constant moment puts an offset difference into the quadratic term. Its H^{1+s} norm costs $C_L\ell\|A\|_{H^{2+s}}$, and its supremum costs $C_L\ell M_1$. No supremum of the second derivative of A is needed.

One coarse inverse difference and the whole-flat-tail contraction CM21 give a uniform H^{2+s} bound on the actual straightened coarse potential and an $O(r)$ nonlinear defect in that norm. The $C^{1,\beta}$ bound can use CM9 and the inverse factor $\ell^{-1-\beta}$; its accumulated $\ell^{1-\beta}$ cost is summable for $\beta<1$.

For the actual positive two-form average $Q^2_{\mu\nu}=Q^0L_\mu L_\nu$, the exact flat cochain identity is

$$d_\ell Q^1 = Q^2 d_a. \tag{MA27}$$

Based-plaquette BCH, with all component offsets retained, has the form

$$F_b(A)=d_bA+[A_\mu,A_\nu]+\mathcal{R}_b(A), \qquad \|\mathcal{R}_b(A)\|_{H^1}\le CbP(M)\|A\|_{H^2}. \tag{MA28}$$

The quadratic remainder contains an offset difference; the higher-degree terms have an explicit factor b. Its derivative uses D^2A in L^2, not in supremum. One-step curvature defects can consequently be summed through the entire Q^2 tail in H^1, giving

$$\|F_r-Q^2_{a\to r}F_a\|_{H^1_r}\le CrP(M)\|A\|_{H^{2+s}_a}. \tag{MA29}$$

For comparing directional currents, differentiate MA29 on the coarse lattice before lifting. Its derivative costs no inverse power of r, because MA29 is an H^1 estimate. The remaining exact coarse derivative of Q^2F_a is a positive line average of D_aF_a. Block lifting that average approximates D_aF_a in L^2 with $O(r^s)$ error, since $D_aF_a\in H^s$. Differentiating a block-held field instead would create false microscopic boundary jumps.

The lifted potential differs in supremum by $O(r)$, which controls the connection factors in covariant differences. The analytic flux map and the specified compact C-sharp correction then give the corresponding $O(r^s)$ comparison of physically tested H and C values, including the shifted-test quadrature. These scalars are exactly gauge invariant, so their actual straightened coarse representative is sufficient for this comparison.

The detailed same-map proof is now written in the stated results, CB1–37. Its CB15–18, CB20–28, and CB31–34 pass this test. In particular the one differentiated curvature defect has the required $O(r)$ bound, and the C-sharp comparison retains the cubic factor M_F. The same two-factor estimates compare the full derivative-curvature tensor and the pointed $(\partial f)F\,DF$ term in CD17a. Thus the route supplies a regular classical bridge on its actual collars and conventions. It is not a new off-shell identification of all original prepared maps.

Two limits remain essential. First, an $O(r^s)$ comparison is adequate for a dimension-six value, but is not adequate for replacing a dimension-four quadratic term after multiplication by r^{-2}. The exact full lower quadratic family must remain, or its separate second-order conversion must be proved. Second, a classical current comparison does not prove the pushforward of the Wilson EOM/contact module through an interacting step.

101.10.1 10.1 The graded intersection and whole completions pass

MT37–44 now consistently distinguishes the same physical own-current pullback from the arbitrary independent-current extension. The quadratic completion is subtracted in its entirety before the cubic kernel is formed, and the cubic completion is also subtracted in its entirety before the quartic remainder is named.

The two-bound minimum

$$a_m^{(d)}(j) = \min\{g_j^{-m},\, \alpha^{d-m} g_j^{-d}\}, \qquad m = 2, 3, \tag{MA30}$$

is correct for these same kernels. Both estimates can be retained separately instead; no interpolation principle is needed. The physical cap proves that the completed source retains its original full-core degree. In particular an H label does not acquire an artificial cubic full-core factor g_j^{-3}.

All entries in MT44 check. For example the H-labelled cubic term has canonical size at most

$$C A_j r^{\beta} (g_k/\alpha)(g_k/g_j)^2 q_\infty^3, \tag{MA31}$$

whereas the C-labelled quartic tail has size

$$C A_j r^2 (g_k/\alpha)(g_k/g_j)^3 q_\infty^4. \tag{MA32}$$

On the full physical core, the current normalization $(g_k/\alpha)^d$ converts every physical-cap bound to the relative factor $(g_k/g_j)^d$, as in MT43. The stated running envelope then makes the strict age weights summable.

These are bounds for an admitted completed regular readout. They are not a proof that an unspecified off-shell preparation belongs to that readout class.

The revised joint center cancellation is also necessary. The quadratic path-area cubic and the local cubic moment can cancel in the total dimension-six projection. Neither must vanish separately. MT45 correctly retains the resulting principal-grade correction at order $g_k(g_k/g_j)^2$.

101.10.2 10.2 The proper-difference lemma is valid with its displayed premise

MT46 uses one map, one physical fluctuation variable, one wall family, and the same determining data for both terms. If MT47 holds throughout its physical supremum disc, then

$$D_{\mathbf{s}}(U, 0) = 0 \quad \text{for every complex wall value.} \tag{MA33}$$

Banach-valued radial Schwarz therefore supplies $g_k \|B\|_\infty/\delta$, with the original age factor still present. Linear wall localization preserves MA33. The one joint positive Gaussian comparison then proves MT48's full-core small factor for the new difference.

This also resolves the generic quartic-tail concern: the deterministic old tail need not be small on its own physical core. It remains the old tag. Only its exact proper fluctuation difference receives a new $O(g_k/\delta)$ birth bound.

The premise must not be weakened to a real-cutoff bound, or to a bound at walls equal to one. Nor does the identity at the physical endpoint identify the retained deterministic tag with the next off-shell prepared source member. That latter identification is expressly left open.

101.10.3 10.3 The running Gaussian packet is an actual common reference

MT49–51 correctly uses the induced Schur precisions. For the positive flat gauge quotient the channel projections are nested and orthogonal. Conditioning on level k fixes the entire retained P_k component; the other channels have zero conditional mean. Hence the scalar channel weights do not change the classical conditional mean.

The composition identity follows exactly:

$$\begin{aligned} H_{i:j}H_{j:k} &= H_i^{-1}(Q^{(j:i)})^{\mathsf{t}}H_jH_j^{-1}(Q^{(k:j)})^{\mathsf{t}}H_k \\ &= H_i^{-1}(Q^{(k:i)})^{\mathsf{t}}H_k = H_{i:k}. \end{aligned} \tag{MA34}$$

One cannot replace H_j by a nearest-neighbor coarse Maxwell precision in this identity.

At first marked order, the quadratic covariance contraction is a retained vacuum/test coefficient. A principal invariant cubic has no scalar linear contraction for a simple color algebra and a color-diagonal flat covariance. These are the limited lower rows proved by MT52. They do not prove the interacting current/contact recurrence or a multiple-mark factorization. Reference-cutoff tail comparisons remain part of the exact transition to the untruncated Gaussian tower.

101.10.4 10.4 The free stable-to-center zero has the required phase scope

For two external momenta and one pointed-test momentum with $\theta + \varphi + \eta = 0$, the complete old-root subcell sum is exactly MT53. A fixed fine anchor contributes a factor depending on η, without altering that root sum. Principal/nonprincipal quadratic aliases thus cancel after this completion.

The induced-precision qualification added to MT53 is important and correct:

$$H_{j:k} = Q^{(j)}H_{0:k}, \qquad d_jQ^{(j)} = Q_2^{(j)}d_0. \tag{MA35}$$

A nonprincipal old-coarse alias receives only nonprincipal fine aliases. Whole-tail L^2 contraction of Q_2 preserves their curvature order at least three. Two such legs have total potential order at least six and cannot contribute to the dimension-six quadratic center.

The principal/principal term has zero center only after the lower quadratic family and the complete degree-four potential moments have been removed. A bare F^2 term can otherwise generate a degree-four potential contribution through principal interpolation. Its actual free evolution must stay in its own lower row.

Similarly, a nonprincipal cubic leg cannot change the leading three-curvature jet. Whole covariant completion corrections and the quadratic-area cancellation remain the explicit graded small terms; they are not separately declared zero.

This proof concerns the completed universal root family. An individual localized phase, a sharp root mask, or a history-dependent coefficient does not have the asserted zero. A smooth partition is summed to one before the free projection, or all its differentiated test terms remain until the corresponding sum.

101.10.5 10.5 Exact final scope

The following statements now follow:

- the clean primitive-column estimate MT32–33;
- the scoped coarse-kernel/anchor/phase lemma of Section 9 and the CB regular classical bridge;
- the two-bound graded envelope MT40–44 for an admitted completed regular member;
- the same-wall proper-difference estimate MT46–48 under the full-disc bound MT47;
- the actual running-Gaussian tower, its first-mark lower rows, and the completed principal free stable-to-center statement MT49–53.

The following are still not proved by those statements:

1. A single global hybrid source construction, with one complete read set and a common admitted wall family, whose old deterministic tag is exactly the member read by the next original prepared step.
2. Preservation of the completed graded class through all actual clean joins and through the entire lower-dimensional, independent current, EOM, and contact recurrence.
3. The live and R-born marked rows with their actual shrinking radii, healing/promotion schedule, and once-owned rare reserve.
4. The resulting all-depth interacting stable-to-center feedback bound, fundamental center cocycle, or complete composite-operator construction.

In particular, no interacting feedback theorem is inferred from the primitive columns. The remaining-interface paragraph after MT53 now states these limitations accurately. No new fatal counterexample was found in MT37–53 under their final displayed hypotheses.

Section 102 / Deterministic source tags, genuine retractions, and independent wall families

This note examines the interface left after MT53. It proves an exact complex-domain retraction identity for the actual classical background maps and a two-family source localization identity. The latter shows that a wallwise retraction semigroup is NOT necessary for the exact source algebra. It also identifies which uniform domain and graded carrier bounds are still needed before these identities become an all-depth marked recursion.

There is no change to the unmarked Wilson kernel, its real cutoffs, coarea factors, or normalization.

Section 102 / Deterministic source tags, genuine retractions, and independent wall families

MT is not modified by this note.

Section 103.1 / Scope and conclusion

At a fixed lattice cutoff and a fixed finite source degree, arbitrary smooth Wilson observables can be carried through the original integrations, coarea coordinates, and matched reference replacements exactly. The correct state is a finite source jet with coefficients that are **unnormalized densities or retained joint kernels**. Transport is linear in this state. It is not a homomorphism from products of fine observables to products of coarse observables.

This distinction resolves the algebraic bookkeeping for lower terms, currents, equation-of-motion terms, and contacts. It does not prove a uniform norm for any of them. In particular:

- a deterministic, fluctuation-independent source polynomial factors from a conditional integral at every finite source degree;
- a source coefficient containing an equation-of-motion insertion transports together with its divergence and contact terms;
- the physical own-current relation can be imposed on an actual physical readout graph, but cannot be imposed on an arbitrary independently weakened paired-field extension;
- field-jet extraction must retain its exact complement. Finite source degree does not make the output a polynomial of finite field degree.

The original Wilson kernels, cutoffs, coarea factors, reference denominators, and old physical source slots are unchanged throughout. The same-fiber reference identity proved in S53–58b and Section 13.7 of the source note is the original-map input for the reference move. The statements below prove its finite-source and differential bookkeeping consequences; they do not replace its coordinate verification.

The main new consequence for the hybrid old-tag construction is in Section 4. An all-walls nested-minimum identity is unnecessary for exact factorization. The genuine unweakened zero-fluctuation tag and the proper fluctuation difference may be localized separately, provided each localization has the correct unweakened endpoint. Their quantitative norms still have to be proved on their actual domains.

103.7.1 6.1 Smooth deterministic pushforward

Let $q : \mathcal{M}_a \to \mathcal{N}$ be smooth on the relevant support. Write vector-valued measures in local coordinates on $\mathcal{N}$, or equivalently by their action on smooth one-forms. Define

$$\mathbf{J}_F = q_*(\rho F\, Dq\, X\, dU).$$

Then XP21 and the chain rule give

$$q_*(\rho\, \mathsf{D}^\dagger_{\rho,X} F\, dU) = -\operatorname{div}_{\mathcal{N}} \mathbf{J}_F. \tag{XP25}$$

This is an identity of distributions. No conditional expectation of $Dq\, X$ has been estimated.

If $r : \mathcal{N} \to \mathcal{P}$ is another smooth observation, the current itself obeys

$$\mathbf{J}_{r\circ q} = r_*(Dr\, \mathbf{J}_q). \tag{XP26}$$

Thus divergence transport and composition form an exact commuting diagram when interpreted as currents and densities.

Higher ordered EOM factors produce higher currents and lower currents together. For example,

$$\begin{aligned} q_*(\rho\, \mathsf{D}^\dagger_{\rho,X_i} \mathsf{D}^\dagger_{\rho,X_j} F) &= \partial_\alpha \partial_\beta T^{\alpha\beta} - \partial_\alpha U^\alpha, \\ T^{\alpha\beta} &= q_*(\rho F\, (X_j q^\alpha)(X_i q^\beta)), \\ U^\alpha &= q_*(\rho F\, X_j X_i q^\alpha). \end{aligned} \tag{XP27}$$

Indeed $X_j X_i (g \circ q)$ is the sum of the Hessian term of g and the displayed first-derivative term. At general order the ordinary finite chain rule gives all higher and lower tensors. Retaining them, rather than replacing each fine current by a selected coarse current, makes the diagram functorial.

103.7.2 6.2 Cutoff boundaries are part of the transported contact

With a smooth cutoff χ, the exact formula is

$$q_*(\chi\rho\, \mathsf{D}^\dagger_{\rho,X} F) = -\operatorname{div} q_*(\chi\rho F\, Dq\, X) + q_*(\rho F\, X\chi). \tag{XP28}$$

A sharp cutoff has the corresponding distributional boundary term. A complete partition $\sum_\omega \chi_\omega = 1$ cancels the sum of these terms. It does not make them zero in each selected history.

In coarea coordinates, derivatives of the Jacobian and of every pulled-back boundary occur in precisely this identity. Keeping the full density is enough to retain them. Differentiating only the displayed Wilson action is not enough.

103.7.3 6.3 General history kernels

For a kernel $k(y, U)$, deterministic or not, the always applicable form is

$$\mathcal{K}(\rho\, \mathsf{D}^\dagger_{\rho,X} F)(y) = \int \rho(U) F(U)\, X_U k(y, U)\, dU. \tag{XP29}$$

If k has selectors or constraints, its derivative is understood as a distribution. The equality follows directly by testing in y and integration by parts in U. Higher ordered factors give $X_r \cdots X_1 k$, in the same order as XP23.

Equation XP29 is particularly useful for a reference replacement. It does not assume that the reference coordinate is a deterministic function of the old field. It differentiates the actual kernel, including its conditional denominator if that denominator depends on the differentiated retained data.

Under composition of two kernels, the derivative acts on the actual fine-input factor and Fubini gives the same composite distribution. This is the kernel version of XP26. Rewriting it as a local differential operator on the new reference variable is an additional assertion, not part of the exact transport theorem.

For nonsmooth selectors, the safe object is the ordered derivative of the **whole kernel distribution**, paired with the smooth original coefficient ρF. These formulas do not define arbitrary products of coincident surface delta distributions. In particular, after exposing a boundary contact one cannot multiply it by another independently exposed boundary contact unless a separate distributional product rule has been proved. Retaining the original smooth observable and differentiating the complete selected kernel avoids that issue.

103.7.4 6.4 The generated effective action cannot be omitted

Suppose on a regular output patch

$$\rho'(v) = q_*\rho(v) = e^{-S_{\mathrm{ref}}(v)+V(v)}, \tag{XP30}$$

where V includes every generated action, coarea, and normalization term assigned to this density. For a smooth output vector field Y,

$$\mathsf{D}^{\dagger}_{\rho',Y} F = \big(Y S_{\mathrm{ref}} - YV - \operatorname{div} Y\big)F - YF. \tag{XP31}$$

Using the reference Wilson current alone omits the term $-YV$.

If X is projectable, $Dq\, X = Y \circ q$, XP25 intertwines the fine and coarse weighted adjoints with this **full** density. Otherwise the transported vector measure depends on F and must be retained; it need not factor as one Y times the transported scalar coefficient.

A simple noncompact diagnostic, with rapidly decaying densities so that the same integrations by parts apply, is

$$\rho(x,y) = \exp[-(x^2 + y^2 + \lambda x^2 y^2)/2], \quad q(x,y) = x, \quad \lambda > 0. \tag{XP32}$$

Its marginal is proportional to $e^{-x^2/2}(1+\lambda x^2)^{-1/2}$. The conditional fine x-score is

$$x + \frac{\lambda x}{1+\lambda x^2}, \tag{XP33}$$

not the reference score x. This is not a Wilson counterexample. It demonstrates the exact missing term in a rule that pushes an EOM insertion forward and then replaces the full effective current by the bare reference current.

Section 103.10 / Exact conclusion for the original fixed-kernel iteration

Fix a finite lattice, its actual source-zero Wilson construction, a finite source degree, and a finite family of globally smooth physical observables. Retain every old variable until its prescribed integration; use the matched reference kernel XP13; and carry all source coefficients, exact completions, and remainders.

Then:

1. Every finite composition of the original operations transports the original source jet exactly by XP6–XP7. Coarea and reference operations preserve the old physical marks by XP12–XP14.
2. Any fluctuation-independent part of the complete source exponent factors by XP16, at all finite source degrees. Independent deterministic/proper-difference localizations with the correct all-one endpoint preserve the same marked integral.
3. Lower and contact coefficients generated by products are exactly the partition sums XP4 and XP9. EOM reductions are the contact-completed differential identities XP20–XP29, including all kernel, effective-action, and cutoff terms.
4. A physical own-current relation is used only on its proved readout graph. It need not hold on all independently weakened paired inputs. Physical field-jet extraction is an exact splitting only when its complement is kept.
5. Terminal moments and connected coefficients are those of the original Wilson measure, with the original global source-zero denominator. No sourced live-island inverse is required.

This closes the exact finite-degree bookkeeping interface. The remaining operator-construction problem is quantitative: prove the joint marked regular/live estimates, the admitted-core propagation of the chosen coefficients, and the required degree dependence. It is not an additional unexplained algebraic identification of the old physical source with a reference observable.

Section 104.1 / Verdict

The final HI and XP notes establish the stated **exact finite-cutoff identities**, with the explicit common-domain, fixed-action, and kernel-composition qualifications.

They remove the demand for an all-wall nested-minimum identity and close the finite-source algebraic bookkeeping. They do not prove the uniform bounded source-preparation lemma, the full graded marked recurrence, or the continuum operator construction. The distinction between these conclusions is maintained in both notes.

The analysis versions:

- hybrid source interface, HI1–HI17, including HI13a and the final joint-wall and shrinking-R-disc qualifications;
- exact marked pushforward, XP1–XP42, including XP35a;
- scalar volume comparison, SV1–SV20, with its stability citation clarified to S74–S81, equivalently only the first inequality of S102.

Section 104.2 / Fixed-action retractions and the two source-wall families

HI3 is correct on its displayed common complex admitted domain. If $C_k = C_{k:j} \circ C_j$, preservation of C_j implies preservation of C_k. Conversely, a critical point for the common action under the C_k constraint is critical under the more restrictive $C_j = C_j W$ constraint. Admitted branch uniqueness therefore gives

$$P_j P_k \simeq P_k P_j \simeq P_k. \tag{HA1}$$

Section 104.2 / Fixed-action retractions and the two source-wall families

This argument needs the same fine action, fixed exterior, and genuinely nested determining maps. It is not a theorem for two independently chosen mixed-history windows or two different effective actions.

The primary verification is appropriate: Bałaban I (0.21), (1.1), and (2.1)–(2.3) uses the classical fine functional for the background minimum while retaining the inherited effective interaction separately. The latter does not silently change the minimum being composed. article.

HI7 now explicitly uses the actual completed level-k minimum readout at physical inputs. Thus applying the earlier physical pullback is the genuine nested critical evaluation, rather than an arbitrary projection of a fluctuating fine field. HI8 follows throughout the common complex domain by HA1 and gauge covariance of the physical current.

The two-family construction is algebraically sound. Define the proper difference by subtracting the zero-fluctuation value of the SAME localized family:

$$D_{\mathbf{s}}(U,b) = F_{\mathbf{s}}(U,b) - F_{\mathbf{s}}(U,0). \tag{HA2}$$

Its wall coefficients vanish at $b = 0$. Localize the direct deterministic tag independently as $T_{\mathbf{t}}$, and require the exact complete endpoint

$$F_{\mathbf{1}}(U,b) = T_{\mathbf{1}}(U) + D_{\mathbf{1}}(U,b). \tag{HA3}$$

There is no need for $F_{\mathbf{s}}(U,0) = T_{\mathbf{t}}(U)$ at arbitrary wall values. One must start from HA3; localizing F first and then discarding an unmatched zero-value coefficient would be wrong. HI makes this distinction explicit.

The final joint-domain qualification is essential. If unmarked action, coordinate, or propagator walls enter a source readout, the range constants and physical fluctuation disc must hold on their joint domain. Separate-family holomorphy does not establish this. The alternative ordering of exact source expansion followed by unmarked localization is valid only while its source coefficient functions and fluctuation coordinates are retained as declared; a later coordinate substitution needs its own domain check.

HI16 implies HI17 by radial Schwarz and linear wall localization. The bound is conditional on the full complex fluctuation disc, not just the real cutoff or walls equal to one. For recent R births, the available radius may be δ_j, and the factor is g_k/δ_j. The final note correctly refrains from replacing this by a fixed-disc factor before the actual delayed admission.

Section 104.3 / Finite source degree and deterministic factorization

The squarefree source algebra and XP4's set-partition coefficients are correct. Coefficientwise transport is linear on unnormalized densities or retained kernels. Its composition is exact for the actual finite integrals, and for infinite localized representations only after the stated absolute convergence justification.

XP16 is valid at every finite source degree:

$$\int K(U, dB) e^{V(U,B)+T(\eta,U)+D(\eta,U,B)} = e^{T(\eta,U)} \int K(U, dB) e^{V(U,B)+D(\eta,U,B)}. \tag{HA4}$$

Here T contains all mixed deterministic source coefficients and is independent of every variable integrated by THIS operation. If it depends on an old variable integrated later in a live island, it stays inside that later joint kernel. A separate source inverse or a product of separately integrated marked masses is not introduced.

Section 104.3 / Finite source degree and deterministic factorization

XP18 includes the mixed deterministic contact and the joint covariance. The new exact recurrence XP40 also has the correct combinatorics:

$$A'_I = T_I + \sum_{\pi \in \Pi(I)} \kappa^{(0)}_{|\pi|}(D_B : B \in \pi). \tag{HA5}$$

The ordered-block multiplicity cancels the factorial in the cumulant series. XP41 is explicitly an additional graded multilinear estimate; HA5 does not prove it.

Finite source degree does not imply finite field degree, a bounded carrier norm, a source-dependent zero-free disc, or uniformity as the cutoff is removed. XP consistently retains the coefficient functions, their field-jet complements, and their actual kernels.

The adjoint sign in XP20 is correct:

$$\mathsf{D}^{\dagger}_{\rho,X} F = -\rho^{-1} \operatorname{div}(\rho F X). \tag{HA6}$$

Its contact multiplication rule is XP22, and repeated adjoints produce the reversed order $X_r \cdots X_1$ on a test or kernel. The second-order formula XP27 correctly contains BOTH the tensor $q_*(\rho F\, X_j q^\alpha X_i q^\beta)$ and the lower current $q_*(\rho F\, X_j X_i q^\alpha)$, with signs $\partial_\alpha \partial_\beta T^{\alpha\beta} - \partial_\alpha U^\alpha$.

The cutoff identity also has the correct sign:

$$q_*(\chi\rho\, \mathsf{D}^{\dagger}_{\rho,X} F) = -\operatorname{div} q_*(\chi\rho F\, Dq\, X) + q_*(\rho F\, X\chi). \tag{HA7}$$

Selected histories retain this boundary term. Only the complete partition cancels its sum. The final text keeps sharp-selector contacts as ordered derivatives of the WHOLE kernel distribution, paired with the smooth original coefficient. It does not multiply coincident surface delta distributions without a separately defined product.

XP31 correctly includes the derivative of the generated effective action, all coarea terms, and the chosen output reference density. A transported fine current need not become one selected coarse Wilson current. If the fine vector field is not projectable, its transported vector measure must remain.

XP35a supplies the necessary distinction for physical own-current relations. Equality on $J = J_{\mathrm{phys}}(U)$ preserves derivatives of the physical pullback, whose tangent lift is

$$\widetilde{X} = X_U + (D_U J_{\mathrm{phys}} X) \cdot \partial_J. \tag{HA8}$$

Repeated derivatives also differentiate this lift's coefficients. Separate ambient U- and J-jets are not identified by equality on the physical graph. This prevents replacing the full EOM/contact packet by a quotient which is not preserved by subsequent joins.

Section 104.5 / Quantitative scope and final freeze

XP39's one-energy majorant is valid under XP38. For a partition with total degree $2q$, the inequality $(1+E)^q e^{-\theta E} \le (1+q/\theta)^q$ pays its moments using one reserve. There is exactly one rare mass B_ω, even if several labels occupy the same live island. Neither coefficient extraction nor the set-partition sum creates additional rarity.

The remaining assertions in the analysis are unchanged:

Section 104.5 / Quantitative scope and final freeze

Interface	The analysis scope
O1: common-domain graded preparation with all active walls	Still required; HI's exact split does not supply its norm or its invariant class
O2: complete marked clean/live/R recurrence	Exact finite-source and EOM/contact bookkeeping is proved; the required graded cumulant, joint-kernel, radius, promotion, and reserve estimates remain
O3: full physical operator normalization	Not implied by finite-source transport or by replacing full-core g_j^{-d} bounds with fixed-canonical $O(1)$ bounds
O4: source-zero intertorus comparison S94	Still open; SV proves the exact gluing identity and its divergent elementary bound
O5: one common continuum field hierarchy	Cutoff limits, tensor covariance, distribution growth, and compatible reflection forms remain; HA1 is only a finite-cutoff classical statement on an identified nested determining hierarchy
O6: Yang–Mills identification	Not established by HI, XP, or SV

The SV citation was clarified: its original vacuum stability comes from S74–S81, equivalently the first inequality of S102. No second $\widetilde{V}$ construction is used or asserted.

Section 105.1 / The precise periodic complex-domain gap is closed

For the actual same-target CK critical branch on a fixed physical torus, CC proves

$$\|F_a(u)\|_\infty \le C\big(\|F_a(A)\|_\infty + M^2\big), \qquad M = C\|A\|_{W_a^{1,p}}, \quad p > 8. \tag{ST1}$$

The common input ball uses $\|A\|_{W^{1,p}} + \|d_a A\|_\infty$, with constants independent of fine spacing, retained spacing, and blocking depth within the specified fixed volume. Flat and reducible references are included. The proof uses the actual Ward equation to estimate the longitudinal multiplier in a negative Sobolev norm before applying a coarse inverse inequality. It keeps the nonlinear force in the form of a divergence of a Hölder-controlled tensor plus an L^p error, and uses the endpoint curl estimate FE.

Section 105.1 / The precise periodic complex-domain gap is closed

These facts replace the earlier irreducibility-dependent attempt to control a separate complex gauge bridge.

WB supplies the required weak multiplier and fractional estimates without assuming the conclusion of ST1. RC gives the compatible version with actual Euler and constraint residuals. The result does not assert a uniform curvature supremum for every small $W^{1,p}$ input, nor arbitrary high spatial derivatives of a constrained background. The common curvature input in ST1 is a real hypothesis.

Section 105.2 / Nonlinear localization has an actual geometric construction

BI gives the complete five-type boxed inverse, including its negative Sobolev rows and the current term of the actual Wilson Hessian. CE constructs a local periodic extension from the actual connection and its transports. Its covariance is an exact group identity. Rough tree gauges are handled in covariant norms; their ordinary derivatives are never declared bounded.

CL defines a genuine nonlinear relative constraint from the original raw mean and endpoint transports. At a mixed interface, the prefix gauge at a node is taken at that node's own retained depth. The outward row includes the extra fine endpoint factor. Its flat derivative is the triangularly corrected mixed boxed row, rather than an incorrectly uniform-depth row.

The weak nonlinear vertices are localized **before** variation. This is necessary because multiplying a preformed negative-Sobolev force by a sharp wall is not a legitimate substitute. On protected regions the actual Euler–Noether equations give

$$F \in W^{1+s,p}, \qquad J_W \in W^{s,p}, \qquad 0 < s < 1/p, \tag{ST2}$$

as well as the original lower regularity bounds. The high auxiliary scalar norm is asserted only in the final controlled analysis gauge. No high-order norm is transported through an insufficiently regular gauge transformation.

MB extends the weak inverse to the specified separated hierarchy with the original fixed bare boundary variables. An adaptive partition sees at most two adjacent retained scales on a complete local inverse carrier. Every unknown and equation has its own scaling factor. The global parametrix has a contraction of order R_{loc}^{-1}, not a product of constants over the number of scales. At the bare boundary, the exact identity constraints give the explicit inverse MB32 on the actual action graph. No fictitious periodic boundary is substituted there.

There are three distinct statements here. An inverse for an individual connection, holomorphy on a fixed Banach tube of connections, and invariance of an all-depth sourced activity norm are different requirements. The first does not imply either of the other two merely because all finite matrices depend analytically on their entries. The fixed-tube nonlinear construction must retain the regular complex perturbation norm specified in its theorem. MC supplies that fixed-tube construction on the separated mixed hierarchy: the complex perturbation is controlled in the adapted first-order norm relative to a fixed real regular reference, and the own-scale complex curvature bound is imposed separately. Its contraction is uniform in the number of determining levels. The global scalar onto estimate eliminates the auxiliary force, and the actual nodal gauge restores the raw target. This closes the mixed classical construction in that stated domain; it does not close the third, marked-activity invariance statement.

On an actual critical graph the unweakened new readout satisfies $W_{\mathbf{1}}(U) = U$ in the input gauge. PS explains how this identity can be used in a source slot while retaining the original Gaussian variables, action, cutoffs, and all old stored fields. It applies to the actual inner minimum of a nested constraint problem. An arbitrary fluctuating connection $e^{gB}U$ is not declared to be such a minimum. The new

Section 105.2 / Nonlinear localization has an actual geometric construction

family also is not identified with the original map on every independent off-shell pair (U, J); NL records why that stronger assertion is false.

Section 105.3 / The ordinary scalar source now stays in the same Wilson integral

The scalar route in S uses the original source-free maps and their published domains. It does not need to replace them by the new CL family. Its source profiles have the explicit domain

$$J(z)=\sum_i z_i f_i, \qquad \sum_i r_i\|f_i\|_{C^\alpha}\le \rho, \quad |z_i|<r_i, \quad 0<\alpha<1. \tag{ST3}$$

Every physical insertion normalization belongs in this budget. This is not an arbitrary fine-plaquette product disc whose radius is uniform in the number of microscopic sites.

The large-field reference operation keeps the old marked numerator in its old variables. Its reference field is fixed on the new output fiber, and the numerator and denominator retain the same original reference cutoff. The partition of new selector labels is summed before its mass identity is used. An individual selected history need not be normalized. The old observable is never silently evaluated on the replacement coordinate.

For a regular reconstruction on an own-scale layer of length ℓ, the published low regularity bounds imply

$$|\psi(U_p)|\le C(a/\ell)^4\epsilon_*^2, \qquad \#\{p \text{ assigned to the layer}\}\le C(\ell/a)^4 N_\ell. \tag{ST4}$$

Thus the primitive scalar charge is bounded by $C\rho N_\ell$. Own-spacing stored compact variables use their separate compact plaquette bound. The ancestry accounting in S19 assigns each original primitive once; a primitive that has become a completed analytic member is not charged a second time. Gauge invariance of the scalar trace permits a controlled representative for this estimate without a bound on an unrelated complex conjugating matrix.

The positive cold estimate is strengthened by retaining fixed unused energy and volume fractions in the same integral:

$$\int e^{\theta\mathscr{E}_\omega+\tau N_\omega}\,d\nu_\omega\le B'_\omega. \tag{ST5}$$

It supplies the ordinary complex source bound, its linear excess, and proper joining of source packets sharing that component. There is one joint kernel and one rarity charge, not a product of two masses for overlapping pieces. The homogeneous source disc is constructed by the finite bootstrap S44a–c before it is differentiated in the heterogeneous freezing argument.

The scalar construction requires two necessary qualifications. The primitive heterogeneous initialization is $M_0^{\mathrm{het}}\le C\rho a^\alpha$, obtained by the actual anchor freezing in S42a. A bound merely of order ρ would not close its scale recurrence. The regular readout estimate also retains the carrier polynomial and the original near/remote split. Its large-carrier tail is used only after the old function has been placed in its actual birth domain. A shrinking large-field-origin radius is not replaced by a fixed clean radius on administrative promotion; its admission delay and radius-explicit tagged kernel remain in the estimate.

The source-zero vacuum is the published original vacuum by definition; only source-excess scalar births are newly extracted. At source zero every kernel, Haar factor, Gaussian normalization, selector,

and denominator is the original one. The lower bound is for the full positive terminal density on its actual regular Haar core, not a selected signed cluster contribution. Consequently the complete source-vacuum bound follows for this same Wilson functional.

105.3.1 A strict local normalizer, with its actual source norm

S87–S93 turns the complete analytic vacuum into a local functional of the source jet through order four. On a carrier of physical diameter r, the cutoff Taylor remainder obeys

$$\|\chi(J - T_y^4 J)\|_{C^\alpha} \le C r^{5-\alpha} \|J\|_{C^5}. \tag{ST6}$$

The four-dimensional root count contributes ℓ^{-4}, so the remaining geometric scale factor is $\ell^{1-\alpha}$, which is summable. Large and wrapping carriers use the exponential locality tail. This includes all clean, Gaussian, and healed large-field vacuum types.

The resulting fixed-radius estimate on each compatible torus is

$$\left| e^{-\mathcal{L}_{a,T}(J)} \frac{Z_{a,T}[J]}{Z_{a,T}[0]} \right| \le e^{C \operatorname{Vol}(T)}, \qquad \|J\|_{C^5} < r_0. \tag{ST7}$$

The normalizer is strictly local, holomorphic in the finite jet, and can be made reflection and translation covariant by finite symmetry averaging. For positive-side profiles the normalized reflection matrix is an exact diagonal congruence of the original Wilson reflection matrix. Its positivity is therefore preserved. No common zero-free domain for the full partition function is inferred from ST7.

The remaining intertorus qualification is narrower than equality of sourced families. S22 constructs the common C^α source normalizer entirely on one reference torus and transports its bound by the physical reflected slab inequality. Its only extra cross-volume numerical input is the one-sided source-zero vacuum-density comparison S94. Exact nonwrapping equality of the original source-zero family would prove that comparison. For it one must verify the background-dependent elementary typed local-factor definitions, old slots, artificial boundary conventions, and marginal extraction on isomorphic complete carriers. Dependence only on the field restricted to a carrier does not exclude an explicit ambient-volume constant. The relevant original CMP99 (3.107) factor definition is identified in S22, but has not been fully verified in this attempt. The flat local definition alone is insufficient because source-zero integration still samples nonflat fluctuations.

Conditional on S94, the reference-cell argument gives the scalar covariance transfer rate $d^{4-2\alpha}$. It needs no intertorus equality of sourced coefficients. The strictly local C^5 estimate ST7 remains valid within each torus without S94, but does not by itself supply that sharper rate. A second canonical decomposition with a genuine source-zero stability bound for the same partition function would be an alternative way to prove S94, as shown in S102–104; such a second decomposition has not been supplied.

SV checks a direct alternative for the same Wilson measure and the same bare coupling. Reinterpret one product Haar space as either q^4 small periodic tori or the large torus, and let ΔS be the difference between their plaquette sums. Then

$$\frac{Z_{a,qT_*}}{Z_{a,T_*}^{q^4}} = \mathbb{E}_{\mu_{a,T_*}^{\otimes q^4}} e^{-\beta_a \Delta S}.$$

The elementary seam estimate costs $O(\beta_a a^{-3}/\ell_*)$ per physical volume. Independent block gauge symmetry shows why small original plaquette energy does not remove this cost in Jensen's inequality.

Section 105.3 / The ordinary scalar source now stays in the same Wilson integral

Reflection positivity supplies the sign of the pressure difference, not its required uniform upper bound. The exact remaining signed gluing estimate is SV11. The normalized positive transfer-trace bound SV16 is another sufficient condition. Neither has been proved here. SV uses the original vacuum stability from S74–81; it does not assume that the second normalization proposed in S102 has been constructed.

Section 105.4 / What is imported in that scalar statement

The scalar theorem remains conditional on the hypotheses of the published source-free construction, including the tuned weak-coupling trajectory. Its imported inputs are the source-free Gaussian and wall estimates, the original low regularity/range theorems for the actual paired readouts, the original preparation identities, and the positive cold estimate with its strict parameter margins. S states where each is used. It neither reinterprets the auxiliary current as a physical source nor assumes a previously unproved ordinary-source theorem as an input.

The source-free theorem, the earlier clean homogeneous history estimate, and the new marked extension must keep the same coupling and vacuum conventions. Even with the scalar estimate above, the physical ultraviolet leading-term comparison and the separate continuum synthesis have their own hypotheses. A local finite-cutoff source estimate is not by itself a proof of scalar continuum convergence, rotational covariance, or interacting Yang–Mills dynamics.

Section 105.5 / Arbitrary insertions: exact progress and the estimate still needed

AI proves the contact-completed Haar equation-of-motion identity for the original Wilson law,

$$\mathscr{E}_a[V] = a^4 \sum_{b,A} (J_b^A V_b^A - \kappa_a X_b^A V_b^A), \qquad \langle \mathscr{E}_a[V] G \rangle = \kappa_a a^4 \sum_{b,A} \langle V_b^A X_b^A G \rangle. \tag{ST8}$$

It holds at every finite cutoff. Its ordered all-degree form keeps all noncommuting derivatives and all contact terms. For disjoint determining bonds the right side is exactly zero. For coincident products it is not zero; the product rule creates the displayed derivative of the other insertion.

In dimension six the exact scalar identity reduces the derivative-curvature square to a selected charge-even cubic completion, a quadratic descendant, the explicit Casimir lower counterterm, and the completed EOM term. The hypercubic quadratic remainder H is a genuine independent direction. The complex null-momentum symbol check prevents eliminating it by an unsupported rotational identity. Thus even this small scalar sector needs a coupled $(C^\sharp, H)$ mixing problem, together with its lower sectors and descendants.

DN proves that the free positive-time Gram matrix of these two directions is nondegenerate. It also proves a noncommuting matrix-product bound under an explicit one-step transfer estimate and the actual lower running tube. MT additionally computes the actual free quotient transfer, $T_0 = L^{-2} I_2$. The nonprincipal aliases of the directional-current leg have offset L^2 norm $O(|\theta|^3)$; hence they cannot change the fourth potential jet of H. The calculation uses two independent external momenta, so it retains pointed tests and descendants that would be invisible in a constant-test diagonal energy. The interacting transfer estimate for the full evolving marked state and a uniform comparison of its physical Gram matrix with the free one have not been proved. The clean primitive-column estimate below is a separate, narrower result. Finite dimensionality and a free nonzero form factor do not supply the full comparison.

The actual-minimizer example HJ is an important limitation, not a failure of the desired quantum theory. It is a family of genuine Wilson constrained minima satisfying the admitted low bounds for which a dimension-eight second-derivative curvature square grows like a^{-1}. One therefore cannot complete the proof by appending arbitrary high conditional-background regularity to CL or CC. Full Gaussian recombination removes some artificial conditioning singularities, but interaction attachments near a physical mark still have to be estimated after the cancellations.

CD supplies a further regular readout estimate on the actual lattice. For a reflected pair of covariant paths it retains the complete quadratic derivative tensor, the test-function descendant, and the signed-area curvature cubic, with a remainder of order r^{2+s}. A coordinate path has the stronger r^{2+2s} remainder by an exact transported variance identity. After the old curvature factors, four-dimensional root count, and dimension-six normalization are all included, the first estimate gives the strict age factor r^s. Its application to a marked birth assumes the displayed coefficient-kernel norm, completed translation/path pairing, and actual protected core. A lone block-grid sum is not reindexed by an arbitrary fine shift, and a path through a wild region is outside that regular estimate. CD by itself does not prove invariance of the required kernel class under the interacting insertion transport.

CB closes a further regular interface for the actual centered nonlinear mean. Its one-step cancellation upgrades to H^{2+s} potential control; the exact positive two-cochain average gives a curvature error of order r in H^1. Lifting an already differentiated coarse field then gives an r^s comparison of the actual fine and coarse directional currents. It does not differentiate a discontinuous block hold. The tested H, $C^\sharp$, full derivative-curvature tensor, pointed $(\partial f)F\,DF$ terms, and quadratic descendants are controlled. The cubic comparison retains its cubic coupling grade. The actual raw mean is related to the analyzed representative by exact gauge covariance.

The marked-transfer argument gives the companion coefficient lemma for a universal regular physical pullback. Cauchy is taken in its finitely many coarse coordinates; the source-anchor norm is taken carrierwise in ℓ^1. Fixed-degree coefficient and path-moment sums therefore cost only carrier polynomials. Full coarse-root sums, reflected partners, and the actual fine test offsets give the required CD pairing. With CB this closes that regular kernel premise on the stated protected windows. It does not cover arbitrary history-dependent coefficients, rough crossings, or the complete independent-current/contact transport. The dimension-four quadratic state is retained exactly: its error cannot be multiplied by a dimension-six factor r^{-2}.

105.5.1 Primitive norms and interactions attached to a finite mark

PA identifies completed original minimum slots in which a source-only MC repair has exact physical value, pointwise in the original Gaussian variables. It uses the very same determining problem, including any pinned fine boundary and stored exterior. The entire primitive collar stays protected before the global source map is localized. A separate local saddle agreeing only on the inner input would not prove that identity.

The canonical moment argument now uses an explicit flat-data family: all determining field coordinates are scaled, so the readout is the identity at parameter zero. Its analytic radius is controlled by q_∞, a supremum of canonical inputs. Only after Cauchy's estimate is proved is that supremum bounded by a finite-carrier Gaussian energy. A pointwise cutoff does not control the total energy norm on a carrier of arbitrary size, and a frozen nonflat exterior is outside this flat-data expansion.

For field valuation $d = 2$ for H and $d = 3$ for $C^\sharp$, the two distinct conclusions are a full physical-core bound with an explicit polynomial g^{-d} cost, and a canonical nonlinear difference of order g times a fixed Gaussian polynomial on a fixed canonical external-field ball. The full-core cost is unavoidable

even on smooth actual minima, which MC leaves unchanged. Neither a physical graph identity nor an original domain inclusion removes it.

MT proves the finite-mark formation lemma under a single positive quadratic reserve. It uses a finite nilpotent source algebra rather than an unjustified ordinary exponential-source disc for a cubic mark. For several marks in one component the polynomial is bounded using that component's joint energy. The coupling-small first-difference estimate additionally requires an incoming polynomial bound for the actual unmarked interaction.

IV verifies that bound for the complete inherited clean interaction. The original marginal counterterm is paired before localization. Each resulting original coefficient vanishes at zero fluctuation for every wall value and is holomorphic on $\|B\|_\infty < \epsilon_1/g_k$. Schwarz's lemma gives

$$|V_k'(Y;U,J,B)| \le CE_0 g_k \|B\|_{\infty,Y} e^{-\kappa d_k(Y)}. \tag{ST9}$$

The old-age sum is already present in the published coefficient bound. This estimate is uniform on the original physical paired core and does not require a flat exterior. It closes a genuine incoming interaction estimate needed near a mark. It does not place a normalized marked activity with its g_j^{-d} norm into the unmarked clean E_0 class, and it does not prove the full marked recurrence.

For the admitted primitive minimum slots, the fresh Wilson/coordinate vertices and the actual current-coordinate Gaussian comparison have also been checked. The Gaussian derivative is taken in the same real variables with the same reference cutoff; its trace normalization is retained. MT32–33 then gives the actual clean primitive-column bound

$$T_j^{\mathrm{cl}} = L^{-2}(I_2 + E_j^{\mathrm{cl}}), \qquad \|E_j^{\mathrm{cl}}\| \le Cg_j + C\,\mathrm{tail}_j. \tag{ST10}$$

It does not identify this primitive matrix with the successive fundamental-matrix ratio after every old stable and live marked activity has fed back. The intersection of canonical low-jet norms and the full physical-core norm, the completed regular readout, and the exact lower/contact state must be retained for that step. Relative coupling powers can be paid by the strict geometric age factor; an arbitrary absolute g_j^{-d} loss at a fixed age cannot.

MT40–45 makes the two-norm requirement quantitative. If the full physical-core norm and canonical low-jet norm are both retained, the quadratic and cubic moment kernels obey

$$\|K_{m,j}\|_{\mathrm{mom}} \le CA_j \min\{g_j^{-m}, \alpha^{d-m} g_j^{-d}\}, \qquad m = 2,3, \quad d = 2,3. \tag{ST11}$$

The remainder is the physical own-current pullback after subtraction of the whole chosen covariant quadratic and cubic completions. In particular, the cubic contribution of the quadratic completion is included before the total cubic center is removed. Its fourth-order tail, at current canonical inputs, has coefficient

$$r^2 \alpha^{d-4} g_k^{4-d} (g_k/g_j)^d, \qquad r = L^{j-k}. \tag{ST12}$$

This supplies a current positive power of the coupling, while strict age decay pays the relative coupling ratio. It does not give a small bound for an arbitrary fourth-order tail on its whole physical core.

For a common admitted readout and complex wall family, the proper difference $R_j(\Psi_{\mathbf{s}}(U,b)) - R_j(\Psi_{\mathbf{s}}(U,0))$ vanishes at zero fluctuation for every wall value. Schwarz on the fixed physical fluctuation disc gives an additional $g_k\|B\|_\infty/\delta$, including for the full-core quartic tail. The deterministic term remains an old tag. This is a local lemma under the whole-disc hypothesis MT47; the exact construction of the old tag and its subsequent localization must also be specified.

MT49–53 identifies the actual free reference through its running nested projection precision. The induced conditional means compose exactly, and their curvature aliases obey the exact cochain identity.

Free quadratic contractions remain in the vacuum/test row; a principal invariant cubic has no invariant linear contraction for a simple gauge algebra. Stable-to-center alias cancellation is valid after the complete old-root and subcell-phase sum, with the dimension-four family retained in its own free evolution. It is not an identity for an individual history-dependent localized member.

105.5.2 Exact old tags and finite-degree contact transport

HI and XP supersede the algebraic part of the remaining-interface wording after MT53 and in its analysis. They do not supersede the quantitative invariant-class and live estimates.

For the same fine Wilson action, action graph, boundary data, and genuine nested determining maps $C_k = C_{k:j} \circ C_j$, define $P_j(U) = U_j(C_j U)$. On their common admitted complex domain,

$$P_j P_k \simeq P_k P_j \simeq P_k, \qquad j < k. \tag{ST13}$$

Equality is up to the prescribed gauge. One order follows from exact target preservation. For the other, a level-k critical field is critical under every variation preserving C_j, by the constraint chain rule; uniqueness of the admitted small branch identifies it with the level-j reconstruction. The original classical backgrounds minimize the same Wilson functional, so running coefficients and inherited effective interactions do not silently change the objective used in this argument. Changing the window boundary or action graph is outside its scope.

The proper fluctuation difference and the direct deterministic old tag can now be localized in separate source-wall families. At all walls one, ST13 identifies the old tag on the full common complex domain. Away from that endpoint, the proper difference is formed from two evaluations of the same chosen readout and therefore remains zero at zero fluctuation. No equality of the two families' individual wall coefficients is needed. A counterexample in HI shows why a wallwise retraction tower would have been an unjustified stronger demand.

XP proves the finite-source version. Let $A(\eta, U, B)$ be the entire source exponent in a finite nilpotent source algebra, including mixed and contact coefficients. With $T = A(\eta, U, 0)$ and $D = A - T$,

$$\int K_U(dB) e^{V(U,B)+A(\eta,U,B)} = e^{T(\eta,U)} \int K_U(dB) e^{V(U,B)+D(\eta,U,B)}. \tag{ST14}$$

This is an unnormalized exact identity for the original kernel. If T depends on an old variable retained at this step, it stays inside any later integral which integrates that variable. There is no sourced island inverse. Products of marks are joined before their common integration; finite source degree does not imply finite field degree.

For $\rho = e^{-S}$ and a Haar vector field X, write $\mathsf{D}^\dagger_{\rho,X} F = -\rho^{-1} \operatorname{div}(\rho F X)$. The EOM/contact packet is transported as a density and a current. On a smooth observation patch, with an actual selector χ,

$$q_*(\chi \rho \mathsf{D}^\dagger_{\rho,X} F) = -\operatorname{div} q_*(\chi \rho F\, Dq\, X) + q_*(\rho F X \chi). \tag{ST15}$$

Sharp selectors have their distributional boundary term. General history and reference kernels use XP29 rather than a fictitious smooth physical quotient. Full partitions cancel their selector terms only after summing. The complete joint selector/kernel is differentiated before its distributional pieces are split; undefined products of separate boundary distributions are not introduced. The effective output current includes derivatives of every generated interaction and normalization assigned to its density. EOM packets cannot be discarded as a multiplicative ideal, and independent-current derivatives cannot be treated as derivatives tangent to the physical own-current graph.

These statements close the exact finite-degree bookkeeping interface. They do not prove bounds for the coefficient densities, their wall derivatives, or the boundary currents. The required bounded source preparation is stated in HI: every actual incoming member must have an admitted recent/prepared or old/direct representation on the common domain, with summable complete carriers and the MT47 graded estimate on the whole fluctuation disc. The newly formed physical pullback must return to that class after all lower and contact joins. For recent R births the physical disc may shrink with the birth radius; the factor is then g_k/δ_j, not automatically $O(g_k)$ with a fixed denominator. No fixed clean disc is assigned to those recent rare members by fiat.

105.5.3 The unresolved local mixing theorem

For every fixed engineering dimension D, one needs a physically fixed finite list of lattice completions, lower-dimensional counterterms, descendants, and contact representatives. Its marked transport must be the transport through the same original integration kernels. Its local extraction must leave a strict, summable remainder in a **completed** norm that retains all conditioning-color, cutoff-boundary, and contact cancellations before absolute values.

In particular, the still-unproved assertions are:

1. a quantitative near-insertion expansion with coefficients fixed by one physical normalization, independent of the other insertion positions and of the auxiliary analysis grid;
2. a bounded and uniformly invertible center evolution in each retained dimension, with the actual lower-sector mixing and a strict remainder;
3. joint connected bounds of finite factorial order for arbitrarily many fixed-dimensional marks, including overlaps, rare histories, and covariant path or derivative commutators;
4. compatibility of these constructions as the dimension increases, so that all mixed correlations and reflection forms belong to one field family.

AI's local conditional Taylor theorem applies to the actual Wilson measure and gives a strict finite-jet remainder for a bounded physical boundary perturbation. Its coefficients still depend on the actual exterior field. The rough-boundary SU(2) example proves why this does not imply a fixed finite-dimensional approximation uniformly on all exteriors. The positive large-field measure and the fixed physical mixing must enter the missing estimate. Likewise the full Wilson EOM in ST8 cannot be discarded pointwise in an effective action that has generated additional interactions; its exact pushforward includes their boundary and attachment contributions.

Section 105.6 / Remaining common-domain and reconstruction obligations

For a source induction that uses the new nonlinear extension rather than the original scalar readouts, the fixed common input tube must also carry every inherited analytic activity in its original birth core. A bound of the form C times the input radius is not automatically an invariant radius under arbitrarily many compositions. Direct old-to-current readouts, the original preparation inclusion, and strict age gains are the relevant quantities to check; a primitive polynomial alone does not test this issue. The two-family construction ST13–14 avoids iterating repairs into the deterministic old function. It removes an unnecessary wallwise identity, but it does not by itself supply those inherited norms.

DA proves direct admission for a bounded contractible old carrier lying inside a protected regular window. A local holomorphic gauge removes the potential's constant value at the old root; its potential and first differences in old units are $O_D(r^2\epsilon)$. The actual Wilson current rescales exactly as $r^3 J$. The

Section 105.6 / Remaining common-domain and reconstruction obligations

original smaller-core inclusion then applies under the explicit birth-radius inequalities DA8. A fixed finite age delay supplies a strict margin for polynomially varying radius ratios. Recent ages still need a proved norm of the same marked member on an admitted core; large and boundary carriers still need an admitted original representation. Neither an unspecified large blocking factor nor an exponential coefficient justifies evaluating a member outside its domain.

Beyond the local estimates, the proof must obtain one cutoff sequence and one compatible limiting hierarchy, full Euclidean tensor covariance, the regularity and growth hypotheses needed for reconstruction, and a physical vacuum sector with the required Yang–Mills short-distance identification. The earlier scalar angular and phase arguments are conditional implications with those stated inputs. They do not extend automatically from a scalar to every gauge-invariant tensor field.

Nonzero scalar covariance, a positive reflected norm, and a non-Gaussian composite are insufficient to identify interacting Yang–Mills. The operator relations and the chosen gauge group's dynamics must also be established. The mass gap has not been attempted in the construction.

The argument therefore reduces several concrete local and measure matching gaps and closes the exact finite-source/current bookkeeping, but the arbitrary-insertion mixing and common physical hierarchy remain substantive open obligations of this attempt. They are not listed as assumptions of a claimed complete proof.

Section 106 / Regular marked coefficients, original domains, and coarse response kernels

This note continues the analysis in the stated results. It does not assert a construction of four-dimensional Yang–Mills theory. Its subject is the regular part of O1–O2 and, in particular, what is needed to turn MT36 and XP41 into an induction for coefficients which have actually been formed.

The original regular minimum construction has a genuine enlarged-input-core inclusion, which prevents a repeated radius loss in that construction. Exponential locality of its first coarse response, including the physical current, gives the physical coefficient moments without counting microscopic coordinates. Sections 13–15 then provide a current-index representative for an old clean scalar tag, its protected fractional regularity, a same-equation residual comparison, and a test-functional norm in which the strict-age scalar row is stable under actual clean formation.

Several qualifications remain important. A physical kernel norm does not control an arbitrary independently extended current coefficient. The regular state therefore retains a canonical paired norm as well as its physical kernels. A selected history need not have the translation pairing of a completed universal clean family. The smooth-test norm used after regular cancellation does not control a newly selected sharp contact diagonal. The final section records these outstanding all-history and all-contact obligations; the clean scalar result is not a complete Yang–Mills proof.

Section 106.4 / Physical coarse derivatives without a microscopic-coordinate sum

Fix a genuine regular minimum on one specified action graph, with the actual determining hierarchy and fixed exterior. In own coarse units let

$$P_j(v) = U_j(e^v), \qquad f_{j,X}(v) = E_{j,X}(P_j(v), J_{\mathrm{phys}}(P_j(v))). \tag{RG12}$$

For the canonical application the reference determining field is flat. For local comparisons the reference may be a specified regular physical minimum. The common domain is a small **local supremum** coarse polydisc. No global L^p volume factor is included in its radius.

106.4.1 The first coarse response

The original variational paper, Proposition 9 and (182)–(190), gives exponential decay of the derivative of the Landau-gauge minimum with respect to a determining coordinate, including its first-difference, Hölder first-difference, $D^*_{U_0} D_{U_0}$, and Laplacian rows. In a uniform hierarchy, after rescaling the retained spacing to one, these are uniform local estimates. Let h be a coarse bond and let X^+ include the complete response and current collars. Their relevant consequence is

$$\|\partial_{v_h} P_j\|_{C^1(X^+)} + \|\partial_{v_h} J_{\mathrm{phys}}(P_j)\|_{\infty,X^+} \le C e^{-c\,\mathrm{dist}(h,X^+)}. \tag{RG13}$$

The current term must not be inferred from a generic C^1 field bound. There are two legitimate routes to it.

- Differentiate the genuine KKT equation $J(P_j) = Q_j(P_j)^*\lambda_j$. The localized augmented inverse bounds the response of both the field and multiplier. The actual averaging adjoint and its derivative have bounded own-unit row sums on their complete prefix carriers. Thus $\partial J = (\partial Q_j^*)\lambda_j + Q_j^*\partial\lambda_j$ has the same exponential decay.
- Use the strong second-order response rows in (190) and the exact local Wilson-force derivative. On a controlled chart $U = e^{aA}$, that derivative satisfies

$$\|D_A J_a(A)[H]\|_\infty \le C(\|d^* dH\|_\infty + \|DH\|_\infty + \|H\|_\infty). \tag{RG14}$$

  A direct verification expands the analytic plaquette flux on a controlled chart as

$$X_a(A) = d_a A + [A_\mu, A_\nu] + a\mathcal{R}_a(A, D_a A), \tag{RG14a}$$

  where $\mathcal{R}_a$ and its first field derivative are uniformly bounded from the small C^1 chart to C^0. Shifted connection factors in the quadratic term differ from their base values by aD_aA. Moreover the actual covariant incidence adjoint is the flat incidence adjoint plus bounded connection and shift operators. Consequently the nonlinear current remainder and its first field derivative use only A, D_aA, H, D_aH: the apparently dangerous term satisfies $\|a\partial_a^* D\mathcal{R}_a[H]\|_\infty \le 2\|D\mathcal{R}_a[H]\|_\infty$, up to the fixed number of orientations. This gives RG14. The covariant Maxwell row in (190) converts to the flat one by the same first-order connection terms. The left-logarithmic chart conversion differs from the identity by $O(aA)$; its second-difference terms use $aD_a^2H \le 2D_aH$ and $aD_a^2A \le 2D_aA$. Absolute matrix norms make the argument valid on the admitted complex chart. Complete current stencils are essential. This calculation has also been stated large-field note.

The first route avoids expanding the Wilson force and is preferable whenever the full original mixed KKT inverse is already retained. Neither route supplies an unprotected rough-collar current estimate.

106.4.2 Higher coarse derivatives by Cauchy

Let b_X be the full paired ℓ^1-anchor bound on a slightly larger domain. Cauchy for $E_{j,X}$ in the paired Banach chart and RG13 give

$$\|\partial_{v_h} f_{j,X}(v)\|_{\ell^1} \le C b_X R^{-1} e^{-c\,\mathrm{dist}(h,X^+)} \tag{RG15}$$

on a fixed smaller coarse polydisc of radius R. The radius is chosen so that the **same physical minimum and its actual current** stay strictly inside the old paired domain. This follows from the original minimum range estimates; a sufficiently small coarse ball is chosen once.

For fixed $m \le 4$, differentiate in another $m-1$ coarse coordinates by coordinate Cauchy on a still smaller fixed polydisc. The exponential in RG15 remains unchanged. Symmetry of the derivatives lets one select the farthest of the m coordinates. Hence

$$\|\partial_{v_{h_1}} \cdots \partial_{v_{h_m}} f_{j,X}(v)\|_{\ell^1} \le C_m b_X R^{-m} e^{-c\max_i \mathrm{dist}(h_i,X^+)}. \tag{RG16}$$

For a repeated coordinate use the ordinary one-variable derivative Cauchy formula; its factorial is part of C_m. There is no dependence on the total number of coarse bonds in the torus. In four dimensions the number of choices within distance R_0 is bounded by $C|X^+|(1+R_0)^4$; therefore the sum over $m \le 4$ coordinates and every fixed polynomial moment is bounded by a fixed polynomial in $|X^+|, \operatorname{diam} X^+$. Spare carrier locality pays this polynomial.

Apply the argument both to the fixed physical coarse ball and to the coarse canonical ball $v = g_j z$. On the latter, choose the fixed z-radius small enough that the physical pair lies inside the canonical paired ball in RG6. The resulting physical quadratic and cubic kernels obey

$$\|K_{m,j}\|_{\text{moments}} \le C A_j \min\{g_j^{-m}, \alpha^{d-m} g_j^{-d}\}, \qquad m = 2, 3. \tag{RG17}$$

This obtains the two bounds in MT41 from the original paired coefficient and the actual physical response. It does not require treating every microscopic link as an independent Cauchy variable. It also identifies a sufficient replacement for MA20's stronger assumption that a fully localized physical pullback has already been constructed.

106.4.3 The anchor norm is retained before the derivative estimate

Use the nilpotent Banach algebra $\mathbb{C} \oplus \ell^1(\mathcal{A}_j)$, where $\mathcal{A}_j$ is the set of original source anchors. Its norm is $|c| + \|a\|_1$. The source formation proof S8–S11 and the finite-mark reserve proof apply in this algebra. Consequently b_X in RG15 is an actual sum of absolute anchor coefficients, not a bound for one test chosen after the polymer sum.

The primitive anchor weights include the physical integration weights. Their sum on one own-unit carrier is bounded independently of the number of microscopic plaquettes. Original anchors and determining bonds remain in the coefficient, including those which produce an exact contact later.

Section 106.9 / Finite sources require a larger grade than the one-label D6 norm

For a finite source label set I, put $D_I = \sum_{i\in I} d_i$, before accounting for any explicitly retained differentiated-current grade. A full physical bound of size $(\alpha/g)^{D_I}$ together with only quadratic/cubic canonical jets is insufficient when $D_I \geq 4$. Its quartic remainder on canonical fields has the coefficient g^{4-D_I}, which is not uniformly bounded for $D_I > 4$.

A concrete analytic counterexample is

$$F_g(z) = \alpha g^{-D_I} z^{D_I-1}, \qquad D_I \geq 5. \tag{RG27}$$

On a fixed small physical ball it has exactly the allowed full-core size, and its quadratic and cubic jets vanish when $D_I \geq 5$. But $F_g(gz) = \alpha g^{-1} z^{D_I-1}$. This shows why the one-label two/three-jet class must not be silently reused for arbitrary source joins.

For each fixed I, keep the canonical paired ball norm, equivalently all ordinary canonical jets through degree D_I together with a physical remainder of order $D_I + 1$. The analogue of RG11 then has a

Section 106.9 / Finite sources require a larger grade than the one-label D6 norm

polynomial of degree $D_I + 1$, with a nonnegative extra power of the current coupling in the remainder. The one-reserve formation estimate absorbs the total polynomial degree of each set partition in XP40.

For n marked factors with envelopes $a_\nu(1 + \mathscr{E})^{q_\nu/2}$, the same original connected expansion gives a bound of the form

$$\|\kappa^{(0)}_{k,n}(F_1, \dots, F_n)\|_{\mathrm{out}} \le C_{n,Q,\theta} \prod_{\nu=1}^{n} a_\nu, \qquad Q = \sum_{\nu} q_\nu, \tag{RG28}$$

provided their actual common carrier/domain envelope and the locality inequalities above hold. The constant comes from one Gaussian reserve for the total degree, finite connected partitions, and the same carrier covering; it does not use independent rarity factors for overlapping sources. Tensor-product anchor ℓ^1 norms are submultiplicative, so overlap causes no loss of exact anchor labels.

RG28 is the quantitative local implication behind XP41. Its repeated application to a complete marked hierarchy still requires the exact graded domain for all differentiated current and contact coefficients. Derivatives can lower field valuation. They must be assigned the grade they actually have, including cutoff boundary currents, rather than the original grade of the undifferentiated source.

Section 106.11 / What the regular attachment estimate now reduces to

For an actual completed universal regular birth with RG6, the actual coarse responses in Section 4, and the same common regular readout problem, Sections 5–6 supply the coefficient/moment class required by CD and MT40. The completed quadratic and cubic readout terms give their strict r^s, r^β factors; the fourth-order term is RG22. Hence the incoming canonical envelope has the form

$$CA_j \left[r^s \rho_{j,k}^2 q_Y^2 + r^\beta a_{3;j,k} q_Y^3 + r^2 \alpha^{d-4} g_k^{4-d} \rho_{j,k}^d q_Y^4 \right], \tag{RG31}$$

where $a_{3;j,k} = \rho_{j,k}^3$ for $d = 3$, and the sharper physical cap gives $a_{3;j,k} = (g_k/\alpha)\rho_{j,k}^2$ for $d = 2$. The total C-center combines the quadratic path-area cubic and the cubic kernel moment before absolute values, exactly as in MT40–MT45.

Attach one distinguished actual unmarked vertex. IV and MT12–MT13 supply its factor Cg_k on the same joint kernel. The finite-mark moment estimate then gives MT36 on this regular branch, with the three explicit weights in RG31. MT34–MT35 sum the relative coupling factors after the weak floor is chosen.

This is stronger than assuming an unexplained incoming physical kernel class: the coarse kernels follow from the actual paired coefficient and the original first physical response. However, use of RG31 at a **particular localized history member** still requires identifying its literal readout with the common regular branch used in deriving those kernels. Equality at all walls one alone is insufficient. A source localization using HI's two separate families remains exact, but every active proper-family member must have the admitted protected range, and all tail members must keep their original representation and grade.

In particular, common-field pairing is arranged at the current packet level. First file the **entire** translated old family using the smooth test $f\chi_Q$. Then choose one enlarged current determining problem D_0 and one source-wall field for that packet. Choosing a separate field for each old root would not justify CD's root reindexing. Short old carriers and paths lie in the common protected collar; long and boundary-crossing members retain their original tail representation.

When comparing this current-window problem with a birth pullback defined on a larger graph, the relevant estimate is a remote-data comparison for the **same** $E_{j,X}$. It is not a comparison of coefficients produced on different physical tori. One can restrict a global minimum to the same local mixed hierarchy with the literal bare pinned collar: its new finer targets are its actual traces, and the nested-constraint KKT argument makes it critical for that local problem. A comparison with another local minimum is then a variation of remote targets and exterior action data. If the two data sets have a common admitted interpolation, the localized mixed inverse gives an off-support factor $e^{-c\,\mathrm{dist}_{\mathrm{old}}(X,\partial D_0)}$. Its correct norm is the weighted mixed operator norm; summing unweighted microscopic boundary columns is not valid. The existence and range of this literal interpolation must be checked in each boundary filing. The generic symbol $e^{-c/r}$ is not a substitute for that check.

106.11.1 The fractional regularity distinction

The ordinary bounds I (3.31) and II (1.18)–(1.22) alone do **not** supply the H^{2+s} potential or H^{1+s} curvature used by CD. A bounded independently extended current can enter an elliptic response as an L^∞ coefficient without any positive fractional regularity. It is therefore incorrect to append the CD age factor to every original paired-wall bound RG11 merely because the original inner minimum is admitted.

The paired canonical norm and the physical strict-age norm have distinct uses. The former controls an original off-graph tail. The latter must use a source-only physical family with the protected fractional estimates, or an explicitly proved comparison to one. The next section gives a concrete classical construction of such a family. It does not silently identify it with every original wall extension.

Section 106.12 / A source-only physical-target hull and its exact scope

The MC/CL augmented equations permit a prescribed small coarse target. Fix the same actual action graph, mixed determining hierarchy, bare pinned boundary, and relative endpoint convention as in

MC. Replace its last equation by

$$q_U(b) = \theta, \tag{RG32}$$

where $\exp(r\theta)C_kU$ is the **actual original coarse target**, expressed in the original frozen fluctuation coordinates. The notation includes the precise endpoint transport, rather than a substituted averaging map. The forcing in MC10 becomes

$$(-J_{\Omega_0}(U), 0, 0, 0, \theta). \tag{RG33}$$

The coarse target is measured in the existing assigned-depth row norm. It is zero on the fixed bare pinning rows; a change of prescribed exterior values is a separate background parameter, not an unauthorized change of those fixed rows. On a current unit lattice, a bounded target has the fixed finite fractional norms used in the protected bootstrap; finer mixed rows retain their own length factors.

The same uniformly local contraction proves a jointly holomorphic solution on the common target and wall discs. On protected carriers the same-carrier parametrix identity gives the actual local augmented equations. Thus MC13–MC18 give the physical $F \in W^{1+s,p}$, the auxiliary scalar $\omega \in W^{2+s,p}$, and the actual current bounds there. This step uses one coherent typed inverse and weak-vertex localization. Independently weakened propagator identities are not assumed.

At all walls one, the gauge-elimination proof still works. Its test variation satisfies $\mathcal{S}_{U,W}\zeta = 0$, so $Dq_U(b)d_W\zeta = 0$ regardless of the nonzero target θ. Hence $\omega = 0$. Incorporating the coarse logarithmic chart derivative and $\kappa\theta + \mu$ into the physical multiplier gives the true Wilson KKT equation. The coarse Ward identity removes the vertical endpoint-gauge differential. Uniqueness of the admitted original branch identifies the resulting field with the original minimum for that same coarse target, modulo the allowed gauge. Therefore a gauge-invariant source has its original value for **every** admitted original fluctuation B. No Gaussian, real selector, coarea density or unmarked paired extension has been changed.

106.12.1 A physical pre-reconstruction preserves the endpoint

For a source extension one may first use the actual $P_{k+1}(U)$ and its actual current, on a fixed extra collar. The physical pairs explicitly included in I (1.15)–(1.16) supply their regular range. At a physical input U_{k+1}, the same-action nested identity fixes this input. Thus replacing the **marked** off-graph extension this way preserves the original physical B-integrand. The unmarked kernel continues to see its original independent pair (U, J).

Equality as a function of the full original coarse variables also preserves its finite derivatives. This is the contact naturality statement spelled out in LM46a: the physical-current derivative includes the full DJ_{phys} chain rule, and equality multiplied by unchanged selectors and kernels remains equality of the corresponding distributions. Original fine Haar vectors must first be transported with the full XP tensor/vector-density chain rule; this statement does not make an arbitrary fine vector projectable or discard derivatives in retained fluctuation variables. It also does not assert equality of arbitrary independent-current extensions.

106.12.2 A protected wall perturbation can fit a pre-existing core margin

Here is the local estimate needed to avoid multiplying analytic radii by a fixed loss at every birth. Let all changed inverse and nonlinear-vertex carriers be at distance at least H from a packet Q, in

current own units. Let x_s be the target solution and x_1 its physical all-one solution. The exponentially localized typed inverse and local Lipschitz bounds used in MC12 imply

$$\|x_s - x_1\|_{X,\mathrm{near}\,Q} \le C\varepsilon e^{-cH}. \tag{RG34}$$

For clarity, use the uniformly local norm weighted by $e^{-\tau\,\mathrm{dist}(\cdot,Q)}$, with $0 < \tau < c$. Every forcing difference contains either a vertex outside the protected collar or an inverse walk visiting it. Its weighted norm is at most $C\varepsilon e^{-\tau H}$. The difference of the nonlinear fixed-point equations has Lipschitz constant below one half in this norm. Absorption proves RG34. All complete prefix and scalar-row carriers are included in the distance condition.

This weak difference alone is not a current supremum estimate. Nor may one subtract potentials placed in two unrelated final Coulomb gauges. There is a covariant argument for the curvature component. In the common reference chart, RG34 gives a small connection difference in $W^{1,p}$, hence in supremum, and a small curvature difference in L^p. Each curvature has the protected covariant $W^{1+s,p}$ bound. Transfer both to the connection of W_s using $D_{W_s} - D_{W_1} = [A_s - A_1, \cdot]$. The $W^{1,p}$ multiplier estimates at $p > 8$, $s < 1/p$, bound the transferred difference uniformly in $W^{1+s,p}$. Interpolate to an order t with $4/p < t < 1 + s$, and use the local Sobolev embedding. This gives an exponentially small curvature supremum difference, with a reduced positive exponential rate. For the current use the actual protected equation

$$\begin{aligned} J(W_s) - J(W_1) = {} & - d_{W_s}\omega_s - \mathcal{Q}_s^*(\lambda_s - \lambda_1) \\ & - (\mathcal{Q}_s^* - \mathcal{Q}_1^*)\lambda_1, \qquad \omega_1 = 0. \end{aligned} \tag{RG35}$$

The coarse multiplier difference is small by RG34; on a current-unit coarse mesh its supremum follows with a fixed local inverse-inequality constant. CL17 and CL46 bound the complete-prefix averaging adjoint. Its field-variation bound follows by analytic Cauchy on the same strictly smaller physical supremum chart; it is applied to the already small common-reference connection difference. Finally ω_s is small in L^p by RG34 and uniformly bounded in its own covariant $W^{2+s,p}$ norm by MC5. Interpolate to an order t with $1 + 4/p < t < 2 + s$ to bound $d_{W_s}\omega_s$ in supremum. Consequently, after a fixed shrinkage of the protected collar,

$$\mathrm{dist}_{\text{physical pair,Q}}(W_s, W_1) \le C\varepsilon e^{-c'H}. \tag{RG36}$$

Mixed finer rows retain the scale factors in MC3; they cannot be treated by the current-unit inverse inequality without those factors. RG36 controls the curvature/current conditions of the paired core. If an ordinary potential norm is also imposed in a prescribed frame, use the common controlled gauge supplied by the local small-gauge implicit-function theorem, or check that norm in the literal original response chart. It is not supplied by subtracting the two separate analysis gauges.

If the actual all-one readout has a fixed strict margin in the original paired core, choose H once so that RG36 lies within that margin. The nested-minimum clauses are checked with their original uniform minimum response estimates, not dropped. This approach needs no estimate of the form $C_L L^{-2} < 1$. It uses the actual strict all-one inclusion and exponential insensitivity to remote source walls.

106.12.3 Scope of this classical construction

RG32–RG36 by themselves do not identify a complete marked polymer decomposition. Sections 13–15 below give a concrete current-index filing for the completed pure-clean scalar packet and a remote-residual comparison to its old physical kernels. That construction retains the original inner

Section 106.12 / A source-only physical-target hull and its exact scope

minimum and avoids imposing fractional regularity on the outer weakened response. It is stronger than merely claiming an exponential boundary error, and it does not truncate coarse coordinates in a way that destroys the constraints.

Those later sections still do not identify every mixed-history source slot or every independently formed contact row. Such terms must retain their literal old data, current window, exact anchor shifts, and admitted boundary/live representative. Their full recurrence remains outside the clean scalar conclusion.

Section 106.13 / Carry an old label by a current local minimum

There is a more direct way to obtain the protected fractional regularity required by a clean scalar carrying term. The crucial choice is the **index of the carrying function in the next original source localization**. A birth label and the index of its current local representative are different data.

Fix a smooth current partition member $f\chi_Q$. Choose one cube D_0 of the type actually allowed in I (1.15), (3.25), and Lemma 4, whose protected interior contains Q, all short old carriers used below, and their complete prefix read sets. Shortness means a bound $D_{\mathrm{rel}} \le cH/r$ on the complete relative carrier/path/source-offset lengths, with a fixed sufficiently small c. This condition is translation and reflection covariant; it is not a root-dependent mask that tests containment in an absolute window. Since the test support is in Q, the condition ensures that every root reindexing and path used by CD stays inside the common protected hull. Choose the original block/collar parameter once sufficiently large for this fixed geometry. An arbitrary new boundary problem is not being added to the original domain. Write

$$P_k^{D_0}(U) = U_k(D_0, M'(U)), \qquad \Pi_k^{D_0}(U) = (P_k^{D_0}(U), J_{\mathrm{phys}}(P_k^{D_0}(U))). \tag{RG37}$$

The notation M' includes the actual determining hierarchy and the literal prescribed boundary data. All these variables are included in the carrier D_0.

Let $R_{j,X}$ be an old local paired coefficient, with its engineering factors, source anchor and birth label j retained. For the completed short-carrier part of the packet define

$$T_{j,k}^{Q}(U, J; f) = \sum_{X \text{ short}} R_{j,X}\big(\Pi_k^{D_0}(U); f\chi_Q\big), \qquad j \le k. \tag{RG38}$$

The scale conversions in this formula are the original physical conversion to birth units; in particular a current curvature and current become r^2F and r^3J, where $r = L^{j-k}$. The integrated old-root count and dimension-six normalization are the same $r^{-4}r^{-2}$ as in MT39. Formula RG38 is not a redefinition of those factors.

This is a local **current-index** k paired function on D_0, and it may ignore the independently supplied J. I (1.15)–(1.16) gives the physical reconstructed pair on this allowed domain. The nested clauses give its admission in each old pair core occurring in RG38. The complete source packet uses one and the same $P_k^{D_0}$; a different background for each old root would invalidate the root pairing in CD.

On an actual current minimum $U_k(V)$, restriction to D_0 with its actual traces is again that same constrained minimum. Uniqueness on the original branch gives $P_k^{D_0}(U_k(V)) = U_k(V)$ on the protected read sets. Thus RG38 agrees there with the old source, as a function of **all** retained coarse variables V. This is an identity of the original graph, not merely agreement at one flat field. LM46a therefore applies to its physical derivatives and retained-variable contact distributions. An arbitrary fine Haar derivative still uses the full XP pushforward rule.

At the next formation, apply the original I/II localization to the current function $T^{Q}_{j,k}$ at index k. Its inner field is then a genuine k-level minimum on every source wall. The outer weakened field $H_{k,s}$ only supplies its target and prescribed exterior. The old label j is not restarted. This is the reason the following current-unit regularity lemma applies even though the outer off-graph response itself need not have a fractional current bound.

106.13.1 The multiplier of the original current minimum

Let $W = P_k^{D_0}(U)$ on an admitted source wall. In physical tangent coordinates its true Wilson equation is

$$J_{D_0}(W) + Q_{\mathrm{raw,k}}(W)^{\mathsf{t}}\lambda_k = 0. \tag{RG39}$$

Use the actual action graph, with the original plaquettes and boundary convention. The sign of λ_k is fixed by RG39.

For a retained determining coordinate c, let $H_{k,c}$ be the derivative of this original minimum when that coordinate is varied and all other determining and prescribed data are held fixed. Target preservation gives $Q_{\mathrm{raw,k}}H_{k,c} = e_c$, with the actual logarithmic endpoint factors included. Consequently

$$\langle \lambda_k, e_c\rangle_k = -\langle J_{D_0}(W), H_{k,c}\rangle_a. \tag{RG40}$$

For a current-unit row, the coarse volume factor in this identity is one. Proposition 9 (190) bounds the right side uniformly. More explicitly, in an adaptive cell of length ℓ, the original regular bounds give $J = O(\ell^{-3}\epsilon)$, and the response to a unit-depth coarse coordinate is $H_{k,c} = O(\ell^{-1}e^{-cd_{\mathrm{ad}}})$. The physical fine volume of the cell is $O(\ell^4)$. These three powers cancel before summing the original exponential cell majorant. The resulting estimate is

$$\|\lambda_k\|_{\ell^\infty(\text{current-unit rows})} + \|\lambda_k\|_{\ell^p_{\mathrm{uloc}}} \le C\epsilon. \tag{RG41}$$

This argument includes the literal fine bare collar through the adapted weights. It does not assume an unscaled uniform current supremum on that collar. A row of another assigned depth keeps its corresponding length factors. The column-sum majorant used here is the one in the original response theorem; it is not a count of all fine bonds in the physical volume.

The raw-to-compatible step is now legitimate. Fine gauge invariance of the actual action and the exact coarse gauge covariance of the raw mean imply the coarse Ward equation $\delta_{V_k}\lambda_k = 0$ on the protected current node rows. Surjectivity of the actual node gauge variations gives this equation before any estimate. At the reference W, the difference between the compatible differential A_k and $Q_{\mathrm{raw,k}}$ is the derivative of an endpoint gauge, of the form $d_{V_k}\rho$, with its convention-dependent sign. Therefore

$$A_k(W)^{\mathsf{t}}\lambda_k = Q_{\mathrm{raw,k}}(W)^{\mathsf{t}}\lambda_k. \tag{RG42}$$

No such identity is asserted for an arbitrary negative-Sobolev force or for a multiplier not obeying this Ward row. This is precisely the distinction required by the raw-row obstruction FI.

There is also an independent proof of RG41 after this Ward conversion. Regard the actual regular W as the reference of the already established full MB five-type inverse. Its linear matrix sends $(0,0,0,0,\lambda_k)$ exactly to $(-J(W),0,0,0,0)$. The proper adapted weak-current norm is bounded by $C\epsilon$. Applying the inverse therefore bounds the multiplier in its assigned-depth coarse ℓ^p row, and the current-unit local inverse inequality gives its supremum. This argument uses the same W and its actual compatible row; it neither replaces the minimum nor applies the raw row to a general negative force.

106.13.2 Fractional regularity of this original inner field

Apply CL17–18 to the complete fine-to-current prefix, and CL46 if a single admitted top transition is present. On a protected region of current retained depth one, RG41–42 give

$$\|J(W)\|_{W^{s,p}} \le C\|\lambda_k\|_{\ell^p_{\mathrm{uloc}}} \le C\epsilon, \qquad p > 8, \quad 0 < s < 1/p. \tag{RG43}$$

The fractional restriction is real: a held coarse multiplier is not assigned an arbitrarily high spatial Sobolev norm. The row sums and fractional estimate include the entire actual prefix and its endpoint convention.

The original inner minimum already has the admitted lower regularity and current/curvature supremum. Put it in the controlled local Coulomb analysis chart on a smaller protected collar. The Wilson expansion and interior elliptic bootstrap in MC16–18, now with auxiliary scalar $\omega = 0$, give

$$\|A_W\|_{W^{2+s,p}} + |F(W)\|_{W^{1+s,p}} + \|A_W\|_{C^{1,\beta}} + \|J(W)\|_{\infty} \le C\epsilon, \qquad s < \beta < 1 + s - 4/p. \tag{RG44}$$

The higher norms here belong to that controlled analysis chart, or their declared covariant versions. They are not transported as ordinary derivatives through a rough tree gauge. The constants depend on the once-fixed protected collar and original regular parameters, and not on microscopic depth.

This proves the needed fractional estimate for the **actual current inner minimum**. An independently varied outer J enters the finite current targets or remote prescribed data of this problem; it does not appear as an arbitrary fine current on the right side of its interior Euler equation. Finer mixed rows adjacent to a live component are outside this uniform current-depth conclusion unless their own-scale factors and protected geometry are separately checked.

Section 107.1 / Hypotheses and conclusions

This note continues the marked part particularly O1–O2 of its analysis.
It does not assert the missing all-depth marked recurrence. It proves several of its inputs on actual Wilson variables:

1. a joint polynomial-energy bound for the dimension-six compact primitives and their finite canonical Haar jets, including the changes of field degree made by contact differentiation;
2. a local discrete div–curl estimate that controls the derivative primitive on an entire protected collar from curvature and the **actual** Wilson current, also on a common admitted complex collar;
3. an explicit payment of fixed engineering, radius, and contact costs along an actual live history by the original R-II birth/renewal bank;
4. preservation of one positive reserve through a finite source coefficient, and the resulting bounds at the first complete healing of the compact sector;
5. a precise distinction between the resulting R birth's joint analytic jets and the still-needed physical graded jets and stable feedback estimate.

The uncovered cases are listed in Section 8. In particular, a low-regularity range theorem for a prepared unmarked coefficient is not silently extended to every derivative insertion on every marked wall. A common domain is needed before a norm on that domain is used.

Section 107.1 / Hypotheses and conclusions

All references to S, MT, HI, XP and AI are to the argument. References to numbered sections of the stated results are explicitly identified. The only new external input scope here is the original source-zero R-II construction; none of the marked inequalities below is attributed to that paper.

Section 107.2 / Compact primitive bounds with their physical dimensions intact

Let the actual stored real group variables live on a lattice of spacing ℓ. For a fixed finite carrier Y, enlarged once to include every plaquette, transport and difference used below, put

$$e_Y(U) = \sum_{p \subset Y^+} s(P_p), \qquad \mathscr{E}_Y = \beta\, e_Y, \qquad c_- g^{-2} \le \beta \le c_+ g^{-2}. \tag{LM1}$$

Here $s(P) = N - \operatorname{Re}\operatorname{tr} P$. The constants in $\beta \asymp g^{-2}$ are fixed convention constants. The proof uses the positive energy of these same plaquettes on these same stored variables. It does not borrow an energy assigned to a different reconstructed field.

Unitarity and orthogonal projection give

$$\|P - I\|_F^2 = 2s(P), \qquad \|X_{\mu\nu}\| \le C\ell^{-2} e_Y^{1/2}, \qquad \|D_\rho X_{\mu\nu}\| \le C\ell^{-3} e_Y^{1/2}. \tag{LM2}$$

The last estimate is just the difference of two unitarily transported fluxes. It requires no regularity or small plaquette assumption. Consequently the local integrated complementary derivative insertion of AI33–35 and the exact cubic completion of AI32 satisfy

$$\begin{aligned} |H_{\ell,Y}| &\le C\ell^{-2} e_Y, \\ |C^\sharp_{\ell,Y}| &\le C\ell^{-2} e_Y^{3/2}. \end{aligned} \tag{LM3}$$

The second line includes the cube defect, its transport correction, and the divergence in AI32b. Indeed the cube defect contains at least two factors $P - I$; multiplication by a flux supplies the third factor; a microscopic difference costs ℓ^{-1}. Higher products of plaquette defects are bounded by a constant times the cubic power on this fixed compact carrier. Thus the local dimensionless canonical packets

$$M_{2,Y} = \ell^2 g^{-2} H_{\ell,Y}, \qquad M_{3,Y} = \ell^2 g^{-3} C^\sharp_{\ell,Y} \tag{LM4}$$

obey

$$|M_{d,Y}| \le C_d \mathscr{E}_Y^{d/2}, \qquad d = 2, 3. \tag{LM5}$$

This proves the incoming polynomial-energy envelope for these actual own-spacing primitives. The factor ℓ^{-2} in LM3 has not disappeared: LM4 records exactly where it was placed.

There is also a useful unconditional fallback for a compact slot whose Wilson action was already canceled in an exact reference comparison. Since $e_Y \le C_Y$,

$$|M_{d,Y}| \le C_Y g^{-d}. \tag{LM5a}$$

This is a supremum estimate on its actual compact variables and uses no retained energy. The polynomial $g^{-d} = x^{d/2}$ is charged once to that stored record's actual birth/renewal bank. Its physical spacing still appears as in LM4 and LM20. Thus an energy from an unrelated field is unnecessary even in this case. The fallback does not control an old regular coefficient entering a new component after an uncontrolled long regular age.

If a primitive is summed with a bounded physical test over a larger set, apply LM3 at each root. For a fixed number of insertion roots, enlarge their joint carrier and keep their common energy. For an arbitrary extensive sum, retain the rooted sum and its explicit cell count; replacing it by one fixed-carrier constant would be incorrect.

107.2.1 Finite Haar jets and their lowered field degree

Write X_b^A for the left bond derivative of AI3 and $Y_b^A = gX_b^A$ for its canonical version. For any fixed ordered list $\boldsymbol{b}$ of m such derivatives, on an enlarged but still fixed carrier,

$$|Y_{\boldsymbol{b}} M_{d,Y}| \le C_{d,m} \begin{cases} \mathscr{E}_{Y^+}^{(d-m)/2}, & 0 \le m \le d, \\ g^{m-d}, & m \ge d. \end{cases} \tag{LM6}$$

In particular these are uniform polynomial-energy bounds for $0 < g \le 1$. The statements for $m = d$ agree.

Proof. Expand the compact observables as a finite sum of words in plaquette defects, inverse plaquette defects, and compact transports. For H, each word has at least two defect factors. For $C^\sharp$, AI24–27 and AI32b give at least three. A bond derivative can remove at most one of those vanishing factors. Derivatives of transports and of any remaining factors have bounded matrix norm on the compact group. After $m \le d$ derivatives, every term therefore contains at least $d - m$ defect factors and is bounded by $Ce_{Y^+}^{(d-m)/2}$. Extra defect factors are harmless on a fixed compact carrier. For $m \ge d$ all derivatives are bounded by a fixed constant. Multiplication by g^{m-d} and LM1 prove the assertion. The proof concerns these explicit compact words. A general function with zero jets at the trivial configuration would not follow from this argument. □

LM6 is useful precisely because it does **not** preserve the original degree after differentiation. A first derivative of a cubic packet may have degree two; a second derivative of a quadratic packet may have degree zero. These are the lower/contact rows already retained by XP and AI. They cannot be assigned degree three or two merely because the original source label had that degree.

107.2.2 Wilson score and exact EOM packets

For the unnormalized Wilson density $\rho = e^{-\beta S_0}$ with Haar reference measure, the adjoint of a constant canonical bond vector field is

$$\mathsf{D}_{\rho,Y}^\dagger F = -YF + \sigma_Y F, \qquad \sigma_Y = Y(\beta S_0). \tag{LM7}$$

It uses the original source-zero action. Locality of a bond derivative and the first-order vanishing of the Wilson force give

$$|\sigma_Y| \le C\mathscr{E}_{\mathrm{star}(b)}^{1/2}, \qquad |Y_1 \cdots Y_m \sigma_Y| \le C_m \quad (m \ge 1). \tag{LM8}$$

For the second inequality, $m + 1$ ordinary derivatives of the local Wilson action are bounded; the prefactor is $\beta g^{m+1} = O(g^{m-1})$. Hence an ordered packet of n adjoints applied to one of LM4 obeys

$$\left| \mathsf{D}_{\rho,Y_n}^\dagger \cdots \mathsf{D}_{\rho,Y_1}^\dagger M_{d,Y} \right| \le C_{d,n}(1 + \mathscr{E}_{Y^{+n}})^{(d+n)/2}. \tag{LM9}$$

Here Y^{+n} denotes the complete union of the original carrier, all the differentiated bond stars, and their finite transport stencils. If a chosen bond is distant, its star is retained in that union; it is not silently placed in a fixed geometric neighborhood of the old root. To prove LM9, expand the ordered product by the Leibniz rule. Each term is a derivative of the compact packet multiplied by derivatives of local scores. LM6 and LM8 bound each term; there are finitely many terms at a fixed order. This expansion also preserves their exact signs and contact supports. A variable vector-field coefficient is included by retaining its derivatives as additional factors, with their own bounds; it is not covered by calling the vector field constant.

Section 107.2 / Compact primitive bounds with their physical dimensions intact

These statements settle finite primitive Haar/EOM packets on the original compact Wilson variables. They do not establish bounds for derivatives of a nonlinear reconstruction or for derivatives of an individual sharp selector. Those cases are treated separately in Sections 3 and 7.

Section 107.4 / Rough crossings cannot be silently put in the protected case

There are three different uses of a primitive.

Section 107.4 / Rough crossings cannot be silently put in the protected case

Actual slot	Estimate available here	Scale and domain that must be retained
Real own-spacing stored compact links	LM3–9, without small-field assumptions	Their own spacing and the positive Wilson energy assigned to those same variables
A reconstruction with a complete admitted protected collar	LM12 and LM17–19	The reconstruction layer ℓ, actual current, and common complex collar
A stencil crossing a genuinely rough interface without such a collar	Only its literal formula, or LM3 if its full stencil is actually stored at one compact spacing	No replacement by a regular-layer estimate is justified

The distinction has a direct quantitative cause. A jump of size $O(\epsilon\ell^{-2})$ in the flux across one fine edge costs $O(a^{-1}\epsilon\ell^{-2})$ in its difference. The scalar curvature bound and the fine plaquette count do not cancel that a^{-1}. If the entire interface is represented by actual fine compact records, LM3 charges their fine spacing. If it is a synthetic interface between reconstructed layers, an estimate on its actual current or a valid collar decomposition is required.

Even a uniform Hölder curvature bound with exponent less than one does not supply the missing estimate. In a Cartan direction on a unit-scale periodic window take $A_2(x) = ca^{1+\zeta}(-1)^{x_1/a}T$, with other components zero and $0 < \zeta < 1$. Its curvature has size $O(a^{\zeta})$ and bounded discrete C^{ζ} seminorm, whereas its directional current has size $O(a^{\zeta-1})$. The integrated derivative square diverges like $a^{2\zeta-2}$. This is a test of the proposed implication from a weak norm, not a counterexample to an actual constrained minimizer. An actual minimizer can avoid it precisely through its physical Euler/current equation and a valid interior collar.

107.5.1 The scale conversion is an explicit factor

For an integrated operator of engineering dimension D, define its dimensionless own-scale amplitude with the factor ℓ_j^{D-4}. If its physical insertion is carried without being integrated or renormalized from ℓ_j to $\ell_k = L^{k-j}\ell_j$, reexpressing that same insertion in current dimensionless units multiplies it by

$$L^{(D-4)(k-j)}. \tag{LM20}$$

For LM3 this is $L^{2(k-j)}$. This formula holds while keeping the birth coupling normalization fixed. If it too is changed from g_j^{-d} to g_k^{-d}, the additional exact factor is $(g_j/g_k)^d$. A different ratio arising from a canonical field readout is a different operation and is not conflated with this one.

For finitely many marks in one joint component, the worst engineering growth is $\exp(c_* \sum_s n_s)$, where n_s is the number of carried marks at that actual live step and c_* depends on their finite maximum dimension. It is a per-step geometry charge, not a repeated use of an old Wilson or Gaussian energy. A fixed polynomial Cauchy cost α_j^{-q}, with $\alpha_j \asymp g_j(\log x_j)^r$, has logarithm at most $C_q \log x_j$ and is charged once at its actual birth.

107.5.2 What the primary R-II calculation actually supplies

The published source has more structure than a small final value of $T_k 1$. Before the last summation it retains a factor $\exp[-\kappa_j(Z) - 2p_0(g_{j(Z)})]$, with an original birth index $j(Z)$. Its no-new-failure evolution is the geometric enlargement S; the future waiting bank is (1.80), and (1.81) bounds its total cost. New failed tests and component mergers are separate updates, (1.83)–(1.88). The final mass estimate is obtained only afterwards. These facts are used here at their source-zero values. [R-II, printed pp.384–387, (1.79)–(1.89).]

For clarity use the notation marked bridges Section 8.3, where $P_0(x) = (\log x)^{p_0}$, $R_j \asymp (\log x_j)^{r_0}$, and $d'_j(Z)$ is the original rescaled component size. The original geometric argument bounds the future sum of a local charge

$$w_j(Z) = C_w M^4 R_j^5 (d'_j(Z) + 1) \tag{LM21}$$

by

$$\sum_{n=j+1}^{j+K_j(Z)} w_n(S^{n-j}Z) \le C C_w M^4 L^5 R_j^6 (d'_j(Z) + 1). \tag{LM22}$$

This is the dimension-four form of that numerical argument; the added $+1$ pays the final bounded-size waiting period. In particular, an extra fixed cost per live step is included by increasing C_w once.

The strict hierarchy

$$p_0 > 6r_0, \qquad 2p_1 - 9r_0 > p_0 \tag{LM23}$$

pays the resulting birth bank, renewal bank, and merger overlap cost after a fixed fraction of the positive action is set aside. This is the same comparison of powers as Section 8.3. The primary text explicitly retains the strict renewal comparison $2p_1 - (d+5)r_0 > p_0$ before weakening the failed-test factor.

Thus any **fixed finite** collection of the engineering factors LM20, polynomial birth-radius costs, and finite compact-jet costs can be paid in this numerical induction. This conclusion has two qualifications. First, it applies to steps for which the marked object is actually in the live component governed by that bank. It does not pay an arbitrarily long regular age before the object enters that component. Second, the constants may depend on the finite source order and finite operator list. No all-orders growth theorem is obtained by hiding that dependence.

107.5.3 The three updates preserve the ownership rule

In a no-new-failure step, subtract the enlarged local charge LM21 from the existing future bank. At a renewal, use its new failed-test credit for the new future bank and retain one birth reserve. At a merger, union the actual records and marked atoms, add child banks and new failure credits, subtract the new local charge and keep one reserve for the joint component. The same overlap estimate used by R-II pays the extra future charge; a tree of binary merges handles more components.

The positive energy of a stored record is not spent again at a later age. The additional engineering payment is in the numerical waiting bank. Two marks on overlapping records do not receive two copies of a rare mass. Identical action-completion and inverse-completion atoms are canceled before absolute values. The old deterministic zero-fluctuation piece keeps its tag; only its actual proper difference is a new input. These are precisely the exact transport rules S59–62 and XP, with the extra finite costs now displayed.

107.5.4 An older compact origin is not reset at a new failure

For a compact mark originally stored at scale j, keep both $x_j^{d/2}$ from LM5a and $L^{(D-4)(k-j)}$ from LM20 until its actual lineage has been identified. If that variable remained unintegrated, its earlier live history and its earlier banks remain part of the joint record. A new failure or merger does not change the old origin coupling or authorize a second independent rare factor. If intervening regular integrations removed the variable, the surviving source is an inherited coefficient and must be treated by L2, rather than by a compact bound on a variable that is no longer stored.

There is a simple diagnostic for a wrong payment. With x_k fixed and $x_j \asymp x_k + c(k-j)$,

$$x_j^q L^{\nu(k-j)} e^{-c' P_0(x_k)} \longrightarrow \infty \qquad (j \text{ recedes}). \tag{LM23a}$$

Thus the final component's current-coupling rarity alone cannot pay these factors. Nor does $e^{-c' P_0(x_j)}$ alone dominate arbitrary exponential dwell. The future waiting bank, a large initial spatial size, or the intervening renewal credits are the indispensable additional parts of the original R-II calculation. Section 4.2 uses those parts; it does not replace them by a final scalar rare bound.

The new results do not close the stable marked-birth feedback row They isolate its live requirements more sharply.

107.9.1 L1. Protected coverage at every primitive slot

For every actual reconstructed primitive or source-wall term, prove that its full readset has either an own-spacing compact representation or a common protected collar satisfying LM10 and LM17. This includes the actual current bound and the simultaneous original and source walls. LM12 removes a higher-curvature-derivative requirement but not this coverage requirement. A rough crossing lacking both descriptions remains unbounded by the present estimates. Section 9 verifies the actual-current and curvature inputs for the original completed regular inner branch, including its source walls and a radius-restricted healed R readout.

107.9.2 L2. Inherited marked coefficients entering a live component

Given a previously generated coefficient with its original radius and complete contact module, prove on the actual prepared readout either its joint envelope LM24 or its already-known analytic norm bound. R-II (1.65) does place **its original effective-action class** in its prepared domains. To apply it to a modified marked class one must identify the marked coefficient's complete argument and birth domain with those variables. One cannot infer this merely from gauge invariance or from having removed finitely many field jets.

In addition, a coefficient with a long regular age before live entry needs the regular graded age estimate at that entry. The future live bank pays LM20 during live residence; it does not erase an earlier uncontrolled factor $L^{2(s-j)}g_j^{-q}$. This is exactly where a claimed closure based on small final rarity alone would be circular.

107.9.3 L3. Physical jets of the healed output and marked feedback

Section 9 transfers LM27 to physical coarse jets in operator norm on a smaller explicit R birth ball for the original completed regular branch. The needed completed kernel and source-anchor moments, the full quadratic covariant completion, and all lower rows must still be controlled. Together they must

give the quantitative incoming class needed for the one-interaction attachment

$$\left\| \mathsf{P}_6 \mathcal{L}^{(2)}_{V_s}(R_j \circ \Phi_{j,s}, V_s) \right\| \le C g_s \, W_{s,j} \, \|R_j\|_{\mathrm{graded}}, \tag{LM30}$$

with the actual recent-birth radii in $W_{s,j}$, the proper-difference factor on an admitted complex disc, and a summable age kernel. LM25 controls the integral once this incoming bound holds. It does not prove LM30 for an unknown generated coefficient.

107.9.4 L4. Nonlinear cutoff/current/contact packets

For every whole-kernel distribution created by the exact XP transport, bound the derivatives on the right side of LM28 after pullback through the actual nonlinear maps, including the source-zero effective-action score and the tangent physical-current lift. The compact initial packets satisfy LM9 and LM29. An arbitrary inherited pullback does not follow from those initial estimates.

These are analytical obligations, not alternative definitions of the same Wilson measure. The finite-source algebra, exact compensation pairs, first-healing ownership rule, and contact multiplication formulas remain unchanged. No source-zero intertorus comparison is proved by this note; that separate argument obligation remains as stated there.

Section 107.11 / The actual current response

The current row used in RG13 can be checked directly from the Wilson force. This is useful because a first-derivative bound for a connection alone does not bound the first derivative of its current. Work in one complete chart and current stencil, in units where the retained spacing is one, and let the fine spacing be $a \le 1$. Suppose $U_b = \exp(aA_b)$, with A in a fixed small complex C^1 ball. Fixed factors of i depend on the Hermitian convention and do not affect the norm statement. The analytic plaquette expansion has the form

$$X_a(A) = d_a A + [A_\mu, A_\nu] + a\mathcal{R}_a(A, \partial_a A), \tag{LM38}$$

where shifted factors in the quadratic term have been replaced by their values at the base point. Their differences contain $a\partial_a A$, and so belong to the remainder. On this chart $\mathcal{R}_a$ and its first field derivative are uniformly bounded from C^1 to C^0. This follows by Taylor expansion of the finite plaquette word:

division by a^2 leaves the linear curl, the quadratic commutator, and an analytic remainder with an explicit factor a. There is no sum over a growing set of bonds in this calculation.

The adjoint incidence defining the force is the flat incidence adjoint plus bounded connection and shift operators. The nonlinear Wilson force therefore satisfies

$$J_a(A) = d_a^* d_a A + \mathcal{N}_a(A), \qquad \|D\mathcal{N}_a(A)[H]\|_\infty \le C(\|\partial_a H\|_\infty + \|H\|_\infty). \tag{LM39}$$

For the remainder in LM38 the potentially troublesome divergence is bounded by

$$\|a\partial_a^* D\mathcal{R}_a(A)[H]\|_\infty \le 2\|D\mathcal{R}_a(A)[H]\|_\infty \tag{LM40}$$

per orientation. For the commutator term, the discrete Leibniz rule uses only first differences of A and H. The bounded connection corrections have the same property. This proves LM39, and hence

$$\|D_A J_a(A)[H]\|_\infty \le C(\|d_a^* d_a H\|_\infty + \|\partial_a H\|_\infty + \|H\|_\infty). \tag{LM41}$$

The original variational response is written in a relative chart, $U = \exp(aH_0)\exp(aA_0) = \exp(aA)$. The local analytic Baker–Campbell–Hausdorff map gives $A = A_0 + H_0 + aS_a(A_0, H_0)$. Its derivative is $h_A = h_0 + aT_a(A_0, H_0)h_0$. The second differences of the extra term are harmless here: on a lattice $a\|\partial_a^2 h\|_\infty \le 2\|\partial_a h\|_\infty$, and the same inequality applies to the coefficients of T_a. The discrete product rule therefore bounds the chart correction by $C(\|\partial_a h_0\|_\infty + \|h_0\|_\infty)$. Passing from the covariant Maxwell row at U_0 to the flat one has only the same first-order connection corrections. Proposition 9 and (190) of the 1985 variational paper retain both the covariant Maxwell and Laplacian rows needed for this step. Thus their exponentially localized response also controls the derivative of the **actual** Wilson current.

This verification is valid in the common complex chart because it uses absolute matrix norms and analytic finite words, not positivity of a complex bilinear form. It proves the current derivative used in the coarse Cauchy argument on an admitted protected collar. It does not produce that collar, extend a minimum across a rough boundary, or give a fractional derivative bound from a generic independent current in L^∞.

Section 107.12 / First entry of an old regular mark: exact coordinates and the residual gate

There is a useful distinction between an old compact source and an old regular coefficient. In the latter case the fine variable on which the original source was evaluated may already have been integrated out. Its current representative must then be used. Restoring a bound on the original fine primitive would reverse an exact integration and lose the regular age gain.

Let j be the birth scale and k the first step at which the current complete marked carrier enters a live component. At the unweakened physical endpoint, HI and XP give the deterministic current representative

$$T_{j,k}(V) = R_j\big(U_j(U_k(V)), J_{\mathrm{phys}}(U_j(U_k(V)))\big), \tag{LM42}$$

together with its separately born proper differences and lower/contact rows. The notation refers to the original nested determining constraints and the same Wilson action, with all fixed exterior data retained. Equation LM42 is not asserted for an arbitrary independently weakened pair (U, J). If the birth has already had field jets removed, every removed exact projection and its free companion remain in the state.

107.12.1 What the original large-field map actually retains

R-II (1.50), printed p.370, writes the old minimum in terms of the old determining set, retained V'', and the actual fluctuation $g_k B$. Outside the modified interior the determining data agree with the corresponding old ones. Its subsequent expansion (1.51) first separates the zero-B value. Equations (1.59)–(1.60) retain precisely this deterministic-plus-difference organization for the original member functions. Thus there is no algebraic need to replace LM42 by a newly sampled reference variable.

The old selectors also remain present. In (1.70) they are still χ_k, χ; printed p.378 removes them on a branch only after the new small-field tests imply they are identically one. In the remaining large-field branches they are retained with the corresponding old gauge constraints. In particular, the new failure flag alone does not erase every old regularity condition.

These facts identify a legitimate place to apply a **previously proved** regular age bound to LM42: before the joint live integration, on its actual retained current variables. They do not by themselves establish the bound on every complex localized branch. The original range statement (1.65) maps to the mixed class U'_k defined in IV, and the explanatory paragraph explicitly invokes the older determining set's weaker regularity conditions on the changed layer. It guarantees the domain of the original effective-action members. It is not an identity with a smaller physical and canonical marked core.

107.12.2 A rigorous restricted carrying sector

Suppose a complete current representative $T_{j,k,I}(\xi)$ depends only on the variables actually integrated or still retained in the one joint live record ξ, and not on its outer complex pair. Suppose that on the exact real selector support a regular estimate has already been proved in the form

$$|T_{j,k,I}(\xi)| \le A_{j,I} W_{k,j} (1 + \mathscr{E}_\omega(\xi))^{D_I/2}. \tag{LM43}$$

Then this old mark can be carried through that joint R operation without an additional analytic source-wall extension of $T_{j,k,I}$. The R-II kernel is holomorphic in its outer pair, the carrying function is unchanged, and its same-kernel absolute majorant with an unused energy reserve gives

$$\left| \int K_\omega(U, J; d\xi) T_{j,k,I}(\xi) \right| \le B_\omega A_{j,I} W_{k,j} C_\theta(D_I). \tag{LM44}$$

For a joint finite-source coefficient, replace the right side by the partition sum LM25 with the factors $A_{j,I} W_{k,j}$. There is one B_ω for the joint integral. Dominated holomorphic integration proves outer analyticity. This is exactly the arbitrary-carrying-function case of R-II (1.73), with an energy majorant inserted once. No mass of an individual sourced component is inverted.

The assumption that the carrying function is independent of the outer pair is substantive. A localized old physical pullback generally depends on determining variables outside a live component. Even if its original fine carrier meets that component, its complete coarse-response and prefix readset need not lie inside the retained integration. Such a mark does not satisfy this restricted lemma merely by being assigned a live label. Enlarging its carrier records those dependencies; it does not integrate those variables or create a rare factor for them.

The same point applies to contacts. For a whole-kernel derivative, the weak identity LM28 transfers the derivative to the full test. If its finite ordered derivatives through order m obey LM43 with the corresponding lowered grades and finite constants, LM44 holds for those weak contact rows as well. Original source-degree truncation and the exact XP partition/contact coefficients are unchanged. This statement does not estimate the nonlinear map derivatives which occur in a straddling pullback.

107.12.3 A source-only reconstruction and its exact endpoint

For a fixed actual old action graph let q_{old} be its entire determining map, including its bare boundary data, and let $P_{\mathrm{old}}(v)$ be its admitted Wilson minimum. Consider the source-only replacement

$$\widehat{\Gamma}_{\omega,\mathbf{s}} = P_{\mathrm{old}}(q_{\mathrm{old}}(\Gamma_{\omega,\mathbf{s}})), \qquad \widehat{J}_{\omega,\mathbf{s}} = J_{\mathrm{phys}}(\widehat{\Gamma}_{\omega,\mathbf{s}}). \tag{LM45}$$

The original kernel and its independent current are left untouched. If the original unweakened old readout is its genuine admitted minimum, uniqueness gives

$$(\widehat{\Gamma}_{\omega,\mathbf{1}}, \widehat{J}_{\omega,\mathbf{1}}) = (\Gamma_{\omega,\mathbf{1}}, J_{\mathrm{phys}}(\Gamma_{\omega,\mathbf{1}})). \tag{LM46}$$

Thus LM45 is an exact possible source extension on every common domain where it exists. It does not change the physical source at all walls one. All root terms in a tested packet must use the same old determining problem and the same reconstructed field; root-dependent local minima cannot be combined as though they were one field in the reflected quadratic pairing.

The endpoint also preserves the complete finite physical differential module. If $v \mapsto \Gamma_{\omega,\mathbf{1}}(v)$ is the actual old minimum on an open admitted set of current variables, the two source functions obtained from LM45 and from the old physical graph are equal as functions of v. Consequently every fixed ordered derivative in those actual variables agrees. For a paired coefficient, the first derivative along that graph is

$$DF(U, J_{\mathrm{phys}}(U))[h, DJ_{\mathrm{phys}}(U)[h]], \tag{LM46a}$$

and its higher derivatives include the corresponding higher physical current chain-rule terms. They are not independent-current partial derivatives. Multiplying the equal physical source functions by the same complete kernel and its same sharp selectors gives equal distributions; all their whole-kernel weak derivatives therefore agree as well. This proves preservation of the existing XP contact identities at the endpoint. It supplies no off-endpoint norm for those derivatives, and it does not discard a selector boundary term.

The missing quantitative assertion for the general first-entry sector is now precise. With $\mathfrak{C}^{\mathrm{spare}}_{j,k}$ denoting the actual physical/canonical domain on which the old graded age estimate was proved, one needs

$$(\widehat{\Gamma}_{\omega,\mathbf{s}}, \widehat{J}_{\omega,\mathbf{s}}) \in \mathfrak{C}^{\mathrm{spare}}_{j,k} \tag{LM47}$$

uniformly on the old real selector support, the full original outer pair domain used at this entry, the simultaneous preparation and source walls, and the complete packet collars. The canonical derivatives must also satisfy the fixed finite grades and anchor sums used in the old norm. LM47 is not a consequence of LM46 alone or of member analyticity in (1.65).

There is no microscopic link-count obstruction to comparing the target data in LM45. Along a path of retained length ℓ, write $U_b = e^{aH_b}\bar{U}_b$ relative to a real unitary reference. Telescoping the product and using $\|H\|_\infty \le \eta/\ell$ gives

$$\|U(\gamma)\bar{U}(\gamma)^{-1} - I\| \le C\eta e^{C\eta}. \tag{LM48}$$

Normalized local averages and their fixed logarithmic charts inherit the same bound. The factors a in each link variation cancel the number $O(\ell/a)$ of links. Thus a verified own-scale first-order perturbation of the actual readout would give a uniform target perturbation. The outstanding problem is establishing that perturbation with a strict margin on every original mixed entry and proving its fractional/current rows where the age norm uses them. LM48 does not supply these hypotheses from a bare small-plaquette test.

Finally, a real retained small-field cutoff is of order $g_k(\log g_k^{-2})^{p_0}$, whereas an R-origin analytic radius may be $g_j(\log g_j^{-2})^{q}$. Their comparison must use the actual old-to-current readout and its age factors.

Section 107.12 / First entry of an old regular mark: exact coordinates and the residual gate

A current failed-test bank cannot pay an earlier unsupported regular age. The delay and one-time origin payment in the feedback-resolvent chapter handle the numerical cost once LM47 and the corresponding transition estimate are known; they do not prove LM47.

Section 108.1 / Scope

The exact current and contact identities XP20–29 can be equipped with a finite-order weak norm. For actual deterministic coarse targets this gives quantitative transport without differentiating a sharp selector. The nonlinear target derivatives are derivatives in field variables, not high spatial derivatives of a physical background. For the original normalized path averages these derivatives have bounds with no microscopic link count.

This does not make an arbitrary selected large-field kernel bounded on the same weak space. Section 6 gives an explicit obstruction to that inference. Section 8 proves a stronger, model-specific bound for the outer regular-pair variables of the original R kernel. In particular, a selected physical EOM coefficient and the derivative of its whole selected kernel are different distributions until the boundary contacts have been included. The distinction prevents a false closure of the all-history current module.

All source labels, original differentiation anchors, full determining readsets, and once-owned energy records are retained. Products of distributional currents are never introduced. A joint source coefficient is formed in the original smooth finite source algebra before its weak representation is taken.

Section 108.4 / Actual normalized path averages have no microscopic jet count

The one-step gauge averages in I (0.4), (0.11)–(0.12) are formed from finite path holonomies, normalized sums over block sites and paths, and analytic group logarithms and exponentials on their admitted domains. For the contour mean defined by I (0.10), the implicit derivative is uniformly invertible in a fixed small neighborhood; its constants do not grow with the number of averaged entries when the equation is normalized by that number.

Fix an output coarse bond. Every underlying contour has bounded repetition multiplicity for each input bond. The multiplicity is a geometric constant for the shortest-coordinate paths in that definition, even if the path contains many fine bonds. For a fixed ordered list of r input bond derivatives, differentiating its holonomy therefore creates at most C_r words. Each undifferentiated transport is unitary on real configurations, and each differentiated factor has a fixed matrix norm. It follows that

$$\|X_{b_r}^{A_r} \cdots X_{b_1}^{A_1} U(\gamma)\| \le C_r. \tag{WCT11}$$

The constant does not contain the total number of links in γ. For controlled complex configurations the same statement has an additional fixed factor bounding all partial transports in the admitted chart. A physical potential bound of order η/ℓ on a path of length $O(\ell)$ gives such a factor $e^{C\eta}$.

Normalized averaging preserves WCT11 by the triangle inequality. The finite chain rule through the actual logarithms, exponentials, and contour mean preserves it on a fixed strictly admitted domain. A fixed input bond belongs to the determining region of only a bounded number of neighboring output bonds. Therefore

$$\Gamma_r(q) \le C_r, \qquad 1 \le r \le m, \tag{WCT12}$$

Section 108.4 / Actual normalized path averages have no microscopic jet count

for this actual one-step target, independently of the ultraviolet mesh and the total volume. The complete prefix and contour readsets are included. A derivative vanishes if its anchor does not occur in that readset.

This proof also applies to a literal raw macro-average if that is the map in a given minimum constraint and it has the same normalized path formula with uniform repetition and output-incidence bounds. It does not identify a composite of nonlinear blocking maps with a different raw macro-average. That identity, when used, must come from the actual constraint construction. Nor does WCT12 assert that a logarithmic average is smooth through a rough configuration where its chart is not admitted. Such a configuration remains a retained compact record until the original construction gives a regular target.

At a mixed boundary, each target and input record keeps its own scale and coupling. WCT11 is still a path statement, but the output-incidence bound and the canonical ratios in WCT8 must be checked for the actual mixed readset. A single input anchor cannot be assigned simultaneously to arbitrarily many ancestral target layers without counting those layers.

Section 108.9 / Exact scope of the weak-contact route

The following pieces are now quantitative: a represented finite weak norm; bounded smooth multiplication with exact ordered contacts; deterministic pushforward with explicit finite target-jet constants;

Section 108.9 / Exact scope of the weak-contact route

ultraviolet-independent one-step jet bounds for the actual admitted normalized path target; and a one-reserve joint energy estimate for the resulting tensor packet. The original normalized R kernel also has the outer regular-pair derivative bound WCT22–24 on its admitted spare core; its original source-zero denominator is included in that estimate. All of these avoid any requirement for arbitrary high spatial derivatives of a reconstructed field. WCT27–28 also control the relative logarithm of that same original single-component reference body as its outer parameters vary, while retaining its absolute base scalar and every actual component count.

Three additional facts are still needed for an all-depth current/contact induction. First, the actual composite or successive target constants must fit the graded age norm; a fixed constant per step is not enough. Second, the internal retained-coordinate and cutoff-boundary rows of a selected R operation need their dual bound or a direct complete-contact estimate. They are not covered by the now-proved outer regular-pair row. Third, the scalar straddling readout must satisfy LM47 on its actual common domain and complete collars. The weak formulation does not replace that scalar domain requirement.

Consequently this chapter closes a deterministic and primitive-contact part of the module. It does not close the full MT36 transition row or turn the Yang-Mills construction into a complete proof.

Section 109 / A noncircular resolvent for marked birth feedback

This chapter addresses the algebraic closure after the actual marked rows have been bounded. It does not supply a missing Wilson-measure row by an abstract hypothesis. Its purpose is to remove a second possible circularity: normalizing an old birth by the inverse of a center transfer matrix before that full transfer matrix has been proved invertible. The hypotheses below are explicit, and their model-specific scope is recorded separately.

Section 109.2 / Uniform closure and invertibility

Proposition FR. For constants in FR2 fixed, there is $g_* > 0$, independent of the number of steps, such that every C_k is invertible. There are constants H, D, also independent of the number of steps, for which

$$\|B_k C_k^{-1}\| \le H g_k, \qquad C_{k+1} = T_k C_k, \qquad \|T_k - I\| \le D g_k \le \tfrac{1}{2}. \tag{FR3}$$

Consequently

$$\|C_n C_m^{-1}\| + \|C_m C_n^{-1}\| \le 2\exp\left(2D \sum_{k=m}^{n-1} g_k\right), \qquad 0 \le m < n. \tag{FR4}$$

In particular, normalization by the full center evolution is legitimate *after* this proof, not as an input to the marked estimate.

Proof. Choose $H = 2b$ when $b > 0$; if $b = 0$ all births vanish and the assertion reduces to a product of $I + E_k$. Put $D = a + 1$ and

$$M = \sum_{n \ge 1} w_* e^{-\sigma n/2} < \infty. \tag{FR5}$$

Choose g_* small enough that

$$2Dg_* \le \sigma/2, \qquad Dg_* \le 1/2, \qquad dg_* M \le 1/2, \qquad fHg_* M \le 1. \tag{FR6}$$

The zero constants in these inequalities impose no restriction.

At $k = 0$, $B_0 = J_0$ and $T_0 = I + E_0$, so FR3 holds. Suppose it has been proved through the preceding step and that C_k is invertible. For $j < k$ the already constructed inverse factors give

$$\|C_j C_k^{-1}\| \le \prod_{\ell=j}^{k-1} (1 - Dg_\ell)^{-1} \le e^{2Dg_*(k-j)}. \tag{FR7}$$

This uses only previously proved factors. Define $H_j = B_j C_j^{-1}$. Multiplying the first equation of FR1 on the right by C_k^{-1}, then using FR2, FR5–7, gives

$$\begin{aligned} \|B_k C_k^{-1}\| &\le b g_k + d g_k \sum_{j<k} w_{k-j}\, H g_j \|C_j C_k^{-1}\| \\ &\le b g_k + d H M g_* g_k \le H g_k. \end{aligned} \tag{FR8}$$

The second equation defines

$$T_k = I + E_k + \sum_{j<k} F_{k,j} H_j C_j C_k^{-1}. \tag{FR9}$$

Its last sum has norm at most $f H M g_* g_k$, so $\|T_k - I\| \le D g_k$. The Neumann bound proves that T_k and then $C_{k+1} = T_k C_k$ are invertible. This completes the induction. Products of the factors and their inverses give FR4. Neither the matrices nor the feedback maps need commute. □

109.2.1 The feedback itself is quadratic in the current coupling

If the actual running tube additionally gives $g_j \le c_g g_k$ for $j < k$, FR9 has the stronger estimate

$$\|T_k - I - E_k\| \le f H c_g M g_k^2. \tag{FR10}$$

Indeed use that inequality in the sum in FR8–9 instead of $g_j \le g_*$. The bounded partial-sum lower envelope $x_j \ge x_k + c(k-j) - C_0$, with $x_k \ge h > C_0$, implies $c_g = (1 - C_0/h)^{-1/2}$. It is not necessary to assume monotonicity of every individual step. This does not improve a primitive $E_k = O(g_k)$ to $O(g_k^2)$; it concerns the stable feedback term alone.

109.2.2 A bounded initial stable packet

A primitive insertion can contain an initial stable lattice remainder whose graded norm is bounded but not $O(g_0)$. Suppose this packet is $R_{-1} : C \to B_{-1}$, with norm at most r_*, and its contributions at step k are bounded by $d g_k w_{k+1} r_*$ and $f g_k w_{k+1} r_*$, respectively. Include those terms in FR1. The same induction works after increasing H, D by constants depending on r_* and decreasing g_*. For example put $W_0 = w_* e^{-\sigma}$, $H = 2(b + d r_* W_0)$, and $D = a + 1 + f r_* W_0$, and then impose FR6 with these constants. The case $H = 0$ is treated directly, as above. The inverse factor between its initial center and step k costs at most $e^{2Dg_* k}$, which is paid by w_{k+1}. New births still satisfy $\|B_k C_k^{-1}\| \le H g_k$. The center receives an additional $O(g_k e^{-\sigma k/2})$ transient. This term is not incorrectly assigned an extra factor g_k.

Section 109.3 / What this does and does not settle for

FR proves that a correctly established, uniformly graded MT36 row is sufficient to close the associated first-source memory and normalize its full center evolution. No assumption about an already normalized center ratio is needed. Its proof does not follow merely from the primitive columns MT32–33: those provide the E_k entry, not all of FR2.

The actual J, K, F must be formed after exact source-zero subtraction, the exact deterministic-tag split, and the correct physical lower/current/ contact filing. In particular the exact free dimension-four family may not be replaced by its nearest-neighbor representative. A selected history with missing translation phases does not satisfy a symmetry hypothesis by being placed in a Banach space.

Section 109.3 / What this does and does not settle for

The spaces B_j permit different birth radii. They do not remove those radii from a proof of FR2. If an R-origin packet has an inverse radius δ_j^{-1}, its cost must first be paid by its actual origin-tagged rare/age bank, or retained in the weight and proved summable. Similarly, a one-joint-integral moment bound supplies K, F only when the actual incoming envelopes have been established for that same integral. Replacing these statements by definitions would be circular.

At source degree greater than one, XP40 is an exact triangular hierarchy with forcing from set partitions of lower degree. FR applies to its linear part only after the corresponding multilinear forcing norms and the current/contact modules have been bounded. Fixed finite source degree alone does not prove those quantitative bounds or control the growth of constants with source degree.

Section 109.5 / One-time radius costs and the actual origin of an R bank

Here is the numerical payment that can precede FR12. Let o be the actual origin of one large-field record, and $j \geq o$ its completed regular birth. Suppose a proved bound for that record contains an unused factor $e^{-cP_0(x_o)}$, where $P_0(x) = (\log x)^p$ and $p > 1$. Suppose also that its declared radii satisfy, for each radius used in a fixed finite coefficient,

$$\rho_j \geq c_\rho x_j^{-u} (\log x_j)^{-v}, \qquad \gamma_j \geq c_\gamma x_j^{-u'} (\log x_j)^{-v'}, \tag{FR17}$$

with fixed nonnegative exponents; replacing a positive logarithmic power by its reciprocal only weakens such a lower bound. By FR11, $x_j \leq x_o + C_0$.

The direct admission conditions $CL^{-2n}\alpha \leq \rho_j/2$ and $CL^{-3n}\alpha \leq \gamma_j/2$ are satisfied once

$$n \geq \max\left\{0, \frac{\log(2C\alpha/\rho_j)}{2\log L}, \frac{\log(2C\alpha/\gamma_j)}{3\log L}\right\}. \tag{FR18}$$

The ceiling of the right-hand side is bounded by $A + B\log(2 + x_j)$. In particular a radius of order $g_j(\log x_j)^q$ generally gives a delay of order $\log x_j$, not merely $\log\log x_j$.

Any fixed product of inverse powers of these radii, coupling powers, and a factor $e^{\nu h_j}$ has logarithm at most $C + M\log(2 + x_o) + M'\log\log(2 + x_o)$. Hence, after increasing the weak floor,

$$x_j^M (\log x_j)^{M'} e^{\nu h_j} e^{-cP_0(x_o)} \leq C\, e^{-cP_0(x_o)/2}. \tag{FR19}$$

The same unused remainder can dominate any specified positive power of g_j, by allocating another fixed fraction of the original exponent. Fixed carrier polynomials are paid separately from spare exponential carrier decay. This is a one-time cost for this actual record and finite coefficient; it is not repeated at every descendant or every source label.

LM identifies such unused origin and waiting reserves in the actual R-II calculation. Combining FR19 with that ledger is legitimate only for costs belonging to the record in question. An analytic coefficient born at a much earlier regular scale $i < o$ can have an inverse g_i cost which is unbounded with x_o fixed. FR19 does not pay it. That coefficient must first be controlled through its regular history in the graded norm. Likewise arbitrary live engineering dwell is paid by LM's actual future/renewal banks, not by a fresh use of FR19.

These numerical lemmas settle how verified radius and delay factors can enter a stable feedback proof. They do not establish the incoming graded envelope or the exact readout identity of an otherwise uncontrolled marked coefficient.

Section 110.1 / Result and its precise scope

This note proves a finite-cutoff identification of the Gaussian covariance on the **actual nested-minimum graph**. The proof uses the same action and the actual composition of constraint maps. It

Section 110.1 / Result and its precise scope

includes both nonlinear-constraint multiplier corrections, the actual independent-coordinate selector, and the coarse soft gauge quadratic. No new Yang–Mills measure is introduced.

The result provides an algebraic substitute, on that graph, for deriving the covariance-compression identity by quoting the unavailable background-propagator formula (3.185). Sections 10–13 then identify a compatible local augmented realization, prove uniform current-scale spectral bounds, and obtain an exponentially small torus defect for the **complete clean physical Gaussian normalization**, including the original pivot and soft-gauge factors. These statements hold on the admitted clean regular physical core and retain the same Wilson action and target coordinates.

The note does **not** identify the independent-current extension on the full paired (U, J) domain or prove the complete nonlinear and large-field source-zero torus comparison S94. An independent-current identity is not a necessary premise of the proved direct physical Gaussian comparison. The outstanding complete-formation estimates are recorded in Section 14.

All derivatives below are taken in the actual relative coordinate charts. In particular a nonlinear block average must not be replaced by a different direct mean, and an endpoint or gauge correction to its differential must not be dropped.

Section 110.8 / Identification of the physical Gaussian in Ba·aban I

Here f is the original Wilson action, q is the actual map to the current coarse configuration, and r is the next actual averaging step. Their composition is the one defining the next constrained minimum. The previously established nesting identity PS1–PS3 uses precisely this action and these maps. A direct substitute mean would not satisfy the same argument without a separate fiber identity.

The source formulas in I, printed pp.265–268, is particularly useful. The saddle is chosen from the constrained minimum of the Wilson action plus the gauge term, and the new minimum is related to the old average by (2.3). Equations (2.6)–(2.8) expand the Wilson action on the actual inner minimizing

graph. The nonlinear outer constraint is then linearized using its actual pivot right inverse in the paragraph following (2.10). The polynomial called $C^{(2)}$ there is the quadratic part of that nonlinear constraint, while its linear differential is LQ. The gauge quadratic is separated in (2.5).

These exact operations identify the physical-graph precision as

$$\Delta^{(k)}\Big|_{\ker R} = (F'' - B + K_g)\big|_{\ker R}\,. \tag{HC30}$$

This equality is a statement about the quadratic form on its actual constraint space. It does not require equality of arbitrary ambient matrix extensions of that quadratic form. One may add a form vanishing on $\ker R$ without changing the covariance after applying C.

The quadratic-current subtraction in this identification is fixed by (HC20)–(HC22): in the notation of the exact inner lift, $DF(0)b = \langle H_1 b, J\rangle$, so the precision correction is $-2\langle H_1 h C^{(2)}(b), J\rangle$. This supplies the sign and factor directly from the actual change of variables rather than relying on

The earlier correction already included in Δ_1 is the inner correction in A. Both corrections are retained.

The actual elimination in I, p.268, takes the remaining current bond variables as B, writes the constrained bond field as CB, and defines its covariance by $(C^*\Delta^{(k)}C)^{-1}$. With the actual selector L_B, equations (HC18) and (HC30) therefore give the physical-graph identity

$$\boxed{C^{(k)}(U_{k+1}) = L_B Q\,\Gamma_{\mathrm{fine,g}}(U_{k+1}) Q^{\mathsf{t}} L_B^{\mathsf{t}}.} \tag{HC31}$$

Here $\Gamma_{\mathrm{fine,g}}$ is the Gaussian of the same composed constrained Wilson Lagrangian, with the actual coarse soft gauge pulled back and a legitimate normalized inner gauge fixing as in Section 6. It is not a covariance chosen from an unrelated boxed action.

The inherited effective interaction stays in the remainder integral in (2.12)–(2.13). It is not included in f for this Gaussian identification. Likewise the nonlinear Haar, gauge, and pivot-change densities are not silently promoted to, or removed from, the quadratic precision. Their constants and remainders retain their original placement.

Thus (HC31) is a genuine identity on the actual finite-cutoff physical graph. It resolves the algebraic covariance-identification part of the proposed compression route there.

Section 111.1 / Hypotheses and conclusions

This note continues S94 and Section 22 of the stated results, with the same Wilson coupling, normalized Haar measure, and source-zero vacuum convention. It does **not** establish S94 for the complete interacting construction. It proves a quantitative replacement for the Gaussian part of the missing local-factor comparison and identifies exactly what is still needed to insert that replacement into the original nonlinear regular and R histories.

The new proved statements are the following.

1. An invertible finite-range matrix, including an uneliminated local constraint system, has an exact convergent polynomial expansion for its inverse and log absolute determinant. Uniform singular value bounds give an explicit exponentially small torus defect.
2. The expansion has a common holomorphic neighborhood independent of the number of cells. It applies to nonflat backgrounds whenever the corresponding real matrices have the specified uniform bounds.

3. For a constrained Gaussian integral the determinant of the full saddle matrix includes the coarea normalization exactly. Thus the scalar inverse and projector may be left uneliminated; no unidentified determinant can silently be dropped.
4. The primitive Wilson Hessian and every specified finite path constraint are literally natural under copying their full local data, at arbitrary nonflat backgrounds. This is an exact statement about their matrix entries, not a finite Taylor-jet comparison.
5. The resulting current-scale wrapping bound is ultraviolet summable in four dimensions. Fixed polynomial losses in the inverse coupling do not change that conclusion.
6. For one fixed-L layer, the flat box-and-line rows together with specified local tree gauges have an explicit bounded right inverse and a uniform curl Poincare inequality. Their constrained Wilson saddle remains uniformly invertible in a small nonflat chart. The separate original-gauge identification is not asserted.
7. the stated results proves the exact physical-graph covariance compression for the actual nested Wilson minima, including both nonlinear constraint multipliers, the coarse soft gauge, and the actual pivot coordinates. Its HC32–49 also proves a compatible local augmented realization, uniform current spectral bounds, and the complete clean physical Gaussian torus comparison with the original pivot and soft-gauge normalizations. It does not identify the full independent-current extension, and does not require that identity for this physical comparison.

None of these statements identifies a new localization with every actual nonlinear/R coefficient or proves its source-zero formation and terminal estimates. The last distinction is material: a bound for the Gaussian factors alone is not a bound for the original Wilson partition function after all nonlinear integrations.

For reference, the target remains

$$\frac{V_{a,T_*}}{\mathrm{Vol}(T_*)} - \frac{V_{a,qT_*}}{\mathrm{Vol}(qT_*)} \le C, \qquad a \downarrow 0, \tag{V6.1}$$

with one and the same bare coefficient β_a on both tori. The original stability bound S74–S81 already yields

$$|\log Z_{a,T}[0] - V_{a,T}| \le C_{\mathrm{st}}\,\mathrm{Vol}(T). \tag{V6.2}$$

It does not bound the pressure difference in (V6.1) without an additional torus comparison. This note does not replace (V6.2) by a stability assertion for a newly defined vacuum.

Section 111.2 / Finite-range matrix theorem

Let $\Gamma_n = (\mathbb{Z}/n\mathbb{Z})^4$. Give each cell at most d_0 real coordinate types, with a fixed periodic type pattern. The number d_0 is a current-scale count; it must not grow with the ultraviolet depth. A matrix K_n has range at most R if its entry between two coordinates is zero whenever the cell distance exceeds R. All distances in this section are in current-scale cells. We include constraint variables and multiplier variables among the coordinate types when needed.

Suppose

$$\|K_n\| \le M, \qquad \|K_n^{-1}\| \le m^{-1}, \qquad 0 < m \le M, \tag{V6.3}$$

in the ordinary Euclidean coordinate norm, with m, M, R, d_0 independent of n. The matrix need not be symmetric. Set

$$A_n = K_n^{\mathsf{t}} K_n, \qquad B_n = I - M^{-2} A_n, \qquad \rho = 1 - m^2/M^2 < 1. \tag{V6.4}$$

For real matrices, A_n is positive definite and $\|B_n\| \le \rho$. Matrix B_n has range at most $2R$.

111.2.1 Inverse

The exact operator-norm convergent expansion is

$$K_n^{-1} = M^{-2} \sum_{k=0}^{\infty} B_n^k K_n^{\mathsf{t}}. \tag{V6.5}$$

Indeed the geometric series is A_n^{-1}, and $A_n^{-1} K_n^{\mathsf{t}} = K_n^{-1}$. Its truncation after $k = p-1$ has range at most $(2p-1)R$, and

$$\left\| K_n^{-1} - M^{-2} \sum_{k=0}^{p-1} B_n^k K_n^{\mathsf{t}} \right\| \le \frac{\rho^p}{M(1-\rho)}. \tag{V6.6}$$

Thus finite polynomial pieces depend only on the literal entries of the local matrix; nested scalar/coarse inverse conventions do not enter their definition.

111.2.2 Determinant, with its branch fixed

If D_n is the total matrix dimension, then

$$\log |\det K_n| = D_n \log M - \frac{1}{2} \sum_{k=1}^{\infty} \frac{\operatorname{Tr} B_n^k}{k}. \tag{V6.7}$$

This follows by diagonalizing the positive matrix A_n, using $\log\det A_n = 2\log|\det K_n|$, and expanding $\log(I - B_n)$. Absolute convergence follows from $|\operatorname{Tr} B_n^k| \le D_n \rho^k$.

Assign each coordinate to its cell and write tr_x for the trace over that cell's assigned coordinates. The determinant density at a root is

$$h_n(x) = d_x \log M - \frac{1}{2} \sum_{k=1}^{\infty} \frac{\operatorname{tr}_x B_n^k}{k}. \tag{V6.8}$$

Every degree-k term in (V6.8) is a sum of closed matrix walks whose entire carrier lies within distance $2kR$ of x. The factor $1/k$, matrix trace, and all coordinate sums are retained. This is a convergent full determinant identity, not an asymptotic expansion in the coupling or background field.

111.2.3 Exact comparison on two periodic lattices

Let K_n and K_{qn} be obtained from one local matrix recipe, with the same local parameters and a background on the large torus obtained by repeating the small-torus background. Here the repeated background can be nonflat. Every original local stencil, reference path, and coordinate type must be copied. Let

$$k_0 = 1 + \left\lfloor \frac{n-1}{4R} \right\rfloor. \tag{V6.9}$$

The harmless restriction $R \ge 1$ can be imposed by increasing the declared range. For $1 \le k < k_0$, a rooted matrix walk has a common injective lift: $2kR < n/2$. Consequently

$$\frac{1}{|\Gamma_n|} \operatorname{Tr} B_n^k = \frac{1}{|\Gamma_{qn}|} \operatorname{Tr} B_{qn}^k \qquad (k < k_0). \tag{V6.10}$$

This equality holds before any norm bound or root sum. Coordinate counts also agree per cell. Subtracting (V6.7) and retaining only its two tails gives

$$\left|\frac{\log|\det K_n|}{|\Gamma_n|}-\frac{\log|\det K_{qn}|}{|\Gamma_{qn}|}\right|\le d_0\sum_{k\ge k_0}\frac{\rho^k}{k}\le\frac{d_0\rho^{k_0}}{k_0(1-\rho)}. \tag{V6.11}$$

The constant is independent of q. If $\rho=0$, the right-hand side is interpreted as zero. When n is smaller than the fixed range, the estimate is just a uniform current-cell bound; no ultraviolet claim is made at such a terminal geometry.

For inverse entries on a common lifted neighborhood, the same argument applied to (V6.5) gives a difference at most $2\rho^p/[M(1-\rho)]$ once the two truncated polynomials have the same complete determining neighborhood. This also compares a fixed finite collection of inverse blocks.

What is required here. Equality of matrix recipes is not inferred from off-diagonal decay. A family $K_n=c_nI$ has perfect off-diagonal locality while its determinant density depends on the arbitrary parameter c_n. The local entries, including every zero-order coefficient and coordinate measure, must themselves be the same functions of the copied data.

Section 111.8 / Why the complete same-Wilson assertion remains open

There are now three logically different remaining routes.

Direct defect route for the original state. Use the proved clean physical Gaussian comparison HC32–49 as a primitive input, keep the original current variables and normalizers, and prove TD2 or TD5 for the complete nonlinear and R state. HC50–51 lets the clean current cutoffs remain fixed under the Gaussian comparison. The unproved part is the same-domain comparison of inherited nonlinear readouts and every original rough preparation, carrying integration, and healing operation, including its scalar vacuum increment. TD11–12 then gives S94 in the original convention. This route requires neither exact equality of nonwrapping original coefficients nor a second source-zero stability theorem.

Original-factor route. Verify the actual non-Abelian typed factors of Bałaban's generalized random-walk expansion (3.107), including local scalar/coarse inverses, artificial boundaries, background extensions, gauge rows, and the normalization measures. With those actual definitions, the complete-carrier induction in marked-bridges Section 18 and S22.2 can be checked. The new matrix theorem is unnecessary if literal naturality holds for all those original factors.

Replacement route. The clean physical covariance and Gaussian normalization are now identified in HC. To construct a completely new localization, use (V6.5), (V6.7) as its new localization, and then prove the nonlinear formation/large-field estimates on the entire common domain for every generated source-zero activity. Finally prove either that the resulting scalar vacuum agrees with the original convention up to a physical-volume bound, or independently establish the second stability inequality in S102 for the same $Z_{a,T}[0]$. An algebraic all-one identity for the Gaussian covariance does not by itself perform these nonlinear integrations.

The unresolved leaves are therefore precise.

Leaf	Established here	Still required for S94
Raw Wilson and specified finite path entries	Exact nonflat copied-data equality, including analytic vertices	Identification of every primitive used by the original factor recipe
Clean physical Gaussian	HC33 identifies the minimized action; HC34–44 gives the local typed realization and current spectrum; HC45–49 compares the original Gaussian including pivot and soft-gauge factors	Complete nonlinear and R sensitivity, not an additional off-pair identity for this Gaussian leaf
General finite local augmented inverse	Exact polynomial inverse and determinant expansion, with its coarea factors	Any use for a different mixed/R integral must identify that integral's actual covariance and full measure

Section 111.8 / Why the complete same-Wilson assertion remains open

Leaf	Established here	Still required for S94
Complex Gaussian neighborhood	One dimension-independent tube under (V6.13); HC Section 13 gives the clean physical current-operator tube using the explicit MC curvature condition	Admission of all original inherited nonlinear and R arguments into their own common analytic cores
Finite nonlinear Gaussian readout	Full-function comparison under the explicit smoothness or nondegeneracy assumptions	Uniform weighted derivative/density bounds for every actual inherited readout and its cutoffs
Regular/R vacuum births	Ultraviolet sum proved if (V6.33) holds	Identification and bound (V6.33) for the complete original birth family
Same Wilson partition function	Original stability (V6.2) and original clean Gaussian convention retained	Prove actual TD2/TD5 and the original nonlinear vacuum-increment estimate, or complete the alternative S102 route

Thus the new result is a usable nonflat Gaussian comparison theorem, with the scalar/gauge normalizations exposed, and not a claimed solution of O4. The source-zero torus obligation remains on the analysis.

Section 111.9 / Physical covariance compression and the unweakened endpoint

The exact elementary matrix and Gaussian identities in this chapter refer to the physical constraint graph. Balaban I, pp. 267?268 after (2.11), identifies the current covariance as the inverse of the constrained precision formed with the local pivot-elimination map C. Balaban II, p. 13, (2.7), uses a resolvent representation of its square root; its reference to the background propagator (3.185) includes an additional nonnegative local quadratic term.\n\nThe algebraic compression in (HC17)?(HC31) is derived on the physical graph. Equations (HC32)?(HC49) construct a compatible augmented realization and the clean physical Gaussian comparison in the original normalization. These formulas do not identify an arbitrary off-pair extension. The propagator substitutions allowed by Balaban II (1.5)?(1.7) require the stated regularity, bounds and identification of the unweakened endpoint.\n\nFor the extraction and source-zero factors, see Balaban I (1.20)?(1.22), II (1.6)?(1.10), III (3.16)?(3.20) and the paragraph after (3.47), and R-II (1.71)?(1.72), (1.90)?(1.98). A choice of Gaussian normalization by itself does not prove the missing intertorus comparison.

Section 112 / Propagation of a local torus defect

The remaining source-zero comparison S94 concerns the original finite-torus vacuum convention. Exact equality of every nonwrapping coefficient is a sufficient condition, but it is stronger than necessary. This note proves that a controlled defect of the original one-step map would also suffice. It does not assume that this defect estimate has already been proved for the nonlinear Wilson and large-field maps.

Section 112.6 / Scope for the actual Wilson comparison

The local resolvent and constrained-Hessian calculations in the argument 6 chapters give concrete candidate primitive errors of order e^{-cd_k}, subject to their stated original-operator and domain checks. TD-L and TD-H show how such errors could propagate without an exact nonwrapping identity. This removes a potentially unnecessary requirement of contraction or full Lipschitz dependence, but it does not establish the following model-specific input:

> The complete original regular and large-field one-step formation, including its fixed source-zero normalization, is uniformly Hölder with exponent at least $1/L$ on a compatible common local state domain, and all primitive intertorus changes on that domain have the displayed exponentially small rooted norm.

This input includes nonlinear readout, every old activity, the exact selector support and boundary mass, reference denominators, gauge and Gaussian factors, and the required running-coupling rows. No selected history is assumed to be a positive normalized measure. Until this input, or the alternative literal original-factor comparison, is verified, O4 and S94 remain unproved. The numerical propagation and the linear-probe selector lemma are established here with their displayed hypotheses; they are not a claimed solution of O4.

Section 113 / Conditions on classical and quantum responses

A bound for a specified classical reconstruction, a deterministic target or a numerical recurrence applies to the Wilson integral only after identifying the same object for every coefficient under consideration. The following sections retain this distinction for the regular and large-field kernels, their weak-contact terms, the feedback recurrence and the torus defect.

Section 113.1 / Numerical feedback and torus propagation

The delayed feedback argument FR11–24 is consistent. The estimate of the previous center inverses is made before the next factor is defined; there is no use of the inverse whose existence the induction is proving. The logarithmic admission interval has length $O(1+\log x)$, where $x = g^{-2}$. The running tube therefore makes its contribution $O(g[1+\log x])$, which tends to zero at a fixed sufficiently weak floor. Generated stable feedback is $O(g^2[1+\log x])$. The initial packet in FR20–24 contributes a separate delayed $O(g)$ transient. It cannot be silently counted as a newly generated $O(g)$ birth with a second coupling. The origin-reserve argument FR19 pays only costs owned by that actual large-field record. It does not pay an arbitrary inverse coupling from an earlier regular birth.

TD-L follows by direct iteration. The supergeometric distance $L^n d_k$ dominates a fixed amplification C^n. TD-H also holds at its borderline $\theta L = 1$: its one-step induction, rather than an independently summed forcing bound, supplies the uniform constant. The polynomial variant TD9–10 and the four-dimensional root count TD11–12 are correct under their stated scale hypotheses. The finite linear-probe estimate TD14–15 uses only the one-dimensional Gaussian density bound and Cauchy–Schwarz; independence of the probes is unnecessary.

These conclusions require a defect for the **complete** state and the original vacuum increment. None supplies the missing Wilson estimate for nonlinear selectors, old retained variables, reference denominators, or all generated activities. In particular the linear Gaussian probe calculation is not an anti-concentration theorem for every original curvature cutoff. The exact original source-zero normalization must remain part of the comparison.

Section 113.2 / What the original regular slot does establish

RG1–4 use a specific feature of the original construction: the inner coefficient already contains a true minimum, while the outer fluctuation response is the part subsequently weakened. Thus the inner minimum in that slot is physical on every admitted original source wall. Original I Lemma 4 supplies its enlarged-input-core inclusion. This is stronger than a generic composition of two small analytic balls and does not require an unproved inequality $C_L L^{-2} < 1$.

The paired canonical norm remains necessary. RG5 is a valid counterexample to recovering an independently extended coefficient from its physical pullback. The two-radius bounds RG8–11 follow by ordinary Banach-space Cauchy estimates. RG27 correctly shows why the one-label quadratic/cubic grade is insufficient for an arbitrary finite-source join; the required canonical jet order must increase with the total source degree.

The free cancellation RG29–30a is valid for the declared scalar polynomial class and actual centered, color-invariant flat kernel. Gaussian contraction lowers a polynomial of degree at most three to degree at most one in the proper difference. For a compact semisimple group the invariant scalar linear color tensor vanishes. The constant, all lower coefficients, and explicit current/contact rows remain. RG30b then gives the stated extra coupling **if** the incoming discrepancy jets already have the stated bound and the actual response and density comparison estimates hold. It is not a proof of that incoming grade for every generated history member.

CD pairing must use one field for the entire current packet, including its translated old roots and source-anchor reindexing. Separate local minima chosen for separate roots do not support that identity.

Section 113.2 / What the original regular slot does establish

A remote comparison must retain the actual action and target data. RG47–50 now does this for the protected pure-clean scalar packet by a fixed-reference residual argument; no interpolation between exterior sectors is needed there. This does not authorize replacing a physical torus or a bare pinned boundary in an unrelated slot.

Section 113.3 / Fractional regularity and the current supremum norm

The distinction made in regular Section 11.1 is essential. Uniform $C^{1,\beta}$ and L^∞ Laplacian bounds do not imply positive fractional second regularity. On a fixed periodic box, for example,

$$u_N(x) = N^{-2}\sin(Nx_1)$$

has uniformly bounded $C^{1,\beta}$ norm for $\beta < 1$ and uniformly bounded Laplacian in supremum, whereas its H^{2+s} norm grows like N^s. Therefore the ordinary original paired-wall estimates do not by themselves supply CD's strict age factor.

Likewise the range $0 < s < 1/p$ used by CL/MC has $s < 4/p$ in four dimensions. A $W^{s,p}$ current bound does **not** embed into a current supremum bound. The valid route is the actual protected equation

$$J(W_s) = -d_{W_s}\omega_s - \mathcal{Q}_s^*\lambda_s.$$

CL17 and CL46 give the complete-prefix adjoint bound in L^∞; analytic differentiation on the same smaller physical supremum chart gives the corresponding field-variation bound. The auxiliary scalar is controlled in $W^{2+s,p}$, which does control its first covariant derivative in supremum. An actual physical inner minimum has $\omega = 0$, but the true coarse multiplier must still be bounded in its assigned-depth norm before the adjoint lift is used. A current-unit multiplier and a finer retained row have different physical factors.

There is a further frame issue in applying RG34. Its weak difference is measured in a common reference frame, whereas MC's higher norms are proved in a final analysis gauge. One cannot subtract potentials in two unrelated Coulomb gauges and interpolate them. A covariant repair is available for the curvature/current conditions: the common $W^{1,p}$ difference gives a small curvature difference in L^p and a small connection difference in supremum. Transfer the two curvature $W^{1+s,p}$ estimates to one covariant connection using $D_s - D_1 = [A_s - A_1, \cdot]$ and the $W^{1,p}$ multiplier bounds at $s < 1/p$. Interpolation to an order between $4/p$ and $1 + s$ then gives a small curvature supremum. Interpolating the small L^p norm of ω_s with its own covariant $W^{2+s,p}$ bound to an order above $1 + 4/p$ controls $d_{W_s}\omega_s$. The $\mathcal{Q}^*$ difference uses the common-reference supremum variation. Any additional ordinary potential condition requires an explicit common gauge alignment.

Section 113.4 / Physical endpoint identities and genuine localization

The target-forced reconstruction RG32–33 has the correct endpoint mechanism on its admitted graph. In the all-one gauge-elimination test, $S\zeta = 0$ makes $Dq\, d_W\zeta = 0$ even when $q = \theta$ is nonzero. The actual Wilson KKT equation remains after the auxiliary scalar is eliminated. Uniqueness identifies the admitted physical orbit with the original minimum for that same target, boundary, and frozen fluctuation B.

This proves equality of gauge-invariant values for every admitted B, and hence of physical derivatives as functions of the original coarse variables. Those derivatives include DJ_{phys} and the full finite chain

rule. It proves neither equality of arbitrary independent-current partials nor projectability of every original fine Haar vector. XP's retained-variable derivatives and lower/contact rows still have to be carried. Multiplication by the same complete kernel and selectors preserves the resulting distributional equality.

An off-graph extension must be one fixed holomorphic, equivariant function of its declared inputs. Choosing an arbitrary real comparison reference separately at each complex input does not define such a function. CE is appropriate for this purpose because its tree, cutoff, and extension box are fixed geometric choices and its extension depends on the actual links. The real reference in CE1 is only a device for bounding transports. A fixed enlarged packet box must include every tree, target prefix, and collar read set in its carrier. Endpoint equality and a local classical estimate alone do not yet file every old composite into that carrier with its required age and anchor norm.

113.4.1 The current-index representative and remote-residual comparison

RG37–51 make a further valid improvement for a completed contractible pure-clean scalar packet. The old birth label remains j, while the function passed to the next original source localization has the current representative index k. Its unweakened inner field is therefore the actual current-unit minimum on every source wall. The original current multiplier, the coarse Ward row, and the compatible adjoint lift give its protected fractional regularity. A direct proof of the multiplier bound is available from HC's five-type inverse: input $(Q^*\lambda, 0, 0, 0, 0)$ has output $(0, 0, 0, 0, \lambda)$, so the weak Wilson-current estimate bounds the current-unit multiplier row.

The residual method RG47–50 correctly compares two fields by solving one global old equation. The approximate vector has only its multiplier component; its only residual is the first KKT row. That residual is zero on the complete packet collar and supported at old distance cH/r. The weighted inverse and contraction therefore give the displayed exponential local difference. No interpolation of unrelated exterior sectors is used. When using RG46, the intermediate raw adjoint must first be replaced on λ_k by the compatible adjoint; the current coarse Ward equation kills its endpoint term exactly.

The canonical part of this comparison has an additional elementary inclusion worth making explicit. If $n = k - j$, the running tube gives

$$r\frac{g_k}{g_j} \le L^{-n}\left(1 + \frac{b_+ n + C_1}{h}\right)^{1/2} \le C, \qquad r = L^{-n}.$$

The current canonical input ball can consequently be chosen once smaller by this constant and the fixed CE constant. The old field and its reconstructed comparison then belong to a spare old canonical ball. Applying canonical Cauchy there, and physical Cauchy on the larger old ball separately, avoids an absolute inverse birth-coupling loss in RG51. The finite recent-age window keeps its original admitted representative.

For the quartic estimate, root gauges must be compared in CE's already small canonical representative, or aligned first by a constant root conjugation. Two tree gauges of an arbitrary rough pure gauge need not have a transition near the identity. After this alignment the small transition and its bounded first difference permit the stated outer cutoff without redefining the common extension. With these points retained, the new argument supplies the protected pure-clean scalar readout; it does not identify a selected nonsymmetric history, an unhealed boundary read set, or the whole finite contact module with that readout.

113.4.2 The smooth-test grade closes the clean scalar source-moment row

RG52–57 replace a strict-age claim in the raw microscopic anchor ℓ^1 norm by a justified estimate in $\mathfrak{T} = (C^{2,s})^*$. The exact finite anchor coefficients and labels are still retained. Banach-valued coordinate Cauchy and the field-kernel moment estimates apply in this operator norm.

The source Taylor calculation RG53–54 is valid in the full $C^{2,s}$ norm. For $f(x + rz)$, the second-order remainder and each of its first two old-variable derivatives are bounded by $Cr^{2+s}(1 + D)^{2+s}\|f\|_{C^{2,s}}$; its second-derivative Hölder seminorm has the same factor. A source moment of order m already supplies r^m, leaving only $2 - m$ field/path orders before the quadratic dimension-six extraction. This is why an order-two test functional does not require four test derivatives at the next birth. The quadratic source remainder has the exact scaling $r^{2+s}r^4r^{-6} = r^s$. The cubic and quartic rows retain their separate field grades.

The integration estimate RG56 follows directly from the operator norm, and the original one-reserve connected formation preserves it after actual source admission. These calculations justify the stated completed pure-clean scalar row RG57 with its free companion and lower rows. The short/tail cutoff must be imposed on covariant relative carrier, path, and source-offset lengths, so it survives the exact reflection/root reindexing. With source support in Q, a threshold proportional to H/r then puts the entire short family in the one common protected hull. An unrelated hard containment mask would not justify the same cancellation.

The diagonal-contact exclusion in regular Section 15.3 is mathematically necessary. For example, the finite two-anchor functional

$$\Lambda_\varepsilon(f, g) = \varepsilon^{-1}[f(0) - f(\varepsilon)]g(0)$$

has uniformly bounded norm on two $C^{2,s}$ test slots. Keeping only coincident anchors removes its off-diagonal term and leaves $\varepsilon^{-1}f(0)g(0)$, whose norm is unbounded. Thus bounded smooth multiplication controls an already formed contact; it does not bound creation of that contact by a diagonal mask on an arbitrary weak coefficient. Actual XP contact equations, internal cutoff currents, and higher engineering-dimension projections remain separate quantitative rows.

Section 113.5 / Large-field entry and weak currents

LM34a settles the ordinary completed-readout cutoff-versus-radius comparison by an admissible once-fixed choice of the original logarithmic exponents. LM43–44 also give a genuine carried-coefficient estimate when the old representative depends only on the retained real live variables and already has its declared graded envelope there. One joint positive reserve pays the total finite polynomial degree.

The straddling situation LM47 is different. The original R-II formulas retain the old determining data, the old fluctuation, and the old selectors. The weaker mixed $\mathcal{U}'$ admission in R-II (1.65) does not show that the same old representative lies in its original spare physical and canonical cores for all real old cutoffs and every outer complex preparation wall. Neither the ordinary logarithmic-radius choice nor a new independent periodic minimum proves that inclusion. The complete old read set and one common packet field must remain visible.

LM50–52 give a distinct exact option: retain the whole real packet before entry and preserve the age already gained there. This avoids evaluating that packet at a straddling complex preparation. A blocked output still has its actual delta constraints, however. To promote this option into a local analytic R transition, LM53 requires the genuine mixed output/restoration map, and LM54 requires the same-Wilson bound for the extra good halo. The identity-row classical inverse does not prove that quantum measure identity. A numerical halo cost fitting the available bank is not a proof that the

modified kernel has that cost.

WCT1–15 give a quantitative finite-order weak norm and deterministic pushforward estimate with the exact ordered target derivatives. WCT11–12 avoid a microscopic link count for the actual normalized path target. These are useful contact estimates. They do not bound an arbitrary selected normalized integration kernel on the same weak space. The example WCT17–18 demonstrates that a positive mass-one kernel can have an arbitrarily large first dual derivative. For the original R component, however, the stronger R-II (1.73)–(1.75) estimate does give the outer regular-pair row: WCT21–24 prove total-variation Cauchy bounds with canonical cost $(g_k/\rho_k)^m$ on the admitted spare core. This includes the original source-zero reference denominator. Its frozen internal selectors are not differentiated in that row. A literal current-target chart must still map the stated disc into that same paired core.

There is no omitted free component-size factor in this **single original component** assertion. R-II p.379 bounds its entire comparison error by $Cg_k\mathcal{P}(\log g_k^{-2})$; p.388 explicitly uses $|\sigma| \le 1$ after (1.92). A larger fixed weak floor gives $\sigma_* \le 1/4$. A product of n component kernels instead has the retained bound $\exp(3\sum_i \sigma_{*,i})$, at most $e^{3n/4}$; that count belongs to its carrier/activity bookkeeping. It cannot be absorbed into a constant independent of geometry, nor does this conclusion extend to the unproved buffered operator.

The same small comparison gives the relative-log-body estimate WCT27–28 for the outer parameters of one original component with one fixed real reference. The ratio lies within distance $e^{\sigma_*} - 1 < 1$ of one, so its logarithm has a bounded holomorphic branch and Cauchy derivatives. Its absolute $\log Z(z_0)$ is retained. This supplies no comparison of that base scalar between physical tori; product logs still sum over the actual component count.

WCT19 identifies the omitted cutoff-boundary current if a selected physical EOM is replaced by the derivative of its whole selected kernel. Primitive EOM identities may be formed before selecting histories, but later effective EOM and source-contact rows cannot be discarded that way. The remaining R derivative issue concerns the internal old coordinates and their moving selector boundaries, together with any unbounded actual target-jet composition. The partition identity WCT26 cancels whole derivatives before a common readout, but different history-dependent readouts require their own boundary compatibility proof.

Section 113.6 / Leaves still requiring a proof for the same Wilson measure

1. A complete common-packet localization of every previously generated marked coefficient, with its literal old/current determining maps, protected physical domain, canonical grade, source anchors, and tails. LM47 is a concrete first-large-field-entry instance of this leaf; the buffered alternative has the distinct map and healing obligations LM53–54.
2. Uniform transport of the entire finite current/contact module through every selected regular and large-field operation. This includes the actual target compositions, multiplier grades, cutoff boundary currents, internal retained-data reference and normalization derivatives, and one consistent reserve. The outer regular-pair R derivative row WCT24 is already available and need not be assumed again.
3. The resulting all-history marked bounds with the verified carrier constants needed to instantiate FR's otherwise proved feedback induction. A local $O(g)$ birth estimate is insufficient without these incoming envelopes and identities.
4. The complete source-zero state comparison and actual vacuum-increment estimate on different physical tori, or an alternative literal-factor proof of S94. TD's propagation and finite-probe lemma isolate useful sufficient estimates but do not establish this Wilson input.

Section 113.6 / Leaves still requiring a proof for the same Wilson measure

No one of these leaves is merely a missing standard algebraic step. They are quantitative model-specific assertions about the original integration and its generated coefficients. The complete interacting four-dimensional existence proof therefore remains incomplete.

Section 114.1 / What is now quantitative in the regular marked calculation

The regular chapter RG retains two bounds on each born coefficient: its full paired physical core and its canonical field/current grade. This is necessary because a term proportional to the difference between an independent current and the physical Wilson current can vanish on the physical graph while remaining nonzero on an independently weakened pair. The physical pullback alone does not bound that extension.

For the original completed inner minimum, I Lemma 4 and II's actual source-wall range give a fixed enlarged-input inclusion. This avoids an unsupported repeated loss of analytic radius. The first physical coarse response is controlled by the original variational Proposition 9, including its Maxwell and Laplacian rows. LM38–41 independently verifies the needed current derivative directly from the Wilson plaquette flux:

$$\|DJ_a(A)[H]\|_\infty \le C(\|d_a^* d_a H\|_\infty + \|D_a H\|_\infty + \|H\|_\infty). \tag{R6.1}$$

The apparent second-difference error in the nonlinear flux has a factor of the fine spacing; that factor is used before taking the supremum. There is no assumption of bounded spatial derivatives of arbitrary order. Coordinate Cauchy estimates on coarse determining variables then give fixed-order kernel moments with exponential distance decay. The estimate counts coarse coordinates and their fixed polynomial moments, not independent microscopic bonds.

On a verified common current field, the old dimension-six remainder has the three distinct gains: a fractional quadratic gain, a cubic gain, and a quartic gain. With $r = L^{j-k}$ and $\varrho_{j,k} = g_k/g_j$, its canonical envelope has the form

$$CA_j \left[r^s \varrho_{j,k}^2 q_Y^2 + r^\beta a_{3;j,k} q_Y^3 + r^2 \alpha^{d-4} g_k^{4-d} \varrho_{j,k}^d q_Y^4 \right], \qquad d = 2, 3. \tag{R6.2}$$

The cubic coefficient retains the sharper cap for a degree-two label. The fourth-order term follows from an integral remainder with only four coarse field derivatives. A genuine interaction attachment supplies the additional $O(g_k)$ from the actual inherited unmarked fluctuation vertex and its same-coordinate density comparison. Its free companion uses the exact Gaussian mean/covariance tower and the full lower quadratic family. A centered color-invariant Gaussian contracts a proper polynomial of degree at most three to a constant and a scalar linear color tensor; the latter vanishes for a compact semisimple group. The constant, lower terms, and current/contact terms are retained.

Two regularity qualifications are essential. First, the original $C^{1,\beta}$ wall bound alone does not supply the H^{2+s} potential used by CD. The relevant fractional estimate must come from the genuine constrained minimum and its protected Euler–Noether equations. Second, the entire translated old packet must use one current field. Independently chosen fields at different roots do not justify the reflected pairing or its source shifts.

RG37–51 constructs a local current-packet representation to meet those requirements on protected contractible regular carriers. Its internal field is the genuine current-index minimum, including on the admitted outer source walls. One fixed covariant extension is used as an old reference. The old nested multiplier is cut off only outside the protected packet; all constraint and gauge residual rows vanish exactly, and the remaining Euler residual is separated from the packet. One weighted compatible inverse then compares this field with the original old physical minimum. No interpolation between different boundary problems is assumed. The top Ward identity converts the raw multiplier adjoint before its norm is estimated.

Canonical admission is checked independently of physical admission: the old-core inclusion requires the scaled ratio $C_H r(g_k/g_j)$ to fit its spare canonical radius. The running tube supplies a fixed age delay for this protected comparison. The exact physical endpoint retains the original fluctuation variables. Long, winding, and boundary members retain their admitted tail representation. This construction does not admit arbitrary mixed large-field carriers.

114.1.1 Source grades and smooth tests

For a finite label set I, the canonical field grade must extend at least through $D_I = \sum_{i\in I} d_i$, with a physical remainder of order $D_I + 1$. Keeping only the quadratic and cubic jets fails once the total source degree grows; RG27 is an explicit analytic counterexample. One positive reserve absorbs the total polynomial degree of a joint source partition. It is not a separate reserve for each overlapping label.

The strict CD age gain is a bound in the declared smooth-test and pointed-descendant module. It is not a gain in the absolute norm of arbitrary unsmoothed anchor coefficients. RG52–57 now closes this scalar clean row using the dual of the local $C^{2,s}$ test norm. For a source functional Λ, its moments of order zero, one, and two are retained exactly. Taylor expansion of $f(x + rz)$ has a remainder bounded in the full old-variable $C^{2,s}$ norm by $Cr^{2+s}(1 + D)^{2+s}\|f\|_{C^{2,s}}$, where D is the old carrier diameter. A moment of order m leaves only $2 - m$ spatial orders to be extracted. Thus the combined source/field calculation uses the same test order at each birth; it does not consume two new test derivatives each time.

The actual clean connected formation is bounded in this Banach-valued norm. Together with the actual interaction attachment, the free companion, the protected comparison, and explicit carrier recycling, this gives the scalar clean instance of MT36 in RG57. Exact original anchors remain as algebraic data. Disjoint finite label joins are bounded in the corresponding multilinear test norm, but creation of diagonal contact rows is not supplied by this norm. Higher engineering dimensions

require their own finite joint moment calculation. These test and field grades are also separate from the derivative order of an equation-of-motion current.

Section 114.2 / The feedback algebra no longer assumes its own normalization

FR proves a noncircular closure theorem for the actual linear first-source system *once its quantitative transition rows have been established.* After removing the known engineering factor, write

$$b_k = J_k c_k + \sum_{j<k} K_{k,j} b_j, \qquad c_{k+1} = (I + E_k)c_k + \sum_{j<k} F_{k,j} b_j. \tag{R6.3}$$

Each birth retains its own Banach space, analytic radii, grade, anchors, and origin. The deterministic old term stays under its old tag. Under the displayed $O(g_k)$ transition estimates with a summable age kernel, FR constructs the complete center response C_k and proves its invertibility by induction. The bound for $B_k C_k^{-1}$ uses only previously constructed inverse factors. Thus the center normalization is a conclusion, not an input to its own stable-feedback bound.

The result permits the actual logarithmic admission delay. Put $x_k = g_k^{-2}$, and retain the running-tube bounds FR11. If a birth cannot enter the regular core until $h_j \le A + B\log(2 + x_j)$, the age weight may be

$$w_{k,j} \le w_* e^{-\sigma[(k-j)-h_j]_+}. \tag{R6.4}$$

FR16 proves the uniform weighted sum

$$\sum_{j<k} w_{k,j} g_j \|C_j C_k^{-1}\| \le C \frac{1 + \log(2 + x_k)}{\sqrt{x_k}}. \tag{R6.5}$$

Consequently the feedback from generated stable births is $O(g_k^2[1 + \log(2 + x_k)])$. A bounded initial stable packet has its separate delayed $O(g_k)$ transient, proved in FR20–24. There is no false extra coupling assigned to that packet. The primitive center correction may still be only $O(g_k)$.

A radius of order $g_j(\log g_j^{-2})^q$ generally requires a delay of order $\log g_j^{-2}$, not merely a logarithm of that logarithm. FR17–19 pays finite inverse-radius and delay costs from a specified unused part of the **actual** large-field origin bank. It does not pay an uncontrolled inverse coupling from an earlier regular birth, reset an origin at healing, or spend the same reserve on every descendant.

These are proved summation and inverse-product statements. The missing all-history instances of J, K, F are not supplied by naming the spaces in R6.3. O2 and the model-specific part of O3 therefore remain.

Section 114.3 / Large fields: actual energy, radii, and the remaining entry problem

LM proves own-spacing compact Wilson envelopes for the primitive dimension-six marks and their finite canonical Haar jets. Its complex discrete div–curl estimate controls the protected directional curvature derivative using the actual physical current, a curvature norm, and the exact Bianchi defect. It uses Hermitian absolute norms for estimates and does not claim positivity of a complex bilinear energy.

Section 114.3 / Large fields: actual energy, radii, and the remaining entry problem

The original R-II completed coefficient returns to the paired domain specified in III. An explicit once-chosen auxiliary hierarchy $q_0 = q_1 = q > p_0$, with the required fixed separation of its radius constants, gives

$$\frac{g_k P_k}{\rho_k} \le C(\log x_k)^{p_0 - q} \longrightarrow 0. \tag{R6.6}$$

This places the entire original frozen ordinary fluctuation cutoff strictly inside the completed coefficient's radius. It is a choice in the construction of the original analytic extension; it does not extend an arbitrary previously given function by fiat. The actual complex preparation errors remain a small coupling times a fixed logarithmic polynomial.

The first-entry coordinate argument keeps the original old determining data, old fluctuation variables, and old selectors. The previously earned regular age bound belongs to the current representative before that integration. It is lost if one instead starts estimating the initial fine primitive again.

For a complete read set inside one retained real joint integral, LM43–44 gives the carried-coefficient bound with that saved age and one positive reserve. The original outer kernel supplies its analyticity. A packet straddling that integral is not covered. The precise remaining original-map assertion is LM47: the source-only old-target reconstruction must lie in the same spare physical and canonical cores over all old real cutoffs, simultaneous complex preparations, and complete collars. The weaker mixed member domain in R-II (1.65) is not a proof of that statement.

LM50–54 also examines retaining a whole real packet and a good halo. The joint same-measure pushforward identity is exact, but the old coarse-target delta constraints remain. To use it as an analytic live operator one must construct and bound the actual mixed output map, its restoration to the original targets, and its positive shell and reference normalization. A bare pinned classical inverse does not prove this quantum integration estimate. The enlarged component must also keep the original failed-seed geometry. The existing numerical bank can pay the stated finite shell cost if that cost is proved; it does not prove the cost or authorize a new failure credit.

Section 114.4 / Exact contacts now have some quantitative transport rows

XP's finite-source algebra, ordered Haar identities, and full whole-kernel contacts are unchanged. WCT equips finite current representations with a dual smooth-test norm. For deterministic actual targets it proves transport by the ordered chain rule, with the exact finite target-jet constants and lower current terms. The original normalized path target has one-step bounds without a microscopic link count. A fixed constant per step is not yet an all-depth bound; the composite or graded accumulation must still be controlled.

The original R-II estimate is stronger than a positive mass bound. With the internal coordinates and their real selectors frozen, its normalized source-zero kernel is a measure-valued holomorphic function of the admitted outer pair. Cauchy gives the finite outer derivative cost

$$\|(g_k\partial)^m \nu_z\|_{\mathrm{TV}} \le C_m M_\sigma (g_k/\rho_k)^m, \qquad M_\sigma = e^{3\sigma_*}. \tag{R6.7}$$

The original reference denominator is included. No sourced island inverse has been introduced. WCT21–25 proves this row in an admitted current-scale chart and its unnormalized version with one history bank. It does not apply a coarse radius to an arbitrary microscopic bond direction. The displayed comparison factor is retained exactly; R-II's original admissible single-component estimate bounds σ_* by a coupling times a fixed logarithmic polynomial. One stronger weak floor therefore gives

$M_\sigma \le e^{3/4}$. For a product of n component kernels the count remains explicit: the bound is at most $e^{3n/4}$, not the same single-component constant. The unnormalized version uses its actual complex majorant B_ω once, so origin and locality costs remain visible in an all-history application.

WCT27–28 also supplies a scalar normalization row. For one fixed real comparison pair z_0, positivity and the exact exponential comparison give

$$\frac{Z(z)}{Z(z_0)} = \int e^{\sigma(z,\xi)}\, d\nu_0(\xi), \qquad \left|\frac{Z(z)}{Z(z_0)} - 1\right| \le e^{1/4} - 1 < 1. \tag{R6.7a}$$

The relative logarithm consequently has one bounded holomorphic branch and finite canonical derivatives of order m bounded by $C_m(g_k/\rho_k)^m$ on the smaller original disc. Its absolute base value $\log Z(z_0)$ is retained. This bounds outer-parameter variation of the actual component body; it does not compare two different physical-torus reference bodies. Logarithms of a product sum over the actual component count.

Internal old-coordinate derivatives and moving selector boundaries remain distinct. The distribution $\chi \operatorname{div} \nu$ is not the distribution $\operatorname{div}(\chi\nu)$; their boundary current must be retained. A partition cancels whole derivatives only against the appropriate common test. Different history readouts require equality of the necessary interface values and derivatives, or a direct bound on those boundary currents. WCT's positive Markov-kernel example shows why a mass-one bound alone cannot replace this derivative estimate. Those internal rows remain part of O2.

Section 114.5 / The clean physical Gaussian torus comparison is now identified

HC proves the constrained-Hessian identity for the actual nested Wilson minima. If q is the inner constraint and r the next constraint, the fine composed Hessian contains both corrections

$$A = f'' - \lambda q'', \qquad B = \Lambda r'', \qquad H_{\text{composed}} = A - Q^{\mathsf{t}} B Q. \tag{R6.8}$$

The coarse precision is the original $C^{\mathsf{t}}(F''-B+K_g)C$, including the actual pivot injection and soft gauge. The covariance is the corresponding compression of the constrained fine Gaussian. Normalization factors which cancel from a covariance are kept when computing the vacuum.

The raw differential is not inserted into MB's compatible inverse. Instead HC32–38 proves equality of the raw and compatible minimized action as a function of the actual nearby target, using both gauge fiber inclusions. The compatible multiplier Hessian is an explicit small local perturbation. Its bound uses the geometrically summable one-step lengths, not a product of constants over the blocking depth. The resulting exact Schur readout and the fixed-step cochain estimate give uniform lower and upper spectral bounds for the **current** precision. No bounded Euclidean condition number is asserted for the large hidden fine matrix.

On a copied physical target, the base minimum and multiplier are literal periodic copies by the local equations and regular-branch uniqueness. The typed inverse and its cutoff commutators compare the current Hessian, covariance, and diagonal logarithm. HC48–49 then includes the actual pivot, current measure powers, and the original root-fixed Haar soft-gauge insertion. It introduces no global prime-determinant convention. On the admitted clean physical core,

$$\left| \frac{\log Z_{G,T}}{\mathrm{Vol}(T)} - \frac{\log Z_{G,T'}}{\mathrm{Vol}(T')} \right| \le C \ell_k^{-4} e^{-cT_*/\ell_k}. \tag{R6.9}$$

Section 114.5 / The clean physical Gaussian torus comparison is now identified

This closes a model-specific Gaussian part of the torus comparison. It includes nonflat regular backgrounds in the stated domain. The complex physical tube still requires the separate curvature bound already specified by CC/MC. It is not a theorem about arbitrary independent-current extensions or every large-field history.

HC50–51 compares two clean Gaussians in the same original current coordinates. A covariance change is a quadratic density perturbation; the identical measurable real cutoff need not be moved by a different square root. This removes an unnecessary selector-comparison issue for that clean operation. It does not identify every retained old or large-field test.

Section 114.6 / What is still missing from the source-zero comparison S94

The nonlinear normalizer N_k, inherited activities, and complete R histories remain in the original source-zero vacuum. Neither R6.9 nor the single-component outer-parameter logarithm R6.7a compares all those quantities. In particular, no arbitrary torus-dependent scalar is removed by a new normalization convention. Smallness in the coupling does not justify dropping these vacuum terms: an unmatched fixed power of g_k per current cell is not summable against ℓ_k^{-4} in four dimensions. Even the usual $\exp[-c(\log g_k^{-2})^p]$ rarity is not, by itself, an ℓ_k^4 factor along a logarithmic weak trajectory. The comparison must retain the cancellation of the local vacuum terms or derive the required extra dependence on the torus distance.

TD supplies a second route besides exact nonwrapping equality. A uniform Lipschitz defect recurrence for the complete original state, with forcing e^{-cT_*/ℓ_k}, remains exponentially small even if its step constant exceeds one. A fixed Hölder exponent $\theta \geq 1/L$, with $\theta < 1$, is also sufficient; the borderline follows by an explicit barrier induction. Fixed current scale powers are allowed. The four-dimensional root sum is then uniform in the ultraviolet cutoff.

These propagation statements are proved. Their application still requires the actual same-vacuum nonlinear/R formation estimate on matched local state domains, including the complete history, reference normalization, and any genuinely moving tests. The finite-Gaussian linear-probe example in TD is not a universal anti-concentration theorem for those tests. Thus S94 and O4 remain unproved, although the clean physical Gaussian input is no longer an unidentified conditional covariance.

Section 115.1 / Hypotheses and conclusions

This chapter continues LM42–54 and RG37–58 without changing their old birth labels, determining records, selectors, or Wilson measure. It does **not** prove LM47 for every R history. It makes four more concrete advances.

1. It proves that exact locality in the background field is insufficient for the proposed finite-current-variable retention argument. The obstruction occurs in the flat linearization of the same Wilson constrained minimum and has an explicitly computable exponential tail.
2. It constructs a nonredundant mixed target for a genuine finite set of current integration variables, using the original one-step averages and their original crossing pivots. Its restoration map and Haar Jacobian are explicit; the original normalization is preserved exactly.
3. It gives an exact exponentially convergent current-target localization in a small commuting Wilson sector. This verifies that a localization expansion can repair strict locality in that sector. It does not identify that sector with the arbitrary nonabelian mixed R domains.
4. It proves an all-degree connected target-carrier expansion for the actual noncommuting regular KKT problem on one fixed reference, avoiding the volume cost of radial shells. The weak source lifts WH make the two original primitive sources legitimate roots of that expansion, with their quadratic and cubic grades retained. A final restricted theorem also localizes the reference potential and its physical multiplier on one declared global small weak chart.

The distinction between an exact change of coordinates and a bounded analytic R kernel is maintained throughout. In particular, a holomorphic coordinate chart does not make a sharply selected density holomorphic.

Section 115.2 / What the primary R map does and does not identify

The relevant primary $_{\text{the analysis}}$ is R-II, printed pp. 373–375, (1.61)–(1.65). The boundary gauge in (1.61) is retained because two determining representations meet there. The successive localizations preserve that boundary information. The stated range of (1.65) is the old space defined in R-I (1.64)–(1.69), and the argument uses weaker old conditions on the changed layer. R-II also explicitly distinguishes localization in a background field from locality in retained integration variables before its last localization. These facts do not identify the range with the spare marked physical/canonical cores in LM47.

The available copy of R-II is the original paper. A public bibliographic records and later notes, but no original R-I that lists R-I as a source contains or text of (1.64)–(1.69). Its

Section 115.2 / What the primary R map does and does not identify

The more restrictive marked-core inclusion is therefore not being attributed to a primary theorem that has not been scope.

For reference, the still required inclusion is

$$(P_{\mathrm{old}}(q_{\mathrm{old}}(\Gamma_{\omega,\mathbf{s}})), J_{\mathrm{phys}}(P_{\mathrm{old}}(q_{\mathrm{old}}(\Gamma_{\omega,\mathbf{s}})))) \in \mathfrak{C}_{j,k}^{\mathrm{spare}}. \tag{ME1}$$

It must hold on the simultaneous original preparation and source wall domain, with every determining prefix, junction row, bare boundary value, and real selector required by that history. At the physical endpoint, LM46 gives equality with the original marked coefficient. That endpoint identity alone neither gives ME1 nor bounds a complex disc around it.

The retained real constraints are indispensable to that equality. They are not replaced below by an unconstrained periodic problem. The new construction is instead an exact factorization inside the same finite-cutoff real measure, where its hypotheses on current variables are actually satisfied.

Section 115.4 / An explicit nonredundant mixed target for a genuine current-variable halo

This section concerns the original **one-step** current-to-coarse average. It does not claim the same pivot property for an arbitrary stack of older raw constraints or for the matched-reference variables introduced by R. Those records remain in the underlying measure.

Let E be the oriented current bonds, C the oriented coarse bonds, and

$$q : \mathcal{D} \subset G^E \longrightarrow G^C \tag{ME14}$$

the original averaging map on its actual branch. The domain $\mathcal{D}$ is the original domain of this step; a rough configuration outside that domain is not assigned a logarithmic average by this notation. For a coarse bond c, let $S(c)$ be its complete readset, including the original rooted contour averages and the central straight path. Let $p(c) = b_0(c)$ be its original central crossing bond.

I (0.11)–(0.12) supplies the readset, and the discussion immediately before I (2.11) identifies this pivot and the right inverse supported on these pivots. The following geometric property follows directly from those formulas:

$$S(c) \cap \{p(c') : c' \in C\} = \{p(c)\}. \tag{ME15}$$

Indeed, rooted paths remain inside either endpoint block; a straight path between translated endpoints crosses only the common face of those two blocks. A different coarse bond has a pivot on a different block face. The central path has the same property. This argument keeps every original path, rather than replacing the mean by a single line product.

115.4.1 Pivot closure for an arbitrary finite readset

For a genuine current-bond readset $H \subset E$, put

$$\begin{aligned} I_H &= \{c : p(c) \in H\}, \qquad O_H = C \setminus I_H, \\ H^+ &= H \cup \bigcup_{c \in I_H} S(c). \end{aligned} \tag{ME16}$$

ME15 shows that this closes after one enlargement:

$$\{c : p(c) \in H^+\} = I_H, \qquad p(c) \notin H^+ \quad (c \in O_H). \tag{ME17}$$

In particular the complete readset of each omitted row lies in the retained halo, and every kept row still has its own unretained pivot. At fixed blocking factor L,

$$|H^+| \le (1 + CL^4)|H|. \tag{ME18}$$

Only current bonds are counted here. No microscopic bond count at an earlier cutoff enters ME18. The support lies in the original halo plus a bounded number of neighboring current-scale blocks.

Define

$$\begin{aligned} M_H(v) &= (v_{H^+}, q_{O_H}(v)), \\ r_H(h,z)_c &= \begin{cases} q_c(h), & c \in I_H, \\ z_c, & c \in O_H. \end{cases} \end{aligned} \tag{ME19}$$

Every argument of $q_c(h)$ is present by construction. Consequently

$$q = r_H \circ M_H \tag{ME20}$$

is an exact identity of the original nonlinear maps, without a curvature expansion or an error term.

115.4.2 A block-shaped version

An arbitrary pivot-closed halo need not itself have the geometry of an allowed R history. There is, however, a useful block-shaped version of ME19. Choose a finite buffered union $\mathcal{R}$ of coarse L-blocks containing the endpoints of every bond in the original H. Retain every current bond with both endpoints in $\mathcal{R}$. Let I consist of coarse rows whose two endpoint blocks lie in $\mathcal{R}$, and keep the coarse rows in O, which have at least one endpoint block outside $\mathcal{R}$.

The omitted rows have their entire readset inside the retained region. A kept row's central crossing pivot is unretained, including when the row crosses the interface. Thus ME19–20 apply again. This is the familiar two-scale determining pattern with no additionally retained inward rows. Complete scalar/gauge records must accompany it if a gauge slice is subsequently chosen.

This verifies the geometric mixed target, not membership in the original R induction. The added buffer is source bookkeeping; it is not a newly failed large-field test and does not acquire an origin bank. The original seed geometry, S mergers, and duration bounds still have to be checked if the buffer is made part of an R output domain.

Section 115.8 / Exact noncommuting target localization on a fixed regular reference

The two-target theorem V7N27–30 supplies a noncommuting version of the elementary shell construction. Fix one actual regular reference $\bar{U}$, its raw target $\bar{V}$, multiplier, boundary conventions, and full compatible prefix. These are fixed parameters; they are not reselected as the target changes. Let $x(\theta)$ be the five-type solution of the actual shifted Wilson equations in the same small complex target ball. If two targets agree in a collar of a protected readout, V7N30 bounds the difference there by $C\eta e^{-cR}$, where R is the collar width and $\|\theta\|_\infty \le \eta$.

Truncating θ outside successive current balls therefore gives an exact finite-current-target shell expansion of this local solution, with exponentially small increments. Every truncation is an actual target of the same constrained Wilson problem. All five rows, including the scalar and gauge rows, are kept. This is not a restriction of the fine action to a smaller graph.

The field components of x are in the declared common compatible reference frame. Equality with a raw-target physical source uses HC33 and the original coherent node-gauge restoration. A joint old record linking colors or transporters at different depths must retain that full restoration; separate gauge choices for tree vertices are not authorized by this expansion.

There is, however, a quantitative reason not to use that shell expansion directly in the original polymer norm. In four dimensions, a current ball of radius R has order R^4 cells. Its tree size touching all those cells also has that order. A radial tail bound alone therefore gives the estimate

$$d_M(B_R) \asymp R^4, \qquad e^{\kappa d_M(B_R)} e^{-cR} \geq \exp(c_1 \kappa R^4 - cR) \longrightarrow \infty. \tag{ME48}$$

The sequence e^{-cR} itself obeys the proposed radial bound and fails this weighted estimate. This proves failure of the inference, without claiming that every actual shell coefficient saturates the radial bound. An exponential-in-volume retention cost causes the same problem. Exponential locality in radius is insufficient for declaring a whole ball to be one polymer.

115.8.1 The actual weak equation has a connected-carrier expansion

The following construction avoids that loss. It applies to the actual uniform regular problem of V7N3–6, on its fixed reference and fixed small complex ball. It also applies coefficientwise to a finite source algebra, without altering a source label.

Tile the current lattice by the once-fixed cells used in the typed inverse. Use their graph distance $d(u, v)$. Denote the actual multiplier-corrected inverse by $G = \mathbb{K}_{\bar{\lambda}}^{-1}$, and write its fixed local block decomposition as G_{uv}. The proved inverse majorant has the form

$$\|G_{uv}\|_{Y_v \to X_u} \leq C_G e^{-m_G d(u,v)}. \tag{ME49}$$

The norms include all five types and the recorded weak coefficients in the negative rows. More explicitly, use the bounded smooth cell restriction and reconstruction maps of MB/MC, with fixed overlap. Their reconstruction after restriction is the identity on the global field space. In a negative row, apply the partition to its actual weak coefficient representation; keep the local cutoff commutators needed when a derivative crosses the partition. Do not choose an unrelated representative of a quotient negative norm. Then G_{uv} is the global inverse preceded and followed by these literal reconstruction/restriction maps. A fixed point in the cell slots is consistent because it is the restriction of that global inverse; reconstructing it gives the global equation. The fine cutoff does not enter C_G, m_G. The actual equation is

$$x = GI\theta - G\mathcal{N}(x), \qquad \mathcal{N}(0) = D\mathcal{N}(0) = 0. \tag{ME50}$$

Here I puts the physical current target into its actual constraint row. Choose C_I to bound the sum of the norms of its finitely many target-coordinate columns in one current cell, not merely its operator norm. The current cell, type and color dimensions are fixed. This choice pays the absolute target-kernel count in ME57.

The local nonlinear vertices are taken **before** their weak coefficient bounds are summed into a global negative norm. Each output cell v reads a connected set C_v of at most v_0 current cells. This set includes its actual plaquettes, complete mean prefixes, scalar rows, endpoint factors and every incidence collar. Its size is fixed independently of the number of fine links in a cell. The local holomorphic expansion can be written

$$\mathcal{N}_v(x) = \sum_{q \geq 2} \sum_{v_1, \ldots, v_q \in C_v} N^{(q)}_{v; v_1 \ldots v_q}[x_{v_1}, \ldots, x_{v_q}], \qquad \sum_{v_1, \ldots, v_q} \|N^{(q)}_{v; v_1 \ldots v_q}\| \leq C_N r_N^{-q}. \tag{ME51}$$

The Taylor coefficient includes $1/q!$. To obtain the second bound, first use the uniform holomorphic weak-vertex ball of V7N6/MC on the complete local carrier. Insert its fixed cell partition into each input slot. The at most v_0^q resulting choices, the fixed reconstruction-map norms in these slots, and

the Banach polarization bound $q^q/q! \le e^q$ are absorbed by reducing r_N once. The field vertex is written in divergence form, and its actual Sobolev product bounds are used there; no unrestricted point evaluation on a negative space is made. The mean and shifted-multiplier vertices use the actual CL compatible adjoint and endpoint logarithm. Thus ME51 is a positive local coefficient majorant for the declared weak equations. It is not inferred from an arbitrary unlocalized Banach-operator estimate.

Choose one nearest-neighbor current-cell path P_{uv} joining u to v. It has at most $1 + d(u, v)$ cells. For $0 < \kappa < m_G$,

$$C_\kappa := \sup_u \sum_v e^{\kappa|P_{uv}|}\|G_{uv}\| < \infty. \tag{ME52}$$

Indeed the number of cells at distance n is bounded by $C(1 + n)^3$, and the remaining summand decays as $e^{-(m_G-\kappa)n}$. This estimate is uniform on compatible finite tori. The chosen path is a carrier assignment, not a new gauge transport or a replacement for a determining prefix.

Expand ME50 recursively. A leaf is an actual local target coordinate. A propagation edge is an actual block G_{uv}. An internal vertex is one of the local tensors in ME51. Assign to a tree the union of all propagation paths and all local vertex carriers. That union is connected, contains the output cell, and contains every variable target read by that tree. Coefficients and signs are the literal Taylor coefficients of ME50.

For an activity family $x_{u,X}(\theta_X)$, define its target-carrier norm by

$$\|x\|_{\kappa,\eta} = \sup_u \sum_{\substack{X \ni u \\ X \text{ connected}}} e^{\kappa|X|} \sup_{\|\theta_X\|_\infty \le \eta} \|x_{u,X}(\theta_X)\|_{X_u}. \tag{ME53}$$

The word target is essential: $\bar{U}$, $\bar{\lambda}$ and the reference gauges are fixed parameters of every coefficient. Their dependence is discussed in Section 7.3.

The union bound for carrier sizes makes products submultiplicative. ME51–52 therefore give the scalar majorant

$$z \;\longmapsto\; A\eta + B\sum_{q\ge 2}(z/r_N)^q, \quad A = C_\kappa C_I, \quad B = C_\kappa C_N e^{\kappa v_0}. \tag{ME54}$$

All spatial summation and tree branching occur in this majorant. There is no further volume-dependent tree count. Choose the target radius small enough that it lies in the physical solution ball and satisfies

$$2A\eta \le \min\{r_N/4,\ r_N^2/(8B)\}. \tag{ME55}$$

When $B = 0$, omit its second restriction. On $0 \le z \le 2A\eta$, the nonlinear derivative of ME54 is at most $4Bz/r_N^2 \le 1/2$. The map preserves that interval. Its positive tree series consequently converges, and its value is at most $2A\eta$. Equivalently, the same contraction constructs the activity-valued solution of ME50. Forgetting the carrier labels gives a solution of the original small weak equation; uniqueness identifies it with the original constrained Wilson solution.

Thus one has the **exact**, absolutely convergent expansion

$$x_u(\theta) = \sum_{X \ni u} x_{u,X}(\theta_X), \qquad \|x\|_{\kappa,\eta} \le 2A\eta. \tag{ME56}$$

Each activity reads finitely many actual current targets, even though the complete solution is not finitely local. Every component of the augmented solution and every original determining equation is represented by the same expansion. In particular, neither the scalar nor the gauge rows have been discarded to obtain locality. Individual activities are not separately constrained minima: only their full sum solves the five rows. No original delta constraint is split into independent constraints for those activities.

The convergence is an all-orders assertion on a nonzero complex target ball. It is stronger than a collection of fixed-order derivative estimates. On the half-radius ball, the positive majorant also yields, for each fixed r,

$$\sup_{u} \sum_{X \ni u} e^{\kappa|X|} \|D^r x_{u,X}\|_{\eta/2} \le r!\,(2/\eta)^r\, 2A\eta. \tag{ME57}$$

The same bound holds with absolute sums of the target-coordinate Taylor kernels: differentiation chooses ordered leaves of the positive tree majorant. No hidden fine-coordinate count appears. Fixed polynomial factors in the carrier size can be paid by choosing a strictly smaller output exponent than κ. The cell paths make each carrier connected, so its original coarse tree size is at most a fixed multiple of $|X|$. ME56 therefore supplies a volume-weighted target-polymer bound after the corresponding fixed choice of exponent.

115.8.2 Local analytic readouts and the source interface

Let $\mathcal{O}$ be a local analytic readout of the five weak unknowns on one fixed connected protected current carrier X_0. Assume its actual weak-coordinate holomorphic radius is $r_O > 0$ and its coefficient majorant is C_O. Expand it at the fixed reference and substitute ME56. Attach X_0 to every tree. After reducing η so that the resulting input norm is below $r_O/2$, the same positive series gives

$$\mathcal{O}(x(\theta)) = \mathcal{O}(0) + \sum_{X \supset X_0} \mathcal{O}_X(\theta_X), \qquad \sum_X e^{\kappa|X|} \|\mathcal{O}_X\|_\eta \le C_O e^{\kappa|X_0|} \frac{C_{X_0}\, 2A\eta/r_O}{1 - C_{X_0}\, 2A\eta/r_O}. \tag{ME58}$$

The fixed constant C_{X_0} accounts for the finite current-cell and type slots of this readout, not its number of microscopic links. The constant term $\mathcal{O}(0)$ is retained. The same argument works for a Banach-valued finite-source coefficient with its genuine declared norm and radius; coefficients, contacts and birth labels are not projected out during the tree expansion.

Local link and bounded-length holonomy readouts are examples: their dependence on b is analytic with uniform weak-coordinate bounds. The physical H/Csharp weak source lifts WH provide more substantive examples, as follows.

Use one fixed protected current box Ω, with the source cutoff and every complete compatible prefix from WH5 contained in it. WH12 constructs an auxiliary massive two-form readout on that box. Its exact endpoint identity WH14 and the directional derivative identity WH15 identify WH18 with the actual quadratic primitive at every admitted physical target. WH20 is the exact cubic completion, including the test derivative, cube defect and quartic completion terms. WH16 and WH19–22 give uniform weak analytic bounds for both roots. The only extra carrier count is the fixed number of current cells in Ω and those complete prefixes, not the number of fine bonds in them.

At a flat zero-multiplier reference these roots have valuation two and three, respectively. The solution ME56 starts at target degree one. Consequently its tree expansion preserves those valuations, and for a sufficiently small common target ball,

$$\begin{aligned}\sum_X e^{\kappa|X|}\|\widehat{H}_X\|_\eta &\le C_H\|f\|_\infty\eta^2,\\ \sum_X e^{\kappa|X|}\|\widehat{C}^\sharp_X\|_\eta &\le C_C\|f\|_{C^1}\eta^3.\end{aligned}\tag{ME59}$$

For example, if a root has degree d, its local weak Taylor coefficients on a spare radius r_O are bounded by a geometric majorant starting at degree d. Substitution of the positive tree majorant $z \le 2A\eta$ gives $Cz^d/(1 - C_{X_0}z/r_O)$, proving ME59. The finite cell-slot and polarization constants are included in C_H, C_C and the once-small target radius. This proves the all-degree envelope before taking any fixed jet.

At a nonflat fixed reference the zero and lower target rows must remain. If t_{ref} includes its physical potential and reference multiplier norms, the corresponding full bounds are $C(t_{\mathrm{ref}}+\eta)^2$ and $C(t_{\mathrm{ref}}+\eta)^3$, with the displayed test norms and fixed carrier factor. A canonical conclusion of order g^dP^d requires **all** of that physical strength, including the reference data, to be $O(gP)$. Smallness of the varying target alone does not supply that conclusion.

Every equality here is an equality of the original primitive evaluated on the same Wilson minimum. A fixed finite joint scalar polynomial, already formed with its prescribed source labels and contacts, can be expanded coefficientwise in the same algebra. The degrees add before the single joint estimate. The original birth coupling, engineering factor and anchor are kept; this corollary does not restart an old birth or remove its prior scale conversion. The auxiliary Hodge inverse changes only the source readout away from the physical graph, and changes no integration kernel or target constraint. A fresh use of a fine Haar contact still needs its actual XP/LM transport; coarse target Cauchy derivatives are not relabeled as arbitrary fine Haar derivatives.

A general inherited paired coefficient is **not** automatically an example. Its original domain may require a curvature supremum and an independently scaled current. A weak $W^{1,p}$ ball is not such a domain, as Section 8 shows. One must either identify an actual weak analytic source lift preserving that coefficient, or prove its full physical-pair pullback with its original source grade and radius. Exponential comparison of two true physical pairs, when available from V7N33–37, helps bound a source comparison on an admitted paired core. It does not identify an arbitrary nonphysical weak Taylor root with that source.

115.8.3 What this repairs and what it does not

ME56 repairs the radius-versus-volume defect in ME48 for the fixed regular target problem. A long inverse propagation pays its own path length, and a high branching order pays its target powers. It does not replace all dependencies by a solid ball of radius equal to the farthest target.

Three model-specific checks are still necessary before this can be used as the all-history mixed entry construction.

First, the reference remains a parameter. If $\bar{U}$ or its multiplier was chosen as a function of the full current field or of the outer complex pair, that dependence remains in $x_{u,X}$ and $\mathcal{O}_X$. It is not removed by their target locality. In particular these are outer-independent carrying functions of LM43 only when their fixed reference data are themselves legitimately fixed or retained in that record. A full original polymer carrier must also localize the reference dependence, using the actual preparation and walls. A path assigned to G_{uv} does not assert that its coefficient depends only on the reference inside that path.

Second, every target used in this expansion belongs to the same small complex regular ball. A rough retained component, a new mixed seam, or an old shrinking R birth radius is not admitted merely because the series converges for the clean reference. The missing domain check is still ME1, or its literal alternative with the same real selectors and source wall data.

Third, the exponential norm does not pay an unspecified retention cost. If a new kernel estimate costs $e^{\gamma|X|}$, ME56 can absorb it from a strict exponent reserve only when γ is below that reserve. A factor $e^{C(1+\log g^{-2})|X|}$ is not automatically of this form uniformly as $g \to 0$. The original canonical coordinate normalizations may cancel such powers, or the existing origin bank may pay a bounded recorded enlargement, but that has to be proved in the actual kernel. ME29 preserves its normalizer exactly; it does not prove the needed conditional density estimate.

Thus the new finite-readset expansion is a quantitative same-Wilson construction on an identified common regular chart. The reference, mixed-domain, source-grade and live-kernel obligations remain visible instead of being included in the definition of its norm.

115.8.4 A joint reference-block and target expansion on one global weak chart

There is a further limited improvement of the reference-locality statement. Suppose the reference itself is supplied in one fixed global potential chart,

$$U = e^{aw}, \qquad \sup_v \|w\|_{W^{1,p}(C_v^+)} \le \epsilon, \qquad \|\lambda\|_{\ell^\infty} \le \epsilon, \qquad p > 8. \tag{ME60}$$

Use a single prescribed gauge and common logarithmic branch on the whole compatible periodic graph. The collars in ME60 have the fixed size needed by the complete original prefixes. This hypothesis does not construct such a gauge for an arbitrary regular configuration or identify different holonomy sectors. The assertion below is uniform in volume, fine cutoff and prefix depth for fields already in this chart.

Write $\zeta = (w, \lambda)$, and let $\mathbb{K}(\zeta)$ be the actual five-type linear matrix with the relative Wilson Hessian and constraint-multiplier Hessian, as in HC36–37. It is an analytic matrix family even when U and λ do not satisfy a stationary equation. Set

$$\mathbb{K}(\zeta) = \mathbb{K}_0 + \mathbb{P}(\zeta), \qquad G_0 = \mathbb{K}_0^{-1}. \tag{ME61}$$

Here $\mathbb{K}_0$ is the full flat typed matrix, including its scalar, gauge, coarse constraint, constant and harmonic coordinates. No gauge row is omitted to obtain an inverse. The constant and harmonic modes are modes of its spatial fields, not additional global variables with a separately assigned local kernel. The site average S retains the scalar cell means, and Q retains the harmonic one-forms. For example, at zero momentum and constant κ, BI1 gives exactly $A = B,\ \phi = k,\ \psi = h,\ \nu = g,\ \mu = f - \kappa B$. The weighted AL19 estimate acts on the full local five-type matrix on these unrestricted spaces. No independent $1/\operatorname{Vol}(T)$ projector is declared exponentially local.

The required perturbation bound is supplied by specific previous estimates. BI17 controls all covariant differential changes in their typed rows, with outer derivatives left in weak form. BI19–22 controls the actual raw intermediate reference fields and the compatible mean rows and adjoints; its bootstrap sums positive powers of the physical step lengths. BI23–25 controls the actual relative Wilson Hessian, including its chart-current term, directly on the weak potential ball. HC34–35 controls the multiplier-Hessian term by $C\|\lambda\|_\infty$. For this use its complete-prefix bounds are the weak-chart BI19–22/CL17 bounds; a curvature supremum is not inferred from ME60.

Section 115.8 / Exact noncommuting target localization on a fixed regular reference

Each of these operators has finite current range once its complete prefix is retained. Their literal coefficient expressions are holomorphic on a common weak ball: for the differential and Wilson terms this is the weak local product calculation in BI16–17 and BI24–25, and for the mean terms it is the convergent actual finite-step expansion with the uniform intermediate-field bootstrap. The number of fine vertices is not summed as independent scalar coefficients. It is controlled by the physical Sobolev and coefficient norms in those estimates.

Use the same restriction and reconstruction convention as in ME49–51. Finite current range, bounded overlap and these local estimates give both absolute typed block sums:

$$\sup_u \sum_v \|\mathbb{P}_{uv}(\zeta)\|_{X_v \to Y_u} + \sup_v \sum_u \|\mathbb{P}_{uv}(\zeta)\|_{X_v \to Y_u} \le C\epsilon. \tag{ME62}$$

These are sums over current cells and five field types. The negative output rows are the recorded coefficient rows, not arbitrary representatives of their quotient norms. A block $\mathbb{P}_{uv}$ reads reference data only in one fixed complete carrier joining its input and output cells. The row and column sums contain only a bounded number of such blocks. Localizing a weak input with the fixed smooth cell maps changes its norm by a fixed constant, which is already included in ME62.

Taylor expansion in the **reference block Banach variables** therefore has the positive coefficient majorant

$$P_\kappa(\epsilon) \le C_P e^{\kappa v_P} \sum_{r \ge 1} (\epsilon / r_P)^r. \tag{ME63}$$

Here v_P is a fixed current-carrier size. The radius r_P includes the finite cell-slot, reconstruction and Banach-polarization losses, just as in ME51. In particular ME63 is uniform in carrier size and microscopic depth. It does not assert an absolute sum of derivatives with respect to all individual fine-link reference coordinates.

Let $C_{0,\kappa}$ be the weighted path row and column majorant of G_0, defined as in ME52. Decrease the weak ceiling once so that

$$C_{0,\kappa} P_\kappa(\epsilon) < \tfrac{1}{2}. \tag{ME64}$$

Then the actual inverse has the exact convergent expansion

$$G(\zeta) = \sum_{n \ge 0} (-G_0 \mathbb{P}(\zeta))^n G_0, \qquad \|G\|^{\mathrm{kernel}}_{\kappa,\epsilon} \le \frac{C_{0,\kappa}}{1 - C_{0,\kappa} P_\kappa(\epsilon)}. \tag{ME65}$$

The kernel norm sums over the forcing cell and over connected carriers, at a fixed output cell, and includes the corresponding column norm. A carrier contains the output and forcing cells, all flat propagation paths, and every reference block appearing in its perturbation vertices. Its coefficient is analytic in only those reference blocks. The bound follows by multiplying the positive path and local-vertex majorants. This proves reference locality of the inverse on the specified chart, rather than assigning a path to a still reference-global coefficient.

Substitute ME65 into the target trees. The local nonlinear vertex coefficients are jointly analytic in ζ and x on the same fixed weak balls. Their Taylor majorant is a product geometric series in the reference and unknown radii, starting at degree two in x. Choosing ϵ as in ME64 and then a smaller η gives the same contraction as ME54–55, with the inverse constant in ME65. Carrier unions remain

connected. This yields an all-degree expansion jointly local in reference blocks and actual current targets.

There is a necessary physical-identity qualification. For arbitrary independent (w, λ), the actual shifted equations have a constant residual $J(U) - Q(U)^{\mathsf{t}}\lambda$ in their first row. The centered target equation ME50 is then an analytic extension obtained by subtracting that residual. On a physical reference satisfying $J(U) = Q(U)^{\mathsf{t}}\lambda$, the residual is exactly zero and the joint expansion equals the original target problem. If the literal uncentered equations are used for independent reference variables, retain their constant residual as an additional local insertion; the same argument gives a bound of order $\eta + \epsilon$. No stationary equation for an arbitrary weak reference is claimed, and no such residual is discarded in a physical integral.

115.8.5 Eliminating the physical reference multiplier in this same chart

In this particular uniform chart the multiplier need not remain a separately global parameter. A smooth, local flat row lift provides an explicit analytic extension.

Let Q_0 be the actual flat compatible full-prefix average, and choose its uniform-grid lift E_0 from MB7–12. This is the construction with smooth interior profiles and their exact two longitudinal moments. It is **not** a discontinuous block hold. In the uniform case no mixed seam correction is needed. The exact flat composition identifies this prefix with its boxed box-and-line average; the moment calculation applies to that same Q_0, not to a substituted nonlinear macro-average. Its columns have fixed current carriers, bounded overlap, and

$$\begin{gathered} Q_0 E_0 = I, \qquad \|E_0\|_{\ell^q \to W^{1,q}} \le C_q \quad (1 < q < \infty), \\ \|E_0\|_{\ell^1 \to L^1} + \|E_0\|_{\ell^\infty \to L^\infty} \le C. \end{gathered} \tag{ME66}$$

The endpoint bounds follow directly from the profiles' uniform supremum, fixed physical support and bounded overlap; they do not require an endpoint Sobolev theorem. The finite exceptional meshes in MB are included. In particular each column is a uniformly bounded $W^{1,p'}$ test form on its fixed carrier. Consequently the actual physical pairing

$$j_0(w) = E_0^{\mathsf{t}} J(e^{aw}) \tag{ME67}$$

is legitimate for the local $W^{-1,p}$ Wilson current. The transpose includes the original fine physical volume weight. BI's divergence-plus-zero-order current representation, paired with the smooth column and its first differences, proves a local analytic bound $\|j_0(w)\|_\infty \le C\epsilon$. It introduces no derivative of a discontinuous box and no inverse fine-mesh factor.

Let $Q(U) = Dq_U(0)$ be the actual compatible physical row at the reference, in current units. Define

$$Q(U)E_0 = I + R(w). \tag{ME68}$$

BI22, the physical L^1/L^∞ lift bounds, and the finite complete-prefix carriers give absolute current row and column sums $O(\epsilon)$ for $R(w)$. Its coefficients read only the reference inside those carriers. Its Taylor series has the same kind of block majorant as ME63. Thus, for a once-smaller weak ceiling,

$$\lambda_{\mathrm{rec}}(w) = (I + R(w))^{-\mathsf{t}} j_0(w) = \sum_{n \ge 0} (-R(w)^{\mathsf{t}})^n j_0(w) \tag{ME69}$$

has a convergent connected-reference-carrier expansion and a uniform norm bounded by $C\epsilon$. The transpose series uses the **column** bound on R; an unproved row-to-column implication is not used.

At an actual physical regular reference the full gauge and Noether rows give the compatible KKT identity $J(U) = Q(U)^{\mathsf{t}}\lambda$, with the HC sign convention. Therefore

$$j_0(w) = (I + R(w))^{\mathsf{t}}\lambda, \qquad \lambda_{\mathrm{rec}}(w) = \lambda. \tag{ME70}$$

This is exact. Away from that physical constraint equation, ME69 is only an analytic multiplier readout; it neither imposes stationarity nor proves a Ward identity. Substituting ME69 into the joint expansion ME65 preserves convergence after reducing the reference radius once, and removes the separately supplied multiplier from its reference carriers on the actual physical branch. The fixed flat lift E_0 and the declared global gauge define this off-graph extension. No off-graph gauge covariance of λ_{rec}, or equivariant gluing of these extensions across different local gauges, is claimed. Physical covariance follows only where its value has been identified with the actual multiplier.

The result narrows the reference-locality gap in one well-defined regular setting. It does not provide a global weak gauge for arbitrary original R records. If the input potential itself is subsequently substituted by a global minimum depending on other retained variables, that new dependence must still be composed with an admitted expansion; ME65 does not make it finitely local by itself. Nor do the reference block Banach bounds give absolute fine-link contact columns, old canonical ages, shrinking R birth domains, selector-compatible mixed preparation, or the normalization bound for an enlarged live kernel. Those remain separate obligations of the same Wilson construction.

Section 116.1 / Purpose and exact scope

The tree localization in the mixed-entry chapter applies to an analytic readout of the weak KKT unknowns. A generic coefficient defined only on a curvature-supremum/current core is not such a readout. The counterexample ME46–47 makes that distinction necessary. This chapter constructs a uniform weak analytic readout for two particular physical observables: the complementary directional-current square H in AI33–35 and the exact cubic completion $C^\sharp$ in AI32a–d. These are the actual bare primitives, with their chosen lattice completions.

The construction is a source-only extension. It changes neither a Wilson action nor a target constraint nor a quantum integration kernel. On every admitted physical KKT solution it agrees exactly with the original observable. Off that graph it retains the compatible multiplier as an additional variable. It is not identified with an arbitrary old independent-current extension merely because the two agree physically.

The main device is a local covariant two-form equation. Its right-hand side uses the actual current equation and the exact lattice Bianchi defect. A mass term makes its auxiliary inverse uniformly bounded. This auxiliary equation reconstructs derivatives of the curvature only for the purpose of evaluating the source. It is not a new gauge-field minimum and imposes no additional constraint on the Wilson measure.

Section 116.10 / Dependencies and independent check

AI4–6, AI16–27 and AI32a–35 give the exact lattice observables and their holomorphic flux/cube identities. CL17 supplies the actual compatible adjoint estimate; V7N3 and MC20 give the physical current identity with the original multiplier and endpoint factors. LM13 supplies the discrete flat div–curl identity used in WH9. The auxiliary source equation WH8–16 and its use for WH18 are new

The mixed-entry and contact the analysis stated the zero-exterior boundary convention, the exact identities WH13–15, the complex inverse argument, the L^4 cube-defect estimate, and the distinction between this physical-equivalent source lift and an arbitrary old independent-current extension.

Section 117.1 / Scope and results

This chapter continues the current/contact obligations It does not assert all-history MT36. It gives three quantitative improvements and two exact alternative representations.

First, a complete sharp-selector boundary current can be estimated by ordinary measures in the complementary failure sector. No surface density, nondegeneracy of a cutoff probe, or multiplication of boundary delta distributions is needed. The estimate applies directly to actual finite Wilson/Haar packets with the LM energy bounds. It also applies to a completed clean analytic packet on a fixed larger fluctuation domain, with its original failed-test reserve. The larger domain and the density being differentiated must be identified; an arbitrary mixed reference density does not acquire these properties from positivity alone.

Second, bare contact joins have an exact engineering and canonical grading. A join of engineering dimensions Δ_E, Δ_j has dimension $\Delta_E + \Delta_j - 4$; its explicit β^{-1} lowers the canonical field grade by two when $\beta = c_\beta g^{-2}$, with its literal fixed convention factor c_β^{-1} retained. Every ordered primitive lower term and actual shifted source anchor remains present. The resulting compact words have a joint energy bound before any weak diagonal operation.

Third, the actual compatible regular averaging prefix has uniformly bounded fixed-order derivative columns, with exponential decay in its depth. The physical-volume factor in CL17 becomes a strict column contraction in dimensionless link coordinates. This closes WCT's one-step-versus-all-depth

derivative issue for this specific forward map. Its equality with a raw physical observation is used only for an admitted terminal gauge-invariant test. A stack of gauge-coupled old records needs its full restoration data.

Fourth, fixed finite bare EOM contacts can be formed before any sharp partition and then carried as unreduced scalar coefficients. This gives an exact all-history scalar realization of those finite identities. It avoids representation-induced selector derivatives, but it does not justify deleting later EOM directions or normalizing them by the old two-dimensional physical center.

Finally, an ordinary scalar measure paired with a whole-current representation can be transported together through a genuine smooth map. This permits failure-sector boundary bounds even for constraint-singular measures, without writing an internal effective density score. The required failure envelopes for both partners and the exact treatment of coordinate-dependent reference factors remain quantitative obligations at an unknown mixed transition.

All delta constraints, source-zero reference bodies, original source anchors, and actual component counts remain part of the record. There is no sourced-island denominator in the arguments below.

117.3.1 Compact Wilson packets

For the bare Wilson density $\rho = e^{-\beta S_0}$ and canonical Haar fields $Y = gX$, LM6–9 gives the following facts for every fixed finite compact packet and ordered derivative list:

$$\|F\| + \|S\| \le C_{I,r} A_I (1 + \mathscr{E})^{D_{I,r}/2}. \tag{C7.7}$$

Here the degree includes every actual score factor and every lowered contact degree. The complete carrier includes the differentiated bond stars, including a distant star if the derivative is distant. Finite source labels are joined before the estimate. These are explicit compact word bounds, not a general claim that an analytic zero jet has an energy envelope.

Suppose an original failed sector has its still-unused positive reserve

$$\int \eta_i \rho e^{\theta \mathscr{E}} \, d\lambda \le B_i. \tag{C7.8}$$

The original Wilson/Gaussian failed-test argument supplies such reserves with its own geometry and normalizations, as recorded in LM and S13. Then C7.5–8 gives

$$\mathcal{M}_r(\mathcal{B}_{\chi,\boldsymbol{Y}}) \le C_{I,r,\theta} A_I \sum_i B_i. \tag{C7.9}$$

No new inverse conditional mass is introduced. If the relevant Wilson action has already been canceled in a reference comparison, C7.7 must instead use the actual compact supremum cost, charged once to its original record as in LM5a. One cannot use a canceled action as the reserve in C7.8.

This proves the weak bound for complete primitive cutoff-boundary packets at their original Wilson coordinates. It is stronger than an assertion that their partition sum cancels: each displayed failure sector has its own quantitative represented bound.

117.3.2 Exact engineering and canonical grades of a bare contact join

There is also a concrete graded contact-creation statement before any weak anchor norm is taken. Work at one common original spacing a and the actual master coupling convention $\beta = c_\beta g^{-2}$. The fixed positive constant c_β is retained; for example a convention for the Wilson plaquette coefficient

may give $c_\beta = 2$. Setting $g = \beta^{-1/2}$ instead would be a different canonical-coordinate convention, not a change to the Wilson measure. Keep AI3 exactly:

$$J_b^A = a^{-3} X_b^A S_0, \qquad \kappa_a = \beta^{-1} a^{-3}.$$

Let the EOM insertion have declared engineering dimension Δ_E, and write its canonically normalized coefficient as

$$V_{E,b}^A = g^{-d_E} a^{3-\Delta_E} v_{E,b}^A(U; f_E). \tag{C7.9a}$$

The finite compact word v_E contains at least $d_E - 1$ plaquette-defect factors, with $d_E \geq 1$. All its test weights, transports and actual discrete derivative conventions are included. Let another integrated bare insertion be

$$O_j(f_j) = g^{-d_j} a^4 \sum_x a^{-\Delta_j} f_j(x) o_{j,x}(U), \tag{C7.9b}$$

where o_j is a fixed finite word family with at least d_j defect factors. These formulas declare every spacing power; a hidden inverse spacing in a coefficient or an unrecorded test derivative would not meet their hypotheses.

AI8–9 gives its exact contact with the EOM insertion:

$$\begin{aligned} \mathcal{C}_{E,j} &= \kappa_a a^4 \sum_{b,A} V_{E,b}^A X_b^A O_j \\ &= a^4 \sum_b a^{-(\Delta_E+\Delta_j-4)} \beta^{-1} g^{-(d_E+d_j)} \sum_{\substack{x,A \\ b \in \operatorname{supp} o_{j,x}}} v_{E,b}^A f_j(x) X_b^A o_{j,x}. \end{aligned} \tag{C7.9c}$$

In particular its merged engineering dimension and its canonical coefficient power are exactly

$$\Delta_{\text{merge}} = \Delta_E + \Delta_j - 4, \qquad \beta^{-1} g^{-(d_E+d_j)} = c_\beta^{-1} g^{-(d_E+d_j-2)}. \tag{C7.9d}$$

The first equality comes from the original two physical sums and κ_a; no informal normalization of a contact delta is used. The literal factor c_β^{-1} belongs to this same original measure and source normalization.

For each fixed b, only a fixed number of roots x occur: the word has a fixed microscopic determining stencil. Their actual offsets from b are $O(a)$, with the fixed stencil constant included. All of these displaced test arguments stay in C7.9c. The product rule removes at most one defect from o_j, so its new compact word has valuation at least $\max(d_E + d_j - 2, 0)$. Counting only finite local incidences and using LM2 then gives a polynomial-energy bound for that exact dimensionless cell coefficient. It uses the energy of the union of its original plaquettes and derivative stars. Translation and finite products of the prescribed smooth tests have their ordinary bounded finite-derivative norms; no diagonal restriction of a previously weakened two-anchor coefficient has occurred.

The same calculation can be iterated for a fixed finite ordered bare contact forest. If r joins merge its participating labels,

$$\Delta = \sum_i \Delta_i - 4r. \tag{C7.9e}$$

Every ordered adjoint lower term must still be retained. Such an internal completion can contribute another factor β^{-1} and reduce the canonical valuation by two without changing the engineering dimension. For a term with t such additional factors, its explicit canonical power and defect bound are

$$c_\beta^{-(r+t)} g^{-d_*}, \qquad d_* = \sum_i d_i - 2(r+t), \qquad \text{val(word)} \geq \max(d_*, 0). \tag{C7.9f}$$

The convention factor is written in the exact coefficient, although its fixed finite power can be absorbed in a numerical estimate. This is a lower bound on valuation, not permission to replace a higher actual valuation by an equality. Haar commutators, scores, signs, original test shifts and lower rows remain ordered exactly as in AI10–12. The term count is finite at fixed label set and order. If $d_* < 0$, its displayed coefficient is a positive power of g, and the compact supremum bound suffices.

For each resulting cell word c, with complete carrier Y_c,

$$|c_\beta^{-(r+t)} g^{-d_*} c(U)| \le C(1 + \mathscr{E}_{Y_c})^{\max(d_*,0)/2}. \tag{C7.9g}$$

The original dimension and physical root weights in C7.9c–e remain outside this dimensionless estimate. Finite canonical adjoints are treated by LM7–9 with their full additional stars. Their joint polynomial is paid by one original positive reserve on the union of the actual records, as in C7.8. Neither each contact edge nor each label receives an independent bank.

This proves bare contact creation grades for the indicated compact word family, including every ordered primitive lower term. It is not a proof that a generated effective coefficient has such a word representation, or that its internal effective score has the required bound. Mixed stored spacings retain their separate original scale factors and are not replaced by the common-a calculation above.

117.3.3 The clean fluctuation density on its full spare tube

On a completed clean current chart use the same real canonical coordinates as the original Gaussian integral. Its smooth, unselected density has the form

$$\rho(B) = c \exp\{-\tfrac{1}{2}(B, DB) + W(B)\}, \tag{C7.10}$$

where the original coordinate, Haar, gauge, and inherited analytic remainders are included in W, and their normalizers remain in c. The constant c is independent of the integrated B. HC supplies the current Gaussian absolute row bounds; MT27–31 and IV supply the actual fresh and inherited local analytic remainder bounds. These statements retain their source-free weak-trajectory hypotheses. For complex admitted external parameters use the original absolute Gaussian majorant in the failure estimate; positivity of a complex density is neither needed nor claimed.

Here is a direct bound which avoids a hidden extensive energy tilt. Fix the whole ordered list of canonical derivative anchors and a finite maximal order r. Use the actual free pivot coordinates B, the original maps C, h, D, and

$$B' = g_k C B - hD(g_k C B). \tag{C7.10a}$$

The original ordinary fluctuation tests imply $\|B\|_\infty \le C_0 P_k$ on their joint success set. In the free pivot coordinates this follows from their free-coordinate entries $B'_{\text{free}} = g_k B$; a fixed alternative block convention only changes C_0. Choose a product extension ψ equal to one on a neighborhood of that set and supported in $\|B\|_\infty \le 2C_0 P_k + 2r + 2$. Its fixed-order coordinate derivatives are bounded independently of the number of variables. All its derivatives are retained.

The entire support of ψ, together with the unit coordinate polydiscs needed for the r derivatives, lies in the same admitted analytic fluctuation domain after the weak floor is chosen once. For an ordinary completed clean readout this follows from its fixed physical spare radius and $g_k P_k \to 0$. For an already completed regular readout with the original R radii, the corresponding margin is LM34a, $g_k P_k/\rho_k \to 0$. This is a margin statement only; it does not identify an uncompleted mixed readout with that domain.

Suppose the incoming completed scalar packet has its already proved graded bound on this whole enlarged tube, with saved amplitude $A_I W_{k,j}$. This condition is verified for RG's admitted completed

contractible pure-clean one-label D6 packet: RG44–55 holds on the full physical core, and its physical/canonical decomposition gives its polynomial evaluation on this tube. Unit canonical coordinate Cauchy may therefore be applied at *every* point in the enlarged tube, not only at $B = 0$. It preserves $A_I W_{k,j}$ in every fixed canonical jet, at the cost of a fixed polynomial in P_k. Exact products of already bounded admitted packets have the same property with their actual product amplitude. This argument does not supply the missing graded bound for an arbitrary previously generated multi-label or contact coefficient.

For completeness the score of C7.10 has the same uniform polynomial bound. The first logarithmic derivative is $-(DB)_a + \partial_a W$. On the support of ψ,

$$|(DB)_a| \le \left(\sum_b |D_{ab}|\right) \|B\|_\infty \le C(1 + P_k).$$

No total energy of all current bonds has been introduced. Write the actual analytic remainder as its original localized carrier sum. A member differentiated at a is zero unless its complete carrier contains that anchor. MT27 and IV10 bound that member by a fixed g_k-weighted polynomial in the local canonical supremum, with its original exponential carrier weight. Unit-disc Cauchy on the same enlarged tube bounds every fixed derivative by the same type of polynomial. The rooted carrier sum at a is uniformly finite. Higher logarithmic derivatives additionally contain only the quadratic precision and these same rooted derivatives of W. The finite ordered product rule, including every derivative of ψ, consequently gives

$$\sup_{\operatorname{supp}\psi} \left\{ \|\psi F_I\| + \left\| \rho^{-1} \mathscr{D}_{\boldsymbol{\partial}} (\psi \rho F_I) \right\| \right\} \le C_{I,r} A_I W_{k,j} (1 + P_k)^{d_{I,r}}. \tag{C7.11}$$

The adjoint is viewed here as its scalar density. Its degree and constant are fixed by the actual finite packet and derivative order. A sum over derivative-anchor components uses its already retained absolute tensor majorant. An extensive source sum keeps its explicit root count.

Apply C7.6a before estimating a boundary. Each elementary ordinary test in C7.10a has the original finite one-step determining stencil. The pivot independence in ME15–25 gives the same locality for its implicit straightening correction; it is not a nonlocal inverse minimum. A fixed list of r derivative anchors therefore meets at most $C_{r,L}$ such probes. Their exact first-failure sectors retain every other original selector as a fixed factor. This argument applies to these ordinary current fluctuation tests. An older selector depending on a nonlocal reconstructed field is not silently included.

Use the original Gaussian failed-probe estimate on this same spare chart, with one unused positive reserve. For definiteness if its unspent part gives

$$\int_{\operatorname{supp}\psi} \chi_{\mathrm{far}} \eta_i |\rho| \, dB \le B_i \, e^{-\delta P_k^2}, \qquad \delta > 0, \tag{C7.11a}$$

where B_i includes every original normalizer, history weight and component count, then

$$(1 + P_k)^{d_{I,r}} e^{-\delta P_k^2} \le C_{I,r,\delta} e^{-\delta P_k^2/2}.$$

A formulation in terms of an original positive quadratic tilt gives the same conclusion. Only the existing reserve for these failed tests is split; a derivative order or source label does not create another reserve. Combining C7.4, C7.11, and this elementary bound gives

$$\mathcal{M}_r(\mathcal{B}_{\chi,I}) \le C_{I,r} A_I W_{k,j} \sum_i B_i e^{-\delta P_k^2/2}. \tag{C7.12}$$

Section 117.3 / Actual Wilson and clean Gaussian instantiations

The original failed-test proof is applied to the actual extended chart density, where its Gaussian lower quadratic bound and the small g_k-weighted interaction bounds still hold. The support of ψ only restricts this integral. Its collar supplies no additional rarity and is charged to the same failed probe.

Thus the saved regular age is preserved when creating these clean whole cutoff-boundary packets. The number of noise coordinates per probe is fixed by L; the number of independent rare components is the original one. The bound does not contain an ambient torus dimension from a global Gaussian energy moment or a global first failure.

The proof does not estimate the internal score of an arbitrary still-live reference kernel in old compact coordinates. Such a density may still contain delta constraints, history-dependent reference denominators, and readouts not admitted on the extension collar. C7.10 cannot be assigned to that kernel by notation. A Gaussian-looking reference numerator with an unproved internal adjoint is outside C7.12, even if WCT has proved its genuine outer field derivatives.

Section 117.7 / What the same argument gives on an entirely clean history

There is an exact filtered identity, but it should not be confused with a closed graded estimate. Fix a finite source label set and form all its primitive joint contacts in the actual bare algebra first. Represent one resulting joint coefficient as a finite sum of whole currents,

$$\mathcal{C}_I = \sum_{r=0}^{m_I} \mathscr{D}^{(r)} \nu_{I,r}. \tag{C7.25}$$

All measures and tensors in this formula contain the original joint Wilson density, constraints, selectors and normalization factors. The derivative anchors and source anchors remain part of the data. This representation involves no product of distributions belonging to separately estimated label blocks.

Let q be one actual smooth admitted regular observation map. The finite chain rule gives tensors $\mathcal{J}_{r,s}(q)$ such that

$$q_*\mathcal{C}_I = \sum_{s=0}^{m_I} \mathscr{D}^{(s)} q_* \left[\sum_{r=s}^{m_I} \mathcal{J}_{r,s}(q) \nu_{I,r} \right]. \tag{C7.26}$$

In ordinary coordinates these are exactly the finite set-partition terms of Faà di Bruno; ordered Haar coordinates retain their invariant-frame and commutator terms. Canonical-coordinate changes retain the literal coupling ratios. In every convention, $\mathcal{J}_{r,0} = 0$ for $r > 0$, because differentiating a constant test gives zero. Thus a positive-order current can lower its derivative order, but cannot create an order-zero row under this whole deterministic pushforward. Original scalar contacts already present in $\nu_{I,0}$ remain scalar rows.

Composition of the actual observation maps proves the same identity through a literal clean prefix. The label set is unchanged and the maximal derivative order does not increase. If a cutoff commutator is exposed, its entire C7.3–5 completion is retained. This gives a finite filtration by label and derivative order, not a finite-dimensional space of coefficient functions. It also does not make the represented weak current norm into a multiplicative algebra for newly formed connected or coincident coefficients.

Here is a concrete reason that C7.16 alone does not close the higher engineering grades, even before any mixed entry is encountered. Two dimension-six labels joined once have dimension eight by C7.9e. In the engineering convention of C7.20, a single merged dimension-eight row carries L^{4n}. The positive-current estimate currently available from C7.19 gives only

$$L^{4n}\lambda^n = (2L)^n, \qquad \lambda = 2L^{-3}. \tag{C7.27}$$

This upper bound gives no age gain. Increasing the incoming derivative order does not by itself improve λ^n to λ^{rn}: nonlinear chain terms can lower the output derivative order.

That distinction is sharp for the analytic information used in the composition theorem. Set $\alpha = L^{-3}$ and $f(u) = \alpha u + cu^2$, with $c \neq 0$. On a sufficiently small fixed disc it has the first-derivative bound 2α and fixed higher-jet bounds of the required kind. Its literal iterates satisfy

$$(f^{\circ n})'(0) = \alpha^n, \qquad (f^{\circ n})''(0) = 2c\,\alpha^{n-1}\frac{1-\alpha^n}{1-\alpha}. \tag{C7.28}$$

Indeed the second derivative obeys $h_{n+1} = \alpha h_n + 2c\alpha^{2n}$, $h_0 = 0$. In the pullback of a second derivative of a test, the second-order output term has coefficient α^{2n}; the first-order output term has coefficient C7.28, of order α^n. Its dimension-eight engineering conversion grows like L^n. An additional fixed small value of c does not alter that conclusion for arbitrary depth.

The same diagnostic survives an exact oddness condition and the absence of a quadratic jet. For $f(u) = \alpha u + cu^3$, its second derivative at zero vanishes at every iterate, but

$$(f^{\circ n})'''(0) = 6c\,\alpha^{n-1}\frac{1-\alpha^{2n}}{1-\alpha^2}. \tag{C7.28a}$$

This follows from $h_{n+1} = \alpha h_n + 6c\alpha^{3n}$, $h_0 = 0$. A sufficiently small fixed disc again gives the derivative ceiling 2α. Thus excluding the quadratic rank-lowering term does not force a higher-order current to acquire one contraction for every derivative. This generic odd-map example also makes no assertion of a noncancellation for the actual centered Wilson mean.

Section 117.7 / What the same argument gives on an entirely clean history

This example diagnoses the insufficiency of the available norm estimate. It is not a counterexample to a cancellation in the actual Wilson contact module. To close that module one must show that its actual rank-lowered coefficients acquire the additional valuation, source moments or spatial moments required by their engineering rows, or establish an exact Ward-compatible projection which places them in the correct lower rows with quantitative bounds. A finite bare contact forest and a triangular derivative-order identity do not supply that multigraded estimate. The primitive weak source lifts WH do not claim it for previously generated effective coefficients either.

Exact momentum moments could repair an offending row if they yield an additional factor $L^{-\gamma n}$ with $L^4\lambda L^{-\gamma} < 1$, or if an exact Ward decomposition places that row in a lower engineering block. The needed moments cannot be assigned solely from its bare contact label. For example, take constant smearings, the actual EOM coefficient $V = J$, and the dimension-six scalar $P = a^4 \sum_b |J_b|^2$. In ordinary dimensionless link coordinates u near the flat field, write $\nabla S_0(u) = H_0 u + O(u^2)$. AI8's contact then has leading term

$$\kappa_a a^4 \sum_b J_b X_b P = \beta^{-1} a^{-4} \left[2\langle u, H_0^3 u \rangle + O(u^3) \right] \tag{C7.28b}$$

at fixed cutoff. This Taylor remainder is not an additional uniform old-age estimate. On a nonzero transverse mode the quadratic term is nonzero. Thus a universal vanishing at zero *source* momentum is false even for an actual bare D8 contact. Its field-momentum factors H_0^3 are still present and could be used in a stronger current/Ward estimate. This calculation neither discards those factors nor proves their uniform nonlinear transport; that is precisely the additional graded statement which is required.

Section 117.8 / A paired-measure alternative to an internal density score

The failure-complement argument has a more general form which can be useful on constrained measures. Let τ and $\nu_1, \dots, \nu_J$ be ordinary finite signed or Banach-valued measures on one actual realized configuration space. The latter may have tensor indices and derivative-anchor records. Suppose

$$\tau = \sum_{j=1}^{J} \mathscr{D}_{\boldsymbol{Y}_j} \nu_j \quad \text{as distributions,} \qquad r_j \geq 1. \tag{C7.29}$$

Each $\boldsymbol{Y}_j$ is a fixed ordered canonical derivative list, with all actual coefficient costs included if another frame is used. Absolute continuity is unnecessary. In particular, the measures may be supported on original constraint manifolds.

For any measurable $0 \leq \chi \leq 1$, define the whole selected residual by operations on ordinary measures:

$$\begin{aligned} \mathcal{B}_\chi &:= \chi\tau - \sum_j \mathscr{D}_{\boldsymbol{Y}_j}(\chi\nu_j) \\ &= -\left[(1-\chi)\tau - \sum_j \mathscr{D}_{\boldsymbol{Y}_j}((1-\chi)\nu_j)\right]. \end{aligned} \tag{C7.30}$$

The second line follows simply by subtracting C7.29. If $m = \max_j r_j$, its represented norm obeys

$$\mathcal{M}_m(\mathcal{B}_\chi) \leq \int (1-\chi)\, d|\tau| + \sum_j \int (1-\chi)\, d|\nu_j|. \tag{C7.31}$$

There is no product of a singular distribution with χ: the scalar companion τ is an ordinary measure, and the other multiplications occur before their distributional derivatives. C7.31 needs failure envelopes for both partners. It does not infer such envelopes from the equality C7.29.

117.8.1 The pair is preserved by a genuine smooth pushforward

Let q be an actual smooth map on a neighborhood of the joint support, with the required jet-weighted variations finite. The chain rule C7.26 gives ordinary output tensor measures ν'_s, $s \geq 1$, for which

$$q_*\tau = \sum_{s\geq 1} \mathscr{D}^{(s)}\nu'_s, \qquad \nu'_s = \sum_j q_*\big[\mathcal{J}_{r_j,s}(q)\nu_j\big]. \tag{C7.32}$$

Thus the scalar companion can be carried along with the current representation without dividing by a density or differentiating an effective logarithm. This is the ordinary distributional pushforward identity; constraint-singular measures cause no additional step.

An output selector must be the same function on the same realized output variables for both partners. Its failure variations satisfy

$$\begin{aligned} \int (1-\chi)\, d|q_*\tau| &\leq \int (1-\chi\circ q)\, d|\tau|, \\ \int (1-\chi)\, d|\nu'_s| &\leq \sum_j \int (1-\chi\circ q)\|\mathcal{J}_{r_j,s}(q)\|\, d|\nu_j|. \end{aligned} \tag{C7.33}$$

If an original selector also reads a retained gauge, constraint, or older variable, that variable must remain in the joint realization. A selector depending on a discarded integration variable is not replaced by a function of the coarse target without an exact identity.

After selection the packet is $\chi\tau = \sum_j \mathscr{D}_{\boldsymbol{Y}_j}(\chi\nu_j) + \mathcal{B}_\chi$. The residual in C7.30 has an ordinary order-zero failure companion as well as its current terms. It is therefore incorrect to say that the selected packet remains a pure divergence. This order-zero residual is retained through every later pushforward, with its actual failure reserve.

117.8.2 Normalization and auxiliary kernels still have to commute correctly

Original source-zero factors must be carried consistently. A constant factor, or an external parameter independent of every displayed derivative, can multiply both partners in C7.29. A coordinate-dependent factor generally cannot. At order one, the exact identity in the present adjoint convention is

$$h\tau = \mathscr{D}_Y(h\nu) + (Yh)\nu \qquad \text{when } \tau = \mathscr{D}_Y\nu. \tag{C7.34}$$

For higher orders the ordered product rule gives the full lower current and scalar terms. A denominator depending on differentiated variables requires those terms and their quantitative bounds; it does not disappear because it is source zero.

Likewise, merely adjoining a normalized conditional kernel $K(x, dz)$ need not preserve C7.29 with partners $\tau(dx)K(x, dz)$ and $\nu_j(dx)K(x, dz)$. Its dependence on x produces the same product-rule commutators. Unit total mass is insufficient, as in WCT's Markov-kernel example. An explicit admitted smooth deterministic realization using independent auxiliary noise would preserve the pair by C7.32, but its actual map and jet bounds would then have to be proved. No such realization is being assigned to an arbitrary live R kernel here.

The alternative is nevertheless substantive. A literal coarea pushforward of a known paired measure needs no internally differentiated scalar density. One can instead preserve its ordinary scalar companion and tensor measures, together with the actual source-zero normalization and all selection residuals. Closing the mixed estimate by this route requires quantitative incoming failure envelopes for **both** partners, the actual smooth joint readout and its admission, and correct transport of every reference factor. These are an alternative to deriving a new internal effective score, not already proved bounds for the unknown mixed partners.

Section 118.1 / Exact target and scope

The target is the one-sided comparison S94 for the original source-zero vacuum,

$$\frac{V_{a,T_*}}{\mathrm{Vol}(T_*)} - \frac{V_{a,qT_*}}{\mathrm{Vol}(qT_*)} \le C, \tag{V7N1}$$

with the same bare Wilson coupling and normalized Haar convention on both physical tori. The original two-sided stability estimate relates this vacuum to the logarithm of that same partition function. Neither a new choice of vacuum scalar nor a comparison only of quadratic approximations proves V7N1.

This chapter proves a further physical primitive comparison. It compares the full nonlinear constrained Wilson action for a current target perturbation on a common protected carrier. The comparison is uniform in the hidden fine cutoff and blocking depth. It then gives the actual nonlinear classical fluctuation-vertex defect, including its coupling power, and a relative integral estimate for common real selectors. It does not assert that every inherited or large-field integrand has already been placed in this common representation. The full S94 obligation remains open at the end of the chapter.

The input reconstruction results used here are the actual compatible maps and typed inverses CL/MB/MC, the physical equality of minimized functions HC32–33, and the actual multiplier-Hessian inverse HC34–38. The argument does not replace the raw differential by the compatible differential on arbitrary forcing. It also does not use the independent current extension of a coefficient away from the physical graph.

Let the target perturbation be supported on current coordinates in Y. Use the weighted local norms

$$\|x\|_{X,\gamma,Y} = \sup_{Q} e^{\gamma \operatorname{dist}(Q,Y)} \|x\|_{X(Q)}, \tag{V7N7}$$

and the corresponding coefficient-localized negative norms for Y. A sufficiently small fixed $\gamma > 0$ leaves the inverse majorant summable. Since the nonlinear vertices have range R_0, one input factor may carry the weight and all other factors use their unweighted bounds. V7N6 consequently gives

$$\|\mathcal{N}_T(x)\|_{Y,\gamma,Y} \le C\|x\|_{X,\mathrm{uloc}}\|x\|_{X,\gamma,Y}. \tag{V7N8}$$

There is only a fixed $e^{\gamma R_0}$ cost. It is independent of the diameter and number of sites of Y. The contraction therefore proves

$$\|x_T(\theta)\|_{X(Q)} \le C\|\theta\|_\infty e^{-\gamma \operatorname{dist}(Q,Y)}. \tag{V7N9}$$

This is a bound for the full nonlinear solution on its fixed complex ball, not just for finitely many Taylor coefficients at zero.

118.3.1 Comparing different physical tori without changing the graph

Suppose Y and its surrounding collar of width $D \ge D_0$ have a common injective lift in T and T'. The collar must contain every complete prefix and incidence carrier; its width is measured in current units. Choose a smooth scalar cutoff equal to one near Y, supported inside this common collar, with its transition at distance at least cD from Y. Its derivatives through the fixed orders used in the typed equations are bounded uniformly. Use its values on the appropriate fine, node, and row coordinates to cut off **all five components** of $x_{T'}(\theta)$, and transplant the result to T. Denote it by x_{app}.

Section 118.3 / Exponential locality of the full nonlinear minimum

The actual equations and their reference coefficients agree on every common complete carrier. Outside the cutoff support the approximate unknown is zero, which is the exact base solution. Near Y it is the original solution with the same target. Therefore its residual is supported in the cutoff annulus, enlarged by R_0.

All rows of this residual must be kept. In particular, $q_T(\chi b) - \theta$, $S_T(\chi\phi)$, and the scalar and gauge commutators are generally nonzero there. The MB weak commutator estimates, V7N6, and V7N9 give

$$\|\mathcal{F}_T(x_{\mathrm{app}};\theta)\|_{Y,\mathrm{uloc}} \le C\|\theta\|_\infty e^{-cD}. \tag{V7N10}$$

For example, a mean row reads only its complete bounded-diameter prefix carrier, and every input in an annulus row has the small bound V7N9. Differential cutoff terms are bounded in their assigned weak types; they are not estimated by a count of fine point values. The nonlinear mean need not commute with the cutoff. Its failure to commute is precisely one of the bounded residual terms.

The original action graph on T has not been excised or given a new boundary condition. Solve the small residual with the same $\mathbb{K}_{\bar\lambda,T}^{-1}$ and nonlinear contraction. Uniqueness of the small solution gives

$$\|x_T(\theta) - x_{\mathrm{app}}\|_{X,\mathrm{uloc}} \le C\|\theta\|_\infty e^{-cD}. \tag{V7N11}$$

In the protected region $x_{\mathrm{app}} = x_{T'}(\theta)$. In particular the current multiplier coordinates satisfy

$$\sup_{y\in Y} |\mu_T(\theta;y) - \mu_{T'}(\theta;y)| \le C\|\theta\|_\infty e^{-cD}. \tag{V7N12}$$

The constants are independent of the cover, fine spacing, and hidden prefix depth. This proof uses one common reference and a controlled residual, rather than interpolating between unrelated action boundaries or identifying two ambient inverses by fiat.

Section 118.4 / The full constrained Wilson action and its nonlinear fluctuation vertex

Let

$$F_T(\theta) = \mathscr{W}_{a_f}\big(U_{k,T}(e^{\theta}\bar{V}_T)\big) \tag{V7N13}$$

be the actual minimized unscaled Wilson action on the admitted branch. The compatible fiber has the same value. The Lagrange multiplier identity in the target logarithm coordinates is exact:

$$D_\theta F_T(\theta) = -\kappa\theta - \bar{\mu} - \mu_T(\theta). \tag{V7N14}$$

This identity follows by differentiating the constraint and the stationary action, so no fine-coordinate action trace is introduced. The signs agree with HC38. At zero it gives $DF_T(0) = \bar{\lambda}$, which is copied to the cover. The multiplier in this identity is in logarithmic target coordinates. A raw coarse Ward multiplier has an additional endpoint-logarithm transpose. HC33 permits differentiation of the common minimized value directly, so that conversion is not silently omitted.

Subtract the base values, and compare the same compactly supported θ on the two tori. Integrating V7N14 on the line $t\theta$, $0 \le t \le 1$, and using V7N12 yields

$$\begin{aligned}\big|[F_T(\theta) - F_T(0)] - [F_{T'}(\theta) - F_{T'}(0)]\big| \\ \le C n_Y \|\theta\|_\infty^2 e^{-cD},\end{aligned} \tag{V7N15}$$

where n_Y is the number of actual current target coordinates in Y, at most a fixed number per current cell. The base values are retained; when the whole background is copied their densities agree exactly

by the local Wilson sum. The comparison in V7N15 is a local change of action, not a comparison of extensive base actions without their volume factors.

The estimate holds uniformly on a common complex target ball. In particular, Cauchy gives full finite target jets of the difference, with constants determined by that fixed ball and with the same exponential factor. This is not the finite-jet matching argument excluded by S100–101: V7N15 already controls the entire function.

118.4.1 The coupling power after subtracting the quadratic form

Write the action remainder in this current chart as

$$R_T(\theta) = F_T(\theta) - F_T(0) - DF_T(0)[\theta] - \tfrac{1}{2}D^2F_T(0)[\theta,\theta]. \tag{V7N16}$$

The analytic difference $R_T - R_{T'}$ has a zero of order three. Apply the one-variable power-series remainder to $t\theta$ on the fixed target ball used in V7N15. For $\|\theta\|_\infty \le \eta_0/2$,

$$|R_T(\theta) - R_{T'}(\theta)| \le C\eta_0^{-1} n_Y \|\theta\|_\infty^3 e^{-cD}. \tag{V7N17}$$

Let B'_{can} denote the canonical current fluctuation before accounting for the local constraint straightening. The physical fluctuation substitution is $\theta = g_k B'_{\mathrm{can}}$. The actual Wilson prefactor in I (0.22), (2.1) is g_k^{-2}. Consequently

$$\left| g_k^{-2}[R_T(g_kB'_{\mathrm{can}}) - R_{T'}(g_kB'_{\mathrm{can}})] \right| \le C g_k n_Y \|B'_{\mathrm{can}}\|_\infty^3 e^{-cD}. \tag{V7N18}$$

There is no inverse bare-coupling cost in this displayed nonlinear fluctuation defect.

The order of the original changes of variables matters. In I pp. 267–268 the unscaled constraint is straightened first, the remaining linear variable is scaled by g_k, and the linear pivots are then eliminated. In the present notation the actual physical target logarithm is

$$\theta_g(B) = g_kCB - hD(g_kCB) = g_k\Theta_{g_k,\bar V}(B), \qquad \Theta_{g,\bar V}(B) = CB - g^{-1}hD(gCB). \tag{V7N18a}$$

Here C is the original linear pivot embedding, h is the original one-step pivot right inverse, and $D(z) = O(z^2)$ is the original analytic straightening correction. The quotient extends analytically at $g = 0$. The factors i are absorbed in the anti-Hermitian Lie algebra convention used throughout this chapter. This notation distinguishes our B'_{can} from the unscaled B' used earlier in I.

On copied nonwrapping current data the two maps, all their fixed-step paths, and their Haar/pivot Jacobians are identical. Their output obeys $\|\Theta_{g,\bar V}(B)\|_\infty \le C\|B\|_\infty$ whenever the original cutoff lies in $g_k\|B\|_\infty \le c\eta_0$. For the frozen logarithmic fluctuation bound $P_k \le C(\log g_k^{-2})^p$ used in LM, this inclusion holds on the entire original cutoff once the common weak-coupling ceiling is fixed sufficiently small: $g_kP_k \le C(\log x_k)^p/\sqrt{x_k} \to 0$. No selector is restricted or renormalized in applying the estimate. Composition in V7N18 therefore keeps the stated form, with a fixed enlargement of Y.

If the straightening has nonlinear terms, first and second target terms must be composed before extracting the quadratic form in B. The copied linear gradient contributes the same expression on both tori. The difference of Hessians supplies exactly the corresponding quadratic Gaussian correction already retained in HC; the remaining composed difference still starts at degree three and obeys V7N18 by the same analytic remainder argument. Thus no outer constraint-Hessian contribution is lost.

The soft-gauge higher terms, Haar density, and the local straightening Jacobian are literal common fixed-step functions on this carrier. They do not create an additional ambient-volume discrepancy.

Section 118.4 / The full constrained Wilson action and its nonlinear fluctuation vertex

They remain in the original integral. This verifies a full nonlinear **classical** clean fluctuation-vertex comparison for the same Wilson action, beyond the clean Gaussian comparison

It does not compare an arbitrary inherited E_k or a live R operator. Those are separate functions of old data inside the integral, even when their local classical reference is the one just compared.

Section 118.8 / Exact naturality of an original finite reference primitive

One part of the original R reference construction can be compared exactly, without a large-body sensitivity estimate. This is a finite-graph statement about the actual reference integral.

In R-II (1.3), the relaxed reference U_0 is the regular minimum of the literal Wilson action on Z, with its actual raw determining constraints fixed outside Λ. The small axial-gauge branch is the one analyzed in (1.12)–(1.13). Retain the complete record $\mathfrak{R}$: dynamic bonds and plaquettes of Z, its action boundary and fixed exterior values, the complete determining set $B_k(Z)$, targets, node and gauge records, Λ, G_0, the action cutoff ζ_0, the original selector, coupling, and normalized Haar convention.

Suppose a second nonwrapping record $\mathfrak{R}'$ is obtained by copying every item through a finite-coordinate bijection ι. This is equality of the actual graphs and constraints, not merely equality of a background on an interior subregion. The original plaquette action and the original finite path averages then satisfy exactly

$$\mathscr{A}_{\mathfrak{R}'}(\iota U) = \mathscr{A}_{\mathfrak{R}}(U), \qquad M_{\mathfrak{R}'}(\iota U) = \iota M_{\mathfrak{R}}(U). \tag{V7N38}$$

For the second identity, apply the same one-step averaging formula successively on the copied complete prefixes. All boundary terms, branches, and endpoint paths are among the copied data. Thus the

constrained variational fibers are in bijection. Uniqueness in the prescribed small regular axial-gauge branch gives

$$U_0' = \iota U_0. \tag{V7N39}$$

The same equality holds for its admitted analytic continuation: the two local analytic equations and their initial branch agree. No new inverse or global gauge choice is being inferred from field-locality alone.

Apply this also to the constrained readout $U_{k,Z}(V'V_0)$ appearing in the original denominator (1.2). Its finite graph, target, and reference are copied. Changing variables $V' \mapsto \iota V'$ in that actual integral preserves the product Haar measure, every gauge delta, the selector, and the action integrand. Consequently its original reference factor and logarithm satisfy

$$I_{\mathfrak{R}'}(\iota b) = I_{\mathfrak{R}}(b), \qquad \log I_{\mathfrak{R}'}(\iota b) = \log I_{\mathfrak{R}}(b). \tag{V7N40}$$

Here b denotes all retained parameters and old coordinates of this reference primitive. The real logarithm is fixed by positivity; its admitted analytic branch is continued from that same value. The explicit scaling, Haar and gauge factors extracted in (1.10) have the same actual numbers of bonds and gauge rows. They are retained, so the logarithmic equality does not discard an extensive constant. On an integer cover of degree q^4, a copied nonwrapping component has exactly q^4 lifts; the log of their product and the physical volume both acquire that factor.

This establishes naturality for an original finite reference primitive wherever its complete record is literally copied and admitted by the original reference construction. It does not require a lower bound on its small mass. It also does not assert that two arbitrarily prepared R bodies have identical records.

118.8.1 What the later localized formulas still contain

The distinction matters for the proposed full structural induction. R-II explicitly states immediately after (1.70) that $E_k(Z)$ collects normalization and vacuum terms whose localizations meet Z. This scalar is retained in the component operation (1.71), together with the inherited potential $V(X^-)$ and the older operations from R-I (1.100). It is not eliminated by saying that the field arguments are restricted to X.

The final R-II operations do preserve naturality once their inputs are natural: copied finite sums and integrals, the displayed component factorization, and the absolutely convergent connected series have termwise coordinate bijections. The scalar bookkeeping is included in that statement. This proves propagation of equality; it does not prove equality of an unidentified primitive input.

The scope II formula (1.6) imports its typed random-walk factors from the background-field paper's (3.107). It states their dependence on the local background and their boundary support. It does not display the full finite-domain inverse, padding, background extension and auxiliary scalar/coarse normalization rules which define those factors. The interior identity following III (3.47) compares the restricted construction with the whole-lattice construction at the same ambient convention. It does not independently supply that missing intertorus seed identification. R-II also imports the preparatory operation R-I (1.100), whose complete formula has not been scope here.

Accordingly V7N38–40 verifies a concrete original R primitive, while the following statement remains unproved: every original zero-wall typed factor and every preparatory retained operation is the same finite-graph function under the proposed copied record, with all inherited scalar atoms attached. Once that statement is established, the subsequent nonlinear and connected formation introduces no new obstruction to structural induction. Without it, declaring $E_k(Z)$ or an old normalized operation identical would assume part of the vacuum comparison being sought.

118.8.2 The original determinant formula confirms the scalar counts

II's final Gaussian-normalization discussion points to formula (63) of the 1985 three-dimensional stability paper. The relevant paragraphs, printed pp.271–272, have now been scope. They first eliminate the original constraints and retain a term $-\sum_c \log\det S(V^{(k)}, b_0(c))$, where S is the actual one-step pivot coefficient. They then write the remaining logarithm as $-\frac{1}{2}\operatorname{Tr}\log(C^{\mathrm{t}}\Delta_k C)$, and localize its resolvent. The absolute Gaussian constant cancels in that paper's relative expression $\log Z(U) - \log Z(1)$.

For the present absolute vacuum comparison that constant must remain. In actual current free coordinates, with their original pivot determinant denoted by J_P, the finite-dimensional identity is

$$\log Z_G = -\log J_P + \frac{n_{\mathrm{free}}}{2}\log(2\pi) - \frac{1}{2}\operatorname{Tr}\log D. \tag{V7N41}$$

It agrees with HC48. The one-step pivot matrices are identical on copied complete mean carriers, and the free-coordinate count is copied with its original row/gauge convention. Thus neither factor has an unidentified ambient constant. All coupling and Haar powers previously extracted from these coordinates remain as their own original count factors.

The determinant identity itself is independent of dimension. For example, if $0 < cI \le D \le CI$ and $\alpha > C$, then

$$-\frac{1}{2}\operatorname{Tr}\log D = -\frac{n_{\mathrm{free}}}{2}\log\alpha + \frac{1}{2}\sum_{r\ge 1}\frac{1}{r}\operatorname{Tr}(I - D/\alpha)^r. \tag{V7N42}$$

The series converges absolutely because $\|I - D/\alpha\| \le 1 - c/\alpha < 1$. It commutes with every finite-coordinate identification. Equivalently one can use the positive resolvent formula of the original paper. These are statements about the specified quadratic form, not an application of its three-dimensional stability theorem in four dimensions.

The original resolvent paragraph still invokes the background propagator formulas (3.183), (3.185) and the corresponding walk expansion. Its local seeds are not explicitly defined there. This new primary check therefore confirms the normalization and pivot part it does not fill the remaining original typed-factor identification in Section 8.1.

Section 118.10 / A direct source-free stability route

A second route would avoid comparing every transition if it supplied a single real scalar density c_a, with exactly the same bare Wilson coupling and normalized Haar convention, such that

$$\left|\frac{\log Z_{a,T}}{\mathrm{Vol}(T)} - c_a\right| \le C_{\mathrm{can}} \quad \text{for } T = qT_*, \tag{V7N43}$$

uniformly in the cutoff and the integer cover q. No source construction would be needed for this estimate. To check precisely what is available, write

$$p_{a,T} = \frac{\log Z_{a,T}}{\mathrm{Vol}(T)}, \qquad v_{a,T} = \frac{V_{a,T}^{\mathrm{pub}}}{\mathrm{Vol}(T)}.$$

On the stated tuned weak trajectory, at a fixed terminal physical spacing, the original upper density bound and the actual terminal Haar-core lower bound S77–80 give

$$|p_{a,T} - v_{a,T}| \le C_{\mathrm{st}}. \tag{V7N44}$$

The original density bound is conditional; V7N44 also uses the admitted core and a cutoff-uniform bound on the terminal inverse coupling. A weak upper ceiling $g_j \le \gamma$, by itself, does not prevent the terminal coupling from tending to zero and does not give that uniform core integral.

Combining V7N43 and V7N44 would give, directly,

$$|v_{a,T_*} - v_{a,qT_*}| \le 2(C_{\mathrm{st}} + C_{\mathrm{can}}). \tag{V7N45}$$

Thus the proposed route is mathematically sufficient. The issue is whether its new common-vacuum assertion V7N43 is among the proved inputs.

It is not the normalization assertion made in the scope original papers. Immediately before Theorem 1 in III, printed p.262, the initial scalar E is defined as the sum of the one-step vacuum expressions when each actual full lattice $T^{(j)}$ is taken as the small-field region. The remaining E_k is obtained by subtracting the expressions generated in the actual small-field regions. This specifies the ambient-torus scalar already retained as $V^{\mathrm{pub}}_{a,T}$. Translation invariance makes its value per volume constant on each torus; it does not show that this value is independent of the ambient torus. The full integral defining N_k, its Gaussian determinant, and the nonlinear and healed vacuum births all remain part of that scalar. The shared infinite-volume beta convention in I (1.20)–(1.22) fixes the running coupling, not the vacuum density.

R-II, printed p.355, proves its density theorem under the assumption that the entire effective coupling sequence remains in its stated small interval. The introduction explicitly retains that assumption because the proof invoked to remove it had not then been published. Neither the argument nor V7N44 upgrades that theorem to an unconditional four-dimensional construction. The issue with the direct route is also not an absence of a positive lower density estimate: R-II (0.1) supplies one, and S77–80 explains its integration on the admitted tuned core. The missing additional statement is the common-vacuum comparison, together with the trajectory and terminal conditions just specified.

Two possible meanings of "infinite-volume counterterm" need to be separated.

- If c_a is a sum of infinite-volume local RG vacuum births, one must construct those births for the same Wilson measure and prove that replacing the finite-torus births changes their total by at most $C\,\mathrm{Vol}(T)$, uniformly in cutoff and covers. This is precisely the nonlinear and reference-normalizer locality issue in Sections 8–9, or the independent second same-measure stability construction S102–104. Defining the symbol c_a does not prove this comparison.
- If $c_a = p_{a,\infty}$ is instead the actual infinite-volume Wilson pressure at each fixed cutoff, the source-zero reflection inequality gives $p_{a,\infty} \le p_{a,qT_*} \le p_{a,T_*}$ along the compatible cover family. In this case it would suffice to prove the single-reference-torus estimate $p_{a,T_*} - p_{a,\infty} \le C$ uniformly in a. This is a weaker sufficient target than V7N43, but it is not supplied by the separate ambient-normalized stability estimates.

For comparison, the direct bare-action estimate has the wrong cutoff dependence. Index the bond variables of a qT_* torus and of q^4 independent periodic T_* tori by the same origins and directions, and use the same product normalized Haar measure. Only plaquettes crossing the block seams have different coordinate products. If the side of T_* is t_*, there are at most $Cq^4(t_*/a)^3$ such plaquettes. The compact-group Wilson plaquette function is bounded, so positive reweighting gives

$$|p_{a,qT_*} - p_{a,T_*}| \le \frac{Cg_0(a)^{-2}}{a^3 t_*}. \tag{V7N46}$$

This proves no ultraviolet-uniform bound. Taking a thermodynamic limit at each fixed a preserves that defect. The clean Gaussian torus estimate repairs its quadratic sector, but cannot by itself replace the full nonlinear estimate.

Finally, bounds of the form V7N43 with a separate C_T for every fixed torus do not yield the cover-uniform S94 unless the relevant constants are uniformly bounded over the covers, or the pressure

Section 118.10 / A direct source-free stability route

ordering in the second case supplies the needed lower comparison. For one specified pair of tori those fixed constants would suffice; S94 asks for more. This direct-route $_{\text{the analysis}}$ therefore gives a precise alternative sufficient target, but does not close S94 or eliminate the outstanding model-specific comparison.

Section 119 / Quantitative comparison with the complete integration

This chapter gives a quantitative part of the source-zero comparison that can be proved without identifying two different physical-torus reference bodies. It concerns the original positive Wilson/R integral on one fixed complete record, its dependence on its actual coefficients, and the subsequent original scalar polymer expansion. The record contains every retained compact variable, Gaussian variable, target constraint, sharp selector, gauge factor, and normalization multiplicity. Its equality in a cross-torus application remains a mathematical assertion to verify. No change of variables or source-dependent denominator is hidden in the comparison.

The results below complement the nonlinear physical minimum comparison in the argument 7 source-zero chapter. They do not turn the clean Gaussian result into a complete proof of

Section 119 / Quantitative comparison with the complete integration

S94.

Section 119.2 / The bound for the actual retained Wilson and Gaussian terms

The exponent in an original R component contains own-scale Wilson actions, Gaussian quadratic forms, analytic inherited terms, and scalar normalization terms. R-II (1.71)–(1.75) gives a positive reference integral with its original selectors fixed. The comparison immediately before (1.73) treats outer-parameter dependence of its quadratic and analytic terms; WCT21–28 already records that particular comparison. Here the same fixed-record calculation permits simultaneous changes in the displayed inherited coefficients as well.

At level j, retain N_j as an upper bound for the actual number of relevant current coordinates or cells, with the fixed group and local-incidence constants displayed through c_G. Suppose the differences of the two exponents have the following pieces:

- a Wilson coefficient difference $\delta\beta_j$;
- a Gaussian precision difference δD_j, in the same original current Gaussian coordinates;
- an inherited analytic-potential difference δV_j;
- the original scalar normalization differences, with their actual multiplicities.

For the particular original scheme used in S94, the first difference vanishes. I (1.20)–(1.22) defines its beta function from the infinite-volume polarization tensor, and I (2.15) uses that function. The same bare Wilson coupling therefore generates the same running sequence on both finite physical tori. With that convention retained, $\delta\beta_j = 0$, and the numerical cutoff thresholds determined by that sequence are also the same. RC8 below records coefficient sensitivity for completeness; it does not introduce a spurious finite-torus coupling extraction into the original construction. The dependence of a cutoff's argument on its actual reference or constraint coordinates is a separate issue.

On a frozen fluctuation cutoff write

$$|B_j|^2 \le N_j P_j^2, \qquad \|\delta D_j\|_{\ell^2\to\ell^2} \le \eta_j. \tag{RC7}$$

There is no choice of a new covariance square root in this formula. If two Gaussian measures were originally normalized, their logarithmic determinant ratio is included in the scalar normalization difference.

For the defining representation of $SU(N)$, the actual real Wilson plaquette term $s(P) = N - \operatorname{Re}\operatorname{tr} P$ obeys $0 \le s(P) \le 2N$. The number of incident plaquettes per lattice cell is fixed. Therefore

$$|\delta\beta_j S_{0,j}| \le c_G N_j |\delta\beta_j|, \qquad \frac{1}{2}|\langle B_j, \delta D_j B_j\rangle| \le \frac{1}{2} N_j P_j^2 \eta_j. \tag{RC8}$$

These are bounds on the original compact fields and real fluctuation variables, including fields not close to the identity. They do not subtract two Wilson actions and then replace each action separately by a factor of size $g_j^{-2} N_j$.

Assume the inherited difference is specified in its actual rooted coefficient norm on this record:

$$\sup_x \sum_{X\ni x} \sup_{\xi\in\operatorname{supp}\lambda_\omega} |\delta V_{j,X}(\xi)| \le \varepsilon_j. \tag{RC9}$$

Every member has a nonempty carrier and all of its determining data are retained. Summing first over its carrier gives the literal estimate $|\sum_X \delta V_{j,X}| \le N_j \varepsilon_j$, up to the fixed number of root types. Empty-carrier constants are instead assigned to the scalar normalization row; they are not erased.

Let ζ_j bound the total absolute scalar difference per current cell, including every original Gaussian, gauge, vacuum, and reference-normalization factor placed at this level. More generally one can retain

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the sum of their separate multiplicities rather than combine them into $N_j\zeta_j$. Equations RC7–9 prove the pointwise bound

$$|W(\xi)| \le E_\omega, \qquad E_\omega = \sum_j N_j \left(c_G|\delta\beta_j| + \tfrac{1}{2}P_j^2\eta_j + \varepsilon_j + \zeta_j \right). \tag{RC10}$$

RC3–6 now apply to this same Wilson integral. Thus the quantitative dependence on these coefficients is established, with actual scale counts and without an inverse reference mass. The estimates do not assume an input bound on a new conditional sourced measure.

RC10 is an implication for differences of the specified input state. It does not prove that two tori have those input differences. Nor does it compare two different constraint fibers or two different cutoff sets. If such records differ, their equality, a controlled change of variables, or an additional measure comparison is still required.

Section 120.1 / Domains of the mixed and marked estimates

The mixed-entry and weak-source arguments use the domains and normalizations specified in ME60?70 and C7.28a?b. The higher-contact diagnostic (C7.27?28) and the paired-measure alternative (C7.29?34) concern different source and measure hypotheses. The uniform original mixed-domain estimate (LM47), the marked transition on every history (MT36), and the complete torus comparison (S94) require additional estimates.

The delicate identity is WH14, not merely its estimate. The right-hand sides in WH11 satisfy, on a physical KKT solution,

$$r_\delta = \delta_U(\chi F), \qquad r_d = d_U(\chi F).$$

Consequently the right-hand side in WH12 is the restriction of $(d_U\delta_U + \delta_U d_U + 1)(\chi F)$. The mass term χF in WH12 is necessary and is present. The zero-exterior realization, the support collar, and the complete differentiated stencils identify this with the operator in WH8. Its inverse then gives WH14 exactly. This calculation neither assumes $d_U^2 = 0$ nor identifies different physical boundary-value problems.

The norms used to construct the inverse are consistent. At the flat reference, zero extension gives the massive discrete div–curl identity. The changes in both d and its algebraic transpose are zeroth-order maps controlled by the potential supremum. Their weak-form perturbation is therefore small from H_a^1 to H_a^{-1}; differentiating the reference potential in an uncontrolled supremum norm is unnecessary. Complex invertibility is a Neumann-series statement, not an assertion that a complex quadratic form is positive. The ordered-form pairing specified in Section 2 fixes the transpose and flat-Laplacian signs.

WH2 also has the advertised fine-mesh behavior. Its linear four-link term is precisely the lattice curl. Every remaining word has at least two potentials and coefficient a^{n-2}, which yields the displayed L^p bound by the potential supremum and translated physical L^p norm. Thus it does not silently require a curvature supremum. WH4 uses the forward and inverse cube identities and the actual transported plaquette deviations. The dimensional factor is a: two dimensionless deviations give a^4, while the additive physical cube difference divides by a^3.

The compatible current in WH5 retains the multiplier and the endpoint tangent factors. It agrees with the physical current only after the physical KKT equations and elimination of ϕ. This distinction is essential. The auxiliary reconstruction then provides enough H^1 control to evaluate the separate directional currents in WH17. It does not replace their complementary square by the square of their sum. The cubic completion WH20 retains its quartic and test-derivative terms.

Limit: the conclusion is a weak analytic extension of the specified bare sources and the expressly listed descendants. It does not identify an arbitrary inherited coefficient away from the physical graph, does not preserve its previously gained age without a further estimate, and does not construct weak extensions of every new contact density. Canonical g^2/g^3 conclusions require the entire reference-plus-fluctuation strength, including the reference multiplier, to have the stated gP scale.

Section 120.3 / Connected-carrier expansion: the absolute derivative issue

The target-tree construction ME49–59 uses an actual fixed-reference KKT inverse and actual local nonlinear weak maps. Assigning an inverse matrix block a connecting path does not make that block a function of reference data on that path. The text preserves this distinction: it localizes the varied target carrier, while the reference remains fixed.

The potentially stronger assertion is the absolute target-kernel sum in ME57. Generic Banach-space Cauchy estimates alone would not bound an absolute sum of all coordinate coefficients. Here the positive tree majorant, the absolute finite-current-coordinate leaf bound, the local multilinear coefficient bounds, and the finite overlap reconstruction supply that stronger estimate. For a nonnegative degree majorant $\sum_d c_d \eta^d$, the elementary inequality

Section 120.3 / Connected-carrier expansion: the absolute derivative issue

$$(d)_r(\eta/2)^{d-r} \le r!(2/\eta)^r \eta^d$$

controls the marked derivative coefficient sum. This justifies the claimed fixed-order absolute estimate without a fine-coordinate count or an unjustified inference from an operator norm. Negative vertices have to be assigned to weak-form carriers before this coefficient estimate; the construction explicitly does so.

The expansion sums to the original nonlinear solution by the same contraction and uniqueness argument. Individual activities are not separate constrained minima, and need not themselves satisfy the original nonlinear constraints. The text does not make that invalid substitution.

Limit: this proves neither joint locality in a changing global reference nor admission of a straddling old source or its cutoff collar. It supplies useful actual primitive roots through WH, not the missing all-history LM47 assertion. The further restricted joint-reference theorem ME60–70 is the analysis in Section 8 below; it does not remove the global-chart prerequisite.

Section 120.4 / Boundary currents and bare contact grades

The whole boundary identity C7.4 is correct as a distributional identity. It follows by subtracting the full unselected adjoint identity and represents the current by an order-zero measure and one whole ordered differential current on the failure sector. C7.5 is their total-variation estimate. It does not multiply undefined derivatives of sharp selectors. The first-failure partition is exact and does not create additional positive reserves.

The reduction to dependent probes in C7.6a requires exact independence from every differentiated coordinate. It cannot be used merely because a remote dependence is exponentially small. The extension cutoff retains its own derivatives in the scalar adjoint packet. Its spare domain and support outside the original success set are what permit the same failed-sector reserve to pay for the extension. Neither this operation nor the fixed-anchor coordinate cutoff introduces an ambient coordinate count.

The bare-contact powers check directly. In C7.9c the two physical sums, the coefficient $a^{3-\Delta_E}$, and $\kappa_a = \beta^{-1}a^{-3}$ give

$$a^4 a^{-(\Delta_E+\Delta_j-4)}\beta^{-1}g^{-(d_E+d_j)}.$$

With the actual convention $\beta = c_\beta g^{-2}$, this contains c_β^{-1} and the canonical power $g^{-(d_E+d_j-2)}$. The factor is explicitly retained in the revised formulas. Derivatives remove at most one defect per hit, so the valuation inequalities are lower bounds, not equalities. Actual shifted test arguments, lower adjoint rows, commutators, and one joint positive reserve remain necessary. The finite bare forest calculation is not a diagonal restriction of an already weakened multi-anchor coefficient.

Limit: C7.9 and C7.12 apply to the actual compact Wilson and admitted completed-clean densities described there. They give no internal score bound for a still-live mixed constrained reference body. The latter's deltas, normalizer, and previously generated marked coefficients cannot be removed by the boundary identity.

Section 120.5 / Prefix jets: valid fixed-order bound, insufficient higher grades

The all-depth estimate must use absolute kernel columns, not only an abstract ℓ^1-operator norm with matrix-valued blocks. C7.13 now explicitly uses the CL entrywise coefficient comparison with the

Section 120.5 / Prefix jets: valid fixed-order bound, insufficient higher grades

positive flat box-line average. Fixed color dimension is absorbed in the perturbation before choosing the strict $\lambda = 2L^{-3} < 1$ margin. The partition recurrence then gives $\Gamma_r(n) \le A_r\lambda^n$ for each fixed order: its nonlinear forcing contains products of at least two lower-order decays and is summable against the linear recurrence.

Endpoint chart and Haar changes occur once at input and output. The physical coupling factors in C7.19, the raw/compatible gauge relation, and the restriction to actual coherent clean prefixes are indispensable. In particular, a mixed identity-plus-crossing map is not automatically contractive in this same column norm.

The added C7.27–28 identifies an actual insufficiency in the available implication. A dimension-eight contact requires an engineering factor L^{4n}, whereas the proved positive current bound supplies only λ^n. The resulting $(2L)^n$ is not an age gain. For $f(u) = \alpha u + cu^2$,

$$(f^{\circ n})''(0) = 2c\alpha^{n-1}\frac{1-\alpha^n}{1-\alpha}$$

is the correct solution of the iterate recurrence. It demonstrates why an incoming second derivative can feed an output first derivative with only first-order decay. Increasing the maximum derivative order does not repair this loss.

This is an obstruction to a proposed inference from the existing jet norm, not a counterexample to the Wilson model. A successful higher-contact argument needs additional actual graded cancellations, moments, valuations, or a quantitative Ward-compatible lower-row projection. It cannot cite the existing prefix theorem alone, even on an entirely clean history.

The added odd-map identity C7.28a also checks: its third derivative obeys $h_{n+1} = \alpha h_n + 6c\alpha^{3n}$, hence has only α^n-order decay. An exact absence of the quadratic jet therefore does not by itself repair the argument. The actual bare contact C7.28b has the stated powers as well. Since $P = a^{-2}\langle u, H_0^2 u\rangle + O(u^3)$ at fixed cutoff and $J = a^{-3}H_0 u + O(u^2)$, its contact is $2\beta^{-1}a^{-4}\langle u, H_0^3 u\rangle$ at quadratic order. It is nonzero on a transverse mode with constant source tests. This rules out a universal zero-source-momentum moment; it does not remove the displayed field-momentum factors or prove their nonlinear graded transport.

Section 120.6 / Nonlinear torus primitive and actual scalar records

The shifted KKT comparison retains the constraint-Hessian term and all five equation rows. Cutting off all unknowns leaves annular residuals which are corrected by the original operator; it does not replace the physical problem by a problem on a cut graph. The multiplier formula for the target derivative gives the quadratic smallness and the original Wilson prefactor in the action comparison.

The composition with the actual finite-current straightening requires a check when the background action gradient is nonzero. Its contributions cancel in the intertorus difference because the reference gradient and the entire straightening map are literally copied. The quadratic term remains in the Gaussian comparison; the remaining normalized nonlinear defect has the claimed cubic coupling order. This would not hold merely by dropping each torus's linear term independently.

The boundary-data extension compares solutions in the same fixed reference and retains the actual restored gauges. Its curvature comparison uses the physical current equation and the exact cube defect after the weak solution comparison. It does not infer a complex curvature supremum from an arbitrary weak potential norm. The target-log endpoint factor in the true current is required and is present in V7N33.

Section 120.6 / Nonlinear torus primitive and actual scalar records

For the finite relaxed reference problem, equality of objectives, fibers, denominator integrals and component counts follows under a literal copy of the complete graph and record. This is a valid change-of-variables conclusion within that stated premise, not an identification of arbitrary reference bodies on different tori. The Gaussian normalization V7N41 retains the pivot Jacobian and the $n_{\mathrm{free}} \log(2\pi)/2$ constant. Its logarithm series V7N42 is the correct finite-dimensional determinant identity. Neither statement determines the missing original nonlinear reference vacuum scalar.

RC's positive fixed-record comparison has no inverse small-mass loss because its bounds hold pointwise before integration against the same positive measure. Its complex variant separately needs the stated smallness to prevent zeros. The old-level sum RC13 is paid by $e^{-cL^n d}$, which dominates the actual L^{4n} coordinate growth and fixed logarithmic losses. An adaptive hierarchy needs the corresponding actual distance and count information; current-sized records cannot be assumed.

RC18 is a valid rooted cluster derivative estimate under the common convergence majorant. It includes the vacuum logarithm. The added weighted criterion is essential for stronger nonconstant locality moments; ordinary RC15 alone does not give those moments. Different torus index sets must be matched on nonwrapping lifted carriers with their multiplicities, with unmatched carriers controlled separately.

Section 120.7 / Paired constrained measures and normalization commutators

C7.29–34 gives a valid alternative to assuming a smooth internal effective density. If the scalar companion τ and the tensor partners ν_j are ordinary finite measures and $\tau = \sum_j \mathscr{D}_j \nu_j$ as distributions, their common selection has the exact complementary residual C7.30. Only ordinary measures are multiplied by the measurable selector. The represented estimate C7.31 consequently follows by total variation, even when the measures are supported on constraints. The equality alone gives no failed-sector bound for either partner.

A genuine smooth joint map preserves this identity by the full chain rule, with its required jet-weighted variations finite. The weighted pushforward variation inequalities C7.33 have the right direction. A selector which reads old, gauge or constraint variables requires those variables in the same joint realization. After selection, the complementary residual contains an ordinary order-zero failure measure. It cannot be omitted on the ground that the original unselected packet was a pure divergence.

The normalization warning is necessary and its sign is correct. In the convention $\langle \mathscr{D}_Y \nu, \varphi \rangle = \int Y \varphi \, d\nu$,

$$\langle h\tau, \varphi \rangle = \int Y(h\varphi)\, d\nu = \langle \mathscr{D}_Y(h\nu) + (Yh)\nu, \varphi \rangle.$$

Thus a source-zero denominator depending on differentiated coordinates still produces a commutator. A normalized conditional kernel depending on those coordinates produces analogous terms; unit total mass does not remove them. An independent-noise smooth realization is a possible exact transport mechanism only after its actual map, admission and jets are established. The text does not assign such a realization to an arbitrary original R kernel.

The analysis now retains this alternative explicitly: the missing mixed step needs either controlled internal adjoints or a normalization-preserving paired representation with both quantitative failure envelopes. It does not incorrectly declare an internal scalar score the only possible route, and it does not claim that the paired route is already closed.

Section 120.8 / Authoritative scope and claims which remain unproved

Two precision issues identified in the initial the analysis scope were corrected The absolute prefix column sum is now attributed to the actual box-line coefficient comparison, rather than to an abstract matrix-valued L^1 norm alone. The scalar diagnostic now uses $\alpha = L^{-3}$, $\lambda = 2\alpha$, and $f(u) = \alpha u + cu^2$, so it really satisfies the stated first-derivative bound on a nonzero small complex disc. The iterate formula has the same α throughout. These corrections make the summary consistent with the detailed chapter.

The authoritative chapter preserves the fixed-reference scope of target localization, the primitive-only source lift, actual constraint/normalizer records, higher-contact limitation and common-measure premise of the scalar comparison. Its opening and the analysis correctly retain the incomplete scope.

Obligation	What the new arguments do not provide
LM47	Admission of every previously generated marked readout and its finite jets on the actual common mixed entry, failure collar and complex domain, with its saved grade.
All-history MT36	Either quantitative internal mixed adjoints or a normalization-preserving paired representation with both failure envelopes; joint jets through every actual mixed transition; a multigraded higher-source/contact/current system that closes under rank lowering and physical quotient operations.
S94	A comparison of the complete inherited and relaxed records on different physical tori, including changed constraints/selectors when present, all reference denominators, original vacuum increments and actual component counts.

The new results narrow these obligations and settle useful primitive and fixed-record implications. They do not collectively imply the missing model-specific bounds. The analysis must therefore continue to state that the interacting four-dimensional existence proof is incomplete.

Section 120.9 / Further the joint-reference extension ME60-70

The additional theorem has a stronger locality conclusion than ME49–59 under a stronger, explicit chart hypothesis: the entire periodic reference is already represented by one prescribed global small potential chart, with uniformly local physical $W^{1,p}$ norm and a small coarse multiplier. It does not construct that chart for arbitrary histories. Within this scope, the new derivation checks as follows.

120.9.1 The flat inverse includes the zero modes correctly

AL9–19 and BI1–8 concern the full local five-type matrix on unrestricted spatial field spaces. Constants and harmonic modes are included as modes of those fields; they are not extra global variables to which

a local kernel is assigned. The site constraints retain scalar means in every current cell, while the box-and-line constraint controls the harmonic one-forms. With constant coarse mass, the zero-momentum equations explicitly give

$$A = B, \qquad \phi = k, \qquad \psi = h, \qquad \nu = g, \qquad \mu = f - \kappa B.$$

Thus no singular zero-momentum scalar inverse is being used. AL19 conjugates the full local matrix on these unrestricted spaces by one common smooth spatial weight. Its uniformly small commutator and the full energy inverse give the exponential majorant, and BI's local regularity argument gives the typed Banach majorant. The same pointwise distance bound gives both current row and column sums. No separate $1/\operatorname{Vol}(T)$ projector is declared to decay exponentially. Keeping the full system, rather than independently localizing a mean-zero inverse and its constant projector, is essential here.

120.9.2 The reference perturbation has actual local block bounds

The new perturbation is the sum of the actual covariant differential changes in BI17, complete compatible averaging changes in BI19–22, the relative Wilson Hessian in BI23–25, and the constraint-multiplier Hessian in HC34–35. The weak-chart prefix estimates replace any need to infer a curvature supremum from the weak potential norm. The latter Hessian is evaluated at the diagonal reference input, precisely the scope of HC34–35.

Their actual coefficient representations have bounded current range when complete prefixes are retained. Filing the negative rows in coefficient form before localization, and using the fixed smooth restriction/reassembly maps, gives the individual typed block bounds. Each block has only a fixed number of current neighbors, so both absolute block sums in ME62 follow. This is not a conversion of a global Banach norm into an absolute sum over microscopic coordinates.

The holomorphic coefficient estimates give the reference-block Taylor majorant ME63 with the finite cell-slot and polarization losses included. Combining it with the flat path majorant gives the convergent Neumann series ME65 after the stated one-time decrease in the weak ceiling. Every reference dependence then occurs in an explicitly carried local perturbation vertex. The flat propagation paths add no hidden varying reference data.

The nonlinear substitution is also correctly centered. For independent reference data the residual $J(U) - Q(U)^{\mathsf{t}}\lambda$ need not vanish. The new text expressly identifies the centered equation as an off-graph analytic extension and retains the residual as a local insertion if the actual uncentered equation is used. The equality with the original physical target problem is asserted only when that residual vanishes. This avoids treating an arbitrary weak reference as a stationary point.

120.9.3 The multiplier readout uses a legitimate smooth test lift

The uniform-grid specialization of MB7–12 supplies a right inverse E_0 of the same full flat boxed prefix. Its smooth profiles have fixed physical support, uniformly bounded first differences and bounded overlap, including the finitely many exceptional fine meshes. Their two moment identities give $Q_0 E_0 = I$. In particular the columns are admissible $W^{1,p'}$ tests for the actual local weak current.

Consequently $j_0 = E_0^{\mathsf{t}} J(U)$ is a legitimate analytic weak pairing with the original fine volume weight. It has no discontinuous-hold derivative loss. The profile supremum, support and overlap also give the displayed physical L^1 and L^∞ endpoint bounds. Together with BI22 these prove that

Section 120.9 / Further the joint-reference extension ME60-70

$R = (Q(U) - Q_0)E_0$ has finite current range and absolute row and column sums of size $O(\epsilon)$. Thus its transpose Neumann series has the needed carrier majorant.

At a physical reference, testing the actual compatible equation $J(U) = Q(U)^{\mathsf{t}}\lambda$ against E_0 gives exactly

$$E_0^{\mathsf{t}} J(U) = (Q(U)E_0)^{\mathsf{t}}\lambda, \qquad \lambda_{\mathrm{rec}} = (I + R)^{-\mathsf{t}} j_0 = \lambda.$$

The relative chart is evaluated at its zero tangent input, so the physical tangent conventions in these formulas agree; no unrestricted raw differential is substituted for the compatible row. The reconstruction has only an off-graph analytic-readout meaning away from that equation. The text now explicitly disclaims off-graph gauge covariance and equivariant gluing across different analysis gauges, since the flat lift and global gauge define this extension.

Conclusion of this additional the analysis: ME60–70 proves the stated joint reference-block and target-carrier expansion, and the multiplier identification, on the declared global weak chart using the cited earlier estimates. It does not supply a global gauge for arbitrary mixed records, absolute fine-link contact columns, saved source age, original R birth domains, selector-compatible preparation or enlarged-kernel normalization. Substituting a reference which is itself an unspecified global function of old retained variables still requires a valid composition estimate. LM47, all-history MT36 and complete S94 consequently remain open.

121.1.1 A false locality shortcut is now ruled out inside the Wilson model

RG37's representative is local in a background connection, including its actual boundary trace. That does not imply a finite read set in the current variables which determine the background minimum. ME6–12 computes this distinction in a transverse Cartan sector of the same flat Wilson Hessian.

With literal block means v_m, the crossing curvature satisfies

$$A_L g_{m-1} + B_L g_m + A_L g_{m+1} = v_{m+1} - v_m, \quad A_L = \frac{L^2-1}{6L}, \quad B_L = \frac{2L^2+1}{3L}. \tag{R7.1}$$

Its inverse kernel is a nonzero alternating exponential. On a compatible even torus the local curvature responds to a target at every distance. The corresponding plaquette energy has a nonzero second derivative with respect to that remote target. Thus strict finite current-variable locality cannot be inferred from background locality. This is an exact obstruction to a proposed shortcut, not a disproof of LM47.

121.1.2 The finite-variable mixed map itself is now explicit

For a genuine finite current read set, the original one-step averages have the pivot property ME15: a row reads its own central crossing pivot and no other row's pivot. Closing the retained set under the omitted row read sets terminates after one enlargement. With that literal closure H^+, ME19–20 gives

$$M_H(v) = (v_{H^+}, q_{O_H}(v)), \qquad q = r_H \circ M_H. \tag{R7.2}$$

There is a block-shaped version with identity rows inside a buffered region and coarse rows crossing or lying outside it.

At flat field the own-pivot derivative in group coordinates is $L^{-3}I$. On its complete regular chart the outside-pivot matrix stays block diagonal and uniformly invertible. ME24–31 therefore gives a genuine fixed-step coarea chart with the original Haar Jacobian. For the original history measure,

$$\nu_H = (M_H)_* \mu, \qquad (r_H)_* \nu_H = q_* \mu, \qquad \nu_H(1) = \mu(1). \tag{R7.3}$$

The omitted inside target constraints remain as actual deltas until they are integrated. Every old selector, gauge factor and reference normalization is carried along. No new island probability or sourced denominator is introduced.

This closes the formal restoration-map question for the specified one-step geometry. It does not turn that chart into an original analytic R kernel. A sharp factor becomes $\chi(\Psi_H(h, z, w))$, and an analytic coordinate change does not remove its boundary. A useful live-kernel estimate still needs the actual selector preparation and the Wilson/Gaussian action cancellation.

121.1.3 Connected carriers replace an insufficient spherical expansion

The new V7N boundary-data theorem gives exponential influence decay for the actual nonlinear regular minimum. An exact telescoping expansion in target balls follows. But a radius-R ball in four dimensions contains order R^4 cells, so a bound e^{-cR} does not pay a carrier weight $e^{\kappa|X|}$. ME48 records this quantitative failure.

ME49–57 instead expands the actual five-type weak equation into connected carriers. Each inverse block is assigned a current-cell path; each nonlinear vertex retains its complete local read set. The negative rows are localized in their actual weak coefficient representations before taking norms. Bounded restriction and reassembly, the finite number of current-cell slots per vertex, and the Banach polarization factor are all included.

The positive all-order majorant is

$$z \le A\eta + B \sum_{q \ge 2} (z/r_N)^q. \tag{R7.4}$$

For the explicit small radius in ME55 it has a fixed point at most $2A\eta$. Uniqueness of the original weak solution then gives

$$x_u(\theta) = \sum_{X \ni u} x_{u,X}(\theta_X), \qquad \sup_u \sum_{X \ni u} e^{\kappa|X|} \sup_{\|\theta_X\| \le \eta} \|x_{u,X}\| \le 2A\eta. \tag{R7.5}$$

This is an exact convergent expansion on a nonzero complex target ball, uniform in hidden fine cutoff and prefix depth. Fixed-order target jets have the corresponding absolute majorant.

Its domain is one fixed admitted regular reference. Its carriers describe the varied targets. The inverse coefficients can still depend on the whole fixed reference and its multiplier. Assigning a path to an inverse matrix element does not localize that reference dependence. Nor does the target norm absorb an unspecified cost $e^{C(1+\log g^{-2})|X|}$ of a new retained kernel.

121.1.4 Reference dependence is also localized on one common small weak chart

ME60–70 proves a further, more restricted result. Suppose the whole reference has already been supplied in one common prescribed gauge with a small uniformly local $W^{1,p}$ potential and a small compatible multiplier. The actual five-type matrix is $\mathbb{K}_0 + \mathbb{P}(w, \lambda)$. BI and HC give the finite-current-range weak coefficient bounds for $\mathbb{P}$, including the constraint-multiplier Hessian and every scalar row. Its local reference-block Taylor majorant is $P_\kappa(\epsilon)$. If the weak ceiling is decreased once so that $C_{0,\kappa} P_\kappa(\epsilon) < 1/2$, the actual inverse is the convergent Neumann series

$$G(w, \lambda) = \sum_{n \ge 0} (-G_0 \mathbb{P}(w, \lambda))^n G_0.$$

Each term has a connected carrier containing its propagation paths and all reference blocks which its vertices read. Substitution in the nonlinear target trees gives joint reference-block and target locality on that chart.

The physical reference multiplier can also be reconstructed there. The smooth local flat row lift E_0 has uniform weak test norms, and $Q(U)E_0 = I + R(w)$ has finite-range row and column bounds. Thus $(I + R(w))^{-\mathsf{t}} E_0^{\mathsf{t}} J(U)$ has a convergent local reference expansion and equals the original multiplier whenever $J(U) = Q(U)^{\mathsf{t}} \lambda$. Off that physical graph it is only a chart-analytic readout.

This closes reference locality in the specified global weak chart. It does not construct that chart for arbitrary original mixed records, establish compatibility between different gauge patches, or admit a rough seam or old shrinking source domain. Reference block Banach derivatives are also not absolute fine-link contact columns. Those restrictions remain part of LM47 and MT36.

Section 121.2 / The two primitive sources have a genuine weak analytic root

The WH chapter supplies a missing interface for using R7.5 with the actual quadratic H and cubic completion $C^\sharp$. It does not infer a curvature supremum from an arbitrary weak potential norm.

On one fixed protected box, let $F(V)$ be the exact plaquette flux and $B(V) = d_V F(V)$ its exact cube defect. On a weak $W^{1,p}$ connection ball, $p > 8$, the actual plaquette word series gives

$$\|F(V)\|_{L^p} + \|a^{-2}(P(V)^{\pm 1} - 1)\|_{L^p} \le C\epsilon, \qquad \|B(V)\|_{L^{p/2}} \le Ca\epsilon^2. \tag{R7.6}$$

The current proxy is the original compatible adjoint with every logarithmic tangent factor retained:

$$\mathcal{J}_U(b,\mu) = -E(b)^{-\mathsf{t}} Dq_U(b)^{\mathsf{t}}(\kappa q_U(b) + \bar{\mu} + \mu). \tag{R7.7}$$

It equals the physical Wilson current at an actual KKT solution. Away from that graph it is an independent analytic source readout.

WH11 constructs the exact divergence and curl data for $\chi F(V)$, using R7.7 and the retained cube defect. WH12 solves a massive covariant two-form equation with componentwise zero-exterior Dirichlet data. Its precise mass term is included. At a physical KKT point the resulting form is exactly $\chi F(V)$; on the source carrier its differentiated flux is therefore the original one. On arbitrary weak inputs it has a uniform H^1 bound. The complex inverse is obtained by a small bilinear-operator perturbation, not by assigning positivity to a complex Wilson action.

The reconstructed directional current pieces give a weak analytic extension of the actual H, preserving its separate hypercubic direction. The exact smeared AI32a expression gives the weak analytic $C^\sharp$ directly, including all cubic, quartic, cube and test-derivative terms. At the flat reference their valuations are at least two and three. In that flat-reference case ME59 gives

$$\sum_X e^{\kappa|X|}\|\widehat{H}_X\|_\eta \le C\eta^2, \qquad \sum_X e^{\kappa|X|}\|\widehat{C}^\sharp_X\|_\eta \le C\eta^3, \tag{R7.8}$$

with the displayed physical test norms and original source normalizations restored.

The canonical conclusion requires the entire reference and multiplier strength, as well as the fluctuation, to be $O(gP)$. A reference of fixed strength ϵ can still contribute ϵ^2/g^2 to a normalized quadratic source. The physical and canonical cores remain different.

This construction proves a primitive source lift and its exact physical identity. It does not prove that every previously generated paired coefficient has a weak lift with its old age, nor that all higher contact densities admit the same representation. It introduces no change to a Wilson action, averaging constraint, or integration kernel.

Section 121.3 / Contacts: creation grades, sharp boundaries, and deep transport

121.3.1 Primitive contact creation now has explicit grades

For the actual bare EOM identity AI8, let the EOM label and another label have engineering dimensions Δ_E, Δ_j and canonical field grades d_E, d_j. C7.9a–g keeps the coefficient $\kappa = (\beta a^3)^{-1}$, the local word incidence count, and every shifted source test. The merged contact has

$$\Delta_{\mathrm{merge}} = \Delta_E + \Delta_j - 4, \qquad d_{\mathrm{merge}} = d_E + d_j - 2. \tag{R7.9}$$

For a fixed ordered contact forest with r joins, its dimension is $\sum_i \Delta_i - 4r$. Internal adjoint completion factors lower the canonical grade separately and keep their exact coefficients. If $\beta = c_\beta g^{-2}$, the factor c_β^{-1} accompanies every corresponding β^{-1}; it is not removed by a change of measure.

The actual compact word is formed before a weak source norm is taken. Its finite-source energy estimate uses the one original reserve for the joint packet. No bounded operation of taking new contact diagonals in an arbitrary weak norm is presumed.

121.3.2 A complete sharp-boundary current is paid by the failed sector

Let the whole ordered adjoint satisfy $\mathscr{D}_{\boldsymbol{Y}}(\rho F\, d\lambda) = \rho S\, d\lambda$. For any sharp selector χ, the full commutator has the exact complementary representation

$$\begin{aligned} \mathcal{B}_{\chi,\boldsymbol{Y}} &= \chi\rho S\, d\lambda - \mathscr{D}_{\boldsymbol{Y}}(\chi\rho F\, d\lambda) \\ &= -\big[(1-\chi)\rho S\, d\lambda - \mathscr{D}_{\boldsymbol{Y}}((1-\chi)\rho F\, d\lambda)\big]. \end{aligned} \tag{R7.10}$$

Thus its represented weak mass is bounded by the ordinary failure-sector integral of $|\rho|(\|S\|+\|F\|)$. The ordered lower terms stay in S. There is no product of surface delta distributions and no requirement that a cutoff probe have a nonvanishing derivative.

C7 verifies this for the actual compact Wilson packets and for the completed clean scalar packet whose incoming RG bounds are already established. In the clean case it checks the full enlarged canonical tube, finite jets around every point of that tube, the internal Gaussian/analytic score, and the original unused failed-probe reserve. Cutoff probes invariant under the derivative anchors remain fixed in the measure; only the genuinely affected probes enter the first-failure partition. This avoids a spurious ambient-volume count. All derivatives of the auxiliary smooth extension are charged to that same original failure sector.

An arbitrary live mixed density may contain unresolved target deltas and reference ratios. Its internal score has not been identified with the clean score, and the new argument does not supply it from an outer-parameter Cauchy estimate.

C7.29–34 supplies a second exact route which does not assume a smooth density. If ordinary signed or Banach-valued measures obey $\tau = \sum_j \mathscr{D}_{\boldsymbol{Y}_j}\nu_j$, the same complementary identity holds with scalar partner τ and current partners ν_j. It bounds the boundary by the sum of their original failure-sector variations. An actual smooth joint pushforward preserves the pair through its full jet rule, even when the original measures are supported on constraints. The selected residual has an order-zero failure companion; it is not declared to remain a pure divergence.

This alternative requires quantitative failure envelopes for both partners and the actual admitted joint map. It also has to carry the source-zero factors correctly. For a coordinate-dependent normalization h, the order-one identity is $h\tau = \mathscr{D}_Y(h\nu) + (Yh)\nu$, with the higher ordered commutators retained in higher degree. Merely adjoining an x-dependent kernel of total mass one does not preserve the pair. Thus the remaining mixed obligation can be formulated as either an internal adjoint bound or this normalization-preserving paired-envelope construction; neither has been established for all original mixed transitions.

121.3.3 The compatible regular prefix has all-depth derivative decay

CL17's absolute comparison with the positive flat box-line coefficients gives, after conversion to dimensionless link coordinates, a strict one-step absolute column bound

$$\sup_b \sum_c \|(D\Phi_i)_{cb}\| \le L^{-3}(1 + C_L a_{i+1} M) \le \lambda = 2L^{-3} < 1. \tag{R7.11}$$

Fixed higher one-step jets have finite absolute kernels. The partition form of the chain rule proves for every fixed order

$$\Gamma_r(n) \le A_r \lambda^n. \tag{R7.12}$$

Input and output invariant-frame conversions occur once, with their lower terms retained. C7.19 gives the full current bound with its canonical coupling ratios. A packet having exactly the D6 engineering conversion L^{2n} obtains the gain $(2/L)^n$. The compatible stack is bounded by a geometric sum. Raw terminal gauge-invariant tests use the exact physical identity; gauge-linked raw records at several depths require their coherent joint restoration.

This is not the higher finite-source transport theorem. Two D6 labels can create a dimension-eight contact, whose engineering factor is L^{4n}. The bound R7.12 alone gives no decay against that factor. Higher derivative order does not by itself supply the missing engineering decay. For $\alpha = L^{-3}$, $\lambda = 2\alpha$,

Section 121.3 / Contacts: creation grades, sharp boundaries, and deep transport

the scalar map $f(u) = \alpha u + cu^2$ has $|f'| \le \lambda$ on a sufficiently small complex disc and satisfies

$$(f^{\circ n})''(0) = 2c\alpha^{n-1}\frac{1-\alpha^n}{1-\alpha}. \tag{R7.13}$$

This diagnoses an insufficient implication; it is not a counterexample to a special Wilson cancellation. The remaining graded induction must prove the additional field valuation, source moments or exact contact projection that the higher engineering rows require.

C7.28a repeats the diagnostic for an odd cubic map, whose quadratic jets vanish identically: its third iterated jet still has only first-order scale decay. The actual bare constant-test contact C7.28b also has a nonzero quadratic term proportional to $\langle u, H_0^3 u\rangle$ on a transverse flat Wilson mode. Thus a universal zero source-test-momentum moment cannot be assigned to every D8 contact. Its field-momentum factors remain available, but their required nonlinear transport has not been proved.

Bare joint contacts may also be formed before the history partition and carried as unreduced scalar coefficients. That exact alternative avoids representation-induced selector derivatives. It leaves an enlarged finite jet system containing null/contact directions; the old two-dimensional physical normalization cannot simply be assigned to that larger system.

Section 121.4 / Source zero: the physical nonlinear comparison is now quantitative

On copied admitted physical references, V7N3–18 compares the actual nonlinear constrained Wilson minimum. Its shifted KKT matrix includes the reference multiplier's constraint Hessian. The torus transplant cuts off all five unknowns and retains every target, scalar, node, gauge and field residual.

For a common protected target perturbation θ,

$$|[F_T(\theta) - F_T(0)] - [F_{T'}(\theta) - F_{T'}(0)]| \le C n_Y \|\theta\|_\infty^2 e^{-cD}. \tag{R7.14}$$

The proof integrates the actual current multiplier. The count n_Y is the number of current target coordinates, not the number of microscopic bonds. The whole analytic function is compared. This is stronger than matching any finite collection of Taylor coefficients.

After subtraction of the quadratic form and the original substitution $\theta_g(B) = g_k CB - hD(g_k CB)$, the nonlinear classical fluctuation defect is

$$C g_k n_Y \|B\|_\infty^3 e^{-cD}. \tag{R7.15}$$

The composed linear and Hessian terms, pivot embedding, Haar Jacobian and local soft-gauge factors are retained. A genuinely common real selector can then be kept fixed while the positive integrands are compared. No inverse of a tiny selected mass is needed.

V7N27–37 also compares arbitrary small target arrays which agree near an observation. It proves exponential differences of the five weak unknowns, the actual current in supremum, and the actual curvature in $W^{1,p}$ and supremum. Both endpoints retain their separate complex physical curvature bound. The proof subtracts the true current and multiplicative cube equations, with their fine-spacing factors before norms are taken.

An actually admitted copied mixed hierarchy has the corresponding statement with its physical row weights r_c^4, dimensionless target $r_c\theta_c$, multiplier $r_c^3\mu_c$, and adaptive distance. It does not admit an arbitrary R history. Moreover an old row can use a path through coarser cells. Its adaptive distance is not automatically the birth-level torus distance T_*/ℓ_j.

Section 121.5 / Absolute normalization, component counts, and the remaining S94 input

V7N38–40 proves equality of one original finite R reference primitive under a literal copy of its complete record. It uses R-II's actual relaxed Wilson minimum and actual denominator integral. The action boundary, raw constraints, axial-gauge branch, selectors, coupling, Haar factors, and all coordinate counts are copied. Uniqueness gives the same reference; a change of those exact integration variables gives the same denominator and prescribed logarithm. A degree-q^4 cover has exactly q^4 lifts of a nonwrapping component.

This is more than a comparison of normalized outer derivatives, but less than equality of every completed R body. R-II keeps the scalar $E_k(Z)$, the sum of normalization and vacuum atoms meeting Z, in its component operation. The older prepared operations and inherited potentials remain there as well.

RC proves the useful quantitative alternative when the actual record is the same and its coefficients differ. For a positive reference measure with all its constraints and selectors, a real perturbation $|W| \le E$ gives

$$|\log Z_1 - \log Z_0| \le E. \tag{R7.16}$$

No lower bound on Z_0 is used beyond positivity. The actual exponent defect includes every Wilson coefficient, current precision, inherited potential and scalar normalization with its own level and component multiplicity. For the original coupling convention the Wilson coefficient differences vanish: I defines the beta function through an infinite-volume polarization tensor, so the same bare coupling gives one volume-independent running sequence.

If a birth-level defect is bounded by $\mathrm{poly}(g_j^{-2}, T_*/\ell_j)e^{-cT_*/\ell_j}$, RC13 sums the actual old counts $N_j \le CN_kL^{4(k-j)}$. It retains every independent copy and normalization multiplicity. An estimate only in the latest scale's adaptive distance does not have that implication.

The completed scalar connected expansion has the explicit Lipschitz estimate

$$|\log \Xi(z_1) - \log \Xi(z_0)| \le \sum_X |z_1(X) - z_0(X)| e^{a(X)} \tag{R7.17}$$

on a common majorized convergence ball. The vacuum logarithm is included. A stronger local-output norm requires the corresponding weighted majorant and spare locality margin; the unweighted hypothesis is not used to manufacture such a margin.

The scope original Gaussian determinant formula confirms the pivot determinant, the actual number of free Gaussian coordinates, and $-\frac{1}{2}\operatorname{Tr}\log D$. Only a relative logarithm permits its common scalar Gaussian constant to cancel. Its absolute vacuum contribution remains in S94.

The remaining source-zero task is to identify or compare all original localized inverse factors and preparatory R records, then propagate the estimates for every inherited coefficient and absolute scalar. The source definitions imported from the 1985 background-propagator paper and R-I have not all been scope or identified here. The finite-reference identity and later formation estimates do not establish those missing primitive identities. Even with the primary definitions available, their required matched domains and scale-sensitive estimates would still need proof.

V7N43–46 also checks a direct source-free stability route. The original density bounds, on their stated trajectory and terminal core, compare the physical pressure with the actual ambient-torus vacuum scalar. III defines that scalar using the full finite lattices at the successive scales. It does not identify it with a common infinite-volume vacuum density. A cutoff-uniform comparison to such a common density would suffice, but remains an additional premise. If one uses the true infinite-volume Wilson pressure instead, reflection reduces the sufficient input to a uniform finite-size pressure bound on the

Section 121.5 / Absolute normalization, component counts, and the remaining S94 input

one reference torus. Direct same-Haar seam comparison gives only $Cg_0(a)^{-2}/(a^3t_*)$, which diverges in the ultraviolet. This alternative therefore does not close the gap. Separate constants for each fixed torus also do not establish the required uniformity in the covering degree.

Therefore the required inequality

$$\frac{V_{a,T_*}}{\mathrm{Vol}(T_*)} - \frac{V_{a,qT_*}}{\mathrm{Vol}(qT_*)} \le C \tag{R7.18}$$

is still open for the complete original construction.

Section 122.1 / Scope and the remaining mixed-history distinction

This chapter continues LM47 after ME60–70. It constructs a weak analytic source extension for an already generated local paired coefficient on a protected current packet. The extension uses the same Wilson action, actual raw determining maps, and a literal pinned fine collar. It does not require a curvature supremum for an arbitrary off-shell weak input.

The construction is source-only. It does not replace an R integral, selector, reference denominator, or physical target. It agrees with the original source on the admitted physical critical graph. For a completed regular scalar packet whose graded estimate was already established by RG, its actual current minimum permits that same estimate to be used on the weak extension, without restarting the birth label.

This is not yet LM47 on every original mixed R domain. The source extension is constructed on a specified local weak chart. Admission of the original prepared readout to that chart, compatible record gluing, and the live kernel bounds remain separate. In particular no previously unproved higher-contact grade is supplied by the new source map.

Section 122.3 / Weak inputs and a reference confined to the fine collar

Let b be a complex fine potential on the complete fixed record, measured in one prescribed flat analysis chart, with

$$\|b\|_{W_a^{1,p}(D^+)} \le t, \qquad p > 8, \qquad 0 < t < t_0. \tag{M8.5}$$

Uniformly local norms can be used on a fixed bounded collection of current boxes. Since the current diameter is fixed, the constants below are independent of a and N. The discrete Sobolev estimate gives $\|b\|_\infty \le C_D t$.

The input need not have uniformly bounded physical curvature. Put $V(b) = e^{ab}$. Choose an auxiliary reference $U_b = e^{a\chi b}$, where $\chi = 1$ on the actual prescribed exterior and **every bare identity row with its complete stencil**, and $\chi = 0$ in the interior. The bare region and the transition have thickness at most a fixed multiple of La. The transition is confined to fine cells and their first adjacent scale; its first differences are at most C/a. Every longer prefix is separated from that transition by the chosen collars. This reference is used only to solve the auxiliary source problem. When $N = 0$, take $U_b = V(b)$ on the entire unit-mesh record; all estimates below are then fixed finite-stencil estimates.

The crucial point is that its sharp fine transition occurs at assigned length a, not at current length one. Every transition plaquette has a finite number of factors $e^{a\chi b}$, so

$$a^2\|F_a(U_b)\|_\infty \le Ca\|b\|_\infty \le C_D at. \tag{M8.6}$$

A first adjacent tile has length at most La, which changes the constant by a fixed power of L. The scaled first differences of its logarithmic potential have the same bound. In the interior the reference is flat. Thus the *adapted* MC reference conditions hold with size $C_D t$, even though the unmodified input $V(b)$ may have no current-scale curvature supremum bound.

The exterior parameters can vary holomorphically with b. During each auxiliary minimization they are held fixed. Their occurrence in a boundary plaquette is a literal finite word, so the bare-tile bounds of MB32 remain uniform. This is a family of the same local Wilson boundary problems, not a variation of an integration measure.

The mixed target is the actual target of $V(b)$, expressed relative to U_b by the MC nodal relative convention. Thus it includes $G_{m,U_b,V(b)}$, every actual endpoint transport, and the prescribed raw data. Its physical row logarithm is, literally,

$$\theta_m(b) = q_{m,U_b}(w_b), \qquad w_b = a^{-1}\log[V(b)U_b^{-1}] = (1-\chi)b. \tag{M8.6a}$$

The equality on the right is bondwise: both exponential factors on that bond are scalar multiples of its same Lie algebra element. There is no independent arbitrary target added here.

The relative potential has $\|w_b\|_{\mathcal{A}^{1,p}_{U_b}} \le C_D t$. Away from the transition this is the constant rescaling M8.8 below. On a transition tile of length $\sigma \asymp a$, the additional derivative is $\sigma^2(D_a\chi)b$, bounded by $C_D at$ after rescaling. The uniformly small reference makes its fixed covariant and analysis norms equivalent. Apply the actual mixed full-space estimate CL17/CL46 to q_{m,U_b} on this adapted weak ball. For a row of assigned length σ, it gives

$$\|\sigma\theta_m(b)_c\| \le C_D t, \qquad \|(0,0,0,0,\theta_m(b))\|_{Y_m,\mathrm{uloc}} \le C_D t. \tag{M8.7}$$

The second norm is the **full five-type forcing norm** MC3, with the target in its fifth row. The second, third and fourth forcing rows remain exactly zero. In particular every site or inward-junction constraint remains its original homogeneous scalar equation; it is not replaced by an independently prescribed node target. The nodal relative factors used to form the fifth row are the actual same prefix factors of CL17/CL46, with their uniformly bounded site-map derivatives. Thus the assertion is stronger than a stand-alone pointwise row bound. On bare identity rows the fifth-row target is zero because $U_b = V(b)$ on their complete stencils. On a nonbare prefix meeting the fine transition, σ is comparable to a, and its actual finite-word estimate proves M8.7. All longer prefixes are protected from that transition by the chosen separation. Their input is controlled by the scaled weak norm; for a subcell of length $\sigma \le 1$,

$$\|\sigma b(\sigma\cdot)\|_{W^{1,p}} \le C\{\sigma\|b\|_\infty + \sigma^{2-4/p}\|D_a b\|_{L^p}\} \le C_D t. \tag{M8.8}$$

The positive power $2-4/p$ prevents a fine-scale loss. All fixed prefix and color constants are included before taking the weak ceiling. M8.7 is a norm bound over the actual local row blocks; it is not a sum over all fine target coordinates.

The maps in M8.6–8 are holomorphic on the fixed weak ball. In particular their complex logarithms remain on one branch after choosing t_0 small once. Their fixed-order Banach derivatives are uniform on a smaller ball. An absolute sum of fine-link derivatives is not asserted.

Solve the inhomogeneous mixed augmented system at the actual target $\theta_m(b)$, using reference U_b and the literal action and boundary just specified. The proof is the MC contraction with its row right-hand side retained, as in RG32–33 and V7N22–26. The norm of the residual and inhomogeneous data is at most $C_D t$ by M8.6–8. The actual mixed inverse, with its prescribed bare rows, gives one fixed contraction ball and therefore a holomorphic solution.

At all reconstruction walls equal to one, the MC Noether elimination still applies to nonzero target data. The chosen test variation preserves the actual compatible row; hence the auxiliary gradient vanishes. The fine nodal gauge is then restored to the literal raw target $M_m(V(b))$. Both endpoint gauge factors are included: the target was formed with $G_{m,U_b,V(b)}$, while the solution has $G_{m,U_b,W}$. Their quotient is the actual nodal correction. On bare nodes both factors are the identity, so the correction fixes the prescribed fine collar.

Call the resulting physical critical field $\mathscr{P}_m(b)$. It obeys

$$\begin{aligned} M_m(\mathscr{P}_m(b)) &= M_m(V(b)),\\ \mathscr{P}_m(b)|_{\text{fixed collar}} &= V(b)|_{\text{fixed collar}},\\ J(\mathscr{P}_m(b)) &= Q_{\mathrm{raw,m}}(\mathscr{P}_m(b))^{\mathsf{t}}\lambda_m(b) \end{aligned} \tag{M8.9}$$

with the declared multiplier sign convention. Its source values may be evaluated before the final gauge restoration when the source is jointly gauge invariant. Gauge covariance of the physical boundary problem and uniqueness give the same source on overlapping admitted gauge charts. This does not construct an admitted chart for an arbitrary original R record.

On the protected current region all determining rows have assigned length one. The actual current equation, its complete prefix adjoint, and the MC interior bootstrap give

$$\|A_{\mathscr{P}_m}\|_{W^{2+s,p}} + \|F_a(\mathscr{P}_m)\|_{W^{1+s,p}} + \|J(\mathscr{P}_m)\|_\infty \le C_D t, \qquad 0 < s < 1/p. \tag{M8.10}$$

The potential norm is in the controlled local analysis gauge. The curvature/current versions use their actual covariant transports. The fine collar itself is not assigned this current-scale bound.

M8.10 uses the physical equation of the newly constructed source readout. It is not a curvature estimate for the arbitrary weak input $V(b)$. The complex estimate is uniform on the same smaller open weak ball because the contraction, the actual current equation and the protected elliptic argument all apply there. At finite cutoff all maps are holomorphic; the displayed uniform bound makes their strong-output Cauchy estimates uniform as well.

122.4.1 A target-preserving scalar slice

The critical-input identification needs a small augmented solution, not merely an appeal to criticality. It can be obtained in the reference $U = U_b$, without an additional curvature hypothesis on the weak input $V = V(b)$.

For an allowed fine gauge h, let $\mathfrak{s}_{U,V}(h)$ be the actual relative nodal cocycle:

$$\mathcal{B}_{m,U}(V^h) = \mathcal{B}_{m,U}(V)^{\mathfrak{s}_{U,V}(h)}. \tag{M8.10a}$$

This includes the two relative prefix factors and the actual raw node evaluation of h, in their original order. Its derivative at the identity is $\mathcal{S}_{U,V}$. On bare nodes the prefix factors are the identity, so the

condition $\mathfrak{s}_{U,V}(h) = I$ literally imposes $h = I$ there. Fixed exterior gauge values are excluded from the unknowns.

Write $w^h = a^{-1}\log(V^h U^{-1})$. Solve

$$\begin{aligned} \Delta_U \psi + S_U^{\mathsf{t}} \nu &= 0, \\ -\delta_U w^h + \psi &= 0, \\ \mathfrak{s}_{U,V}(h) &= I. \end{aligned} \tag{M8.10b}$$

The unknown $\zeta = \log h$ has adapted $W^{2,p}$ type, ψ has the L^p type and scale factor ℓ^2, and ν has the actual node type and scale factor ℓ^4. The three forcing types are respectively $W^{-2,p}$, L^p, and the original node row. At $V = U$ the derivative is the scalar constrained biharmonic saddle system of MB36, with its actual S_U, fixed exterior and bare rows. Eliminating ψ gives its literal fourth-order scalar block.

For $\|w\|_{\mathcal{A}^{1,p}} \le C_D t$, the derivative differs from that block by $O(t)$ in these same types. The derivative of the physical gauge action is d_V followed by the original logarithmic chart derivative; the connection difference and chart factor give the required $W^{2,p} \to W^{1,p}$ perturbation. The relative node perturbation is the full CL17/CL46 site-cocycle bound. Products are bounded after each constant adaptive rescaling because $p > 8$. The nonlinear remainder has the same small Lipschitz bound. These are the MB scalar inverse and the MC weak products, applied to the displayed three equations; no point-sampled continuum gauge parameter is introduced.

The forcing at $h = I$ is $(0, -\delta_U w, 0)$, of size $C_D t$. Hence the scalar contraction gives one holomorphic solution with

$$\|\log h\|_{W^{2,p}_{\mathrm{ad}}} + \|\psi\|_{L^p_{\mathrm{ad}}} + \|\nu\|_{\mathrm{node,ad}} + \|w^h\|_{\mathcal{A}^{1,p}} \le C_D t. \tag{M8.10c}$$

Its cocycle equation preserves exactly the compatible target $q_U(w^h) = q_U(w) = \theta_m(b)$, and its bare values preserve the literal prescribed fine collar. This is a slice through a small physical orbit, not a change of the target fiber.

122.4.2 Exactness on the original current critical graph

Suppose $V(b)$ is itself an admitted critical field for the original current constraint C_N and the same full Wilson action. Freeze its actual exterior record. By M8.4, any allowed variation preserving M_m preserves C_N. Therefore $V(b)$ is critical for the auxiliary constrained boundary problem.

Apply the scalar slice M8.10b to this critical input. The resulting V^h is still critical for its actual raw mixed target. To check constraint qualification, note that $A_{w^h} = Dq_U(w^h)$ is onto: replacing the compatible row and its adjoint in the known MB matrix by their values at w^h is an $O(t)$ typed perturbation. If a raw row covector annihilates $DM_m(V^h)$, the exact raw gauge chain and algebraic surjectivity of allowed node evaluation give its raw nodal Ward identity. That identity removes the endpoint-gauge differential, leaving a covector in the kernel of $A^{\mathsf{t}}_{w^h}$. Surjectivity makes it zero. Thus the raw differential has full row rank on its actual dynamic target rows at finite cutoff, and an actual raw multiplier exists.

Wilson gauge invariance gives that multiplier's nodal Ward equation. Transport it by the actual relative endpoint factors and the actual target logarithm chart derivative. Call the resulting compatible multiplier ℓ; with the declared sign it satisfies

$$E_a(w^h)^{\mathsf{t}} J(V^h) + A^{\mathsf{t}}_{w^h} \ell = 0. \tag{M8.10d}$$

All those row-chart factors and their inverses are uniformly bounded on M8.7. In the weak negative row, $\|E_a(w^h)^{\mathsf{t}} J(V^h)\| \le C_D t$: this is the actual Wilson weak-gradient bound BI23, with the literal bare finite plaquettes. It does not require an unscaled current or curvature supremum for V^h.

Apply the known inverse K_U^{-1} to the vector whose only nonzero component is the compatible multiplier. Since its first row is exactly $A_U^{\mathsf{t}}\ell$,

$$\begin{aligned} \|\ell\|_{\mathrm{row,ad}} &\le C\|A_U^{\mathsf{t}}\ell\|_{W_{\mathrm{ad}}^{-1,p}} \\ &\le C_D t + Ct\|\ell\|_{\mathrm{row,ad}} \le 2C_D t. \end{aligned} \tag{M8.10e}$$

The second inequality uses the full adjoint bound for $A_U - A_{w^h}$; the last follows after decreasing the fixed weak ceiling. In particular the physical multiplier cannot leave the admitted ball while the weak field remains in this slice.

The explicit augmented candidate is now

$$x_{\mathrm{cand}} = (w^h, \psi, 0, \nu, \ell - \kappa\theta_m(b)). \tag{M8.10f}$$

Here MC9's sign is used, so M8.10d plus $\kappa q + \mu = \ell$ is its first row. If the physical KKT multiplier is defined with the opposite sign, both occurrences of ℓ change consistently. The second and third rows are M8.10b, the fourth is $\phi = 0$, and the last is the actual nonzero target. The row mass satisfies $\ell_{\mathrm{cell}}^3\kappa\theta = O(\ell_{\mathrm{cell}}\theta)$, so every component of this candidate is $O(t)$ in MC3. It therefore lies in the same once-enlarged inhomogeneous contraction ball as the auxiliary solution. Uniqueness and the literal raw target restoration give

$$\mathscr{P}_m(b) \simeq V(b) \quad \text{on the protected read sets.} \tag{M8.11}$$

The equivalence is only the prescribed allowed fine gauge, which is fixed on the literal boundary. Every gauge-invariant paired source has exactly the same value. Since the identity holds as a function of all admitted original current variables, its fixed physical derivatives also agree, including all derivatives of the own-current relation. Multiplication by the unchanged kernel and selectors preserves the whole distributional identities. This does not remove any normalization commutator or fine-vector pushforward term.

Section 122.8 / Carrier accounting and why this is not a general finite-halo theorem

The packet window is chosen before the source norm is taken. Shortness is the original bound on the complete relative carrier, including boundary traces, source offsets, transport paths and every prefix. With the source test supported in a current cell and $rD_{\mathrm{rel}} \le cH_0$, all those read sets lie in one fixed protected current box. The auxiliary source readout therefore has fixed current size independently of the number of old microscopic cells inside it.

This does not put an arbitrary inherited polymer in that box. The long-carrier part remains the original exponentially weighted tail, with its full old carrier and every original radius cost. No solid ball of volume D_{rel}^4 is assigned to it. In particular, calling a long polymer a single current root would lose the carrier weight and invalidate a later R normalization count.

The precise original long-tail estimates and their admission conditions are unchanged. A complete first-entry theorem must handle these tails in the original mixed domain or by a genuine controlled retained kernel. The new bounded-packet construction does not prove that step. It also does not authorize a different minimum for each term in a completed quadratic root pairing; the common packet field is essential.

Section 122.9 / Original mixed entry and normalization: the unresolved identification

The source map M8.12 can be used on an actual mixed branch only after its literal input and full record are identified. The currently proved first-entry sector LM43–44 is outer-independent: the whole incoming coefficient and its retained source/current partners must be functions of variables present in that one original joint integral. Its norm must hold on the actual real selector support. The source-only construction above supplies a bounded current readout in its own weak chart, but does not by itself establish that each original R preparation supplies that input or keeps the complete carrier inside its retained variables.

If the whole localized current read set is genuinely retained, its saved grade can be carried as before. For a joint finite-source packet with polynomial envelope of total degree D, the original one-reserve estimate gives

$$\left\| \int K_\omega(z; d\xi) F_{j,e}(\xi) \right\| \le B_\omega \, C_{D,\vartheta} A_j W_{e,j}. \tag{M8.20}$$

Here B_ω is the original joint source-zero majorant and $W_{e,j}$ is the literal saved regular factor. The statement uses no sourced denominator and no newly granted failure reserve. M8.20 is applied only after retained-variable independence is checked; merely intersecting a large-field component is not that check.

Constraint-supported scalar/current partners are kept together. If a source-zero reference factor depends on a differentiated variable, the exact ordered normalization commutators remain, beginning with

$$h\tau = \mathscr{D}_Y(h\nu) + (Yh)\nu. \tag{M8.21}$$

M8.11 preserves the physical endpoint identity as a distribution, but it does not remove the second term in M8.21 or bound an unknown internal mixed derivative of the reference factor. A bound on outer derivatives of a normalized R kernel can be used only when the transported direction is actually one of those outer coordinates.

The original R domain may be described by several local gauges and may contain nontrivial holonomy or a rough retained seam. The present one-chart proof neither produces a common chart there nor makes a block-Banach derivative into a fine-link derivative. These are actual remaining parts of LM47.

Section 122.9 / Original mixed entry and normalization: the unresolved identification

The original selectors and their common complex failure collar must still be checked; a real small-field test alone is not used to assert this inclusion.

Section 122.10 / Conclusion and exact dependency list

The new estimates are M8.6–10 (weak input to a strong protected physical source pair), M8.11–15 (a local weak lift of a previously generated paired coefficient with exact physical identity), M8.18 (the old-radius margin for that explicit readout), and M8.19 (the existing completed regular scalar age bound on that weak root).

Their model-specific inputs are the actual raw mean composition, the MB separated mixed geometry and literal bare inverse, the MC physical Noether elimination and protected bootstrap, its nonzero target extension RG32–33/V7N22–26, and the established RG scalar kernel and source-moment calculation. The complete source and boundary records are retained in every application. The new auxiliary map changes only the source extension away from its original physical graph.

The all-history LM47 assertion, the higher-contact parts of MT36, and the full Yang–Mills construction remain incomplete. A source weak root for an already controlled regular packet is an additional proved input to those problems, not a substitute for original mixed admission, invariant higher grades, or quantum reference estimates.

Section 123.1 / Result of the targeted check

The auxiliary source lift M8 does not yet imply admission at actual first mixed entry. This note isolates two independent missing assertions: a quantitative estimate for the **actual prepared field**, and an exact identification of the source-only readout with the original source at the appropriate endpoint. Neither is replaced by the analyticity of the original unmarked R kernel.

The new calculation is a necessary, gauge-invariant curvature test for the proposed weak-chart admission. It gives the exact assigned-scale weights that an actual selector/preparation estimate must control. No literal configuration satisfying every original first-entry selector and violating that test has been constructed. Thus this is a precise remaining quantitative implication, not a counterexample to LM47.

Section 123.4 / Exact conversion of the assigned-scale curvature moments

Partition the plaquettes read in D into actual primary cells Δ; assign a plaquette on a common face to one cell once. Let $\sigma_\Delta \le 1$ be its original assigned length in current units. Write

$$c_\Delta(\Xi) := \left\| \sigma_\Delta^2 F_a(\Xi)(\sigma_\Delta \cdot) \right\|_{L^p_{a/\sigma_\Delta}(\widehat{\Delta})}, \qquad \widehat{\Delta} = \sigma_\Delta^{-1}\Delta. \tag{AE8.6}$$

Section 123.4 / Exact conversion of the assigned-scale curvature moments

These are the moments of the **actual field** Ξ, not free parameters or upper bounds substituted for it.

The lattice volume factor is explicit:

$$\begin{aligned} c_\Delta(\Xi)^p &= \sigma_\Delta^{2p-4} \sum_{P\in\Delta} a^4 |F_a(\Xi)_P|^p, \\ \|F_a(\Xi)\|^p_{L^p_a(D)} &= \sum_{\Delta\subset D} \sigma_\Delta^{4-2p} c_\Delta(\Xi)^p. \end{aligned} \tag{AE8.7}$$

A bounded-overlap cover gives the corresponding two inequalities with its actual overlap constant. No number of fine cells has been suppressed.

Let $\mathcal{D}_\omega$ be the original proposed outer/source wall domain, specified independently of the desired field bound, and let τ be its proposed weak ceiling. The exact necessary selector/preparation estimate is

$$\sup_{\substack{\xi \text{ in the actual old selector support} \\ (z,\mathbf{s})\in\mathcal{D}_\omega}} \sum_{\Delta\subset D} \sigma_\Delta^{4-2p} c_\Delta\Big(\Gamma^R_{\omega,\mathbf{s}}(z,\xi)\Big)^p \ \le C_D \tau^p, \tag{AE8.8}$$

if that composite is the proposed input. For the original real cutoff one seeks $\tau = C_D g_e P_e$; for a larger complex outer/source ball one uses its actual chosen weak radius, such as $c\rho_e$. These are separate bounds on separately specified domains. If the proposed input is AE8.2, the moments of AE8.2 must instead be estimated. There is no automatic implication from one of these estimates to the other.

M8's strict old-core condition asks additionally for $C_D r t < \rho_j$, $r = L^{j-e}$. The established M8.18 proves this numerical comparison when the actual input estimate has size $t \le C_D g_e P_e$, and its analogous complex-radius comparison is also available. The unresolved step is the estimate for the actual input, not that final numerical comparison.

Section 123.6 / The separate critical-graph and source identity obligation

M8.11 identifies its source retraction with an input on the actual current physical critical graph, for the same action, constraints and boundary. A prepared field at simultaneous localization walls need not be on that graph. Therefore even a proof of AE8.3 would not by itself identify the old source with

$$E_{j,X}(\mathscr{P}_m(\Gamma^R), J_{\mathrm{phys}}(\mathscr{P}_m(\Gamma^R))). \tag{AE8.10}$$

One must first prove the exact endpoint identity using the original current representative and its complete determining records. One must then construct and bound the source-only wall extension whose endpoint is that same representative. Equality of arbitrary off-graph extensions is neither needed nor claimed. The chosen extension must remain compatible with the unchanged prepared kernel, selectors, relative gauge data and normalizing factors. All inherited lower/contact terms and ordered normalization commutators remain in that comparison.

The conclusion of this check is consequently limited but explicit. The actual R field has been identified, the necessary current-scale curvature moment has been calculated with every assigned-scale weight retained, and the remaining source-identity question has been separated from norm admission. The scope range statement and density comparison do not yet prove either remaining assertion. No new LM47 or all-history grade is claimed.

Section 123.7 / The finite-preparation-depth route is not yet verified

R-II p. 381 places failed preparatory tests at $j - N \le m \le j$ for components under R-I (i),(ii). Its p. 384 genealogy argument, before (1.80), uses those conditions with $N = R_j$. These statements do not supply a uniformly fixed source preparation depth.

The interval labels **failure tests**. It is not, as written, a lower bound on the assigned lengths of every primary cell read by the prepared source. In particular, the current field composite and the old arrays still present in T_X must be distinguished. The latter can remain in the original measure without belonging to the source's current field record. Their presence does not disprove a lower length bound for the field; their test labels also do not prove it.

Here is the exact useful implication if the missing geometric statement can be established. Suppose all actual primary cells meeting the complete proposed field record have

$$\sigma_\Delta \ge \sigma_* := L^{-CN_e}, \qquad c_\Delta(\Xi) \le C_0\epsilon_e. \tag{AE8.11}$$

Assume their rescaled shapes have the stated fixed volume bounds and bounded overlap. Since $\sum_\Delta \sigma_\Delta^4 \le C_D$, AE8.7 gives

$$\begin{aligned} \|F_a(\Xi)\|^p_{L^p_a(D)} &\le C\epsilon_e^p \sigma_*^{-2p} \sum_\Delta \sigma_\Delta^4, \\ \|F_a(\Xi)\|_{L^p_a(D)} &\le C_D L^{2CN_e}\epsilon_e. \end{aligned} \tag{AE8.12}$$

Section 123.7 / The finite-preparation-depth route is not yet verified

If the actual own-scale **potential** charts are also given in one coherently glued allowed gauge, the identical calculation for A and $D_a A$ gives $\|A\|_{W_a^{1,p}(D^+)} \le C_D L^{2CN_e}\epsilon_e$. The gauge premise is separate; AE8.12 alone is a curvature estimate.

For a fixed $N_e \le N_*$, this is indeed only a fixed cost. If the original selectors provide $\epsilon_e \le C g_e P_e$, it can be paid by M8's once-chosen weak floor and strict birth-radius comparison. Thus this route would close the **norm part** for the corresponding restricted class if its actual geometric, selector and gauge statements were verified.

The dependence on N_e cannot be discarded for recent shrinking births. The bound just obtained would require, already for $j = e$,

$$L^{2CN_e}(\log x_e)^{p_0-q} \ll 1, \qquad \rho_e \asymp g_e(\log x_e)^q. \tag{AE8.13}$$

Bounded N_e satisfies this after raising the weak floor. If $N_e \le \kappa \log\log x_e$, a larger once-fixed logarithmic exponent $q > p_0 + 2C\kappa \log L$ also suffices, provided that the original source-domain construction permits this choice before its birth cores are fixed. It cannot enlarge an already specified core by a numerical declaration. If instead N_e is comparable to $(\log x_e)^r$ for any fixed $r > 0$, this particular upper bound cannot close AE8.13 for any fixed q. Additional decay of the actual prepared field would then be needed. This last statement concerns the strength of the available bound; it does not assert that a permitted field attains it or violates its source core.

The exact original clauses needed to complete this route are now:

- The beginning of R-I Section 1, conditions (i),(ii) and the choice of N, must identify the preparation depth for the current first-entry record. An index attached to a historical factor is not silently identified with the current entry scale.
- The determining-set construction underlying R-II (1.61)–(1.63) must imply a lower assigned-depth bound on the complete D^+ record through all three joins $B'_k(Y_1)$, $B_1(Y_2)$, and $B_k(Y_3)$, including the actual source boundary and junction traces. Internal microscopic steps of a complete prefix are not assigned primary-cell depths; even a uniformly current-scale minimum has such fine steps.
- R-I (1.64)–(1.69), at the actual final preparation index $n = N$, must supply the needed own-scale bounds and their allowed common gauge on that record, on the old selector support and common complex domain. A failed-test event gives a lower threshold, not the upper source-input estimate required here.

None of those full source-record statements has been verified from the accessible formulas. In particular, no bound $\sigma_\Delta \ge L^{-CN_*}$ with fixed N_* is claimed for actual first mixed entry. The separate endpoint and all-wall source identity obligation in Section 6 also remains.

Section 124 / An actual D8 Ward reduction, local contact projection,

This chapter continues C7.27–34. It does not assume that a higher derivative current contracts once per derivative. The new results are an exact reduction of a specified Wilson D8 contact, a quantitative local projection of its scalar part, an actual regular-minimum test, and its complete flat conditional Gaussian calculation.

The scalar remainder after the stated projection has a genuine fourth-order engineering gain relative to its own word mesh on the admitted physical curvature core. The projected low-loop-degree terms

Section 124 / An actual D8 Ward reduction, local contact projection,

and the whole EOM current are retained. They have not been identified with an already closed lower-dimensional operator center. In particular, this result does not establish all-history MT36.

Section 124.5 / An exact Wilson regular minimum where the unprojected contact diverges

The HJ construction provides a direct check on CP21. It is an actual regular constrained Wilson minimum, with all raw target and Noether conditions retained.

Section 124.5 / An exact Wilson regular minimum where the unprojected contact diverges

Use the HJ transverse Cartan configuration on a fixed even torus of side T, retained spacing one, and fine spacing $a = 1/n$ with n odd. Its current is exactly

$$J_2(x) = j_a(x_1)H, \qquad j_a(x_1) = \epsilon(-1)^{y_1} \quad \text{on the centered block at } y_1, \tag{CP28}$$

and all other current components vanish. There are T jump planes, each with jump magnitude 2ϵ. Its projected plaquette flux is $Y_a H$ with $\|Y_a\|_\infty \le C\epsilon$. The logarithmic curvature satisfies

$$\frac{\sqrt{2}}{a^2}\sin(a^2 F_a/\sqrt{2}) = Y_a. \tag{CP29}$$

The potential, curvature and current lie in the admitted uniform real regular core described in HJ.

Take $f = h = 1$ in CP4. At this Cartan configuration the current-directed variation stays in the Cartan subgroup. The exact directional Hessian of the Wilson current is

$$D^*[c_a\, D(\,\cdot\,)]\,, \qquad c_a(x) = \cos(a^2 F_a(x)/\sqrt{2}) = \sqrt{1 - a^4 Y_a(x)^2/2}. \tag{CP30}$$

Accounting for the conversion from a dimensionless link derivative to a physical connection derivative gives

$$C_{1,1}(U_a) = 2\beta^{-1} a^4 \sum_x c_a(x) |D j_a(x_1)|^2. \tag{CP31}$$

Every factor can also be obtained directly from $\kappa a^4 \sum_i J_i X_i P_1$.

Since $D j_a$ has magnitude $2\epsilon/a$ at each jump plane and is zero elsewhere, literal fine-lattice counting gives

$$\frac{8\beta^{-1}\epsilon^2 T^4}{a}(1 - Ca^4\epsilon^2) \;\le C_{1,1}(U_a) \le\; \frac{8\beta^{-1}\epsilon^2 T^4}{a}. \tag{CP32}$$

This is an exact nonlinear Wilson example, not only the quadratic Taylor diagnostic C7.28b. It rules out a uniform unprojected physical-background contact bound on that regular core.

For canonically normalized labels one may put $\epsilon = g_{\mathrm{W}} P$ in this same bare Wilson family: the factor $\beta^{-1} g_{\mathrm{W}}^{-4} \epsilon^2$ is $c_\beta^{-1} P^2$. The a^{-1} interface growth therefore survives that canonical substitution.

CP23 has a bounded right side on the same fields, because their curvature is uniformly bounded. Thus the interface growth in CP32 is located in the exact EOM and low-loop-degree sector of CP21. It cannot be assigned to the quartic scalar remainder or removed by claiming a source-test momentum zero.

Section 124.7 / Paired measures, selection, and normalization

CP21 is an equality of globally smooth fine observables. Multiplying by the original Wilson measure gives

$$C_{f,h}\,\mu = \tau_{f,h} + (L_{f,h} + R^{(8)}_{f,h})\mu, \qquad \tau_{f,h} = \sum_i \mathscr{D}_{X_i}\nu_i, \quad \nu_i = \kappa a^4 V_i^{f,h}\mu. \tag{CP40}$$

This is the paired representation in C7.29. The scalar partner $\tau_{f,h} = \mathscr{E}[V^{f,h}]\mu$ and every vector partner are kept.

A genuine smooth physical coarea pushforward carries CP40 by its exact derivative rule. Original constraint records can be retained in the joint realization. For example, the map $U \mapsto (U, q(U))$ transports each derivative direction with its actual Dq component. Merely appending $\delta(q(U) - y)$ and leaving the derivative directions unchanged would omit the constraint current and is not this construction. If an actual selector χ is introduced, the whole selected pair has the complementary failure residual C7.30, including its order-zero companion. CP27 can be used for the specified scalar

Section 124.7 / Paired measures, selection, and normalization

remainder's failure envelope. The vector coefficients and projected scalar sector need their own actual envelopes; the quartic bound is not assigned to them.

If a source-zero factor h_{norm} depends on differentiated variables, the order-one identity remains

$$h_{\mathrm{norm}}\tau_{f,h} = \sum_i \mathscr{D}_{X_i}(h_{\mathrm{norm}}\nu_i) + \sum_i (X_i h_{\mathrm{norm}})\nu_i. \tag{CP41}$$

Higher ordered currents retain all corresponding subword commutators.

Thus the exact projection does not commute for free with coordinate-dependent reference normalization. The original denominator, its actual differentiated variables, and the scalar commutators remain part of the quantitative state. A bound on pure outer derivatives does not control an internal old-coordinate derivative absent an exact joint identity.

Section 124.9 / Scope and dependencies

The following additions are established in this chapter:

1. The exact actual Wilson D8 reduction CP6 and its invariant finite-incidence representation CP12.
2. The globally smooth local loop-jet projection CP17, its idempotence, and the quantitative fourth-order scalar remainder CP23–27, with the exact physical quotient CP27a–c and weak analytic root CP27d–e.
3. The exact nonlinear regular-minimum divergence CP32, which identifies a real interface obstruction to an unprojected background estimate.
4. The complete constrained Gaussian Ward formula CP36–39, including its vacuum trace and divergent coarse current matrix.
5. The exact finite-source paired-measure bookkeeping CP40–42 and the limited finite-word extension CP43–44.

The open quantitative tasks remain the engineering classification and transition of the projected low-loop-degree sector, the current partners and their normalization commutators on every actual mixed domain, and the admission and saved-gain estimates for previously generated coefficients. These are necessary to instantiate the stable marked-feedback theorem. All-history MT36 and the full existence proof are therefore still incomplete.

The exact Wilson and Haar identities used here are AI1–15. C7 supplies the finite current filtration, the failure-complement identity and its normalization qualification. HJ supplies the actual constrained Cartan background, including its raw gauge/Noether identity and regular-minimum identification. The Gaussian calculation in Section 6 is proved directly with the actual constrained covariance and requires no new source-dependent denominator.

Section 125 / Exact reduction of the low scalar sector of the Wilson

This chapter strengthens CP6 for its specified bare Wilson contact. For SU(N), the complete Hessian-square scalar can be evaluated far enough to remove the ambiguity between loop degree and engineering dimension. The result is an exact equality consisting of a complete Ward current, an identity term, a weighted single-plaquette Wilson action, and a remainder which is fourth order in the actual plaquette defects. No continuum equation of motion, source-momentum moment assumption, or replacement of the source anchors is used.

The assertion is restricted to the explicitly defined contact with two color-independent external tests. It is not a theorem for arbitrary previously generated marked coefficients or for an effective action whose current differs from the original Wilson current. Those restrictions are material for all-history MT36.

Section 125.5 / Quantitative bounds on actual domains

Let Ω be the union of all actual plaquette stars occurring with nonzero test factors. The defining polynomials and the bounded incidence number give the global real compact estimate

$$|\mathcal{Q}_{f,h}| \le C_N \|f\|_\infty \|h\|_\infty \sum_{p \subset \Omega} \|P_p - I\|^4. \tag{CQ24}$$

The same estimate holds in a common small complex plaquette domain with the bounded complex transports in K_b retained. In the real compact case the constants depend only on the fixed group and lattice dimension. In the complex case they also depend on the once-fixed bound for these actual prefix transports and plaquette inverses. Small plaquette defects alone do not bound arbitrary complex pure-gauge transports. The weak small-link chart used below supplies the required common bounds. There is no mesh or volume dependence in the constants once that chart is fixed. No inverse fluctuation operator is used to prove CQ24.

Put $F_{a,p} = a^{-2}(P_p - I)$ and use the actual lattice norm $\|F_a\|^4_{L^4_a(\Omega)} = a^4 \sum_{p \subset \Omega} \|F_{a,p}\|^4$. Then

$$|\mathcal{R}^{(8)}_{f,h}| \le C_N \beta^{-2} \|f\|_\infty \|h\|_\infty \|F_a\|^4_{L^4_a(\Omega)}. \tag{CQ25}$$

This is a bound on the actual nonlinear matrix plaquette, not on a formal curvature expansion.

For a carrier of physical four-volume at most $C\ell^4$ satisfying the actual curvature core bound $\|F_a\|_\infty \le \epsilon_\ell \ell^{-2}$, CQ25 gives

$$a^4 |\mathcal{R}^{(8)}_{f,h}| \le C_N \beta^{-2} \|f\|_\infty \|h\|_\infty (a/\ell)^4 \epsilon_\ell^4. \tag{CQ26}$$

Thus the remainder really supplies the fourth-power scale gain which generic regular-prefix jet contraction alone failed to supply in C7. The factor a^4 is the engineering factor for this word's own mesh, namely the specified bare Wilson mesh in CQ4. It cannot be reset to an arbitrary later birth scale. With the coupling convention of the same measure, $\beta = c_\beta g_W^{-2}$,

$$g_W^{-4} \beta^{-2} = c_\beta^{-2}. \tag{CQ27}$$

Every original finite-source label and saved gain therefore remains attached when CQ26 is used. Applying the calculation to an effective coarse scale requires that scale's full action and current, including all inherited terms.

The lower action row has a quantitative bare mixing coefficient as well. Define the own-mesh canonical action insertion $\mathsf{S}_W(\psi) = g_W^{-2} \sum_p \psi_p s(P_p)$. Then exactly

$$g_W^{-4} a^4 \sum_p k_p s(P_p) = g_W^2 \, \mathsf{S}_W \left(2c_\beta^{-2} \Lambda - C_F c_\beta^{-3} g_W^2 D \right). \tag{CQ27a}$$

Because each plaquette has four edges and twelve ordered distinct edge pairs, the weight in parentheses has supremum at most $C_{N,c_\beta}(1 + g_W^2)\|f\|_\infty \|h\|_\infty$. This proves an actual $O(g_W^2)$ lower-action contact coefficient for this family. The scalar coefficient has a different size:

$$g_W^{-4} a^4 k_0 = 2c_\beta^{-2} A_0 + N C_F c_\beta^{-3} g_W^2 \sum_p D_p. \tag{CQ27b}$$

Its full original bond and plaquette counts remain in the finite-source vacuum bookkeeping. Neither estimate is a bound for later inherited scalar increments or for arbitrary effective transition coefficients.

Section 125.5 / Quantitative bounds on actual domains

There is also an actual weak analytic root. On the fixed physical box and common weak chart of WH, $U = e^{aA}$ with $\|A\|_{W^{1,p}} \le \epsilon$, $p > 8$, the established nonlinear plaquette estimate gives $\|F_a\|_{L^p_a} \le C\epsilon$, hence

$$|\mathcal{R}^{(8)}_{f,h}(e^{aA})| \le C_N \beta^{-2} \|f\|_\infty \|h\|_\infty \epsilon^4. \tag{CQ28}$$

Small $W^{1,p}$ norm gives small link and fixed microscopic loop words, so the same holomorphic identity holds on the weak chart. This provides the flat valuation four and fixed weak Cauchy bounds for this explicit remainder. It does not prove admission of an arbitrary old coefficient on an unrelated mixed domain.

For the large-field bookkeeping use $\|I - P\|^2_{\mathrm{HS}} = 2s(P)$. Each local summand obeys

$$|\mathcal{Q}_{\mathrm{local}}| \le C_N \|f\|_\infty \|h\|_\infty \left(\sum_{p \subset \Omega_{\mathrm{local}}} s(P_p) \right)^2 = C_N \|f\|_\infty \|h\|_\infty \beta^{-2} E_\Omega^2, \quad E_\Omega = \beta \sum_{p \subset \Omega_{\mathrm{local}}} s(P_p). \tag{CQ29}$$

One already assigned failure-energy reserve absorbs this fixed polynomial in E_Ω. It supplies no additional reserve for a new label and no unproved smallness for a previously generated coefficient. The explicit prefactor of the scalar remainder in CQ22 must still be carried through the large-field transition.

Section 126 / An exact projection of the source-difference

This supplement evaluates the additional row $\mathcal{B}_{f,h}$ in CQ41–46. It uses the same bare Wilson measure, invariant derivatives, couplings, source anchors, and oriented fluxes. It proves an exact decomposition into a complete Ward current, a source-second-difference weighted flux square, a weighted physical directional-current square, and an explicit identity-plus-Wilson-action term.

The directional weight is generally first order in $aD_a f$. No second-order cancellation is assumed. That remaining physical tensor row is retained with its actual weight and its own weak analytic root. Sections 7–8 then identify a particular pointed completion already defined in AI4.1 for which the literal endpoint weights do give an exact second-order source difference and a full bare D8 remainder estimate.

Section 126.6 / What is gained and what remains

For common scalar tests $f_b = f(x)$ and $h_b = h(x)$, the exact weights satisfy

$$\begin{aligned} \|\chi\|_\infty &\le C\{\|h\|_\infty \|D_a^2 f\|_\infty + \|D_a h\|_\infty \|D_a f\|_\infty\}, \\ \|\omega\|_\infty &\le Ca\|D_a f\|_\infty. \end{aligned} \tag{CS18}$$

These are actual test differences. The second estimate cannot generally be improved to $Ca^2\|D_a^2 f\|_\infty$. At a source jet for which f is affine on the complete local stencil,

$$\omega_{\nu;\mu} = a(\partial_\nu f - \partial_\mu f) \tag{CS19}$$

exactly. For instance $f = x_\nu$ on that stencil gives $\omega_{\nu;\mu} = a$. A smooth periodic test can have such a local stencil inside its support; global affine periodicity is not assumed.

The WH reconstruction supplies an explicit weak extension of $\mathcal{H}_{\omega,h}$: replace every $Q_{b;\mu}$ by its WH17 reconstructed value. Its physical-graph equality and weak bound are

$$|\widehat{\mathcal{H}}_{\omega,h}| \le Ca\|h\|_\infty \|D_a f\|_\infty \big(\epsilon + \|\bar{\mu}\|_{\mathrm{row}} + \|\mu\|_{\mathrm{row}}\big)^2. \tag{CS20}$$

The flux row uses the actual weak-chart flux and CS18. Hence both rows now have explicit analytic roots without taking a higher field derivative. At the flat reference with $\bar{\mu} = 0$, their joint weak valuation in the perturbation and multiplier variables is at least two. Nonflat-reference lower Taylor rows are retained.

The remaining $\mathcal{H}_{\omega,h}$ is a physical directional tensor row. Decomposing its three weights into their average and their zero-sum part gives an H row and weighted traceless directional rows. The latter have not been identified with an already closed scalar center. Their actual $aD_a f$ factor is saved; it is not replaced by a fourth-power D8 remainder bound. The algebraic constraint $\sum_\mu Q_{b;\mu} = 0$ alone does not erase these weighted squares: the values $Q = (K, -K, 0)$ and zero-sum weights $(1, 0, -1)$ give $|K|^2$. This check is not a new EOM-cohomology or quantum nondegeneracy theorem.

If a finite hypercubic/root symmetrization is already part of the specified source, CS14 is applied to every term with its complete transformed anchors. No independent new averaging is introduced. An improvement of CS19 would require an explicit cancellation in that actual finite sum, including its shifted field representatives. It does not follow by calling the source a scalar.

Finally, all complete currents in CS14–15 retain their original constraint-coordinate derivatives, selector-boundary currents, and paired-normalization commutators under transport. These identities evaluate a further bare contact and preserve its grades and source differences. They do not prove the remaining all-history current estimates or replace an inherited effective score by the Wilson score.

Section 127.1 / Scope: the original component and its actual outer chart

This chapter sharpens WCT21–28 for the original component operation R-II (1.71). It does not replace a live mixed component by an arbitrary normalized Markov kernel. It uses precisely the original fixed real integration coordinates, selectors, history record, and admitted outer pair chart. An old internal-variable derivative is covered only after its actual map into this outer chart has been proved. LM47 is not a consequence of this calculation.

There are two new quantitative conclusions. Every positive-order outer derivative of the normalized kernel has the small comparison factor itself, rather than only the constant total-variation ceiling. The exact lower-current and scalar commutators created by adjoining this kernel inherit that factor, with the original normalization included. Their zero fiber mass is retained algebraically but is not used to erase them under a fiber-dependent readout.

Let z_0 be the fixed original real comparison pair, let ξ denote all original integration coordinates of this component, and write its finite positive body as T_0. On the same complex outer chart as WCT27, the original comparison has the form

$$T_z(d\xi) = e^{\sigma(z,\xi)} T_0(d\xi), \qquad \sigma(z_0,\xi) = 0, \qquad \sup_{z,\xi} |\sigma(z,\xi)| \le \varepsilon < \tfrac{1}{4}. \tag{NC1}$$

The original constraints and fixed selectors are in T_0. They are not removed or replaced by independent Haar coordinates. Put

$$\begin{gathered} Z_0 = T_0(1) > 0, \qquad K_0 = Z_0^{-1} T_0, \qquad m(z) = K_0(e^{\sigma(z,\cdot)}), \\ Z(z) = Z_0 m(z), \qquad K_z = m(z)^{-1} e^{\sigma(z,\cdot)} K_0. \end{gathered} \tag{NC2}$$

Section 127.1 / Scope: the original component and its actual outer chart

These are the actual source-zero component body and kernel. The scalar Z_0, which may be very small, remains part of the physical effective density. No selected sourced mass is introduced.

R-II's displayed comparison preceding (1.73), together with the original admissible component geometry, supplies the bound recorded in WCT21a:

$$\varepsilon_{k,X} \le C g_k \mathcal{P}(\log g_k^{-2}). \tag{NC3}$$

Here $\mathcal{P}$ is one fixed polynomial for the once-chosen construction parameters. This applies to that original component, not to an arbitrary enlargement or uncounted product of components. All statements below may instead retain the exact $\varepsilon_{k,X}$, without simplifying it by NC3.

Section 127.6 / Products of original components and one available reserve

Suppose an original factorization has exactly n component kernels. Retain that count. For independent kernel coordinates and a shared admitted outer chart,

$$K_z^{\mathrm{prod}} = \bigotimes_{i=1}^{n} K_{i,z}, \qquad E = \sum_{i=1}^{n} \varepsilon_i. \tag{NC16}$$

The telescoping tensor identity, NC5, and $\|K_{i,z}\| \le 1 + C\varepsilon_i$ give

$$\|K_z^{\mathrm{prod}} - K_0^{\mathrm{prod}}\|_{\mathrm{TV}} \le CEe^{CE}. \tag{NC17}$$

Cauchy then gives $C_r E e^{CE} \rho^{-r}$ on a common smaller chart. The relative logarithms add exactly, so their r-th derivatives have the sum of the individual NC8 bounds. Neither result is a component-independent constant when n is unbounded. If components share internal variables, NC16 is not their kernel; their original joint integral is retained instead.

There is a simple legitimate payment when an already proved original unnormalized weight has an unused large-field reserve. Write $x = g^{-2}$, and suppose that same weight contains $e^{-cR(x)}$, where the actual reserve satisfies $R(x) \geq c_1 (\log x)^{\alpha}$ for some fixed $\alpha > 0$. For every fixed polynomial $\mathcal{Q}$ and positive $\theta < c$,

$$\mathcal{Q}(\log x) e^{-cR(x)} \leq C_{\mathcal{Q},c,\theta} e^{-(c-\theta)R(x)}. \tag{NC18}$$

This follows by maximizing the left polynomial times $e^{-\theta R}$. Thus an NC3 commutator multiplied by that particular original reserve is $O(g)$ with a smaller reserve. If a weaker or different reserve is available, the corresponding explicit maximization must be used instead. No reserve is created by normalization, and the same reserve may not be spent independently for each mark in a joint packet.

For fixed source degree, all finitely many derivative and product-rule polynomial costs can be combined before NC18 is applied once. This preserves the one-reserve bookkeeping. It does not pay an unproved earlier age estimate, a new $e^{Cg^{-2}|X|}$ retention cost, or an uncounted array of old component normalizers.

127.6.1 The actual charged component count can be paid with VC

There is a more precise joint payment for the component factor NC17. Suppose its independent original components belong to the charged record ledger of VC3, with every component assigned to its literal record and a fixed finite ownership multiplicity. Write $w_j = 1 + \log(1 + g_j^{-2})$. A finite set of original logarithmic polynomials can be enlarged to one fixed integer power $p \geq 1$, so that

$$\varepsilon_i \leq C g_{j_i} w_{j_i}^{p}, \qquad \sum_i w_{j_i} \leq C_{\mathrm{own}} N_{\omega}^{\log}. \tag{NC19}$$

This is a condition on the components already present in that ledger. It does not create a charge for an exterior clean component, a second copy of an old variable, or an extra source normalizer.

Let $g_{\max}$ be the maximum of the running couplings over all charged scales that occur in these admissible records, and put

$$\eta_* = C C_{\mathrm{own}} \sup_{x \geq x_*} x^{-1/2} \big[1 + \log(1 + x)\big]^{p-1}. \tag{NC20}$$

Then $\eta_* \to 0$ as the once-chosen weak floor $x_* \to \infty$. Since $p \geq 1$ and $A_\omega \geq N_\omega^{\log}$, the same literal sum $E = \sum_i \varepsilon_i$ obeys both estimates

$$E \leq \eta_* A_\omega, \qquad E \leq C g_{\max} A_\omega^{p}. \tag{NC21}$$

The second estimate uses $\sum_i w_{j_i}^{p} \leq (\sum_i w_{j_i})^{p}$; it does not replace the old component count by the current count. The first estimate makes the exponential factor in NC17 affordable, and the second retains an explicit coupling gain.

Allocate a fraction $r_{\mathrm{NC}} > 0$ of the one reserve VC6a before making this estimate. If $C\eta_* \leq r_{\mathrm{NC}}/2$, then

$$C E e^{CE} \leq C_p g_{\max} A_\omega^{p} e^{r_{\mathrm{NC}} A_\omega / 2} \leq C_{p, r_{\mathrm{NC}}} g_{\max} e^{r_{\mathrm{NC}} A_\omega}. \tag{NC22}$$

The last constant follows by maximizing $u^p e^{-r_{\mathrm{NC}} u/2}$ on $[0, \infty)$. Keeping all other already assigned source and comparison weights in the same integral, VC6 therefore yields

$$\sum_{\omega} \int \| K_z^{\mathrm{prod}} - K_0^{\mathrm{prod}} \|_{\mathrm{TV}} e^{|F_{\mathrm{other}}|} \, d\nu_\omega \leq C_{p, r_{\mathrm{NC}}} g_{\max} B_X, \quad |F_{\mathrm{other}}| \leq (r_{\mathrm{total}} - r_{\mathrm{NC}}) A_\omega. \tag{NC23}$$

Section 127.6 / Products of original components and one available reserve

Here the product comparison is a pointwise bound on the actual conditionally fixed record; its integration variables are not adjoined a second time. If the original trajectory gives $g_{j_i} \le C g_k$, the right side is $C g_k B_X$. The literal old couplings were retained until this displayed inequality.

For positive-order outer derivatives, the same argument applies to the NC17 Cauchy estimate with its literal factor $C_r \rho^{-r}$, or its stated canonical-direction factors, still present. It does not prove those factors are bounded for an unidentified internal direction. It does prove that, once the outer directions and charged components have been identified, the component count and its exponential amplification do not remove the small coupling factor. This is a joint consequence of the actual NC comparison and the restricted VC reserve, not a component-independent bound on normalized kernels before that reserve is used.

Section 127.7 / What this closes and what it leaves open

The new result is a quantitative normalization-commutator row for the original admitted outer R component. Every proper derivative term has the original small comparison factor. Its exact current lift includes all scalar and lower-current terms, and an available original reserve can pay its finite logarithmic costs once. This is a checked model-specific consequence of the original comparison, not a hypothesis assigned to a generic Markov kernel.

The following are not consequences of it:

1. LM47 for an old mixed readout and all its source walls.
2. An internal-old-variable derivative estimate on a live constrained body whose selectors or reference records move.
3. Higher-engineering contact decay. NC13 does not improve the regular-prefix factor λ^n, and it does not pay the dimension-eight conversion L^{4n}.
4. A bound on the absolute base vacuum $\log Z_0$, or a comparison of two different torus records.
5. All-history FR/MT36 closure. The other quantitative rows, component counts, and physical projection remain necessary.

Section 128.1 / Hypotheses and conclusions

The complete source-zero comparison S94 is still unproved. This chapter does not replace the original finite-ambient-torus vacuum by an infinite-volume counterterm. It retains the original Wilson measure, its preparation denominators, and the scalar part of its completed R formation.

Section 128.1 / Hypotheses and conclusions

There is a useful additional quantitative result. The fixed-record comparison in RC1–10 extends through the **sum of internal histories** and the **completed original R cluster formation**, without dividing by the possibly very small mass of a live component. The appropriate input norm charges exponent differences to the unused positive energy and to actual old-scale records. The record charge may include the logarithms already present in the original scalar normalizers. Thus a uniform small difference per original record is sufficient for this formation row even when the number of old records is large.

This is a comparison theorem for the displayed original integral on matched records. It does not establish that the two physical tori supply those matched records and small input differences. In particular, the original preparation denominator is not canceled by the argument below. Only an optional factorization of an already formed component into its mass and its probability kernel is recombined.

The new statements use the weak-trajectory and admitted-range hypotheses of the original source-zero R calculation. They add no claim that the original weak trajectory has been proved unconditionally.

Section 128.2 / The part of the original formula needed here

The primary source is T. Balaban, *Large field renormalization. II. Localization, exponentiation, and bounds for the R operation*, Communications in Mathematical Physics **122** (1989), 355–392:

Equations (1.71)–(1.72) retain the earlier preparation inside each component operator; its history sum also acts on the external interaction. Equation (1.73) is a positive-measure bound for an arbitrary carrying function. The scalar costs in the argument before (1.76) include the logarithmic normalizers at their own scales. History entropy is paid before (1.79), and (1.80)–(1.89) provide the age and size bank. Finally (1.90)–(1.98) form scalar activities by applying these operators to joint Mayer products. The empty Mayer family includes the component mass; it is not subtracted. These facts identify the operator placement used below. No inaccessible definition from R-I is reconstructed by inference from the word “localized.”

Fix a component X at level k. Unfold all original operations stored in it, but do not integrate any copy of a retained variable a second time. On a fixed admissible real comparison background, write the resulting disjoint-history measure as

$$T_{0,X}F = \sum_{\omega\in\mathfrak{H}_X} \int F(\omega,\xi)\, d\mu_{0,X,\omega}(\xi). \tag{VC1}$$

Here a history includes its original old field slots, constraints, target values, gauge choices, selectors, and scalar multiplicities. The measure includes the original reference denominators. It is not an unconstrained product Haar measure. A delta constraint remains part of the record until its original integration is carried out.

On the original admitted complex domain, the actual integrand may be complex. The positive dominating measure from S12 is denoted

$$d\nu_{X,\omega} = e^{|\sigma_{X,\omega}|}\, d\mu^{\mathrm{real}}_{X,\omega}, \qquad |d\mu_{0,X,\omega}| \le d\nu_{X,\omega}. \tag{VC2}$$

Already spent action, spatial, and age fractions are included in this definition. A subsequent estimate

Section 128.2 / The part of the original formula needed here

must use the remaining reserve; it cannot reuse the original unspent numerical bound.

Let $\mathcal{R}_\omega$ be the actual integration and retention records **charged in the original cold positive estimate** for this history. This excludes exterior clean arrays, external fixed data, and auxiliary fine coordinates introduced only to describe a constraint. The Gaussian arrays included here are precisely the ones whose integration costs occur in that positive estimate; their sizes are counted in their own units. An inherited atom may read other fixed data, but that reading creates no new chargeable array. Each $R \in \mathcal{R}_\omega$ has a scale $j(R)$, a number n_R of its own-scale cells or independent coordinates of the declared type, and an assigned nonnegative energy $\mathscr{E}_R$. The fixed number of group and coordinate types can be included in n_R. Re-reading an old variable through a reconstruction does not create a new record. Set

$$\begin{aligned} x_j &= g_j^{-2}, \qquad w_j = 1 + \log(1 + x_j), \\ N_\omega^{\log} &= \sum_{R \in \mathcal{R}_\omega} w_{j(R)} n_R, \qquad \mathscr{E}_\omega = \sum_{R \in \mathcal{R}_\omega} \mathscr{E}_R, \qquad A_\omega = N_\omega^{\log} + \mathscr{E}_\omega. \end{aligned} \tag{VC3}$$

Every energy is owned once. Distinct stored layers at the same physical location are still distinct records. In four dimensions their old-scale count can be as large as

$$N_j \le C N_k L^{4(k-j)}. \tag{VC4}$$

Nothing in the following argument replaces the right side by CN_k.

128.3.1 Why the original positive calculation permits this weight

S86 retains a fixed positive fraction of the original Wilson and Gaussian energies and a fixed per-record entropy reserve. Its proof can retain the logarithmic record reserve in VC3 as well. The reason is specific to the original cost calculation: its normalization and coordinate factors already contribute a fixed constant times

$$\sum_j (1 + \log(1 + x_j)) |Z_j| \tag{VC5}$$

at the corresponding original scales. More explicitly, the cost ledger contains the group/Haar integrations, the original Gaussian arrays on $Z_j \cap \Omega_j$, scalar normalization multiplicities, and the admissible-history entropy. The compact integrals cost at most their normalized Haar mass; Gaussian coordinates cost a fixed exponential in their actual number; the scalar costs add at most $Cw_j|Z_j|$; and the history count adds $C(MR_j)^{-4}|Z_j|$. Thus every record counted in VC3 is bounded by a fixed multiple of these old-scale counts, with the declared finite coordinate types included. This assertion does not include an independent clean array outside this ledger.

For clarity, a safe geometric cost majorant, retaining components, is

$$\mathcal{C}_\omega = \sum_j \sum_{Z \in \mathrm{Comp}(Z_j)} M^4 R_j^5 (d_j'(Z) + 1). \tag{VC5a}$$

The original cover counting bounds $w_j|Z_j|$ by a fixed multiple of this quantity: the number of occupied MR_j cells in a connected component is at most a fixed multiple of its spanning-tree length plus one, and the original logarithmic factor is paid by one further R_j. The $+1$ is retained for **each** component. The source-free positive proof bounds its nonenergy costs by $C_0\mathcal{C}_\omega$; its birth and age bank pays this cost with a strict margin. Multiplication by $e^{rN_\omega^{\log}}$ increases this same cost coefficient by a fixed amount rC_1. It neither changes the scale power in VC5a nor introduces an extra history array. In the original age induction, the future-payment sum is linear in this cost coefficient. Replacing C_0 by $C_0 + rC_1$ therefore changes only that coefficient in the same future-payment bound. The original strict dominance of the birth bank over this geometric cost persists after a fixed reduction of its available energy fraction and, if necessary, an increase of the once-chosen weak floor. This is the ledger justification for the reserve, rather than an inference from boundedness of the final component sum.

The original failed-test and age-bank inequalities are strict. First choose a sufficiently small fixed energy fraction and a sufficiently small fixed increase of the cost coefficient; then choose the weak floor and geometric constants as in the original argument. Reducing a positive Gaussian coefficient by that fraction changes its integral by a fixed exponential per actual coordinate. Reducing a Wilson failed-test energy changes its rare exponent by a fixed fraction. Increasing the coefficient in VC5 is absorbed by the same strict bank. This gives, for some fixed $r > 0$,

$$\sum_{\omega\in\mathfrak{H}_X} \int e^{rA_\omega}\, d\nu_{X,\omega} \le B_X. \tag{VC6}$$

Here B_X is the **weakened remaining** original rare and spatial majorant. In particular it is not asserted to equal $T_{0,X}1$. The original history sum and its entropy are included in VC6. Equivalently one can retain record-level majorants $B_{X,\omega}$ through that sum; their original positive combinatorial summation produces B_X.

This proof establishes VC6 only for the records and scalar costs actually present in the original construction and for a fixed increase of their coefficient. It does not permit an arbitrary number of newly introduced normalizers. It also does not claim a reserve with arbitrary powers of $\log x_j$, arbitrary inverse powers of g_j, or a prescribed exhaustion of the original rare exponent. Such changes require a calculation of the bank.

Most importantly, VC6 has been obtained by modifying an upper-bound calculation for the same positive integral. No Wilson coupling, cutoff, constraint, reference denominator, or physical measure has been modified.

128.3.2 One reserve when several estimates are combined

VC6 is one budget. If source weights, the NC normalization-current costs, and a vacuum comparison are present together, choose positive fractions before applying any of them:

$$r_{\text{source}} + r_{\text{NC}} + r_{\text{vacuum}} + r_{\text{spare}} \le r_{\text{total}}. \tag{VC6a}$$

Use VC6 with r_{total} and replace r in VC9–11 by r_{vacuum}. All previously inserted weights remain inside the same positive integral and must consume at most the other assigned fractions. For example, if their combined absolute exponent is at most $(r_{\text{source}} + r_{\text{NC}})A_\omega$, then

$$\sum_{\omega} \int e^{r_{\text{vacuum}}A_\omega} e^{|F_{\text{source}}|+|F_{\text{NC}}|}\, d\nu_\omega \le B_X. \tag{VC6b}$$

Section 128.3 / The logarithmic record reserve

A fixed-order polynomial in the joint record count is paid once from r_{spare}, using $A^m \le m! r_{\mathrm{spare}}^{-m} e^{r_{\mathrm{spare}} A}$. The original NC polynomial-in-logarithms costs can likewise be combined at that fixed order before paying one assigned fraction of the original rare bank. A mark, contact, or newly written normalization factor does not create a second independent bank.

The following classifications are statements about differences of specified input data, not a proof that the two tori have such differences.

128.5.1 Gaussian forms can be charged directly to their own energy

For a positive original Gaussian quadratic form $Q_0(B) = \frac{1}{2}\langle B, D_0 B\rangle$ on its actual gauge and constraint fiber, suppose

$$\left\| D_0^{-1/2}(D_1 - D_0)D_0^{-1/2} \right\| \le \eta. \tag{VC13}$$

The inverses here are on that fiber; no omitted harmonic or gauge mode is silently removed. Then

$$|Q_1(B) - Q_0(B)| \le \eta Q_0(B). \tag{VC14}$$

This is a pointwise quadratic-form inequality. If Q_0 is one of the energies assigned in VC3, its comparison consumes the energy reserve. Unlike the crude cutoff bound in RC8, VC14 does not multiply the defect by $P_j^2 N_j$. Any determinant used to normalize this Gaussian remains in the scalar row below.

The clean physical-current Hessian comparison supplies instances of this inequality after its uniform spectral lower bound has been checked. VC13 has not thereby been checked for every mixed primitive or for an unidentified coordinate conversion.

128.5.2 Old analytic atoms retain their original owners

Suppose, at scale j, the difference of the inherited analytic atoms is bounded on the actual common readout domain by

$$\sup_u \sum_{Y \ni u} \sup |\Delta V_{j,Y}| \le \delta w_j. \tag{VC15}$$

Assign every atom moved into a live exponent to its first intersected actual record and an intersected own-scale cell, as in S59–62. Summing first over the atoms rooted at that cell gives

$$\left| \sum_{j,Y \text{ in the live body}} \Delta V_{j,Y} \right| \le C\delta N_\omega^{\log}. \tag{VC16}$$

An atom that already has an owner keeps that owner through a merger. Exact completion terms with opposite signs are canceled before absolute values. A copy of the same variable in a later readout does not create an additional allowance in VC16.

128.5.3 Scalar atoms are included by absolute value

Write each scalar normalization and vacuum atom with its actual multiplicity $m_{j,\alpha}(\omega)$. A sufficient input bound is

$$\sum_\alpha m_{j,\alpha}(\omega)|\Delta c_{j,\alpha}| \le \delta w_j N_j(\omega). \tag{VC17}$$

The constants include Haar-coordinate factors, Gaussian normalizations, gauge factors, retained reference normalizations, and exact vacuum-completion terms. Empty-carrier constants are not deleted. If a reference denominator is $D_{i,b} > 0$ on the real record, its contribution to W is

$$-\log D_{1,b} + \log D_{0,b} \tag{VC18}$$

with the original multiplicity. Establishing a small bound for VC18 is part of establishing VC17. A small bound for a normalized outer kernel does not establish it.

The original action cancellation is performed before these absolute bounds. A raw denominator may contain a large classical Wilson exponential whose compensating numerator term has the opposite sign. Those two terms remain paired in the exact difference W until their original cancellation is made. The logarithmic cost ledger in VC5 concerns the remaining normalizing cost, not an isolated absolute bound on that large raw Wilson exponential. If the two tori do not have a matched cancellation, the resulting additional defect must be proved and included; it has not been absorbed by calling it a scalar normalizer.

On a literal copied finite reference body, V7N38–40 already gives exact equality of the original raw denominators. In that case VC18 is zero. Otherwise it remains an input to prove. For complex domains the relevant original analytic logarithm or reciprocal must have been admitted before using VC18; this chapter does not deduce a global zero-free property from the positive reserve.

With a common running coupling, the old compact Wilson actions in a matched R record are the same functions of the same retained real variables. They contribute no coefficient defect. If an actual coefficient changes in a different application, its relative change can instead be charged to that action's once-owned energy. One must not manufacture a change of an old compact Wilson action by evaluating it on a newly chosen smooth minimum.

128.5.4 A limited improvement concerning old counts and distance

Suppose the input bounds in VC13, VC15 and VC17 are all bounded by a common

$$\delta_k = Ae^{-cD_k}. \tag{VC19}$$

Section 128.5 / Which actual input differences meet this norm

Equations VC14–18 put their total defect into VC7. Hence VC10 gives an unnormalized component difference of order $\delta_k B_X$, even if D_k is only a common adaptive distance. The large old count is paid **inside the same positive integral** by VC6. It has not been reset to a current-scale count.

This improvement is restricted to defects that are inside a live R body with this unused reserve. It cannot be used for an uncanceled logarithm of a small body, an external scalar increment, a clean-volume sum, or a reference factor whose comparison is not known. For those rows the birth-distance summation RC13, or an equivalent restoring estimate, is still needed. Nor does VC19 follow merely from exponential decay of a mixed inverse: the changes of the actual readouts and their domains must first have been bounded.

Section 129.1 / Domains and scope of the mixed-entry conclusions

The argument checks the new chapters M8, AE8, CP, CQ, CS, NC, and VC against their displayed hypotheses and the earlier MB, MC, RG, WH, and RC interfaces. It independently checks the principal algebra, scale powers, constraint placement, and normalization identities. It also checks the specific R-II cost ledger used by VC against the primary paper, printed pp. 379–387. It is not a new independent proof of all previously imported reconstruction lemmas.

The new results are useful but restricted. The auxiliary weak source map does not admit every original mixed R readout. The explicit Wilson contact has an exact Ward projection and a controlled quartic remainder; the subsequent SU(N) calculation below further reduces its scalar low-loop sector to an identity and a weighted actual plaquette action. Those exact lower rows and the currents still require transport. The original R formation can be compared on matched records with a weighted exponent defect; the actual two physical tori have not been shown to meet every comparison input. Thus LM47, all-history MT36, and complete S94 remain unproved.

Two clarifications were requested during the analysis and have been applied. The OS statement protects the complete projected packet carrier, including any tree and symmetry enlargement. M8's birth-core inclusion refers to the complete stated RG paired-core inequalities, rather than to an arbitrary analytic domain described only by a radius. The same-branch proof and the fixed-clean-radius scope were also made explicit as detailed below.

129.2.1 Raw target restoration and the original variational problem

The geometric restoration argument is the nonlinear version of MB2. At each level an original coarse row is either already retained or its two endpoint blocks are refined. In the second case its actual input rows and connecting prefixes are retained. Repeated evaluation of the original one-step raw means gives the factorization

$$C_N = \mathcal{R}_m M_m. \tag{PA8.1}$$

This is not an assertion that two newly chosen averaging prescriptions are associative. The original maps and their literal composition are used. Its differential consequence is exactly $\ker DM_m \subseteq \ker DC_N$.

The boundary functional must include every original plaquette involving a dynamic variable. Plaquettes with only fixed variables contribute constants. On that literal graph, a critical field for the full current constraint is critical for the more restrictive mixed fiber. A periodic replacement, an omitted boundary plaquette, or a new target would invalidate this implication.

This variational implication alone does not identify a solution. The original field must also lie in the same auxiliary small uniqueness branch. The revised M8.10a–f now checks this with a target-preserving scalar slice in the admissible reference U_b. Its unknowns have the actual $W^{2,p}$, L^p, and node types of

MB's scalar saddle. The relative nodal cocycle is constrained to the identity; on a bare node this is literally the original pinned gauge condition. Its forcing is $(0, -\delta_U w, 0)$, bounded in the same adapted types. The derivative is an $O(t)$ perturbation of the actual scalar block, with no fine-mesh loss.

The raw constraint qualification and the physical Ward equation then produce a compatible multiplier ℓ. Its bound follows by applying the known mixed inverse to $A_U^{\mathsf{t}}\ell$ and absorbing $(A_U - A_{w^h})^{\mathsf{t}}\ell$. The weak Wilson gradient bound is sufficient. The explicit candidate $(w^h, \psi, 0, \nu, \ell - \kappa\theta)$ satisfies all five inhomogeneous rows, with the displayed multiplier convention. Each component is $O(t)$, since the target mass has the exact scaled factor $\ell_{\mathrm{cell}}^3 \kappa\theta = O(\ell_{\mathrm{cell}}\theta)$. One fixed enlargement of the contraction ball, followed by a smaller weak ceiling, includes this candidate. This repairs the possible gap between criticality and same-branch uniqueness without imposing an extra curvature supremum on the original weak input.

129.2.2 The fine collar and the complex curvature condition

M8 chooses its auxiliary reference to equal the input on all pinned fine collar rows and to be flat inside. A transition plaquette contains only a bounded number of factors $e^{a\chi b}$, hence

$$a^2 |F_a(U_b)| \le Ca\|b\|_\infty. \tag{PA8.2}$$

Because the transition and the necessary neighboring stencils are assigned length comparable to a, this is the adapted curvature estimate needed by MC. It would not be a current-scale curvature estimate. On these tiles the scaled mesh is one, and their fixed finite differences have the same uniform bound. Outside the transition the reference is flat. Thus the construction checks the separate complex curvature hypothesis for this specific reference, rather than silently deriving it from a general weak Sobolev norm.

The boundary data remain fixed in each minimization, but can depend holomorphically on the input parameter. The finite action words at that boundary then give a holomorphic family of the same boundary problems. Fixed exterior variables must remain excluded from the unknowns.

For a row of length $\sigma \le 1$, the rescaling used in M8 is

$$\|\sigma b(\sigma\cdot)\|_{W^{1,p}} \le C\big(\sigma\|b\|_\infty + \sigma^{2-4/p}\|D_a b\|_{L^p}\big). \tag{PA8.3}$$

The power $2 - 4/p$ is correct in four dimensions. Row-block norms have bounded own-scale coordinate counts; no absolute sum over all fine target coordinates follows. The complete-prefix separation is needed to prevent a long row from reading the sharp reference transition.

With those checks, the inhomogeneous MC contraction and its physical Noether elimination give a strong protected source pair on the stated fixed weak chart. These conclusions depend on the previously proved MB/MC reconstruction results, with their exact geometry and domains.

129.2.3 Birth-core inclusion and saved grades

The source inclusion is legitimate for the original RG paired core whose complete inequalities are controlled by the potential, curvature, current, nested-criticality, and gauge comparisons stated in M8. Analyticity on an unspecified domain alone would not prove it. With those actual inequalities and fixed comparison constants, rescaling to the birth units gives potential, curvature, and current factors r, r^2, r^3, respectively. The conservative radius test $Crt < \rho_j$ is therefore valid for that specified core.

The relative-coupling calculation is independently correct. With $n = e - j$, $r = L^{-n}$, and the declared weak-trajectory bound,

$$\frac{r g_e P_e}{\rho_j} \le C L^{-n} \sqrt{1+n}\, (\log x_e)^{p_0 - q}. \tag{PA8.4}$$

The geometric supremum is finite and $q > p_0$, so one weak floor pays a strict margin for all ages. The common analytic-radius comparison has the same bounded geometric factor. This uses the declared trajectory; it does not prove that trajectory.

RG55 is a bound for a completed pure-clean scalar D6 packet in its test-functional norm. Its proof uses a common protected current field, the exact nested old constraint equation, the remote-residual comparison on the same old graph, and the complete source-moment and symmetry pairings. M8 supplies the required protected current field for its bounded packet. Under the same old coefficient/core and complete carrier hypotheses, that existing bound can therefore be evaluated on the new source-only weak readout. Each old factor r^s, r^β, r^2 is retained. This is not a proof of a new higher-contact grade, and is not a bound for a selected nonsymmetric history family.

The current M8.19 correctly keeps the fixed clean α of the established RG55 theorem. Its separate shrinking-domain admission theorem does not supply a saved RG grade for an arbitrary R-generated coefficient. A future theorem on a shrinking α_j would have to retain that actual radius in its remainder estimate. The canonical Cauchy calculation M8.18a is also correct: it bounds the literal factor $g_e/t_{j,e}$ without changing the old coupling powers already present in the coefficient bound. Polarization in the full multilinear Banach derivative is now explicitly paid by e^m.

The long-carrier exclusion is essential. A fixed current box contains the short complete relative carrier, not every inherited polymer. Long tails must retain their old carrier weights, radius costs, and actual mixed-domain or retained-kernel admission. Nor does this source extension identify the original R-prepared input with the weak chart. Those are substantive remaining portions of LM47.

129.2.4 AE8 isolates a necessary condition on the actual prepared field

The actual-entry note correctly separates the prepared composite Γ^R, a raw retained field, and the further completed old source minimum. The field proposed as the M8 input must be named before its norm is estimated. Its allowed gauge must also retain every boundary, prefix and junction record.

The plaquette weak estimate shows that a current-unit $W^{1,p}$ bound t implies an L^p curvature bound Ct, on the admitted small branch and with the fixed complex adjoint bounds. For the actual assigned-scale cell moments,

$$c_\Delta^p = \sigma_\Delta^{2p-4} \sum_{P \in \Delta} a^4 |F_P|^p, \qquad \|F\|^p_{L^p_a(D)} = \sum_\Delta \sigma_\Delta^{4-2p} c_\Delta^p. \tag{PA8.18}$$

This conversion is exact with the lattice volume weights. One cell of scale L^{-n} contributes $c_\Delta L^{(2-4/p)n}$ to the corresponding norm bound. An own-scale bound alone has not paid this factor. The actual selectors and preparation must control the weighted sum on the separately stated real and complex domains.

The note correctly treats this as a necessary condition, not a sufficient construction of a holomorphic gauge and not an actual counterexample to LM47. It explicitly has not constructed a field satisfying all original first-entry selectors and violating the condition. A bound on this moment also does not

establish M8's critical-endpoint source identity at arbitrary localization walls. AE8.10 correctly retains that separate endpoint and extension obligation.

129.2.5 The finite-preparation-depth shortcut remains conditional

The follow-up AE8.11–13 checks a possible route to the necessary bound. If the actual complete source record has $\sigma_\Delta \geq \sigma_* = L^{-CN_e}$ and every own-scale moment is at most $C\epsilon_e$, its fixed cell-shape and overlap assumptions give $\sum_\Delta \sigma_\Delta^4 \leq C_D$. Consequently

$$\sum_\Delta \sigma_\Delta^{4-2p} c_\Delta^p \leq C\epsilon_e^p \sigma_*^{-2p} \sum_\Delta \sigma_\Delta^4, \qquad \|F_a(\Xi)\|_p \leq C_D L^{2CN_e} \epsilon_e. \tag{PA8.22}$$

This is the correct worst-case population cost; it is larger than the contribution of just one finest cell. In a coherently glued allowed potential gauge, the corresponding potential and derivative costs are σ_*^{-1} and σ_*^{-2}. The stated $W^{1,p}$ bound with the latter factor is therefore valid. Curvature control by itself does not supply that gauge premise.

For a recent birth $j = e$, the resulting numerical source-core condition is exactly $L^{2CN_e}(\log x_e)^{p_0-q} \ll 1$. Bounded N_e is harmless after the weak-floor choice. A bound $N_e \leq \kappa \log\log x_e$ is paid numerically when $q > p_0 + 2C\kappa \log L$, provided that exponent is actually permitted by the original source-domain construction. One may not enlarge an already specified old core merely by choosing a new symbol q. If N_e is comparable to $(\log x_e)^r$, $r > 0$, no fixed logarithmic power pays this particular upper bound. That statement does not assert that an actual prepared field attains the upper bound.

The R-II failure-test interval and the genealogy conditions with $N = R_j$ do not identify the assigned depths of the complete prepared source record. Microscopic internal steps of a full prefix also are not primary-cell depths. AE8 correctly leaves these actual geometric and selector implications, and the source identity, open.

129.3.1 The two Ward steps

Starting from CP2, the first step leaves $\kappa^2 a^4 \sum_i f_i X_i^2 P_h$. Expansion gives

$$2\kappa^2 a^8 \sum_{ij} f_i h_j ((X_i J_j)^2 + J_j X_i^2 J_j). \tag{PA8.5}$$

The second term becomes another complete EOM coefficient plus $2\kappa^3 a^8 \sum_{ij} f_i h_j X_j X_i^2 J_j$. Using $J_j = a^{-3} X_j S_0$ gives the exact scalar coefficients $2\beta^{-2} a^{-4}$ and $2\beta^{-3} a^{-4}$ in CP6. The derivative order $X_j X_i^2 X_j$ is preserved. No commuting Haar-frame approximation was used.

For constant tests, $\Delta_H S_0 = 4C_F \sum_p \mathrm{Re}\,\mathrm{tr}\, P_p$ and Casimir commutation give $\Delta_H J = -4C_F J$. The scalar fourth-derivative term in CP10 is consequently $-32 C_F^2 \beta^{-3} a^{-4} \sum_p \mathrm{Re}\,\mathrm{tr}\, P_p$, with the displayed sign.

The color contractions in CP11 are gauge invariant and have bounded local incidence. For a repeated bond, all plaquettes in its full star remain. The constants are uniform only because the local carrier and word family have fixed finite size.

129.3.2 The loop projector is a genuine projection

The finite logarithm coordinate has the same exponential jet as the identity through degree three. It follows that $jR = I$, where j is the intrinsic third-jet map and R is the chosen smooth extension. Hence $(Rj)^2 = Rj$. Averaging right inverses of the same intrinsic jet map preserves this relation. This is the reason the symmetry-averaged operator remains idempotent; arbitrary averages of projections need not be idempotent.

Tree gauge fixing is used only as a pointwise coordinate description of an observable. It is not a new measure or a removal of constraint variables. In a contractible fixed carrier, the transported plaquette product gives the required loop-defect bound. A wrapping flat loop would defeat that bound and is correctly excluded unless its actual holonomy record is retained.

The fourth-order Taylor estimate and the explicit prefactor in CP20 give

$$a^{-4} \times (a^2 M)^4 = a^4 M^4. \tag{PA8.6}$$

Counting actual fine incidences in a physical volume $O(\ell^4)$ therefore yields CP25. Its left normalization is the own word mesh a^4, bare in this calculation. It cannot be changed to an arbitrary later birth length.

On a weak $W^{1,p}$ chart, the finite overlap sum instead uses the plaquette L^4 norm. This verifies CP27e without a curvature supremum. The remainder starts at field degree four, but the retained quadratic and cubic loop jets can encode higher engineering derivatives through their finite offsets. A low-loop-degree label does not identify them with a closed low-engineering center.

129.3.3 Physical quotient and its carrier condition

For a complete projected packet strictly in positive time, the reflected EOM current differentiates only negative-time variables. Its pairing with every positive-time observable is zero by the exact Ward identity. CP27b is consequently an equality in the linear OS quotient.

The required complete packet includes the projector's trees and any symmetry-averaged enlargement. Protecting only the original visible contact support would not suffice. This support qualification was added during the analysis.

The quotient is not multiplicative: the exact term $\kappa a^4 \sum_i V_i X_i P$ in CP27c survives a later product. Thus the current record cannot be discarded before future source joins. Nor is a background bound on the quartic remainder its full quantum OS-norm estimate.

129.3.4 Nonlinear interface and constrained Gaussian checks

There are T jump planes in the HJ example, with $(T/a)^3$ transverse fine sites per plane and jump derivative $2\epsilon/a$. Their physical sum is

$$a^4\, T(T/a)^3(2\epsilon/a)^2 = 4\epsilon^2 T^4/a. \tag{PA8.7}$$

Multiplication by the contact's $2\beta^{-1}$ and its exact cosine coefficient gives CP32. This disproves a cutoff-independent bound for the unscaled physical-background contact. It does not disprove every own-mesh graded estimate: multiplying this example by its bare a^4 gives an a^3 quantity. The distinction matters when contacts are born at later scales.

For the constrained Gaussian, $HK = Q^{\mathsf{t}}\Delta_c$ gives the mean term in CP36. The covariance difference is

$$\operatorname{Tr}(H^3C) - \operatorname{Tr} H^2 = -\operatorname{Tr}(QHQ^{\mathsf{t}}\Delta_c). \tag{PA8.8}$$

The divergence of $2\beta^{-2}QHQ^{\mathsf{t}}\Delta_c B$ is exactly the scalar trace companion in CP36. Thus full fluctuation completion gives a complete coarse Ward current, with its interface sensitive matrix and scalar partner. Estimating only its quadratic part would miss the cancellation. The positive slice and treatment of harmonic variables in CP33 are necessary for this calculation.

129.3.5 The analysis of the stronger SU(N) low-sector reduction

The following reduction concerns this bare family. The following calculation independently checks its constants. Use an orthonormal anti-Hermitian basis with $-\operatorname{Tr}(T_AT_B) = \delta_{AB}$, and let $d_G = \dim \mathfrak{su}(N)$. Write C_F, C_A in this same normalization. For one plaquette set

$$R(P) = I - \tfrac{1}{2}(P + P^{-1}), \quad A(P) = \tfrac{1}{2}(P - P^{-1}), \quad K(P)_{AB} = -\tfrac{1}{2}\operatorname{Tr}(\{T_A, T_B\}R(P)). \tag{PA8.11}$$

All plaquettes incident to a bond are based in that bond's actual tangent frame. Let K_b be the sum of these matrices and z_b the literal plaquette valence. The symmetric part of the same-bond Hessian is $z_bI - K_b$. Its antisymmetric part is one half the structure-constant contraction of the bond gradient. Their bilinear orthogonality therefore gives

$$\sum_{A,B}(X_b^A X_b^B S_0)^2 = z_b^2 d_G - 2z_bC_F\sum_{p\ni b} s_p + \operatorname{Tr}_{\mathfrak{g}} K_b^2 + \tfrac{C_A}{4}|\nabla_b S_0|^2. \tag{PA8.12}$$

For two distinct bonds in one plaquette, rotating the two color bases after differentiation makes the mixed Hessian norm the norm of $-\operatorname{Re}\operatorname{Tr}(T_AT_BP)$. Its square is $d_G - 2C_Fs + \operatorname{Tr}_{\mathfrak{g}} K^2 + (C_A/4)|A_0|^2$. The exact matrix identity $A^2 = -2R + R^2$ then gives

$$\begin{aligned}\sum_{A,B}(X_b^A X_c^B S_0)^2 &= d_G + (-2C_F + C_A/2)s_p + Q(P_p),\\ Q(P) &= \operatorname{Tr}_{\mathfrak{g}} K(P)^2 - \tfrac{C_A}{4}\operatorname{Tr} R(P)^2 + \tfrac{C_A}{4N}(\operatorname{Tr} A(P))^2.\end{aligned} \tag{PA8.13}$$

The ordinary local lattice must have a unique common plaquette for such distinct bonds. Tiny wrapping periodic graphs with multiple common plaquettes require their literal incidence expansion instead. The inversion and conjugation invariance of the displayed expression allows the actual orientations and base transports. Since $R(P) = O((P-I)^2)$ and $\operatorname{Tr} A(e^{\xi}) = O(\xi^3)$ for traceless ξ, Q is genuinely quartic at the identity. This is stronger than a mere statement that it is even.

Each link Casimir commutes with its own invariant derivatives and with the other link derivatives. Thus the weighted fourth-derivative term is exactly

$$\sum_{b,c,A,B} f_bh_cX_c^B(X_b^A)^2X_c^BS_0 = -C_F^2\sum_p F_pH_p\operatorname{Re}\operatorname{Tr}P_p, \quad F_p = \sum_{b\in p} f_b, \quad H_p = \sum_{b\in p} h_b. \tag{PA8.14}$$

There is one final current-square term in PA8.12. Put $w_b = f_b h_b$, $W_p = \sum_{b\in p} w_b$, and $P_w = a^4 \sum_{b,A} w_b J_{b,A}^2$. Its coefficient in the CP6 scalar is $q = (C_A/2)\beta^{-2}a^{-2}$. The exact Ward step

$$P_w = \mathscr{E}[wJ] + \beta^{-1}a^{-2}C_F \sum_p W_p \operatorname{Re}\operatorname{Tr} P_p \tag{PA8.15}$$

keeps that current and adds the required scalar companion.

For a full explicit check define

$$\begin{aligned} D_0 &= d_G \left\{ \sum_b z_b^2 w_b + \sum_p (F_p H_p - W_p) \right\}, \\ A_p^{\mathrm{sc}} &= -2C_F \sum_{b\in p} z_b w_b + (-2C_F + C_A/2)(F_p H_p - W_p), \\ B_p^{\mathrm{sc}} &= -2C_F^2 F_p H_p + (C_A C_F/2) W_p, \\ Q_{f,h} &= \sum_b w_b \operatorname{Tr}_{\mathfrak{g}} K_b^2 + \sum_p (F_p H_p - W_p) Q(P_p). \end{aligned} \tag{PA8.16}$$

Then the original contact is exactly the EOM with vector $V^{f,h} + q\, wJ$, plus the following three scalar rows:

$$\begin{aligned} I_{f,h} &= a^{-4} \left\{ 2\beta^{-2} D_0 + N\beta^{-3} \sum_p B_p^{\mathrm{sc}} \right\}, \\ A_{f,h}^{\mathrm{W}} &= a^{-4} \sum_p (2\beta^{-2} A_p^{\mathrm{sc}} - \beta^{-3} B_p^{\mathrm{sc}}) s_p, \\ R_{f,h}^{\mathrm{W}} &= 2\beta^{-2} a^{-4} Q_{f,h}. \end{aligned} \tag{PA8.17}$$

No source offset has disappeared: the products $F_p H_p$ keep all ordered original bond-anchor pairs, and W_p keeps precisely the diagonal ones. The literal valences also remain. This calculation was checked directly for the SU(2) one-plaquette Hessian and, as an additional algebraic check, for central SU(3) plaquettes.

The remainder has the same quartic physical and weak estimates as CP, now with a single β^{-2} prefactor. The identity and weighted actual Wilson action are exact lower rows, so a generic unresolved point-split quadratic loop remainder is no longer needed for this fixed bare family. On a complex domain the constants also retain the once-fixed bound on the actual prefix transports and inverses. Small plaquette defects alone would not bound an arbitrary nonunitary pure gauge. The weak small-link chart used in CQ28 supplies the required common transport bound. All-history source normalization of the exact lower rows and transport of their currents remain necessary. A field-dependent old coefficient multiplying the contact generates derivatives of that coefficient under these Ward steps; their LM47 estimates have not been supplied by PA8.11–17. The physical quotient remains linear, with the same complete positive-time carrier condition and the same later-product contacts.

129.3.6 Canonical lower mixing and the physical directional-current contact

The bare canonical coefficients in CQ27a–b check by direct substitution of $\beta = c_\beta g_W^{-2}$. Multiplication of the action row by $g_W^{-4}a^4$ is g_W^2 times the stated canonical action insertion. Its coefficient retains the

exact weight $2c_\beta^{-2}\Lambda - C_F c_\beta^{-3} g_W^2 D$. The identity coefficient instead retains its full bond and plaquette counts. A small lower-action mixing coefficient does not make that identity row small or remove it from finite-source normalization.

The new CQ35–46 calculation also checks for the exact physical directional-current-square primitive H of AI33–35. For each target bond $b = (x, \nu)$, the plane indices have matrix $T = I - \mathbf{1}\mathbf{1}^{\mathsf{t}}/3$. Its partial same-bond Hessian has symmetric part $2I - K_{b;\mu}$. In the $T_{\mu\rho}$ contraction, both constant and linear terms vanish because $T\mathbf{1} = 0$. The remaining antisymmetric contraction is $(C_A/4)\sum_{\mu,\rho} T_{\mu\rho}\langle a^3 J_{b;\mu}, a^3 J_{b;\rho}\rangle$.

For an off-diagonal Hessian pair only the unique shared plaquette plane occurs, so its weight is $T_{\mu\mu} = 2/3$. With a constant first test, the three other edges give exactly the factor 2 in CQ37. The nonconstant version gives the ordered pair weight $(2/3)t_p$. Thus the first Ward completion contributes the coefficients $4\beta^{-2}a^{-4}$ in CQ38 and $(4/3)\beta^{-2}a^{-4}$ in CQ42.

The other product-rule term is fixed by the actual weighted Casimir:

$$\Delta_f J_{b;\mu} = -C_F J^F_{b;\mu}, \qquad J^F_{b;\mu} = 4 f_b J_{b;\mu} + W^f_{b;\mu}. \tag{PA8.19}$$

It therefore contributes exactly $-2C_F\beta^{-2}a^{-2}(4H_{fh} + \mathcal{B}_{f,h})$. Adding the diagonal antisymmetric term gives the stated lower coefficient $(C_A/2 - 8C_F)\beta^{-2}a^{-2}H_{fh}$. For $f = 1$, $W^f = 0$ and $t_p = 3h_p$, so CQ42 reduces to CQ38 with all factors intact. The canonical conversion to $g_W^{-2}a^2 H$ yields the exact $O(g_W^2)$ coefficient in CQ39.

For the common scalar test placed at each bond's initial point, listing the four actual edges of the two adjacent plaquettes gives precisely the two differences q_+, q_- in CQ43. In particular,

$$q_+/a = D_\mu f + D_\nu f, \qquad q_-/a = 3D^*_\mu f + D_\nu f(x - a\hat{\mu}). \tag{PA8.20}$$

The exact flux formula AI6 then gives CQ44. Its L^2 estimate contains $\|D_a f\|_\infty$, rather than a fine inverse power. This is an actual source difference; it is not deduced from a zero of field momentum. Arbitrary orientation-dependent bond tests do not satisfy this bound without their own difference control.

Finally, replacing just the first directional current in $\mathcal{B}$ by the WH reconstruction leaves its actual weak-chart flux untouched. WH14–15 gives exact equality on the physical KKT graph. WH16 and CQ45 yield the ordinary physical L^2-pairing estimate

$$|\widehat{\mathcal{B}}_{f,h}| \le C\|h\|_\infty \|D_a f\|_\infty \, \epsilon(\epsilon + \|\bar{\mu}\|_{\mathrm{row}} + \|\mu\|_{\mathrm{row}}). \tag{PA8.21}$$

No second derivative of an arbitrary weak field has been bounded. The flat valuation is at least two when the reference is flat, $\bar{\mu} = 0$, and b, μ are scaled together. At a nonflat reference its full strengths remain, and possible lower Taylor rows cannot be discarded. The new source-difference row has not been identified with a closed scalar or tensor center; its later transitions and all current, selector, and normalization companions remain necessary.

129.3.7 CS: summation by parts and the retained directional tensor

CS1–20 further evaluates the source-difference row without changing its original tests or measure. The algebraic polarization in CS4 has the correct signs. Since $q_- - q_+ = aD^*_\mu F^f$ and $|T_\mu^{-1}X|^2 - |X|^2 = aD^*_\mu |X|^2$, the two explicit inverse mesh factors cancel. On the periodic scalar pairing, $(D^*_\mu)^{\mathsf{t}} = D_\mu$, so its summation by parts gives the positive coefficient

$$\chi_{\nu;\mu} = \tfrac{1}{2} D_\mu (h_\nu D_\mu^* F^f_{\mu\nu}). \tag{PA8.23}$$

The forward shift of h is included. Thus the remaining derivatives are actual source differences, not estimates of additional field derivatives.

The total-current completion has the negative scalar coefficient $-r_p^{f,h}/3$, with $r_p^{f,h} = F_p^f \sum h_b - 4 \sum f_b h_b$. The exact identity $r_p = -\frac{1}{2} \sum_{b,c} (f_b - f_c)(h_b - h_c)$ checks both its sign and its ordered-edge multiplicity. It is of order a^2 for the stated common smooth scalar tests, and vanishes for constant h.

Expanding $Q_\mu = J_\mu - J/3$ gives exactly the vector $Y = h\{(2/3) \sum \omega_\mu J_\mu - (1/9)(\sum \omega_\mu) J\}$. Its divergence is the literal plaquette coefficient u_p in CS12. Hence the total vector $\mathcal{V} = Y - h W^{f,\mathrm{tot}}/3$ and $v_p = u_p - r_p/3$ give CS14 with all scalar terms retained. Multiplying by CQ42's coefficient gives the stated $-2C_F^2 \beta^{-3} a^{-4} \sum_p v_p \operatorname{Re} \operatorname{tr} P_p$ in CS15. No conditional mass has been divided out.

The flux square is the same exact projected anti-Hermitian square used in CQ8. Thus CS16 and its quartic Q_F are correct. Combining the two ordered plane directions at their common base gives $\Xi_p = \chi_{\nu;\mu} + \chi_{\mu;\nu}$; there is no additional factor two in CS17. The action and quartic rows retain their complete source differences and original plaquette counts.

For clarity, the full bare canonical prefactor on this new contribution is exactly

$$\begin{aligned} g_W^{-4} a^4 [-2C_F \beta^{-2} a^{-2} \mathcal{B}_{f,h}] = & -2C_F c_\beta^{-2} a^2 \{\mathscr{E}[\mathcal{V}] + \mathcal{D}_{f,h} + \mathcal{H}_{\omega,h}\} \\ & - 2C_F^2 c_\beta^{-3} g_W^2 \sum_p v_p \operatorname{Re} \operatorname{tr} P_p. \end{aligned} \tag{PA8.24}$$

Splitting the last real trace retains both its identity and action parts. A later birth scale or a later effective score is not used in this formula.

There is a further exact count cancellation for the complete periodic sum. Set $M = \sum_b h_b \sum_\mu \omega_{b;\mu}$. Each bond has two plaquettes in each plane and total valence six, so $\sum_p u_p = (4/3 - 6/9) M = 2M/3$. From CS1, $\sum_\mu \omega_{b;\mu} = \frac{1}{2} \sum_{p \ni b} F_p^f - 12 f_b$, and hence $\sum_p r_p = 2M$. Therefore $\sum_p v_p = 0$. The complete identity contribution $N \sum_p v_p$ vanishes at this stage. This does not erase individual localized scalar atoms before their exact sum, or give the cancellation to an arbitrary selected history.

CS18's bounds follow from the literal scalar difference product rule. In particular, for a locally affine scalar test, $\omega_{\nu;\mu} = a(\partial_\nu f - \partial_\mu f)$. Its weight is generally first order in $a D_a f$, with no uniform second-order replacement. WH's reconstructed directional currents therefore give exactly CS20 by a bounded-weight physical L^2-square estimate. This is a weak analytic root on the declared WH chart, with its reference strengths and fixed volume; its joint flat valuation does not remove nonflat lower rows.

The weighted physical Q-square is not erased by the algebraic relation $\sum_\mu Q_\mu = 0$. For example, the algebraic data $Q = (K, -K, 0)$ and traceless weights $(1, 0, -1)$ give $|K|^2$. This limited calculation does not constitute a new equation-of-motion cohomology theorem or quantum nondegeneracy proof. The average and traceless directional rows still need their actual all-history transport. Likewise, any improvement from a previously specified hypercubic symmetrization must be proved with its shifted field representatives; it is not supplied by the word "scalar."

129.3.8 The identified pointed completion has a stronger fourth-power remainder

CS21–31 is restricted to the pointed signed-coordinate completions defined in AI4.1 and retained in WH. It does not replace an old initial-point representative or average a selected history anew. For a reversed bond every partial physical gradient has the same minus-adjoint transformation. Their quadratic contractions and total-current subtraction are therefore scalar functions of the unoriented bond. The per-bond EOM scalar is also invariant: its divergence is the exact link-Casimir trace in CS21. Thus the pointed average of either outgoing-bond scalar is precisely the average over its outgoing and incoming bonds.

Smearing gives the endpoint weight $f^c_\nu(x) = [f(x) + f(x + a\hat{\nu})]/2$, and similarly for h. The two original operator labels must be completed independently. Their bilinear contact then has weights f^c, h^c. A diagonal group average of an already formed contact would not in general give that product. CS correctly keeps this distinction.

The four-edge sum of endpoint weights is the four-vertex scalar sum. Direct evaluation of the two original flux weights gives CS24a, including its first-difference bound. This direct argument is needed because f^c is orientation-dependent as a bond test and hence does not satisfy CQ45's initial-point hypothesis. Averaging the two flux weights now gives the exact identity

$$\omega^c_{\nu;\mu} = \frac{a^2}{2}\{\Delta^{\mathrm{cen}}_\mu f(x) + \Delta^{\mathrm{cen}}_\mu f(x + a\hat{\nu})\}. \tag{PA8.25}$$

The raw affine obstruction CS19 still applies to its different initial-point representative. No contradiction or unproved scalar symmetry is used.

The product correction CS26 also has its stated sign and factor: $f^c h^c = (fh)^c - (a^2/4) D_\nu f D_\nu h$. Substitution of this identity and CS14/17 into CQ42 verifies every term of CS28–29. In particular the added EOM vector has coefficient $-2C_F\beta^{-2}a^{-2}$, the action coefficients of Ξ and v have signs minus and plus respectively, and the H_η remainder has coefficient $-\alpha_H\beta^{-2}/4$. The lower row is exactly the same pointed primitive smeared by fh, plus the explicit correction; the two are not identified without that correction.

The estimate CS30 requires that the packet and its full read sets occupy a fixed bounded number of physical ℓ-boxes, of volume at most $C\ell^4$, as in the WH theorem. Otherwise the restored L^p-to-L^4 and current norms retain a volume factor. Under that premise, the four remainders, after multiplication by their own-mesh a^4, have the following checked powers, apart from the common $C\beta^{-2}$ and the displayed test norms:

Exact summand	Scale bound
$2\beta^{-2}a^{-4}\mathcal{Q}^H$	$(a/\ell)^4\epsilon_\ell^4$
$-2C_F\beta^{-2}a^{-2}\sum \Xi Q_F$	$(a/\ell)^6\epsilon_\ell^4$
$-2C_F\beta^{-2}a^{-2}\mathcal{H}_{\omega^c,h^c}$	$(a/\ell)^4\mathcal{M}_\ell^2$
$-\alpha_H\beta^{-2}H_\eta/4$	$(a/\ell)^4\mathcal{M}_\ell^2$

The last two bounds use genuine source differences and the WH physical L^2 current-square bound. They require no higher field derivative. They have joint flat field valuation two, with two source differences, and are correctly not called quartic field remainders. The original canonical factor is exactly $g_W^{-4}\beta^{-2} = c_\beta^{-2}$, proving CS31 with no new coupling or birth-mesh substitution.

This establishes the complete bare scalar projection and the stated fourth-power weak remainder for the identified independently completed packet. It leaves its complete current, subsequent contacts,

Section 129.3 / CP: independent algebra and projection checks

normalization derivatives, and actual mixed-history admission to be proved. It is not an all-history invariant tensor-center theorem.

The density of $K_z - K_0$ relative to K_0 gives the stated total-variation bound directly. Cauchy applied to this difference, rather than to K_z, retains ε in every positive derivative. This is a valid strengthening of a constant kernel norm ceiling on the same admitted chart. Its scalar analogue controls $\log Z(z) - \log Z_0$, not the absolute $\log Z_0$.

The ordered subword identities NC10, NC14, and NC15 are correct for noncommuting analytic vector fields. The product rule never reorders the derivatives assigned to one factor. In NC10 the empty derivative subword is an ordinary scalar companion; omitting it would be an error. Each differentiated kernel has zero fiber mass, but its pairing with a fiber-dependent readout need not vanish.

These identities concern currents already written in the admitted outer variables. They do not turn an internally integrated old Haar derivative into an outer derivative, nor freeze a selector moved by that old variable. The original joint realization and its derivative map remain necessary. An absolute sum of microscopic derivative coordinates also requires its own tensor norm.

The tensor-product estimate retains $E = \sum_i \varepsilon_i$, so it has not hidden the component count. It applies only when the original factorization has independent component coordinates. Shared coordinates require one joint body. The logarithmic payment NC18 is valid for one actual unused reserve and fixed total derivative order; it creates no reserve through normalization.

There is also a precise compatibility with VC's charged record reserve. If each component in that actual tensor factorization is owned by at least one of VC's already charged records at its true scale, then for any fixed $\eta > 0$ a once-large weak floor gives

$$\varepsilon_i \le C x_{j_i}^{-1/2} \mathcal{P}(\log x_{j_i}) \le \eta w_{j_i}, \qquad E \le \eta N_\omega^{\log}. \tag{PA8.10}$$

Thus the exponential e^{CE} in NC17 can consume one assigned fraction of the same VC reserve. Fixed powers of the count consume the declared spare fraction. This requires the actual ownership and factorization; an uncharged clean component or a second copy of an old record has no such allowance. It does not estimate an internal old derivative merely by paying its count.

NC19–23 makes a stronger correct use of this observation. On the same record it retains both $E \le \eta_* A_\omega$ and $E \le C g_{\max} A_\omega^p$, with a fixed integer $p \ge 1$. The second follows from $\sum_i w_i^p \le (\sum_i w_i)^p$; it does not discard the old couplings. Half of the assigned NC reserve pays e^{CE}, and the other half pays A_ω^p. Integration in the same VC body therefore gives $C g_{\max} B_X$, without a remaining logarithmic factor. Literal outer Cauchy and canonical-direction factors remain. This is a charged-record, common-chart statement, not a bound on arbitrary normalized kernel products.

Section 129.5 / VC: weighted comparison and its actual model input

The primary R-II text explicitly has logarithmic normalization costs at each original scale, Gaussian coordinate costs on the displayed old arrays, and admissible-history entropy before (1.79). Its (1.80)–(1.88) induction pays a fixed cost coefficient using creation and renewal factors, with a strict small-coupling margin. Thus adding a sufficiently small fixed multiple of the same charged logarithmic record count changes that cost coefficient, rather than the scale power or the collection of integration records.

Section 129.5 / VC: weighted comparison and its actual model input

This supports VC6 on the restricted record set of VC3, after shrinking the available reserve and making the original weak-floor choice. The +1 per component in VC5a is needed even when its spanning length vanishes. Uncharged exterior clean arrays and artificially duplicated coordinates have no such allowance. The energy part must be the once-owned positive energy retained in the original bound after its numerator/denominator cancellations, not an arbitrary raw Wilson action added a second time.

Given VC6 and the matched-support bound $|W_\omega| \le \delta A_\omega$, the moment argument is exact:

$$\frac{1}{m!}\int |W_\omega|^m d\nu_\omega \le (\delta/r)^m \int e^{rA_\omega} d\nu_\omega. \tag{PA8.9}$$

Summation proves VC9–11 before any division by a component mass. Its old coordinate counts are paid inside the same positive integral. For a Gaussian form difference, the relative form norm in VC13 indeed bounds the defect by its actual Gaussian energy. Changing the constraint fiber would require a separately proved coordinate map.

The analytic formation argument uses the original common strict Mayer/connected-expansion ball. Interpolating the original bodies and external interactions inside that ball gives the stated Cauchy Lipschitz estimate, including the scalar evaluation. The actual history sum acts jointly on all incident Mayer factors. No product of separately integrated marks is substituted for it.

The cancellation $ZK = T$ concerns an optional factorization of a formed body. Its exact finite-difference identity is correct. It does not cancel the genuine preparation denominator already inside T, and it does not remove an independent external scalar increment. VC23–24 correctly shows why a weighted absolute estimate does not imply a uniform relative logarithm bound for a tiny mass.

Finally, a cross-torus application still needs matched constraints, selectors, reference records, full typed seeds, readout domains, and the small exponent defects for all genuine normalizers. The formation theorem supplies the passage from such inputs to a completed output; it does not establish those inputs. A common adaptive distance is sufficient only for defects inside a reserve-bearing live body. External scalar and clean-volume rows retain their original birth-distance summation requirements.

Section 129.6 / Hypotheses and conclusions

Interface	Verified addition	Remaining model-specific obligation
Old scalar source root	Fixed-window source-only smoothing, quantitative radius margin, and preservation of an already proved completed RG grade	Identify every actual R-prepared readout and complete long carrier with an admitted domain and its true retained variables

Section 129.6 / Hypotheses and conclusions

Interface	Verified addition	Remaining model-specific obligation
D8 contact	Exact SU(N) Wilson/current-square reduction, physical H contact and source-difference projection, with a fourth-power weak remainder for the identified independently pointed-completed H packet; linear physical quotient	Transport the exact lower rows and currents; control general generated tensor families, later joins, actual mixed admission and full quantum remainders
Normalization currents	Small positive-order outer kernel jets, all ordered lower-current/scalar commutators	Transport internal old directions and moving selectors to an actual common chart with their correct normalization factors
Source-zero comparison	Weighted old-record reserve and Lipschitz completed original R formation, including its scalar value	Prove all cross-torus seed, support, denominator, and independent scalar comparisons

No the analysis result upgrades the existence attempt to a complete proof. The positive results above should be retained with these precise scopes when the analysis is updated.

The R8.1–26 the analysis was also checked against these scopes. It correctly separates the constructed M8 source chart from actual mixed entry, shrinking-domain admission from saved-grade control, the bare contact decomposition from full effective-action transport, and matched-input R formation from complete S94. The later narrow H and source-difference additions do not alter those unproved obligations.

130.1.1 The same action and the actual nonlinear determining rows

M8 works on a fixed current-size contractible box with a complete exterior record and pinned fine collar. Its mixed shells are separated by a fixed multiple of their own cell sizes. Their total width is bounded independently of microscopic depth. The actual raw means can be filled recursively, giving

$$C_N = \mathcal{R}_m M_m, \qquad \ker DM_m \subseteq \ker DC_N. \tag{R8.1}$$

Every chosen row keeps its original averaging prefix, endpoint gauge, junction node and prescribed exterior values. The variational functional includes all original Wilson plaquettes involving a dynamic variable. No periodic boundary action is substituted for this literal restriction.

For a small complex weak input b in one specified analysis chart, the auxiliary reference equals e^{ab} on all pinned fine rows and is flat inside. Its sharp transition is assigned fine scale. The direct finite-word estimate $a^2|F_a(U_b)| \le Ca\|b\|_\infty$ therefore proves the needed adapted complex curvature condition for that reference. It does not claim current-scale curvature control for an arbitrary weak input.

The target is the actual mixed target of e^{ab}, expressed relative to that reference with its full compatible nodal convention. M8.7 bounds the entire inhomogeneous five-type forcing vector. The previous MB/MC inverse and contraction then give a physical local source readout $\mathscr{P}_m(b)$. On the protected current region,

$$\|A_{\mathscr{P}_m}\|_{W^{2+s,p}} + \|F_a(\mathscr{P}_m)\|_{W^{1+s,p}} + \|J(\mathscr{P}_m)\|_\infty \le C\|b\|_{W^{1,p}}, \qquad p > 8, \quad 0 < s < 1/p. \tag{R8.2}$$

This is a common complex bound on the smaller admitted weak ball. The strong estimate is a consequence of the constructed physical equation on the protected region. It is not inferred from weak Sobolev regularity alone, and is not assigned to the sharp fine collar.

130.1.2 Criticality and uniqueness are joined by an explicit slice

R8.1 makes an original current-critical input critical for the more restrictive auxiliary fiber. That fact alone would not identify the two solutions. M8.10a–f supplies the missing same-branch argument.

The actual relative nodal cocycle defines a target-preserving scalar gauge slice. Its three equations use the constrained scalar biharmonic block of MB, including the bare gauge pins. A small perturbation of that known inverse gives the gauge and scalar auxiliary variables. The physical Ward identity then

converts the raw multiplier to the compatible multiplier ℓ, with every endpoint and logarithmic target factor retained. The known inverse at the auxiliary reference gives

$$\|\ell\|_{\mathrm{row,ad}} \le Ct + Ct\|\ell\|_{\mathrm{row,ad}}, \qquad t = \|b\|_{W^{1,p}}. \tag{R8.3}$$

Absorption bounds the multiplier by $2Ct$. The explicit candidate $(w^h, \psi, 0, \nu, \ell - \kappa\theta)$ then satisfies all five inhomogeneous equations in the same contraction ball. Uniqueness proves the exact gauge-invariant source identity on the original physical critical graph. No extra curvature supremum for the weak critical input is required in this argument.

130.1.3 The birth core and the saved grade are different assertions

An already generated coefficient may be composed with this readout only if its **complete stated paired core** is covered: potential, curvature, current, nested criticality, boundary traces, gauges, prefixes and determining collars. Analyticity on an otherwise unspecified ball does not establish that inclusion. M8.13 checks the actual RG core inequalities for the bounded protected packet.

With old birth j, current level e, and $r = L^{j-e}$, the potential, curvature and current scale as r, r^2, r^3. For the stated shrinking birth radius $\rho_j \asymp g_j(\log g_j^{-2})^q$ and real cutoff $g_e P_e$, the declared weak trajectory gives

$$\frac{r g_e P_e}{\rho_j} \le C L^{-(e-j)} \sqrt{1 + e - j}\,(\log g_e^{-2})^{p_0 - q}. \tag{R8.4}$$

Since $q > p_0$, one weak floor gives a strict margin at every age. M8.18a keeps the actual canonical differentiation factor $g_e/t_{j,e}$; it does not introduce a new unrecorded inverse birth coupling. The weak trajectory used here remains an explicit input, not a theorem proved by the radius comparison.

The **saved-age** conclusion is narrower. For the completed, contractible, pure-clean one-label D6 scalar packet already covered by RG55, the common protected field permits that same estimate:

$$\|\widehat{R}_{j\to e}(b)\|_{\mathfrak{T}} \le C A_j \left\{ r^s a_2^{(d)}(j) t^2 + r^\beta a_3^{(d)}(j) t^3 + r^2 \alpha^{d-4} g_j^{-d} t^4 \right\}, \qquad d = 2, 3. \tag{R8.5}$$

The clean radius α, original dimension-four row, pointed descendants, source moments and exact completed symmetry family remain the ones in RG55. The norm is its test-functional norm, not an absolute microscopic-anchor norm with an invented decay. Admission to a shrinking R-origin core does not by itself give R8.5 to an R-generated or higher-contact coefficient.

The weak source map has one fixed current-size packet. Long inherited carriers retain their original tails, radii and admission requirements. M8 does not replace them by a solid ball or declare them one bounded root. Most importantly, it does not establish that every original R-prepared argument, with every simultaneous selector and source wall, lies in its weak chart. That is still part of LM47.

130.1.4 The actual entry check retains the assigned-scale weights

AE8 identifies the nested prepared R-II field, including its boundary gauge and original determining maps, separately from both the raw integration variable and the further old physical reconstruction. A proposed weak-chart input must be specified as one of those actual maps. The original range statement (1.65) and the small complex density comparison do not themselves give its current-unit weak bound.

For the actual proposed input Ξ, a bounded small weak chart necessarily gives $\|F_a(\Xi)\|_{L^p} \le C\tau$. Partitioning its plaquettes among cells of actual assigned length σ_Δ gives the exact identity

$$\|F_a(\Xi)\|^p_{L^p_a(D)} = \sum_{\Delta \subset D} \sigma_\Delta^{4-2p} \|\sigma_\Delta^2 F_a(\Xi)(\sigma_\Delta \cdot)\|^p_{L^p_{a/\sigma_\Delta}(\widehat{\Delta})}. \tag{R8.5a}$$

Section 130.1 / An inherited source now has a weak root on a specified chart

One cell of scale L^{-n} converts its own-scale moment by $L^{(2-4/p)n}$. Thus a uniform small own-scale ceiling does not establish the required current-unit estimate. The original selectors and simultaneous complex walls must control the weighted sum for the actual prepared field. No permitted first-entry configuration violating that estimate has been constructed; AE8 is a necessary test, not a counterexample to LM47.

Even that norm estimate would not identify arbitrary prepared wall fields with M8's physical critical graph. The exact endpoint source identity and its chosen bounded wall extension remain a separate obligation with the original action, constraints and normalizers.

130.2.1 The full Ward identity comes before the estimate

Let

$$S_0(U) = \sum_p (N - \operatorname{Re}\operatorname{tr} P_p), \quad d\mu = Z^{-1} e^{-\beta S_0} \prod_b dU_b, \quad J_i = a^{-3} X_i S_0, \quad \kappa = (\beta a^3)^{-1}. \tag{R8.6}$$

The index i includes an independent bond and a color generator. Define

$$\mathscr{E}[V] = a^4 \sum_i (J_i V_i - \kappa X_i V_i), \quad P_h = a^4 \sum_i h_i J_i^2, \quad C_{f,h} = \kappa a^4 \sum_i f_i J_i X_i P_h. \tag{R8.7}$$

Here f_b, h_b are field-independent, color-scalar external tests, with all original bond locations retained. The exact Wilson Ward identity is $\int \mathscr{E}[V] G \, d\mu = \kappa a^4 \sum_i \int V_i X_i G \, d\mu$.

Two Ward steps give CP6:

$$C_{f,h} = \mathscr{E}[V^{f,h}] + 2\beta^{-2} a^{-4} \sum_{i,j} f_i h_j (X_i X_j S_0)^2 + 2\beta^{-3} a^{-4} \sum_{i,j} f_i h_j X_j X_i^2 X_j S_0. \tag{R8.8}$$

The noncommuting order of the derivatives is unchanged. Initially CP projected the finite scalar Wilson words through loop degree three. That construction is a genuine idempotent projection, but a low loop degree does not identify an engineering dimension. CQ now evaluates the entire low sector explicitly for this family.

130.2.2 The color calculation closes this scalar projection

Normalize the generators by $-\operatorname{tr}(T_A T_B) = \delta_{AB}$, so $C_F = (N^2-1)/N$, $C_A = 2N$, and $d_G = N^2 - 1$. For one plaquette set

$$R(P) = I - \tfrac{1}{2}(P + P^{-1}), \qquad A(P) = \tfrac{1}{2}(P - P^{-1}), \qquad K(P)_{AB} = -\tfrac{1}{2} \operatorname{tr}(\{T_A, T_B\} R(P)). \tag{R8.9}$$

The symmetric Hessian part is $I - K$. Its antisymmetric part is half the Lie commutator applied to the plaquette gradient. Completeness gives

$$\sum_{A,B} (\operatorname{Re}\operatorname{tr}(T_A T_B P))^2 = d_G + (-2C_F + C_A/2) s(P) + Q(P), \tag{R8.10}$$

where

$$Q(P) = \operatorname{tr}_{\mathfrak{g}} K(P)^2 - \tfrac{C_A}{4} \operatorname{tr} R(P)^2 + \tfrac{C_A}{4N} (\operatorname{tr} A(P))^2. \tag{R8.11}$$

This is an exact compact-group identity. Since $R = \frac{1}{2}(I-P)(I-P^{-1})$, its first two remainder terms are quartic in plaquette defects. Near the identity, $\operatorname{tr} A = \operatorname{tr}\sinh\xi = O(\xi^3)$ for $P = e^{\xi} \in \mathrm{SU}(N)$. The last term starts at order six. Compactness extends the quartic bound over the real group.

For a same-bond Hessian block the symmetric part is $zI - \sum_{p\ni b} K(P_{p,b})$; its antisymmetric square is $(C_A/4)a^6|J_b|^2$. The based plaquette words and their transports are retained. On the specified periodic lattice with each circumference at least three fine spacings, two distinct bonds share at most one plaquette, so R8.10 evaluates every off-diagonal block. The bond valence is $z = 6$.

The fourth-derivative term has an independent exact simplification:

$$\sum_{A,B} X_c^B\,(X_b^A)^2 X_c^B S_0 = \Delta_b\Delta_c S_0 = -C_F^2 \sum_{p\ni b,c} \operatorname{Re}\operatorname{tr} P_p. \tag{R8.12}$$

Casimir commutation justifies this equation, including $b = c$. The remaining same-bond J_b^2 term is reduced by one more complete Ward identity. No higher spatial derivative of the background is assumed in any of these steps.

130.2.3 The exact result retains identity, action, and source offsets

CQ16–22 gives explicit coefficients k_0, k_p and the additional current. With those literal coefficients,

$$\boxed{C_{f,h} = \mathscr{E}[\widetilde{V}^{f,h}] + k_0 + \sum_p k_p s(P_p) + \mathcal{R}^{(8)}_{f,h}.} \tag{R8.13}$$

For example, k_p retains separately $\sum_{b\in\partial p} f_b h_b$ and $\sum_{b\neq c\in\partial p} f_b h_c$, with the latter an ordered pair sum. No source pair is moved to a common plaquette center. The extra vector is $(C_A/2)\beta^{-2}a^{-2}(fh)J$, in addition to the complete vector from R8.8. The identity coefficient includes the original bond and plaquette counts. Neither it nor the action row can be discarded from finite-source logarithms or normalization formulas.

The real compact bound is

$$|\mathcal{R}^{(8)}_{f,h}| \le C_N\beta^{-2}\|f\|_\infty\|h\|_\infty\|a^{-2}(P-I)\|^4_{L^4_a(\Omega)}. \tag{R8.14}$$

Its complex version also retains a fixed bound for the actual prefix transports and their inverses. Small complex plaquette defects alone do not bound an arbitrary complex pure gauge. The common weak small-link chart used by WH supplies these transport bounds.

On a physical carrier of volume at most $C\ell^4$, if its actual curvature satisfies $\|a^{-2}(P-I)\|_\infty \le \epsilon_\ell \ell^{-2}$, then

$$a^4|\mathcal{R}^{(8)}_{f,h}| \le C_N\beta^{-2}\|f\|_\infty\|h\|_\infty (a/\ell)^4\epsilon_\ell^4. \tag{R8.15}$$

This is the missing fourth-power gain for the remainder of this particular contact. The factor a^4 is its **own word-mesh** engineering normalization, bare here. It is not an arbitrary later source birth length. If $\beta = c_\beta g_W^{-2}$, its original canonical prefactor is exact: $g_W^{-4}\beta^{-2} = c_\beta^{-2}$.

The lower action coefficient has a useful actual canonical size as well. If $\mathsf{S}_W(\psi) = g_W^{-2}\sum_p \psi_p s(P_p)$, then CQ27a gives

$$g_W^{-4}a^4\sum_p k_p s(P_p) = g_W^2\mathsf{S}_W(2c_\beta^{-2}\Lambda - C_F c_\beta^{-3} g_W^2 D). \tag{R8.15a}$$

The weight is bounded by $C(1+g_W^2)\|f\|_\infty\|h\|_\infty$ using the actual four edges and twelve ordered distinct pairs. This proves an $O(g_W^2)$ lower-action mixing coefficient for this bare family. Its identity coefficient still contains every fine bond and plaquette count; it has not been assigned the same small bound.

On the stated weak $W^{1,p}$ ball, the WH plaquette estimate instead proves

$$|\mathcal{R}^{(8)}_{f,h}(e^{aA})| \le C_N\beta^{-2}\|f\|_\infty\|h\|_\infty\epsilon^4. \tag{R8.16}$$

Thus this explicit remainder has a quartic weak analytic root, which can enter the existing connected localization on its admitted chart. The failure-sector estimate is a fixed polynomial in the original local Wilson energy and uses one already assigned reserve.

130.2.4 The current is null only in the stated linear quotient

If the whole packet, including its determining records, based plaquettes and vector coefficients, lies strictly in positive time, the Ward identity gives

$$[C_{f,h} - k_0 - \sum_p k_p s(P_p)]_{\mathrm{OS}} = [\mathcal{R}^{(8)}_{f,h}]_{\mathrm{OS}}. \tag{R8.17}$$

This proves the required null direction for this packet without an enlarged Gram inverse. It does not prove a bound for the full quantum norm of the remaining operator. Nor is the quotient multiplicative: a later insertion G leaves exactly

$$G\mathscr{E}[\widetilde{V}] = \mathscr{E}[G\widetilde{V}] + \kappa a^4 \sum_i \widetilde{V}_i X_i G. \tag{R8.18}$$

Those descendants keep their original label union and source anchors. After conditioning, the current also has components in the actual constraint coordinates. Selector derivatives and derivatives of every normalization factor remain in its paired representation.

Field-dependent inherited test coefficients produce further derivatives in the Ward calculation. At a later effective scale, the score is that of the full inherited action. Substituting only its Wilson term would omit the derivatives of its inherited interaction. CQ23 is not such an omission: its scope is explicitly the original bare measure and the displayed external tests.

130.2.5 The interface calculation explains why the completion matters

CP28–32 evaluates the unprojected contact on the actual nonlinear Cartan constrained minimum from HJ. Its block-step current gives $C_{1,1}(U_a) \asymp \beta^{-1}\epsilon^2 T^4/a$, with an explicit cosine correction. This refutes a uniform **unscaled** physical background bound for that contact. It does not refute every own-mesh graded bound: multiplication by its bare a^4 gives an a^3 quantity in this example.

The completed Gaussian computation independently keeps the same issue visible. With $\Delta_c = (QH^{-1}Q^{\mathrm{t}})^{-1}$ and $R_a = QHQ^{\mathrm{t}}$, CP36 gives

$$\mathbb{E}\Big[2\beta^{-1}\langle A, H^3A\rangle - 2\beta^{-2}\operatorname{Tr}H^2 \mid QA = B\Big] = 2\beta^{-1}\Big\{\langle\Delta_cB, R_a\Delta_cB\rangle - \beta^{-1}\operatorname{Tr}(R_a\Delta_c)\Big\}. \tag{R8.19}$$

The right side is a complete coarse Ward divergence. Its matrix can have the block-interface divergence. The quadratic term and its scalar trace must therefore stay together. Generic regular jet contraction alone supplies neither this cancellation nor the fourth-power remainder in R8.15.

130.2.6 The physical directional-current square has a further checked row

CQ35–46 also treats the physical quadratic direction from AI33–35,

$$H_h = a^4\sum_b h_b \sum_{\mu,\rho\neq\nu} (\delta_{\mu\rho} - \tfrac{1}{3})\langle J_{b;\mu}, J_{b;\rho}\rangle. \tag{R8.19a}$$

This is kept distinct from the total-current square. With constant first test, its contact has an exact complete current, identity and plaquette-action rows, an explicit lower H_h row, and a quartic plaquette remainder. The same-bond constant and linear Hessian terms cancel because the directional projection annihilates the constant three-vector. The lower H_h coefficient is $(C_A/2 - 8C_F)\beta^{-2}a^{-2}$. Relative to its own canonical normalization, it is exactly

$$(C_A/2 - 8C_F)c_\beta^{-2}g_W^2. \tag{R8.19b}$$

Thus a second actual bare mixing row has a small coupling factor.

For varying first test, CQ42 retains the additional exact row $-2C_F\beta^{-2}a^{-2}\mathcal{B}_{f,h}$. Its coefficient contains the literal differences $\sum_{d\in\partial p} f_d - 4f_b$, with all bond locations unchanged. For a common scalar smooth test these are bounded by $Ca\|D_a f\|_\infty$. The exact flux formula and WH's reconstructed directional currents give a uniform weak analytic root for $\mathcal{B}_{f,h}$. The derivative of the test is responsible for this estimate; no additional derivative of a weak field is assumed.

CS1–20 further reduces that source-difference row exactly into a complete current, source-second-difference weighted plaquette terms, and a weighted physical directional-current tensor. For raw initial-point tests the last weight is only $O(aD_a f)$; an affine test proves that a second-order gain cannot simply be asserted. Every scalar completion and shifted source anchor is retained. The identity $\sum_p v_p = 0$ cancels one newly displayed scalar trace contribution only after its complete original periodic sum; it does not erase each localized scalar atom.

These raw tensor rows have not all been assigned closed mixed-history grades. The following narrower completed-packet result does supply the required scalar remainder gain. Flat valuation statements still require the flat reference and simultaneous scaling of the field and multiplier; nonflat reference strengths and lower Taylor rows are retained.

130.2.7 The identified pointed completion has the full fourth-power remainder

CS21–31 identifies precisely the pointed signed-coordinate completion already defined in AI4.1. Reversing a bond's own direction transforms all its partial forces by the same minus-adjoint map to the adjacent positive bond. Their scalar squares and the complete EOM density are unchanged. The rooted signed-direction average therefore gives the literal endpoint test weights

$$f_\nu^c(x) = \tfrac{1}{2}\{f(x) + f(x + a\hat{\nu})\}, \qquad h_\nu^c(x) = \tfrac{1}{2}\{h(x) + h(x + a\hat{\nu})\}. \tag{R8.19c}$$

The two operator labels are averaged independently, as their identified representatives require. A diagonal average of one already formed contact would generally give different correlated weights. No arbitrary inherited history is newly averaged.

For this exact packet, the directional weight and the lower-source product correction are

$$\omega_{\nu;\mu}^c = a^2\Delta_\mu^{\mathrm{cen}} f_\nu^c, \qquad f_\nu^c h_\nu^c = (fh)_\nu^c - \tfrac{a^2}{4}(D_\nu f)(D_\nu h). \tag{R8.19d}$$

Both are exact source identities with their original endpoints. The quadratic tensor terms consequently have two source differences. They have not been discarded or called quartic field words.

CS28–29 gives the complete contact as its full Ward current, explicit identity and weighted Wilson-action terms, the original pointed lower primitive $(C_A/2 - 8C_F)\beta^{-2}a^{-2}H_{(fh)^c}$, and the specified remainder $\mathcal{R}_H^c$. On a complete protected packet of volume at most $C\ell^4$, with the admitted WH source domain, its weak analytic extension satisfies

$$g_W^{-4}a^4|\widehat{\mathcal{R}}_H^c| \le C_N c_\beta^{-2}(a/\ell)^4\|f\|_{C^2_{\ell,a}}\|h\|_{C^2_{\ell,a}}\{\epsilon_\ell^4 + \mathcal{M}_\ell^2\}. \tag{R8.19e}$$

Section 130.2 / The specified bare D8 contact has an exact lower-sector reduction

Here ϵ_ℓ is the dimensionless weak flux strength, $\mathcal{M}_\ell$ includes the complete current and reference multiplier strengths in WH, and the test norm retains all required shifted differences. The flux quartic correction actually has a sixth-power gain. The two quadratic tensor terms obtain their fourth-power gain from R8.19d and the already controlled directional current L^2 norm. No higher field jet is assumed.

On the physical compatible KKT graph this source extension equals the actual remainder. Thus the bare pointed H-contact packet now has a complete scalar projection and the quantitative remainder that the regular-prefix estimate alone did not supply. This leaves the full current, future contacts, cutoff-boundary currents, normalization commutators and all-history admission to be bounded on their actual mixed domains. The raw initial-point statements and their more limited gain are unchanged.

Section 130.3 / Original outer normalization commutators retain a small factor

NC uses the actual fixed-coordinate original R component on its admitted outer pair chart. Write $T_z = e^{\sigma(z,\xi)}T_0$, with T_0 positive and $\sup|\sigma| \le \varepsilon < 1/4$. The original R-II comparison gives $\varepsilon \le Cg_k\mathcal{P}(\log g_k^{-2})$ for that component. The original constraints, selectors and history records remain in T_0.

For $K_z = T_z/T_z1$, the direct density calculation and Cauchy on $K_z - K_0$ give

$$\|K_z - K_0\|_{\mathrm{TV}} \le C\varepsilon, \qquad \|D^r K_z\|_{\mathrm{TV}} \le C_r\varepsilon\rho^{-r}, \qquad r \ge 1. \tag{R8.20}$$

Every positive derivative retains the small comparison factor. The corresponding relative logarithm is controlled, but the absolute scalar $\log T_0(1)$ is not bounded by R8.20.

For an actual outer current $\tau = \mathscr{D}_{Y_1\cdots Y_m}\nu$, the exact lift is

$$\tau \otimes K = \sum_{A\subseteq[m]} \mathscr{D}_{Y_A}(\nu \otimes (Y_{A^c}K)). \tag{R8.21}$$

Subwords retain their original order. In particular the empty subword is an actual scalar correction, not an omitted current. Every proper correction has at least one differentiated kernel and receives R8.20's small factor. Its zero fiber mass does not make it vanish under a fiber-dependent physical readout.

There is also a quantitative answer to the component-count issue. For independent **original charged** components, let $E = \sum_i \varepsilon_i$. Their product difference is bounded by CEe^{CE}. With the VC record weight A_ω, fixed ownership and the original logarithmic powers, NC19–23 proves

$$E \le \eta_* A_\omega, \qquad E \le Cg_{\max}A_\omega^p, \qquad CEe^{CE} \le Cg_{\max}e^{r_{\mathrm{NC}}A_\omega}. \tag{R8.22}$$

Here $\eta_* \to 0$ at the chosen weak floor and p is fixed. One assigned reserve fraction pays both the exponential count and the polynomial. Integration gives $Cg_{\max}B_X$, and the original trajectory permits $g_{\max} \le Cg_k$ when its actual charged scales are no later than k.

Literal Cauchy radii and canonical direction factors remain. This does not identify an internal old derivative as an outer one, pay an uncharged clean array, or estimate an unknown moving-selector derivative. It supplies a checked small outer commutator row once those coordinates and records are actually the ones in the original component.

130.4.1 A reserve for the actual old records

VC strengthens RC's fixed-record estimate using the original positive R-II cost ledger. Its Gaussian counts, scalar normalizers and history entropy are charged at their original scales. The scalar costs already contain $(1+\log g_j^{-2})|Z_j|$. Their geometric bound retains the number of connected components, including a separate $+1$ for each component.

Increasing the coefficient of those already present costs by a fixed small amount is paid by the same strict age bank after one weak-floor choice. This proves the restricted moment reserve

$$A_\omega = \mathscr{E}_\omega + \sum_{R\in\mathcal{R}_\omega} [1+\log(1+g_{j(R)}^{-2})]n_R, \qquad \sum_\omega \int e^{rA_\omega}\, d\nu_\omega \le B_X. \tag{R8.23}$$

Only records charged in that actual ledger belong to $\mathcal{R}_\omega$. Exterior clean arrays and auxiliary coordinates do not become chargeable by appearing in a new description. Each energy and record is owned once. Source, normalization-current, comparison and spare fractions are all allocated from this one reserve.

130.4.2 Exponent differences, denominators and absolute scalar output

Suppose the two literal common records satisfy $d\mu_{1,\omega} = e^{W_\omega} d\mu_{0,\omega}$ and $|W_\omega| \le \delta A_\omega$. Then

$$\|(T_1 - T_0)F\| \le \frac{\delta}{r-\delta} B_X \|F\|_\infty, \qquad 0 \le \delta < r. \tag{R8.24}$$

The proof bounds the m-th coefficient by $B_X\|F\|(\delta/r)^m$ before summing the original internal histories. A relative Gaussian-form defect is charged directly to its own energy; actual inherited analytic atoms and remaining logarithmic scalar costs are charged to their original owners. Large old counts need not be moved outside this positive integral.

The comparison includes absolute scalar atoms. A genuine preparation denominator contributes $-\log D_1 + \log D_0$ to the defect. The original large classical numerator/denominator action cancellation is made before taking absolute values; an unmatched classical term is an additional defect to prove. It is not covered merely by calling it a logarithmic normalizer.

An optional later factorization $T_i = Z_i K_i$ can be recombined exactly before estimates. This cancels its own product-rule normalization terms. It does not cancel an original denominator already inside T_i, nor an independent external vacuum increment. VC's one-point example also shows that a moment reserve for absolute masses does not give a uniform relative logarithm for a very small component mass.

The original R formula applies each unnormalized component operator once to the **joint** Mayer product. VC25–29 retains that placement and the internal history dependence. Its analytic comparison on a common strict formation ball gives

$$\|\mathcal{F}_X(1) - \mathcal{F}_X(0)\| \le \frac{\delta}{r_1 - \delta} M_X. \tag{R8.25}$$

Evaluation at the trivial external pair preserves this bound for the absolute R vacuum increment. The empty Mayer family, the one-polymer connected term, and every actual component multiplicity remain present.

130.4.3 What is still required for S94

R8.24–25 are quantitative bounds for the original formation on **matched inputs**. The full two-torus assertion still requires those inputs to be the actual original ones at every level: typed inverse factors, boundary prescriptions, prepared readouts, constraints, selector supports, genuine reference denominators, inherited potentials and external scalar increments.

For records already inside a reserve-bearing live R body, R8.23 can pay the old count even if their common defect uses an adaptive distance. This improvement does not apply to uncharged clean-volume sums, an uncanceled relative logarithm, or an independent external scalar. Their earlier birth-distance or equivalent restoring estimate remains necessary.

After actual complete nonwrapping records are matched, a degree-q^4 cover has q^4 copies, and division by physical volume cancels that factor. Every wrapped tail needs its complete carrier estimate. A visible local support alone is insufficient: the carrier includes its original gauge, determining-prefix and normalizer read sets. These complete matching and tail assertions have not been established for the full inherited construction.

Consequently the required complete comparison

$$\frac{V_{a,T_*}}{\mathrm{Vol}(T_*)} - \frac{V_{a,qT_*}}{\mathrm{Vol}(qT_*)} \le C \tag{R8.26}$$

remains unproved uniformly in fine cutoff and covering degree. The already proved nonlinear physical minimum and Gaussian comparisons are inputs to this problem. They are not silently identified with its full vacuum construction.

Section 131.1 / Scope and the distinction specified

The weighted actual-entry assertion AE8.8 has not been proved for every original prepared R field and every simultaneous complex wall. The R-I range clauses (1.64)–(1.69), its initial preparation conditions (i),(ii), and the complete marked version of those clauses have not been scope in primary full text. R-II's citation of them is not a substitute for that inspection.

There is a useful new model-specific estimate. The exact scalar remainders CQ23 and CS29 can be bounded directly on an assigned-scale cell family, without first putting the whole mixed field into one unweighted current-unit weak chart. The required flux and current moments have powers four and two, respectively. The flux requirement is weaker than AE8's full p-moment requirement on a family with the fixed total geometry specified in AE9.6b below. The combined flux/current condition is a different sufficient condition on arbitrary fields: a weak potential bound does not itself control the actual directional current. This is proved below for the actual nonlinear Wilson words, retaining their original mesh, all shifted test values, and their complete Ward and lower scalar partners.

This does not prove LM47 for a general previously generated analytic coefficient. It also does not control the full current in CS28 or its later contacts. The revised sufficient norm for the two explicit scalar remainders is not silently substituted for the stated LM47 core. The actual selector estimate needed to use it is stated separately.

Section 131.2 / The original prepared argument is unchanged

Write the literal R-II(1.61)–(1.63) composite as

$$\Gamma^{R}_{\omega,\mathbf{s}}(z,\xi). \tag{AE9.1}$$

The notation has the same meaning as AE8.1: the fixed original history ω retains the actual boundary gauge, joined determining sets, old retained/integrated variables ξ, outer pair z, and simultaneous preparation/source walls $\mathbf{s}$. It does not replace that composite by an independent interpolation.

There are still three different fields:

$$U_{\mathrm{raw}}, \qquad \Gamma^{R}_{\omega,\mathbf{s}}, \qquad P_{\mathrm{old}}\big(q_{\mathrm{old}}(\Gamma^{R}_{\omega,\mathbf{s}})\big). \tag{AE9.2}$$

An estimate for one is not assigned to another. In particular, the configuration example in Section 6 below is an actual Wilson configuration, but is not claimed to be the output of AE9.1.

The primary R-II text retains the gauge in (1.61) because different determining representations meet at its boundary layer. Its subsequent joins lead to the range statement (1.65). The range is the original

Section 131.2 / The original prepared argument is unchanged

space defined in R-I, and the argument discusses admission of old effective-action terms into their old analyticity domains. It does not identify that space with the stronger unweighted weak ball introduced for M8. The exact sources are listed in Section 9.

Section 131.3 / Cellwise flux and current moments for the actual contact words

Fix the original bare Wilson mesh a, a physical test/readout length $\ell \geq a$, and the same $SU(N)$ conventions as CQ and CS. The word mesh a remains fixed throughout this chapter. It is not replaced by the current assigned length of a cell.

Take the actual assigned primary cells meeting the complete local read sets. Assign each plaquette once and each bond-current once to a cell, keeping separate incidence lists if necessary. Denote their assigned lengths by

$$a \leq \sigma_\Delta \leq \ell. \tag{AE9.3}$$

The assigned plaquettes and bonds in a cell have total lattice four-volume at most $C_{\mathrm{sh}}\sigma_\Delta^4$. A bounded-overlap cover gives the same estimates with its actual overlap constant. The finite incidence constants of the plaquette stars are included once. No count of primary cells or connected components is suppressed.

Use the actual matrix flux

$$F_{a,p} = a^{-2}(P_p - I).$$

For $p > 8$, define

$$\begin{aligned} \epsilon_\Delta(U) &= \sigma_\Delta^{2-4/p}\|F_a(U)\|_{L_a^p(\Delta)}, \\ \mathcal{M}_\Delta(U)^2 &= \sigma_\Delta^2 \sum_\nu \sum_{\mu\neq\nu} \|Q_{\nu;\mu}(U)\|_{L_a^2(\Delta)}^2, \qquad Q_{\nu;\mu} = J_{\nu;\mu} - J_\nu/3. \end{aligned} \tag{AE9.4}$$

These are moments of the specific actual field being evaluated. They are not assumed selector ceilings. In particular, the current in AE9.4 is the physical bare Wilson directional current of that field; it is not an arbitrary independent current coordinate from an outer analytic pair.

For complex fields, use the positive matrix norm only in estimates; the defining Wilson identities retain their bilinear holomorphic pairings. Assume that the actual finite prefix transports, their inverses, and the plaquette inverses occurring in CQ/CS are bounded by the fixed chart constant B. This is precisely the extra qualification in CQ24. It is not inferred from small plaquette defects alone. If those transport bounds are cell dependent, their actual powers must remain in the sums below instead of being absorbed into one constant.

Cellwise Hölder gives

$$\begin{aligned} \|F_a\|_{L_a^4(\Delta)}^4 &\leq |\Delta|_a^{1-4/p}\|F_a\|_{L_a^p(\Delta)}^4 \\ &\leq C_{\mathrm{sh}}\sigma_\Delta^{-4}\epsilon_\Delta^4, \\ \sum_{\nu,\mu\neq\nu} \|Q_{\nu;\mu}\|_{L_a^2(\Delta)}^2 &= \sigma_\Delta^{-2}\mathcal{M}_\Delta^2. \end{aligned} \tag{AE9.5}$$

All powers follow from physical volume: the flux has dimension two, and the current dimension three. This is a direct lattice inequality at every cutoff, not a continuum substitution.

Section 131.3 / Cellwise flux and current moments for the actual contact words

Define the two dimensionless mixed moments

$$\begin{aligned}\mathfrak{F}_4(U;\ell) &= \sum_{\Delta}(\ell/\sigma_\Delta)^4\epsilon_\Delta(U)^4,\\ \mathfrak{J}_2(U;\ell) &= \sum_{\Delta}(\ell/\sigma_\Delta)^2\mathcal{M}_\Delta(U)^2.\end{aligned}\tag{AE9.6}$$

The sums are literal. Their size and the ownership of their terms still have to be estimated on the original history. For a disjoint assignment the current moment is simply

$$\mathfrak{J}_2(U;\ell) = \ell^2\sum_{\nu,\mu\neq\nu}\|Q_{\nu;\mu}(U)\|_{L^2_a}^2.\tag{AE9.6a}$$

Likewise $\ell^4\|F_a\|_{L^4_a}^4 \le C\mathfrak{F}_4$. Thus these are the already relevant physical L^4-flux and L^2-current controls expressed with the actual assigned-scale data. Introducing their cellwise notation supplies no new regularity theorem.

For comparison with an unweighted flux bound, one must also keep the total geometry. On a disjoint assignment, discrete Hölder in the cell index proves

$$\mathfrak{F}_4(U;\ell) \le \ell^4\left(\sum_{\Delta}\sigma_\Delta^4\right)^{1-4/p}\|F_a\|_{L^p_a}^4.\tag{AE9.6b}$$

A bounded overlap contributes its recorded constant. Consequently the flux requirement is weaker at fixed $\sum_\Delta\sigma_\Delta^4 \le C\ell^4$, but the individual cell-volume upper bounds alone do not prove that total bound. No such comparison is needed for the direct scalar inequalities below.

131.4.1 The current-square contact

The pointwise identity CQ23 is unchanged:

$$C_{f,h} = \mathscr{E}[\widetilde{V}^{f,h}] + k_0 + \sum_p k_p s(P_p) + \mathcal{R}^{(8)}_{f,h}. \tag{AE9.7}$$

Its coefficients retain the two original source anchors and all ordered bond incidences. Applying CQ24 and AE9.5 gives

$$\begin{aligned} g_W^{-4} a^4 |\mathcal{R}^{(8)}_{f,h}(U)| &\le C_{N,B} c_\beta^{-2} \|f\|_\infty \|h\|_\infty \sum_\Delta (a/\sigma_\Delta)^4 \epsilon_\Delta(U)^4 \\ &= C_{N,B} c_\beta^{-2} (a/\ell)^4 \|f\|_\infty \|h\|_\infty \mathfrak{F}_4(U;\ell). \end{aligned} \tag{AE9.8}$$

Here $\beta = c_\beta g_W^{-2}$ is the coupling of this same Wilson measure. This uses the actual matrix plaquettes and the original bare a, including on cells assigned a much larger length.

The estimate does not require a global $W^{1,p}$ chart for U. It requires the displayed actual flux moment and, in the complex case, the actual common transport bounds. It controls only the explicit scalar remainder. The full Ward current, lower action coefficient, and absolute scalar coefficient in AE9.7 remain present.

131.4.2 The independently pointed-completed H contact

Use exactly the identified completed source representatives from CS21–31. Their endpoint weights satisfy

$$\omega^c_{\nu;\mu} = a^2\Delta^{\mathrm{cen}}_\mu f^c_\nu, \qquad f^c_\nu h^c_\nu = (fh)^c_\nu - \frac{a^2}{4} D_\nu f D_\nu h. \tag{AE9.9}$$

They are not newly averaged for a selected history. Write

$$T_\ell(f,h) = \|f\|_{C^2_{\ell,a}}\|h\|_{C^2_{\ell,a}}, \qquad \|f\|_{C^2_{\ell,a}} = \max_{r\le 2}\ell^r\|D^r_a f\|_\infty, \tag{AE9.10}$$

with the same mixed and shifted differences as in CS.

Retain the four summands of the exact remainder CS29:

$$\begin{aligned}\mathcal{R}^c_H = {} & 2\beta^{-2}a^{-4}\mathcal{Q}^H - 2C_F\beta^{-2}a^{-2}\sum_p \Xi_p Q_F(P_p) \\ & - 2C_F\beta^{-2}a^{-2}\mathcal{H}_{\omega^c,h^c} - \frac{\alpha_H}{4}\beta^{-2}H_\eta.\end{aligned} \tag{AE9.11}$$

The source identities give

$$\begin{aligned}\|\Xi\|_\infty &\le C\ell^{-2}T_\ell(f,h),\\ \|h^c\omega^c\|_\infty &\le Ca^2\ell^{-2}T_\ell(f,h),\\ \|\eta\|_\infty &\le C\ell^{-2}T_\ell(f,h).\end{aligned} \tag{AE9.12}$$

No derivative of the field occurs in AE9.12.

The quartic polynomial bounds for $\mathcal{Q}^H$ and Q_F, AE9.5, and the actual L^2 current sum imply respectively

$$\begin{aligned}&\text{first summand:} && Cc_\beta^{-2}T_\ell(f,h)(a/\ell)^4\mathfrak{F}_4,\\ &\text{second summand:} && Cc_\beta^{-2}T_\ell(f,h)(a/\ell)^6\mathfrak{F}_4,\\ &\text{third summand:} && Cc_\beta^{-2}T_\ell(f,h)(a/\ell)^4\mathfrak{J}_2,\\ &\text{fourth summand:} && Cc_\beta^{-2}T_\ell(f,h)(a/\ell)^4\mathfrak{J}_2,\end{aligned} \tag{AE9.13}$$

after multiplication by the common canonical factor $g_W^{-4}a^4$. For example, the third line is exactly bounded by

$$Cc_\beta^{-2}a^2(a^2\ell^{-2}T_\ell(f,h))\sum_\Delta \sigma_\Delta^{-2}\mathcal{M}^2_\Delta = Cc_\beta^{-2}T_\ell(f,h)(a/\ell)^4\mathfrak{J}_2. \tag{AE9.14}$$

Consequently, since $a/\ell \le 1$,

$$\boxed{g_W^{-4}a^4|\mathcal{R}^c_H(U)| \le C_{N,B}c_\beta^{-2}(a/\ell)^4T_\ell(f,h)\{\mathfrak{F}_4(U;\ell)+\mathfrak{J}_2(U;\ell)\}.} \tag{AE9.15}$$

This is an estimate for the actual nonlinear bare scalar expression at U. It is valid on any common complex domain where the stated matrix and moment bounds hold. A WH source reconstruction may be inserted only where its exact identity with this physical expression has already been established. The theorem itself does not extend that identity to an arbitrary wall field.

The source derivatives remain measured at ℓ. Thus the current-square moment has weight $(\ell/\sigma)^2$, not a needlessly enlarged fourth power obtained by treating the test as if it varied at every smaller assigned scale. This improvement would be lost if the exact endpoint cancellation were replaced by a generic higher-field-jet estimate.

Section 131.5 / The sharper selector obligations and saved factors

For these two scalar remainders, a sufficient actual mixed estimate is

$$\sup_{\substack{\xi \text{ in the original selector support} \\ (z,\mathbf{s}) \text{ in the simultaneous domain}}} \{\mathfrak{F}_4(\Xi;\ell) + \mathfrak{J}_2(\Xi;\ell)\} \le C_{\mathrm{packet}}, \tag{AE9.16}$$

where Ξ is stated literally as one of AE9.2. No new cutoff is introduced to force AE9.16. For the J^2 remainder only the flux moment is needed. All genuine denominator and paired-measure factors are untouched by this pointwise estimate.

If a previously extracted finite-source factor is G_j, multiplying AE9.8 or AE9.15 by G_j leaves G_j visible on the right:

$$|G_j|\, C(a/\ell)^4\{\mathfrak{F}_4 + \mathfrak{J}_2\}. \tag{AE9.17}$$

This elementary fact records what a successful atlas estimate must preserve. It does not prove that an arbitrary old coefficient is bounded by G_j on the new domain. In particular, if f or h is replaced by an inherited field-dependent coefficient, the exact Ward identities acquire its field derivatives; AE9.15 cannot be applied while omitting them.

The full unweighted AE8 bound remains a sufficient input route for the M8 reconstruction, whose physical output receives a separate current estimate. On arbitrary inputs it does not imply AE9.16, because it does not control the directional current in AE9.4. For the flux requirement alone, its curvature requirement is of order

$$\sigma^{4/p-2}\epsilon_\Delta \lesssim \tau, \tag{AE9.18}$$

whereas the flux contribution to AE9.16 is $\sigma^{-4}\epsilon_\Delta^4$ in units $\ell = 1$. The next section exhibits actual Wilson configurations satisfying the combined atlas condition while violating the unweighted curvature condition. The analysis also gives a reverse high-frequency example: a small weak potential and small curvature L^p norm can have an unbounded directional-current L^2 norm. Thus the two combined conditions are not ordered on arbitrary fields. On the separately verified physical reconstruction, the WH current bound is an additional input and the scalar estimates apply there.

Neither AE9.16 nor the general marked-core inclusion has been verified for every original prepared history. The advance is the explicit model-specific quantity that suffices for the already reduced bare scalar rows; it is not a declaration that the original selectors bound that quantity.

Section 131.7 / What a legitimate depth payment would require

Suppose a specified subfamily of the original histories has the following *verified literal records*: a deepest retained scale $\sigma_{\min} = \ell L^{-n_*}$; at least one distinct original charged record for every step of that retained chain; and at most $C_{\mathrm{own}} \sum_R n_R$ assigned source cells. These conditions refer to the records in VC3, not to extra analysis coordinates.

If each chain step has $n_R \geq 1$, with

$$w_* = \inf_R [1 + \log(1 + g_{j(R)}^{-2})],$$

then the exact record sum gives

$$n_* \le C_{\mathrm{own}} A_\omega / w_*, \qquad \#\{\Delta\} \le C_{\mathrm{own}} A_\omega / w_*. \tag{AE9.25}$$

For a fixed finite power q, own-scale bounds $\epsilon_\Delta + \mathcal{M}_\Delta \le C_*$ would therefore imply

$$(\mathfrak{F}_4 + \mathfrak{J}_2)^q \le C_q (1 + A_\omega)^q \exp\left(\frac{4qC_{\mathrm{own}} \log L}{w_*} A_\omega \right). \tag{AE9.26}$$

Choose a source fraction r_{source} of the one reserve in VC6a. If its existing weak floor permits $4qC_{\mathrm{own}} \log L / w_* \le r_{\mathrm{source}}/2$, the polynomial is paid from the other half. Then

$$\sum_\omega \int (\mathfrak{F}_4 + \mathfrak{J}_2)^q e^{F_{\mathrm{other}}}\, d\nu_\omega \le C_q B_X, \quad |F_{\mathrm{other}}| \le (r_{\mathrm{total}} - r_{\mathrm{source}}) A_\omega. \tag{AE9.27}$$

This calculation preserves any explicit old factor such as AE9.17. It pays the entire finite packet jointly. It creates no reserve, changes no Wilson measure, and does not pay every mark independently. The floor condition depends on the fixed source power q. This argument does not furnish one floor uniformly over arbitrarily high source degree, or a common all-order continuum hierarchy.

The hypothesis of AE9.25 has **not** been verified for all fields in AE9.1. The original R-II age bank does pay future component costs, but an age-bank number is not an independent stored array at each formal sublevel. In particular:

- A preparation depth N within one current step is not by itself a list of N charged arrays.
- An auxiliary fine collar or a long determining prefix creates no new original record.
- An exterior clean array is not chargeable merely because an inherited source reads it.
- A regular interval after a region has healed need not retain the earlier R coordinates.

For a literal persistent chain with its actual arrays, AE9.25–27 are a valid quantitative payment. They are not an all-history solution of AE8. The exact missing model statement is the charge assignment for the complete actual source records, or a different preparation bound that avoids needing that payment.

Section 131.9 / Primary source specified precise conclusion

The original paper scope online is T. Bałaban, *Large field renormalization. II. Localization, exponentiation, and bounds for the R operation*, Communications in Mathematical Physics 122 (1989), 355–392:

The relevant the published source are the actual joins and retained boundary gauge at (1.61)–(1.63), the range discussion (1.65), and the literal cost/age genealogy (1.79)–(1.85). The Rutgers primary bibliographic record for R-I confirms the article identity but does not provide its range clauses:

No later purported Yang–Mills proof or source-cache description is used as mathematical evidence. The zero-byte local placeholders named for the 1987 and 1988 papers are not readable primary copies.

The new established result is the direct mixed-domain bound AE9.8/AE9.15 for the actual CQ/CS scalar remainders, with its exact own-scale weights and original word mesh. AE9.19–24 proves why this quantity is distinct from the unweighted AE8 requirement and why energy-only reasoning does not supply it. AE9.25–27 gives a valid use of the existing reserve only on literally verified charged chains.

AE8 on every original prepared history, LM47, and all-history MT36 remain unproved. No general mixed-entry counterexample has been established either.

Section 132 / Whole-pair density commutators, later contacts, and the unresolved current grade

This chapter continues NC and the current part of the authoritative the argument ledger. It proves a quantitative density-commutator estimate which does not differentiate the density perturbation in internal variables. It includes sharp selectors, every later smooth source contact, and the original normalization factors as ordinary coefficients before a differential distribution is formed.

The result applies to a complete scalar/current pair on one identified reference realization. It does not manufacture that pair on a different constraint fiber. Its model-specific application uses the actual small complex density perturbation in R-II (1.71)–(1.75). The old partner norms and saved factors remain explicit.

A separate calculation below shows why the scalar remainder's fourth-power grade cannot be assigned to the whole CQ or CS Ward pair. At the trivial field its scalar companion has an extensive identity term. Even after the full conditional Gaussian cancellation, a bounded canonical test detects a third-power, rather than fourth-power, contribution for the actual step-average example. These statements do not disprove MT36 with a correctly separated current/contact sector. They rule out a particular unjustified grade assignment.

Section 132.1 / Complete paired distributions on the actual reference space

Fix one actual finite-dimensional coordinate realization M, including all retained variables and prescribed exterior records. Let $\mathcal{W}$ be a finite list of nonempty ordered words in specified smooth vector fields. The vector fields carry their original canonical couplings, physical units, anchor data,

and coordinate coefficients. For a word $w = (Y_1, \dots, Y_r)$, use the convention

$$\langle \mathscr{D}_w \nu, \varphi \rangle = \int Y_1 \cdots Y_r \varphi \, d\nu. \tag{CT9.1}$$

This fixes the order independently of alternative adjoint conventions. No word is reordered.

The input is an ordinary finite scalar measure τ and ordinary finite coefficient measures ν_w such that

$$\tau = \sum_{w \in \mathcal{W}} \mathscr{D}_w \nu_w \quad \text{as distributions on } M. \tag{CT9.2}$$

Measures can be signed, complex, or valued in a fixed source test-functional Banach space. They may be singular and supported on the actual constraint graph. Define their represented pair measure and mass by

$$m_{\mathcal{P}} = |\tau| + \sum_{w} |\nu_w|, \qquad \|\mathcal{P}\|_{\mathrm{pair}} = m_{\mathcal{P}}(M). \tag{CT9.3}$$

For an output distribution, $\mathfrak{M}_{\mathcal{W}}$ denotes the infimum of the sums of variations of its representing coefficient measures in the allowed words and the empty word. Any explicit representation provides an upper bound for this quantity. A physical weight assigned to a word is included in its coefficient measure, rather than changed by the following identities.

CT9.2 is substantive. If an actual smooth graph map $U \mapsto (U, q(U))$ is used, its pushed vector field contains $(Y, Dq\, Y)$. Leaving the second component out generally creates a normal derivative of the delta constraint and violates CT9.2. Likewise, a selected scalar numerator is not automatically the directional derivative of its selected vector coefficient; its previous boundary packet must first be retained.

All bounds below require a common realization for the two measures being compared. For the original R density comparison it is enough that $m_{\mathcal{P}}$ be absolutely continuous with respect to its one reference body T_0, with a proved coefficient envelope there. More generally, a direct bound for the density ratio on $m_{\mathcal{P}}$-almost every point suffices. A bound stated only T_0-almost everywhere cannot be used for unrelated singular coefficient measures.

Section 132.3 / Sharp selectors and their actual failure reserve

Let $0 \le \chi \le 1$ be an original measurable selector. The selected identity is

$$\chi h\tau = \sum_{w} \mathscr{D}_w(\chi h \nu_w) + \mathcal{C}_{\chi h}(\mathcal{P}). \tag{CT9.9}$$

Linearity in the multiplier gives

$$\mathcal{C}_{\chi h}(\mathcal{P}) = \mathcal{C}_h(\mathcal{P}) - \mathcal{C}_{(1-\chi)h}(\mathcal{P}), \tag{CT9.10}$$

and therefore

$$\mathfrak{M}_{\mathcal{W}}(\mathcal{C}_{\chi h}(\mathcal{P})) \le \int |h-1|\, dm_{\mathcal{P}} + \int (1-\chi)|h|\, dm_{\mathcal{P}}. \tag{CT9.11}$$

The first term is the small density commutator. The second is the complete original failure packet, including its order-zero companion. There is no exposed boundary delta distribution.

For many original tests, use their existing first-failure record or the exact disjoint identity

$$1 - \prod_{i=1}^{n} \chi_i = \sum_{i=1}^{n} \left(\prod_{j<i} \chi_j \right) (1-\chi_i). \tag{CT9.12}$$

The right side is paid by the one actual failed-sector reserve. An auxiliary ordering changes no rarity credit.

If a far selector is independent of every original vector field in every word, it may be included in all measures before CT9.2 is used. Only the genuinely dependent probes are then assigned a near failure. Small influence is not independence. An old reconstruction-dependent selector retains its entire determining set.

Section 132.6 / One shared reserve pays the entire relative perturbation

For the independent original components actually charged in the VC ledger, let $E = \sum_i \varepsilon_i$. The product ratio satisfies $\|h - 1\|_\infty \le CEe^{CE}$, both for the unnormalized product and for the product of the actual normalized ratios, with a changed fixed constant. NC19–22 gives

$$E \le \eta_* A_\omega, \qquad E \le C g_{\max} A_\omega^p. \tag{CT9.21}$$

Suppose the complete pair's pointwise coefficient envelope on that same original positive body is

$$dm_{\mathcal{P},\omega} \le A_I W_{e,j} (1 + A_\omega)^d \, d\nu_\omega. \tag{CT9.22}$$

Here every actual scalar and current coefficient occurs on the left; CT9.22 is not supplied by a bound for one projected scalar remainder. For the bare fixed compact packets below it follows from the explicit word calculation. For a general inherited mixed packet it remains an input to be proved.

Allocate one fraction $r_{\mathrm{CT}} > 0$ of the existing VC reserve to this complete expression. Choose the weak floor so that $C\eta_* < r_{\mathrm{CT}}/2$. Polynomial maximization gives

$$(1 + A_\omega)^d CEe^{CE} \le C_{d,p,r_{\mathrm{CT}}} g_{\max} e^{r_{\mathrm{CT}} A_\omega}. \tag{CT9.23}$$

Keeping every other assigned weight in the same integral then proves

$$\sum_\omega \mathfrak{M}(\mathcal{C}_{h_\omega}(\mathcal{P}_\omega)) \le C_{I,r} A_I W_{e,j} g_{\max} B_X. \tag{CT9.24}$$

For later sources use the total polynomial degree of CT9.17 in the single maximization CT9.23. A contact, derivative, or source label does not receive a separate reserve. The original source-zero component count and ownership multiplicities remain those in VC. Actual failed-sector terms in CT9.11 are included using their remaining assigned failure fraction, rather than charged again to each label.

Section 132.10 / Updated current-transport conclusion

The proved advance is CT9.6, together with CT9.13–24 for the actual original relative density perturbation. It carries complete current partners, all later smooth finite-source contacts, sharp failure packets, and normalization factors without requiring separate internal derivatives of that perturbation. It retains the original small g_k-times-logarithms factor, and one previously assigned reserve pays the charged component count and finite polynomial cost.

Its precise remaining model-specific inputs are:

1. A complete ordinary scalar/current pair on the literal original reference realization, with all constraint and previous boundary partners retained.
2. Its joint envelope CT9.22, including the actual old mesh and saved grades of every partner, on every simultaneous source wall.
3. The derivative bounds of later inherited source coefficients in CT9.17 on that same support.
4. The actual mixed readout jets in CT9.36, or a verified two-slot representation for those terms to which a stronger first-jet contraction is assigned.

Bare creation in item 2 is verified by CT9.25–28. The full mixed-history versions of these four statements have not been proved here. CT9.30–35 show why the scalar CQ/CS remainder estimate cannot replace them. Thus all-history MT36 remains open.

Section 132.11 / Sources and exact dependencies

The original model input is T. Bałaban, *Large field renormalization. II. Localization, exponentiation, and bounds for the R operation*, Communications in Mathematical Physics 122 (1989), 355–392, (1.71)–(1.75), printed pp.378–380:

The paper's positive reference density and bounded complex perturbation are used in CT9.18–20. The new paired-current inequalities are proved above and are not attributed to that paper. NC supplies the exact relative-normalization calculation and charged component estimate; VC supplies the restricted

Section 132.11 / Sources and exact dependencies

single reserve. XP supplies the finite-source coefficient convention and exact same-kernel transport. C7 supplies the failure-sector interpretation and regular-prefix jet bounds. CQ and CS supply the exact bare contact vectors and scalar coefficients used in CT9.25–31. CP33–39 supplies the complete conditional Gaussian setup used in CT9.32–35.

Section 134 / A closed Gaussian Ward family and its exact engineering

This chapter strengthens CT9.32–35 by proving the matching upper bound and an exact recursion for the complete conditional Gaussian current. It also compares that bound with the source normalization actually used in MT1 and MT39. A positive power obtained after inserting an additional a^4 is not automatically a bound for a source which did not contain that factor.

The Gaussian theory here is the actual flat quadratic Wilson reference on the physical slice used in CP33. Its original coupling $\beta = c_\beta g_j^{-2}$ is fixed through this conditional Gaussian tower. A later g_e is retained as the actual canonical coordinate normalization. This does not assert that a nonlinear running effective density remains Gaussian.

Section 134.1 / The actual block-and-line adjoint has a sharp face-energy bound

Use the direct flat average $Q_{a,\ell}$ of FW and HJ. Its physical adjoint holds a coarse component constant on its centered ℓ-blocks, then applies the normalized line average in that component's direction. Write

$$Q_{a,\ell}^{*} = L^{*}P_{0}. \tag{GG9.1}$$

The own-direction line filter can differ between vector components. Each is an L^{2} contraction and commutes with fine translation differences. The physical block hold P_{0} is an L^{2} isometry.

For a scalar component, exact counting of the one fine layer crossing every coarse face gives

$$\|D_{\nu,a}P_{0}u\|_{2,a}^{2} = \frac{\ell}{a}\|D_{\nu,\ell}u\|_{2,\ell}^{2}. \tag{GG9.2}$$

Indeed a face has $(\ell/a)^{3}$ fine crossings. Each contributes $a^{4}|u(y+\ell\hat{\nu})-u(y)|^{2}/a^{2}$, whereas its coarse difference contributes $\ell^{4}|u(y+\ell\hat{\nu})-u(y)|^{2}/\ell^{2}$. The ratio is ℓ/a. All original faces and component multiplicities remain present.

For the flat Maxwell Hessian $H = d_{a}^{*}d_{a}$ on the transverse slice, or its equivalent positive physical realization in CP33, the curl form is bounded by the full-gradient form. Applying GG9.2 and the line contractions componentwise therefore proves

$$\begin{aligned}\langle u, R_{a,\ell}u\rangle_{\ell} &:= \langle Q_{a,\ell}^{*}u, HQ_{a,\ell}^{*}u\rangle_{a} \\ &\le C\frac{\ell}{a}\|\nabla_{\ell}u\|_{2,\ell}^{2}, \qquad R_{a,\ell} = Q_{a,\ell}HQ_{a,\ell}^{*}.\end{aligned} \tag{GG9.3}$$

Harmonic variables are fixed or retained separately as in CP33. No inverse is assigned to the kernel of H. The transposes are on the actual physical row spaces: a fine transverse projection does not change the curl energy, and a pure-gradient raw endpoint term pairs to zero against the co-closed coarse multiplier. Thus the box-and-line expression computes this physical form without discarding a nonzero gauge component of an unrestricted row.

Rescale the retained spacing to one, write $r = a/\ell$, and put $\widehat{R} = \ell^{2}R_{a,\ell}$. Then

$$0 \le \widehat{R} \le Cr^{-1}(-\Delta_{1}), \qquad \|\widehat{R}\|_{\mathrm{op}} \le Cr^{-1}. \tag{GG9.4}$$

The ambient box-and-line form has a fixed coarse range because both original adjoint stencils have bounded physical diameter $C\ell$ and H has its actual nearest-plaquette range. Restriction to a physical transverse row space, especially in a nonlocal coordinate basis, need not leave a finite-range matrix.

Section 134.1 / The actual block-and-line adjoint has a sharp face-energy bound

The operator and trace estimates below do not require locality of that restricted matrix. The lower bound CT9.35 is sharp in its power: the HJ alternating transverse field has energy $4\epsilon^2 T^4/a$ in unit retained spacing.

This is a bound on the compensated form QHQ^*. Bounding $\|H\|_{\mathrm{op}} = O(a^{-2})$ first would lose a power.

Section 134.4 / Which age gain the original source actually receives

There are two distinct quantities in the construction. CT9.34 and GG9.7 keep the original-word factor a^4 explicitly. MT1's physical dimension-six sources are instead $g^{-2}H_a$ and $g^{-3}C_a^\sharp$; they do not

contain the additional a^2 used in LM4's dimensionless single-cell energy estimate. Their dimension-eight contact likewise cannot be assigned a new a^4 merely to make its bound decay.

At retained spacing one, the exact Gaussian calculation gives the following comparison, with the same coarse count and relative coupling factors in every row:

Actual recorded coefficient multiplying $g_j^{-4}\mathcal{W}_{a,1}$	Net bound
1	Ca^{-1}
An already proved a^2 coefficient	Ca
The original-word factor a^4, if it is genuinely part of that packet	Ca^3

These are one formula:

$$a^p g_j^{-4}\mathcal{W}_{a,1} = a^{p-4}\mathcal{S}_a, \qquad \|a^p g_j^{-4}\mathcal{W}_{a,1}\| \le Ca^{p-1}. \tag{GG9.16}$$

The bounded test CT9.34 shows the corresponding lower power for this Gaussian reference, so a uniform improved exponent cannot be inferred by a norm change which still controls that test.

If a complete packet genuinely has the own-word a^4 factor, GG9.10 supplies $L^{-3(e-j)}$ times the recorded relative coupling polynomial. MT34–35 then makes that kernel summable. This is a valid separately graded Gaussian sector. If a stable dimension-six birth has an actual additional a^2 multiplying this dimension-eight current, its $L^{-(e-j)}$ kernel is also summable. The required prefactor must already have been proved in that birth's exact decomposition.

In contrast, the physical contact with no such prefactor still has a^{-1} growth. Replacing the preserved word normalization by the current-size factor ℓ^4 multiplies GG9.7 by $(\ell/a)^4$, giving exactly that growth in retained units. This explains the apparent discrepancy with the generic $L^{4n}\lambda^n$ warning in C7: the extra engineering conversion and the input coefficient mass must be specified. Neither result gives a power gain to an unnormalized physical contact.

MT39 has already included the old-root count r^{-4} and the dimension-six normalization r^{-2} in its scalar stable row. GG9.6 likewise already includes all original fine roots. Appending another r^{-4} root count to GG9.7 would double count them. Undoing a factor a^4 absent from the actual source, however, is a necessary normalization correction. Thus the closed Gaussian row does not itself prove MT36 for the physical source normalization currently in MT1. It supplies the row for exactly those packets whose recorded coefficient puts them in the summable cases of GG9.16.

Section 134.5 / Scope of the positive result

The sharp face estimate GG9.3, complete matrix recursion GG9.13–14, coarse scalar/vector bounds GG9.9–11, and later retained contact identity GG9.15 are established for the actual flat constrained Gaussian reference. Their own-word-normalized grade is three, with literal coarse counts and relative couplings retained.

For a nonlinear Wilson transition, the Hessian coefficient, nonlinear averaging response, non-Gaussian conditional covariance, effective density, and the resulting first-order drift must all be controlled. A flat Gaussian cancellation cannot be assigned to those terms by perturbative notation. The companion nonlinear Hessian calculation NH9 isolates a principal tensor estimate and the precise unbounded-by-current-results drift obligation.

Section 135 / The exact Wilson Hessian tensor and its compensated

The complete constant-test CQ current has an exact second-current representation. Its principal tensor can be bounded on the admitted nonlinear regular chart using the actual prefix adjoint and the Wilson plaquette Hessian form. The first-order drift created by the nonlinear observation remains a distinct obligation. No generic higher-jet estimate is used to claim its missing grade.

Throughout this chapter the contact word keeps its original mesh a and bare coupling g_j, with $\beta = c_\beta g_j^{-2}$. First rescale the retained length to one. The fine/retained ratio is then $a = r$.

Section 135.6 / Established result and remaining scope

The exact second-current representation NH9.5, the actual principal/drift split NH9.8, the compensated nonlinear Hessian bound NH9.13, and the canonical principal estimate NH9.17 are proved on their stated original regular chart. The strict principal exponent $2 + 2s > 2$ uses actual spatial Hessian form control and the already proved fractional prefix-adjoint bound. It is not a claim of extra decay for generic higher field jets.

The full contracted drift estimate NH9.18, its effective-density partners, actual mixed-domain admission, and their inherited current grades remain unproved. The closed flat Gaussian sector GG9 does not supply those nonlinear identities. Consequently these results strengthen the current module without establishing all-history MT36.

The subsequent CD9 chapter sharpens this scope in a restricted direction. CD9.1–3 bound the once-only fixed-reference terminal frame contribution by the principal estimate above. CD9.7–28 prove a complete centered constant-test response bound of order $a\|F\|_\infty$ on a real buffered rectangular regular prefix, with its original fine count and an exact invariant-test interpretation. That strong real diagonal result does not supply the off-diagonal complex mixed-history row, source-graded later partners, or scalar total-variation bound asserted nowhere here.

Section 136 / Terminal frames and a centered real-prefix curvature

This chapter separates a controlled final-coordinate term from the intrinsic response drift. It then proves a curvature-sized estimate for the complete contracted second response of a genuinely centered regular prefix, on a real protected curvature chart. The proof keeps the transported Wilson Hessian

tensor, its color defect, the initial logarithmic frame term, and the actual intermediate coordinate counts.

The centered result has a precise use on invariant terminal tests. It does not identify a fixed-reference off-diagonal current with that centered family, does not estimate every later marked derivative, and does not bound the ordinary scalar companion in total variation. Consequently neither CT9's full pair envelope nor all-history MT36 is closed by this chapter.

The Wilson Hessian row throughout is the constant-test CQ identity NH9.5, with $f = h = 1$. Nonconstant source tests introduce their actual additional partners; this chapter does not derive estimates for all of those partners.

Section 136.2 / A fixed logarithmic reference can have drift at zero curvature

The following calculation rules out a curvature-only bound for the uncorrected coordinate drift at a fixed off-diagonal reference. It does not rule out such an estimate for the remaining intrinsic term after CD9.2, nor for a complete invariant-test current.

Use the construction convention $-\operatorname{tr}(T_A T_B) = \delta_{AB}$, so $C_A = 2N$, and let $z = 6$ be the actual number of plaquettes incident to one bond. At any flat connection the same-bond Wilson Hessian is

$$B_{(b,A),(b,B)} = z\delta_{AB}. \tag{CD9.4}$$

Each incident plaquette contributes one δ_{AB}: after varying that bond its plaquette is a conjugate of $e^{sT_A}e^{tT_B}$, and the trace is conjugation invariant.

For x in the group logarithm chart, BCH gives

$$\left.\frac{d^2}{dt^2}\log(e^{tT_A}e^{x})\right|_{t=0} = \frac{1}{6}[T_A,[T_A,x]] + O(\|x\|^2).$$

The compact adjoint Casimir is $\sum_A [T_A,[T_A,x]] = -C_A x$. The summed expression is odd in x: inversion commutes with the bi-invariant Haar Laplacian, while $\log(g^{-1}) = -\log g$. Thus the summed quadratic term vanishes and

$$\Delta_H \log(e^{x}) = -\frac{C_A}{6}x + O(\|x\|^3). \tag{CD9.5}$$

Take the actual zero-step endpoint chart at retained spacing one, with fixed reference $U = I$, and a small smooth nonconstant pure gauge $V_b = g(x)g(x')^{-1}$. Such a field lies in a weak regular chart when the chosen gauge is sufficiently small and smooth; arbitrary high-frequency pure gauges are not being declared admissible. Every plaquette of V is the identity, but some $x_b = \log V_b$ is nonzero. For $q_{\log,b}(V) = \log V_b$, different-bond second derivatives are zero. Therefore

$$\sum_{i,j} B_{ij}X_jX_iq_{\log,b}(V) = -\frac{zC_A}{6}x_b + O(\|x_b\|^3) = -C_Ax_b + O(\|x_b\|^3). \tag{CD9.6}$$

Both the curvature and J vanish. The first drift summand in NH9.18 is $2g_j^2/g_e$ times CD9.6; its lower score summand is zero. Thus a standalone estimate of that coordinate coefficient by $C\|F(V)\|$ is false.

For the identity observation the intrinsic response is the constant $\widehat{A} = (g_j/g_e)I$. The intrinsic term in CD9.2 is then exactly zero: CD9.6 is entirely a terminal-frame contribution. CD9.3 controls that contribution in the proper principal norm. This example is therefore a check on the coordinate bookkeeping, not a counterexample to the complete physical current.

Section 137.1 / Scope and the original conditional object

Complete S94 remains unproved. This chapter makes two of its matching requirements more precise and proves a quantitative comparison for the nonlinear physical action inside a common reference fiber. It does not replace the actual R construction, its weak-trajectory assumptions, or its inherited vacuum by a new normalization.

The source-free reference insertion identified in S53–S58b is the starting point. Fix the full retained datum b, including every old variable on which the reference depends. Its original relative tree fiber has measure $d\lambda_b(C)$, its original reference cutoff is $\chi_b(C)$, and its positive reference density is $r_i(C;b)$. The index $i = 0, 1$ denotes the two actual choices being compared. When the two tree fibers and cutoff functions have been identified by their literal coordinate map, put

$$D_i(b) = \int \chi_b(C) r_i(C;b)\, d\lambda_b(C), \qquad K_i(dC \mid b) = \frac{\chi_b(C) r_i(C;b)}{D_i(b)} d\lambda_b(C). \tag{VM9.1}$$

These are the **original preparation denominators**, not the optional mass/kernel factorization of an already formed R component. The fiber contains the original independent output variables and gauge forest. All constraints and any older integrations remain in their original positions. In particular b is not assumed to be geometric exterior data alone.

On this real fiber $K_i 1 = 1$, even when D_i is very small. The proof of S54 gives Haar Jacobian exactly

one for its relative coordinates. The reference is fixed before the independent output is introduced. This is the source-free preparation identified from R-II (1.3), its variational uniqueness argument, the new relative variable in (1.34), and the integrated output-selector decomposition on printed p.378. A later removed gauge delta is used only under the original integral on which its removal is licensed.

There is no assertion here that arbitrary records on two physical tori already have the same λ_b, χ_b, target data, or reference readout. VM9.1 specifies the actual common object required for the following exact identities.

Section 137.2 / Only the reference-coordinate oscillation enters its normalized kernel

Suppose the two positive densities on that fiber are related by

$$\log \frac{r_1(C;b)}{r_0(C;b)} = c(b) + v(C;b). \tag{VM9.2}$$

The scalar $c(b)$ can be arbitrarily large. It may depend on all retained old data; it is required only to be independent of this integration variable C. Direct integration, before any bound, gives

$$\begin{aligned} \log D_1 - \log D_0 &= c + \log K_0(e^v), \\ \frac{dK_1}{dK_0} &= h := \frac{e^v}{K_0(e^v)}, \qquad K_0(h) = 1. \end{aligned} \tag{VM9.3}$$

Thus the complete reference numerator/denominator pair cancels its own C-independent difference exactly. This is a cancellation inside the genuine reference insertion. It does not infer a bound for $\log D_1 - \log D_0$ from an absolute mass bound, and it does not delete the scalar c from an independently displayed vacuum term.

For real v, let $\operatorname{osc}_C(v) = \sup v - \inf v$. Then

$$e^{-\operatorname{osc}_C(v)} \le h \le e^{\operatorname{osc}_C(v)}, \qquad \|K_1 - K_0\|_{\mathrm{TV}} \le e^{\operatorname{osc}_C(v)} - 1. \tag{VM9.4}$$

Here total variation is the norm of the signed measure, without the alternative factor one half. The estimate follows because $e^{\inf v} \le K_0(e^v) \le e^{\sup v}$. No lower bound for D_0 or D_1 enters.

On a complex comparison chart, assume K_0 is the fixed positive real reference and v is holomorphic with $\sup |v| \le \varepsilon < 1/4$. Then $|K_0(e^v) - 1| \le e^{\varepsilon} - 1$, so the actual denominator is nonzero on that smaller chart and

$$\|h - 1\|_\infty \le C\varepsilon, \quad \|K_1 - K_0\|_{\mathrm{TV}} \le C\varepsilon, \quad |\log K_0(e^v)| \le C\varepsilon. \tag{VM9.5}$$

The logarithm is the branch continued from zero. This does not prove zero-freeness of an unrelated large component integral. A reference comparison between two complex kernels can use the same fixed positive base only after both of their representations and radii are established.

For the real source-zero endpoints there is a stronger useful form which does not require a small total oscillation for a large component. The genuine logarithmic kernel ratio satisfies

$$W_{\mathrm{ref}} := \log h = v - \log K_0(e^v), \qquad |W_{\mathrm{ref}}| \le \operatorname{osc}_C(v). \tag{VM9.5a}$$

If the already proved oscillation bound is $\operatorname{osc}_C(v) \le \delta n_b$, and n_b is an original charged count with $n_b \le CA_\omega$, this is the actual reference contribution $|W_{\mathrm{ref}}| \le C\delta A_\omega$ required by VC7. The comparison interpolation can now be defined as $e^{zW_{\mathrm{ref}}}K_0$. Its intermediate values need not be normalized reference kernels. VC's one assigned exponential reserve controls it, without requiring a complex zero-free disc for the separately interpolated denominator $D(z)$ on a large body. Thus neither a large original count nor a tiny reference mass forces division by that mass. The count inclusion and the oscillation estimate must be proved on the actual record; auxiliary coordinates or an uncharged external array do not satisfy that inclusion by name.

For a bounded Banach-valued carrying function F and any $a(b)$ independent of C, the exact centered formula is

$$(K_1 - K_0)F = K_0\big[(h-1)(F-a)\big]. \tag{VM9.6}$$

It follows that

$$\|(K_1 - K_0)F\| \le C\varepsilon \inf_{a \text{ independent of } C} K_0\|F - a\|. \tag{VM9.7}$$

The constant part of the actual carrying function disappears in this kernel difference. Any dependence on older variables is retained by taking the infimum conditionally at fixed full b. If the common old numerator measure is $d\mu(b)$, then integration of VM9.7 uses that one original measure and its one majorant. It creates no second copy of a cold component mass.

Positive-order outer derivatives of this comparison retain its small factor by Cauchy on $h-1$ or $K_1 - K_0$, with the actual common radius. The full ordered scalar/current commutators in CT9 must still be included. A derivative of an old internal coordinate has not been identified with such an outer derivative by VM9.5.

137.2.1 Original action completion must be undone before using the cancellation

The reference insertion is often displayed after a new regular action has been extracted. If its exact conditional density is written as

$$e^{-A_i(C,b)}B_i(C,b)\,d\lambda_b(C) = Q_i(b)\,K_i(dC \mid b), \tag{VM9.8}$$

then the comparison in VM9.3–7 concerns the complete right-hand side or the complete product on the left. A scalar shift $A_i \mapsto A_i + c_i(b)$ requires $B_i \mapsto e^{c_i(b)}B_i$. These two factors cancel in the physical density, but their separate absolute scalar increments do not vanish individually. Every later vacuum completion keeps that pairing until its specified subtraction is made.

Consequently VM9.6 does not assert that the empty Mayer family of the completed R formation has zero change. After action completion its carrying factor can contain the compensating regular action; it need not be the constant function in the original reference coordinate. Nor does it cancel the old numerator Q_i, an original denominator in another conditional insertion, or an independently extracted external vacuum increment.

This is the precise difference from the invalid maneuver of writing an arbitrary completed component as $T = ZK$ and then declaring all of its genuine preparation denominators canceled.

Section 137.3 / Exact common refinement of the original output selectors

Let $\{\pi_{0,\alpha}\}$ and $\{\pi_{1,\beta}\}$ be two original finite nonnegative selector partitions on the **same actual conditional space** (C,ξ,b), with

$$\sum_{\alpha}\pi_{0,\alpha}=1,\qquad \sum_{\beta}\pi_{1,\beta}=1. \tag{VM9.9}$$

They may be sharp characteristic functions. Their common refinement is the collection of weights $\pi_{0,\alpha}\pi_{1,\beta}$. It is again a partition of unity. It is a reindexing of one integral and is not the product of two selected component masses.

For arbitrary carrying functions $F_{0,\alpha}$ and $F_{1,\beta}$, the exact pointwise identity is

$$\sum_{\beta}\pi_{1,\beta}F_{1,\beta}-\sum_{\alpha}\pi_{0,\alpha}F_{0,\alpha}=\sum_{\alpha,\beta}\pi_{0,\alpha}\pi_{1,\beta}(F_{1,\beta}-F_{0,\alpha}). \tag{VM9.10}$$

This requires no derivative or smallness of the selectors. It follows by inserting the other partition of unity into each sum. The actual labels and complete records survive on the refinement cells.

Combining VM9.3 with VM9.10, at fixed full old state, gives

$$\begin{aligned}&K_1\sum_{\beta}\pi_{1,\beta}F_{1,\beta}-K_0\sum_{\alpha}\pi_{0,\alpha}F_{0,\alpha}\\&\quad=K_0\sum_{\alpha,\beta}\pi_{0,\alpha}\pi_{1,\beta}\{(h-1)(F_{1,\beta}-a)+F_{1,\beta}-F_{0,\alpha}\}.\end{aligned} \tag{VM9.11}$$

The omitted scalar term is $aK_0(h-1)=0$, exactly. Here a may depend on all fixed old variables but not on C. In particular, if all representatives are the same actual physical carrying function on each refinement cell, selector movement introduces no extra error.

If the representations use different prepared fields, $F_{i,\alpha}=\Phi_{i,\alpha}\circ q_{i,\alpha}$, the remaining term is their actual difference

$$\Phi_{1,\beta}(q_{1,\beta}(C,\xi,b))-\Phi_{0,\alpha}(q_{0,\alpha}(C,\xi,b)). \tag{VM9.12}$$

It has not vanished because both labels are called histories of the same field. An ordinary scalar observable needs its actual value comparison. A pushed current of order m tests derivatives of $\varphi\circ q_1-\varphi\circ q_0$ through order m, so its quantitative comparison additionally needs those actual map jets, with their own coordinate and scale conventions. No common derivative chart is manufactured by the refinement.

The original R output partition on printed p.378 is an instance of VM9.9 after its fiber is identified as in S58b. Its summation remains inside the component integral and acts on the joint interaction. Thus VM9.10 is an exact available operation for that specified partition. It is not an assertion that two different nonlinear delta-constraint supports are already the same conditional space.

137.3.1 No extra reserve or component count is created

For a positive carrying majorant G,

Section 137.3 / Exact common refinement of the original output selectors

$$\sum_{\alpha,\beta} \int \pi_{0,\alpha}\pi_{1,\beta} G \, d\nu = \int G \, d\nu. \tag{VM9.13}$$

Hence a VC reserve already proved for this $d\nu$ is unchanged by the refinement. The number of pairs (α, β) never appears as a multiplicative entropy factor. Conversely an estimate whose exponent charges *both* descriptions' independent old counts needs a comparison with the one actual charged ledger; VM9.13 does not prove that comparison.

This is particularly relevant if a scale assigned by one history differs from the other's scale. Their old spacings are retained in VM9.12. The refinement neither resets a contact word's mesh nor grants a new age or rarity factor.

Section 137.5 / The single reserve and why it does not create missing ancestry records

The estimates above can be used with VC's existing positive reserve after all input differences have been proved in its actual norm. They introduce no new reserve. A common refinement satisfying VM9.13 preserves that same integral exactly.

Suppose a completed exponent defect satisfies $|W_\omega| \le \delta A_\omega$ on its actual refined record. Then VC10 and VC28 apply with the fraction already assigned to the vacuum comparison. Neither a canceled reference-coordinate constant nor the introduction of a pair of selector labels allows the same old energy to be spent again.

There is an important limitation for proposed scale-weight payments. For a literally persistent live component with a nonempty **original charged record at each of** n intervening scales, VC3 gives

$$A_\omega \ge w_{\min} n, \qquad L^{mn} \le \exp\{(m \log L / w_{\min}) A_\omega\}. \tag{VM9.21}$$

This elementary deduction is valid only after those n records have been identified in the original integral. It cannot be applied to the formal depth of a newly described coordinate atlas. The positive R-II estimate retains actual compact integrations, actual old Gaussian arrays, normalizers, and its history entropy. Its age bank in (1.80)–(1.88) also pays future component costs, but a future payment is not a second independent coordinate record.

In particular, preparatory sublevels inside one R step and clean aging after a healed coefficient do not acquire new charged arrays by VM9.21. The original paper uses $N = R_j$ in the future-geometry argument before (1.80); this does not identify N with a count of independent stored coordinates. A proposed use of VM9.21 for AE8 must prove the ancestry statement and retain one allocated reserve.

Section 137.7 / Primary source specified source boundary

T. Balaban, *Large field renormalization. II. Localization, exponentiation, and bounds for the R operation*, Communications in Mathematical Physics **122** (1989), 355–392:

The p.378 output partition and integrated gauge step, p.379 original component placement and history-dependent carrying factor, and pp.384–387 future-cost induction were checked
The deductions in VM9 are the displayed conditional-fiber algebra and nonlinear estimates, not claims that the primary source already states the complete two-torus comparison.

The full defining passage for the background-field paper's typed expansion (3.107), and the complete R-I retained operation (1.100), have not been scope Bibliographic metadata or a secondary transcription is not used to declare those definitions matched. The already checked finite-reference naturality V7N38–40 remains valid in its stated complete-record scope.

Section 138.2 / The complete finite record and its covering lift

Fix an admissible original primitive record $\mathfrak{R}$. Include the following actual data, without replacing any of them: the dynamic fine bonds and every action plaquette; the prescribed exterior bonds and action boundary rule; all determining rows at their assigned scales; the whole raw averaging prefixes and endpoint paths; junction and gauge pins; the prescribed exterior targets; Z, Λ, G_0; the relative output slots; ζ_0; χ; the coupling; the Haar convention; and every finite choice or averaging multiplicity attached to the primitive. Every old datum read by one of these objects is part of the record.

Let $\pi : T' = qT \to T$ be the periodic lattice cover. Suppose the **whole finite record**, including these read sets and paths, has an injective lift to the universal lattice. One sufficient condition is containment of that complete record in a translated open fundamental box. A merely nonwrapping visible action support does not suffice if a determining prefix winds around the torus.

Choose a lift of one root in each connected component of the complete record. Lift its edges by their actual coordinate steps. Because the record has a single-valued injective universal lift, the endpoint is independent of the chosen path: every closed path in the record has zero net winding. Projecting the lift to T' defines a coordinate bijection ι onto its copied record $\mathfrak{R}'$, and

$$\pi \circ \iota = \mathrm{id}_{\mathfrak{R}}. \tag{RN9.2}$$

All action plaquettes, orientation signs, path orderings, row scales, boundary values and gauge roots are copied by this bijection. Define $b' = \iota b$, $\zeta_0' = \iota \zeta_0$, and use the identical scalar cutoff rule in the copied relative coordinates. This constructs a genuine lifted original primitive; it does not declare that an independently generated history on T' selected this same record.

Section 139.1 / Distinct flux and directional-current hypotheses

The identities in AE9, CT9, DC9 and VM9 require a complete scalar/current pair on the same reference space, retention of both slots of the gradient form, a common reference fiber and the original assigned scales. Naming a mixed history does not supply those data.\n\nThe combined flux and directional-current condition in AE9 has a weaker flux requirement on fixed-geometry records and a separate current requirement. As a condition on arbitrary Wilson fields it is not uniformly weaker than an unweighted weak-potential or curvature bound. The two quantitative examples use actual Wilson configurations. Neither is identified as an output of the original prepared R map. The all-history mixed-domain bound, joint marked grades and source-zero matching still require their own hypotheses.

Section 139.2 / Assigned-cell powers and the corrected comparison of norms

For a cell of assigned length σ, AE9 defines $\epsilon = \sigma^{2-4/p}\|F_a\|_{L^p(\Delta)}$. Its flux calculation is

$$|\Delta|_a^{1-4/p}\|F_a\|_{L^p(\Delta)}^4 \le C\sigma^{4-16/p}\sigma^{-8+16/p}\epsilon^4 = C\sigma^{-4}\epsilon^4. \tag{AU9.1}$$

The original word mesh a is not involved in this conversion of the physical norm. It reappears in the actual canonical contact prefactor. Multiplying CQ's quartic polynomial estimate by $g_W^{-4}a^4$ gives exactly $Cc_\beta^{-2}\sum_\Delta (a/\sigma_\Delta)^4\epsilon_\Delta^4$.

The directional-current identity is equally literal:

$$\mathfrak{J}_2 = \sum_\Delta (\ell/\sigma_\Delta)^2\sigma_\Delta^2\|Q\|_{L^2(\Delta)}^2 = \ell^2\|Q\|_{L^2(D)}^2 \tag{AU9.2}$$

for a disjoint assignment. A bounded-overlap version has only its specified overlap constant. No number of source cells disappears.

For CS29 the four canonical contributions have, respectively, the factors

$$(a/\ell)^4\mathfrak{F}_4, \qquad (a/\ell)^6\mathfrak{F}_4, \qquad (a/\ell)^4\mathfrak{J}_2, \qquad (a/\ell)^4\mathfrak{J}_2. \tag{AU9.3}$$

The third factor uses $\omega^c = a^2\Delta^{\mathrm{cen}}f^c$, and the fourth uses the product correction $f^ch^c - (fh)^c = -a^2(Df)(Dh)/4$. The source differences are still measured at ℓ. No extra derivative of a weak field

has entered the argument. Complex estimates retain the original uniform prefix and inverse bounds; small plaquettes alone are not used to supply them.

For a fixed-geometry primary-cell record, the flux norm ordering can be stated precisely. Hölder across the cells gives

$$\begin{aligned}\mathfrak{F}_4 &= \ell^4 \sum_{\Delta} \sigma_\Delta^{4-16/p} \|F_a\|_{L^p(\Delta)}^4 \\ &\le \ell^4 \left(\sum_{\Delta} \sigma_\Delta^4 \right)^{1-4/p} \|F_a\|_{L^p(D)}^4 .\end{aligned} \tag{AU9.4}$$

Thus $\sum \sigma_\Delta^4 \le C\ell^4$, with the actual bounded overlap, makes the global dimensionless curvature norm control $\mathfrak{F}_4$. A per-cell upper bound on its assigned volume alone does not give this conclusion for an arbitrary multiply counted cell list.

AE9's concentrating commuting example also checks. Its exact link integrals, with $a = \sigma^2$, give plaquette angles of size $a^2\sigma^{\alpha-2}$ and adjacent-angle differences of size $a^3\sigma^{\alpha-3}$. The actual Wilson potential has bounded derivatives on this shrinking angle range. Consequently

$$\|F_a\|_{L^p} \asymp \sigma^{\alpha+4/p-2}, \quad \mathfrak{F}_4 = O(\sigma^{4\alpha-4}), \quad \mathfrak{J}_2 = O(\sigma^{2\alpha-2}). \tag{AU9.5}$$

For $1 < \alpha < 2 - 4/p$, both displayed atlas moments tend to zero and the curvature L^p norm diverges. The source is an actual compact commuting configuration; the continuum one-form only specifies its links. This proves one non-implication without identifying a prepared R output. For $0 < \alpha < 1$, its $\beta S_0 \asymp \beta\sigma^{2\alpha}$ and divergent $\mathfrak{F}_4$ also correctly show why energy alone is not a pointwise atlas bound.

139.2.1 A reverse example: weak input does not control the actual current

The missing converse cannot be restored on arbitrary fields. Take a unit four-torus, $a = 1/(2m)$, and the normalized embedded Cartan generator

$$T = \operatorname{diag}(i/\sqrt{2}, -i/\sqrt{2}, 0, \dots, 0), \qquad -\operatorname{tr} T^2 = 1.$$

Let $\zeta_n = (-1)^n$, and put

$$U_1(x) = \exp(\varepsilon a^2 \zeta_{x_2/a} T), \qquad U_2 = U_3 = U_4 = I. \tag{AU9.6}$$

The potential $A_1 = \varepsilon a \zeta T$ has uniformly small $W_a^{1,p}$ norm for fixed small ε. The only nontrivial plane is 12, where

$$P_{12}(x) = \exp(2\varepsilon a^2 \zeta_{x_2/a} T), \qquad \|a^{-2}(P - I)\|_{L^p} \le C\varepsilon. \tag{AU9.7}$$

Use the exact scalar Wilson force $q(t) = \sqrt{2}\sin(t/\sqrt{2})$ in this embedded SU(2) subgroup. The adjacent force difference is

$$J_{1;2}(x) = 2\sqrt{2}\, a^{-3} \sin(\sqrt{2}\varepsilon a^2) \zeta_{x_2/a} T, \tag{AU9.8}$$

up to the irrelevant simultaneous orientation sign. The other partial forces for this bond direction vanish. Hence $\sum_{\mu \ne 1} |Q_{1;\mu}|^2 = (2/3)|J_{1;2}|^2$, and the unit-volume lattice sum gives

$$\mathfrak{J}_2 = \frac{16}{3} a^{-6} \sin^2(\sqrt{2}\varepsilon a^2) \sim \frac{32}{3} \varepsilon^2 a^{-2}. \tag{AU9.9}$$

At the same time $\mathfrak{F}_4 \le C\varepsilon^4$ on the unit primary cell. This is an exact nonlinear example, not merely a continuum derivative estimate.

Section 139.2 / Assigned-cell powers and the corrected comparison of norms

Therefore the combined requirement $\mathfrak{F}_4 + \mathfrak{J}_2 \le C$ and the unweighted weak-potential/curvature requirement are generally incomparable on arbitrary configurations. The flux-only ordering in AU9.4 is valid. On an identified WH physical reconstruction the separate current equation supplies a current bound, and that additional theorem can be used. Weak Sobolev regularity by itself cannot replace it. Neither AU9.6 nor AE9's concentrating example is claimed to obey all original first-entry selectors or to be an actual $\Gamma^{R}_{\omega,\mathbf{s}}$.

Section 139.3 / The full-pair density estimate is algebraically and quantitatively valid

Let $\tau = \sum_w \mathscr{D}_w \nu_w$ be CT9's complete ordinary scalar/current pair on one actual reference realization. For bounded h, the difference

$$h\tau - \sum_w \mathscr{D}_w(h\nu_w) = (h-1)\tau - \sum_w \mathscr{D}_w((h-1)\nu_w) \tag{AU9.10}$$

is a literal representation by ordinary coefficient measures and their original ordered derivatives. Its represented mass is at most $\int |h-1|\,(d|\tau| + \sum_w d|\nu_w|)$. This verifies CT9.5–6. The estimate uses no derivative of h, but includes the ordinary scalar companion. It would not follow from a bound for one projected scalar remainder alone.

For smooth h, the ordered product rule expands AU9.10 into exactly CT9.7. All nonempty subwords retain their order, and the empty subword is a scalar term. For a sharp selector, CT9.4 is still defined because it multiplies only ordinary measures. It does not require an undefined product of the selector with a previous derivative distribution. The complementary-failure split CT9.10–12 is exact, including its scalar partner.

For a later smooth coefficient G, testing CT9.13 against φ expands $Y_w(G\varphi)$. The empty subword is exactly the term subtracted from $G\tau$. The nonempty subwords give CT9.14. Expanding the same product in CT9.4 proves

$$G\mathcal{C}_h(\mathcal{P}) = \mathcal{C}_h(\mathcal{P}_G). \tag{AU9.11}$$

Thus CT9 includes the derivatives of a field-dependent later coefficient and all its scalar/lower-current contacts. Its finite source estimate requires the displayed actual derivative norms of those coefficients; it does not prove them for an unadmitted inherited source. No product of two unrelated derivative distributions appears.

The R-II use is correctly restricted to the actual common comparison body $T_z = e^{\sigma} T_0$. With $|\sigma| \le \varepsilon$, both e^{σ} and $e^{\sigma}/K_0 e^{\sigma}$ differ from one by $C\varepsilon$. This remains independent of the possibly tiny reference mass. The real inherited action remains inside T_0; it has not been called a small perturbation. A complete pair still has to be constructed on that reference space, including its nonlinear constraints, previous cutoff partners, and full graph tangent directions.

Section 139.4 / Bare creation, exact flat counts, and the Gaussian obstruction

The normalization in CT9.25 checks directly from the Ward identity:

$$g_W^{-4}a^4\mathscr{E}[V]\mu=\sum_i \mathscr{D}_{g_W X_i}(g_W^{-5}\kappa a^8 V_i\mu). \tag{AU9.12}$$

The leading vector $\kappa X P_h$ is $\beta^{-1}a^{-5}$ times a first-order vanishing compact word. The other completed vectors have size $\beta^{-2}a^{-5}$ times such words. Their normalized coefficient is therefore exactly of the type $c_\beta^{-2}g_W^{-1}u+c_\beta^{-3}g_W v$. The local force bound and compact Haar derivative bounds give the per-root

coefficient envelope $C\mathscr{E}_Y^{1/2}$ and scalar envelope $C(1+\mathscr{E}_Y)$ in CT9.28. These use the same original plaquettes and their original energy. If that action has already canceled, its energy cannot be borrowed back.

On the full fine torus, the original counts are $4V_a$ bonds and $6V_a$ plaquettes. Therefore

$$A_0 = d_G(36\cdot 4 + 12\cdot 6)V_a = 216 d_G V_a, \qquad D_p = 2C_A - 32C_F. \tag{AU9.13}$$

This verifies CT9.30's leading coefficient $432d_G$. CS28 has $\sum_p t_p = 72V_a$, giving its exact coefficient $96d_G$. The nonzero scalar companion at the flat field is consistent with the per-root creation envelope and inconsistent with assigning the quartic remainder's pointwise grade to the whole extensive pair.

For the completed Gaussian calculation, the original physical adjoints give $\Delta_c = (QH^{-1}Q^{\mathsf{t}})^{-1}$ and $R_a = QHQ^{\mathsf{t}}$. The complete coarse Ward expression is CT9.32, including its trace. Testing it with $1-\cos(t\sqrt{\beta}\langle u, B\rangle)$ and using its divergence representation gives

$$g_W^{-4}a^4 \int W_c \Psi \, d\mu_c = 2c_\beta^{-2} t^2 a^4 \langle u, R_a u\rangle e^{-t^2\langle u,\Delta_c^{-1}u\rangle/2}. \tag{AU9.14}$$

The sign and both powers of β follow by applying $\mathbb{E}[\Delta_c B \sin(t\sqrt{\beta}\langle u,B\rangle)] = t\beta^{-1/2} u e^{-t^2\langle u,\Delta_c^{-1}u\rangle/2}$.

HJ's actual box-and-line average makes the displayed transverse coarse vector into a fine block step. Its own-component line average changes nothing because that vector is constant in its component direction. There are the stated T jump planes, so $\langle u, R_a u\rangle = 4\varepsilon^2 T^4/a$. The step is transverse and mean zero. The fixed physical torus gap bounds $\langle u,\Delta_c^{-1}u\rangle \le CT^2\|Q^{\mathsf{t}}u\|_{L_a^2}^2$, uniformly in a. The test has uniform fixed-order derivatives in $B = g_W b$ coordinates. This proves CT9.35's positive ca^3 lower bound.

It rules out a uniform Ca^4 ordinary smooth-test bound for this complete Gaussian Ward distribution. It is not a nonlinear counterexample along an unproved running Wilson trajectory, and it does not rule out a separately graded current sector. The full trace is necessary even in this diagnostic.

Section 139.5 / Two independent gradient slots genuinely give two first responses

DC9.3–5 follows by applying the first chain rule independently to its two separate functions. In canonical directions it gives

$$\|q_*(A \otimes A\,\nu)\|_{1,\mathrm{TV}} \le \left(\sup_{x,i} \sum_{\alpha} |A_i^{\alpha}(x)|\right)^2 \|\nu\|_{1,\mathrm{TV}}. \tag{AU9.15}$$

No second jet of q occurs. The actual regular first-column bound therefore supplies λ^{2n}. With the declared D8 conversion L^{4n}, this is the stated $(4/L^2)^n$ factor, times the literal canonical ratio and the complete incoming tensor mass.

That last qualification is essential. For the actual diagonal Wilson Dirichlet tensor, the coefficient per fine index is $\kappa a^4 g_W^{-2} = c_\beta^{-1} a$. Its sum on a fixed physical volume has order a^{-3}, before additional source normalization. One cannot discard that original coordinate count and claim a stronger uniform result from two response factors.

The Ward product sign, canonical coefficient, and measure-level flux pushforward in DC9.10–16 check. An inherited smooth action has the exact full score $J + \kappa XW$. A later retained scalar insertion contributes its actual gradient contact. These are valid identities for the specified retained sector.

The first-response matrix is a function of the input configuration, not generally of $q(x)$ alone; its notation must reflect that. The fine Wilson action and the fine current-square source are not constant

Section 139.5 / Two independent gradient slots genuinely give two first responses

on a general coarse fiber. The four terms of DC9.18 must therefore all be retained. Only its retained/retained term has the two-slot estimate without further derivative information. Equality on a critical graph is insufficient to delete the other three terms under a fluctuating integral.

VM9.1 specifies a common original fiber and cutoff. For positive reference densities satisfying $r_1/r_0 = e^{c+v}$, direct integration proves

$$D_1 = e^c D_0 K_0(e^v), \qquad \frac{dK_1}{dK_0} = \frac{e^v}{K_0(e^v)}, \qquad |\log(dK_1/dK_0)| \le \operatorname{osc}_C(v) \tag{AU9.16}$$

at real endpoints. This is a cancellation in the genuine original reference fraction. It requires no lower bound for D_0. The centered carrying-function bound and the complex small-disc bound in VM9.5–7 follow with their stated positive reference. The endpoint logarithm can be interpolated without pretending that a separately interpolated denominator is zero free on a large complex body.

The action-completion qualification in VM9.8 is necessary and correct. Extracting a scalar from the action creates its opposite coefficient factor. Their product is unchanged, but their separately recorded vacuum increments are not zero. In particular the completed R empty-Mayer term need not be constant in the original reference variable. Independent external scalars do not belong to the canceling fraction.

For two partitions on the same actual space, the refinement $\pi_{0,\alpha}\pi_{1,\beta}$ has sum one. VM9.10–13 is therefore exact, including its history-dependent residual. For a positive original majorant G,

$$\sum_{\alpha,\beta} \int \pi_{0,\alpha}\pi_{1,\beta} G \, d\nu = \int G \, d\nu. \tag{AU9.17}$$

There is no extra pair-count factor or second reserve. This does not identify two different nonlinear constraint supports, nor make two differently prepared carrying functions equal. Their value difference remains VM9.12, and a current comparison also needs the actual readout jets.

VM9.17 follows from the established physical multiplier locality and the derivative of the full critical action, with the original target count n_Y, coefficient x_k, and admitted path length t. Its canonical substitution has the stated factor $n_Y P_k(P_k + Q_k)$. It is not an estimate for arbitrary absolute baseline actions.

139.6.1 The original reference has a cutoff-weighted action

The scope early R-II formulas use a cutoff-weighted action $A(\zeta_0, U_{k,Z}(V'V_\Lambda))$. A multiplier identity for the full minimized action is not automatically its derivative. This distinction must be retained when applying VM9 to the genuine original denominator.

On the same already verified physical target sector, a fixed spatial cutoff can be handled by an additional local estimate. For the holomorphic Wilson plaquette function,

$$ds(P)[\dot P] = -\frac{1}{2}\operatorname{tr}[(I - P^{-2})\dot P], \qquad |ds(P)[\dot P]| \le Ca^4 |F_a|\,|\dot F_a|. \tag{AU9.18}$$

The actual factor a^4 permits summing the fine plaquettes as a physical integral. If $\zeta_0 = 1$ on a distance-D neighborhood of the complete changed-target carrier, the established localized response into the support of $1 - \zeta_0$ controls the difference between weighted and full action variations. The

Section 139.6 / Genuine reference cancellation and selector refinement

original cutoff must be fixed along that variation, or its derivative is an additional term. Its complete separation, target domain, reference field, and physical-cell counting must be checked before using this extension. This does not supply the missing actual mixed-fiber identification or inherited interaction comparison.

Section 139.7 / One reserve and the finite-degree quantifiers

CT9's factor CEe^{CE} and its fixed polynomial envelope are correctly paid together. The inequalities $E \le \eta_* A$ and $E \le C g_{\max} A^p$ yield

$$(1+A)^d CEe^{CE} \le C_{d,p,r} g_{\max} e^{rA} \tag{AU9.19}$$

after one allocated reserve fraction and its stated floor choice. The other source, comparison and energy weights remain in that same integral. No separate credit is created by an additional contact label.

The depth payment in AE9.25–27 and VM9.21 is valid only for literal original charged chains. It does not count auxiliary coordinates, formal preparation sublevels, or clean aging as new records. In addition, its condition

$$4qC_{\mathrm{own}} \log L/w_* \le r_{\mathrm{source}}/2 \tag{AU9.20}$$

is for fixed finite source power q. This argument does not produce one weak floor uniform over arbitrarily high source degree. A common all-order field hierarchy still needs its own quantifier and compatibility argument.

Section 139.8 / The extended Gaussian family and its normalization

The new GG9 upper bound matches CT9's lower power. For a literal block hold, counting the fine links crossing each original coarse face gives

$$\|D_{\nu,a}P_0u\|_{2,a}^2 = (\ell/a)\|D_{\nu,\ell}u\|_{2,\ell}^2. \tag{AU9.21}$$

The own-direction line filter is an actual contraction and commutes with fine differences. Applying these facts componentwise to the curl form proves GG9.3. This retains the original physical adjoints; it does not first estimate the fine Hessian by its larger a^{-2} operator norm.

The local box-and-line form has finite range in the ambient coarse coordinates. A transverse restriction or a change to nonlocal independent physical coordinates need not preserve that matrix range. The operator, trace and full coarse-count estimates used in GG9.9–11 do not require a finite-range matrix in such a basis.

The coordinate conversion in GG9.7 also checks. With $b = \ell B/g_e$, $r = a/\ell$, and $\gamma = g_e/g_j$, the coarse score is $c_\beta\gamma^2\widehat{\Delta}b$. Its contraction with $v_r = 2c_\beta^{-2}r^4\widehat{R}\widehat{\Delta}b$, minus the literal divergence of that vector, is exactly the displayed scalar $\mathcal{S}_r$. Therefore

$$\mathcal{S}_r\mu_e = \mathscr{D}_b(v_r\mu_e), \qquad \|\mathcal{S}_r\mu_e\|_{\mathrm{TV}} + \sum_\alpha \|v_r^\alpha\mu_e\|_{\mathrm{TV}} \le Cr^3m_e(1 + g_j/g_e). \tag{AU9.22}$$

The second bound uses the stated flat reconstruction bound on $\widehat{\Delta}$, and its actual four-dimensional covariance trace with harmonic modes removed or retained separately. All m_e independent coarse coordinates are counted. This is a global count on a global packet, not an assertion that a global current becomes one local source root.

Section 139.8 / The extended Gaussian family and its normalization

The second conditioning in GG9.13–14 has its exact covariance. Its first trace cancels the old trace and leaves the new trace. This proves closure for this quadratic Gaussian family with its original initial coupling and mesh. Later smooth retained sources have the stated derivative contacts. Their actual derivative envelopes are still inputs when those sources are not members of a previously controlled family.

The source normalization qualification in GG9.16 is essential:

$$a^p g_j^{-4} \mathcal{W}_{a,1} = a^{p-4} \mathcal{S}_a, \qquad \|a^p g_j^{-4} \mathcal{W}_{a,1}\| \le C a^{p-1}. \tag{AU9.23}$$

Thus the proved positive a^3 row contains an actual a^4 coefficient. A physical D8 source which lacks it still has the a^{-1} behavior detected by CT9's test. This distinction prevents using the new Gaussian result to claim the physical MT36 row. The original fine roots have already occurred in the exact trace cancellation, so an additional fine-root factor would also be incorrect. Both normalization mistakes are avoided by GG9.

The residual global color average of CT9's bounded cosine test preserves its quadratic forms because the free covariance and Hessian are color diagonal. Its positivity and fixed canonical derivative bounds survive. This is a test for the linearized physical gauge symmetry; a uniform nonlinear representative is not asserted.

Section 139.9 / The nonlinear Hessian representation and its principal estimate

For the bare compact Wilson density and constant tests $f = h = 1$, the Casimir calculation gives $\sum_j X_j X_i X_j S_0 = (C_A/2 - 4C_F) X_i S_0$. The bracket sign follows from the compact adjoint Casimir. Substituting this into the complete CQ current yields exactly NH9.5, including $\zeta = 3C_A/2 - 16C_F$. One may integrate its lower row by parts once more to obtain the equivalent identity

$$\begin{aligned} g_j^{-4} a^4 \mathscr{E}[\widetilde{V}]\mu = 2c_\beta^{-3} \sum_{i,j} \mathscr{D}_{(Y_j,Y_i)}((X_i X_j S_0)\mu) \\ + \zeta c_\beta^{-4} g_j^2 \sum_i \mathscr{D}_{(Y_i,Y_i)}\mu. \end{aligned} \tag{AU9.24}$$

The ordered word is (j, i). Interchanging it without its bracket correction would alter the first-order row. This calculation uses the actual Wilson score and has no unproved small-coupling approximation.

In canonical output coordinates, applying the chain rule to that ordered word produces both NH9.8's two-response tensor and its contracted second-response drift. The two responses in the former have the exact physical-volume conversion

$$A_i^\alpha = (g_j/g_e) a^3 (C_U^{\mathsf{t}})_i^\alpha, \qquad \sum_{i,j} (X_i X_j S_0) A_i^\alpha A_j^\gamma = (g_j/g_e)^2 a^4 \mathcal{H}_U(C_U^{\mathsf{t}} e_\alpha, C_U^{\mathsf{t}} e_\gamma). \tag{AU9.25}$$

Here $\mathcal{H}_U(v, w) = a^2 \sum v_i w_j X_i X_j S_0$. The a^3 comes from one dimensionless-to-physical tangent factor and the four-dimensional adjoint volume ratio. No new current word mesh has been chosen.

The local plaquette derivation of NH9.11 is sound. Its symmetric second variation is controlled by the two covariant first curls and a curvature times two-tangent term. The antisymmetric ordered-Hessian correction is confined to one bond and is $Ca \int |J||v||w|$, rather than an unrecorded derivative of curvature. The exact directional-force formula bounds $\|aJ\|_{L^p}$ by the actual weak plaquette norm. The weak-field existence assumptions remain at $p > 8$. Only CL18's adjoint mapping estimate is used with Hilbert exponent two. Its flat prefix condition is then $2s < 1$. For the once-only endpoint factors, choose $1/2 < \alpha < 1 - 4/p$. The physical weak chart gives a uniform C^α bound for those analytic

chart multipliers, so multiplication is bounded on H^s for every $s < 1/2$. This supplies a separate justification beyond CL18's final $W^{s,p}$ paragraph; no weak-field theorem is being applied at $p = 2$. Choosing $2/p < s < 1/2$ therefore gives

$$\begin{aligned} |\mathcal{H}_U(C_U^{\mathsf{t}}\lambda, C_U^{\mathsf{t}}\lambda')| &\le C_s a^{-2+2s}\|\lambda\|_2\|\lambda'\|_2, \\ \sum_{\alpha,\gamma}|\eta^{\alpha\gamma}| &\le C_s c_\beta^{-3}(g_j/g_e)^2 a^{2+2s} m_e\, q_*|\mu|. \end{aligned} \tag{AU9.26}$$

The Sobolev exponent is exactly sufficient for $H^s \subset L^{2p/(p-1)}$. The loss a^{s-1} occurs on each differentiated adjoint. The entrywise sum uses the actual finite support of the nonlinear compatible prefix in its declared local output chart. It does not use a finite-range assertion after an unspecified global gauge projection. A coarse coordinate count is retained.

The explicit lower drift has a separate verified estimate:

$$\begin{aligned} \sum_i g_j X_i S_0 A_i^\alpha &= (g_j^2/g_e)a^2\langle J, C_U^{\mathsf{t}} e_\alpha\rangle_a, \\ \left|\sum_i g_j X_i S_0 A_i^\alpha\right| &\le C(g_j^2/g_e)aM. \end{aligned} \tag{AU9.27}$$

On a verified physical-current core this improves to $C(g_j^2/g_e)a^2 M_J$. Neither version changes its original coupling to a canonical field strength without a separate substitution.

The remaining contracted term is

$$2\sum_{i,j}(X_i X_j S_0)Y_j A_i^\alpha. \tag{AU9.28}$$

The zero of this term at the flat field is valid: the Hessian is color diagonal and the actual compatible logarithmic mean is linear on each one-color commuting subspace. This is only a value at one field. The available generic second-jet estimate still retains its fine coordinate count and one column factor. It does not establish the required uniform nonlinear grade.

All the identities in this section precede restriction to a regular chart. Bounding the principal coefficient measure on that chart is legitimate. Multiplying the whole distribution identity by an original sharp selector also requires its full cutoff-boundary partners. NH9 does not provide those mixed-history partners or the inherited score contributions automatically.

Section 139.10 / The additional primitive-reference naturality argument

RN9 correctly distinguishes two original variational problems. The inner field $U_{k,Z}(CV_\Lambda)$ has its full target array fixed. The relaxed reference U_0 frees the interior target while retaining the exterior rows. The cutoff-weighted Wilson action in the denominator is evaluated on those unweighted minima; it does not define a third weighted minimization problem.

The copying argument is valid for an actual full finite record with an injective universal lift. Copying every action plaquette, ordered averaging path, boundary datum, gauge pin, selector read set, and admitted branch identifies the finite variational problems. Relabeling the actual free group coordinates preserves normalized Haar measure and all forest deltas without a Jacobian. Once the copied inner fields have the same weighted action, it follows directly that

$$D_{\mathfrak{R}'}(\iota b) = D_{\mathfrak{R}}(b). \tag{AU9.29}$$

For this equality, uniqueness of the admitted minimum orbit is enough, since the weighted Wilson action is gauge invariant. Pointwise equality of chosen representatives additionally requires the copied

gauge rule. The original counts and finite gauge-choice average are retained separately, so the extracted coordinate constants and logarithmic remainder agree too. No small denominator is divided by an independently estimated lower bound.

There is a specific source-verification distinction. The scope R-II argument on pp.357–359 establishes the relaxed U_0 branch. It must not be cited as if it also proved uniqueness for every fixed-target inner field. The separate CMP102 primary text has now been scope: equations (1)–(7) and Theorem 1 on pp.277–279 give uniqueness of the critical gauge orbit for its original nested domains, separation and mixed target regularity, with $\epsilon_1 < a_1$ and $B_3\epsilon_1 \le \epsilon_0 \le a_0$. Proposition 9 on p.309 specifies the analytic branch. This supplies the correct source theorem. Applying it to the original localized inner field still requires that field's literal domain, target and branch to satisfy those clauses. A generic copied graph does not prove their admission. Equality of unselected fibers identifies minimizer sets; the scope uniqueness theorem supplies orbit equality on its admitted domains. No typed propagator expansion is needed for that elementary implication, and none has thereby been scope or identified on an arbitrary mixed history.

For a connected complete nonwrapping record the original root has exactly one copy for each of the q^4 deck choices. The log denominator and physical volume both multiply by q^4 only when these copies are the actual disjoint components of the original product. The proof does not assert that separately generated histories on the two tori have selected those copies.

RN9's weighted-action extension is quantitatively correct on its already established physical target sector. The exact derivative AU9.18 supplies a factor a_f^4 before summing fine plaquettes. Cauchy–Schwarz on each current cell, the local curvature response, and the four-dimensional exponential tail give

$$|[A(\zeta_0, W_1) - A(\zeta_0, W_0)] - [A(W_1) - A(W_0)]| \le C n_Y t\eta e^{-cD}. \tag{AU9.30}$$

The cutoff is fixed along the actual target path and equals one on a complete D-neighborhood of its changed-target carrier. Every point of the path remains in the common admitted chart. The field strength includes its reference, and the derivative is of the full constrained solution with all five equation rows. The current-cell tail count is summable uniformly in the fine mesh and covering degree. A moving cutoff would have an extra derivative term, and a mixed geometry would need its actual assigned-scale locality estimate.

Applying AU9.30 on both sides of VM9's verified full-action comparison gives its weighted exponent bound $C x_k n_Y t\eta e^{-cD}$. This repairs the weighted-versus-full action gap only on that identified physical sector. Inherited potentials, separately extracted vacuum scalars, native-history matching and wrapped tails remain outside this primitive argument.

Section 139.11 / The terminal logarithmic drift and its controlled frame part

The proposed zero-step calculation is correct, but its interpretation must retain the chosen coordinates and fixed reference. Let $-\operatorname{tr} T_A T_B = \delta_{AB}$, so the compact adjoint Casimir is $C_A = 2N$. On the small logarithmic chart, BCH gives

$$\sum_A X_A^2 \log(e^x) = \frac{1}{6}\sum_A [T_A, [T_A, x]] + O(\|x\|^3) = -\frac{C_A}{6}x + O(\|x\|^3). \tag{AU9.31}$$

The coefficient comes from twice the t^2 coefficient in $\log(e^{tT_A}e^x)$. The potential quadratic contribution sums to a bracket with x itself and vanishes. Equivalently, bi-invariance of the Haar Casimir under inversion makes the displayed analytic function odd in x.

Section 139.11 / The terminal logarithmic drift and its controlled frame part

An admitted weak pure-gauge example can be chosen without an uncontrolled microscopic gauge: take a smooth periodic real function ϕ, one fixed Cartan generator T, and $g(x) = e^{\varepsilon\phi(x)T}$. Then $V_b = g(x)g(x + a\hat{\nu})^{-1}$ has all plaquettes equal to the identity and $J = 0$, while its small link logarithm x_b is nonzero on some bonds. Its physical potential and weak norm are $O(\varepsilon)$, uniformly in the fine mesh. With the reference held at I, the zero-step observation is $q_b = \log V_b$. On a nondegenerate fine torus, each bond has $z = 6$ incident plaquettes, and $X_b^A X_b^B S_0 = z\delta_{AB}$ at this flat configuration. The observation has no mixed-bond second derivatives. Hence

$$\sum_{i,j}(X_i X_j S_0)X_j X_i q_b = -\frac{zC_A}{6}x_b + O(\|x_b\|^3). \tag{AU9.32}$$

The NH9 canonical contracted drift has the additional literal factor $2c_\beta^{-3}g_j^2/g_e$. Its explicit lower score term vanishes here. This disproves a curvature-only bound for this unseparated fixed logarithmic-coordinate drift, even at zero prefix length. It does not prove an obstruction for the intrinsic drift or for gauge-invariant tests and does not establish any counterexample on an original mixed selector support.

There is a positive, exact resolution of this coordinate term. Let $Z_\alpha = g_e X'_\alpha$ be output invariant directions, let $b^a = \log(WW_{\mathrm{ref}}^{-1})^a/g_e$, and put $M_\alpha^a = Z_\alpha b^a$. Write $\widehat{A}_i^\alpha$ for the intrinsic first response, so $A_i^a = M_\alpha^a(q)\widehat{A}_i^\alpha$. If the terminal reference is fixed under the current directions, the ordered product rule gives

$$\begin{aligned}\sum_{i,j} B_{ij}Y_j A_i^a &= \Theta^{\alpha\gamma}(Z_\gamma M_\alpha^a)(q) + M_\alpha^a(q)\sum_{i,j} B_{ij}Y_j\widehat{A}_i^\alpha,\\ \Theta^{\alpha\gamma} &:= \sum_{i,j} B_{ij}\widehat{A}_i^\alpha\widehat{A}_j^\gamma, \qquad B_{ij} = X_i X_j S_0.\end{aligned} \tag{AU9.33}$$

No symmetry of the ordered Hessian is used. The principal tensor in intrinsic directions is obtained from NH9's coordinate tensor by the two inverse endpoint frames. These are bounded and local in each actual coarse bond. Also $\|Z_\gamma M_\alpha^a\| \le Cg_e$ on the fixed admitted terminal chart. Consequently the full frame-drift coefficient measure obeys

$$\sum_a |\xi_{\mathrm{frame}}^a| \le C_s c_\beta^{-3}(g_j^2/g_e)a^{2+2s}m_e\, q_*|\mu|. \tag{AU9.34}$$

The original word normalization, actual coarse count and once-only terminal frame are retained. As before, removing an a^4 factor absent from a physical source changes the physical bound.

For the zero-step group observation $q(W) = W$, its intrinsic response is the constant $(g_j/g_e)\delta$. The second term in AU9.33 is therefore zero. AU9.32 is entirely the controlled terminal-frame term, and it must not be presented as a counterexample to a corrected intrinsic curvature estimate.

If W_{ref} is itself a live function of a differentiated variable, the product rule also has $(D_{W_{\mathrm{ref}}}M)Y_j W_{\mathrm{ref}}$. The regular calculation with fixed external reference does not prove that extra term's grade. A full joint readout must retain it. The fixed-reference off-diagonal and mixed-history intrinsic drift remains a separate obligation. The next section proves a narrower centered real estimate.

Section 139.12 / The centered strong-curvature prefix row

CD9 supplies a stronger result with explicitly stronger geometry. Each prefix must have a buffered coordinate-rectangular enclosing box containing all actual determining paths, straight-path comparators, Wilson Hessian stars, ribbon fillings and curvature propagation stencils. Its physical curvature

supremum is α, with the stated smallness. A nonwrapping graph with holes is insufficient. All fields and analysis gauges in this argument are real. The reference is held fixed while taking jets, and those jets are evaluated at the diagonal $V = U$.

The native-path comparison checks. PG9 compares each actual raw coarse bond with its straight lower-scale path; PG20 controls the intermediate curvature. Unitarity of the real transports gives

$$e_s \le Le_{s-1} + C_L a_s^2 \alpha, \qquad e_s \le C_L a_s^2 \alpha, \qquad \|U_s^{N^s g} - I\| \le C_L a_s a_t \alpha \quad (s \le t). \tag{AU9.35}$$

The final inequality uses one fine coordinate-tree gauge on the complete scale-a_t box, first giving a fine-link bound $C_L a a_t \alpha$, then telescoping the actual straight products. It invokes no nonlinear associativity. This gauge is chosen for a pointwise estimate after the actual jets have been defined, and it stays fixed during their variations. Its sampled restrictions give the entire intermediate hierarchy in that same gauge.

In that common gauge, CL17 supplies the exact leading column factor L^{-3}. Its relative errors have summable size $C_L a_{s+1} a_t \alpha$. Thus for the actual centered dimensionless prefix response and the transported ordered Wilson Hessian,

$$\begin{aligned} \|S_t\|_{\mathrm{col}} &\le C_L (a/a_t)^3, \\ \|S_t - S_t^0\|_{\mathrm{col}} &\le C_L (a/a_t)^3 a_t^2 \alpha, \\ R_t = S_t H S_t^{\mathsf{T}}, &\qquad \|R_t - R_t^0\|_{\mathrm{ent}} \le C_L a^2 \alpha. \end{aligned} \tag{AU9.36}$$

For a scale-a_t patch the response errors use exactly $(a_t/a)^4 (a/a_t)^6 a_t^2 \alpha = a^2 \alpha$. The fine-Hessian difference contributes $(a_t/a)^4 (a/a_t)^6 a a_t \alpha = a^3 \alpha / a_t \le a^2 \alpha$. The fine Hessian has its original bounded plaquette incidence. Neither a new Wilson Hessian at scale a_t nor an extra root count has been substituted. Using $2L^{-3}$ at every step would lose this proof; CD9 correctly keeps the exact leading factor and sums the errors.

The ordinary relative-logarithm recursion is exact, with the fine derivative order retained:

$$B_{t+1} = A_t B_t + R_t : K_t, \qquad |B_0(b)| \le C_L a^2 \alpha, \qquad |R_t : K_t| \le C_L a^2 \alpha. \tag{AU9.37}$$

Here $K_t = D^2 \Phi_t(0)$. The initial term is the same-bond Lie bracket contracted with the antisymmetric fine Hessian, hence with the original score. It must not be set to zero in this coordinate recursion. For the forcing, R_t^0 is color diagonal and K_t^0 vanishes on any two input directions with one common color, because the actual commuting means are arithmetic. Thus $R_t^0 : K_t^0 = 0$, even though its spatial tensor need not equal a Wilson Hessian. The two differences in the forcing use one and the same enlarged stage-$t+1$ analysis gauge. Changing its radius by the fixed factor L changes only C_L.

The final suffix column from spacing a_s is $C_L a_s^3$, while the actual stage count for a final record of size m_E is at most $C_L m_E a_s^{-4}$. Therefore

$$\sum_{\alpha \in E} |B_n^\alpha| \le C_L m_E a^2 \alpha \left(a^{-1} + \sum_{s=1}^{n} a_s^{-1} \right) \le C_L m_E a \alpha. \tag{AU9.38}$$

Both the initial frame and every intermediate forcing are counted. The canonical contracted-current coefficient retains $2c_\beta^{-3} g_j^2 / g_e$. CD9.27 gives the same $a\alpha$ order for its explicit lower score row, with its separate coefficient ζ. A verified stronger current bound gives that row's stated $a^2 M_J$ improvement.

For the principal coefficient on this same strong real geometry, the exact flat conversion is $R_n^0 = a^4 Q H_{\mathrm{phys}} Q^*$. GG9's face estimate and AU9.36 therefore give

$$\sum_{\alpha, \gamma \in E} |R_n^{\alpha\gamma}| \le C_L m_E (a^3 + a^2 \alpha). \tag{AU9.39}$$

Section 139.12 / The centered strong-curvature prefix row

Its canonical coefficient is $2c_{\beta}^{-3}(g_j/g_e)^2$. This calculation does not infer a weak $W^{1,p}$ bound from an axial gauge. At the centered terminal output, the derivative of its logarithmic frame is a same-bond Lie-bracket tensor; the flat color-diagonal part of R_n^0 contracts to zero. The remaining frame cost is consequently $C_L m_E a^2 \alpha$, with the original g_j^2/g_e conversion. Subtracting it from AU9.38 verifies CD9.28's centered intrinsic drift bound.

The invariant-test consequence is valid. For every fixed reference U, CL15 gives equality of the compatible and raw pullbacks of an individually gauge-invariant terminal function on a neighborhood in V. Differentiating in V with U fixed, then evaluating at $V = U$, gives the actual jet of the fixed raw pullback. It introduces no derivative of a moving reference. At the actual output $\mathcal{M}^n(U)$, the centered ordinary terminal Hessian is the symmetrized invariant Hessian. Thus its pushed second-current coefficient uses the symmetric part of R_n, while B_n retains the initial ordered-bracket contribution. This yields a same-Wilson invariant test-functional row for coefficient measures carried by the stipulated real body.

The original Wilson measure is not asserted to be supported on that body. Restricting the second-current coefficient measures to a selected region is not the same as multiplying the ordinary scalar Ward companion by its selector; the full boundary partners remain. No ordinary scalar total-variation bound, general multi-history invariant identity, complex-wall extension, or marked derivative estimate follows. In addition, removing an a^4 factor absent from a physical source changes $a\alpha$ to $a^{-3}\alpha$. The new centered row does not change the physical normalization by declaration.

Section 139.13 / Hypotheses and conclusions

The valid additions are the direct scalar contact bounds in literal mixed-cell norms; the full-pair relative-density commutator and its later smooth source contacts; the actual bare own-mesh creation envelopes; the separately retained Dirichlet sector; and the genuine reference-fraction and selector-refinement comparisons. The extended Gaussian result gives a sharp closed row with its actual word normalization, and the nonlinear Hessian form gives a strict principal-tensor gain on its admitted regular chart. The terminal coordinate-frame part of the drift has the same principal bound, with its explicit coupling conversion and fixed-reference hypothesis. On the stronger complete real curvature geometry, the centered prefix now has the $a\alpha$ drift estimate and the $a^3 + a^2\alpha$ principal estimate on the specified invariant terminal test class. The full-record primitive copying identity is exact with its branch identification retained, and the weighted-action tail has its correct physical volume sum. The Gaussian test and exact normalization comparison establish a real restriction on the physical current grade.

The unresolved transitions are still model specific. The original prepared fields have not been shown to meet AE8 or the alternative scalar moment condition on every simultaneous wall. General old coefficients have not been admitted with all their saved grades. The complete mixed current pair, later coefficient derivatives and readout jets remain unbounded in the needed induction. In particular, the fixed-reference off-diagonal and mixed-history intrinsic drift, including live-reference partners where present, has not been bounded at the required physical grade. The complete scalar companion and source-graded derivatives remain outside the centered real result. Full reference-fiber matching, inherited oscillations, independent scalar increments and wrapped tails remain unproved. Therefore the argument does not promote LM47, all-history MT36 or S94 to established results.

Section 140.1 / Actual mixed scalar estimates and the unresolved entry theorem

AE9 keeps three different fields separate: the raw integration variable, the original prepared composite $\Gamma^{R}_{\omega,\mathbf{s}}$, and the old physical reconstruction evaluated on that composite. It does not prove a current-unit weak bound for the prepared composite by bounding either of the other fields.

For one specified actual field, let a be the original contact-word mesh, ℓ the physical test length, and σ_Δ the assigned cell lengths. The literal cell moments are

$$\begin{aligned} \epsilon_\Delta &= \sigma_\Delta^{2-4/p}\|a^{-2}(P-I)\|_{L^p_a(\Delta)},\\ \mathcal{M}^2_\Delta &= \sigma_\Delta^2 \sum_{\nu,\mu\neq\nu} \|Q_{\nu;\mu}\|^2_{L^2_a(\Delta)},\\ \mathfrak{F}_4 &= \sum_\Delta (\ell/\sigma_\Delta)^4 \epsilon_\Delta^4, \qquad \mathfrak{J}_2 = \sum_\Delta (\ell/\sigma_\Delta)^2 \mathcal{M}^2_\Delta. \end{aligned} \tag{R9.1}$$

Every plaquette, directional current, source offset and component count has its original owner. Cellwise

Section 140.1 / Actual mixed scalar estimates and the unresolved entry theorem

Hölder and the exact CQ/CS reductions prove

$$\begin{aligned} g_W^{-4}a^4|\mathcal{R}^{(8)}_{f,h}| &\le Cc_\beta^{-2}(a/\ell)^4\|f\|_\infty\|h\|_\infty\mathfrak{F}_4,\\ g_W^{-4}a^4|\mathcal{R}^{c}_{H}| &\le Cc_\beta^{-2}(a/\ell)^4\|f\|_{C^2_{\ell,a}}\|h\|_{C^2_{\ell,a}}(\mathfrak{F}_4+\mathfrak{J}_2). \end{aligned} \tag{R9.2}$$

The four pointed-H contributions have powers $(a/\ell)^4\mathfrak{F}_4$, $(a/\ell)^6\mathfrak{F}_4$, and twice $(a/\ell)^4\mathfrak{J}_2$. The second-power assigned-scale weight of the current moment follows from the exact endpoint test cancellation; replacing it by a generic field-jet estimate would lose that information. Complex versions require bounds for the actual transport words and their inverses on one common domain.

The flux criterion is weaker than a full unweighted flux L^p bound at fixed total cell geometry, by AE9.6b. The combined flux/current criterion is a different condition on arbitrary fields. The analysis explicit commuting period-two Wilson example satisfies

$$\|A\|_{W^{1,p}_a}+\|F_a\|_{L^p_a}\le C\varepsilon, \qquad \mathfrak{J}_2\asymp\varepsilon^2a^{-2}. \tag{R9.3}$$

Conversely, the concentration example in AE9 has bounded or vanishing $\mathfrak{F}_4+\mathfrak{J}_2$ while its unweighted flux L^p norm diverges. Neither example is asserted to be an original prepared R field. They exclude incorrect implications between norms; they do not disprove the required original selector estimate.

For the original prepared fields, the supremum of R9.1 over the original selector support and simultaneous complex walls remains to be bounded. The full AE8 weak-entry route remains unproved as well. The complete old paired core, its physical endpoint identity, its wall extension and its saved grade must also be checked. Multiplying R9.2 by an already established saved factor preserves that factor; this does not establish an old coefficient's bound on a new domain.

AE9's depth payment applies only when the claimed ancestry consists of actual charged records in the existing reserve. Formal substeps, uncounted collars and exterior clean arrays do not become chargeable. For each fixed moment order, a weak floor can absorb the stated depth exponential. No one weak floor uniform over arbitrarily high operator order is proved by that argument.

Section 140.2 / A whole-pair transport row with the internal small factor

CT9 uses ordinary coefficient measures on the same actual realization. With the convention $\langle \mathscr{D}_w \nu, \varphi \rangle = \int Y_w \varphi \, d\nu$, suppose a complete incoming pair satisfies

$$\tau = \sum_w \mathscr{D}_w \nu_w. \tag{R9.4}$$

The scalar companion, every lower current, constraint response and boundary partner are included. For a bounded multiplier h, its completed density commutator is defined using ordinary measures:

$$\begin{aligned} \mathcal{C}_h(\mathcal{P}) &= h\tau - \sum_w \mathscr{D}_w(h\nu_w) \\ &= (h-1)\tau - \sum_w \mathscr{D}_w((h-1)\nu_w). \end{aligned} \tag{R9.5}$$

Consequently its displayed represented-current norm obeys

$$\mathfrak{M}(\mathcal{C}_h) \le \int |h-1| \left(d|\tau| + \sum_w d|\nu_w| \right). \tag{R9.6}$$

Section 140.2 / A whole-pair transport row with the internal small factor

There is no derivative of h on this right side. For smooth h, expansion of R9.5 contains the full score and every ordered lower current. For sharp selectors, it remains a legitimate tuple of ordinary coefficient measures; it does not multiply a discontinuous function by an already differentiated distribution.

For a later smooth coefficient G, CT9.13 constructs the complete product pair $\mathcal{P}_G$, including its last scalar contact, and proves

$$G\mathcal{C}_h(\mathcal{P}) = \mathcal{C}_h(\mathcal{P}_G). \tag{R9.7}$$

The ordered product rule is applied before taking norms. Applying it coefficientwise in the original finite source algebra retains the labels, partitions and canonical factors. The derivatives of an inherited field-dependent G remain part of its required envelope.

On its actual admitted reference chart, the original R-II comparison has $h = e^{\sigma}$, or $h = e^{\sigma}/m$, with $\|h-1\|_{\infty} \le C g_k \mathcal{P}(\log g_k^{-2})$. R9.6 retains this small factor even for internal current directions. This strengthens the outer-Cauchy row It does not assert that the whole inherited real action is small, or differentiate the reference kernel without its complete partners.

For an original selector χ,

$$\mathcal{C}_{\chi h} = \mathcal{C}_h - \mathcal{C}_{(1-\chi)h}. \tag{R9.8}$$

The second term is an original failure-sector packet requiring its own actual envelope. The component sum and polynomial moments use the same VC reserve, with fixed ownership and one allocation of its fractions. If that whole-pair envelope is established, CT9 gives the quantitative small transport row. Establishing the envelope for every inherited packet on its mixed support remains a separate open task.

The bare CQ/CS pair does have a same-Wilson-measure creation bound. For a fixed microscopic source root, its actual canonical coefficients satisfy

$$\sum_i |d\nu_i/d\mu| \le C\mathscr{E}_Y^{1/2}, \qquad |d\tau/d\mu| \le C(1+\mathscr{E}_Y). \tag{R9.9}$$

This preserves the original root count and cannot spend action energy which a previous reference insertion has already canceled. Moreover, the complete scalar companion is nonzero at the flat field, with the explicit constants checked in CT9. Its norm cannot be replaced by the quartic scalar-remainder norm in R9.2.

140.3.1 Separately retained Dirichlet slots

For an ordinary joint tensor measure and two separate functions,

$$\mathcal{B}_\nu(F,G)=\sum_{ij}\int Y_iF\,Y_jG\,d\nu_{ij},\qquad \eta_{\alpha\gamma}=q_*\sum_{ij}A_i^\alpha A_j^\gamma\nu_{ij},\tag{R9.10}$$

DC9 proves exact pushforward and a norm bound by the square of the actual first-column response. With the admitted regular-prefix bound $C\lambda^n$, $\lambda\le 2L^{-3}$, a packet whose already specified D8 conversion is L^{4n} receives $(4/L^2)^n$, times its original canonical ratios and incoming tensor mass. Two derivatives of one composite function do not receive this estimate: they also produce the second derivative of q.

The actual Ward product with $V=X(F\circ q)$ has the Dirichlet contact R9.10. Its identity persists under the smooth inherited density when the full effective score is included. Bare CQ's fine S_0 and P_h, however, are not generally functions of the retained field on a fluctuating fiber. The retained-sector estimate does not dispose of those other slots or their later contacts.

140.3.2 The closed flat Gaussian family and its precise normalization

For the actual block-and-line average, GG9 proves the sharp form bound

$$R_{a,\ell} = Q_{a,\ell} H Q_{a,\ell}^{*} \le C(\ell/a)(-\Delta_{\ell}). \tag{R9.11}$$

The factor comes from counting the fine crossings of each coarse face. The conditional Gaussian current keeps its mean term and its scalar trace. Subsequent conditioning gives exactly

$$\Delta_{\ell'}^{-1} = Q\Delta_{\ell}^{-1}Q^{*}, \qquad R_{a,\ell'} = QR_{a,\ell}Q^{*}, \tag{R9.12}$$

and the conditional covariance cancels the old trace into the new trace. Thus the complete quadratic Ward family closes through its regular Gaussian tower. Its contacts with functions of the currently retained field have the exact GG9.15 identity and the bounds of their actual derivative envelopes. Insertions at different retained levels keep their joint Gaussian record and require its corresponding derivative and polynomial envelope; this is not an automatic multi-level source closure.

With $r = a/\ell$, canonical output $b = \ell B/g_e$, and $\gamma = g_e/g_j$, the scalar normalized by the original word's $g_j^{-4}a^4$ and its vector satisfy, on the actual canonical cutoff $\|b\|_{\infty} \le P_e$,

$$|\mathcal{S}_r| + \sum_{\alpha} |v_r^{\alpha}| \le C r^3 m_e (1 + P_e + \gamma^2 P_e^2). \tag{R9.13}$$

The actual coarse count m_e remains. The fine count has already entered the exact trace calculation; adding it again would double count. There is also a bounded canonical test giving the corresponding lower power in this Gaussian reference.

At retained spacing one the power comparison is, with the same actual coarse count and relative coupling factors as in R9.13 or its full Gaussian expectation bound GG9.11,

$$\|a^p g_j^{-4} \mathcal{W}_{a,1}\| \le C a^{p-1}. \tag{R9.14}$$

Therefore a genuinely present a^4 coefficient gives a^3, and a genuinely present a^2 coefficient gives a. A physical D8 packet with neither coefficient retains a^{-1} growth. MT1's physical normalization does not contain an extra a^2. No mesh factor may be inserted by changing the name of a norm. R9.13 is a closed, separately graded Gaussian sector; it is not all-history MT36 for the existing physical sources.

Section 140.4 / The nonlinear Hessian tensor and the remaining contracted drift

For the bare Wilson measure μ, the complete CQ current with $f = h = 1$, and $Y_i = g_j X_i$, the ordered Casimir calculation gives the exact nonlinear identity:

$$g_j^{-4} a^4 \mathscr{E}[\widetilde{V}]\mu = 2c_\beta^{-3} \sum_{ij} \mathscr{D}_{(Y_j,Y_i)}((X_i X_j S_0)\mu) + \zeta c_\beta^{-3} \sum_i \mathscr{D}_{Y_i}(g_j X_i S_0 \mu), \qquad \zeta = 3C_A/2 - 16C_F. \tag{R9.15}$$

The order (j, i) and the lower Lie-algebra term are essential. Pushing this identity through the actual nonlinear canonical observation separates a principal tensor and a first-order drift.

On its admitted weak regular chart, choose $p > 8$ and $2/p < s < 1/2$. Direct plaquette differentiation and the actual fractional prefix-adjoint estimate prove

$$|\mathcal{H}_U(C_U^{\mathrm{t}}\lambda, C_U^{\mathrm{t}}\lambda')| \le C_s a^{-2+2s} \|\lambda\|_2 \|\lambda'\|_2. \tag{R9.16}$$

The ordered antisymmetric Hessian part is retained and bounded using $\|aJ\|_{L^p} \le C\|F\|_{L^p}$. The endpoint multipliers use the weak field's Hölder regularity; the fractional norm itself is the L^2-based H^s norm. With the exact physical-volume conversion $A_i^\alpha = (g_j/g_e)a^3(C_U^{\mathsf{t}})_i^\alpha$, the principal tensor satisfies

$$\sum_{\alpha,\gamma} |\eta^{\alpha\gamma}| \le C_s c_\beta^{-3}(g_j/g_e)^2 a^{2+2s} m_e\, q_*|\mu|. \tag{R9.17}$$

This positive own-word power is a spatial form estimate, not an assumed improvement for generic higher jets. It bounds the principal coefficient measure on the admitted chart. Applying the whole current identity after an original selector still requires its complete cutoff-boundary partners.

The explicit lower drift also has a bound of order $(g_j^2/g_e)aM$, improving to $(g_j^2/g_e)a^2M_J$ on a verified physical-current core. The remaining contraction is

$$\sum_{ij}(X_iX_jS_0)Y_jA_i^\alpha. \tag{R9.18}$$

At the centered flat field it vanishes by color diagonality and the exact linearity of the actual average on one-color Abelian subspaces. The two first-response factors in R9.17 cannot replace the full second response in R9.18.

One part of R9.18 can nevertheless be estimated exactly. With the terminal reference fixed under the differentiated directions, write the canonical coordinate response as $A_i^a = M_\alpha^a(q)\widehat{A}_i^\alpha$, where $\widehat{A}$ uses the actual output invariant frame $Z_\alpha = g_eX'_\alpha$. The product rule gives

$$\sum_{ij} B_{ij}Y_jA_i^a = \Theta^{\alpha\gamma}(Z_\gamma M_\alpha^a)(q) + M_\alpha^a(q)\sum_{ij} B_{ij}Y_j\widehat{A}_i^\alpha, \quad \Theta^{\alpha\gamma} = \sum_{ij} B_{ij}\widehat{A}_i^\alpha\widehat{A}_j^\gamma. \tag{R9.18a}$$

The first term is the terminal frame drift. Bounded inverse frames, R9.17 and $Z_\gamma M_\alpha^a = O(g_e)$ give its own-word coefficient bound $Cc_\beta^{-3}(g_j^2/g_e)a^{2+2s}m_e\,q_*|\mu|$. The fixed-reference pure-gauge example in CD9/AU9 has nonzero coordinate drift despite zero curvature, but all of that example is this controlled frame contribution. It does not disprove a corrected intrinsic estimate.

140.4.1 A centered real-prefix contraction is now bounded

CD9 proves a further result for the constant-test CQ row on a complete protected real regular region. At each stage, the entire prefix has a controlled buffered coordinate rectangle enclosing every original path, Wilson Hessian star, ribbon filling and curvature-propagation stencil, with radius bounded by a fixed multiple of that stage's spacing. The actual fine curvature has a small physical supremum α. These additional real curvature and geometry hypotheses are not supplied by the weak chart used for R9.16 or by merely knowing that a visible carrier is nonwrapping.

Let $a_t = L^t a$, $a_n = 1$, and use the genuine reference chain $U_{t+1} = \mathcal{M}(U_t)$. Hold that chain fixed while differentiating the compatible variable chain at $V = U$. For each stage-t prefix, one fine axial gauge controls all its intermediate references. Indeed the actual output-versus-straight-path comparison gives

$$e_s \le Le_{s-1} + C_La_s^2\alpha, \qquad e_s \le C_La_s^2\alpha, \qquad \|U_s^{N^sg} - I\| \le C_La_sa_t\alpha \quad (s \le t). \tag{R9.18b}$$

The first recurrence counts the L original collinear subsegments; it does not reassociate a group mean. Real compact transports and the complete ribbon region are essential to this proof.

Writing S_t for the actual centered first-prefix response, the flat leading column is exactly L^{-3} per step. The geometric perturbation sum, without replacing that coefficient by $2L^{-3}$, gives

$$\|S_t\|_{\mathrm{col}} \le C_L(a/a_t)^3, \qquad \|S_t - S_t^0\|_{\mathrm{col}} \le C_L(a/a_t)^3a_t^2\alpha. \tag{R9.18c}$$

Section 140.4 / The nonlinear Hessian tensor and the remaining contracted drift

For the original ordered Wilson Hessian H, put $R_t = S_t H S_t^{\mathsf{T}}$. Each stage patch contains its actual $O((a_t/a)^4)$ fine coordinates. The response comparison and the local Wilson Hessian estimate consequently prove

$$\|R_t - R_t^0\|_{\mathrm{ent}} \le C_L a^2 \alpha, \qquad |R_t : D^2\Phi_t(0)| \le C_L a^2 \alpha. \tag{R9.18d}$$

The second bound uses the exact cancellation of the flat color-scalar tensor against the flat one-step second jet. The nonflat tensor has not been replaced by a new Wilson Hessian.

The full ordered contracted response satisfies $B_{t+1} = D\Phi_t(0)B_t + R_t : D^2\Phi_t(0)$. Its initial same-bond logarithmic bracket is also retained and is bounded by $Ca^2\alpha$. A stage of mesh a_t has $O(m_E a_t^{-4})$ coordinates in the complete final determining region, while the suffix column costs Ca_t^3. Thus

$$\sum_{\gamma\in E} |B_n^\gamma| \le C_L m_E a^2 \alpha \left(a^{-1} + \sum_{t=1}^{n} a_t^{-1} \right) \le C_L m_E a \alpha. \tag{R9.18e}$$

This is a uniform positive own-word power for the centered real contracted response. On the same geometry the principal tensor has the independent bound $C_L m_E(a^3 + a^2\alpha)$. No weak $W^{1,p}$ estimate for the axial representative is assumed to obtain it. At the centered terminal output, the frame derivative is a Lie-bracket tensor and its flat color-diagonal contraction vanishes. The remaining terminal-frame term is therefore bounded by $C_L m_E a^2 \alpha$, with its original coupling conversion. Subtracting it leaves the same $a\alpha$ estimate for the centered intrinsic drift.

For an individually gauge-invariant terminal test, CL15 identifies the compatible and raw pullbacks in a neighborhood with the reference held fixed. Their jets may therefore be evaluated at $V = U$ before integrating the coefficient representation. The first-current coefficient in the original word convention is bounded by $C_L c_\beta^{-3}(g_j^2/g_e) m_E a\alpha$; the principal coefficient keeps $2c_\beta^{-3}(g_j/g_e)^2$. This is a statement about the same raw physical pullback on that invariant test class. It does not identify arbitrary fixed-reference off-diagonal jets with centered ones.

The bound does not say that the whole Wilson measure lies in this strong region. Restricting a current by an original selector retains its complete boundary partners. The represented-current bound does not supply the ordinary scalar companion's total variation needed by CT9. Removing an a^4 absent from a physical source converts the own-word $a\alpha$ into $a^{-3}\alpha$; no missing birth factor has been supplied.

General mixed-domain and live-reference contractions, simultaneous complex walls, higher marked derivatives, nonconstant-source partners and tests linking different histories still require their own bounds. An inherited effective density also retains its full score and normalization partners. They are not removed by the bare Wilson identity R9.15 or by this real centered estimate.

Section 140.5 / Genuine denominator comparison and its remaining matching premise

VM9 keeps the original numerator and reference denominator together. For positive real reference densities on a common actual conditional fiber, if $r_1/r_0 = e^{c(b)+v(C,b)}$, then

$$\log(D_1/D_0) = c + \log K_0(e^v), \qquad \frac{dK_1}{dK_0} = \frac{e^v}{K_0(e^v)}. \tag{R9.19}$$

The normalized real tilt is controlled by the oscillation of v in the actual integrated variable, even when D_0 is tiny. The C-independent factor cancels within this genuine normalized reference fraction. If a separate action-completion factorization is introduced, its compensating scalar partner must also be retained. An independent vacuum scalar stays in the defect. A synthetic interpolation of the endpoint tilt does not prove an external analytic domain which has not been established.

Section 140.5 / Genuine denominator comparison and its remaining matching premise

Different original selector partitions on the same actual fiber can be compared by their common refinement, with no extra entropy factor. Different readouts retain their difference. Different constraint fibers require an actual identifying map; the partition identity does not supply it.

RN9 verifies the primitive in the original formula as a cutoff-weighted pure Wilson reference integral:

$$D_{\mathfrak{R}}(b) = \int dC\, \delta_{G_0}(C)\chi(C)e^{-x_k A_{\mathfrak{R}}(\zeta_0, U_{k,Z}(CV_\Lambda;b))}. \tag{R9.20}$$

Its inner background has all prescribed targets fixed; the reference U_0 instead minimizes with interior targets free. Both are unweighted variational problems. The cutoff ζ_0 weights the action evaluated in R9.20; it does not redefine either minimization.

For a complete nonwrapping record with an injective cover lift, copying every action plaquette, boundary value, raw prefix, constraint, pin, cutoff, gauge choice and count identifies the two original variational fibers. The scope fixed-target theorem supplies uniqueness up to gauge on its stated nested-domain, target-regularity and smallness class. Those hypotheses must be checked for the literal localized record on the full original χ support, including the mixed-layer target-interface conditions. They are not inferred from the relaxed U_0 argument. For a primitive meeting these identification clauses, the minimizer orbits agree under the copy. Gauge invariance of the weighted action and exact Haar/tree relabeling then prove

$$D_{\mathfrak{R}'}(\iota b) = D_{\mathfrak{R}}(b). \tag{R9.21}$$

The extracted classical factor, count constants, residual logarithm and normalized reference kernel agree separately. This is a genuine primitive denominator row, not an estimate of a surrogate Gaussian denominator. Without the branch-applicability verification the fiber argument proves equality of minimizer sets only.

On the verified physical target sector, RN9 also extends the previous classical-action defect to the actual cutoff-weighted action. The extra tail is bounded by $Cn_Y t\eta e^{-cD}$, with its physical fine-volume factor retained. Canonical target and field strengths cancel the displayed inverse coupling in the classical defect exactly as in VM9. All these admission and localization hypotheses remain explicit.

The independent histories on two physical tori still have to be matched to these copied primitives at every level. Inherited clean and R potentials, typed seed factors, later effective densities, external scalar increments and wrapped full read sets remain in that comparison. The completed R-formation comparison then applies once its inputs really match. The argument does not prove that full induction, and hence does not prove S94.

Section 141.1 / Hypotheses and conclusions

AE8.8 has not been established on the actual selector-supported prepared fields of R-I/R-II, nor on their common simultaneous complex domain. The primary R-I range definitions (1.64)–(1.69), initial preparation conditions (i),(ii), and selector (1.75) have still not been recovered in inspectable full text. The original composite $\Gamma^{R}_{\omega,\mathbf{s}}$ of AE9.1 is unchanged.

There is a stronger, fully model-specific obstruction to one proposed way of filling this gap. The scope 1985 mixed variational theorem does not, by itself, imply a current-unit weak bound, even for its actual minimal configurations. Its level-zero constraints retain fine links literally. A small defect on one retained fine bond therefore survives the nonlinear Wilson minimization exactly. Its physical curvature and directional-current moments can diverge while its dimensionless target size and Wilson action tend to zero.

This result concerns the original mixed variational class. It is not claimed to be a counterexample on the more restricted R-I selector support. In particular, its measured domain must contain the retained fine plaquette. It does not refute a protected-interior estimate whose domain excludes that entire fine star and its required read sets.

Section 141.3 / An actual minimal configuration with a pinned plaquette

Work with $SU(n_c)$, $n_c \geq 2$, and fix the Cartan matrix

$$T = \operatorname{diag}(i, -i, 0, \dots, 0). \tag{AE10.5}$$

Set the original mixed target to the identity except on one positively oriented fine bond in the retained collar:

$$V_{b_*} = e^{\delta T}, \qquad V_{b_*^{-1}} = e^{-\delta T}, \qquad V_b = I \quad \text{on every other original target row.} \tag{AE10.6}$$

Here $0 < \delta < \delta_0$, with δ_0 fixed sufficiently small. Every target plaquette wholly in the fine layer is either the identity or $e^{\pm\delta T}$. All interface plaquettes are the identity by the separation just specified. Hence the full target condition (7) holds with

$$\epsilon_1 = K\delta, \tag{AE10.7}$$

for one fixed constant K, chosen below also to contain the elementary fine-field bounds. For $K\delta < a_1$, Theorem 1 therefore gives an actual mixed Wilson minimal configuration $U_a(\delta)$. In fact the fine

configuration with just this one nonidentity bond is already a representative of that minimal orbit. Section 5 proves this exact assertion by positivity of the original Wilson plaquette terms; no unknown minimizer tails need to be bounded.

Choose a plaquette p_* incident to b_*, entirely inside the retained fine collar. All four of its links are level-zero determining rows, so AE10.3 gives the exact nonlinear equality

$$U_a(\delta)(\partial p_*) = e^{\pm \delta T}. \tag{AE10.8}$$

This is an identity for the minimal configuration, not for an arbitrary off-shell comparison field. It does not require an estimate on the solution away from the retained star. Original allowed gauges fix the target. Even after an arbitrary complex gauge conjugation, the two nontrivial eigenvalues of this plaquette remain $e^{i\delta}$ and $e^{-i\delta}$.

Use the physical matrix curvature of AE9, $F_{a,p} = a^{-2}(P_p - I)$, and a matrix norm dominating spectral radius. For $0 < \delta < \delta_0$, AE10.8 proves

$$\|F_a(U_a(\delta))\|_{L^p_a(D)} \geq c\delta a^{4/p-2}, \qquad p > 2. \tag{AE10.9}$$

Indeed one fine plaquette has physical counting volume a^4, and $|e^{i\delta} - 1| \geq c\delta$. This lower bound survives complex gauge conjugation as well; it is not based on assuming that an arbitrary complex gauge has a bounded inverse.

In the own-scale cell notation of AE8, the containing level-zero cell has its original assigned length $\sigma_\Delta \asymp a$, where a fixed big-block convention only changes the constants. Thus

$$c_\Delta(U_a(\delta)) \geq c\delta, \qquad \sigma_\Delta^{4-2p} c_\Delta(U_a(\delta))^p \geq c\delta^p a^{4-2p}. \tag{AE10.10}$$

Thus the exact weighted identity AE8.7 detects this same divergence. An own-scale bound of order ϵ_1 for an actual minimum cannot be converted to an unweighted bound of order ϵ_1 on a current-size domain containing this fine cell. In particular, the necessary curvature estimate for the M8 weak potential chart would force $\tau \geq c\delta a^{4/p-2}$. For a proposed $\tau = C\epsilon_1 = CK\delta$, the ratio still diverges as $a \downarrow 0$. A weak ceiling independent of the cutoff fails as well for the family in AE10.17–18.

Section 141.6 / Complex domains and higher target derivatives

The explicit one-bond configuration also extends holomorphically to complex z, retaining exactly the same determining record, the target $V(z)$, and the complete read sets. Every plaquette meeting a freely varying bond is still the identity, so it solves the complex constrained critical equations as well. This supplies a genuine analytic critical family for this native variational geometry; it is not an invented R wall.

The obstruction is present at real compact fields. Consequently a common complex domain containing these real centers cannot have the claimed uniform current-unit bound merely by analytic continuation. There is also an exact target-coordinate calculation.

Replace δ in AE10.6 by a complex parameter z. Any holomorphic continuation of the same constrained minimum, on any branch that preserves the actual retained rows, necessarily obeys

$$P_{p_*}(z) = e^{\pm zT}. \tag{AE10.19}$$

This follows from the constraint, not from a bound on the continuation. Every derivative at zero of the observed fine plaquette is therefore

$$\partial_z^m F_{a,p_*}(z)\big|_{z=0} = a^{-2}(\pm T)^m, \qquad m \geq 1. \tag{AE10.20}$$

Since $T^m \neq 0$, its physical L_a^p norm has the same factor $a^{4/p-2}$ at each fixed derivative order. A higher target derivative supplies no missing positive mesh power here. Changing to a dimensionful potential coordinate changes the coordinate radius at the same time and cannot change AE10.9.

On a small dimensionless target disc of radius r_a, an unweighted curvature bound of size τ would necessarily require

$$r_a \leq C\tau a^{2-4/p}. \tag{AE10.21}$$

This is a necessary radius restriction on this pinned word. It is not a construction of an R-I complex domain. If an actual preparation or localization wall changes these retained rows, its altered constraint and complete endpoint identity must be tracked instead; AE10.19 cannot be assigned to that different wall by notation alone.

Section 141.7 / Consequences for the outstanding actual-entry problem

The result proves the following non-implication:

Section 141.7 / Consequences for the outstanding actual-entry problem

native mixed variational regularity and small action

$$\not\Longrightarrow \tag{AE10.22}$$

AE8 current-unit weak admission on every measured domain.

It also rules out replacing the physical directional-current bound by an appeal to constrained stationarity on retained fine rows.

It does not establish that the original R-I selectors admit the constructed targets at the relevant prepared stage. They may force the pertinent readout domain to avoid this fine star, remove the fine retained row by an actual transition, or impose an additional quantitative bound. Which of these happens must be decided from the actual definitions and the actual source read set. It cannot be decided from the variational theorem alone.

The same separation must be preserved for the three fields

$$U_{\mathrm{raw}}, \qquad \Gamma^{R}_{\omega,\mathbf{s}}, \qquad P_{\mathrm{old}}\big(q_{\mathrm{old}}(\Gamma^{R}_{\omega,\mathbf{s}})\big). \tag{AE10.23}$$

The example is a native mixed minimum. No equality to the second or third entry in AE10.23 is asserted without a literal determining record and endpoint check.

Conversely, this same example shows why a failure for one entry of AE10.23 cannot simply be assigned to the next. If an old local reconstruction's entire original determining and prescribed boundary record has complete read sets avoiding b_*, all of those raw data are exactly the flat data. Its original local Wilson minimum is then the flat orbit. Thus that old readout is flat even though the larger measured domain in AE10.9 contains a divergent fine curvature. The required hypothesis here includes the full boundary and junction record, not only the interior target rows. Which records an actual history uses remains the original admission problem.

The original mesh factors in AE9's contact inequalities also stay unchanged. On the pinned fine cell, $a^4\mathfrak{F}_4 \geq c\delta^4$; its displayed engineering factor can be consumed entirely by the fine-cell moment. This calculation does not supply an additional a^4 to the physical source normalization. Nor does it establish a lower bound for a specific projected contact: that would require its actual algebraic coefficient and cancellations. The complete Ward partner, scalar companion, and later contacts remain present.

Finally, this construction has no assigned R history and hence no assigned reserve charge. Its geometric depth cannot be charged to the vacuum reserve without proving a literal ownership map to original failed records. No reserve is spent in this chapter.

Section 141.8 / The original R range, preparation index and selector

Three conditions separate the retained-row example from a conclusion on every original prepared history. First one must match the R-I conditions (i), (ii) and the preparation index to the actual history and read sets, including any possible exclusion of retained fine stars. Second, R-I (1.64)?(1.69) at n = N uses its own assigned-scale weights, target rows, gauges and complex radii. Membership in that space does not give membership in the unweighted M8 weak ball. Third, the supremum in (AE8.8) is taken after imposing selector (1.75) on its original support.\n\nThe variational regular minimum and its small action provide the example in this chapter. Its lower bound tests an inference from native mixed regularity and small action alone. Whether the original prepared selector admits that very configuration requires the original range and selector conditions. See Balaban, CMP 102 (1985), 277?309, Theorem 1, and R-I/R-II, CMP 122 (1989), 175?202 and 355?392, especially R-II (1.61)?(1.65).

Section 143 / Exact finite-source current completion and the physical

This chapter addresses the physical marked-current normalization in GG9.16 and the finite-source contact obligation in MT36. It proves an all-order finite-source formula for the actual flat Gaussian reference, including its joint observations, its absolute scalar coefficients, and the contact subtraction that would cancel its redundant current family. It also gives the exact finite-source change-of-variables identity on the original compact Wilson configuration space.

The results do not prove all-history MT36. The original physical source exponential, a Ward-flow-completed source exponential, and a newly compensated Gaussian source family are different objects. Their differences are computed below. No factor of the original word mesh is added to a source. The actual nonlinear selectors, constraints, denominators and live-reference responses remain obligations of any application to the full construction.

Section 143.1 / The finite-source algebra and the exact Wilson Ward flow

Let

$$\mathcal{A} = \mathbb{R}[\eta_1, \dots, \eta_N]/(\eta_1^2, \dots, \eta_N^2) \tag{PC10.1}$$

be the original finite source algebra. Its positive-degree ideal is nilpotent. Source words below keep their original labels; in particular, a parameter denoted ε_I can stand for one already generated word η_I. Products with overlapping labels are zero, rather than additional independent sources.

On the original finite compact link configuration space, write $d\mu = \rho\, d\lambda$, where λ is product Haar measure and $\rho = Z^{-1}\exp(-\beta S_0) > 0$. For a smooth vector field $V = \sum_i v_i X_i$, define

$$D_V u = -\operatorname{div}_\lambda(uV), \qquad \int (D_V u) F\, d\lambda = \int u\, V F\, d\lambda. \tag{PC10.2}$$

This convention agrees with XP20 after dividing by the density. In the CP1 conventions, the vector field associated with $\mathscr{E}[W]$ is $V = \kappa a^4 \sum_i W_i X_i$, and

$$D_V \rho = \mathscr{E}[W]\rho. \tag{PC10.3}$$

Its coefficient includes exactly the mesh and coupling powers already present in the physical current.

For any positive-degree finite-source derivation and its density adjoint,

$$\mathcal{V} = \sum_{I \neq \varnothing} \eta_I V_I, \qquad \mathcal{D} = \sum_{I \neq \varnothing} \eta_I D_{V_I}, \tag{PC10.4}$$

both exponentials below are finite. One has the exact identity

$$\boxed{\int (e^{\mathcal{D}}\rho)\,(e^{-\mathcal{V}}F)\,d\lambda = \int \rho F\,d\lambda.} \tag{PC10.5}$$

Indeed repeated use of PC10.2 moves $e^{\mathcal{D}}$ onto the test as $e^{\mathcal{V}}$, which cancels $e^{-\mathcal{V}}$. Different V_I need not commute. The expansion uses the powers of the one actual operator $\mathcal{V}$, so their ordered products and commutators are retained automatically.

Since the body of ρ is positive, the following finite-source logarithm is well defined at each finite lattice:

$$A_W = \log_{\mathcal{A}}\big((e^{\mathcal{D}}\rho)/\rho\big). \tag{PC10.6}$$

Its first coefficient is the corresponding Ward current. Its higher coefficients are fixed by PC10.6. They are not generally zero, and PC10.6 is not the same source action as the sum of its first-order currents. All coefficients of positive source degree in $\int e^{\mathcal{D}}\rho\,d\lambda$ vanish exactly. This fact concerns the completed density, including every scalar coefficient.

There is an explicit formula for the whole logarithm. Put $t = \rho^{-1}\mathcal{D}\rho$, and introduce an ordinary auxiliary parameter s. Since

$$\mathcal{D}(\rho e^{A}) = \rho e^{A}(t - \mathcal{V}A), \qquad A_s = \log_{\mathcal{A}}\big((e^{s\mathcal{D}}\rho)/\rho\big), \tag{PC10.6a}$$

one has $\partial_s A_s = t - \mathcal{V}A_s$, $A_0 = 0$. Solving this finite polynomial equation gives

$$\boxed{A_W = \sum_{r\geq 1} \frac{(-1)^{r-1}}{r!}\,\mathcal{V}^{r-1}t.} \tag{PC10.6b}$$

Every ordered derivative in this expression acts on the original coefficient to its right. For a derivation generated by two disjoint original source words, $\mathcal{V} = \varepsilon_I V_I + \varepsilon_J V_J$, the coefficient of $\varepsilon_I\varepsilon_J$ in the logarithm is

$$-\tfrac{1}{2}(V_I t_J + V_J t_I). \tag{PC10.6c}$$

In a larger source algebra, PC10.6b sums every contributing ordered partition and any original higher-word coefficient; it does not replace different partitions by one representative contact.

143.1.1 The complete kernel must participate

Let K be the actual joint integration kernel, with its original selector, constraint records, coarea factors, normalizations and reference data. After testing any retained output variables, PC10.5 applies to the whole function or joint distribution KF:

$$\int (e^{\mathcal{D}}\rho)\,e^{-\mathcal{V}}(KF)\,d\lambda = \int \rho KF\,d\lambda. \tag{PC10.7}$$

By contrast, keeping the kernel fixed and transforming only F gives

$$\boxed{\int (e^{\mathcal{D}}\rho)K(e^{-\mathcal{V}}F)\,d\lambda = \int \rho(e^{\mathcal{V}}K)F\,d\lambda.} \tag{PC10.8}$$

The reason is that the exponential of the derivation $\mathcal{V}$ is an algebra automorphism. Formula PC10.8 displays the response that would be lost by deleting an EOM packet while leaving all original constraint records unchanged.

At first order, a smooth denominator $D(U)$ contributes $V(D^{-1}) = -D^{-2}VD$. A live reference contributes its actual response to VK. If K contains a sharp selector or a coarea constraint, the identity is interpreted in the original joint realization, by distributional transposition after output testing. It does not multiply separately differentiated delta distributions or discard boundary currents. The formal identity supplies no norm bound for these responses.

This extends the exact bookkeeping of XP22–31 and CP40–42; it does not replace their unresolved quantitative hypotheses. In particular, one may keep the original Wilson density in PC10.7 and put all inherited densities in K. There is then no assumption that an inherited effective density has the bare Wilson score.

143.1.2 Support and the limited physical quotient

Every application of V_I differentiates the actual variables in its coefficient and in the function on which it acts. If these variables are absent from the latter's determining set, that contact is exactly zero. Iterating PC10.5 therefore retains the original source offsets and the actual finite chains of intersecting read sets. At fixed source order, no additional contacts are created between completely disjoint determining sets by this algebra alone. Inherited coefficients with larger determining sets must keep those sets; they are not assigned the microscopic support of their first bare ancestor.

The separated-support EOM quotient in CP27a–c is compatible with this identity. A whole current with carrier strictly separated from the other insertions can vanish in the physical pairing. Once a later insertion meets its differentiation variables, PC10.5 creates the corresponding contact. Thus this quotient is not a multiplicative ideal. Reflection-form equality follows from the exact identity when the entire observable pair and its kernel are transported with their actual supports. It does not justify an arbitrary subtraction of contact scalars from a fixed original observable family.

Section 143.5 / Sharp repeated-current boundary powers for the actual average

Retain spacing one and write $Q = Q_{a,1}$, with the actual boxed hold and own-direction line average of GG9.1. Put

$$R_r = QH^{2r-1}Q^{*}. \tag{PC10.28}$$

For every fixed $r \geq 1$, positivity, the spectral bound $\|H\| \leq Ca^{-2}$, and GG9.3 give

$$\begin{aligned} 0 \leq R_r &\leq \|H\|^{2r-2} QHQ^*, \\ \|R_r\| &\leq C_r a^{3-4r}. \end{aligned} \tag{PC10.29}$$

This estimate retains the one face-counting gain of GG9.3; using the norm of every Hessian first would lose it.

The power is sharp on the same actual physical rows. Take a periodic torus of side T, with an even integer number of retained unit blocks, and a transverse coarse field whose only nonzero component is u_2, alternating between $+\epsilon$ and $-\epsilon$ in consecutive blocks along direction 1. It is independent of the other coordinates and lies in one fixed normalized color direction. Its fine adjoint $j = Q^*u$ is the corresponding step field: the own-direction line average acts trivially because there is no direction-2 dependence. The field is co-closed and has zero harmonic component.

On this field the Hessian is the one-dimensional lattice Laplacian in direction 1. Thus

$$\langle u, R_r u \rangle = \langle j, H^{2r-1} j \rangle = \|H^{r-1} D_{1,a} j\|_{2,a}^2. \tag{PC10.30}$$

Each original coarse face contributes a jump of magnitude 2ϵ. With $n = r - 1$, applying H^n to the jump layer produces the coefficient vector of $(2 - S_1 - S_1^{-1})^n$, times $2\epsilon a^{-(2r-1)}$. Its squared coefficient sum is

$$\sum_{k=-n}^{n} \binom{2n}{n+k}^2 = \binom{4n}{2n}. \tag{PC10.31}$$

For $a^{-1} > 2(r-1)$, adjacent layers do not overlap. Counting all original fine transverse sites and all original faces gives exactly

$$\boxed{\langle u, R_r u \rangle = 4\epsilon^2 T^4 \binom{4r-4}{2r-2} a^{3-4r}.} \tag{PC10.32}$$

At $r = 1$, this is the CP39/GG9 sharp a^{-1} result. No additional fine-root factor is appended to PC10.32.

For the uniform current family, H^{2r-1} has its actual lattice range $O(ra)$. Before restriction to a possibly nonlocal coordinate basis, the coarse kernel in PC10.28 comes from the original finite stencils of the two readouts and this Hessian power. Thus its contact range is bounded by their actual readout ranges plus $O(ra)$. The same statement holds for a joint stack, block by block. It is not a claim that an arbitrary matrix representation of a transverse projection is local, nor a locality theorem for arbitrary inherited nonlinear coefficients.

143.5.1 What genuine mesh factors would accomplish

Suppose, as a separately proved fact about a packet, that each of these r current factors carries a^p. Then the covariance coefficient in PC10.25 has the sharp power

$$a^{pr+3-4r}. \tag{PC10.33}$$

Consequently:

Section 143.5 / Sharp repeated-current boundary powers for the actual average

Actual factor on each current	Power at current multiplicity r
No additional factor	a^{3-4r}
A proved a^2 on each	a^{3-2r}
A genuine own-word a^4 on each	a^3

The first row begins with the physical a^{-1} obstruction. The second gives a for one current but already a^{-1} for two. Thus the one-current summability statement for a genuine a^2 coefficient in GG9.16 cannot by itself be iterated into a bound on all joint repeated-current contacts. The third row is a valid separately normalized Gaussian family, when those factors are part of the original recorded packets.

This calculation concerns the conditional expectation of the joint fine product. It does not contradict GG9.15, which concerns a single already conditioned packet multiplied by a function of the currently retained field. Conditional expectation is not multiplicative, as XP already records. Using independent one-current copies would miss precisely the higher terms in PC10.25.

Section 143.8 / Scope for the nonlinear same-measure induction

The following results are established here:

1. PC10.5–8 is an exact finite-source Ward-flow identity for the original compact Wilson measure, including the residual when the actual integration kernel is left fixed.
2. PC10.13–17 is the complete finite-source quadratic formula on one actual flat Gaussian, with a joint stack of retained observations, all scalar companions and all ordered source offsets.
3. PC10.18–20 is an explicit all-order Gaussian contact compensator. It defines a completed Gaussian source family and proves its equality to the source-free Gaussian test functional.
4. PC10.29–32 gives matching upper and lower physical boundary powers for every fixed repeated-current multiplicity. PC10.34–37 keeps the higher absolute scalar masses and their fine coordinate count.
5. PC10.42–43 proves the concrete D8 and scalar contact coefficient in the Ward-flow completion of a pure Gaussian D6 total-current source; PC10.43a retains its fixed-constraint response.
6. PC10.44–50 proves the exact nonlinear same-Wilson mixed contact completion for the bare total-current marks and a real-field g_j^2 bound for the scalar-companion response, with its actual physical a^{-4} and its original energy retained.

These results resolve the Gaussian multi-source bookkeeping gap identified in GG9.15, for the specified joint Gaussian realization. They do not turn a non-Gaussian inherited density into that Gaussian. In particular, the following statements remain unproved:

- that the original physical MT source projection supplies exactly the compensating coefficients required by PC10.18 or the full Ward completion, with all nonlinear source offsets;
- uniform bounds for the nonlinear Wilson score, Hessian, averaging response, and all induced contacts in the inherited effective density on the original selector-supported prepared fields;
- the corresponding common complex-wall admission and saved-grade bounds for previously generated coefficients;
- a same-measure estimate for every constraint, selector, normalization and live-reference response in PC10.8;
- control of the remaining nonredundant high-dimensional contact operators after an exact EOM elimination, including their absolute scalar rows and their use of the single shared reserve.

The restricted real centered-current estimate CD9 cannot supply these missing statements. Its own-word drift gain becomes the physical $a^{-3}\alpha$ bound when the absent a^4 is removed. Likewise, the exact Gaussian compensator cannot be assigned to the nonlinear Wilson transition without proving the required identification and response bounds. AE8/LM47, all-history MT36, and the complete S94 comparison therefore remain open in the full construction.

Section 144 / Conditional covariance defects and the ordinary scalar

CT9 bounds a complete density commutator without differentiating the density. It requires the ordinary scalar companion as well as the current coefficient measures on the same realization. A weak bound for a pushed current does not supply that scalar mass. This chapter identifies the exact defect when a density change is moved through conditioning, proves a sharp obstruction in the actual Gaussian reference, and states precisely how scalar mass can be recovered when the missing coefficient

Section 144 / Conditional covariance defects and the ordinary scalar

and score derivatives are available.

These are finite-cutoff statements. The Gaussian obstruction concerns what follows from a bounded outer density comparison alone. Its test tilt is not asserted to be an actual R-II interaction or selector.

Section 145.1 / Result and scope

The complete source-zero comparison S94 remains unproved. This chapter checks a possible simplification of its native-record matching problem. The simplification is valid for a fully summed and integrated reference insertion, but it is not valid for an individual selected history or for an isolated reference denominator.

There are three conclusions.

1. The original normalized reference insertion, with every output selector summed and its complete new fiber integrated, is independent of the preparatory forest choice. Its value is exactly the old numerator, with the old vacuum scalars and all old integrations retained. This is a direct application of the established S53–S58b identity to the native choice average. It is not a new proof of native-history locality.
2. A selected healed component does not enjoy that cancellation merely because its output gauge delta can be removed. Its genuine reference denominator and its inherited carrying function can still depend on the native preparatory choice. An explicit two-plaquette SU(N) calculation below proves that identical normalized Haar conventions, root counts, and free-link counts do not force two such cutoff denominators to agree.
3. The primary passages checked do not finish the literal identification of every fixed-target field $U_{k,Z}(CV_\Lambda)$ on every original mixed cutoff fiber with the hypotheses of the 1985 variational theorem. The distinct relaxed reference, numerator background, and fixed-target denominator field must remain separate. The complete native inherited atoms and their absolute scalar increments are therefore still unmatched.

The SU(N) calculation is a test of an asserted algebraic simplification. It is not a replacement for the original Wilson measure, a proposed RG transition, or a counterexample to S94. No comparison estimate for the actual original forests is inferred from it.

Section 145.3 / The late gauge removal has a smaller scope

For one fixed, jointly gauge-invariant complete integrand F, the finite-graph identity S53–S54 implies

$$\int F(V)\,dV = \int \delta_G(VW^{-1})F(V)\,dV, \tag{NR10.9}$$

provided the permitted gauge vertices and pinned roots are unchanged and the required current variables are integrated. Here W is a reference fixed at the retained datum. The recursive tree solution and normalized Haar invariance give Jacobian one. This equality concerns integrated functions, not pointwise densities.

NR10.9 allows the auxiliary **output gauge factor** to be removed from that fixed complete integrand. It does not imply

Section 145.3 / The late gauge removal has a smaller scope

$$\int \delta_{G_1}(VW_1^{-1})F_{G_1}(V)\,dV = \int \delta_{G_2}(VW_2^{-1})F_{G_2}(V)\,dV \tag{NR10.10}$$

when the native preparatory choice has changed the integrand itself. For example, F_G can contain $D_G(b)^{-1}$, the action completion associated with that denominator, and coefficients inherited from the same reference choice. Gauge removal leaves those factors in place.

The distinction is already implicit in the old S54 and S58b results. This chapter does not relabel S54 as a new native matching theorem. The new calculation below checks, within an exact finite Wilson integral, that the unproved equality of denominators cannot be supplied by root counts or Haar normalization alone.

Section 145.5 / Hypotheses and conclusions

The following are brief locators for the original source-zero operations, not substitutes for the missing definitions.

Passage in R-II	Identified object or operation
Printed pp.357–359, (1.3), (1.10), (1.12)–(1.13)	The relaxed reference has free interior targets; the actual denominator evaluates the fixed-target field. Proposition 1 starts from R-I (1.75).
Printed pp.360–361, (1.19)–(1.21)	The mixed numerator background uses R-I (1.80) and the 1985 theorem; its stated regularity is used on the support of the action cutoff.
Printed p.378	Healed components permit removal of the old relative cutoffs and integrated output gauge fixing; large components retain them.
Printed p.379, (1.71)–(1.72)	The component operation also acts on the inherited final exponential. The local vacuum term remains. Coordinate reflections and permutations have their stated finite average.

These distinctions agree with S58b, VM9, and RN9. They do not identify the following three fields with one another:

$$U_0(b), \qquad U_{k,Z}(CV_\Lambda(b);b), \qquad U_{k,Z}^{\mathrm{old\,mixed}}(B,V'',b). \tag{NR10.26}$$

The first has the relaxed target problem. The second varies over the actual reference fiber. The third retains old integration variables and appears in the numerator expansion. A regularity estimate for one of them has to be transported to the others through its actual defining map; notation alone does not do this.

Section 145.5 / Hypotheses and conclusions

The full defining R-I passages have not been recovered Consequently no claim is made that their hypotheses fail. The narrower statement is that their literal use has not been verified here. In particular, an invocation of R-I in R-II is not A direct proof of the missing determining-set, interface, and selector clauses.

Section 145.7 / Hypotheses and conclusions

Statement	scope
Fully summed original reference insertion, all new fiber variables integrated	Exact NR10.5–NR10.6; all genuine denominators cancel only at the completed integral, and the old absolute scalar survives.
Output gauge removal for one fixed complete invariant integrand	Established S54/NR10.9; no new native-history conclusion.
Equality of cutoff denominators from only common roots, Haar measure, and free-link counts	False as a general inference; NR10.18 and NR10.25 provide exact SU(N) Wilson tests.
Actual fixed-target denominator branch on every original mixed χ support	Not verified
Equality or controlled defect of the complete independently generated native reference records	Not proved.
Complete inherited vacuum atoms, scalar increments, wrapped carriers, and terminal comparison	Not proved.
Uniform source-zero comparison S94	Not proved.

The progress is a precise separation of an exact cancellable native normalization from a genuine reference-mass comparison. It prevents a gauge-coordinate identity from being used to erase the remaining absolute scalar obligation.

Section 146.1 / The retained-bond configuration is a constrained minimum

The original 1985 variational paper, printed pp.277–279, was checked independently, including the image of p.277. Its nesting permits $\Omega_0 \supset \Omega_1$; it does not require their equality. Consequently its mixed determining lattice can contain an outer level-zero collar. The level-zero averaging constraint is the literal fine-link constraint used in AE10.

For the separated geometry specified in AE10, every link in each plaquette incident to the distinguished bond is fixed. The full coarser averaging and mixed-interface read sets avoid that bond. The single-bond field therefore satisfies the actual nonlinear target constraints, and all its unfixed plaquette action terms vanish.

The following elementary argument is sufficient to identify the minimum. Partition the original plaquette terms into the six terms incident to the distinguished bond and their complement. Every admissible compact field has exactly the same first six terms because all their links are fixed. All terms in the complement are nonnegative. The single-bond field makes that complement zero. Thus it attains the absolute constrained minimum. No estimate for a minimizer tail, Euler–Lagrange inversion, or perturbative replacement of a constraint is needed.

For $T = \operatorname{diag}(i, -i, 0, \dots, 0)$, each of those six plaquettes has eigenvalues $e^{i\delta}, e^{-i\delta}, 1, \dots, 1$. In the stated CQ normalization this gives exactly

$$S_0(U) = 6\,[2(1 - \cos\delta)] = 12(1 - \cos\delta). \tag{AU10.1}$$

The small own-scale regularity conditions hold directly: all nonzero plaquette and force defects lie in a fixed fine star, and the coarser regions are flat. Taking the fixed target-size constant large enough puts this field in the small regularity class of the theorem. The theorem's uniqueness of its critical orbit then identifies the native minimal orbit with this explicit representative.

The physical lower bounds also check. One plaquette gives $\|F_a\|_p \geq c\delta a^{4/p-2}$. At the neighboring parallel bond used in AE10, exactly one transverse plane contributes a nonzero force. Its magnitude is a fixed positive multiple of $a^{-3}|\sin\delta|$. Subtracting the average of its three plane components yields

$$(Q_1, Q_2, Q_3) = (2j/3, -j/3, -j/3), \qquad \sum_i |Q_i|^2 = \tfrac{2}{3}|j|^2. \tag{AU10.2}$$

Multiplying by the one-bond physical volume a^4 proves the $c\delta^2 a^{-2}$ current lower bound. The fine-cell flux moment is $c\delta^4 a^{-4}$. No multiplicity growing with the cutoff was required or omitted.

There is no stationary-equation contradiction. The physical force is evaluated in a direction which the retained-link constraint forbids as an admissible tangent. Constrained criticality does not set that force to zero.

The complex statement has the narrower, correct interpretation in AE10. The same explicit configuration is a holomorphic constrained critical family: every plaquette involving a free link is the identity. Positivity proves its real compact minimum only. The plaquette equality forced by the retained rows,

Section 146.1 / The retained-bond configuration is a constrained minimum

and hence its target derivatives, persist on every holomorphic branch preserving those rows. This does not construct the original simultaneous R walls.

The required scope restrictions are essential. The measured domain must contain the retained fine star. The example is a minimum in the original mixed variational class, not an stated R-selector history. It does not refute a protected-interior estimate which excludes that star, and it cannot be assigned a reserve charge without a literal history and ownership record.

PF10.4 is an exact consequence of disintegration after the full nonlinear chain rule has been applied on the same observation map. For any scalar or tensor coefficient c,

$$q_*(hc\,\mu) = \Big(\bar{h}\,\bar{c} + \mathbb{E}[(h-\bar{h})(c-\bar{c}) \mid q]\Big)\, q_*\mu. \tag{AU10.3}$$

Substituting this identity in the scalar term and in every ordered coarse-word coefficient gives PF10.4 with its stated signs. No conditional normalization was performed. If the map or its live reference changes, the additional response terms are outside this identity and must first be retained.

The bound by 2ε times the fine conditional oscillation is valid. It cannot be replaced by the total variation of the already conditioned coefficient: conditional expectation may cancel large fine fluctuations. Taking the multiplier to be a function of the observation is an exact sufficient condition for the covariance packet to vanish.

The Gaussian scalar calculation in PF10.9–13 checks independently. The physical orthonormal mode convention gives

$$T_a = 2c_\beta^{-2}a^4 \sum_k \lambda_k^2(Z_k^2-1), \qquad \operatorname{Var}(T_a) = 8c_\beta^{-4}a^8 \operatorname{Tr} H_a^4. \tag{AU10.4}$$

In four dimensions a fixed positive fraction of modes have $\lambda_k \asymp a^{-2}$, and the number of such modes is $\asymp a^{-4}$. Thus the standard deviation is $\asymp a^{-2}$. The fourth-moment bound $\mathbb{E}X^4 \le 15(\mathbb{E}X^2)^2$, followed by the stated Hölder interpolation, gives the same order for $\mathbb{E}|T_a|$. This verifies an ordinary fine scalar total-variation statement; it does not contradict a smaller signed coefficient after conditioning.

The additional conditional derivative formula PF10.20–25 also checks. The Moore–Penrose inverse acts on the actual compatible observation range. Consequently $QC = 0$, and Gaussian integration by parts on the positive conditional row space gives

$$\operatorname{Cov}(h, t \mid q = y) = 2(Tm_y)^T C\, \mathbb{E}_y \nabla h + \operatorname{Tr}(CTC\, \mathbb{E}_y \nabla^2 h). \tag{AU10.4a}$$

Both derivatives are tangent to the original conditional fiber: the factors of C remove any dependence on an extension of h in its normal directions. The physical D8 specialization retains precisely $2g_j^{-4}\beta^{-3}w_a H_a$ in the Hessian contraction. For a smooth $h = e^\sigma$, the contracted second derivative is $h[H_a : \nabla^2\sigma + (\nabla\sigma)^T H_a \nabla\sigma]$. Thus the formula exhibits the additional derivative information needed; it does not obtain that information from a bound on $\|\sigma\|_\infty$.

For the bounded test tilt in PF10.14, Paley–Zygmund gives $\mathbb{P}(|T_a| \ge \sigma_a/\sqrt{2}) \ge 1/60$. Since $x \tanh x \ge 0$, the lower bound in PF10.17 follows with the displayed positive constant. The normalization mass stays between $1-\varepsilon$ and $1+\varepsilon$. This is a valid counterexample to an implication based only on an externally analytic small density factor. It is not asserted to be an actual R multiplier or to have a fixed complex domain in all internal coordinates.

One boundary precision is required in using PF10.18 at higher word order. Either the coefficient is compactly supported in the chart, or the boundary flux must vanish at every integration-by-parts

Section 146.2 / Conditioning leaves an ordinary covariance packet

stage. Vanishing of the initial coefficient alone does not remove the flux of an intermediate adjoint derivative. With that convention, the reversed adjoint order in PF10.18 and the first-order scalar recovery inequality are correct. Sharp selectors still require their complete boundary packets.

The following independent one-mode check records why a first-current identity does not identify the entire source exponential. Let $Z \sim N(0,1)$, and let $q = m(Z^2 - 1)$. The original source exponential has mass

$$\mathbb{E}e^{\tau q} = e^{-m\tau}(1 - 2m\tau)^{-1/2}, \qquad \log \mathbb{E}e^{\tau q} = m^2\tau^2 + O(\tau^3). \tag{AU10.5}$$

The density transported by the linear Ward flow instead has ratio

$$e^{-m\tau} \exp\Big[\tfrac{1}{2}(1 - e^{-2m\tau})Z^2\Big] = \exp\Big[\tau q - m^2\tau^2 Z^2 + O(\tau^3)\Big] \tag{AU10.6}$$

to the original Gaussian density. Its total mass is exactly one. The second-order correction is already necessary in this single mode. The matrix Gaussian formulas in the physical-current chapter must preserve precisely this distinction.

For a symmetric finite-source quadratic $T(\eta)$, centered against a positive covariance Σ, direct Gaussian integration gives

$$\begin{aligned} \log Z_\eta(J) = & -\operatorname{Tr}(\Sigma T) - \tfrac{1}{2}\operatorname{Tr}\log(I - 2\Sigma T) \\ & + \tfrac{1}{2} JQ(\Sigma^{-1} - 2T)^{-1}Q^* J. \end{aligned} \tag{AU10.7}$$

This identity makes no coarse inverse-covariance assumption. A joint stack of retained observations is allowed. In a finite nilpotent source algebra with zero source-free part in T, all inverse and logarithmic series terminate coefficientwise.

Multiplying the generating function by the inverse of its vacuum factor and covariance change cancels those factors algebraically. The covariance part depends on the probe J. It is therefore an explicit generating-functional contact prescription, whose identification with the original insertion and projection rules requires an additional proof. It cannot by itself supply a missing birth factor in the original source.

146.3.1 The weighted contact and the joint kernel

The weighted coefficient PC10.15 was compared directly with the original CP4 and CP6 formulas. The scalar quadratic form $2\langle A, HfHhHA\rangle$ has symmetric matrix $H(fHh + hHf)H$. Its Gaussian mean is the stated $2\beta^{-1}\operatorname{Tr}(fHhH)$ before the common physical prefactor. Thus both its original source offsets and its constant companion are present.

More generally, on the actual positive physical space, if $T = HKH$ then $\Sigma T = \beta^{-1}KH$. Expanding the covariance and logarithmic determinant therefore gives exactly

$$[\delta C]_r = 2^r \beta^{-(r+1)} QK(HK)^{r-1}Q^*, \qquad [v]_r = \frac{2^{r-1}}{r}\beta^{-r}\operatorname{Tr}((KH)^r). \tag{AU10.7a}$$

For a representation in the full ambient link space, the projections onto that physical space remain at the exposed ends. Nonconstant source multipliers need not preserve the physical subspace. The projection qualification in PC10.16 is therefore necessary.

The joint-kernel identity PC10.7–8 has the correct distributional scope. First regard the original joint kernel as one already defined distribution and test the retained output variables. Smooth vector fields act on that distribution by transposition, and their Leibniz rule holds against smooth coefficient functions. Hence $e^{\mathcal{V}}(Ke^{-\mathcal{V}}F) = (e^{\mathcal{V}}K)F$ coefficientwise. This gives precisely the residual kernel response in PC10.8. It does not require or permit products of separately differentiated delta distributions. It also supplies no estimates for the resulting selector or normalization currents.

146.3.2 Repeated contacts retain one face gain and their scalar masses

For the uniform physical D8 current the covariance coefficient is $4^r g_j^{-4r}\beta^{-(2r+1)}QH^{2r-1}Q^*$. The upper bound follows from $H^{2r-1} \le \|H\|^{2r-2}H$, retaining the original a^{-1} face bound for QHQ^*. It is therefore $O_r(a^{3-4r})$.

The alternating transverse block test in PC10.30–32 attains that power. It is co-closed and harmonic-free, and its own-direction line filter acts as the identity. Each interface jump becomes $2\epsilon a^{-(2r-1)}$ times the coefficient sequence of $(2 - S - S^{-1})^{r-1}$. The squared sequence norm is $\binom{4r-4}{2r-2}$. Multiplying by the physical volume a^4 and the exact transverse/interface count T^4a^{-3} gives

$$\langle u, QH^{2r-1}Q^*u\rangle = 4\epsilon^2T^4\binom{4r-4}{2r-2}a^{3-4r}. \tag{AU10.7b}$$

The even retained torus size and nonoverlapping fine stencils are explicit hypotheses in the chapter. The coefficient is for the joint fine product, so replacing it by a product of separately conditioned one-current expressions would be incorrect.

The logarithmic vacuum coefficient is

$$[\tau^r]v = \frac{2^{2r-1}}{r}g_j^{-4r}\beta^{-2r}\operatorname{Tr}H^{2r}, \qquad r \ge 2. \tag{AU10.7c}$$

The trace has order a^{-4r-4} in fixed physical volume. A genuine a^4 on each current therefore leaves an extensive a^{-4} scalar contact, even while the covariance coefficient has order a^3. These are distinct coefficients of the same generating function. Their counts cannot be exchanged or dropped.

Finally PC10.42–43 agrees with the one-mode check: the D6 Ward-flow logarithm creates $-\tau^2t_8/2 - \tau^2\operatorname{Tr}M_6^2$. At the mixed degree in two distinct source labels, this is $-t_8 - 2\operatorname{Tr}M_6^2$, the negative of the actual derivative contact. The cancellation includes its scalar and is specific to the completed pure total-current family. It does not cancel the entire source effect with the observable held fixed: the source-only Ward flow still has second covariance coefficient $2Q\Sigma M_6^2Q^*$, as PC10.43a records. Exact removal from the full pairing requires the corresponding observable and kernel transport, or the explicitly defined compensator.

146.3.3 The nonlinear Wilson flow coefficients can also be computed exactly

The nonlinear extension uses the original density rather than its quadratic approximation. Put $t = \rho^{-1}D_V\rho$. The exact product identity $D_V(\rho e^A) = \rho e^A(t - VA)$ shows that $A_s = \log(e^{sD_V}\rho/\rho)$ solves $\partial_s A_s = t - VA_s$, $A_0 = 0$. Hence

$$A_W = \sum_{r\ge1}\frac{(-1)^{r-1}}{r!}V^{r-1}t. \tag{AU10.7d}$$

Section 146.3 / The original Gaussian source and a Ward-flow completion differ

The sum terminates in the actual finite-source algebra. This calculation does not commute distinct vector fields: they remain in the ordered expansion of the single source-dependent derivation.

For the Wilson physical total-current source, write $t_f = g_j^{-2}(P_f - B_f)$, with $B_h = \beta^{-1}a^{-2}\sum_j h_j X_j^2 S_0$. Direct differentiation with its original vector field gives

$$V_f t_h = g_j^{-4}(C_{f,h} - L_{f,h}), \qquad L_{f,h} = \beta^{-2}a^{-4}\sum_{i,j} f_i h_j (X_i S_0)(X_i X_j^2 S_0). \tag{AU10.7e}$$

All factors, source positions, and derivative order agree with the CP conventions. In particular the coefficient is $\beta^{-2}a^{-4}$, obtained from the literal product of κa^4, J_i, and $X_i B_h$.

For uniform $h = 1$, the Casimir identity gives $\Delta S_0 = 4C_F \sum_p \mathrm{ReTr}\, P_p$, so $X_i \Delta S_0 = -4C_F X_i S_0$. It follows that $L_{f,1} = -4C_F\beta^{-2}a^{-2}P_f$, with the stated sign. The original scalar companion B_h itself still contains its constant vacuum contribution; that contribution disappears only when its derivative is taken in $L_{f,h}$.

For two distinct source labels, the second flow-logarithm coefficient is exactly $-\frac{1}{2}(V_f t_h + V_h t_f)$, and thus the symmetrized contact in AU10.7e. This verifies actual nonlinear same-Wilson contact algebra. It does not bound the resulting current, its further derivatives, or the responses of a fixed original integration kernel.

The weighted Casimir reduction PC10.47–48 and its real energy bound were checked directly as well. With the four original incidences $h_p = \sum_{b\in\partial p} h_b$, $\sum_j h_j X_j^2 S_0 = C_F \sum_p h_p \,\mathrm{ReTr}\, P_p$. On real compact links, the trace-pairing inequality and $\|P_p - I\|_{\mathrm{HS}}^2 = 2s_p$ give $\sum_A |X_b^A s_p|^2 \le C s_p$. Finite incidence and Cauchy–Schwarz therefore prove

$$g_j^{-4}a^4|L_{f,h}| \le C c_\beta^{-3} g_j^2 \|f\|_\infty \|h\|_\infty E_Y. \tag{AU10.7f}$$

The physical version without the recorded a^4 has the additional a^{-4}. The proof uses the one original union of plaquette stars, the positive real plaquette energy, and its actual finite multiplicities. It is a genuine same-Wilson bare estimate, with no assertion of a complex-wall bound or an inherited effective-score estimate.

Section 146.4 / Complete reference cancellation and cutoff denominator comparison

The cancellation in NR10.5 is exact with its declared hypotheses. At fixed full old datum, summing all the original selector labels first produces one, and integrating the genuine normalized new reference kernel then produces one. The old numerator and its independent scalar mass remain. The result applies to a closed normalization subdiagram, including a finite sequence of such closed insertions. It does not supply a comparison for a selected healed branch or for a new variable still read by an inherited carrying factor.

The two-tree example has no missing graph or Haar factor. Both trees span the six displayed vertices, have five edges, and pin the same five nonroot gauge variables. Their two free-link coordinates give respectively $(P_1, P_2) = (X, Y)$ and $(P_1, P_2) = (X, YX^{-1})$. Every coordinate substitution used here preserves normalized Haar measure. What changes is the physical image of the relative-link cutoff.

Conjugation invariance of the cutoff gives $\int_{\mathcal{B}} U \, d\mu = v\alpha I$; inversion invariance makes the scalar real. Hence $\mathbb{E}t(YX^{-1}) = \alpha^2$, proving exactly

$$\partial_\beta \log(D_1/D_2)|_{\beta=0} = \alpha(1-\alpha) > 0. \tag{AU10.8}$$

The weak-coupling calculation also retains its normalization. With $\|X\|^2 = N^{-1} \operatorname{Tr} X^2$, the quadratic Wilson expansion is $\|X\|^2/2$. Both integrals have the identical factor $\kappa_N^2 g^{2(N^2-1)}$. For a bounded

rescaled ball, the Gaussian mass of that ball is strictly maximized when the Gaussian is centered there. The one-dimensional slicing argument in NR10.24 proves this strict inequality and therefore the positive limiting logarithmic ratio in NR10.25.

These calculations disprove equality inferred solely from common root counts, Haar conventions, and free-link counts. They do not identify the example's cutoff with the original growing cutoff or its two trees with the actual independent preparatory choices. The native reference-record and inherited scalar comparison is still required for S94.

Section 146.5 / The common complex centered-prefix estimate

The revised complex-current chapter was checked after replacement of its intermediate-potential assumption by the coupled argument CD10.5–10. This replacement is substantive: path norms at intermediate levels now follow from the original fine potential and actual mean-versus-straight comparisons.

Let R_s be the same fine straight path used by the original scale-a_s bond, and let e_s control both the link and its inverse relative to $R_s^{\pm 1}$. With $d_s = e_s/a_s^2$, the original fine-potential bound gives

$$\|U_s(\gamma)^{\pm 1}\| \le \exp(C|\gamma|(m + a_s d_s)). \tag{AU10.9}$$

The inverse control is part of the definition of the error and its one-step estimate; it is not inferred from complex unitarity. The actual one-step finite-stencil expansion then gives the coupled curvature and error recurrences recorded in CD10.7. The curvature linear coefficient is one, up to the actual endpoint factor E_s; the normalized path error has the genuine factor L^{-1}.

The displayed bootstrap is valid. The ceiling on M_* leaves room for $L^{-1}E_s < q < 1$. The constants K, D are then fixed before choosing α_*. The logarithms of the curvature multipliers involve only $\sum a_{s+1}$, $\sum a_{s+1}a_s$, and $\sum a_{s+1}^2$, all bounded geometric sums. Thus $f_s \le K\alpha$ and $e_s \le Da_s^2\alpha$ close with strict room. For $\alpha = 0$, both errors vanish directly. The final comparison with the fine straight path keeps every intermediate logarithm on the original small branch.

The next stage uses one fine coordinate-tree analysis gauge for an entire prefix box. Original fine-path lengths bound its adjoint and inverse adjoint by $e^{Cm\ell}$. The ribbon comparison and the already proved path errors give $U_s^g - I = O_{M_*}(a_s\ell\alpha)$. This is one gauge at each prefix comparison, frozen during the incoming-field jets. It does not alter the original gauge pins or determining records.

The tensor identities are exact over the complex field:

$$S_t^g = G_t S_t G_0^{-1}, \qquad R_t^g = G_t R_t G_t^{\mathsf{T}}, \qquad G_s^{\mathsf{T}} G_s = I. \tag{AU10.10}$$

The last equality is for the invariant complex bilinear form; the absolute coefficient bounds still use the separately bounded operator norms. Intermediate gauge factors cancel before norms are taken. The response-column estimate therefore retains its exact leading L^{-3}, rather than incurring a fixed per-level norm loss.

The transported Hessian calculation keeps the original fine count $(a_t/a)^4$. Its two response columns each have order $(a/a_t)^3$. The relative response defect $a_t^2\alpha$ consequently gives $a^2\alpha$, while the ordered Hessian defect gives at most the same order. The flat tensor is scalar in color, and every same-generator second derivative of the flat compatible mean vanishes. Hence the contraction $R_t^0 : K_t^0$ is exactly zero and the nonflat forcing is $O(a^2\alpha)$ per actual output.

The ordered initial-logarithm bracket is retained. A suffix from scale a_s has column norm $O(a_s^3)$, and the determining enlargement contains $O(m_E a_s^{-4})$ coordinates at that scale. Thus propagating the

initial term and every forcing term yields

$$\sum_{\gamma\in E}|\mathscr{B}_n^{\gamma}|\le Cm_Ea^2\alpha\left(a^{-1}+\sum_{s=1}^{n}a_s^{-1}\right)\le Cm_Ea\alpha. \tag{AU10.11}$$

This verifies the claimed depth-uniform centered coefficient gain.

The parameter derivative bounds use Cauchy's formula for the original holomorphic centered coefficient, not derivatives of an analysis-gauge norm. The Banach-valued output ℓ^1 estimate retains exactly $\nu!\prod_r d_r^{-\nu_r}$ with the declared polydisc margins. Live-reference derivatives are included when those original parameters vary; they are not identified with a separate off-diagonal fixed-reference problem.

The cited CC3–5, CL25/28, and M8.10 statements were scope for their application scope. CC supplies the strong input branch. CL25–28 allow the identified exterior walls while every complete protected equation carrier retains weight one. M8.9–10 apply to the physical readout with reconstruction walls all equal to one, while its complex weak input varies on the proved ball. Arbitrary M8 wall derivatives do not follow from that statement. The final CD10 application paragraphs now distinguish these domains explicitly. In each case the complete protected prefix and ribbon closure must lie in the controlled region. These results do not admit an arbitrary original R-prepared field. The physical source conversion still removes any a^4 absent from that source, taking the displayed drift gain to $a^{-3}\alpha$. The complex extension supplies neither a new birth factor nor a whole inherited scalar-current or selector-boundary estimate.

Section 146.6 / Remaining mixed-entry, contact and torus estimates

The retained-bond minimum, covariance defect under conditioning, Gaussian scalar-mass obstruction, quadratic source and contact coefficients, nonlinear Wilson flow-logarithm, centered complex prefix estimate and complete-reference cancellations have the individual domain and normalization hypotheses used in their derivations. The physical M8 and protected exterior-wall hypotheses remain in force.\n\nThe retained-bond example alone cannot decide whether the original selectors admit the defect. The nonlinear current pair requires estimates on the original mixed support. The full native vacuum records still have to be matched on physical tori. These are respectively the missing applications of AE8/LM47, MT36 and S94.

Section 147.1 / A native mixed minimum does not supply actual R admission

AE10 strengthens the previous arbitrary-field tests. The scope 1985 variational problem has literal level-zero fine-link constraints on its outer retained layer. Choose an admissible separated geometry, one fine bond in that layer, and the original mixed target $e^{\delta T}$ on that bond, with all other target links the identity. Every coarser determining prefix and every mixed-interface target plaquette avoids the full defect star. The complete target condition, including its interface clauses, is satisfied.

All links of the six incident plaquettes are pinned by actual level-zero rows. Their Wilson action contributions are therefore fixed throughout the constraint fiber. Every other plaquette action is nonnegative, and the one-bond field makes all of them zero. It is an explicit global constrained minimum. The original small regularity and uniqueness theorem identifies its orbit with the native minimal orbit. This does not use a linearized constraint or an estimated tail of an unspecified minimizer.

For $T = \operatorname{diag}(i, -i, 0, \dots)$, the exact action and the physical lower bounds on a current-size domain containing that fine star are

$$\begin{aligned} S_0(U_a(\delta)) &= 12(1 - \cos\delta), \\ \|F_a(U_a(\delta))\|_{L^p_a} &\geq c\delta a^{4/p-2}, \\ \mathfrak{F}_4(U_a(\delta); 1) &\geq c\delta^4 a^{-4}, \\ \mathfrak{J}_2(U_a(\delta); 1) &\geq c\delta^2 a^{-2}. \end{aligned} \tag{R10.1}$$

One adjacent parallel bond has a force in only one transverse plane; subtracting the three-plane average gives the exact proportions $(2j/3, -j/3, -j/3)$. Its nonzero force is compatible with constrained stationarity because the corresponding tangent is pinned.

Taking $\delta = a^\theta$, $0 < \theta < 1$, makes the target size, own-scale regularity size and action tend to zero while every displayed physical norm diverges for $p > 8$. The action conclusion holds at fixed coupling, and also for a coupling sequence satisfying $\beta(a)a^{2\theta} \to 0$. No running-coupling conclusion is needed for the counterexample.

The retained plaquette is also exactly $e^{\pm zT}$ on its holomorphic constrained critical family. Thus

$$\partial_z^m F_{a,p}(0) = a^{-2}(\pm T)^m, \qquad r_a \leq C\tau a^{2-4/p} \tag{R10.2}$$

Section 147.1 / A native mixed minimum does not supply actual R admission

is a necessary target-disc restriction for a proposed current-unit curvature ceiling τ. Higher target derivative order supplies no extra positive power of the mesh.

The inference from native mixed regularity and small action to AE8 on every measured domain is therefore false. This is a proved restriction on the proposed argument. It is **not** a verified R-selector history and does not refute an estimate on a protected interior excluding the entire star. Nor does it identify the raw field, the actual prepared composite, and the old physical readout. If the full old determining and boundary record avoids the defect, that old reconstruction has flat data and remains flat. The original selectors and read sets must decide which situation occurs.

No R history has been assigned to this example, so no reserve charge has been assigned to its depth. Its construction cannot be paid for by a newly invented failure record.

Section 147.2 / The centered complex regular row and its literal parameter margins

CD10 extends CD9's centered constant-test current estimate on an identified strong regular branch. Its common complex domain has both a small weak potential bound and a small physical curvature supremum on the complete protected neighborhoods. These neighborhoods contain all original averaging prefixes, Hessian stars, coordinate-tree ribbons and curvature-propagation stencils. A small weak potential alone is not the curvature hypothesis.

The complex proof replaces real compact isometries by controlled transport norms. The leading coefficient in the curvature recursion remains one, with a perturbation summable over the actual scale lengths. One fine analysis gauge controls each complete prefix. Intermediate adjoint changes cancel exactly before their endpoint norms are taken. Consequently no fixed complex frame loss is multiplied once at every depth.

For the original ordered Wilson Hessian and the centered compatible map, the resulting bounds have the same powers as the real row:

$$\begin{aligned} \sup_{\mathcal{D}} \sum_{\gamma\in E} \left| \sum_{ij} (X_i X_j S_0) X_j X_i x_n^{\gamma} \right| &\le C m_E a \alpha, \\ \sup_{\mathcal{D}} \sum_{\gamma,\delta\in E} |R_n^{\gamma\delta}| &\le C m_E (a^3 + a^2\alpha). \end{aligned} \tag{R10.3}$$

Here α is the actual common curvature bound, and m_E is the original local coordinate count with its determining enlargement. The initial ordered logarithmic bracket and every intermediate forcing term remain in the first estimate.

For a declared parameter polydisc with actual interior margins d_r, Cauchy's formula for the original holomorphic coefficient gives

$$\sum_{\gamma\in E} |\partial_z^{\nu} \mathscr{B}_n^{\gamma}| \le C m_E a \alpha\, \nu! \prod_r d_r^{-\nu_r}. \tag{R10.4}$$

The pointwise comparison gauge is not substituted into the definition of this coefficient and then differentiated. The actual parameter radii, canonical factors and source labels are retained.

This applies to the constructed CC strong branch. For CL25–28, exterior wall parameters may vary only while every complete protected equation carrier retains weight one. For M8.9–10, the application fixes all physical reconstruction walls at one and varies the complex weak input on its proved ball. Any M8 exterior-wall extension requires the identified protected CL/MC theorem as a further input; analyticity of the map alone is insufficient. Each application needs the complete determining, ribbon

and curvature closure. The statement does not admit arbitrary original mixed R fields or arbitrary reconstruction walls.

The reference is frozen while the incoming jets are taken, then evaluated at the centered field. Parameter derivatives of this centered family retain the family's reference response; they are not an identification with an independently varying reference in an old marked coefficient. The invariant-terminal-test identity also does not cover a joint test which links independent histories.

Finally R10.3 still uses the NH9 word convention containing its original a^4. Removing a factor absent from the physical source turns $a\alpha$ into $a^{-3}\alpha$. The complex extension supplies neither a missing birth factor nor ordinary scalar total variation, selector-boundary estimates, or complete later-contact bounds.

Section 147.3 / An exact same-Wilson contact completion retains the full kernel

PC10 works in the original finite nilpotent source algebra. Let $\mathcal{V}$ be its positive-degree vector-field derivation on the original compact link space and $\mathcal{D}$ its Haar density adjoint. With the original Wilson density ρ,

$$\int (e^{\mathcal{D}}\rho)(e^{-\mathcal{V}}F)\,d\lambda = \int \rho F\,d\lambda. \tag{R10.5}$$

Ordered source words need not commute as differential operators. The one operator exponential retains their order automatically. The exact logarithmic completion is determined by $\log[(e^{\mathcal{D}}\rho)/\rho]$, including every higher contact and every scalar coefficient. It is generally different from the original exponential of its first-order currents.

The logarithm has the explicit finite formula

$$A_W = \sum_{r\geq 1} \frac{(-1)^{r-1}}{r!}\mathcal{V}^{r-1}t, \qquad t = \rho^{-1}\mathcal{D}\rho. \tag{R10.5a}$$

Indeed its time-dependent version solves $\partial_s A_s = t - \mathcal{V}A_s$, $A_0 = 0$. This retains every ordered contact, including all partitions contributing to an original source word.

For the actual Wilson total-current sources $t_f = g_j^{-2}(P_f - B_f)$, with $B_h = \beta^{-1}a^{-2}\sum_j h_j X_j^2 S_0$, the two-label logarithmic coefficient is

$$\begin{aligned}[\varepsilon_f\varepsilon_h]A_W &= -\frac{g_j^{-4}}{2}(C_{f,h} + C_{h,f} - L_{f,h} - L_{h,f}),\\ L_{f,h} &= \beta^{-2}a^{-4}\sum_{i,j} f_i h_j (X_i S_0)(X_i X_j^2 S_0).\end{aligned} \tag{R10.5b}$$

This is a nonlinear same-Wilson identity. It includes the derivative of the original scalar companion, with its exact source offsets and ordered derivatives. The companion's constant vacuum part remains in the original source even though its derivative in L vanishes.

The weighted color Casimir reduces the third derivative in this specific response to first plaquette-action derivatives. On the original real compact carrier, finite incidence and the unitary gradient bound then prove

$$g_j^{-4}|L_{f,h}| \leq C c_\beta^{-3} g_j^2 a^{-4}\|f\|_\infty \|h\|_\infty E_Y, \qquad E_Y = \beta \sum_{p\in Y^+} s_p. \tag{R10.5c}$$

The original carrier energy occurs once. Its physical a^{-4} has not been removed, and no new reserve has been allocated. This real bare-carrier estimate does not admit those coefficients to an inherited mixed complex domain.

Section 147.3 / An exact same-Wilson contact completion retains the full kernel

If the actual integration kernel K is held fixed, the exact residual is

$$\int (e^{\mathcal{D}}\rho)K(e^{-\mathcal{V}}F)\,d\lambda = \int \rho(e^{\mathcal{V}}K)F\,d\lambda. \tag{R10.6}$$

Thus constraint derivatives, cutoff-boundary currents, coarea terms, live-reference responses and genuine denominator derivatives remain. They cancel only when the whole kernel participates in the stated transformation. At a sharp selector the identity is interpreted on the already defined joint realization, after output testing; it does not multiply independently differentiated delta distributions.

The formula identifies the additional cancellation required by a physical current reduction. It does not bound the resulting kernel responses. In particular, an inherited density is not assigned the bare Wilson score merely because its original ancestor was Wilson.

Section 147.4 / Complete Gaussian contacts reveal stronger physical normalization requirements

The actual flat Gaussian reference can now be calculated at every fixed finite source degree, with one joint observation stack Q. This preserves cross-level constraints even when its joint covariance is singular. With $\Sigma = \beta^{-1}H^{-1}$, an actual quadratic source $A^T T(\eta)A + s(\eta)$ gives

$$\begin{aligned} Z_\eta(z) &= \exp\{v(\eta) + \tfrac{1}{2}z^T C_\eta z\}, \\ v(\eta) &= s(\eta) - \tfrac{1}{2}\operatorname{Tr}\log(I - 2\Sigma^{1/2}T\Sigma^{1/2}), \\ C_\eta &= Q(\Sigma^{-1} - 2T)^{-1}Q^*. \end{aligned} \tag{R10.7}$$

All series terminate in the original source algebra. Nonconstant CQ multipliers remain at their actual fine offsets inside ordered matrix words. Physical projections are retained wherever they do not cancel. No inverse of the joint coarse covariance is used.

For the physical constant-test D8 current, with no added mesh factor,

$$\begin{aligned} t_8 &= 2g_j^{-4}\beta^{-1}\langle A, H^3 A\rangle - 2g_j^{-4}\beta^{-2}\operatorname{Tr} H^2, \\ [\tau^r](C_\tau - C_0) &= 4^r g_j^{-4r}\beta^{-(2r+1)}QH^{2r-1}Q^*. \end{aligned} \tag{R10.8}$$

At $\beta = c_\beta g_j^{-2}$, the covariance prefactor retains g_j^2, together with the actual external canonical couplings. It is not a new positive power of a.

The exact original block-and-line adjoint, tested against an alternating transverse retained field on a fixed physical torus, has

$$\langle u, QH^{2r-1}Q^*u\rangle = 4\epsilon^2 T^4 \binom{4r-4}{2r-2} a^{3-4r}. \tag{R10.9}$$

For each fixed multiplicity r, the matching upper bound has the same power. The lattice is sufficiently fine that the finitely many difference layers do not overlap. This is the joint fine product calculation; conditional expectation has not been treated as multiplicative.

The consequences for literal source coefficients are:

Factor actually present on each current	Covariance power at multiplicity r	Higher scalar power
No additional mesh factor	a^{3-4r}	a^{-4r-4}

Section 147.4 / Complete Gaussian contacts reveal stronger physical normalization requirements

Factor actually present on each current	Covariance power at multiplicity r	Higher scalar power
A separately proved a^2	a^{3-2r}	a^{-2r-4}
A genuine recorded a^4	a^3	a^{-4}

The scalar column refers to the connected vacuum coefficient at $r \geq 2$, in fixed physical volume. The first scalar coefficient was centered away. The higher ones contain the actual fine trace $\mathrm{Tr}(H^{2r}) \asymp a^{-4r-4}$, and the coupling cancels from their leading prefactor. Thus even the separately normalized a^4 family retains an extensive scalar contact. A source-zero denominator does not cancel a source-dependent vacuum coefficient.

An explicit Gaussian contact completion is now available:

$$G_\eta^{\mathrm{comp}} = e^{-v(\eta)} e^{-\frac{1}{2}(C_\eta - C_0):D^2} G, \qquad \mathbb{E}_0[e^{A_\eta} G_\eta^{\mathrm{comp}}(QA)] = \mathbb{E}_0[G(QA)]. \tag{R10.10}$$

It includes the last scalar contact of every later differentiated source. This proves a complete Gaussian cancellation for the explicitly defined prescription. Equality of that prescription with the original nonlinear physical source projection is unproved.

There is also a derived D8 subtraction in the pure Gaussian D6 Ward-flow logarithm: $-\tau^2 t_8/2 - \tau^2 \,\mathrm{Tr}\, M_6^2$. For two distinct source labels it cancels precisely the product contact $t_8 + 2\,\mathrm{Tr}\, M_6^2$. A higher Ward current still remains at a fixed observation: its second covariance coefficient is $2Q\Sigma M_6^2 Q^*$, containing the same sharp boundary form. Full cancellation requires R10.6's kernel response or R10.10's explicit test completion. The physical dimension-six family also contains nonredundant sectors outside this pure current.

Section 147.5 / The scalar companion cannot be recovered from a small outer tilt alone

PF10 disintegrates a complete pair after its actual nonlinear chain rule, on one unchanged common realization and observation. Let $\bar{\tau} = q_*(t\mu)$, $\bar{\nu}_w = q_*(c_w\mu)$, and $\rho = q_*\mu$, retaining its absolute mass. For a fine density multiplier h, set $k_t = \mathrm{Cov}(h, t \mid q)$ and $k_w = \mathrm{Cov}(h, c_w \mid q)$. Then exactly

$$q_*\mathcal{C}_h(\mathcal{P}) = \mathcal{C}_{\bar{h}}(\bar{\mathcal{P}}) + k_t\rho - \sum_w \mathscr{D}_w(k_w\rho). \tag{R10.11}$$

All lower words and actual reference responses belong in the initial coefficient list. If h is retained-measurable the covariances vanish. Otherwise the scalar covariance is an ordinary marked scalar atom. A small $\|h-1\|_\infty$ controls it by the fine conditional oscillation, not by the already canceled coarse scalar mass.

In the actual flat Gaussian, the fine ordinary scalar companion of the genuine own-word packet satisfies

$$\|a^4 t_8 \mu_a\|_{\mathrm{TV}} \asymp a^{-2}, \qquad \|t_8\mu_a\|_{\mathrm{TV}} \asymp a^{-6}. \tag{R10.12}$$

The proof computes its variance from $\mathrm{Tr}\, H_a^4$ and compares its first and second absolute moments using the exact fourth moment of a centered Gaussian quadratic. This is compatible with the much smaller signed current after coarse conditioning.

An externally entire multiplier $h_z = 1 + z\varepsilon \tanh(T_a/\sigma_a)$ obeys $\|h_z - 1\|_\infty \le \varepsilon$ on its declared external disc but produces scalar mass at least $c\varepsilon a^{-2}$. This is a test of the small-tilt implication, not an asserted R interaction or a common internal complex chart.

Section 147.5 / The scalar companion cannot be recovered from a small outer tilt alone

The missing derivative cancellation is also explicit in the Gaussian case. Conditional on the complete joint linear observation, let m_y, C be its actual conditional mean and covariance. For a quadratic scalar with matrix T,

$$k_t(y) = 2\langle T m_y, C\,\mathbb{E}_y \nabla h\rangle + \operatorname{Tr}(CTC\,\mathbb{E}_y \nabla^2 h). \tag{R10.13}$$

This follows from two conditional Gaussian integrations by parts. The derivatives are tangent to the original compatible fiber, even when the stacked observation has dependent rows. It identifies precisely the fine multiplier derivatives whose estimates could control the scalar defect. It does not prove those estimates for an inherited nonlinear density or omit the distributional derivatives of an original sharp selector.

An alternative scalar recovery from retained current coefficients uses repeated adjoints $L_i v = (\partial_i S)v - \partial_i v$. It requires derivatives of the same effective score and coefficient on a common domain, and every intermediate boundary flux. The bare Wilson score and a real terminal coefficient bound cannot supply that information for an inherited density.

Section 147.6 / Closed reference normalization and complete S94 remain distinct

NR10 applies the original normalized reference insertion at fixed full old datum b. If all original selector alternatives are summed and every new fiber variable is integrated, then

$$\sum_{\alpha} \int K_G(dC \mid b) \int \pi_{G,\alpha}(C, \xi, b)\, d\rho_b(\xi) = \int d\rho_b(\xi). \tag{R10.14}$$

The genuine denominator in K_G cancels at that complete integral. The identity holds for each admitted preparatory choice, so its original normalized finite average also cancels exactly. An independent old scalar $e^{s(b)}$ survives unchanged. No component-count loss or reserve expenditure is needed.

This also removes a finite closed normalization subdiagram. It requires every new variable to be integrated, every selector alternative summed, and no outside carrier still reading that variable. An inherited factor instead leaves its actual kernel expectation. A selected healed branch leaves its selector. Neither is replaced by one solely through late output gauge removal.

A two-plaquette SU(N) Wilson test verifies why isolated denominators cannot be equated from common roots and Haar counts. Two rooted trees have the same root, five gauge vertices, two free links, identity reference, action and normalized Haar measure. The plaquette coordinates are (X, Y) and (X, YX^{-1}). With the same central small-link cutoff $\mathcal{B}$, let $\alpha = \mathbb{E}_{\mathcal{B}}[\operatorname{Re}\operatorname{Tr} U/N]$. Then

$$\partial_\beta \log(D_1/D_2)\big|_0 = \alpha(1-\alpha) > 0. \tag{R10.15}$$

In a weak-coupling scaling with a fixed finite rescaled cutoff ball, the identical leading Haar/count terms cancel but the logarithmic ratio tends to a strictly positive Gaussian mass ratio. The physical image of the cutoff changes under the chord-coordinate substitution. The test is not asserted to be an original preparatory pair or the original growing cutoff, and is not a counterexample to S94.

The checked primary passages still distinguish three backgrounds: the relaxed reference, the fixed-target denominator field throughout its original cutoff fiber, and the mixed numerator background with old variables retained. Regularity of one is not a verification of the defining hypotheses of the others. Complete native fixed-target records, their common analytic germs and cutoffs, inherited atoms, typed seeds, independent absolute scalars, full wrapped read sets and terminal terms remain to be matched before the completed R-formation comparison can establish S94.

Section 148.1 / Result and scope

This chapter gives a finite-record test that a proposed proof of AE8 must pass. It uses the original bare plaquette mesh, the literal retained identity rows, and the prescribed boundary values. It does not replace the actual R preparation by the native mixed minimum of AE10.

Three statements are proved below.

1. Wherever four fine links of a plaquette are fixed by the original constraints, its holonomy is fixed exactly. A current-scale curvature bound forces the corresponding *weighted target-holonomy* bound. This assertion needs no estimates on a minimizer.
2. If those constraint identities persist on the original common complex domain, the same bound forces every target-holonomy Taylor coefficient to satisfy an original-mesh Cauchy bound. At first order this is a simultaneous-wall operator norm, not a list of separate one-wall estimates. At higher order it detects a commutator response even when the linearized target curl is zero.
3. A literal cut through the original averaging/readout calculation gives an exact source-record restoration identity. It provides a sufficient *geometric certificate* for transporting an existing source bound without changing the source, provided its restored record actually lies in the old domain. Merely excluding a defect from a visible support is insufficient.

These are checked necessary tests and a restricted restoration lemma. They do not verify the missing R-I selector and range hypotheses for an actual mixed history. AE8/LM47 remains unproved. In particular, the complex-wall version below is **not** inferred from an identity known only at the physical wall vertex.

Section 148.2 / Preparation ranges and native scale labels

In the original construction the determining rows at the terminal preparation index carry their own-scale data. A bound on the size of a component provides no lower bound on the assigned scale of each row read by its operations. The space in R-I (1.64)?(1.69) and its selector (1.75) must be matched to the physical row, its wall and the support on which its supremum is taken.\n\nBalaban, CMP 119 (1988), 243?285, p. 246, p. 255 and (2.4), fixes the threshold used in (SG11.0d). The preparation conditions belong to R-I, CMP 122 (1989), 175?202. The construction of the range in R-II uses the original determining region and its preparation labels. Treating those labels as a common isotropic weak norm would require an additional comparison.

Section 148.4 / A pointwise necessary bound on fixed target holonomies

Whenever SG11.2 holds, the defining product gives, without approximation,

$$P_p(\Gamma_\omega^R(z,\xi)) = H_p(z,\xi), \qquad p \in \mathcal{P}. \tag{SG11.4}$$

Consequently

$$\left(\sum_{p\in\mathcal{P}} a^{4-2p_*}\|H_p(z,\xi)-I\|^{p_*}\right)^{1/p_*} \le \|F_a(\Gamma_\omega^R(z,\xi))\|_{L_a^{p_*}(D)}, \qquad p_* > 8. \tag{SG11.5}$$

Here the symbol p_* denotes the Lebesgue exponent, to avoid confusing it with a plaquette label. A fixed incidence constant can be inserted if the curvature convention counts orientations differently. It is independent of the number of selected plaquettes and of a.

Suppose instead that the proposed weak input is controlled only after an allowed complex gauge h, with

$$\|h(x)\|,\ \|h(x)^{-1}\| \le K \quad \text{on all needed plaquette base points.} \tag{SG11.6}$$

Since $P_p(U^h) - I = h(x_p)(P_p(U)-I)h(x_p)^{-1}$, an operator matrix norm gives

$$\left(\sum_{p\in\mathcal{P}} a^{4-2p_*}\|H_p-I\|^{p_*}\right)^{1/p_*} \le K^2\|F_a((\Gamma_\omega^R)^h)\|_{L_a^{p_*}(D)}. \tag{SG11.7}$$

Thus a proposed bound $\|F_a((\Gamma_\omega^R)^h)\|_{p_*} \le \tau$ forces the left side to be at most $K^2\tau$. No selection of a gauge can remove a violation while retaining the stated uniform bound on its adjoint action.

Section 148.4 / A pointwise necessary bound on fixed target holonomies

This proves a necessary condition on the *actual target array*. It does not require a lower bound on every assigned cell length and does not discard fine boundary rows. It is weaker than full AE8: most plaquettes need not have four fixed links, and their curvature remains to be controlled by an actual analytic/variational estimate.

In particular, take one fixed plaquette with $H_p = \exp(\delta T)$, $T \neq 0$ fixed and δ sufficiently small. The convergent exponential series and its nonzero linear term give

$$\|H_p - I\| \geq c_T|\delta|, \qquad |\delta| \leq C_T K^2 \tau\, a^{2-4/p_*}. \tag{SG11.8}$$

The last inequality is forced by admission. It is not supplied by a native ceiling $|\delta| \leq \epsilon$ independent of the displayed mesh power. AE10 supplies a native constrained-minimum realization of this mechanism. Whether an actual R history retains such a plaquette in the complete source record remains a separate question.

Section 148.9 / Mixed-entry, simultaneous-wall and source-zero hypotheses

The missing hypotheses can now be tested without identifying unrelated objects.

A. Actual physical entry. For each relevant original mixed history, identify the complete source record and the exact retained and prescribed rows read by SG11.1. Determine which fine plaquettes, if any, satisfy the endpoint identity SG11.4. Check SG11.5 on the original selector support, and prove the remaining curvature and common-gauge bounds on the unfixed portion of the record.

B. Common complex walls. Prove which original equations persist under the simultaneous R and source walls. For those equations, check SG11.13–SG11.14 with the actual radii and all wall dependencies. A proof on an open real wall box can supply the holomorphic constraint identity; a physical-vertex identity cannot. No enlarged wall domain or repeated loss of old radius is authorized.

C. Readout identity. Verify the literal cut or another exact restoration argument through each actual R determining-set join, including the prescribed boundary, junction rows and all prefixes. Then prove that the restored old record is in the old analytic domain. If no such cut exists, retain the missing rows in the actual map and estimate their response rather than deleting them.

D. Quantitative marked transport. After admission and source identification, prove the complete paired current and scalar-companion estimates with the same transition, genuine denominator, and retained normalization commutators. The preceding deterministic identities do not supply these all-history MT36 estimates.

The original ranges and selectors required to check A–C have not been recovered in full: the missing clauses are R-I's initial preparation conditions (i),(ii), ranges (1.64)–(1.69), selector (1.75), and the mixed-background construction around (1.80). R-II's separately numbered genealogy inequality (1.80) is not that mixed-background definition. No actual R selector is shown to satisfy or violate the new test. In particular, AE10's native mixed example remains an obstruction to the proposed inference from native regularity and small action alone, and is not relabeled as an R history. The exact simultaneous-wall and nonlinear tests sharpen what an actual admission proof must establish; they do not close AE8/LM47.

148.9.1

The mathematical obstruction proved in AE10 remains limited to an inference from native mixed regularity and action alone. The source-level protected reference estimate SG11.0a–c is the positive restricted case checked here.

Section 149.1 / Hypotheses and conclusions

This chapter addresses the complete current packets in all-history MT36. Its starting point is the original finite, compact Wilson measure, with the original finite-source algebra. The normalized auxiliary reference insertions are the matched insertions S58b, including their old numerator slots, original selectors, tree measures, live references, and genuine denominators. No replacement measure or additional source normalization is introduced.

There are three results here.

1. An exact finite-source elimination formula separates the Ward part of an **original physical current exponential** from its unavoidable scalar contact action. The full observable and kernel responses are included. This is a finite identity for the original nonlinear measure, not a claim that the physical source exponential already equals the Ward-completed exponential.

Section 149.1 / Hypotheses and conclusions

2. For the primitive physical Wilson currents PC10.44–50, the surviving contact action has an explicit connected-support rule and a nonlinear, real compact energy bound at every fixed source order. Its original mesh powers remain. Higher current order does not force higher field degree.
3. The second scalar contact is strictly physical: its integrated coefficient is the variance of the primitive current. For the uniform physical D6 current, a conditional one-link argument proves a lower bound of order $T^4 a^{-8}$, uniformly for sufficiently large Wilson β, on a fixed physical four-torus. This lower bound uses the exact nonlinear Wilson law, not a Gaussian substitute.

These results do **not** prove AE8/LM47, all-history MT36, or complete S94. In particular they do not bound the fine jets of an inherited effective density on an actual mixed complex domain. The last section states the remaining uses and limitations precisely.

The conventions below follow PC10. The density-divergence operator is

$$D_V u = -\operatorname{div}_\lambda(uV), \qquad \int (D_V u) F \, d\lambda = \int u \, V F \, d\lambda. \tag{WC11.1}$$

Here λ is product normalized Haar measure on the finite original link manifold. Boundary and selector derivatives in an augmented representation are weak derivatives of the **complete joint kernel**, after testing. No product of separately differentiated delta functions is used.

149.2.1 The finite-source algebra

Let

$$\mathcal{A}_m = \mathbb{R}[\eta_1, \dots, \eta_m]/(\eta_1^2, \dots, \eta_m^2), \qquad \eta_I = \prod_{i \in I} \eta_i. \tag{WC11.2}$$

The sources commute. A coefficient of η_I has no implicit factorial. Let

$$\mathcal{V} = \sum_{I \neq \varnothing} \eta_I V_I, \qquad \mathcal{D} = \sum_{I \neq \varnothing} \eta_I D_{V_I}, \qquad t = \rho^{-1} \mathcal{D} \rho, \tag{WC11.3}$$

where $\rho > 0$ is a smooth real source-zero density. Every exponential in this section is a finite polynomial in the sources. The vector fields may have nonconstant coefficients. Thus every derivative of a previously generated coefficient is included when the formulas are used.

For a nilpotent source action A, define

$$\boxed{\mathcal{B}_{\mathcal{V}}(A) = e^{\mathcal{V}} A - \int_0^1 e^{s\mathcal{V}} t \, ds.} \tag{WC11.4}$$

The integral is just integration of a finite polynomial in s.

Exact physical elimination identity. One has

$$\rho e^{A} = e^{\mathcal{D}} (\rho e^{\mathcal{B}_{\mathcal{V}}(A)}), \qquad \int \rho e^{A} F \, d\lambda = \int \rho e^{\mathcal{B}_{\mathcal{V}}(A)} e^{\mathcal{V}} F \, d\lambda. \tag{WC11.5}$$

The physical action on the left is unchanged. The right-hand side is an exact finite-source identity, including a transformed carrying function.

Proof. Conjugating the density derivative by ρ gives

$$\rho^{-1}\mathcal{D}(\rho f) = (t - \mathcal{V})f.$$

The operator $\mathcal{V}$ is a derivation. Consequently

$$e^{-\mathcal{D}}(\rho e^{A}) = (e^{-\mathcal{D}}\rho)e^{e^{\mathcal{V}}A}.$$

For completeness, if $u_s = e^{-s\mathcal{D}}\rho/\rho$, then

$$\partial_s u_s = (\mathcal{V} - t)u_s, \qquad u_s = \exp\Big[-\int_0^s e^{r\mathcal{V}}t\,dr\Big].$$

Indeed $\mathcal{V}$ of the exponent is $-(e^{s\mathcal{V}}t - t)$, so the displayed expression satisfies the differential equation and its initial condition. Setting $s = 1$ proves the first identity. Taking the adjoint of the finite exponential, using WC11.1 term by term, proves the second. □

Field-independent scalar coefficients pass through WC11.4 unchanged. In particular, an inherited absolute vacuum scalar is not discarded. If $A = t + H$, then

$$\mathcal{B}_{\mathcal{V}}(t + H) = B + e^{\mathcal{V}}H, \qquad B = \sum_{k\geq 1}\frac{k}{(k+1)!}\mathcal{V}^k t. \tag{WC11.6}$$

Thus all contacts with an additional physical source action H are present. If source words overlap, the corresponding products vanish in exactly the original finite-source algebra.

The actual Ward-completed action from PC10 is

$$A_W = \int_0^1 e^{-s\mathcal{V}}t\,ds. \tag{WC11.7}$$

It satisfies $\mathcal{B}_{\mathcal{V}}(A_W) = 0$. In general $A = t$ does not. Therefore replacing e^t by e^{A_W} removes a real contact action and changes the physical source prescription.

149.2.2 Exact coefficient bookkeeping

For primitive single-label fields,

$$\mathcal{V} = \sum_{i=1}^{m}\eta_i V_i, \qquad t = \sum_{i=1}^{m}\eta_i t_i, \qquad t_i = \rho^{-1}D_{V_i}\rho,$$

the coefficient of a set I of $r \geq 2$ distinct labels is

$$\boxed{B_I = \frac{r-1}{r!}\sum_{\pi\in\mathrm{Perm}(I)} V_{\pi_1}\cdots V_{\pi_{r-1}}t_{\pi_r}.} \tag{WC11.8}$$

In particular,

$$B_{\{1,2\}} = \frac{1}{2}(V_1 t_2 + V_2 t_1). \tag{WC11.9}$$

If the incoming fields carry non-singleton source words, the coefficient is instead the exact sum over ordered disjoint nonempty blocks:

$$B_I = \sum_{k\geq 1} \frac{k}{(k+1)!} \sum_{I_0 \sqcup I_1 \sqcup \cdots \sqcup I_k = I} V_{I_1} \cdots V_{I_k} t_{I_0}. \tag{WC11.10}$$

No block is allowed to reuse a label. Original source offsets and field-independent saved scale gains remain inside their original V_J, t_J. Each term in WC11.10 contains the product of the offsets of its actual blocks. A contact is not assigned the mesh of a later observation merely because it is subsequently read there.

Section 149.7 / What has been closed, and what has not

The exact issue raised by complete kernel normalization now has a precise answer. Summing all selector alternatives and integrating a closed native auxiliary construction removes its complete Ward responses when the carrying function is the old physical observable. It does not remove the contact action of the original physical exponential. The surviving action is WC11.6; its second scalar coefficient is WC11.26. For the uniform primitive current its absolute mass has the genuine nonlinear lower bound WC11.41.

This is consistent with the conditional covariance packet calculation Full scalar/current combinations can cancel into fine derivatives of a carrying density and derivatives of its conditional expectation. Those cancellations are stronger than bounding the scalar covariance in isolation. They still require the actual fine derivatives of the inherited carrying function, the original readout jets, and the boundary/reference responses. The present elimination identity does not bound those derivatives.

There is an explicit coefficient test for the proposed additional cancellation. Define the defect from the full Ward action by $\mathcal{R} = A - A_W$. Equations WC11.4 and WC11.7 give

$$\mathcal{B}_{\mathcal{V}}(A) = e^{\mathcal{V}}\mathcal{R}. \tag{WC11.45}$$

For two primitive labels, if the **actual** inherited action coefficient is A_{12}, its degree-two defect is

$$\mathcal{R}_{12} = A_{12} + \tfrac{1}{2}(V_1 t_2 + V_2 t_1). \tag{WC11.46}$$

Every scalar companion, kernel response already absorbed into that action, and live-reference term belongs in the actual A_{12}. Showing $\mathcal{R}_{12} = 0$ on the same augmented realization would establish this particular compensation. Showing only that one current term is centered, or that a reference kernel has unit mass, does not establish it. A nonzero defect can still require a further current projection or cancellation in specified physical pairings; neither is asserted automatically. At higher order the same test uses the exact finite ordered-block formula for A_W, with the original words and their saved gains.

The available statements and outstanding inputs are as follows.

Section 149.7 / What has been closed, and what has not

Statement	scope and exact scope
WC11.5 and WC11.8–10	Proved finite-source identities for the original smooth density and vector fields; all coefficients, offsets, and ordered contacts retained.
WC11.12–17	Proved full joint weak kernel identities on the original normalized augmented realization; actual denominators and complete selector alternatives retained.
WC11.18	Proved cancellation criterion for an exact ancestral current against its actual old physical carrying function. It does not identify every generated coefficient as such a complete ancestral packet.
WC11.22	Proved fixed-order, nonlinear real compact bound for the specified primitive Wilson currents. It retains a^{-2r}, the actual carrier, and one shared reserve.
WC11.24	Proved exact nonlinear Taylor calculation showing quadratic contacts at arbitrarily high source order.
WC11.26–27	Proved physical second-source scalar identity and absolute-mass consequence for the original current exponential.
WC11.29–44b	Proved nonlinear same-Wilson lower and matching absolute upper bounds for the uniform primitive D6 test and bounded compact physical tests with a plateau, on sufficiently large isotropic tori and for all $\beta \geq \beta_0$.
Actual mixed-history identification	Still requires the complete ancestry of every generated packet, including effective density, references, readouts, and later contacts, on the original selected realization.
Quantitative mixed-domain transport	Still requires AE8/LM47 and bounds for all fine jets entering the complete covariance/current response. The outer-only NC bounds do not establish these.
Complex-domain energy/current estimate	Not established by WC11.22 or the real compact conditional argument. The simultaneous original complex walls and genuine denominator bounds remain.
All-history MT36	Unproved. The exact cancellations above cannot be converted into the required graded transition theorem without the missing model-specific estimates and packet identification.
Complete S94	Unproved. WC11.15 supplies no comparison of two different inherited vacuum constructions with crossing read sets, distinct reference records, or unmatched absolute scalar increments.

No statement in this chapter converts the incomplete Yang–Mills existence argument into a proof. Its new quantitative content concerns the original physical contact action and a real nonlinear Wilson scalar mass. The next admissible step is to determine, for each actual generated packet, whether its source action already contains the compensating contacts required by WC11.4, and then to bound the remaining complete fine-derivative response on the same selected mixed domain.

Section 150 / Nonlinear cancellation of the complete conditional

PF10.4 isolated a scalar conditional covariance together with all its current partners. Its direct total-variation estimate can be much larger than the already conditioned current. This chapter gives a different exact representation of that *complete packet*. It replaces the fine current scalar by derivatives of the actual density multiplier and of its conditional mean. It applies to the nonlinear Wilson current as well as to a Gaussian reference. It does not estimate those derivatives on the original mixed-history support.

The distinction between a represented distribution and an ordinary scalar measure remains essential. The following cancellation bounds the former by a new explicit representation; it does not identify the variation of the original scalar covariance with that smaller bound. Every subsequent operation must carry the displayed scalar and current coefficients together, including the boundary term when present.

Section 151.1 / Scope and the new quantitative conclusion

The complete source-zero comparison S94 is still unproved. This chapter does not identify independent native histories, does not supply the missing simultaneous complex domain, and does not estimate an independent absolute vacuum increment by a probability normalization.

There is a positive quantitative result beyond NR10's closed-normalization identity and its two-tree test. A genuine **admitted real reference primitive** has a finite-dimensional coercive fluctuation integral after its original classical and coordinate factors are extracted. This gives Gaussian tails, including polynomial and small exponential energy envelopes. When two choices have been verified to be coordinate forests of the **same complete physical fixed-target record**, their original cutoff measures can be transported to that common record. They agree on their common physical cutoff core. The genuine denominator ratio and the normalized carrier comparison are then controlled by the two core-exit probabilities. The comparison does not divide by the small unextracted denominator.

The hypotheses in the preceding paragraph are substantive. In particular, an original preparatory label can alter more than its forest. The result does not say that its relaxed minimizer, fixed-target map, mixed numerator background, spatial action cutoff, or inherited readout is unchanged. The original R-I clauses needed to check all these facts on actual mixed histories remain unrecovered here.

The tail argument is a real argument. Section 7 gives a further quantitative extension on a genuinely common analytic chart: a uniform complex bound and a uniformly small real difference imply a small difference on a strict complex subchart. Applied to the two original holomorphic denominator expressions, this keeps their original integration cycles. The common chart, its actual radius and uniform real-slice matching are hypotheses; analyticity alone does not establish them on the simultaneous mixed-history walls.

The original cutoff law is now verified from the primary source chain: $\varepsilon_k = g_k p_0(g_k)$ and $R_{\chi,k} = M_0 p_0(g_k)$. Thus the admitted native primitive has a growing rescaled cutoff and a shrinking group radius at small running coupling. This closes the source-level radius input; it does not close mixed-history admission, complete-record matching or physical torus-distance decay.

Section 151.2 / Native denominator and the factors that remain outside its probability law

Fix the complete old datum b, the actual fixed-target constraint record, its allowed vertices and pins, and one original relative forest G. Every retained variable read by the reference is included in b, even if it will be integrated later. The genuine denominator is

$$D_G(b) = \int dC\, \delta_G(C)\chi_G(C) \exp\{-g_k^{-2} A(\zeta_0, U_{k,Z}(CV_\Lambda; b))\}. \tag{RV11.1}$$

The inner field in RV11.1 has its targets fixed at CV_Λ. It is not the relaxed field U_0 and is not the mixed numerator field. The unweighted variational problem defines the minimizers; the spatial weight ζ_0 is used in the evaluated action.

Let q_G be the actual number of free group coordinates in this relative forest, $m = \dim G_{\text{gauge}}$, and $n_G = mq_G$. In the original logarithmic fluctuation variables $C = \exp(ig_k B)$, write the native extraction as

$$\begin{aligned} D_G(b) &= P_G(b)\widehat{D}_G(b), \\ P_G(b) &= \exp\{-g_k^{-2} A(\zeta_0, U_{0,G}; b) + c_{k,G}^{\text{coord}}\}, \\ \widehat{D}_G(b) &= \int_{\mathcal{B}_G(R_{\chi,k})} f_G(B; b)\, dB. \end{aligned} \tag{RV11.2}$$

The coordinate term contains the actual $n_G \log g_k$, Haar normalization terms, and any other scalar explicitly present in the native extraction. It is retained exactly. No unspecified power of the fine mesh is inserted. The logarithmic remainder is defined here by $I_G := \log \widehat{D}_G$, in agreement with RN9.8's explicitly defined positive additive remainder. This uses RN9's convention rather than inferring a sign for the printed symbol.

The primary input is Balaban, *Large field renormalization. II*, printed pp. 356–359, equations (1.2)–(1.11): the reference component is rectangular and lies in a cube of side $100M$; its axial quadratic form has a positive lower bound; and the extracted integral has a lower bound of the form $e^{-C(M)}$. The native linear remainder is bounded using stationarity and separation, and the higher remainder is small. Equation (1.2) gives the original logarithmic cutoff $R_{\chi,k} = M_0 g_k^{-1}\varepsilon_k$. See the primary paper.

Its radius law is fixed in *Convergent renormalization expansions*, CMP **119** (1988): printed p.246, after (1.1), adopts the first cutoff choice, and printed p.255, equation (2.4), gives its indexed version. With fixed positive A_0, M_0, p_0, it is

$$p_0(g) = A_0(\log g^{-2})^{p_0}, \qquad \varepsilon_j = g_j p_0(g_j), \qquad R_{\chi,k} = M_0 p_0(g_k). \tag{RV11.2a}$$

The exponent p_0 is a positive integer in the construction (printed p.244). R-II continues this same sequence: its printed p.362, equation (1.23), uses the $\varepsilon_h/\varepsilon_k$ comparison from III (2.8), and its theorem and concluding statement retain III's Section 2 density class. These are verified source definitions, not an exponent inferred from a Taylor remainder bound. See III, pp.244, 246, 255–256 and R-II, pp.355, 362, 391. Consequently

$$R_{\chi,k} \longrightarrow \infty, \qquad g_k R_{\chi,k} = M_0\varepsilon_k \longrightarrow 0 \qquad (g_k \downarrow 0). \tag{RV11.2b}$$

Section 151.2 / Native denominator and the factors that remain outside its probability law

All constants in this statement, including the block parameter M and the resulting M_0, are held fixed. The exact formulas below retain $R_{\chi,k}$; RV11.2a supplies its native asymptotics.

The lower bound applies to $\widehat{D}_G$, not to D_G. The classical factor and $g_k^{n_G}$ may make D_G arbitrarily small. They cancel inside the probability kernel only because the same factors occur in its genuine numerator and denominator:

$$K_G(dB \mid b) = \frac{1_{\mathcal{B}_G(R_{\chi,k})} f_G(B;b)\, dB}{\widehat{D}_G(b)}. \tag{RV11.3}$$

An independent old scalar $e^{s_G(b)}$ is not part of this cancellation. Neither the estimate for K_G nor the identity $K_G 1 = 1$ estimates s_G.

Section 151.5 / Genuine denominator and carrier comparison on the common physical core

Let Ω_i be the two original cutoff domains in the common physical quotient, and let μ be their common unnormalized physical Wilson measure. This notation is allowed only after the complete-record equality in Section 4 has been verified. Put

$$D_i = \mu(\Omega_i), \quad K_i = 1_{\Omega_i}\mu/D_i, \quad C = \Omega_0 \cap \Omega_1, \quad \tau_i = K_i(C^c). \tag{RV11.15}$$

The intersection is a proof device. It is not substituted for either native cutoff, neither genuine denominator is replaced by its mass, and its failure is not assigned a new original large-field record.

By Section 3 and RV11.14,

$$\tau_i \le C_M \exp\{-c_M (R_{\chi,k}/\kappa_G)^2\}. \tag{RV11.16}$$

For the verified native law RV11.2a, and fixed M and allowed forests, the geometric factor M_0/κ_G can be retained explicitly or included in the constants. Thus

$$\boxed{\tau_i \le C_M \exp\{-c_M p_0 (g_k)^2\}.} \tag{RV11.16a}$$

The weighted carrier tails have the same rate with their explicit degree-dependent constant $C_{M,p}$ and absolute carrier envelope. A constant rescaled radius would leave a constant error, as in NR10; the original growing radius supplies the small-coupling improvement.

For any fixed $p_0 > 1/2$, this improvement is faster than every fixed power of g_k: for every $s > 0$,

$$C_M e^{-c_M p_0 (g_k)^2} = o(g_k^s) \qquad (g_k \downarrow 0). \tag{RV11.16b}$$

Indeed, with $t = \log g_k^{-1}$, the ratio to g_k^s is $C_M \exp\{-c_M A_0^2 (2t)^{2p_0} + st\}$, which tends to zero because $2p_0 > 1$. The source construction's positive integer p_0 satisfies this threshold. This is a rate in the running coupling. No positive power of an original contact mesh, no bound on a running-coupling trajectory, and no physical torus-distance factor are deduced from it. If the coupling at a fixed physical scale is held fixed, this estimate alone gives no new limit with torus size.

The common-core mass is $D_C = D_i(1 - \tau_i)$. Therefore

$$\log(D_1/D_0) = \log(1 - \tau_0) - \log(1 - \tau_1), \qquad |\log(D_1/D_0)| \le 2(\tau_0 + \tau_1) \tag{RV11.17}$$

when $\tau_i \le 1/2$. In particular, this compares the **genuine raw denominators**, despite their possibly tiny size. Equality of unnormalized physical measures on the core was used before taking absolute values. A lower bound for an unrelated normalization could not prove RV11.17.

Let $K_C = 1_C\mu/D_C$ for the following identity only. Writing the two measures as their core and outer parts gives, in the total variation norm without the factor $1/2$,

$$\|K_1 - K_0\|_{\mathrm{TV}} \le |\tau_1 - \tau_0| + \tau_1 + \tau_0 = 2\max(\tau_0, \tau_1). \tag{RV11.18}$$

For complete transported carrying functions F_i, the exact identity is

$$\begin{aligned} K_1F_1 - K_0F_0 = (1-\tau_1)K_C(F_1 - F_0) + (\tau_0 - \tau_1)K_CF_0 \\ + K_1(1_{C^c}F_1) - K_0(1_{C^c}F_0). \end{aligned} \tag{RV11.19}$$

Every function in this formula is the actual function after complete record transport. If $F_1 = F_0$ on the core and both have the envelope RV11.9 with the same old-data constant $a(b)$, subtract that constant first using $K_i1 = 1$. The weighted tail lemma then gives

$$|K_1F_1 - K_0F_0| \le C_{M,p}A(b)\exp\{-c_M(R_{\chi,k}/\kappa_G)^2\}. \tag{RV11.20}$$

If the complete carriers differ on the core, the first term of RV11.19 remains. It contains the actual inherited readout difference, including any reference, observable or conditional-density response. An inherited factor that reads the prepared field cannot be pulled outside K_i merely because an unrelated normalization closes.

The same algebra handles soft original cutoffs by using the common measure $\chi_0\chi_1\mu$ and $\tau_i = K_i(1 - \chi_{1-i})$, provided both original cutoffs equal one on the declared inner cores. No such plateau is assumed for a different selector without checking its original definition.

151.5.1 Scalars remain literal

The extraction RV11.2 implies exactly

$$\Delta \log D = \Delta \log P + \Delta I. \tag{RV11.21}$$

Thus RV11.17 controls this complete combination. It controls ΔI alone only when equality of the actual classical and coordinate prefactors has also been verified. If the two native choices alter their relaxed variational records, those prefactors cannot be deleted from RV11.21.

For separately retained vacuum factors,

$$\Delta \log(e^s D) = \Delta s + \Delta \log D. \tag{RV11.22}$$

There is no estimate for Δs in RV11.17. For a normalized transition $e^{s_i}K_iF_i$, the two terms $e^{s_1}(K_1F_1 - K_0F_0)$ and $(e^{s_1} - e^{s_0})K_0F_0$ must both be bounded. This retains the independent absolute scalar masses required by S94.

151.5.2 Selected labels, components and later integrations

An individual selected label has its actual factor π_i. Unless its transported value agrees on the core, RV11.19 contains $K_C(\pi_1 F_1 - \pi_0 F_0)$. Removing a gauge delta on a healed component does not remove this term. For complete selector partitions, the exact common refinement of VM9 may be used before absolute values; its weights sum to one and do not create a count of pairs. This algebra does not identify different original nonlinear constraint fibers.

For disjoint original reference components, RV11.11 and the tail bound apply with their separate actual dimensions. For dependent successive insertions, a telescoping comparison is permissible only after each entire retained state has been transported, and the estimates are uniform conditional on that state. Its error contains the sum of the actual insertion errors. The number of insertions or components is retained. It must be related to the existing component ledger before the sum is charged to VC28's single reserve; no new charge is created by this tail argument.

Section 151.8 / What this resolves, and what S94 still requires

The mathematical additions are the weighted tail lemma RV11.5, the relative-forest core estimate RV11.13–14, the complete raw-denominator and carrier comparison RV11.17–20, and the exact-Wilson growing-cutoff result RV11.24. The source-level native cutoff input is now verified, giving the rate RV11.16a–b on admitted, matched real primitives. The dimension-independent propagation lemma RV11.27 and the original complex logarithm bounds extend that rate to RV11.32 on a verified common native chart, with the literal Cauchy margins in RV11.33. Their proofs preserve the real native probability law, the actual free-coordinate counts, the independent scalar masses and the carried functions. Where their original-domain hypotheses hold, they supply positive estimates instead of treating a native cutoff denominator as forest independent.

They leave the following obligations open.

1. **Actual mixed support.** Identify every original selector-supported fixed-target reference map and its entire native cutoff with an admitted variational branch. This includes the old readouts

and the distinction between fixed-target, relaxed and mixed fields. RV11.4 on an admitted primitive is not this identification.

2. **Record matching.** Show that the actual records being compared are related by the complete physical transport used in Section 4, or estimate their core difference in RV11.19. Independent native histories cannot be declared identical by matching root counts.
3. **Inherited envelope and accumulation.** Prove the envelopes for complete inherited densities, including their absolute scalar factors, on the same original cutoff support. The radius law itself is now verified by RV11.2a. Any sums over insertions must use the original component and birth/age ledger, with the explicit degree dependence and all saved scale gains retained.
4. **Common complex localization domain.** Construct and identify the original common analytic charts and their simultaneous parameter radii for the actual mixed preparations, with uniform real-slice matching and bounds for complete inherited objects. RV11.32 then supplies the forest-cutoff comparison on a strict subchart. It does not supply the original marked admission, or the required boundary and constraint-current representations on that domain.
5. **Physical separation and wrapped records.** Produce the required decay with physical torus separation, including every original constraint path and inherited readset that wraps. A factor small in coupling but independent of torus size does not prove S94.
6. **Absolute scalar comparison.** Estimate the independently retained increments Δs and retain the complete extraction in RV11.21–22. A probability normalization cancels only its own scalar factor.

Accordingly the complete source-zero construction has not been matched between physical tori. The new real primitive comparison is available for a verified matching row, and its hypotheses and errors are explicit. It is not a replacement for the remaining S94 rows.

Section 152.1 / The native threshold was recovered from a primary definition

The original parameter law was checked visually in *Convergent renormalization expansions for lattice gauge theories*, printed p.246 and p.255, equation (2.4):

$$\varepsilon_j = g_j p_0(g_j), \qquad p_0(g) = A_0(\log g^{-2})^{p_0}. \tag{AU11.1}$$

The same paper's (2.8) gives the scale comparison for these numbers. R-II continues the density class of that paper's Section 2 and uses that parameter comparison on printed p.362, (1.23). The logarithmic cutoff in R-II (1.2), printed pp.356–357, is $M_0 g_k^{-1}\varepsilon_k$. These sources therefore identify its rescaled radius as $M_0 p_0(g_k)$. The physical group radius tends to zero, while the radius in fluctuation coordinates tends to infinity.

The argument directly verifies the source chain needed for the native reference-tail comparison. Earlier parameter discussions already recorded the threshold law; the unverified input in NR10 was its literal

Section 152.1 / The native threshold was recovered from a primary definition

use in this cutoff comparison. It was not inferred from the smallness of a cubic remainder. The exact R-I preparation and selector clauses remain unavailable; recovering a parameter law does not recover those clauses.

The new tail estimate is small as the *current coupling* tends to zero. It is not automatically small in the original source mesh at a fixed current coupling, and it has no physical torus-distance factor. Any conversion to a history grade would require the actual running-coupling and record-count relations used in that grade.

Section 152.2 / Restricted actual-entry information retains its volume and gauge

The source's representation of its relaxed field is $U_0 = e^{i\eta A_0}$, with both potential and first fine differences bounded by $CM^5R_k\varepsilon_k$ on its stated domain. The spacing in SG11 is the same $a = \eta$. The passage to an L^p norm introduces the actual physical volume factor $v_a(D^+)^{1/p}$. That factor is retained; a large determining record is not silently assigned fixed volume.

The plaquette bound follows by expanding the four link factors: the linear part is a finite difference of the potential, and the noncommutative remainder is quadratic. This yields the stated curvature bound on the same complete stencils. SG11.0c then uses the already fixed cell-rescaling identity. No minimum assigned cell size is assumed for that inference.

Two qualifications were made explicit First, the good gauge in the source need not be the preselected anchored gauge of an old marked chart. Every target, boundary value and endpoint frame must be transformed consistently before using that chart. Second, the field is the relaxed reference U_0, not the three-join mixed preparation Γ^R. The source's bound on the former does not identify it with the latter.

The simultaneous-wall tests keep a separate endpoint and common-domain hypothesis. A retained constraint known at one physical vertex cannot be continued by an identity theorem from that single point. The all-wall statement needs the literal equation throughout a real open parameter box, or a separate proof of its preservation. The nonlinear commutator test correctly detects a second derivative even when the first target curl vanishes.

Section 152.3 / Physical exponential and Ward exponential remain different

WC11's elimination formula was checked by conjugating the density derivative:

$$\rho^{-1}\mathcal{D}(\rho f) = (t - \mathcal{V})f. \tag{AU11.2}$$

Solving the finite source-polynomial evolution gives $e^{-\mathcal{D}}\rho = \rho \exp[-\int_0^1 e^{s\mathcal{V}} t\, ds]$. The scalar residual, the sign of the carrying-function transformation, and the factor $(r-1)/r!$ in the distinct-label contact coefficient follow from this identity. No commutation of fine vector fields is used.

The complete auxiliary kernel must transform with the test. Its normalization cancels its responses only after the full selector sum and full auxiliary integration. The same statement does not hold for a selected history or for a carrier that still reads a new fiber. The explicit quotient $(e^{\mathcal{V}}N)/(e^{\mathcal{V}}D)$ contains the genuine old denominator and its derivatives.

For an actual degree-two source coefficient, the residual test is

$$\mathcal{R}_{12} = A_{12} + \tfrac{1}{2}(V_1 t_2 + V_2 t_1). \tag{AU11.3}$$

Section 152.3 / Physical exponential and Ward exponential remain different

Checking that expression requires the original augmented source coefficient A_{12}, including its existing scalar companions. Choosing a different Ward-completed source would set the residual to zero by changing the prescription. WC11 does not do so.

The real compact all-order bound uses an ideal generated by products of two elementary Wilson forces. Each new Ward derivative supplies a force while differentiating the preceding product. The weighted Casimir identity ensures that the first derivative of the scalar companion also enters this ideal. Bounded incidence gives constants depending on the fixed source order, with no hidden volume factor. The resulting a^{-2r} is retained. It is not a positive-grade transition estimate, nor is it asserted on a complex mixed support.

Section 152.4 / Independent check of the exact-Wilson scalar estimate

The key scalar identity is an integration by parts on the original compact link manifold:

$$\mathbb{E}(Vt) = \mathbb{E}(t^2), \qquad \mathbb{E}t = 0. \tag{AU11.4}$$

It gives the genuine degree-two logarithmic source coefficient. The pointwise contact Vt can have either sign, so the distinction between its integrated coefficient and its absolute variation remains.

The nonlinear lower bound was checked independently through each of its steps. Restricting every link to a ball of radius $c\beta^{-1/2}$ gives a partition lower bound uniform in lattice size. Monotonicity of the mean action and torus symmetry give a small mean plaquette action. A fixed finite patch then has small plaquettes with probability at least one half when β is sufficiently large.

A packing at link-graph distance at least five makes the selected link variables conditionally independent and makes the force-square observable additive in those variables. The good-bath event depends only on the conditioned links: it asserts existence of a vertex gauge making those links close to the identity. A small full plaquette patch implies that bath event, even though the event used to prove its probability includes some unconditioned links. This does not invalidate conditional independence or measurability.

On a flat bath the force-square coefficient is $54 = 6^2 + 18$; the eighteen other links in the six incident plaquettes are distinct when the local patch does not wrap. The scalar-companion contribution is $+24C_F\beta^{-1}s(U)$. Its sign follows from $\Delta S_0 = 4C_F(N\#p - S_0)$. The variable part has a uniformly positive Hessian in a fixed small chart, also for sufficiently close baths.

Compactness, the positive Hessian of the conditional one-link action, and its gap outside that chart give a normalized rescaled density bounded below on a fixed Euclidean ball. A compactly supported Laplacian test bounds the variance there. It does not drop a possibly large rescaled linear term. Summing the conditional variances gives the uniform lower bound for the original Wilson measure.

The upper bound uses bounded local Haar Hessian incidence and $\beta\mathbb{E}|\nabla S_0|^2 = \mathbb{E}\Delta S_0$. Thus the scalar size is established nonlinearly, without appealing to a Gaussian limit. For an unnormalized Wilson body the whole absolute variation still carries its original total mass; a separately inherited scalar also remains. The normalized source-logarithm coefficient is the variance in AU11.4.

The lower-bound proof is not an assertion that each selected mixed history has a good bath with the same conditional probability. Nor does it assert that an arbitrary generated current is independently exponentiated without correlated companions. Those would be different statements about the actual marked construction.

Section 152.5 / The nonlinear conditional covariance cancellation

SC11's first- and second-order identities were checked by testing the full fine product rule and subtracting its retained counterpart. With $D = -\partial$, the second-order packet has the sign $+D_\alpha(R_1^\alpha \rho)$. The ordered-word version has order at most $r - 1$, because every product remainder differentiates the multiplier at least once. Its zero-order coefficient is $\mathbb{E}(Lh \mid q) - \bar{L}\bar{h}$; it is not erased.

The boundary correction is exactly $q_*(h\mathcal{B}) - \bar{h} q_* \mathcal{B}$. No product of independently exposed sharp-boundary currents is needed. The conditional-mean derivative formulas contain both genuine denominator derivatives and the third centered cumulant at second order.

The argument gives three explicit restrictions. The conditional mean must have the regularity needed for the product tests; smooth fine data alone do not prove that. The represented norm must allow the displayed target words with their actual physical coefficients. The NH9 operator is identified with its stated scalar companion against the *bare* Wilson body; an inherited body has the full changed effective score. All three restrictions are now present in SC11.

The resulting formulas give a useful cancellation of the complete packet. They do not bound the ordinary scalar covariance in isolation by the smaller represented norm. Subsequent transport must retain the new scalar and lower-current coefficients together.

Section 152.6 / Reference tails, native records and common complex charts

RV11's weighted tail proof divides only by the extracted coercive integral. The classical factor, coordinate powers, and independent absolute scalar are retained outside it. Its constants explicitly depend on the polynomial envelope degree and the fixed block size. It does not give a degree-uniform estimate for arbitrarily many contacts.

The relative forest recursion is valid with a nonflat compact reference target. Unitarity controls the distance of its gauge factors to one. The dimension count includes the actual pins. Equality of quotient Haar measures is used only after the complete nonlinear fixed-target action and record have been matched. A different preparatory choice can change more than a forest.

On that matched physical record, the common-core mass proves the raw denominator identity exactly. It controls $\Delta \log P + \Delta I$; it does not estimate either summand by itself without further information. A separate vacuum increment Δs remains. The core carrier difference is explicit and is not replaced by zero when an inherited field or selector still differs there.

The real-to-complex propagation argument, where used, requires a common complex chart, a bound on the entire holomorphic function on that chart, and a uniformly small real-slice bound. Mere analyticity is insufficient. Its one-variable upper-half-disc proof has a dimension-independent exponent. A saved prefactor is preserved if it appears in both input bounds; one present only in the real bound loses its exponent under interpolation. Cauchy estimates then retain the literal remaining chart radii. None of this constructs the missing common mixed-history domain.

Section 152.7 / Scope conclusion

The scope proves new exact nonlinear cancellations, a sharp physical scalar estimate for the original Wilson law, and additional admitted reference comparisons. It also recovers a concrete native cutoff parameter. AE8/LM47 for the actual prepared mixed histories, the uniform complete MT36 transition estimates, and complete S94 still lack model-specific hypotheses and estimates. The feedback and continuum conclusions are not instantiated by these auxiliary results.

Section 153.1 / What actual entry has gained

SG11 identifies a restricted positive case: the original relaxed reference minimum U_0, with its original exterior targets. In the source's gauge and normalization,

$$U_0 = e^{i\eta A_0}, \qquad \|A_0\|_\infty + \|\nabla_\eta A_0\|_\infty \le CM^5 R_k \varepsilon_k. \tag{R11.1}$$

On a complete record $D^+ \subset Z$, including all required stencils and its original boundary record, this yields

$$\begin{aligned} \|A_0\|_{W_a^{1,p}(D^+)} &\le CM^5 R_k \varepsilon_k \, v_a(D^+)^{1/p}, \\ \sum_{\Delta \subset D} \sigma_\Delta^{4-2p} c_\Delta(U_0)^p &\le (CM^5 R_k \varepsilon_k)^p v_a(D), \qquad a = \eta. \end{aligned} \tag{R11.2}$$

The volume factors and the fixed block parameter are retained. The second statement is a real gauge-invariant curvature-moment bound for this same field. A good representative is not automatically the fixed anchored chart of an old marked coefficient: all constraint rows, boundary values and endpoint frames must be transformed together.

The source chain identifying the threshold in the native reference cutoff has now been checked directly:

$$\varepsilon_k = g_k p_0(g_k), \qquad p_0(g) = A_0(\log g^{-2})^{p_0}. \tag{R11.3}$$

It is defined in III, printed pp.246,255, (2.4), and continued in R-II. Combined with $R_k \le L(\log g_k^{-2})^{r_0}$, it makes R11.2 small at small current coupling on records with a verified polynomial bound on their actual volume. It gives no extra power of the microscopic source mesh.

The original three-join prepared field $\Gamma_\omega^R(z,\xi)$ has not been identified with U_0. The source's extension and range hypotheses for that field are still needed. The exact R-I preparation conditions (i),(ii), ranges (1.64)–(1.69), selector (1.75), and mixed background around (1.80) were not recovered as inspectable full text. This is a limitation of the quoted source hypotheses, not a theorem that its construction fails.

SG11 also gives exact necessary tests on any actual retained identity rows. If the four links of a plaquette are pinned, its prepared holonomy is the original target holonomy. A proposed common complex curvature ceiling must therefore control its simultaneous weighted target derivatives, with the

original $a^{4/p-2}$ weight. These tests require all-wall constraint preservation before they can be used off the physical endpoint. A second-order commutator can remain when the first target curl vanishes. Neither separate one-wall bounds nor native small action alone proves AE8.

Section 153.2 / Exact physical current elimination retains a contact action

For a finite nilpotent source vector field $\mathcal{V}$, let $\mathcal{D}$ be its Haar density adjoint and $t = \rho^{-1}\mathcal{D}\rho$. WC11 proves, for the original physical source action A,

$$\begin{aligned} \mathcal{B}_{\mathcal{V}}(A) &= e^{\mathcal{V}}A - \int_0^1 e^{s\mathcal{V}} t \, ds, \\ \int \rho e^{A} F &= \int \rho e^{\mathcal{B}_{\mathcal{V}}(A)} e^{\mathcal{V}} F. \end{aligned} \tag{R11.4}$$

The vector fields and their coefficient derivatives retain their actual order. All expressions are finite source polynomials. For the original current exponential $A = t$, the residual is

$$B = \sum_{r \geq 1} \frac{r}{(r+1)!} \mathcal{V}^r t. \tag{R11.5}$$

For m distinct primitive labels its coefficient is the exact ordered sum with factor $(m-1)/m!$. The more general disjoint-block formula keeps source-word partitions, original offsets and saved incoming prefactors.

The full auxiliary kernel transforms with the observable. Its response includes selectors, boundary and constraint distributions, coarea factors, live references, effective scores and the genuine denominator. Only after every selector alternative is summed and the entire new fiber is integrated does its normalization cancel these responses against an old physical carrying function. A selected branch or an inherited output-dependent carrier leaves an explicit residual.

Even in that closed case, the physical action R11.5 remains. The Ward-completed exponential has a different contact prescription. For two primitive labels the actual compensation test is

$$\mathcal{R}_{12} = A_{12} + \tfrac{1}{2}(V_1 t_2 + V_2 t_1). \tag{R11.6}$$

It must be evaluated using the actual inherited source coefficient A_{12} on the original augmented realization. Its scalar companion cannot be discarded, and the result is not set to zero by declaring an auxiliary kernel normalized.

For the specified primitive Wilson currents, the residual contacts have connected microscopic anchor support. On the original real compact fields, WC11 proves at every fixed order $r \geq 2$,

$$|B_I| \leq C_{r,N} c_\beta^{-r} (1 + \beta^{-1}) a^{-2r} \left(\prod_{i \in I} \|f_i\|_\infty \right) E_{Y_I}. \tag{R11.7}$$

The bound uses the one original Wilson energy. It retains the physical inverse powers of a. Its quadratic Taylor part can be nonzero at every source order, so increasing current order alone does not give a higher field degree. Field-dependent inherited coefficients produce additional jets outside this primitive estimate.

Section 153.3 / The physical scalar mass is now controlled in the exact nonlinear measure

For a real primitive current $t_V = \rho^{-1} D_V \rho$, compact integration by parts gives

$$\mathbb{E}_\mu t_V = 0, \qquad \mathbb{E}_\mu(V t_V) = \mathbb{E}_\mu(t_V^2). \tag{R11.8}$$

For two distinct labels carrying that same primitive, the second coefficient of the original logarithmic source expectation is $\mathrm{Var}_\mu(t_V)$, and its physical contact is $B_{12} = V t_V$. The integrated coefficient is nonnegative even though the pointwise contact need not be.

For the uniform physical D6 current in the original four-dimensional $SU(N)$ Wilson probability measure on a torus of side T, WC11 proves the sharp nonlinear estimate

$$\boxed{c T^4 a^{-8} \le \mathbb{E}_\mu B_{12} = \mathrm{Var}_\mu(t_6) \le \int |B_{12}|\, d\mu \le C T^4 a^{-8}.} \tag{R11.9}$$

It holds uniformly for sufficiently large β, with $\beta = c_\beta g^{-2}$, and for tori large enough in lattice units to contain the fixed local patches. This is not inferred from a Gaussian approximation.

The lower bound conditions on the complement of a positive-density set of separated links. Their conditional laws factor, and the observable is additive in them. The elementary Wilson energy bound gives a uniform probability of a good finite bath. On that bath the exact one-link observable has a positive Hessian and the rescaled conditional density has a uniform lower bound. A compactly supported Laplacian test bounds the variance even when the rescaled linear term is large. The upper bound uses bounded Haar Hessian incidence and exact Wilson integration by parts.

The calculation also applies locally: a fixed physical real test equal to one on a positive-volume box has the same lower order. For a bounded deterministic test f, the absolute contact has the upper bound

$$\int |V_f t_f|\, d\mu \le C a^{-4} \|f\|_\infty^2 \# \operatorname{supp} f \qquad (\beta \ge \beta_0). \tag{R11.10}$$

Together these give order a^{-8} for a fixed compact physical test with such a plateau, with constants retaining its support and plateau volumes.

R11.9–10 use the stated Wilson probability measure. Against the raw body $e^{-\beta S_0} d\lambda$, the absolute variation carries the genuine factor $Z(\beta)$. An independent inherited scalar e^s multiplies it by $|e^s|$. No lower bound on that raw mass or inverse conditional normalization is inferred.

This resolves the scalar size for a physical primitive test. It does not prove the required small *renormalized* contact bound. Such a bound needs the actual correlated coefficients in R11.6 or a proved further cancellation. R11.9 also does not assert that every generated current is independently exponentiated in the construction.

Section 153.4 / Nonlinear cancellation of the complete scalar covariance packet

SC11 goes beyond the Gaussian cancellation in PF10. Let μ be a positive finite measure, let a complete fine current satisfy $t\mu = L^*\mu$, and let q be the actual observation, with all varied live-reference coordinates. Put $\rho = q_*\mu$; this measure retains its absolute scalar mass. For a first-order operator $L = V$, write $v = Dq\,V$, use bars for conditional means, and set $k_t = \mathrm{Cov}(h, t \mid q)$, $k_v = \mathrm{Cov}(h, v \mid q)$. Then

$$k_t\rho - D(k_v\rho) = \big[\mathbb{E}(Vh \mid q) - \bar{v} \cdot \nabla\bar{h}\big]\rho. \tag{R11.11}$$

Section 153.4 / Nonlinear cancellation of the complete scalar covariance packet

For a second-order fine operator, the same tested product rule gives

$$k_t\rho - D_\alpha D_\beta(k_a^{\alpha\beta}\rho) - D_\alpha(k_b^\alpha\rho) = R_0\rho + D_\alpha(R_1^\alpha\rho), \tag{R11.12}$$

where

$$R_0 = \mathbb{E}(Lh \mid q) - \bar{L}\bar{h}, \qquad R_1 = \mathbb{E}(c_h \mid q) - \bar{c}_{\bar{h}}. \tag{R11.13}$$

The lower current c_h contains both cross terms of the actual ordered product rule. At every finite order there is an analogous reduction to words of order at most one less, with the same scalar remainder. The entire packet is unchanged by subtracting a function of the complete retained observation from h.

This is a useful nonlinear cancellation before taking variation. It does not equate the ordinary scalar covariance mass with the smaller represented-distribution norm. All the new scalar and current coefficients must remain together in later transport. The conditional mean also needs its actual derivative regularity; smooth fine data alone do not establish that regularity.

The derivative of $\bar{h} = N/Z$ retains both numerator and denominator derivatives. On a legitimate differentiable conditional chart its second derivative contains the full conditional score, score derivatives, and the third centered cumulant. The bare Wilson score cannot replace an inherited effective score. If the selected identity contains a boundary distribution $\mathcal{B}$, the extra response is exactly

$$q_*(h\mathcal{B}) - \bar{h}\, q_*\mathcal{B}. \tag{R11.14}$$

For the actual Wilson Hessian current, the fine scalar response is SC11.22: it contains contracted first and second derivatives of the actual fine multiplier, together with its quadratic derivative term. Removing the NH9 word's recorded a^4 multiplies the entire operator by a^{-4}. These identities supply no absent mesh factor. NC's outer-parameter comparison still does not estimate the required internal derivatives on arbitrary old mixed readouts.

Section 153.5 / What the admitted source-zero reference comparison now proves

The native reference radius is now

$$R_{\chi,k} = M_0 p_0(g_k). \tag{R11.15}$$

For the actual admitted real primitive, R-II's extracted coercive integral gives a weighted tail estimate for its original logarithmic fluctuation variable B:

$$K_G\Big[1_{\{\|B\|_\infty > r\}}(1+|B|)^p e^{\theta|B|^2}\Big] \le C_{M,p} e^{-c_M r^2}. \tag{R11.16}$$

The allowable θ is fixed by the actual coercivity. Polynomial-degree dependence and any absolute scalar in a carrier's envelope remain. The proof divides by the extracted integral only; the raw denominator's classical and coordinate factors remain exact.

For two allowed forests of the same complete physical fixed-target record and pins, the relative gauge recursion controls the transformed link distance by $(2\ell+1)$ times its original distance, where ℓ is the actual maximal root-path length. Thus their original cutoffs have a common physical core. After the entire nonlinear record and evaluated action have been matched, let τ_i be the probabilities of leaving that core. The genuine denominators satisfy

$$\begin{aligned} \log(D_1/D_0) &= \log(1-\tau_0) - \log(1-\tau_1), \\ \|K_1 - K_0\|_{\mathrm{TV}} &\le 2\max(\tau_0, \tau_1), \\ \tau_i &\le C_M \exp[-c_M p_0(g_k)^2]. \end{aligned} \tag{R11.17}$$

Section 153.5 / What the admitted source-zero reference comparison now proves

The original positive integer p_0 makes this smaller than every fixed power of g_k as $g_k \to 0$. This is not a fine-mesh or torus-distance estimate at fixed current coupling.

Complete carriers still have their core difference and their tail integrals. A separate vacuum increment remains in $\Delta \log(e^s D) = \Delta s + \Delta \log D$. The actual free coordinate count is $\dim G(|E| - |V| + |P|)$; no root, pin, component or insertion count is dropped. The common core is a comparison device, not a replacement denominator or a new charged failure event.

The real estimate can also be propagated to a smaller complex chart under explicit additional hypotheses. If the entire log-ratio is holomorphic on a common ball, bounded there by B, and uniformly bounded by δ on its real slice, a one-variable upper-half-disc argument gives

$$|f(z)| \le B^{1/3}\delta^{2/3} \tag{R11.18}$$

on the declared smaller chart. This estimate is independent of the number of field coordinates. The native bounded holomorphic remainders can supply B after their common chart and full prefactors have been identified. This route does not move a Haar cycle. It also does not construct a missing common domain. Cauchy derivatives use its actual remaining radii; a saved factor is preserved only when it is present in both real and complex input bounds.

The two-plaquette Wilson test from NR10 also has a positive exact resolution: for a group cutoff radius $\eta(g) \to 0$ with $\eta(g)/g \to \infty$, the two genuine denominators have log-ratio at most $Ce^{-c(\eta/g)^2}$. Its proof uses the nonlinear Wilson action, not a Gaussian replacement. This does not identify independently generated native histories on different physical tori.

Section 154.2 / Primary definitions specified

The domain and join definitions are as follows.

- T. Balaban, *Convergent renormalization expansions for lattice gauge theories*, CMP **119** (1988), 243–285, printed pp.256–259, (2.11)–(2.24), and pp.261–262, (2.34)–(2.42):
- T. Balaban, *Large field renormalization. II. Localization, exponentiation, and bounds for the R operation*, CMP **122** (1989), 355–392, printed pp.370–375, (1.50)–(1.65):
- T. Balaban, *The variational problem and background fields in renormalization group method for lattice gauge theories*, CMP **102** (1985), 277–309, Proposition 4, (97)–(98), and Section G, (172)–(190), Proposition 9:
- T. Balaban, *Renormalization group approach to lattice gauge field theories. I. Generation of effective actions in a small field approximation and a coupling constant renormalization in four dimensions*, CMP **109** (1987), 249–301, printed p.266, (2.6)–(2.8), and pp.269–270, the opening of Section 3:

The last item is the earlier small-field RG paper denoted [I] by R-II. It is different from the unavailable large-field R-I, denoted [IV] in R-II.

The full R-I text was not recovered No statement below fills in its preparation conditions, (1.64)–(1.69) ranges, or (1.75) selector from their labels.

There are two native facts particularly relevant here. First, III (2.11)–(2.12) defines the background as the constrained Wilson minimum for its actual determining set and averaged target data. Second, III (2.19)–(2.21) treats the external variables as retained parameters when a local large-field integration is performed. The factorization below uses this distinction between retained parameters and the actual current fiber variables. It makes no independence claim about the probability distribution of those parameters.

Section 154.5 / Complete conditional covariance cancels in this sector

Let q denote the actual output map, including its live reference and retained record. Suppose its retained coordinates include $v_{\mathcal{C}}$. Then the protected carrying function is

$$h = F(v_{\mathcal{C}}) = \phi \circ q. \tag{PE12.14}$$

For every integrable coefficient t,

$$\operatorname{Cov}(h, t \mid q) = 0. \tag{PE12.15}$$

This holds for the same conditional measure, even if its selector support is complicated or the numerator density has inherited interactions. No comparison measure is introduced.

The differential cancellation in SC11 is also exact. For a first-order operator V, the chain rule gives

$$E(Vh \mid q) = E(Dq\, V \mid q) \cdot D\phi, \tag{PE12.16}$$

so its complete scalar remainder is zero. For a second-order operator L, the full chain rule gives

$$E(Lh \mid q) = \overline{L}\phi. \tag{PE12.17}$$

Its cross-product current companion is likewise exactly the companion of ϕ computed with the same conditional principal coefficients. Thus both remainders R_0, R_1 of SC11 vanish. The same conclusion holds at every finite order by applying the full chain rule to $\phi \circ q$.

An existing boundary distribution $\mathcal{B}$ is retained. Wherever its products with the smooth carrying function are defined,

$$q_*(h\mathcal{B}) = \phi\, q_*\mathcal{B}, \qquad q_*(h\mathcal{B}) - \overline{h}\, q_*\mathcal{B} = 0. \tag{PE12.18}$$

This follows by testing the two distributions on the same smooth function of q. It neither differentiates the logarithm of a sharp selector nor multiplies two undefined delta distributions.

These cancellations are contingent on the **checked factorization** PE12.14. A physically marked coefficient depending on the discarded fluctuation, an internal readout, or an extra reference response is not covered. Nor does PE12.15 say that its own physical quadratic contact action vanishes: a nonzero inherited coefficient A_{12}, for example, remains that coefficient under PE12.8. This sector does not remove the R12 defect studied elsewhere

Section 154.7 / A nonzero-fluctuation native reference estimate

There is a second positive statement, for a different actual field. Let U_0 be the native relaxed reference minimum in R-II (1.3), and let U_C be the native constrained reference field obtained by changing the interior target through $\exp(iC)$, keeping the original exterior target rows fixed. This is the fluctuation family used in the denominator expansion R-II (1.2), with $C = g_k B'$. It is not the numerator U'_k of Section 3.

Fix a complete record D^+ contained in the top-scale part of the native determining geometry $B_k(Z)$, and contained in the region where the R-II good representative for U_0 is defined. Every bond, first-difference stencil and prescribed boundary value used below belongs to this record. Denote its actual fine four-volume by $v_a(D^+)$, with $a = \eta$.

The 1985 Proposition 9 gives the relative Landau potential $\mathcal{H}(C)$, its analytic dependence on C, and the weighted first-difference bound (173),(190). On this top-scale record the assigned length is one in

current units. Consequently the native weighted bounds there are ordinary current-scale bounds:

$$\|\mathcal{H}(C)\|_{\infty,D^+} + \|D_{a,U_0}\mathcal{H}(C)\|_{\infty,D^+} \le C_M\|C\|_\infty. \tag{PE12.19}$$

The norm on the right is the actual finite target-coordinate sup norm; all target directions may vary together. Its dimension is not silently replaced by one. The fixed constant includes the original finite geometric parameter M.

The target ball may be chosen with a fixed sufficiently small radius $r_M > 0$ for this admitted regular primitive. Here is the point in the source needed to justify that choice. Proposition 9 is formulated using a small upper-bound parameter ε_1; (172) gives $\|C\|_\infty < 2C_1\varepsilon_1$. The background must satisfy its regularity conditions with that upper bound. Once it is uniformly below a fixed admissible ceiling, one may take $\varepsilon_1 = \varepsilon_*(M) > 0$ within all the fixed-point smallness conditions. One is not required to set this parameter equal to the smaller running threshold ε_k. The native reference endpoint satisfies such a fixed ceiling for sufficiently small current coupling by SG11. This argument does not apply to a mixed background whose required regularity has not been proved.

Work in the jointly transformed representative used for the source's relative Landau construction and the good real representative of U_0. SG11 gives

$$\|A_0\|_\infty + \|D_aA_0\|_\infty \le C_M R_k\varepsilon_k. \tag{PE12.20}$$

On a sufficiently small common branch, the logarithm of the product $e^{ia\mathcal{H}(C)}e^{iaA_0}$ has an ordinary potential A_C. The convergent finite-dimensional product/logarithm series, its first lattice differences, and $D_{a,U_0} = D_a + O(A_0)$ give

$$\|A_C\|_{W_a^{1,p}(D^+)} \le C_{M,p}(R_k\varepsilon_k + \|C\|_\infty)v_a(D^+)^{1/p}. \tag{PE12.21}$$

No prescribed exterior variable is reset to the identity in this calculation. When the representative changes, its boundary values, frames and constraint targets transform together. PE12.21 is a representative estimate; it does not identify an arbitrary previously fixed anchored M8 chart with that representative.

On a smaller fixed target ball, analyticity and PE12.19 give, for $r \ge 1$,

$$\|D^r_{B'}A_{g_kB'}\|_{W_a^{1,p}(D^+)} \le C_{M,p,r}\, g_k^r\, v_a(D^+)^{1/p}. \tag{PE12.22}$$

Here D^r is measured as a multilinear operator in the actual target sup norm. For coordinatewise contractions, the actual coordinate counts remain in the contraction; PE12.22 does not make them disappear. One can equivalently keep the Cauchy radius explicitly: on $\|g_kB'\|_\infty \le r_M/4$, an order-$r$ coordinate derivative has the bound from a radius of at least $r_M/(2g_k)$ in that coordinate. For a simultaneous direction use the corresponding single complex line in the target sup-norm ball.

For clarity, if the original axial record has q_Λ free group coordinates and group dimension m, let $n_\Lambda = mq_\Lambda$. For a coordinate tensor T, the elementary consequence is

$$\left\| \sum_{i_1,\dots,i_r=1}^{n_\Lambda} T_{i_1\dots i_r}\,\partial_{B'_{i_1}}\cdots\partial_{B'_{i_r}}A_{g_kB'} \right\|_{W_a^{1,p}(D^+)} \le C_{M,p,r}g_k^r v_a(D^+)^{1/p} \sum_{i_1,\dots,i_r} |T_{i_1\dots i_r}|. \tag{PE12.22a}$$

A bound by the largest coordinate of T has the additional factor n_Λ^r. The actual finite coordinate count is not an estimate for a sum of microscopic action terms.

On the real native cutoff $\|B'\|_\infty \le M_0p_0(g_k)$,

$$\|A_{g_kB'}\|_{W_a^{1,p}(D^+)} \le C_{M,p}(R_k + M_0)\varepsilon_k v_a(D^+)^{1/p}, \tag{PE12.23}$$

Section 154.7 / A nonzero-fluctuation native reference estimate

once this cutoff lies in the smaller fixed target ball. This follows from $g_k p_0(g_k) = \varepsilon_k$, with the original logarithmic threshold and the original volume. It is an estimate on an actual nonzero-fluctuation native reference fiber, not a Gaussian replacement.

The corresponding curvature bound in this same controlled representative is

$$\|F_a(e^{iaA_{g_k B'}})\|_{L^p_a(D)} \le C_{M,p}(R_k + M_0)\varepsilon_k \, v_a(D^+)^{1/p}. \tag{PE12.24}$$

For real fields it is gauge invariant up to the fixed convention for the matrix norm. For complex fields a transfer to another gauge needs its actual bounded adjoint action; the estimate is not a declaration that all anchored gauges are uniformly equivalent.

Neither PE12.21 nor PE12.22 includes independent localization walls of the numerator's three-join construction. The top-scale-record restriction is essential: dropping it replaces the current-scale bounds by the native assigned-scale weights, and restores precisely the AE8 summability question. These estimates add no microscopic source-mesh factor.

Section 154.9 / Remaining actual-entry obligations

The progress and the unresolved statements are deliberately separate.

Statement	scope
Equality of common outside-Λ raw targets for the actual numerator and new minimum	Native constraint identity, PE12.1, using R-II (1.50).
Exact transport and zero internal derivatives for coefficients factoring through those completed retained rows	Proved, PE12.3–PE12.8.
Common-wall source representation after exact protected factorization	Proved as PE12.9; it does not assert the off-shell constraint identity PE12.10.
Preserving an already established grade on the same retained analytic domain	Exact, with transport cost one. Inclusion of a different old M8 domain is still a separate premise.
Finite physical word bounds in this protected sector	PE12.13, retaining the original mesh power, actual complex row bound, word length and coefficient count.
Complete conditional covariance and existing boundary-defect cancellation for output-measurable protected carriers	Proved, PE12.15–PE12.18.
Native nonzero-fluctuation reference bound and its internal derivatives on a complete top-scale record	PE12.21–PE12.24 in the source's jointly transformed representative. It is a reference-fiber result.
Uniform weighted AE8 estimate for the numerator's prepared field on every actual mixed history and simultaneous wall domain	Unproved.
Identification of every inherited marked readout with the protected common-row algebra or with an admitted fixed anchored weak chart	Unproved.
All-history estimates for current/observable/kernel responses crossing those joins	Unproved. The native projected-background identity (1.55) is not sufficient by itself.

In particular, the unresolved inference is now narrower in a useful way: a coefficient whose entire saved dependence record stays in the literal unchanged-row algebra needs no new field admission estimate at this R step. The difficult transitions are those for which that factorization fails. For them the actual selector-supported prepared map, its complete inherited boundary record, and its original simultaneous complex walls must still be controlled quantitatively. The factorization must be checked again when a later operation integrates or changes one of the formerly retained rows. Protection at one step is not an all-history assertion.

Section 155.3 / The actual degree-two compensation for these marks

For two labels choose the decompositions in Section 2 and write

$$O_i = t_i + r_i, \quad V_i = V[q_i], \quad t_i = \rho^{-1} D_{V_i} \rho. \tag{PD12.11}$$

For an inherited source action with its actual coefficient A_{12}, WC11.46 gives the defect from the pure Ward action

$$R_{12} = A_{12} + \tfrac{1}{2}(V_1 t_2 + V_2 t_1). \tag{PD12.12}$$

It is essential also to retain the degree-one defect r_1, r_2. The degree-two coefficient **after** the current elimination WC11.4 is

$$\boxed{b_{12} = A_{12} + \tfrac{1}{2}(V_1 t_2 + V_2 t_1) + V_1 r_2 + V_2 r_1.} \tag{PD12.13}$$

For the original bare sources, $A_{12} = 0$. An equivalent form, useful for retaining all cancellations in the physical mark, is

$$b_{12} = V_1 O_2 + V_2 O_1 - \tfrac{1}{2}(V_1 t_2 + V_2 t_1). \tag{PD12.14}$$

PD12.13 is not a newly supplied physical source term. It is the coefficient in an exact representation of the already fixed source PD12.2, accompanied by the transformed carrying function. In particular, setting $A_{12} = -\frac{1}{2}(V_1 t_2 + V_2 t_1)$ in the original source would change that source; it would still leave the mixed ordinary responses in PD12.13 unless they too were accounted for.

The displayed formulas choose only the two first-order current fields in the eliminating derivation. If the actual decomposition also extracts a first-order vector field V_{12} at source degree two, put $t_{12} = \rho^{-1} D_{V_{12}} \rho$. The same expansion instead gives

$$b_{12} = A_{12} - t_{12} + \tfrac{1}{2}(V_1 t_2 + V_2 t_1) + V_1 r_2 + V_2 r_1. \tag{PD12.13a}$$

The transformed carrying function then contains the additional $V_{12}\Phi$ at degree two.

A genuine higher differential word is handled separately. Let L_{12} be its actual ordered differential operator, including coefficient derivatives in their stated positions, and let L_{12}^* be its formal density adjoint. Put $t_{12} = \rho^{-1} L_{12}^* \rho$. At source degree two, inserting $\eta_1 \eta_2 L_{12}^*$ in the density transport gives exactly the same $-t_{12}$ correction in PD12.13a, accompanied by $+L_{12}\Phi$ in the carrying response. Products of this degree-two term with a positive-degree source vanish, so this assertion does not require L_{12} to be a derivation and does not apply the full WC11 derivation formula to an arbitrary higher-order operator. For a complete current word $L_{12} 1 = 0$; if it has a nonzero zeroth-order part, that scalar response must instead be retained explicitly.

The word's own stored normalization and original mesh are present in both $L_{12}^* \rho$ and $L_{12}\Phi$. Leaving it inside the unprojected A_{12} is also exact; discarding it in both places is not.

For the actual H–$C^\sharp$ pair this reads, with the precise q, r above,

$$b_{HC} = V[q_{H,f}] O_{C,a}(h) + V[q_{C,h}] O_{H,a}(f) - \tfrac{1}{2}\{V[q_{H,f}] t_{C,h} + V[q_{C,h}] t_{H,f}\}. \tag{PD12.15}$$

Thus the mixed current–ordinary terms are explicitly fixed, including the $g^{-2} g^{-3}$ normalization, the test-gradient vector in $q_{C,h}$, and the Casimir/descendant terms in t, r. This formula makes no charge-conjugation cancellation: both chosen scalar sources are C even.

155.3.1 An elementary exact scalar cancellation, and what it cancels

For the bare source, Haar adjointness gives

$$\begin{aligned} \mathbb{E}_\mu O_i &= \mathbb{E}_\mu r_i, \\ \mathbb{E}_\mu (O_1 O_2) &= \mathbb{E}_\mu (r_1 r_2 + b_{12}), \\ \mathrm{Cov}_\mu (O_1, O_2) &= \mathrm{Cov}_\mu (r_1, r_2) + \mathbb{E}_\mu b_{12}. \end{aligned} \tag{PD12.16}$$

One can verify the second equality directly. Since $\mathbb{E}(t_i F) = \mathbb{E}(V_i F)$,

$$\mathbb{E} b_{12} = \mathrm{Cov}(t_1, r_2) + \mathrm{Cov}(t_2, r_1) + \mathrm{Cov}(t_1, t_2). \tag{PD12.17}$$

In the alternative representation $r_i = O_i - t_i$, this is

$$\mathbb{E} b_{12} = \mathrm{Cov}(t_1, O_2) + \mathrm{Cov}(t_2, O_1) - \mathrm{Cov}(t_1, t_2). \tag{PD12.18}$$

Section 155.3 / The actual degree-two compensation for these marks

Equations PD12.16–18 are an exact cancellation of the artificial current-square contribution introduced by the decomposition. They do **not** make the physical covariance of O_1, O_2 zero. In particular a lower bound for the isolated R_{12} cannot simply be assigned to b_{12}, whose mixed ordinary terms can have either sign.

For $C_a^\sharp$, this distinction already occurs in the low field jets. AI32b gives $O_C = O(x^3)$ at the trivial connection. Therefore $r_C = -t_C + O(x^3)$. The vector field $V[q_C]$ vanishes at that connection, so it does not lower the ordinary field valuation. For two identical C-sharp labels,

$$(b_{CC})_{\le 2} = -(V[q_C]t_C)_{\le 2}. \tag{PD12.19}$$

Thus the positive pure-current contact has the opposite sign in the complete compensated quadratic contact. Its contribution is accompanied by the covariance of r_C, which contains the opposite low current jet. At the matched purely quadratic Gaussian truncation this is the exact equality $\mathbb{E}(t_C^2 - V[q_C]t_C) = 0$. This observation is a check of the actual decomposition; it is not a nonlinear all-history bound or an assertion that the physical C-sharp variance vanishes.

Section 156 / Transport of the original degree-two defect with complete kernel responses

This chapter concerns the original finite-source prescription XP3. It does not replace that prescription by a Ward-completed exponential. It identifies how the degree-two defect changes under an actual conditional integration, and gives a version before any conditional division. The latter is the version that remains meaningful for a selected history with sharp selectors.

PD12 supplies an application to the actual physical H insertion. The identities here do not assert that its ordinary connected remainder, or a general inherited coefficient, satisfies MT36.

Section 156.4 / What is still exact with selectors and singular observations

For an actual source-free linear kernel $\mathcal{K}$, define the unnormalized coefficient measures

$$\nu = \mathcal{K}\mu, \quad n_i = \mathcal{K}(O_i\mu), \quad n_{12} = \mathcal{K}((A_{12} + O_1O_2)\mu). \tag{DT12.17}$$

These are always the finite coefficients of the original source transport wherever the original integrals are defined. If the measures have legitimate densities and a nonzero body on an admitted chart, one may form

$$a_i = n_i/\nu, \qquad a_{12} = n_{12}/\nu - a_1a_2. \tag{DT12.18}$$

A sharp selected history need not have a differentiable conditional mean or such a chart. DT12.17 and the whole-kernel identity DT12.6 remain the appropriate formulations there.

In particular, for a kernel attached to the old fine measure,

$$D_{V_i}(\mu K) = K\, D_{V_i}\mu - \mu\, V_iK. \tag{DT12.19}$$

Thus a current of the old measure is not automatically the same current of the augmented measure. The omitted term would be the complete kernel response, including its denominator and selector boundaries. Replacing it by a smooth logarithmic score is allowed only where that score exists. There is no such replacement for a sharp indicator on its boundary.

The exact normalization also gives a useful scalar $_{\text{analysis.}}$ For two identical physical marks with bare A_{12}=0, any honest conditional probability realization has

$$a_{12} = \mathrm{Var}(O \mid q) \geq 0. \tag{DT12.20}$$

The terminal connected coefficient is $\mathrm{Var}_\mu(O)$. A local field-jet decomposition can distribute it among contact, lower and ordinary sectors, with signs, but the complete sum is this fixed quantity. For the actual H mark, PD12 proves that it is nonzero with a quantitative ultraviolet lower bound.

Section 157 / Internal derivatives and sharp boundary currents of the admitted native reference

The bulk tail estimate RV11.5 does not, by itself, estimate a derivative of a sharp selector. This chapter proves the boundary trace estimate for the actual finite-dimensional reference cutoff, and gives a representation of all fixed-order cutoff commutators using that trace. It also obtains internal density derivatives from the native nonlinear action on its actual enlarged target chart.

These statements apply to the admitted reference denominator in R-II (1.2), with its fixed-target minimum. They do not identify the three-join numerator with that denominator, and do not supply the missing all-history lift of an old physical current into these coordinates.

Section 157.5 / The signed boundary partner and conditional covariance

For the normalized native law $dK = \chi f\, dB/\widehat{D}$, let V be an actual internal vector field and let $t = f^{-1}D_V f$. The exact identity is

$$t\, dK = D_V(dK) + d\mathcal{B}_V, \qquad d\mathcal{B}_V = -\frac{f}{\widehat{D}}(V \cdot n)\, d\sigma_{\partial\Omega_R}. \tag{NB12.18}$$

The minus sign can be checked by integrating the identity against the constant 1. The ordinary restricted score has mean equal to the negative outward flux.

For a differentiable multiplier h, with the old datum b fixed,

$$\boxed{\mathrm{Cov}_K(h,t) - \left\{ \int h\, d\mathcal{B}_V - K(h) \int d\mathcal{B}_V \right\} = K(Vh).} \tag{NB12.19}$$

This is the SC11 cancellation on this literal internal fiber, with its complete boundary partner. It follows by testing NB12.18 with h and subtracting $K(h)$ times its test with 1.

If the actual derivative and boundary envelopes obey

$$|Vh(B)| \le A(1+|B|)^p e^{\theta|B|^2}, \qquad |(h - Kh)V \cdot n| \le A_\partial (1+|B|)^{p_\partial} e^{\theta|B|^2}, \tag{NB12.20}$$

then the complete scalar on the left of NB12.19 is bounded by $C_{M,p}A$, and its separately retained boundary partner by

$$C_{M,p_\partial} A_\partial q R^{m-1}(1+R)^{p_\partial} e^{-c_M R^2}. \tag{NB12.21}$$

The derivative envelope is essential. A bound for h only as an outer background function is not NB12.20.

Section 157.5 / The signed boundary partner and conditional covariance

There is a nontrivial native multiplier for which this input can actually be checked. UW12 compares two spatial action weights on the same admitted fixed-target branch and proves for their centered action difference w(B)

$$|w(B)| \le a|B| + b|B|^2, \qquad |\nabla_B w(B)| \le a + 2b|B|. \tag{NB12.22}$$

For $h = e^{-w}$, sufficiently small $a + b$, and a coordinate internal direction, the inequality

$$|h - 1| \le |w|e^{|w|}, \qquad |\nabla_B h| \le e^{|w|}(a + 2b|B|) \tag{NB12.23}$$

gives NB12.20 with $A, A_\partial \le C_M(a + b)$, after absorbing $a|B|$ into a small Gaussian weight. The same moment estimate bounds $K(h) - 1$. Thus NB12.19 is quantitatively $O(a + b)$ in this actual changed-weight case, with the boundary partner further bounded by NB12.21. The classical action difference at B=0 and the prefactor P are still outside this centered comparison, exactly as in UW12.

For the verified top-scale changed-weight sector, UW12.20g–i uses this same native $3R$ target chart and the complex localized curvature response to prove every fixed positive-order internal jet of h has $O(a + b)$ weighted moments. In particular the second internal derivative needed for a second current is estimated there. Lower assigned-scale weight cells retain their literal weighted kernel sums; a small bound for those sums has not been inferred from the top-scale case.

Section 158.1 / Hypotheses and conclusions

Complete S94 remains unproved. This chapter proves a comparison beyond RV11's identical-physical-record case: the spatial action weight in a genuine admitted reference primitive may change. The

fixed-target map, its unweighted relaxed minimum, its original constraints, and its original relative integration coordinates are retained. The two weighted reference densities are genuinely different. Their difference is bounded using the exact nonlinear Wilson energy and localized responses of that same fixed-target map, followed by the native fluctuation moments of RV11.

The uniform distance estimate proved below is restricted to a changed-weight region whose complete padded readout lies in the verified top-scale part of the native record. On lower assigned-scale shells the exact score bound is retained with the actual scale-weighted kernel and cell counts. Their uniform mixed-shell summation is not proved. This distinction is needed even for the reference problem; an exponential in an adapted distance does not authorize dropping its assigned-scale curvature weights.

The improvement over a cutoff-supremum comparison is substantive. A fluctuation cutoff radius does not enter as a bound for every occurrence of the reference coordinate. The comparison has the form a linear response multiplied by its actual first moment, plus a quadratic response multiplied by its actual second moment. Higher complete carriers retain their own moments. This supplies a local comparison with physical-distance decay when the differing weights lie at such a distance. Small running coupling by itself supplies no physical-distance decay.

The raw denominator retains an explicit classical scalar difference. Its original action-completion partner cancels that difference where the complete native pair is actually present. An independent absolute scalar increment does not cancel. Different fixed-target maps, moved selectors, unidentified mixed charts, and inherited carriers still require their own comparisons. No conclusion here admits every mixed history or proves the full source-zero construction.

Section 158.3 / Exact nonlinear Wilson energy with the original mesh

Use the actual fine spacing a_f in current units. Assign each oriented plaquette once to an actual cell Q. In a mixed determining domain these cells are subdivided along the original assigned-scale shells. On real fields,

$$s(P) = N - \operatorname{Re}\operatorname{tr} P = \tfrac{1}{2}\|P - I\|_{\mathrm{HS}}^2, \qquad P \in SU(N). \tag{UW12.5}$$

Compare the plaquettes in the common unitary frame supplied by the actual relative-field comparison. This does not change their Wilson energies. Write

$$F_B(p) = a_f^{-2}(P_p(W_B) - I), \qquad \|F\|_{2,Q}^2 = a_f^4 \sum_{p\in Q} \|F(p)\|_{\mathrm{HS}}^2. \tag{UW12.6}$$

Any required comparison-frame conjugation is included in $P_p(W_B)$ in UW12.6. Thus this is a comparison of gauge-invariant energies, not an assumption that two arbitrarily gauged matrix curvatures coincide. For the exact difference $\Delta F = F_B - F_0$, the elementary identity for squared norms gives

$$\begin{aligned} |A(\Delta\zeta, W_B) - A(\Delta\zeta, U_0)| &\le \sum_Q \omega_Q \left(\|F_0\|_{2,Q}\|\Delta F\|_{2,Q} + \tfrac{1}{2}\|\Delta F\|_{2,Q}^2 \right), \\ \omega_Q &= \max_{p\in Q} |\Delta\zeta(p)|. \end{aligned} \tag{UW12.7}$$

The fine factor a_f^4 remains inside each cell norm. A cell's fine plaquette count has not been put into a constant. In dimension four it is exactly the physical area factor squared occurring in the Wilson action. Write $v_Q = a_f^4 N_Q$, where N_Q is the actual number of assigned oriented fine plaquettes in Q. The original assigned scale σ_Q is also retained. For top-scale current cells v_Q is bounded by a fixed incidence constant and $\sigma_Q = 1$; neither assertion is silently extended to all shells.

158.3.1 The native response used here

Let Y be the actual free reference target coordinates, with all their native target paths and gauge/prefix factors. On an admitted original reference branch the localized target derivative yields

$$\|F_B - F_0\|_{2,Q} \le g \sum_{y\in Y} K_{\mathrm{nat}}(Q,y)|B_y|, \qquad K_{\mathrm{nat}}(Q,y) \le C_M v_Q^{1/2} \sigma_Q^{-2} e^{-c_M d_{\mathrm{nat}}(Q,y)}. \tag{UW12.8}$$

Here the free reference targets y are on their original current scale, so their source-row scale is one. The curvature factor σ_Q^{-2} is the output row of the actual bound (190). If a different source row were used, its original row weight would also be needed. No old fine-row source has been substituted for y. In the following formulas K means this full K_{nat}, not its unweighted exponential alone.

The path is the original tB, $0 \le t \le 1$, and stays in the original reference cutoff. One obtains UW12.8 by integrating the derivative of the actual fixed-target minimum along that path. The relative curvature estimate uses both the field and its first spatial derivative, with the original fine weights. The finite Lie algebra and incidence constants are included in K.

More explicitly, the covariant first variation of a plaquette is the oriented sum of its four transported link variations. The leading terms of size a_f cancel in that sum; the retained covariant difference has size a_f^2 times the first spatial derivative of the tangent field, with the bounded background commutator terms. The two corresponding rows of (190) therefore bound $a_f^{-2}\partial_{B_y} P$ by its assigned-scale curvature kernel. Squaring and summing uses $a_f^4 \sum_p$ as in UW12.6 and contributes the literal $v_Q^{1/2}$. This is why the bound in UW12.8 retains the fine plaquette count and the assigned-scale weight. A bound for the tangent potential alone would not justify UW12.8.

This is the native reference use of the variational paper's Proposition 9 and (182)–(190), printed pp.305–309. R-II itself uses that localized response in (1.5)–(1.6) and (1.16)–(1.17). These sources give the response on their specified regular reference branch. They do not identify every prepared mixed field with that branch. In particular UW12.8 has not been obtained by differentiating a weak potential norm or by assuming a common mixed complex wall.

158.3.2 A quantitative bound for the actual unequal-weight score

Set $\varepsilon_Q = \|F_0\|_{2,Q}$, and retain the following concrete nonnegative vector and matrix:

$$\begin{aligned} \alpha_y &= g^{-1} \sum_Q \omega_Q \epsilon_Q K(Q,y), \\ H_{yz} &= \sum_Q \omega_Q K(Q,y) K(Q,z), \\ a &= \|\alpha\|_2, \qquad b = \tfrac{1}{2}\|H\|_{2\to 2}. \end{aligned} \tag{UW12.9}$$

Equations UW12.7–8 prove, on the full original real cutoff,

$$|w(B)| \le a|B| + b|B|^2. \tag{UW12.10}$$

This is a bound for the original nonlinear evaluated Wilson action. No Taylor remainder has been discarded and no Gaussian action has replaced it. In particular the quadratic term in UW12.10 is the square of a genuine nonlinear field response.

Now restrict to changed-weight cells whose complete required padded readouts lie in the verified top-scale region of $B_k(Z)$. There $\sigma_Q = 1$, and the current-cell kernel has uniform summable row and column sums. If $\|\Delta\zeta\|_\infty \le \omega$, $\varepsilon_Q \le \varepsilon_{\mathrm{bg}}$, and every such changed-weight cell has original current

distance at least D from every actual free coordinate in Y, these kernels give, after a fixed reduction of the decay exponent,

$$a \le C_M \omega(\varepsilon_{\mathrm{bg}}/g)\sqrt{n}\, e^{-cD}, \qquad b \le C_M \omega e^{-2cD}. \tag{UW12.11}$$

For completeness, each column sum of the restricted kernel over changed cells is at most Ce^{-cD}. Each row sum over Y has the same bound because every such pair is separated by D. Schur's inequality therefore gives $\|K_{\mathrm{changed}}\|_{2\to 2} \le Ce^{-cD}$ and the displayed bound for H. The first bound follows from the column sum and the actual number of free coordinates. Bounded group dimension converts free-link count to n; it does not remove that count.

The distance is the actual current-cell distance on the native record, with wrapped paths included. If the changed weight lies near one of those paths, UW12.11 is used with that smaller distance. An apparently local visible support does not enlarge it. The background amplitude is called $\varepsilon_{\mathrm{bg}}$ to distinguish it from any symbol used for the original fine mesh.

For a general changed-weight shell the valid result remains the explicit vector and matrix UW12.9 with K_{nat}. For own-scale cells in four dimensions one can have $v_Q^{1/2}\sigma_Q^{-2} = O(1)$, but the number of those cells still appears in the sums defining α and H. No uniform Schur sum for all actual mixed histories follows from that pointwise cancellation. The needed weighted shell sum remains a concrete unproved input outside the stated top-scale sector.

Section 158.7 / The classical scalar, its native completion, and what remains absolute

There is a separate, explicit background cost:

$$|g^{-2}A(\Delta\zeta, U_0)| \le \frac{1}{2g^2}\sum_Q \omega_Q \|F_0\|_{2,Q}^2. \tag{UW12.27}$$

Its sum is over all actual changed-weight cells. It need not decay with their distance from Y. The background at those cells is fixed before the auxiliary fluctuation B is integrated; its energy is not a tail event for B.

A finite Wilson constraint example makes the distinction exact. Take a plaquette whose four links are fixed by the exterior target constraints, so its holonomy is the same P_* for every value of the free reference coordinates. Change only its spatial weight by t. Then

$$\log(D_1/D_0) = -tg^{-2}s(P_*), \qquad K_1 = K_0, \qquad I_1 = I_0. \tag{UW12.28}$$

This is an exact compact Wilson calculation. It diagnoses which scalar is invisible to probability normalization; it is not a claim that this particular finite example is a complete native mixed history in the original construction.

In the actual R action completion, the numerator is factored so as to contain the corresponding $e^{-g^{-2}A(\zeta_i,U_0)}$. Where that complete pair is present, write it without dropping its remaining original carrier $\mathcal{N}_i$:

$$e^{s_i}\frac{e^{-g^{-2}A(\zeta_i,U_0)}\,\mathcal{N}_i}{D_i} = e^{s_i - c_i^{\mathrm{coord}} - I_i}\mathcal{N}_i. \tag{UW12.29}$$

Section 158.7 / The classical scalar, its native completion, and what remains absolute

The classical term cancels exactly in this paired expression, including when the two spatial weights differ. This follows from each native numerator/denominator extraction; it does not require an estimate of the two raw classical actions by a small number. R-II explicitly performs this extraction in (1.10) and in the numerator analysis beginning on printed p.359. The full localization and regular-action remainder belongs in $\mathcal{N}_i$.

Equation UW12.29 therefore leaves the comparison of I_i addressed above, the actual difference of $\mathcal{N}_i$, any unequal coordinate scalar, and the independent absolute increment $s_1 - s_0$. A term moved into the regular action has its original opposite-sign companion in $\mathcal{N}_i$; it is not discarded by the equality. The empty Mayer family can still have a nonconstant $\mathcal{N}_i$, and its absolute mass need not be unchanged. The scalar part of the later connected expansion must still be compared by VC28–29 after its actual hypotheses hold.

Section 158.8 / Physical torus distance, component counts, and the remaining S94 row

For two physical tori, UW12.17 provides a summable input only if the actual differing spatial weights, with their complete read sets, stay in the verified top-scale region and lie at a distance comparable to T_*/ℓ_j from the primitive's free coordinates. If that identification holds and the remaining background prefactor grows at most polynomially in $1/\ell_j$, then

$$\sum_j \ell_j^{-4} C(1+\ell_j^{-1})^r e^{-cT_*/\ell_j} < \infty. \tag{UW12.30}$$

The original block/coupling parameters and lower physical period are fixed. An intrinsic native separation such as $D = R_j$ is a different input. Its small-coupling tail cannot be renamed a physical torus collar. Depending on the actual growth law it may or may not be ultraviolet summable; UW12.30 has not proved such a replacement.

When a complete record has q_ω reference components, the comparison is performed successively on its actual conditional fibers. The errors are summed over those components, with their actual n_α, background energies, and carried scalar factors. Old shared data are held fixed during each step. A common refinement of original selector partitions is made before absolute estimates, as in VM9.10–13; summing its weights does not create a count of all pairs. There is one assigned reserve for the complete record.

For exact nonwrapping copied records the deck-lift count is still q^4 and cancels against physical volume. This chapter has not identified all native histories with those copied records. In particular no new bound was established here for differing fixed-target maps, shifted original constraint targets or minima, inherited effective-density carriers, independent vacuum increments, or terminal terms. Wrapped determining paths and denominator read sets stay in the full carrier.

S94 ingredient	Established here	Still required for the complete construction
Different spatial action weights on an admitted native fixed-target branch	Exact weighted nonlinear score UW12.9–10; top-scale physical-distance bound UW12.11; extracted-log and kernel comparison UW12.14	Identification of every cross-torus differing weight with this branch and its full separated geometry; weighted cell sums for lower assigned-scale shells

Section 158.8 / Physical torus distance, component counts, and the remaining S94 row

S94 ingredient	Established here	Still required for the complete construction
Fluctuation factors in that comparison	Actual moments replace cutoff suprema; original n, mesh, and carrier degree retained	Full inherited carrier envelopes on the same fiber
Raw denominator scalar	Exact classical difference UW12.4 and its native completion UW12.29	Full remainder-carrier and independent absolute scalar comparison
Weight change together with a verified forest change	Core formula UW12.25 and error UW12.26	A common physical fixed-target record, matched pins/counts, and endpoint tail admission
Complex continuation	A nonzero disc for the actual weight interpolation; RV11 propagation available on an existing common outer chart	The simultaneous original mixed outer chart and its curvature bounds
Complete S94	The top-scale spatial-weight input has a concrete nonlinear estimate, with a literal weighted residual on other shells	All-history matching/comparison, weighted mixed-shell sums, absolute scalar increments, complete wrapped tails, and terminal comparison

Section 159.1 / Hypotheses and conclusions

The proposed use of the 1985 variational paper's equation (162) does not, by itself, prove the desired lower-shell Schur estimate for the nonlinear reference derivative in (190). There is an exact exponent

Section 159.1 / Hypotheses and conclusions

mismatch in the published exponents:

Original equation	Exponential rate
(162), printed p.303	$\delta_0/2$ in the summed majorant,
(189), printed p.308	$\delta_0/4$,
(190), printed p.308	$\delta_0/8$.

(NM12.1)

These fractions are read from the printed formulas.

Equation (162) is useful for a kernel with rate at least $\delta_0/2$, with an additional distance factor if its rate is strictly larger. It is not a sum at every positive exponent. No smaller-exponent version was established.

The primary paper is T. Balaban, *The variational problem and background fields in renormalization group method for lattice gauge theories*, CMP **102** (1985), 277–309,

Section 159.2 / The native index set and the scope of the anchor

Printed p.277, before and after (1), specifies the nested native domains. At level j, the block length is $\sigma_j = L^j\eta$, with $\eta = L^{-k}$ in the paper's current normalization. The retained block centers are

$$A_j = \Omega_j^{(j)} \setminus \Omega_{j+1}^{(j)}, \qquad \mathfrak{B}_k = \bigcup_{j=0}^{k} A_j, \qquad B^j(A_j) = \Omega_j \setminus \Omega_{j+1}. \tag{NM12.2}$$

The printed separation assumption is

$$(L^j\eta)^{-1}\,\mathrm{dist}(\Omega_j^c, \Omega_{j+1}) > RM_1, \qquad R \geq R_1. \tag{NM12.3}$$

The fixed constants are chosen so that the earlier native propagator and gauge-fixing results apply. This is not a statement that every old source record has a buffer of the running logarithmic length R_k.

Equation (162) reads

$$B_3 = 72d^3L^3B_0 \sup_{\mathfrak{B}_k} \sup_{y_1} \sum_{y_2 \in \mathfrak{B}_k} e^{-(\delta_0/2)d(y_1,y_2)}\big(d(y_1,y_2)+1\big)(L^{j(y_2)}\eta)^{-1}. \tag{NM12.4}$$

The factor in parentheses is **distance plus one**, not the spatial dimension plus one.

The anchor y_1 belongs to the current-scale setting in the preceding argument. The first case has the unit block $\Delta_0 = B^k(y)$; the other uses the current-size cube described there, within the top and its adjacent native scale. The unqualified typography $\sup_{y_1}$ should not be exported to arbitrary fine-scale anchors without that context. Such an export would already fail on the diagonal: taking $y_2 = y_1$ at assigned length $\sigma_{y_1} \ll 1$ gives a term $\sigma_{y_1}^{-1}$.

Thus a top-scale reference-coordinate anchor is the appropriate one for using NM12.4. A proposed two-sided bound with arbitrary fine anchors requires its own weights or rescaling.

Section 159.3 / What the displayed sum really gives for own-scale cells

For this paragraph take the actual native cells $Q_y = B^{j(y)}(y)$, or a once-assigned bounded enlargement with

$$v_{Q_y} \le \nu\, \sigma_y^4. \tag{NM12.5}$$

Here ν is the actual volume/stencil multiplicity. It must include any chosen cell enlargement; it is not a claim that an arbitrary inherited carrier has bounded volume. Retain separately the number r_{or} of actual curvature orientations and Lie components if they are summed separately. In four dimensions the positive plaquette orientations number six; a convention retaining both orientations has twelve. The Lie-component count is the original group dimension. If a vector norm already includes these components, the corresponding fixed norm-equivalence factor is used instead of counting them twice.

Let S_β denote the sum in NM12.4 without its prefactor, where $\beta = \delta_0/2$. If a kernel on these cells satisfies

$$K(Q_y, x) \le C\, v_{Q_y}^{1/2} \sigma_y^{-2} e^{-cd(x,y)}, \tag{NM12.6}$$

and x is an allowed current-scale anchor, then for $c \ge \beta$

$$\begin{aligned} \sum_{Q_y,\,\text{components}} K(Q_y, x) &\le C r_{\mathrm{or}} \sqrt{\nu} \sum_y e^{-cd(x,y)} \\ &\le C r_{\mathrm{or}} \sqrt{\nu}\, S_\beta. \end{aligned} \tag{NM12.7}$$

The second step uses $\sigma_y \le 1$ and $d(x,y) + 1 \ge 1$. Thus the volume weight does exactly cancel σ_y^{-2} for these own-scale four-cells. It does not cancel the mismatch between different exponential rates.

At $c = \beta$, the displayed distance weight also gives

$$\sum_{d(x,y)\ge R,\,\text{components}} K(Q_y, x) \le C r_{\mathrm{or}} \sqrt{\nu}\, \frac{S_\beta}{1+R}. \tag{NM12.8}$$

If $c > \beta$, the stronger result is

$$\sum_{d(x,y)\ge R,\,\text{components}} K(Q_y, x) \le C r_{\mathrm{or}} \sqrt{\nu}\, S_\beta e^{-(c-\beta)R}. \tag{NM12.9}$$

These are genuine consequences of the original sum, with no assigned cell discarded. Their hypotheses do not hold for the literal rate $c = \delta_0/8$ in (190).

If the input side is the actual finite set of n reference coordinates, an independently available column bound C_{col} and pointwise bound $K \le C_*$ give the elementary Schur bound

$$\|K\|_{\ell^2 \to \ell^2} \le (C_{\mathrm{col}}\, n C_*)^{1/2}. \tag{NM12.10}$$

This retains the original input count. It does not create the missing column estimate at the slower rate.

Section 159.6 / What remains to be specified in the original metric source

The 1985 paper refers to *Propagators and renormalization transformations for lattice gauge theories. II*, CMP **96** (1984), 223–250, DOI 10.1007/BF01240221, for the definition of $d(y, y')$ and for (2.47)–(2.51). It also invokes that paper's Lemma 2.1 in the nonlinear response proof. Its full statement is not used here.

The complete argument would verify, for the unchanged native $B_k(Z)$, a bound such as

$$\sup_{x \text{ top-scale}} \sum_{y \in \mathfrak{B}_k} e^{-\beta' d(x,y)} < \infty, \qquad \beta' < \delta_0/8. \tag{NM12.17}$$

Then the difference $\delta_0/8 - \beta'$ supplies the required distance exponent. One possible proof would combine the actual path metric with the original fixed M_1 collars, if every path to a lower-level cell is forced to cross the required nested collars and the resulting count is summable. Neither that path

Section 159.6 / What remains to be specified in the original metric source

statement nor its constant has been checked from the original metric definition here. No per-history buffer or selector is enlarged to impose it.

Accordingly, the correctly established scope remains: top-scale Schur summability where the geometry directly supplies the ordinary lattice count, the original weighted residual for general native lower shells, and the faster-rate conditional implications NM12.7–NM12.9. The literal (162)/(190) comparison does not close the general native lower-shell bound.

Section 160.1 / The physical source was specified its current decomposition

MT1 and XP3 specify the original H and C-sharp marks and a bare source action with $A_{12} = 0$. The source labels are distinct even when they insert the same observable. The argument gives distinctions.

- The ordinary physical H mark is not the pure current considered in WC11's sharp scalar theorem. PD12 now uses the actual H observable.
- AI34 gives an explicit exact H decomposition, including two opposite copies of its Casimir scalar. It does not specify a unique all-history projection of every generated coefficient.
- Solving AI32d for C-sharp gives the quadratic descendant coefficient $1/4$, with the complete cubic lattice remainder already in C-sharp.
- The correlated coefficient after current elimination contains $V_1 r_2 + V_2 r_1$. Dropping these terms would replace the physical source by a different one.

The H–H formula PD12.22 was recomputed from

$$t_H = -\frac{\sigma}{3}(P - \beta^{-1}\ell), \quad r_H = \sigma(D - (3\beta)^{-1}\ell), \quad V_H = -\frac{\sigma}{3\beta}W. \tag{AU12.1}$$

Its coefficients, including the positive sign of the scalar response in the complete b coefficient, agree. In the isolated pure-current defect that scalar response has the opposite sign. This difference is required by the actual ordinary companion.

A separate correction distinguishes a degree-two source vector field from a genuine higher differential word. For a word L_{12}, its current uses the formal adjoint $L_{12}^{*}\rho$, and its carrying response is $L_{12}(K\Psi)$. The degree-two source nilpotence justifies that correction directly; the exponential-flow derivation formula is not applied to an arbitrary second-order operator. Its zero-order term, if present, remains a scalar.

Section 160.2 / The nonlinear H variance calculation

The exact flat-bath calculation follows directly. The free bond has three identical plane forces, so its H residual vanishes. The other eighteen affected bonds have one nonzero plane force;

each contributes $2|\nabla s|^2/3$. The full variable part is

$$12|\nabla s(U)|^2 + \text{constant}. \tag{AU12.2}$$

It is not the coefficient 54 from the different total-current observable

The remaining steps of the lower bound are as follows:

- The star of an H summand meets at most one selected free link at the stated graph separation, so the full conditional observable is additive and the conditional Wilson law factors.
- The good-bath event is measurable in the conditioned links. It asks for the existence of a gauge on those links; its probability lower bound can be obtained from a full near-flat patch without conditioning on the unknown free-link value.
- Compactness and the positive Hessian at the identity give a uniform minimum, gap and local quadratic bounds for the exact one-link action when the bath is sufficiently close to flat.
- After rescaling around the actual conditional minimum, a density lower bound on a fixed ball and a test-function integration by parts give the variance lower bound. No uncontrolled linear term is dropped.
- The original factors $g^{-2}a^{-2}$, the β^{-2} conditional variance and the number $c(T/a)^4$ of selected links give exactly cT^4a^{-8}.

The identical-label lower bound for a tested insertion requires f to be real. For complex f the connected coefficient is bilinear, $\mathbb{E}[(O-\mathbb{E}O)^2]$, and need not be nonnegative. PD12.56 and the subsequent ordered variance inequalities now explicitly use real tests. A conjugate-label pairing is a different Hermitian statement. The earlier absolute contact estimates may still use complex test weights through their moduli.

The real compact contact bound PD12.27 spends only the displayed original Wilson energy E_Y. The improved absolute expectation uses the exact force identity and the elementary same-measure plaquette estimate. It is a bound for the explicit bare contact, not for the full variance of the ordinary residual. The weaker quadratic-volume variance upper bound is retained with its actual scope.

Section 160.3 / Conditional transport and the scalar/current distinction

The signs in DT12.13 follow from $D = -\operatorname{div}$. The key product rule is

$$\operatorname{Cov}(f, t_i \mid q)\nu = D(\operatorname{Cov}(f, Dq\, V_i \mid q)\nu) + \{\mathbb{E}(V_i f \mid q) - \bar{v}_i \cdot \nabla \mathbb{E}(f \mid q)\}\nu. \tag{AU12.3}$$

Symmetrizing the current–current term and retaining both current–ordinary terms gives the displayed positive divergence companion. The q-constant specialization agrees with the original terminal covariance identity for H.

The whole-kernel formula DT12.6 keeps the first and second responses of the original K and of its carrying test. At sharp boundaries it is used as a tested identity before conditional division. It does not form products of boundary deltas.

A norm hypothesis is needed in DT12.21. Its upper bound is for the represented-distribution norm. An ordinary divergence cannot be bounded in ordinary total variation merely by the mass of its vector coefficient. Absolute scalar claims are confined to the separately proved estimates in PD12 and to the exact positive terminal connected mass for a real test.

Section 160.4 / The native internal and cutoff-boundary estimates

The new NB12 derivative estimate uses a particular native reference primitive. Its complex norm bound is obtained by rerunning the native action estimate on a fixed finite enlargement of its target radius. The complex nonlinear derivative bound, the complex minimum response, and the original summed component estimate are all required. Neither analyticity by itself nor a bound on the coarse coordinate count would prove NB12.5.

The original selector is not enlarged. Its actual product norm balls give q boundary faces, each with the original R^{m-1} surface factor. A pointwise Gaussian density bound on that wall and the lower bound for the extracted denominator give the boundary trace estimate. This is not obtained by assigning bulk probability to a measure-zero boundary.

The higher-order commutator gives an exact telescoping operator identity. It represents the result by derivatives of single finite surface measures. Corners are retained in those distributional derivatives without multiplying delta functions. The score of the interior density has a negative outward-flux partner; the full differentiated cutoff measure has the corresponding positive flux. The covariance signs in NB12.18–19 and UW12.20e agree under these conventions.

The bound remains a bound for the admitted reference coordinates. Converting an old physical current into those coordinates can differentiate constraints, live references, coarea factors and the genuine denominator. The required conversion is not supplied by the derivative estimate.

Section 160.5 / Changed spatial weights and the corrected scale scope

The original unequal-weight density ratio and its raw scalar are distinct quantities. The centered fluctuation w controls $\log(\widehat{D}_1/\widehat{D}_0)$. The raw denominator still contains

$$-g^{-2}A(\Delta\zeta, U_0). \tag{AU12.4}$$

This term cancels only against the explicitly matched original classical numerator factor. An independent old scalar and a difference of complete carriers remain. A probability normalization does not estimate either of them.

An initial overbroad form of the curvature-response kernel was corrected On an assigned-scale cell Q the literal bound retains

$$v_Q^{1/2}\sigma_Q^{-2}e^{-c\,d_{\mathrm{nat}}(Q,y)}. \tag{AU12.5}$$

The verified uniform spatial summation in UW12.11 applies to changed-weight cells whose complete padded records lie in the native top-scale region. The general expression keeps its weighted cell sums. Even when $v_Q^{1/2}\sigma_Q^{-2} = O(1)$ for one own-scale cell in four dimensions, its multiplicity has not been paid.

The actual changed-weight multiplier has proved internal derivatives, including every fixed-order jet on the admitted top-scale native target chart. This does not extend the common complex wall domain of an arbitrary inherited marked coefficient.

The additional primary check NM12 found an exponent mismatch in the printed rates. The geometry sum in the 1985 equation (162) uses rate $\delta_0/2$, whereas the nonlinear response in (190) has rate $\delta_0/8$. The faster weighted sum cannot be substituted for the slower one. NM12 records the exact faster-rate consequences and the missing smaller-exponent metric estimate; it makes no claim that the native geometry fails that estimate.

Section 160.6 / Protected numerator and the obligations for other histories

In PE12 the protected numerator row is constant because it is an exact function of the original constraint, with its selectors retained. Independently localized joins require their own identity. Native reference estimates refer to a separately identified field.\n\nThe graded transitions for inherited physical insertions, the mixed-entry chart and selector/readout match, and the complete S94 comparison across physical tori still require estimates. The lower bound on the physical H variance rules out cancellation of the entire coincident scalar. It is compatible with a specified lower-dimensional or contact subtraction in the original source scheme.

Section 161.1 / The actual original source fixes the mixed coefficient

The physical marks in MT1 are

$$O_H(f) = g^{-2} H_a(f), \qquad O_C(f) = g^{-3} C_a^{\sharp}(f), \qquad \beta = c_\beta g^{-2}. \tag{R12.1}$$

The original finite-source prescription XP3 gives

$$A^{\mathrm{bare}} = \eta_1 O_1 + \eta_2 O_2, \qquad A_{12}^{\mathrm{bare}} = 0. \tag{R12.2}$$

For a genuine clean output of the original source-free operation $\mathcal{T}$, the inherited coefficient is fixed:

$$A_{12}^{\mathrm{out}} = \frac{\mathcal{T}(\rho O_1 O_2)}{\mathcal{T}\rho} - \frac{\mathcal{T}(\rho O_1)\mathcal{T}(\rho O_2)}{(\mathcal{T}\rho)^2}. \tag{R12.3}$$

The original nonzero output body is required for this ratio. Still-live selected histories retain the undivided joint numerator coefficients. Replacing the original kernel or its denominator would change R12.3.

Section 161.1 / The actual original source fixes the mixed coefficient

If an exact source decomposition is $O_i = t_i + r_i$, with $t_i\rho = D_{V_i}\rho$, the degree-two coefficient after current elimination is

$$\boxed{b_{12} = A_{12} + \tfrac{1}{2}(V_1 t_2 + V_2 t_1) + V_1 r_2 + V_2 r_1.} \tag{R12.4}$$

Thus the pure-current defect $\mathcal{R}_{12}$ is accompanied by the mixed ordinary responses for a physical mark with $r_i \neq 0$. A separately extracted degree-two current word has its own formal adjoint and carrying response; it is not discarded or treated as a first-order derivation.

The exact terminal identity is

$$\mathbb{E}A_{12} + \mathrm{Cov}(O_1, O_2) = \mathbb{E}b_{12} + \mathrm{Cov}(r_1, r_2). \tag{R12.5}$$

Complete auxiliary normalization cancels the responses of that auxiliary kernel only after the entire fiber and every selector alternative are included. It does not make the right side of R12.5 zero.

Section 161.2 / A quantitative result for the actual H packet

PD12 uses the literal AI34 identity

$$\begin{aligned} O_H(f) &= g^{-2}(D_a(f) - P_a(f)/3), \\ t_H(f) &= -\tfrac{1}{3}g^{-2}(P_a(f) - B_a(f)), \\ r_H(f) &= g^{-2}(D_a(f) - B_a(f)/3). \end{aligned} \tag{R12.6}$$

The full Casimir scalar B_a, including its constant term and original plaquette incidences, remains in both partners. This is a specified exact decomposition of the original mark; it is not a claim that MT1 chose a unique projection of every future generated coefficient.

Let

$$\sigma = g^{-2}a^{-2}, \quad P[f] = \sum_i f_i F_i^2, \quad D[f] = \sum_i f_i \sum_\mu F_{i;\mu}^2, \quad \ell[f] = \sum_i f_i X_i^2 S_0, \quad W_f = \sum_i f_i F_i X_i. \tag{R12.7}$$

For the bare physical H pair the exact coefficient is

$$\begin{aligned} b_{HH}(f,h) = \frac{\sigma^2}{9\beta}\{&\tfrac{1}{2}(W_f P[h] + W_h P[f]) - 3(W_f D[h] + W_h D[f]) \\ &+ (2\beta)^{-1}(W_f \ell[h] + W_h \ell[f])\}. \end{aligned} \tag{R12.8}$$

The ordinary directional-square response and the last scalar response are essential.

For the literal completed overlap Y, with $E_Y = \beta \sum_{p\in Y} s_p$, the same nonlinear Wilson action gives

$$|b_{HH}(f,h)| \le C c_\beta^{-2} a^{-4}(1+\beta^{-1})\|f\|_\infty \|h\|_\infty E_Y. \tag{R12.9}$$

This is a real compact bound using one original energy reserve. It has no absent microscopic birth factor. In the original isotropic Wilson probability, its absolute mass satisfies

$$\boxed{\mathbb{E}|b_{HH}(f,h)| \le C a^{-4} N_Y \|f\|_\infty \|h\|_\infty [1 + \sqrt{\log(2+\beta)}],} \tag{R12.10}$$

where N_Y is the actual overlap plaquette count. In physical volume this is $Ca^{-8}|Y|_a$ times the displayed test and logarithmic factors. Its raw mass is $Z_\beta \mathbb{E}|b_{HH}|$. An independent scalar contributes its own absolute multiplier.

Section 161.2 / A quantitative result for the actual H packet

For the complete physical source, a different statement is proved:

$$\boxed{\mathrm{Var}_{\mu_\beta}(g^{-2}H_a(1)) \geq cT^4a^{-8}, \qquad \beta \geq \beta_0.} \tag{R12.11}$$

The constants are independent of the cutoff and torus lattice size once the fixed local patches fit. A fixed bounded real test equal to one on a positive-volume physical box satisfies the corresponding lower bound $c_f a^{-8}$.

The proof conditions on a positive-density family of separated links. At a flat bath the complete variable part of the H observable at one free link is exactly

$$12|\nabla s(U)|^2 + \text{constant}. \tag{R12.12}$$

The free link contributes zero to its directional residual; eighteen other links supply the factor 12. A same-Wilson plaquette estimate gives a uniform probability of good baths. The exact one-link action has a uniform minimum, Hessian and gap there. A rescaled test-function estimate proves a conditional variance at least $c\beta^{-2}$, without dropping the linear term at the conditional minimum.

R12.11 is a coincident physical contact statement. It does not preclude renormalized separated correlations. It does preclude an assertion that complete kernel normalization cancels this entire scalar under the unchanged physical prescription. Centering a single insertion does not change its variance. Identical-label positivity is asserted for real tests; a complex test requires the corresponding conjugate pairing for a Hermitian variance.

The remaining ordinary row is now explicit:

$$\mathrm{Var}(O_H) = \mathrm{Var}(r_H) + \mathbb{E}b_{HH}. \tag{R12.13}$$

R12.10 controls the second term. A cutoff-uniform graded bound for the first term through every actual mixed transition has not been proved. The elementary fallback upper bound is only

$$\mathrm{Var}(O_H(1)) \leq Ca^{-4}\,[\#E\log(2+\beta)+1]^2, \tag{R12.14}$$

which retains its quadratic-volume loss and is insufficient for the required induction.

Section 161.3 / Complete conditional transport and actual internal estimates

On an identified smooth conditional chart, DT12 gives a new transport identity for the original compensated coefficient. With $v_i = Dq\, V_i$, bars denoting actual conditional means, and $\nu = q_*\mu$, it is

$$b_{12}^{c}\nu = \mathbb{E}(b_{12} \mid q)\nu + \mathrm{Cov}(r_1, r_2 \mid q)\nu + D(J_{12}\nu), \tag{R12.15}$$

where

$$\begin{aligned} J_{12} = \tfrac{1}{2}[&\mathrm{Cov}(t_2, v_1 \mid q) + \mathrm{Cov}(t_1, v_2 \mid q)] \\ &+ \mathrm{Cov}(r_2, v_1 \mid q) + \mathrm{Cov}(r_1, v_2 \mid q). \end{aligned} \tag{R12.16}$$

The current partner can persist even when a fine pure-current defect vanishes. The unnormalized tested-kernel form retains all responses when selectors or singular observations prevent use of smooth conditional densities.

The quantitative native reference result is more restricted, but genuinely uses the original nonlinear kernel. Let $C = e^{igB}$, $R = M_0 p_0(g)$, and let n=mq be the actual number of free Lie algebra coordinates.

Section 161.3 / Complete conditional transport and actual internal estimates

For the admitted reference of R-II (1.2), the native complex norm bounds may be rerun at a fixed finite enlarged target ceiling while the real selector remains unchanged. This yields, for every fixed r,

$$|\partial_B^\alpha f(B)| \le C_{M,r}(1+|B|)^r f(B), \qquad |\alpha| \le r. \tag{R12.17}$$

The argument uses the original summed native action estimate, the complex nonlinear derivative bound and the actual minimum response. Finite n or analyticity alone would not prove it.

For the original product-norm cutoff Ω_R, the boundary trace satisfies

$$\frac{1}{\widehat{D}} \int_{\partial\Omega_R} (1+|B|)^p e^{\theta|B|^2} f \, d\sigma \le C_{M,p} q R^{m-1} (1+R)^p e^{-c_M R^2}. \tag{R12.18}$$

The q faces, their surface dimension, and the genuine extracted denominator are retained. A bulk tail estimate alone would not establish this boundary trace.

For an actual ordered internal density word $D_w = D_{i_r} \cdots D_{i_1}$,

$$[D_w, \chi] f = \sum_{j=1}^{r} D_{i_r} \cdots D_{i_{j+1}} [(D_{i_j}\chi) D_{i_{j-1}} \cdots D_{i_1} f]. \tag{R12.19}$$

Thus every fixed-order cutoff response is represented by derivatives of single boundary measures, with a polynomially weighted $e^{-c_M R^2}$ bound. It is not called ordinary scalar total variation.

For a native internal field V, the restricted interior score has boundary partner

$$d\mathcal{B}_V = -\widehat{D}^{-1} f (V \cdot n) \, d\sigma_{\partial\Omega_R}. \tag{R12.20}$$

The exact closed-fiber conditional cancellation is

$$\mathrm{Cov}_K(h,t) - \{\mathcal{B}_V(h) - K(h)\mathcal{B}_V(1)\} = K(Vh). \tag{R12.21}$$

UW12 supplies an actual nonconstant multiplier h from a changed spatial Wilson weight whose positive-order internal jets have small weighted moments on the verified native top-scale sector. This provides quantitative first- and second-current inputs in that sector.

It remains necessary to identify every inherited physical direction with its full lift into these coordinates and to estimate its actual effective multiplier. The lift can vary constraints, conditional denominators and live references. R12.17–21 do not perform that conversion. Every old word retains its original mesh and saved coefficient. The native tail at the current coupling does not absorb arbitrary inverse powers of an older mesh.

Section 161.4 / A protected sector of the actual mixed numerator

R-II (1.50) identifies the old and new target rows outside the active domain Λ. With all their prefixes, frames and prescribed boundaries kept, the actual nonzero-fluctuation numerator satisfies

$$M_{\mathcal{C}}(U_k'(B,\xi)) = v_{\mathcal{C}}(\xi) = M_{\mathcal{C}}(U_k(\xi)). \tag{R12.22}$$

Consequently an old coefficient whose whole saved dependence factors through these common rows has

$$F(M_{\mathcal{C}}(U_k')) = F(v_{\mathcal{C}}), \qquad \partial_B^{\alpha} F(M_{\mathcal{C}}(U_k')) = 0 \quad (|\alpha| > 0). \tag{R12.23}$$

It factors before the original localization joins through the same selector-supported fiber and the same genuine denominator. On an unchanged retained analytic domain, its saved grade has transport cost

Section 161.4 / A protected sector of the actual mixed numerator

one. For finite raw words the bound retains the actual complex row norm, word length, coefficient count and original mesh power.

When these rows belong to the output record, the carrying function is a function of that output. Its complete conditional-covariance remainder and pushed boundary defect then vanish exactly. Outer physical currents may still differentiate the kernel and denominator; those responses remain.

This result does not assert that every independently localized three-join field preserves these rows off the physical endpoint. It gives a source-preserving representation for the checked row-factor sector. General previously generated coefficients still need AE8/LM47 on their actual mixed selector support and simultaneous complex wall domain.

Section 161.5 / Vacuum comparison beyond an identical spatial weight

For a fixed admitted native target branch and original coordinates, UW12 compares two different spatial action weights. Define

$$w(B) = g^{-2}[A(\Delta\zeta, W_B) - A(\Delta\zeta, U_0)]. \tag{R12.24}$$

The exact nonlinear Wilson energy and the localized curvature response prove

$$|w(B)| \le a|B| + b|B|^2, \qquad |\nabla_B w(B)| \le a + 2b|B|. \tag{R12.25}$$

Here a and b are explicit sums of the original curvature, weight difference and native response kernels. On assigned-scale cells, the kernels keep $v_Q^{1/2}\sigma_Q^{-2}$ and their actual multiplicities. For changed-weight cells whose complete padded records lie in the verified top-scale region, separation D gives

$$a \le C_M\|\Delta\zeta\|_\infty(\varepsilon_{\mathrm{bg}}/g)\sqrt{n}\, e^{-cD}, \qquad b \le C_M\|\Delta\zeta\|_\infty e^{-2cD}. \tag{R12.26}$$

The full mixed-shell sum is not inferred from this top-scale estimate. NM12 checks the relevant original exponents explicitly: the displayed geometry sum has rate $\delta_0/2$, while the nonlinear response has rate $\delta_0/8$. The faster summed majorant does not supply the required slower sum. A smaller-exponent estimate on the unchanged native metric remains to be proved.

Native moments then yield, for small $a + b$,

$$\left|\log\frac{D_1}{D_0} + g^{-2}A(\Delta\zeta, U_0)\right| \le C_M(a+b), \qquad \|K_1 - K_0\|_{\mathrm{TV}} \le C_M(a+b). \tag{R12.27}$$

A verified relative-forest comparison adds its two original exit probabilities. A complete carrier still has the difference $F_1 - F_0$ in addition to the kernel change; its own weighted moments must be controlled.

The raw classical term in R12.27 is not a fluctuation tail. Where its literal native numerator partner is present, the cancellation is

$$e^{s_i}\,\frac{e^{-g^{-2}A(\zeta_i,U_0)}\mathcal{N}_i}{D_i} = e^{s_i - c_i^{\mathrm{coord}} - I_i}\mathcal{N}_i. \tag{R12.28}$$

The independent increment s_i, actual coordinate scalar c_i^{coord}, and complete completion carrier $\mathcal{N}_i$ remain. No normalized-kernel identity estimates their absolute differences.

Complete S94 still requires a uniform comparison of the entire inherited source-zero construction on physical tori:

$$\frac{V_{a,T_*}}{\mathrm{Vol}(T_*)} - \frac{V_{a,T}}{\mathrm{Vol}(T)} \le C_0. \tag{R12.29}$$

Unmatched target maps, mixed charts, lower-shell weighted sums, old complete carriers, independent absolute increments, wrapped records and terminal terms have not all been estimated. Neither current weak-coupling tails nor a local probability comparison supplies this global statement by itself.